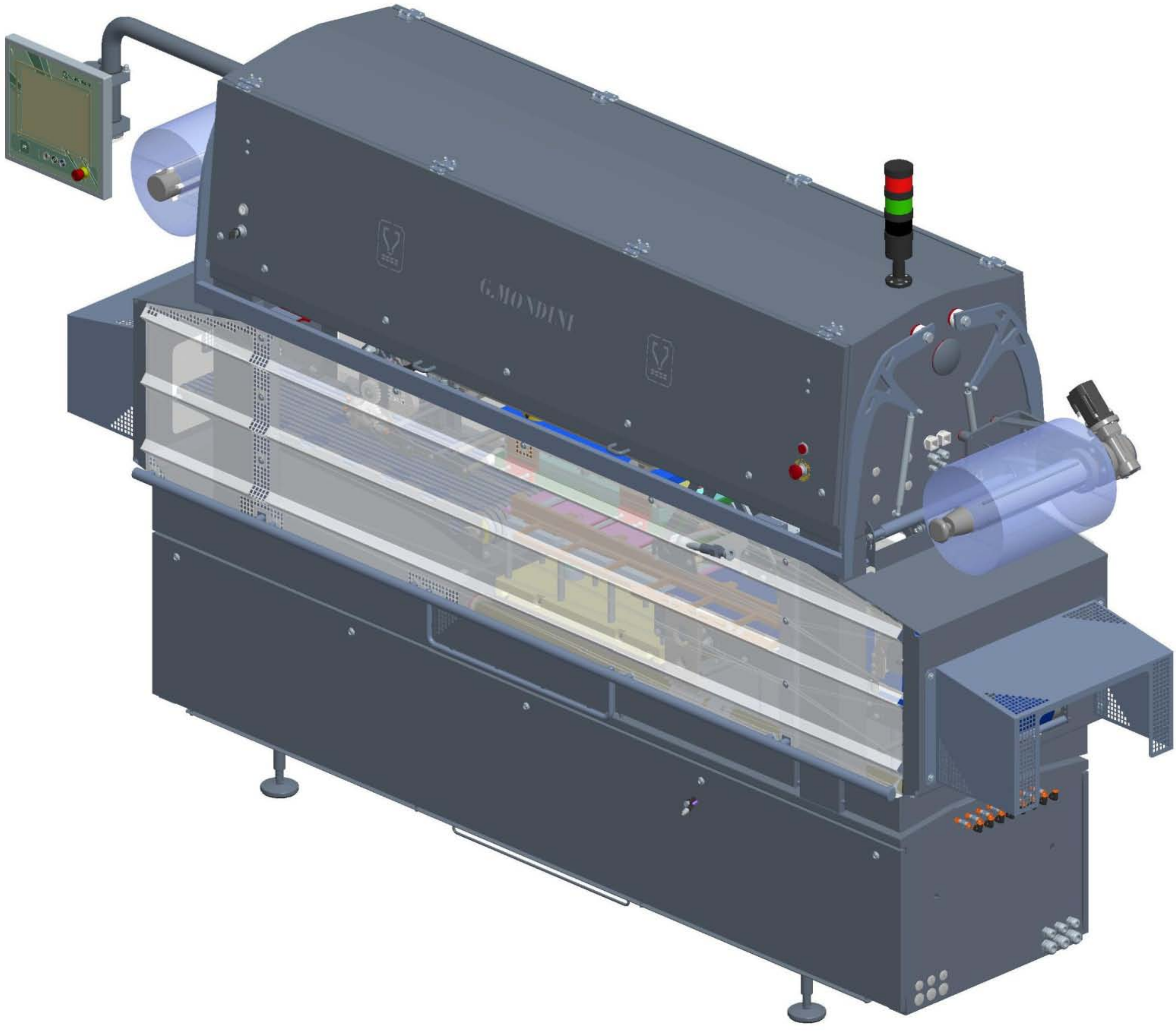
G. MONDINI
DOSATRICI - CONFEZIONATRICI AUTOMATICHE

G. MONDINI S.p.A.
Via Brescia, 5-7
25033 COLOGNE (BS) – ITALIA

CLOSING MACHINE
TRAVE 350 VG



User and maintenance manual

TRANSLATION OF ORIGINAL DOCUMENTS

Construction year: 2018

Serial number: A/18/A-11974

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2 TECHNICAL SPECIFICATIONS

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1 FOREWORD

CONTENTS

1.1 OVERVIEW

This handbook contains all the information needed for correct installation and use, and appropriate maintenance of the line involved.

G.MONDINI S.p.A. insists that this document be read by the persons assigned to the running and the maintenance of the machine, as well as by the persons executing the transport and assembly operations.

This document is the instructions handbook for the **CLOSING MACHINE TRAVE 350 VG** and has been drawn up in conformity with directive 2006/42/EC annex I paragraph 1.7.4. The Use and Maintenance Handbook is to be considered an integral part of the line and is to be kept until the final dismantling. The handbook is to be kept by the person in charge of the system after its installation.

The handbook reflects the technical state at the time of the line marketing and shall not be considered inadequate if later versions are updated on the basis of new experience, furthermore G.MONDINI S.p.A. reserves the right to update production and handbooks with no obligation to update previous production or handbooks.

The drawings and other documents that accompany the system are the property of G.MONDINI S.p.A. that reserves the rights and underlines that these documents are not to be handed over to third parties. It is therefore forbidden to duplicate them in any way, either electronically or mechanically, for any type of use, without first receiving the written permission from G.MONDINI S.p.A.

G.MONDINI S.p.A. shall not be held in any way liable for any direct or indirect damage to persons, goods or animals caused by the use of this documentation and/or the line under any conditions that differ from those foreseen.

1.1.1 Spare parts

It is recommended to use only ORIGINAL SPARE PARTS.

To order spare components send a form like that shown in figure A to G.MONDINI.

The system is marked with a code and a serial number, indicated on the ID plate.

The component description and code, as well as the number of the assembly and the table the component belongs to can be traced in the lists attached to the handbook.

MODULO ORDINAZIONE RICAMBI

Spett.le: **G.MONDINI S.p.A.**

Dosatrici - confezionatrici automatiche
- 25033 COLOGNE (Brescia) ITALIA, Via Brescia 5/7 -
Tel. +39 030 705600 - Fax +39 030 7056250
E-mail: - **info@gmondini.com** - **www.gmondini.com** - **spare_parts@gmondini.com.**

Richiediamo la spedizione dei seguenti ricambi:

Nº MATRICOLA: ____________ **TIPO DI MACCHINA:** ____________

Q.tà	Nº codice pezzo	Nº gruppo	Nº tavola	Descrizione pezzo

DITTA: .... p.I.V.A. Nº:...

Via: .... Nº: ....

Loc.: .... Città: .... C.a.p.:.............. Prov.: ....

INVIO A ½: ☐ Corriere ☐ Corriere espresso ☐ Posta
☐ ....

PAGAMENTO: ☐ Contrassegno ☐ Bonifico bancario (v. ricevuta allegata)
☐ ....

La richiesta è stata compilata da: COGNOME .... NOME ....

In qualità di: ....

Le richieste non saranno evase se mancanti di tutte le indicazioni della ditta richiedente nonché delle generalità complete e di firma del compilatore. La merce viaggia a rischio del committente. Non si accettano restituzioni di materiale in caso di errori di compilazione del modulo.

TIMBRO e FIRMA

Figure A: Spare parts order form.

1.1.2 How to request services

For any type of information or explanations regarding the use, maintenance , installation etc., the G. MONDINI S.p.A. technical department is always at the disposal of the Customer.

With regard to requests, the questions should be set out in clear terms , with reference to this handbook and always indicating the data found on the ID plate of the line.

Any request for after-sales service in the Customer's plant, or questions regarding technical aspects of this document are to be addressed to:

G.MONDINI S.p.A.

Dosatrici – confezionatrici automatiche

25033 COLOGNE (Brescia), ITALIA, Via Brescia 5/7
Tel. +39 030 705600 Fax. +39 030 7056250
E-mail: info@gmondini.com – www.gmondini.com –
spare_parts@gmondini.com

1.1.3 "CE" Declaration of Conformity

"CE" Declaration of Conformity

(in conformity with Machines Directive, annex II letter A)

TRANSLATION OF ORIGINAL DOCUMENTS

The Manufacturer	**G.MONDINI S.p.A.**
Address	Via Brescia, 5-7 – 25033 COLOGNE (BS) – Italy
Tel.	+39-030-705600
Telefax	+39-030-7056250

HEREBY DECLARES THAT THE FOLLOWING MACHINE

Machine type	TRAVE 350 VG C/PS-3+3
Serial number	A/18/A-11974.40 A/18/A-11974.55
Product handled	FOOD PRODUCTS
Type of machine	DOSING AND PACKAGING LINE
G. MONDINI S.p.A. job number	A/18/A-11974
Year of built	2018

COMPLIES WITH THE PROVISIONS OF LAWS IMPLEMENTING MACHINERY DIRECTIVE 2006/42/EC AND SUBSEQUENT AMENDMENTS

The manufacture declares that the machine conforms to these other European directives (where applicable): 2014/35/UE, 2014/30/UE.

The manufacture also declares that the machine conforms to the following other harmonized standards (where applicable): UNI EN ISO 12100:2010, UNI EN ISO 13849-1:2016, EN 60204-1.

The manufacturer strictly prohibits the use of the above machine/fixture in any manner other than that indicated in the Instructions for Use Handbook.

Representative: Patrizio Chiari

Office: Responsible of production **Signature** ....

Cologne, dated 19/12/2018

The technical documentation has been prepared by **G. MONDINI S.p.A.,** a legal entity with head office at 5/7 Via Brescia, 25033 Cologne, Brescia, Italy.

1.1.4 System marking

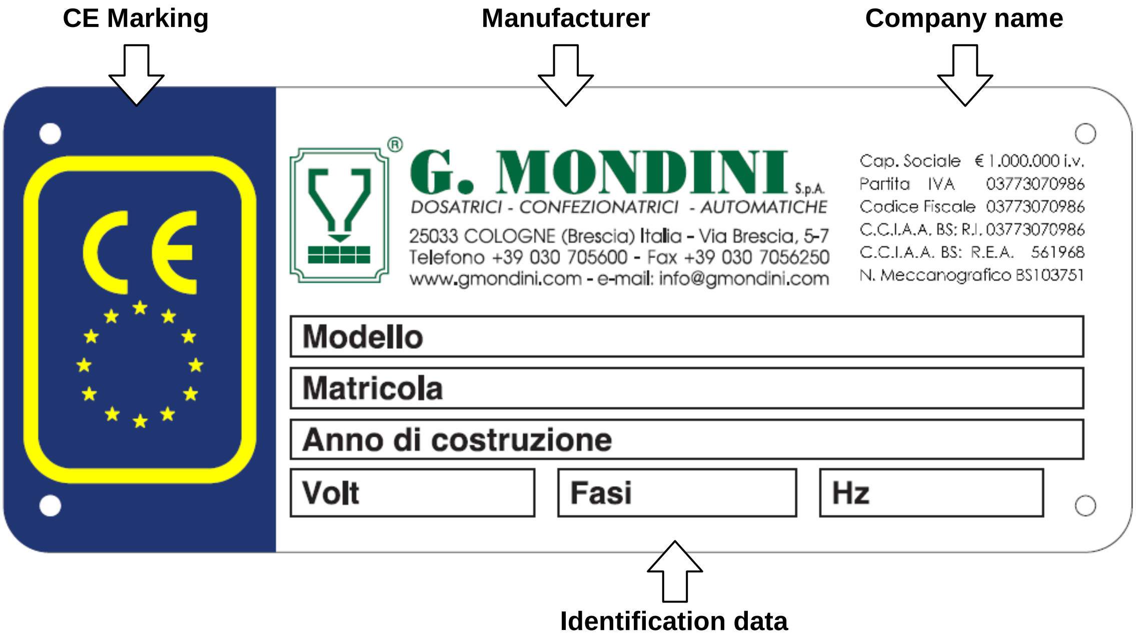


Figure B: CE Plate.

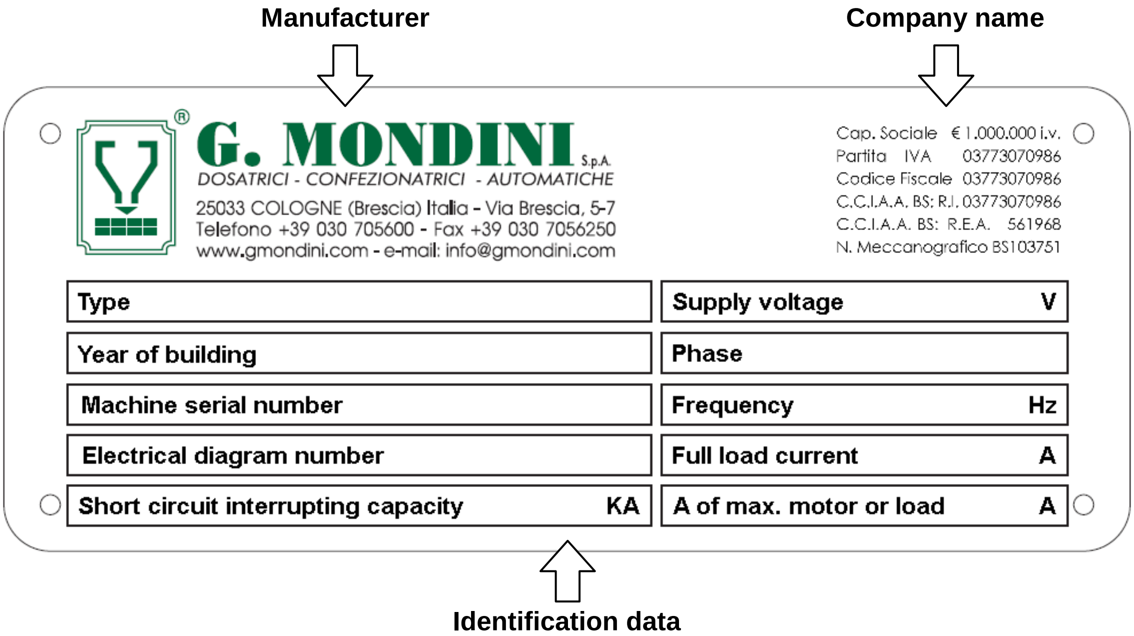


Figure B: USA Plate.

1.1.5 Normative references

DIRECTIVE / STANDARD	EDITION	TITLE
2006/42/CE	DEC. 2009	Machinery Directive
2014/68/EU "PED"	MAY 2014	Approximation of the laws of the Member States concerning pressure equipment.
2014/34/EU "ATEX"	FEB. 2014	Approximation of the laws of the Member States concerning equipment and protective systems intended for use in potentially explosive atmospheres.
1999/92/CE	JAN. 2000	Minimum requirements for improving the safety and health protection of workers potentially at risk from explosive atmospheres.
2014/30/EU	APRIL 2016	Concepts regarding electromagnetic compatibility issues.
2014/35/EU	APRIL 2016	Low-voltage directive
UNI EN 349	NOV. 2008	Safety of machinery. Minimum gaps to avoid crushing of parts of the human body.
UNI EN 415-3	FEB. 2010	Safety of packaging machines. Part 3: Molders, fillers and sealers. Security requirements.
UNI EN 415-10	JAN. 2014	Safety of packaging machines. Part 10: General requirements.
UNI EN 547-1	MAR. 2009	Safety of machinery. Human body measurements. Part 1: Principles for determining the dimensions required for openings for whole body access into machinery.
UNI EN 547-2	MAR. 2009	Safety of machinery. Human body measurements. Part 2: Principles for determining the dimensions required for access openings.
UNI EN 574	NOV. 2008	Safety of machinery. A two-hand control device. Functional aspects. Principles for design.
UNI EN 842	MAR. 2009	Safety of machinery. Visual danger signals. General requirements, design and testing.
UNI EN 981	MAR. 2009	Safety of machinery. System of auditory and visual danger and information signals.
UNI EN 1037	SEPT. 2008	Safety of machinery. Prevention of unexpected start-up.

DIRECTIVE / STANDARD	EDITION	TITLE
UNI EN 1672-2	MAR. 2009	Machines for the foods industry. Basic concepts. Part 2: hygiene requirements.
UNI EN ISO 4413	FEB. 2012	Hydraulic fluid power. General rules and safety requirements for systems and their components.
UNI EN ISO 4414	FEB. 2012	Pneumatic fluid power. General rules and safety requirements for systems and their components.
UNI EN ISO 11161	JULY 2010	Safety of machinery. Integrated manufacturing systems. Basic requirements.
UNI EN ISO 11202	JUNE 2010	Acoustics. Noise issued by machinery and equipment. Determination of emission sound pressure levels at a work station and at other specified positions applying approximate environmental corrections.
UNI EN ISO 12100	NOV. 2010	Safety of machinery. General principles for design. Risk assessment and risk reduction.
UNI EN ISO 13849-1	JUNE 2016	Safety of machinery. Safety-related parts of control systems Part 1: General principles for design.
UNI EN ISO 13849-2	NOV. 2013	Safety of machinery. Safety-related parts of control systems Part 2: Validation.
UNI EN ISO 13850	MAY 2016	Safety of machinery. Emergency stop devices. Functional aspects. Principles for design.
UNI EN ISO 13855	NOV. 2010	Safety of machinery. Positioning of protective equipment in respect of approach speeds of parts of the human body.
UNI EN ISO 13857	MAY 2008	Safety of machinery. Safety distances to prevent hazard zones being reached by upper and lower limbs.
UNI EN ISO 14119	NOV. 2013	Safety of machinery. Interlocking devices associated with guards. Principles for design and selection
UNI EN ISO 14120	JUNE 2016	Machinery safety systems. Guards. General requirements for the design and engineering of fixed and mobile guard systems.
CEI EN 60204-1	MAR. 2018	Safety of machinery. Electrical equipment of machines Part 1: General rules.

1.2 HANDBOOK STRUCTURE

This handbook is divided into 9 chapters.

CHAPTER 1 – GENERAL INFORMATION

This chapter contains the general descriptions regarding the structure of the handbook, the topics dealt with, the symbols, guarantee and responsibilities.

CHAPTER 2 – DESCRIPTION

This chapter contains the description of the system operating principles, the general technical data and the description of the mechanical, electrical and fluidic assemblies that it comprises.

CHAPTER 3 – SAFETY

This chapter contains a description regarding the standards, working environment conditions, ergonomics, accident prevention devices used, residual risks, warning plates on the system and the Manufacturer's Declaration of Conformity .

CHAPTER 4 – INSTALLATION

This chapter contains the instructions for correct packing, handling, transport, unloading, installation in the user's plant, connections to the plant energy sources, checks, controls and any adjustments to be made before the start-up.

CHAPTER 5 – CONTROL SYSTEM

This chapter contains the description of all the controls available for the operator.

CHAPTER 6 –OPERATION

This chapter is addressed to those running the system and contains the instructions for the start-up, use and stopping of the system in the different operating cycles. It also describes all the video pages displayed on the control panel.

CHAPTER 7 –DIAGNOSTICS

This chapter describes all the alarms of the system, the causes that generated them and the procedures to solve them .

CHAPTER 8 – MAINTENANCE

This chapter is for the maintenance technicians and contains the machine maintenance schedule. It includes the warnings, precautions and instructions to carry out the maintenance operations correctly on the system.

CHAPTER 9 – DISPOSAL

This chapter describes how to dispose of the system and how to dismantle it.

1.3 SAFETY STANDARDS CONTAINED IN THIS HANDBOOK

The instructions, indications, standards and safety notes described in the various chapters of the handbook have the purpose of defining how to behave and the obligations to be respected when carrying out the various activities. They form the methods of use for the line, so as to operate under safe conditions for the personnel, the fixtures and the surrounding environment. The safety standards contained are directed to all the skilled persons who are authorised and assigned to carry out the various operations that include:

- transport
- installation
- operation
- use
- management
- maintenance
- cleaning
- putting out of service and dismantling

that constitute the methods of use foreseen for the system.

IMPORTANT NOTE

The system must not be modified in any way and the safety guards must not be removed or disactivated without first informing the Manufacturer. If these instructions are not complied with, the Manufacturer DECLINES ALL RESPONSIBILITY for any situations that might occur. The Manufacturer cannot be held liable for:

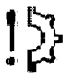
A) THE USER'S SAFETY.
B) PROPER OPERATION of the machine supplied.

- IMPORTANT: If the system supplied is to be linked to an existing installation, the costumer must - unless agreed otherwise - provide adequate protection for the area where the two systems are linked. Mondini S.p.a. declines all liability if damage or injury is caused because this protection has not been provided.

1.4 SYMBOLS USED IN THIS HANDBOOK

This handbook uses certain symbols to call the attention of the reader and point out some particularly important features.

The table below contains a list of the different symbols used in the handbook and describes their meaning.

SYMBOL	MEANING	NOTES
	Danger	Indicates a hazard with accident risk, even death, for the user. Pay very careful attention to the texts indicated by this symbol.
	Warning	A warning of possible downgrading or damage to the line, and/or the fixtures. Pay attention to the texts indicated by this symbol.
	Warning Note	A warning or note regarding key functions or useful information. Pay attention to the texts indicated by this symbol.
	Additional information	Texts that contain additional information are indicated by this symbol. This information does not relate directly to the description of an operation or to the development of a procedure.. It may give reference to other supplementary documentation such as attached instructions for use handbooks, technical documentation or other sections of this handbook.
	Avoid damage to material	Indicates there is a high risk of damaging a part, for example using a wrong tool or assembling following an incorrect procedure
	Special tool	Indicates that for this operation a special tool or fixture is necessary.
	Visual check	Informs the reader that a visual check is to be made. This symbol will also be found in the instructions for use. The user is told to read a measure value, to check an indication, etc.
	Sound check	Informs the reader that a sound check is to be made. This symbol will also be found in the instructions for use. The user is told to listen to an operating noise.
	See the maintenance charts	Indicates that a special maintenance chart is to be consulted.

1.5 RESPONSIBILITIES OF THE MANUFACTURER

G. MONDINI S.p.A. shall not be held liable for situations deriving from the incorrect or improper use of the system described herein or any damage caused by the use of spare parts other than those recommended, by maintenance operations not carried out correctly or by tampering with circuits, components and/or system software.

The responsibility to ensure that the safety instructions indicated in this handbook are applied is assigned to the technical supervisor responsible for the foreseen activities on the line. He shall ascertain that the operatives authorised to carry out the required activities are qualified , that they respect and are aware of the requirements contained in this handbook and the general safety standards applied to the system.

Not observing the safety standards could cause injuries to the personnel and damage to the fixtures.

1.6 GUARANTEE

For the construction of the system G. MONDINI S.p.A. has used materials deemed most suitable in their final judgement.

G. MONDINI S.p.A. guarantees that the system is free of machining flaws and ensures the quality of the material for a period of time and in accordance with the conditions agreed upon when drawing up the contract with the customer.

1.7 LINE MANAGEMENT

The system management is only allowed to authorised operators who have received sufficient instruction, or who have appropriate technical experience.

The operators assigned to the running and the maintenance of the system are to be aware that the knowledge and application of the safety standards is an integrating part of their job.

Operators who are not assigned to activities on the system are not to have access to the operating area and/or the control panels.

Before starting up the system carry out these operations:

- read this handbook carefully
- know which protections and emergency stop devices are installed on the line, where they are located and how they function.

It is forbidden to remove, even only partially, protections and safety devices installed to safeguard the personnel in the system hazardous zones.

The same regulation applies to the warning plates.

It is strictly forbidden to open the doors of the electric cabinet when the system is running and immediately after it has stopped.

The protections and safety devices are to be kept in perfect order to ensure the correct functioning. In the case of malfunctioning or failure on these devices, they are to be immediately repaired or replaced.

The use of commercial components that are not those specified for safety devices and protections could cause malfunctioning or the generation of hazardous situations for the operators working on the line.

1.7.1 Plates on the system

PLATE 1	
	Use gloves to protect from cuts and burns.
PLATE 2	
	Wear safety shoes.
PLATE 3	
	See the use and maintenance handbook.
PLATE 4	
	**Danger!** Mechanical parts in motion.
PLATE 5	
	**Danger!** Hazardous voltages present; Danger of fulguration.
PLATE 6	
	**Warning!** In this zone there are parts that work at high temperatures.

PLATE 7	
	**Warning!** Danger of limbs being crushed.
PLATE 8	
	**Warning!** In this zone of the system parts are operating that could cause deep cuts and serious amputations.
PLATE 9	
	**Warning!** This indication is usually placed near the closing zone of the protective casing, where its closure affects the resumption of production.
PLATE 10	
	**Warning!** Constant hazard pay attention when handling.
PLATE 11	
	Do not remove the safety devices and protections.
PLATE 12	
	Do not operate on moving parts.
PLATE 13	
	This symbol indicates the direction of rotation of the part .

1.7.2 Glossary and symbols

Some terminology and symbols are used in this handbook to call the attention of the reader and to emphasise certain aspects of particular importance .

The tables that follow list them and describe their significance.

Table 1 - Glossary

Term	Description
Emergency stop function	Function to: - prevent, upon occurrence, hazards for persons, damage to machines or machining in progress, or otherwise reduce them. - be activated by a single human action when the normal stop function is inadequate for the purpose. In accordance with this standard, the hazards are those that could arise from: - operation irregularities (machine fault, inadequate processing material, human error, …) - normal operation Note – *Functions such as inversion, limiting movements, deviation, shielding, braking, sectioning, etc, may be part of emergency stop functions. This standard (EN 13850) does not cover these functions.* (Item 3.1 of EN 13850)
Machine actuator	A power mechanism used to start the movement of a machine. (Item 3.3 of EN 13850)
Control circuits	Circuit used to control the machine operation and to protect the power circuits.
Manual control (actuator)	The component of the control device that, when activated, activates the actual control device, and is designed for actuation by a person (see item 4.4.1 of EN 13850). (Item 3.2 of EN 13850)
Customer/ Purchaser	The person who orders and/or purchases.
Machines directive	The machines Directive is the ruling indicated with the Executive Order of the President of the Republic 459/96.
Emergency stop device	Together with the components designed to execute the emergency stop (see fig. 2 of EN 13850, that shows the parts of a machine to which these components may belong). (Item 3.2 of EN 13850)
Enable device (control	Additional control device activated manually and used together with a start command, to permit the machine to operate permanently when activated. (Item 3.26.2 of EN 12100)

Term	Description
Control device	Component of the emergency stop device that generates an emergency stop signal when the associated manual control (actuator) is activated. (Item 3.2 of EN 13850)
Interlock device (interlock)	Mechanical, electrical or other type of device that has the purpose of preventing machine parts operating under certain conditions (usually before the guard is closed). (Item 3.26.1 of EN 12100)
Safety device	Device (not a guard) that eliminates or reduces the risk, either alone or associated to a guard. (Item 3.26 of EN 12100)
Limiting device	Device that prevents the machine or its parts from exceeding the set limit (distance, pressure, etc.). (Item 3.26.8 of EN 12100)
Bill of Materials	List of the components that are part of the mechanical assemblies, fluidic or electrical systems, indicated with the quantity, code and name of supplier.
Manufacturer	Manufacturer of the machine
Supplier	Body (for example, manufacturer, installer agent, system integrator) that supplies the equipment or associated services of the line (the user may also act as his/her own manufacturer).
Automatic operation	Mode in which the entire system executes its operations autonomously, with gates and barriers closed and inserted in the safety circuits.
Manual operation	Operating mode that cuts out automatic running and that allows manual handling activities under the control of the operator.
Installation	Installation is the mechanical and electrical integration of the machine in a production system, in conformity with the Directive safety requirements.
Instructions for use	Safety measures that consist of a set of information, such as texts, words, signs, symbols or diagrams that are used either separately or in combination, to convey the instructions to the user. They are intended for professional and/or non professional users. Note – *Item 6 of EN 12100 deals with instructions for use.* (Item 3.21 of EN 12100)

Term	Description
Machinery/ Machine	A group of parts or components, of which at least one is movable, connected together with appropriate actuators, control and power circuits etc. of the machine, integrally connected for a specific application, in particular for conversion, treatment, handling or packaging of a material. The term "machinery" also covers a group of machines that, to obtain the same result, are arranged and controlled so as to have an integrated operation. Appendix A of EN 12100 contains a general schematic diagram of a machine. (Item 3.1 of EN 12100)
Maintenance and Repair	Maintenance and repair operations are periodical checking activities and/or replacement of mechanical, electrical parts, software or machine components that serve to identify the cause of a fault, that terminate with the machine being returned to the project functioning condition.
Marking	Signs or writing that identify the type of component or equipment, applied by the manufacturer of the component or equipment.
Commissioning or Putting into service	Commissioning is the functional checking activity on the installed system.
Safety measures	A means that eliminates or reduces a hazard.
Operator	Person or persons assigned to the installation, operation, regulation, maintenance , cleaning, repairing or transport of the machine (EN 12100).
Exposed person	Any person that is completely or partially in a hazardous zone (RES 1.1.1 – C of 2006/42/EC).
Skilled person	A person with sufficient technical knowledge or experience to be able to avoid hazards that could be present.
Protections	Safety measures that consist in the use of specific technical measures called protections (guards, safety devices) to protect persons against hazards that cannot be reasonably eliminated or sufficiently limited by design. Note – *Item 5 of EN 12100 deals with protections.* (Item 3.20 of EN 12100)
Contact person	Person responsible for the running of certain operations or assessments that could arise during work or maintenance .
Redundancy	Application of more than one device or system, or part of a device or system, so as to ensure that in the case of a failure in the functioning of one of them, another is available to execute that function.

Term	Description
Guard	A part of a machine used in a specific manner to give protection by means of a physical barrier. According to its construction, a guard may be called hood, cover, shield, screen, door, fencing, enclosure, etc. Note 1 – *A guard may act:* - *alone; it is therefore effective only when it is closed,* - *associated to an interlock device with or without locking of the guard; in this case the protection is insured whatever the position of the guard.* Note 2 – *“Closed” means, for the fixed guard, “kept in position”.* (Item 3.25 of EN 12100)
Fixed guard	Guard kept in position (closed): - either permanently (by welding, etc.), - or by fastening elements (screws, bolts, etc.) that make it impossible to remove/open it without the use of tools. (Item 3.25.1 of EN 12100)
Movable guard	Guard usually mechanically connected to the frame of the machine or to a fixed element nearby (for example by hinges or guides), and that can be opened without the use of tools. (Item 3.25.2 of EN 12100)
Interlocked guard	Guard associated to an interlock device (see item 3.26.1 of EN 12100), so that: - the hazardous functions of the machine “involved” with the guard cannot be carried out until the guard has been closed, - if the guard is opened while machine hazardous operations are running, the order to stop is given, - the closing of the guard allows the machine to run the hazardous operations with which the guard is “involved”, but does not give the start commands. (Item 3.25.4 of EN 12100)
Risk	Combination of probabilities and seriousness of possible injuries or harm to health in a hazardous situation. (Item 3.11 of EN 12100)
Machine safety	Capability of a machine to carry out its functions, to be transported, installed, adjusted, serviced, dismantled and removed under the foreseen use conditions (see Item 3.22 of EN 12100) specified in the instructions handbook (and, in some cases, in a certain time indicated in the handbook) without causing injuries or harm to health. (Item 3.19 of EN 12100)

Term	Description
Dismantling	Dismantling is the demolition and disposal of the parts constituting the machine.
Protected space	Protected space is the area limited off by the protection barriers and assigned for the installation of the machine.
Environment temperature	Temperature of the air or other agent where the equipment is used.
Transport	The series of operations to transfer the work centre from the assembly site of the manufacturer to the final work site of the customer .
Foreseen use of a machine	Use for which a machine has been designed, in conformity with the indications supplied by the manufacturer, or that is considered standard in relation to its design, construction and function. Foreseen use implies also the observance of the technical instructions contained in the instructions handbook (see Item 6.5 of EN 12100), and taking into consideration the incorrect use that it is feasible to foresee. Note – R*egarding incorrect use in the assessment of risks, these types of behaviour should be given particular consideration*: - *incorrect behaviour that results from normal negligence and not a deliberate intention to use the machine improperly,* - *the instinctive reaction of a person during use, in the case of malfunctioning, accidents, failures, etc.* - *behaviour that derives from "the line of least resistance" when performing a task,* - *probable behaviour of some persons, such as children or disabled persons for certain machines (especially those not for professional use).* *See also* Item *3.21 of EN 12100.* (Item 3.22 of EN 12100)
Incorrect or improper use	Use of the machine out of the limits specified in the technical specifications .
User	Body that uses the machine and the associated equipment.
Hazardous zone	Any zone inside and/or near to a machine in which a person is exposed to the risk of injuries or harm to health. Note – *The hazard that causes the risk considered in this definition*: - *permanent presence during the foreseen use of the machine (motion of hazardous moving parts, electric arc during welding, etc.), or* - *that can unexpectedly occur (sudden /unexpected start, etc.)* (Item 3.10 of EN 12100)

2 TECHNICAL FEATURES

CONTENTS

2.1 GENERAL FEATURES

The **CLOSING MACHINE TRAVE 350 VG** has been designed and engineered to seal food packaging containers and trays with an appropriate film.

The sealing process is performed via the following sequence:

1. In-feed of the trays onto the line.
2. Positioning of the trays onto the servomotor-driven inlet grouping belt.
3. Pick-up of the trays by the pusher unit assembly with subsequent placing of the trays into the bottom half tool.
4. Closing of the tool and subsequent sealing.
5. Opening of the tool.
6. Pick-up of the sealed trays by the pusher unit assembly and placing of the containers onto the out-feed belt.

Preparation of the lids for the subsequent sealing cycle.

ADDITIONAL INFORMATION

For the correct operation-sequence that the operator is required to perform when working with the machine, please furthermore consult chapter 6 “MACHINE OPERATIONS” herein.

2.2 MAKE-UP OF THE MACHINE

The **CLOSING MACHINE TRAVE 350 VG** is composed of the following groups of unit assemblies:

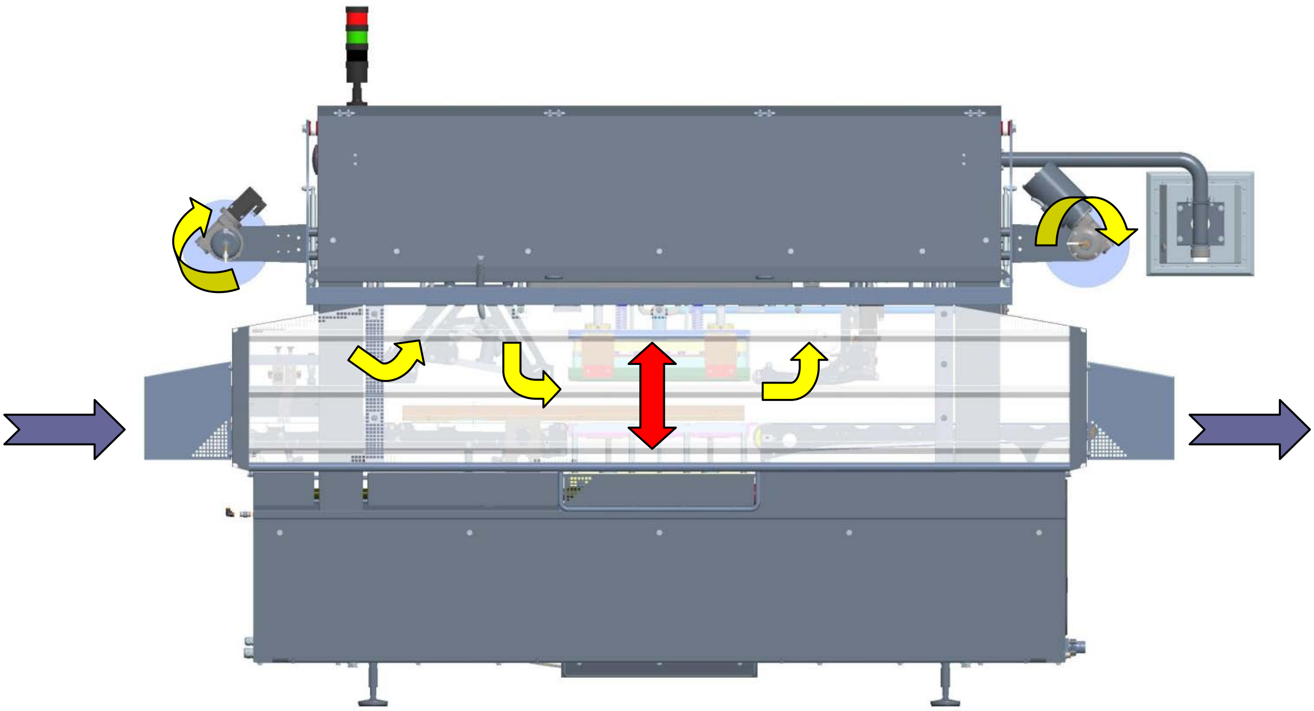
Unit Assembly	Description at section
BASE BED	2.5.1
TOOL CRANK-HANDLE	2.5.2
PUSHER UNIT ASSEMBLY	2.5.3
INLET GROUPING BELT	2.5.4
TIMING DEVICE	2.5.5
EXIT BELT	2.5.6
FRONT MOBILE ROLLER	2.5.7
FILM UNWINDER ROLLER	2.5.8
FILM DRIVE AND PULL ROLLERS AND TRANSMISSIONS	2.5.9
WINDER SHAFT	2.5.10
TOP TOOL PLATE	2.5.11
MOBILE PLATE	2.5.12
COVER CASING	2.5.13
VACUUM CIRCUIT	2.5.14
LUBRICATING SYSTEM	2.5.15
PNEUMATIC SYSTEM	2.5.16
TOOL CARRIER TROLLEY	2.5.17
TOOL	2.5.18

2.3 TECHNICAL FEATURES

The machine's main technical features are listed and specified in the following table:

General features	
Approximate overall/outer dimensions length X width X height	3800 x 965 x 2385
Total machine weight	2850 kg
Vibrations	Not detectable / do NOT constitute a health hazard for machine operators
Noise level	< a 75 dB(A)
Operating temperature	Between 0 and 45°C
Environmental/worksite humidity	Must not be higher than 90%
Production	
Cycle time	4 seconds
Production	2700 trays/h
Production at 80%	2160 trays/h
Pneumatic system	
Hook-up pressure	Max 8 bar
Operating pressure	6 bar
Air consumption	150 Nl/min
connection fitting Ø	10/12 mm
Electrical system	
Power supply voltage	415 Volt – 3 phases + ground – 50 Hz
Auxiliary voltage	24 Volt DC
Socket power supply voltage	220 Volt DC
Total rated power	25 KVA

2.4 FLOW AND MOTION OF THE MACHINE ELEMENTS



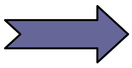
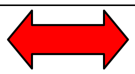
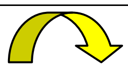
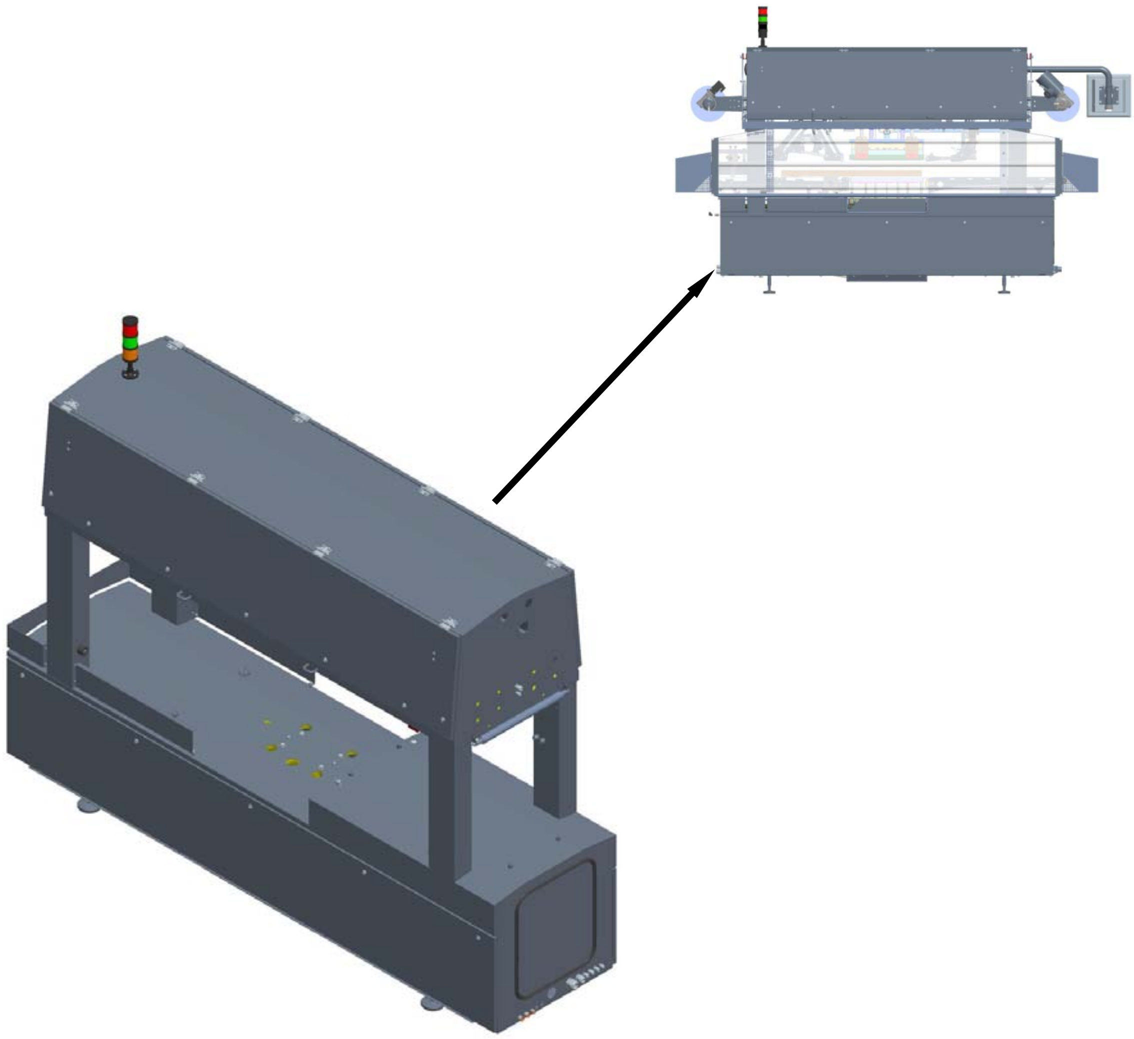
	Tray flow direction
	Tool motion
	Film flow

Figure 2.1 – Element-flow on the TRAVE 350 VG Tray Sealer

2.5 DESCRIPTION OF THE MACHINE UNIT ASSEMBLIES

Following is a description of the various groups of unit assemblies that make up the machine.

2.5.1 Base bed



Function:

Support bed of the overall assembly groups constituting the tray sealer, featuring appropriate, adjustable feet.

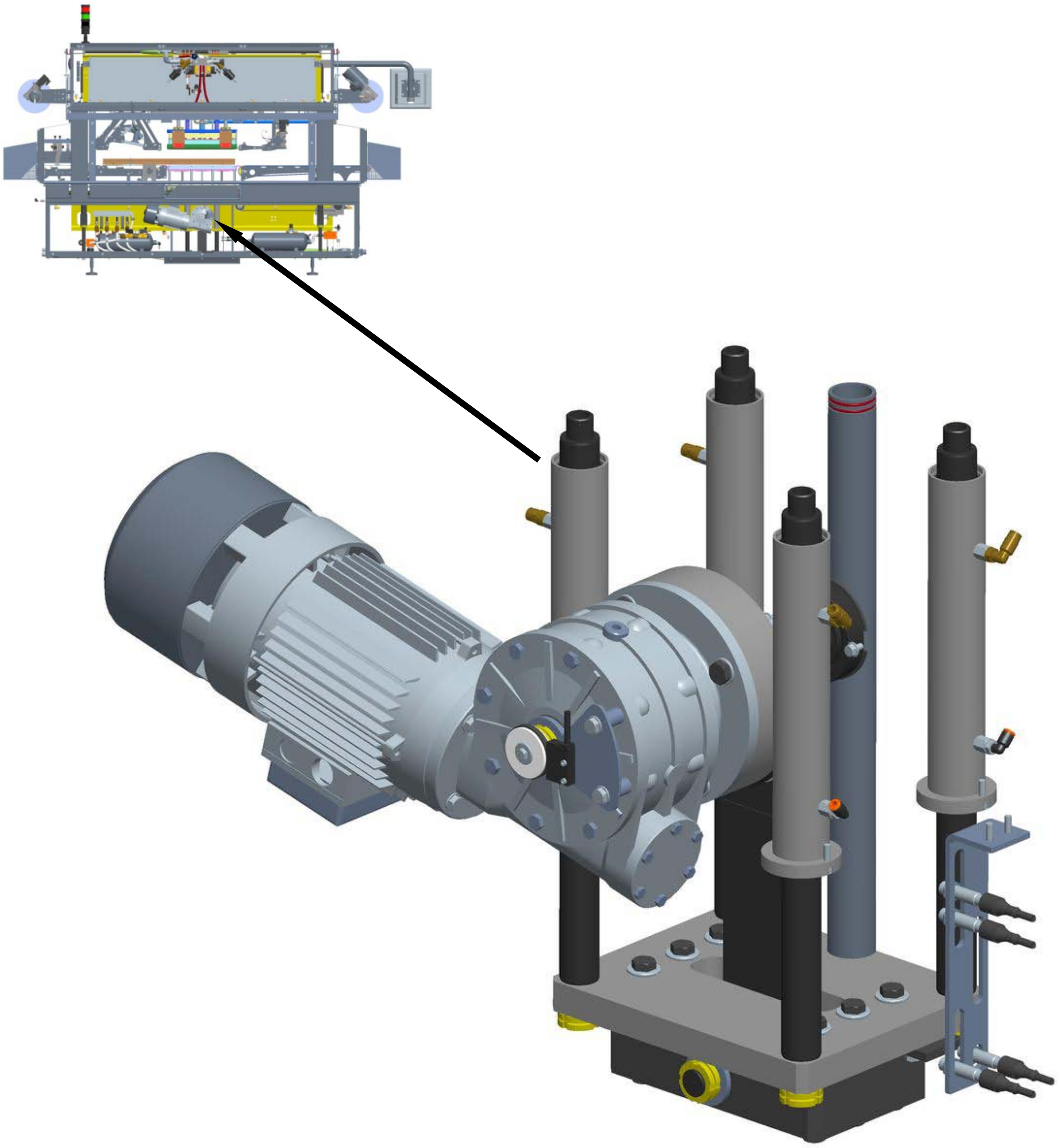
Make-up:

The machine bed in question is composed of a welded steel structure.

The bottom portion of the structure houses the machine’s pneumatic system, as well as the pusher unit assembly and bottom half tool motion and movement elements. The top part of the structure houses the inlet grouping belt assembly group, the tool, the pusher unit assembly and the exit belt.

The whole of the top area is covered by safety guards and covers that are not fitted with safety micro switches because the internal elements must not be accessed for no reason whatsoever, except solely by fully qualified and specialised personnel, for maintenance or adjustment operations.

2.5.2 Tool crank-handle



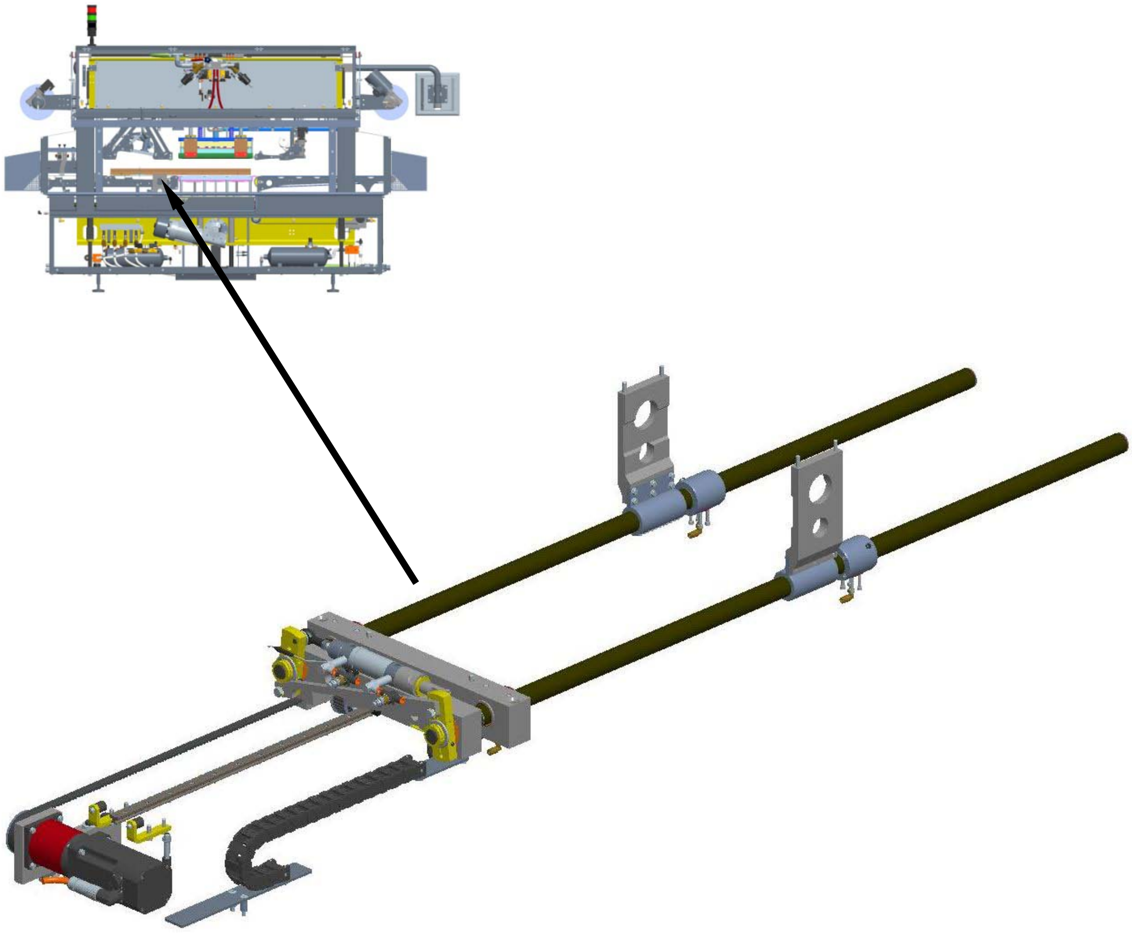
Function:

Motion control of the bottom half tool for tray sealing processes.

Make-up:

This unit assembly is composed of a frame housing 4 columns that, via an appropriate drive system, enable up and down travel of the bottom half tool.

2.5.3 Pusher unit assembly



Function:

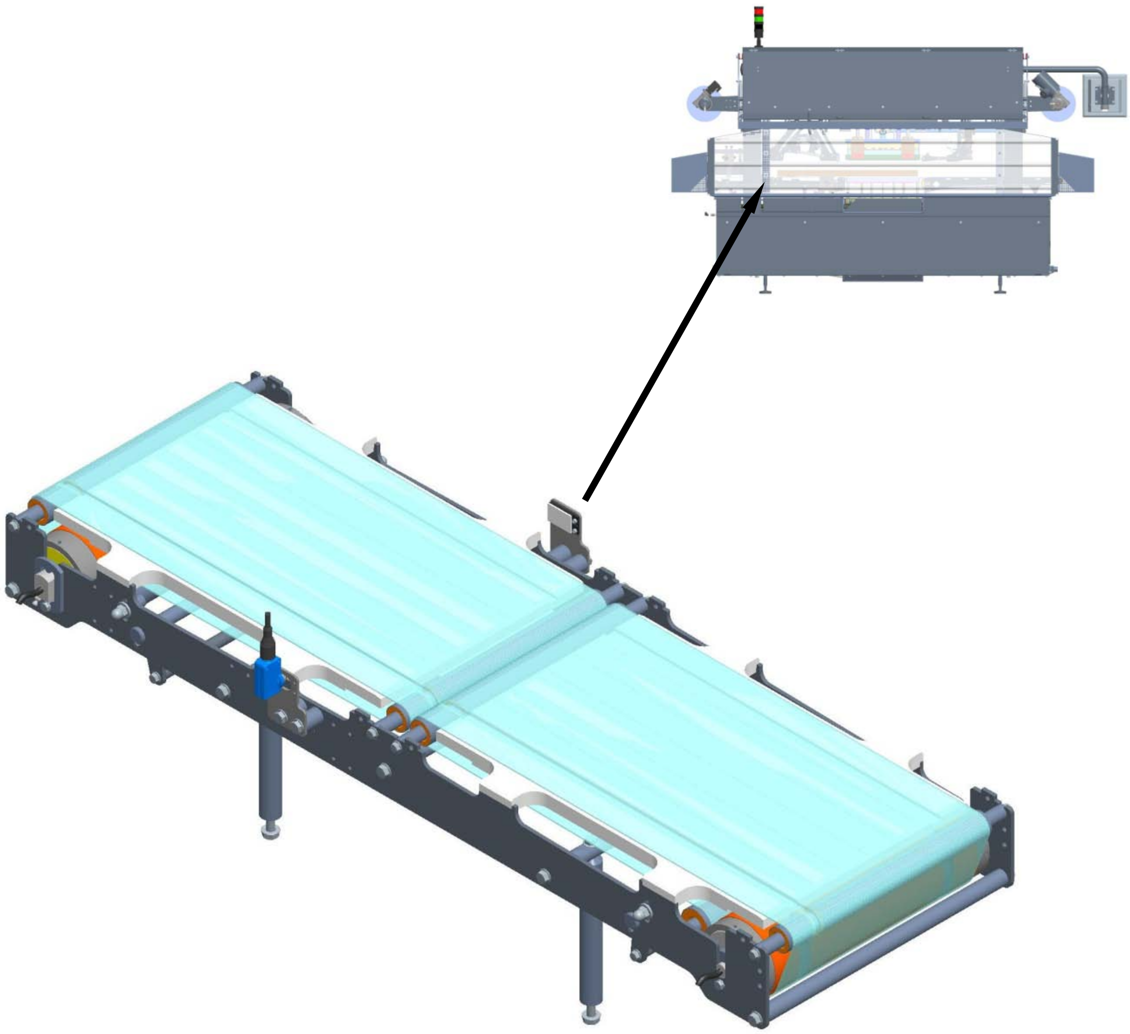
Forward carriage of the trays from the inlet grouping belt to inside the tool, and simultaneous parallel carriage of the trays from inside the tool to the exit belt.

Make-up:

This unit assembly is made up of two horizontal-sliding columns that the pusher units are assembled onto and that physically determine container motion.

It is fulcrum-locked at its bottom end to the machine motorisation setup (that is fixed laterally to the tool carrier) via a specific fork.

2.5.4 Inlet grouping belt



Function:

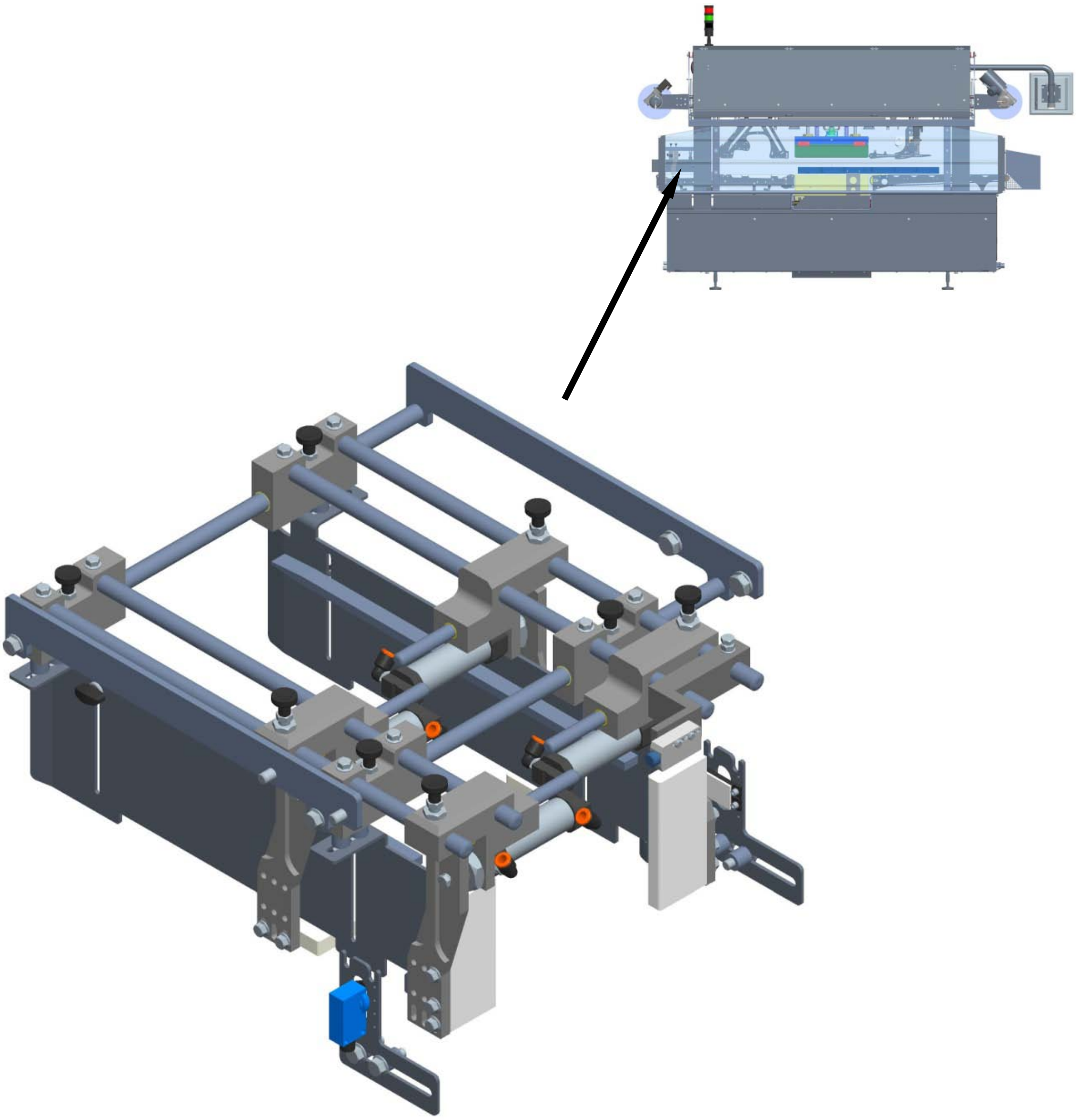
Uploading of the trays and subsequent forward grouping motion towards the pusher unit assembly.

Make-up:

This unit assembly is composed of two belt sections, every belt is controlled by a motor-drum.

Based on the type of process cycle set, the limit switch detects the required number of trays to be fed into the tool and then sends them on to the subsequent belt section that, via a stepper motion, arranges the trays into the right position thereby ensuring correct pick-up by the pusher unit assembly.

2.5.5 Timing device



Function:

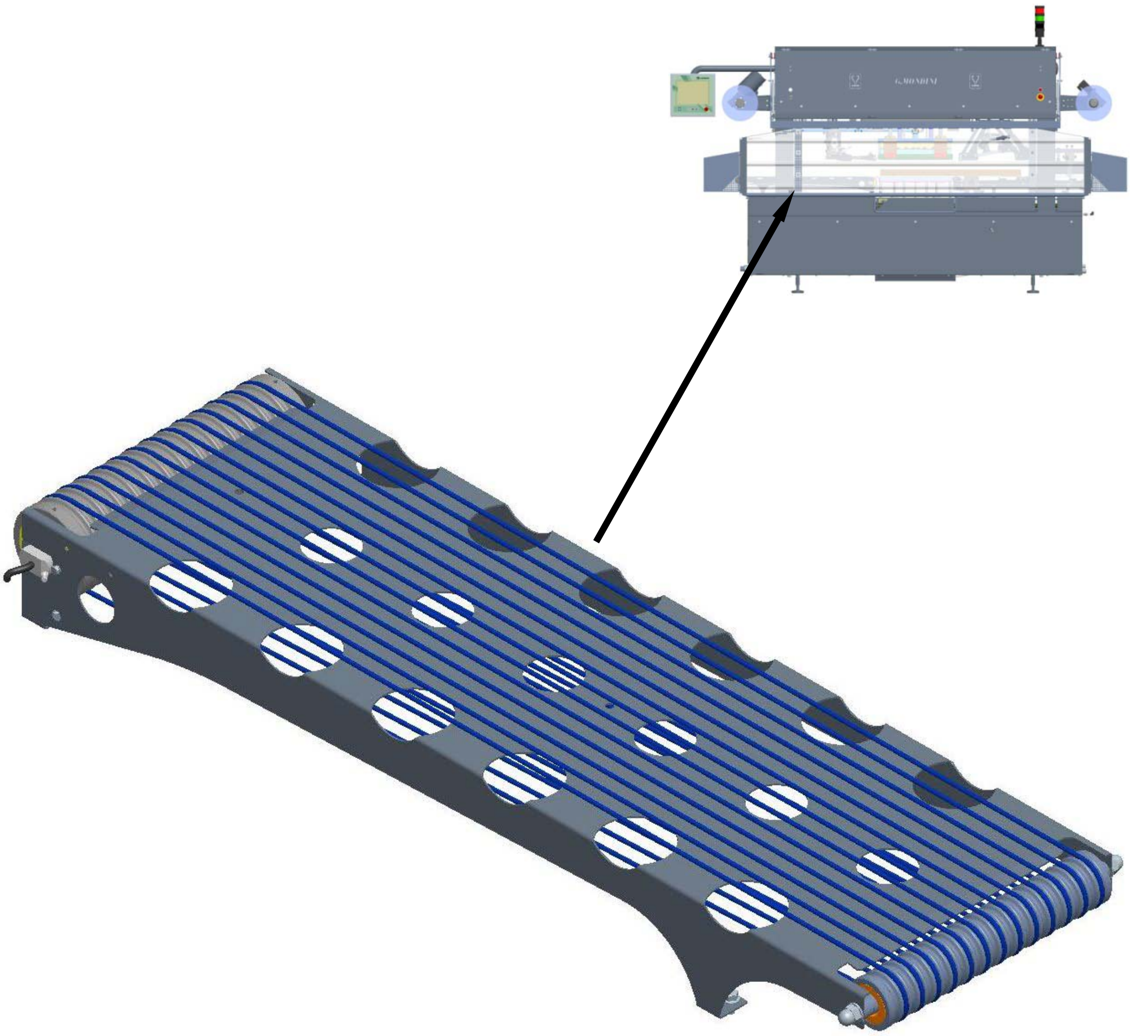
It provides the correct container positioning so as to have the correct rhythm during transfer to the servo-actuated incoming belt.

Composition:

The assembly group in question is made up of four devices.

Combined with the container detection limit switch, this system ensures transfer of the trays to the second section of the belt.

2.5.6 Exit belt



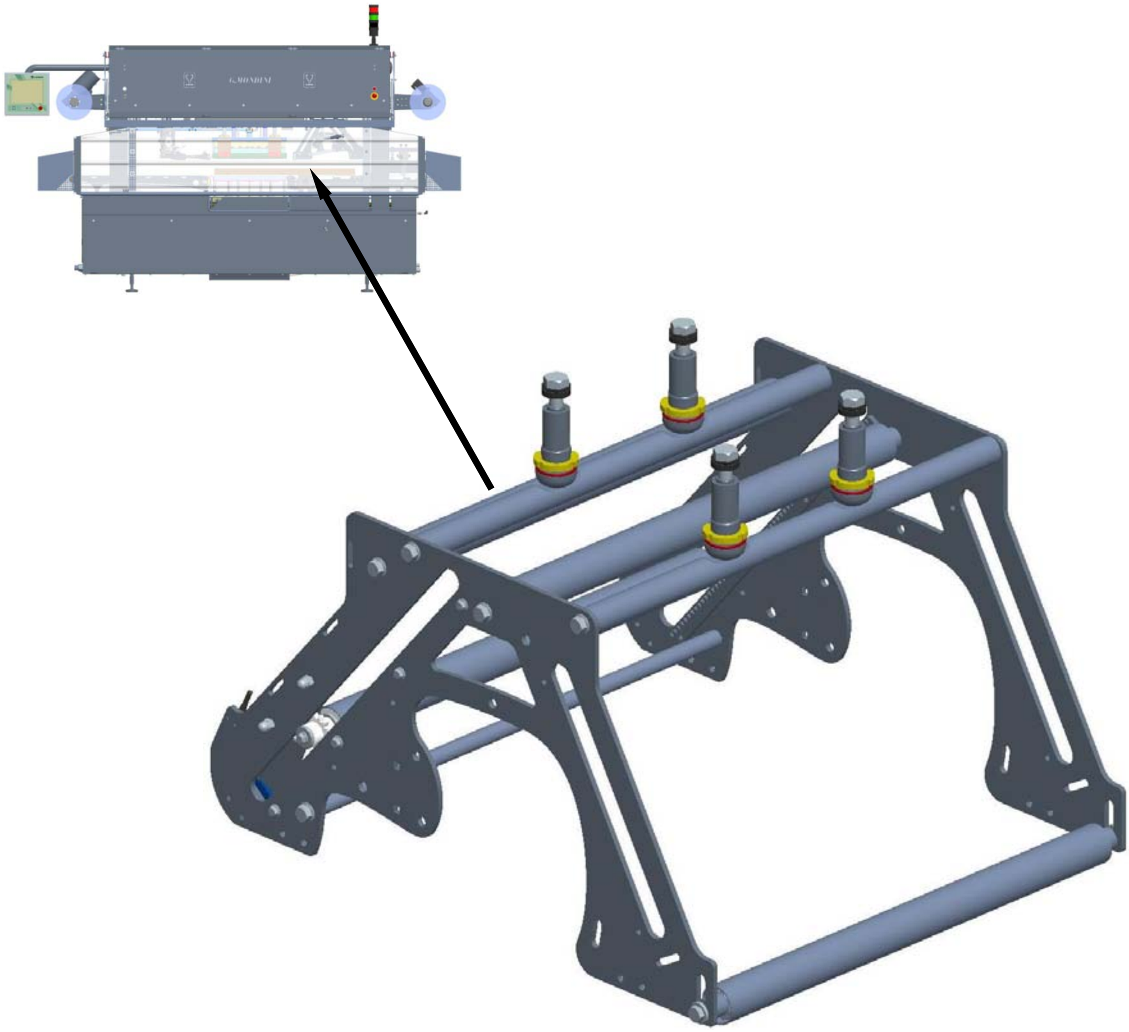
Function:

Carriage of the sealed trays towards the machine exit.

Make-up:

This unit assembly is composed of a motor-drum and of a motor-driven belt that ensure efficient tray evacuation.

2.5.7 Front mobile roller



Function:

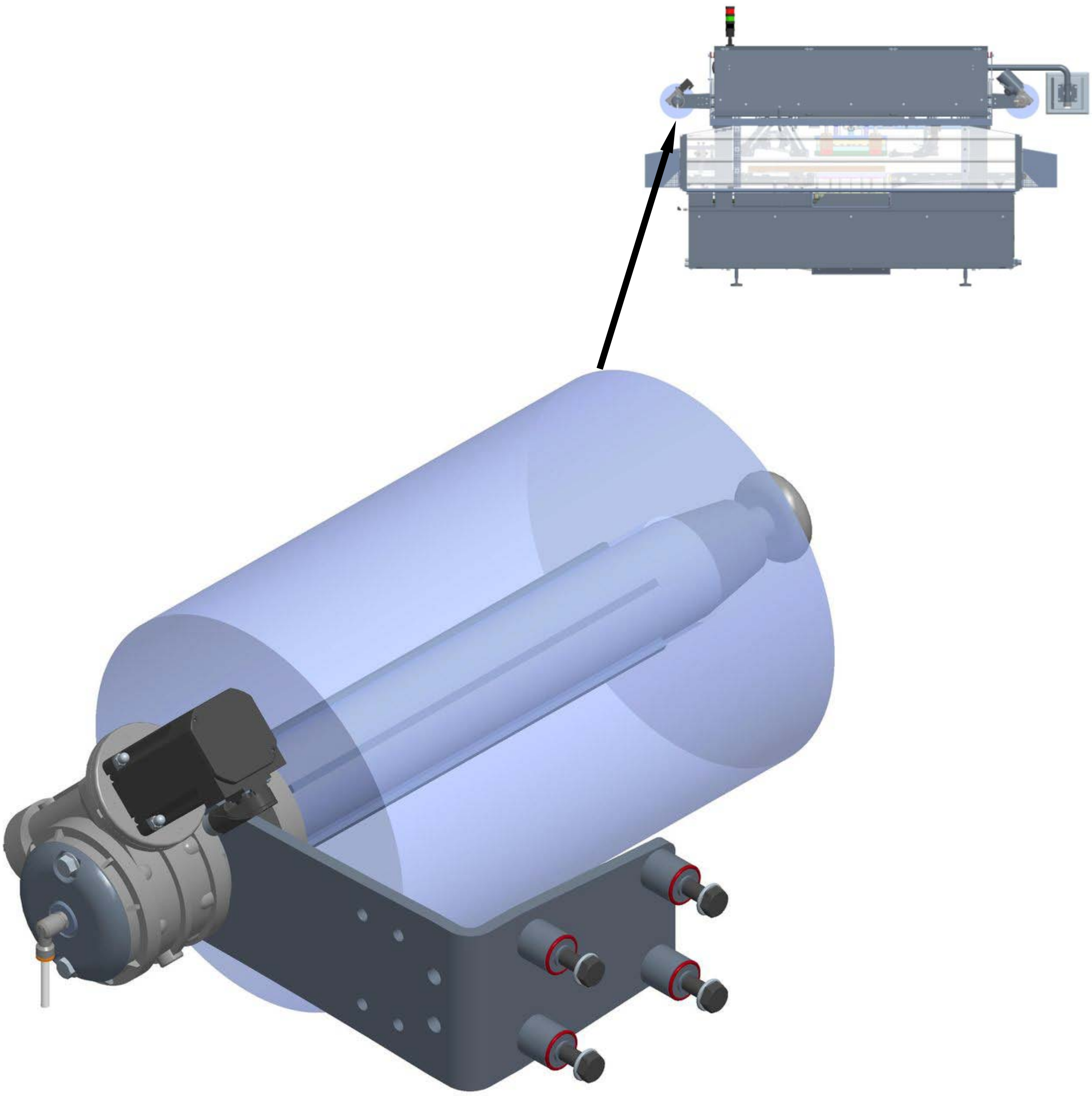
Enables the sealing film to wind over towards the tool area.

Make-up:

This unit assembly is made up of a frame provided with slider rollers that, via a retention system, ensure that the film is unwound correctly onto the film unwinding bobbin.

Appropriate limit switches are provided to detect correct film bobbin centring and when the film bobbin is empty.

2.5.8 Film unwinder roller



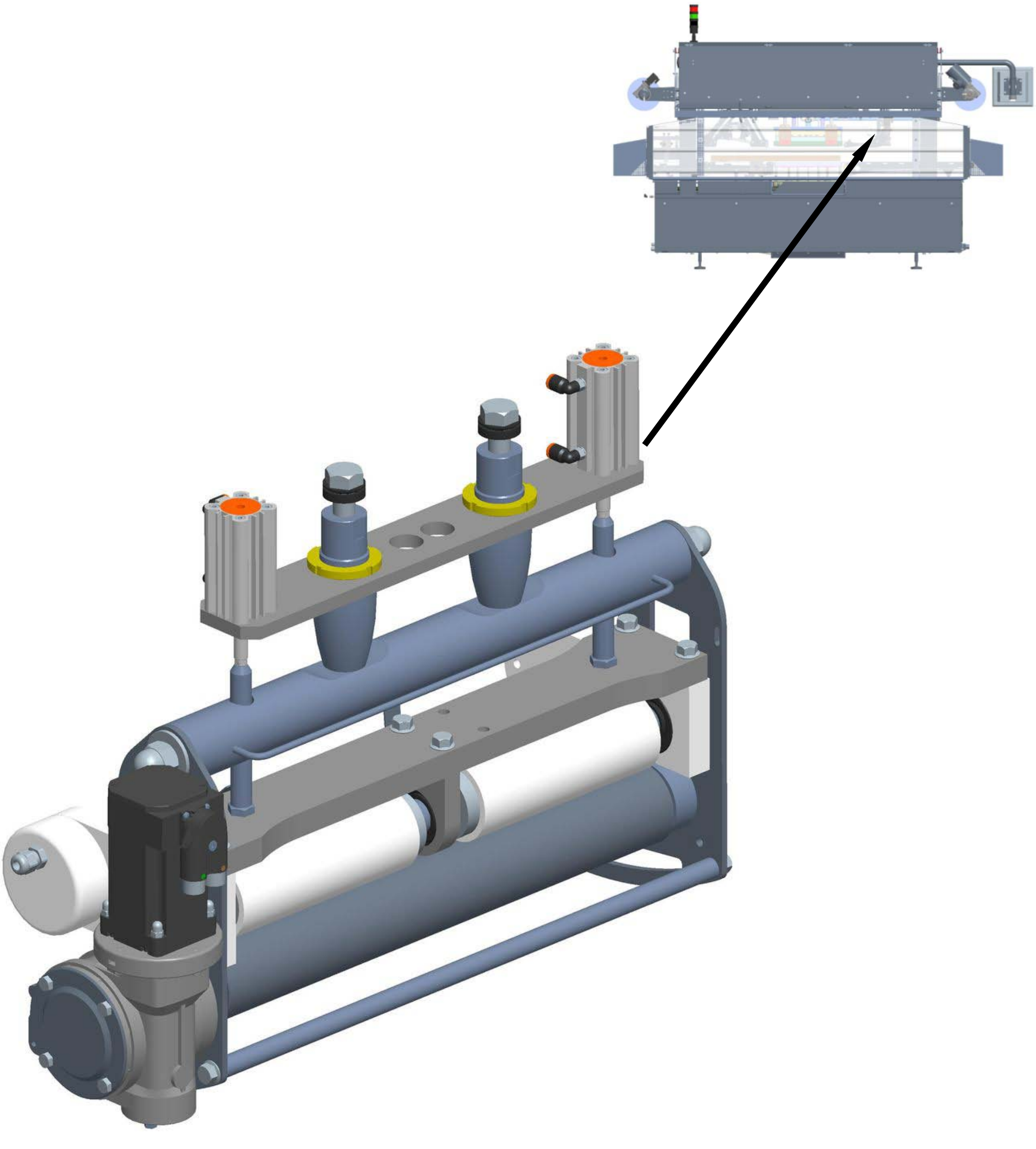
Function:

Enables the sealing film to unwind over towards the tool area.

Make-up:

This unit assembly is composed of two motor-driven shafts that the sealing film bobbins are fitted onto.

2.5.9 Film drive and pull rollers and transmissions



Function:

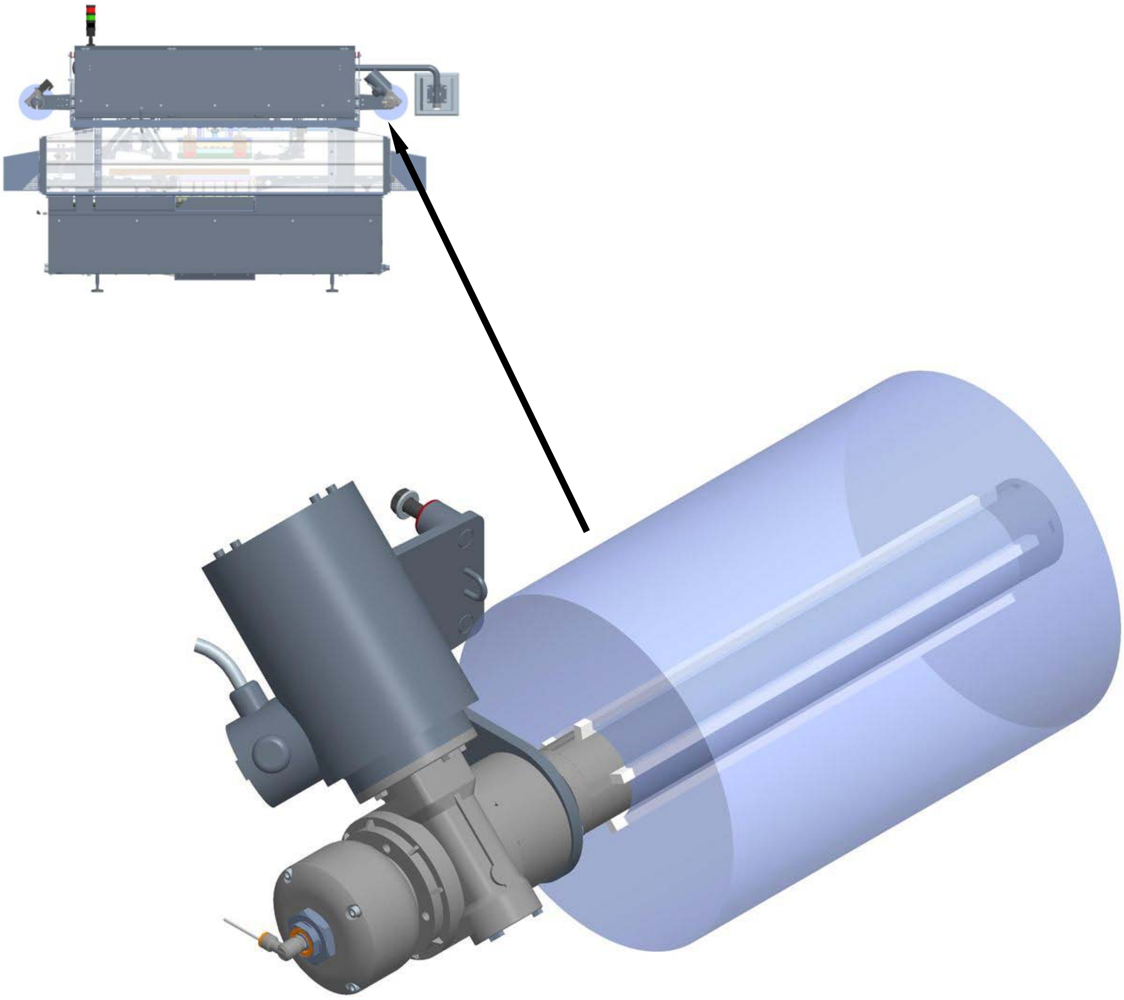
Enables the sealing film waste to slide over from the tool area to the film waste collector shaft.

Make-up:

This unit assembly is made up of a frame fitted with slider rollers that allow for tensioning and wind-up of the sealing film waste.

Monitoring of possible breaks occurring on the sealing film waste is guaranteed by a blade plate system assembled onto one of the slider rollers.

2.5.10 Winder shaft



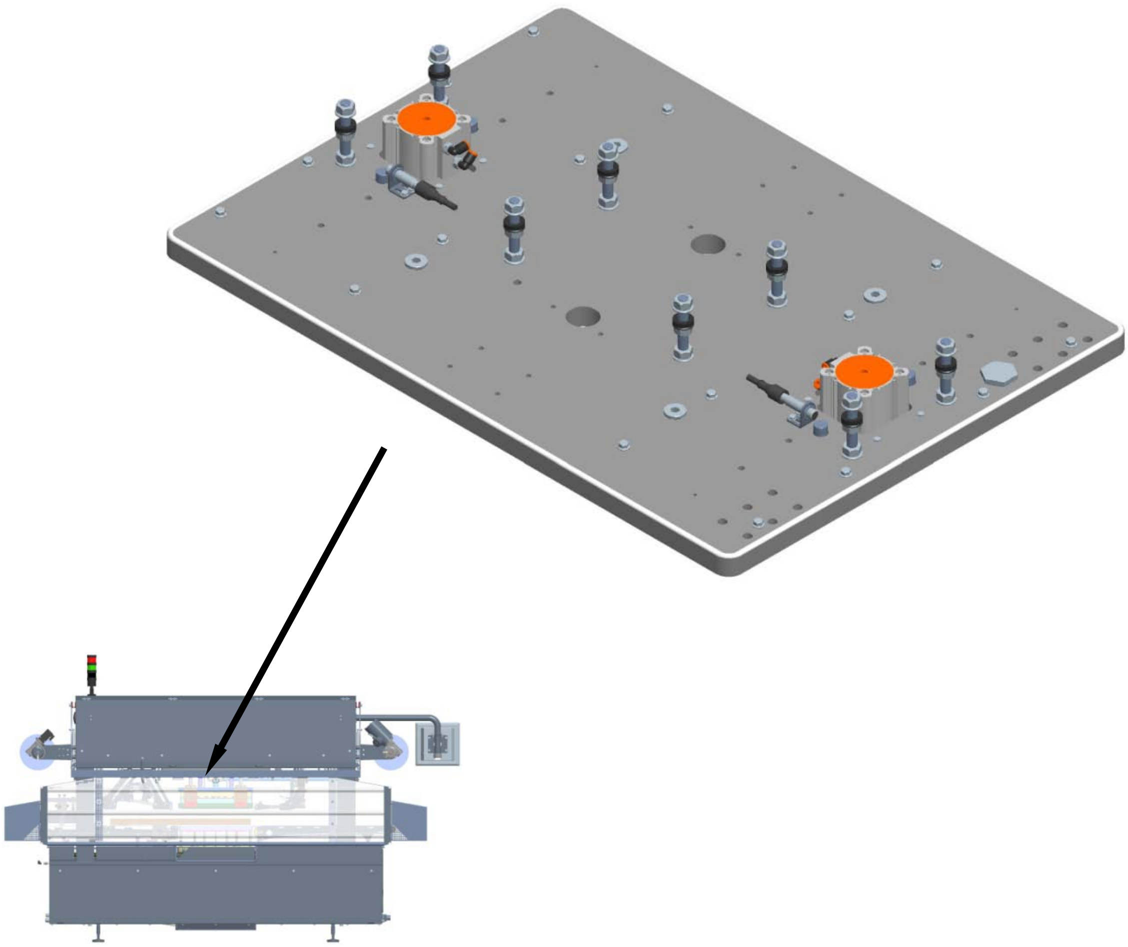
Function:

Allows for pick-up of sealing film waste released by the tool.

Make-up:

This unit assembly is composed of a motorised shaft that the waste is wound onto (i.e. the waste can be with wound directly onto the shaft or onto an appropriate collector cylinder).

2.5.11 Top tool plate



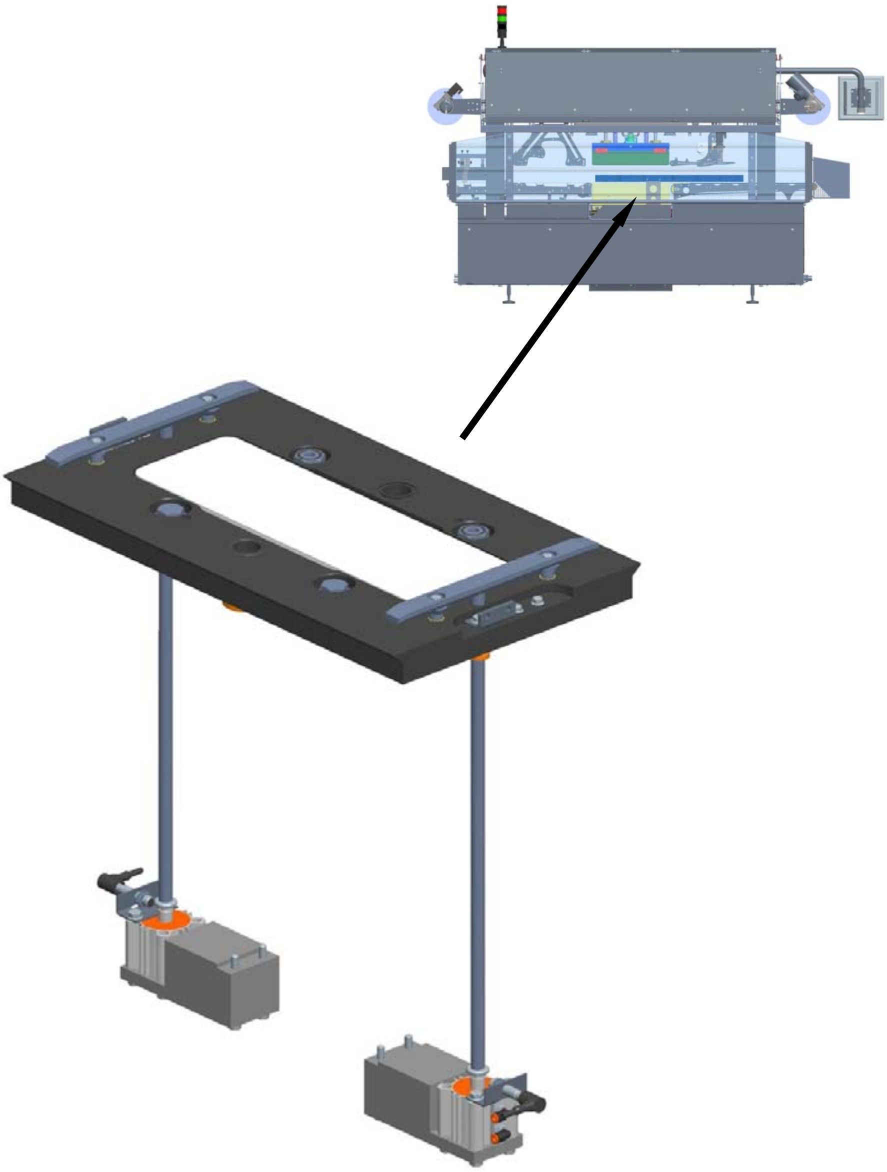
Function:

Enables lock-on of the top half tool.

Make-up:

This unit assembly is composed of a plate fitted with two guides, as well as appropriate centring and stop devices that allow for correct assembly of the top half tool.

2.5.12 Mobile plate



Function:

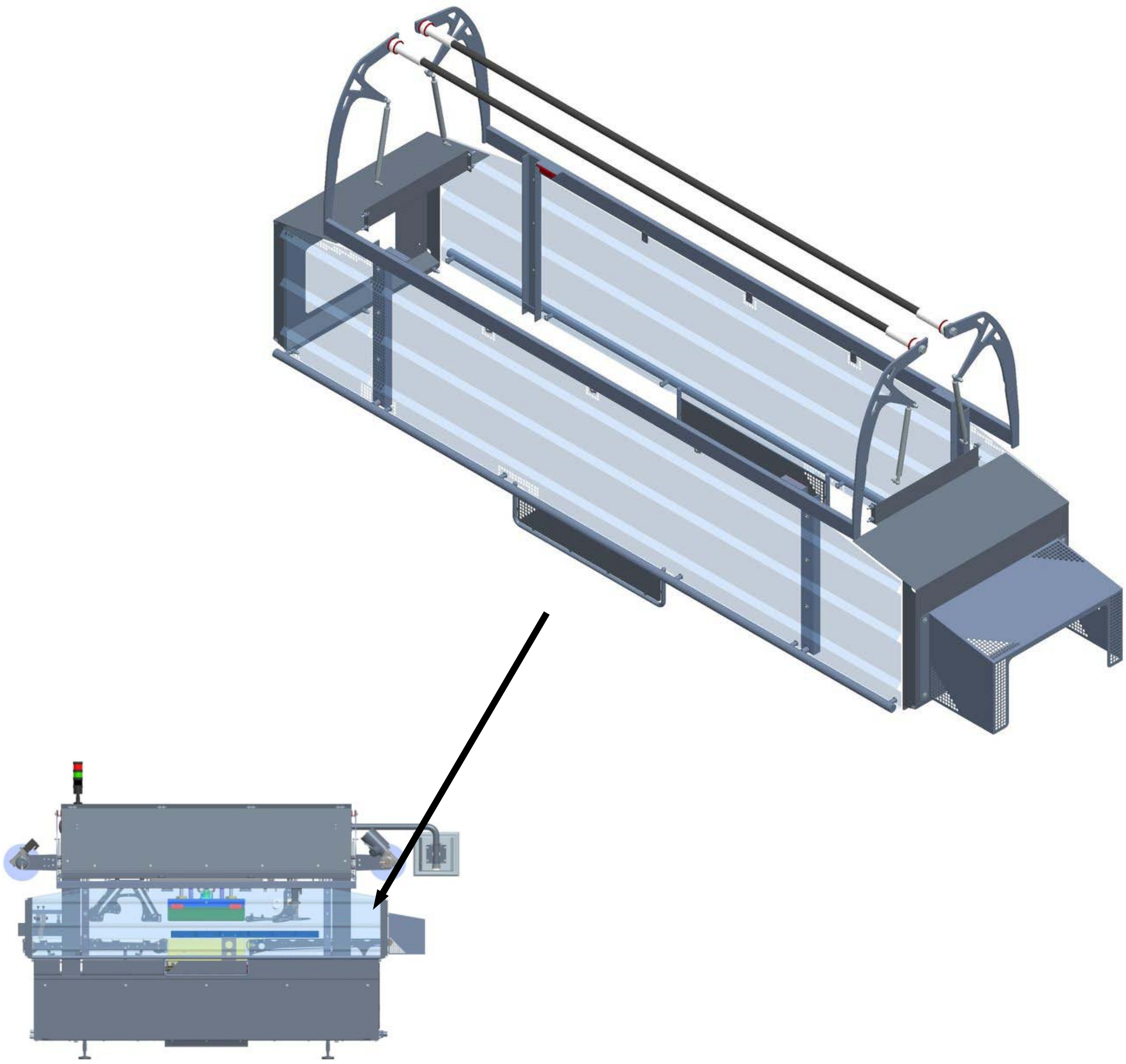
Enables lock-on of the bottom half tool and relative movement of the tool towards the top half.

Make-up:

This unit assembly is composed of a plate fitted with two guides, as well as appropriate centring and stop devices that allow for correct assembly of the bottom half tool.

Furthermore, the sensors for correct plate position detection are also housed here.

2.5.13 Cover casing



Function:

It’s function is to guarantee the safety of the personnel entrusted with machine operations and simultaneously allow for complete machine operation control.

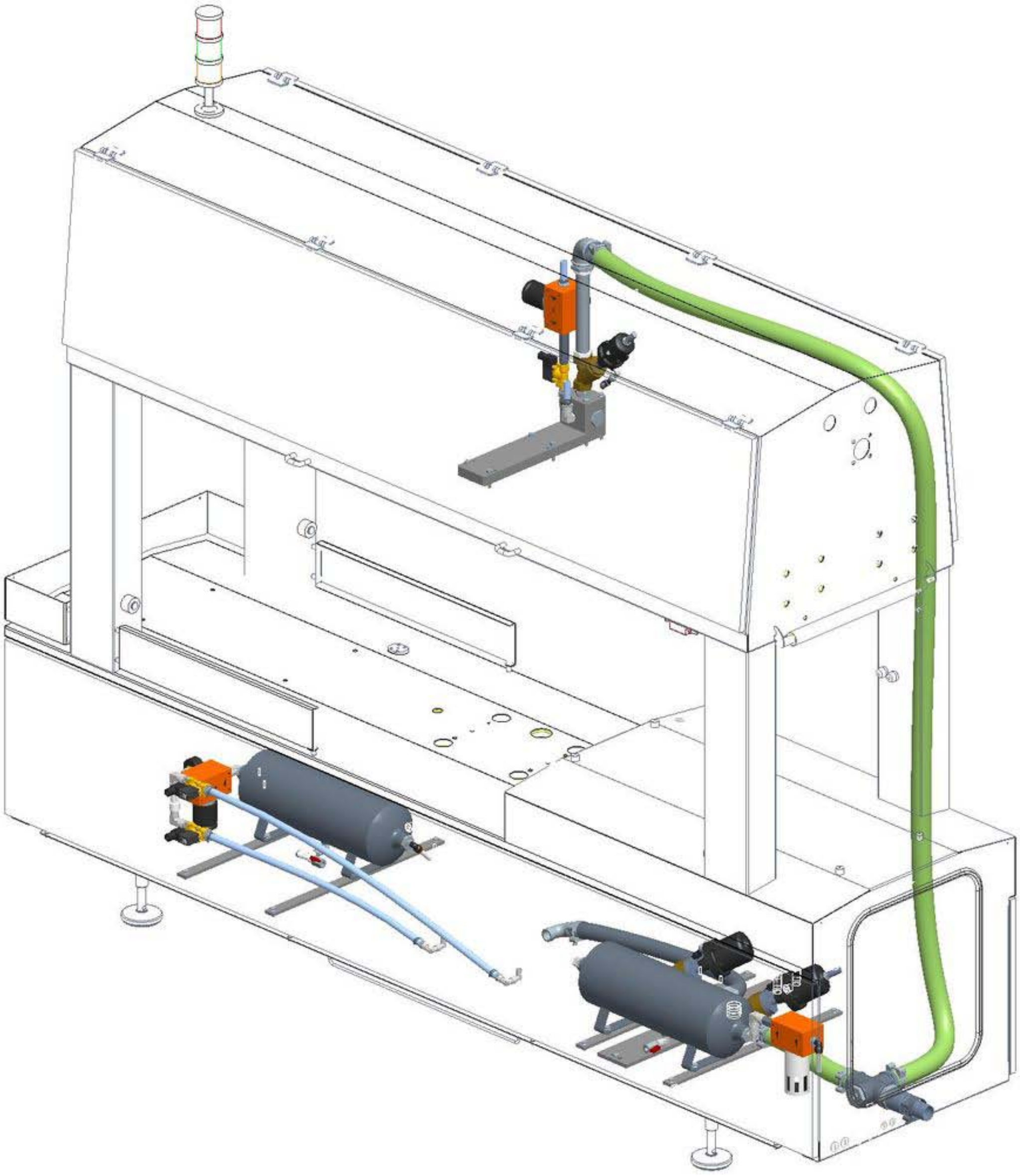
Make-up:

Said protection covering is made up of a stainless steel structure with a lexan panel coating and equipped with safety micro switches.

This machine safety covering entirely covers the tray inlet area and the weld-sealing area (where all the most dangerous mechanical movements occur), as well as the out-feed area.

If one of the safety covering doors are opened whilst the machine is working, the work cycle is instantly stopped and the relative safety covering door open alarm will go off.

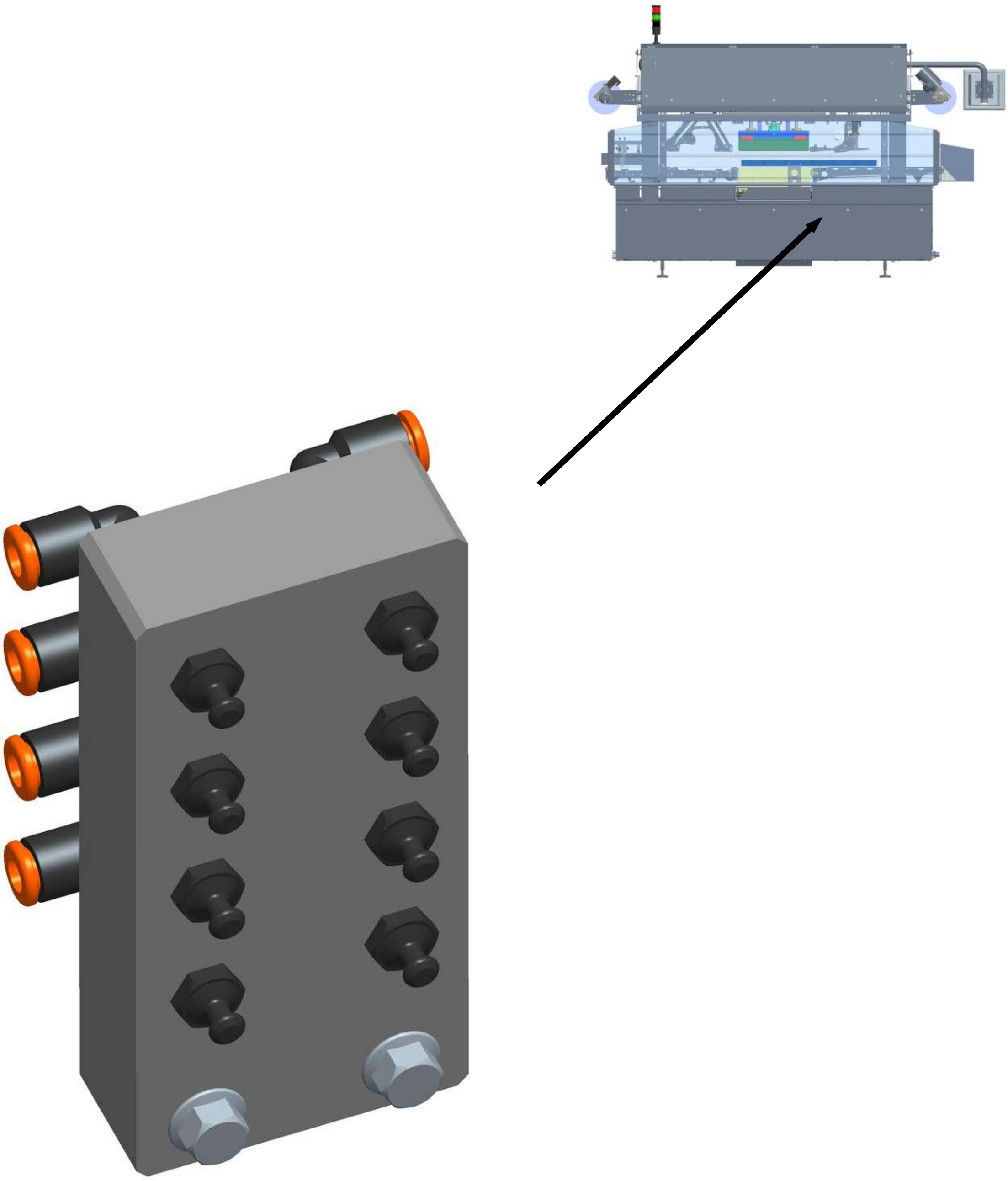
2.5.14 Vacuum circuit



Function:

Provides for vacuum forming internally to the tray during the tool seal-welding phase.

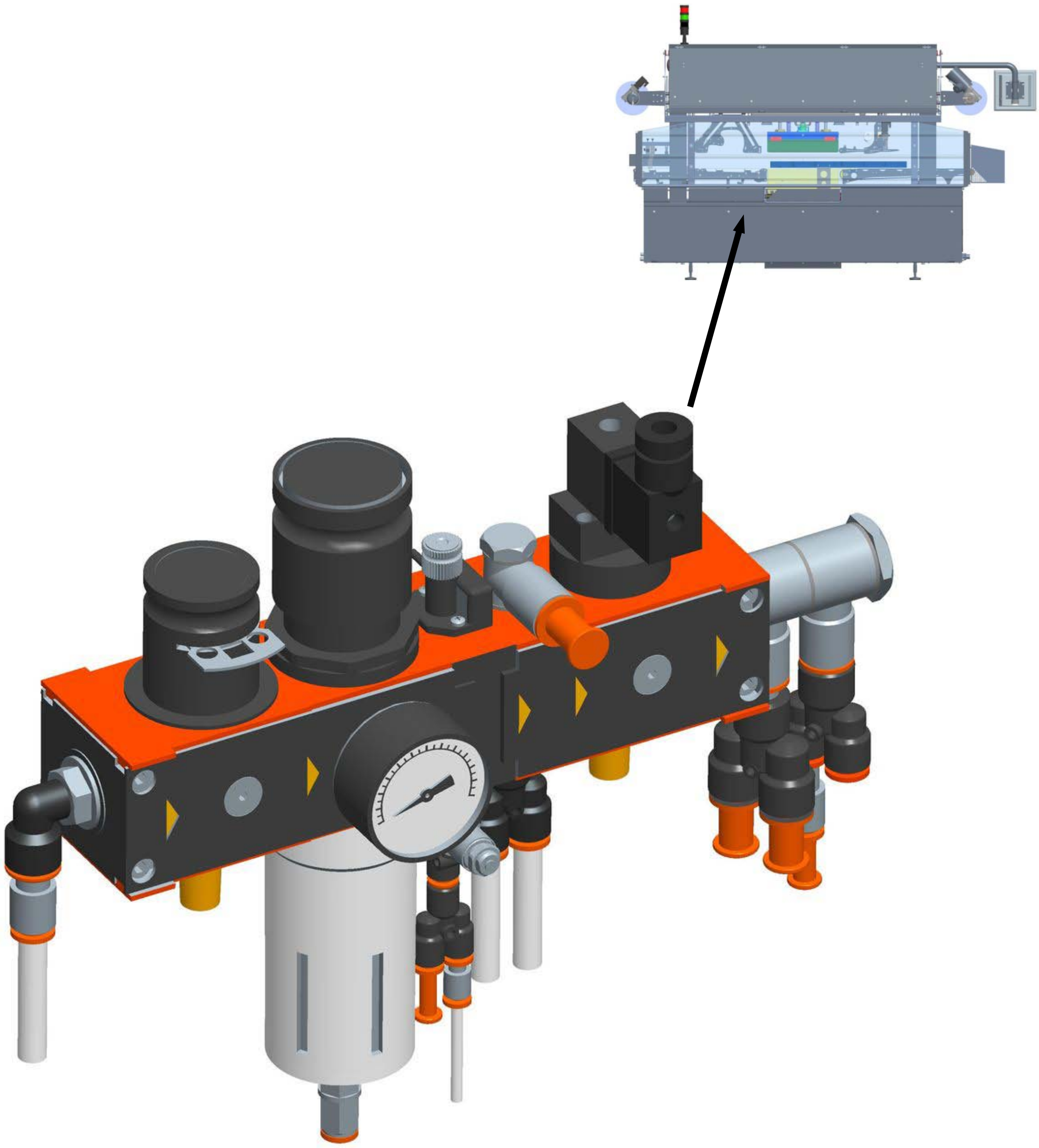
2.5.15 Lubricating system



Function:

Manual lubricates the main components of the machine.

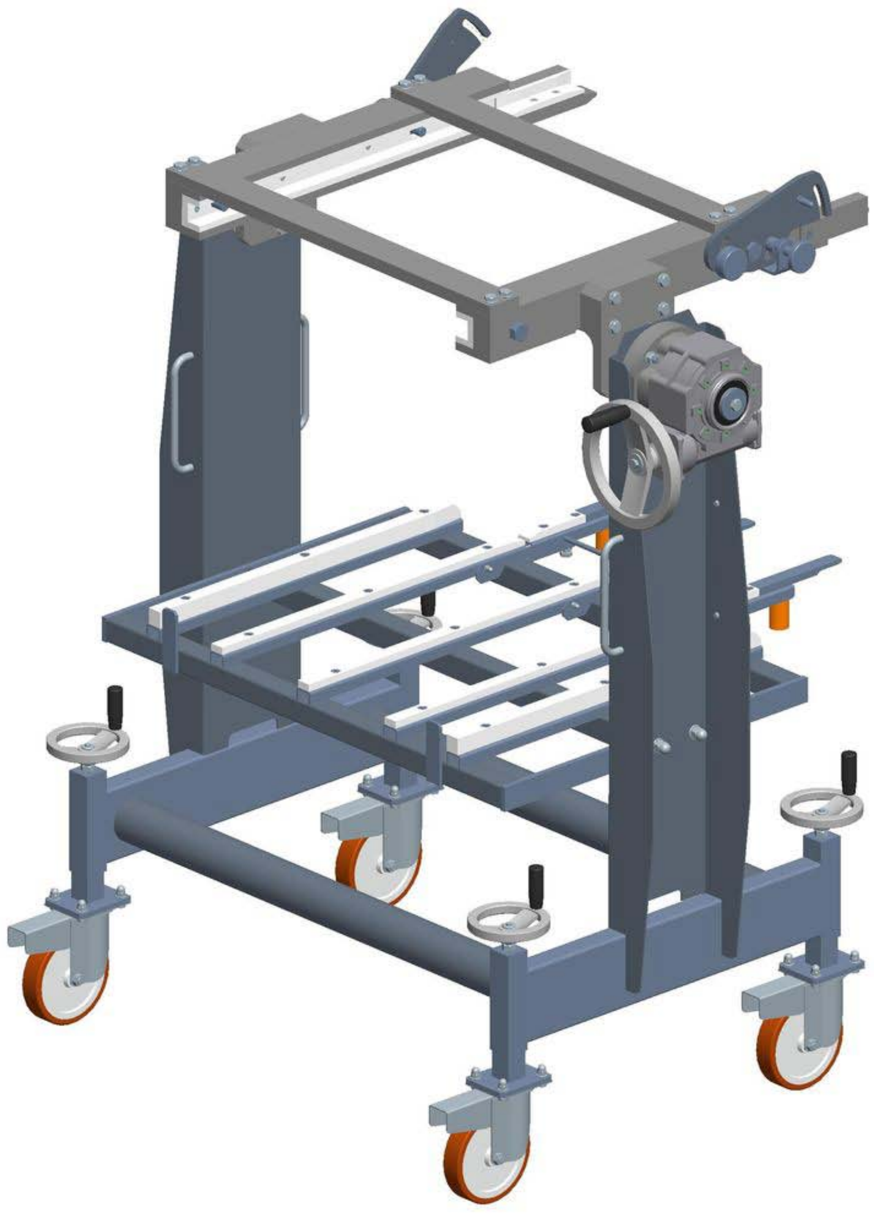
2.5.16 Pneumatic system



Function:

Establishes and regulates operation of the pneumatic actuators present on the machine.

2.5.17 Tool carrier trolley

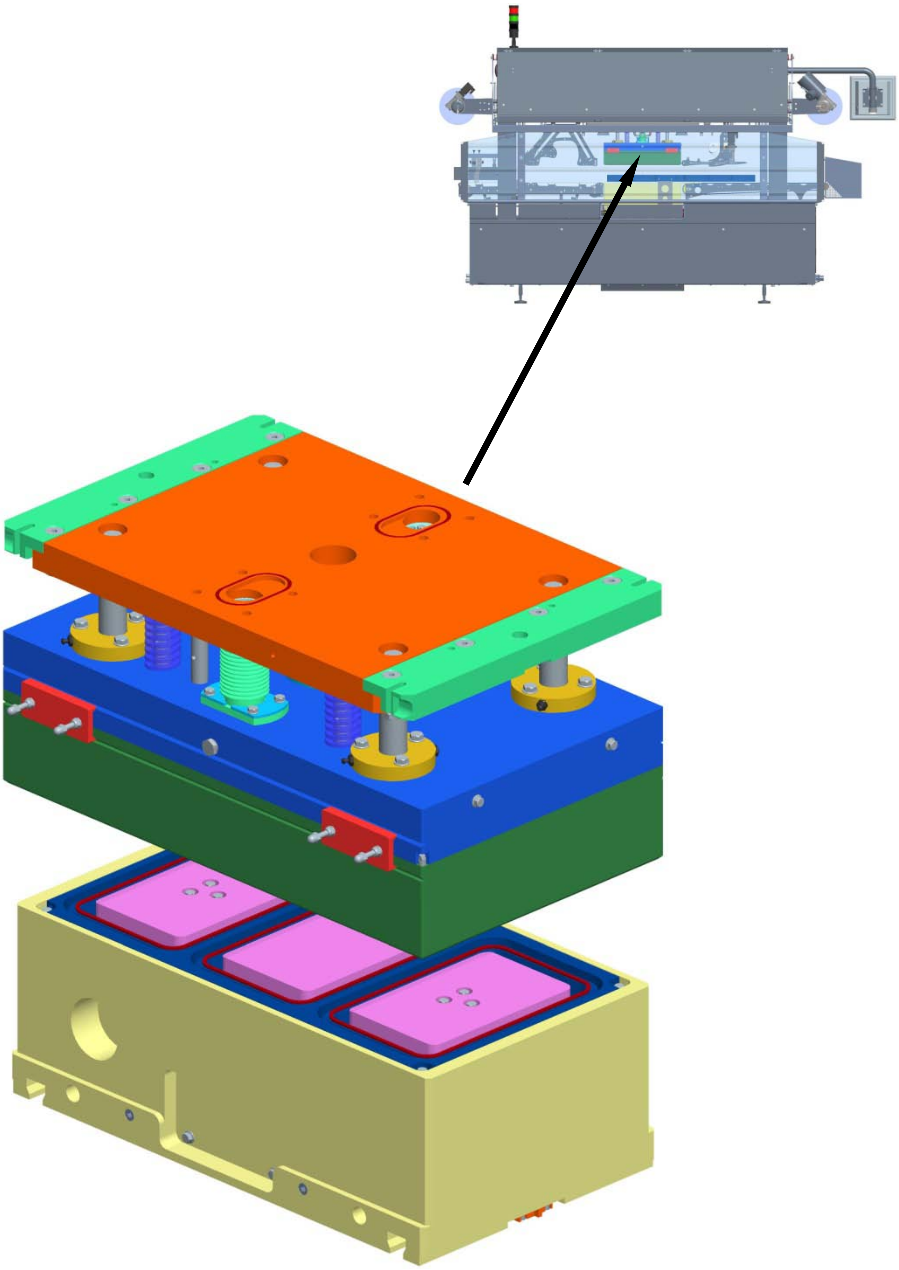


Function:

This trolley is assembled on wheels to provide for easy movement and handling. It allows for all the parts, pieces and tools required for tool change and replacement operations, to be conveniently on hand.

Via a hook-on system to the machine, tools can be removed and installed absolutely safely and with extremely reduced changeover times.

2.5.18 Tool



Function:

This component (that changes based on the type of tray to be sealed) is made up of two parts, i.e. a top tool half and a bottom tool half. It is the part of the machine that carries out the final film-to-container weld-sealing function.

2.6 ENVIRONMENTAL OPERATING CONDITIONS

To provide for the best environmental conditions possible, it is necessary that the **Customer** provides for all the relative utilities and systems to be available on site with the best possible aeration and in full compliance with the IEC EN60204-1 standard provisions. The tray sealer machine must strictly not operate in site environments with atmospheric explosion and / or fire and conflagration hazards.

2.6.1 Lighting

In the sealing machine commissioning or delivery phases it is to be ensured that the existing lighting plant is appropriate and that the area involved does not require specific lighting.

The purchaser must arrange for a specific lighting plant for access to the work areas.

2.6.2 Vibrations

The machine does not generate vibrations having frequencies that may lead to operator safety hazards.

2.6.3 Stability

This tray sealer machine is not provided with any elements that may be subject to precarious stability hazards.

2.6.4 Emissions

Considered that the machine is intended for industrial environments it is hereby confirmed that relative emission levels lie within admitted range limits.

2.6.5 Noise

For noise hazard prevention, G.MONDINI S.p.A. has engineered the sealing machine with the implementation of the best concretely applicable technology. It is hereby guaranteed that the noise level at the operator controls position is < 75 dB(A).

2.6.6 Fume suction

This sealing machine does not have any fume emissions, therefore a fume suction system is not required.

3 SAFETY

CONTENTS

3.1 GENERAL SAFETY STANDARDS

To ensure the maximum working reliability, G. MONDINI S.p.A. has made a careful choice of the materials and components to be used to manufacture the **CLOSING MACHINE TRAVE 350 VG**, subjecting them to inspection before delivery.

Good performance over time also depends on correct use and adequate preventive maintenance, following the instructions contained in this handbook.

The materials are of the best quality and the arrival in the company, storage and use in the workshop are constantly checked to ensure there is no damage, deterioration or malfunctioning.

However, it should be remembered:

- The line is never to be used, nor any intervention be made on it, before carefully reading and fully understanding this handbook, from cover to cover.
- It is forbidden to use the system under conditions, or for a use, that is different to that indicated in the handbook. G. MONDINI S.p.A. shall not be held liable for breakdowns, faults or accidents caused by ignoring this prohibition.

> ⚠ NOTE:
> It is forbidden to tamper with, alter or change, even partially, the fixtures dealt with in this handbook, especially the guards provided for safety of persons. It is also forbidden to operate in any way that does not comply to the indications or to neglect operations that are necessary for safety.

It is the task of the operator to ensure that the line is kept clean and free of foreign matter such as scraps, oil or other.

If air or water is used under pressure for cleaning operations, the operator is to wear protective goggles and mask and make sure that there is nobody near the line that could be hit by material or dust. Inflammable liquids are not to be used for cleaning operations.

Periodically check the condition of the plates and if necessary restore them to the proper condition.

After cleaning the line, check there are no worn or damaged parts (if there are, call immediately for the maintenance engineer) or parts that are not securely fixed (intervening within the competencies of the operator).

Special attention is to be paid to the intactness of flexible hoses and other parts subject to wear.

If these situations are found at the end of the work shift, before leaving, the operator is to place a warning signboard on the control console indicating that the machine is under maintenance and is not to be started up (these signboards can usually be found on the market in the form approved by EU standards).

The operator is to keep the line and the surrounding area free of all loose items that are not integrating parts (for example tools left after a maintenance operation, ropes used to handle parts, personal belongings, etc.).

The operator, the assistant and the maintenance engineer are to wear suitable protective clothing according to the need and the characteristics of the work executed on the line component. It is also important not to wear loose clothes or chains, bracelets or other items that could become entangled in moving mechanical parts. Long hair is to be gathered in a net to avoid becoming caught up in the machinery.

Line protections and safety devices are not to be removed, except where necessary for repair and/or maintenance.

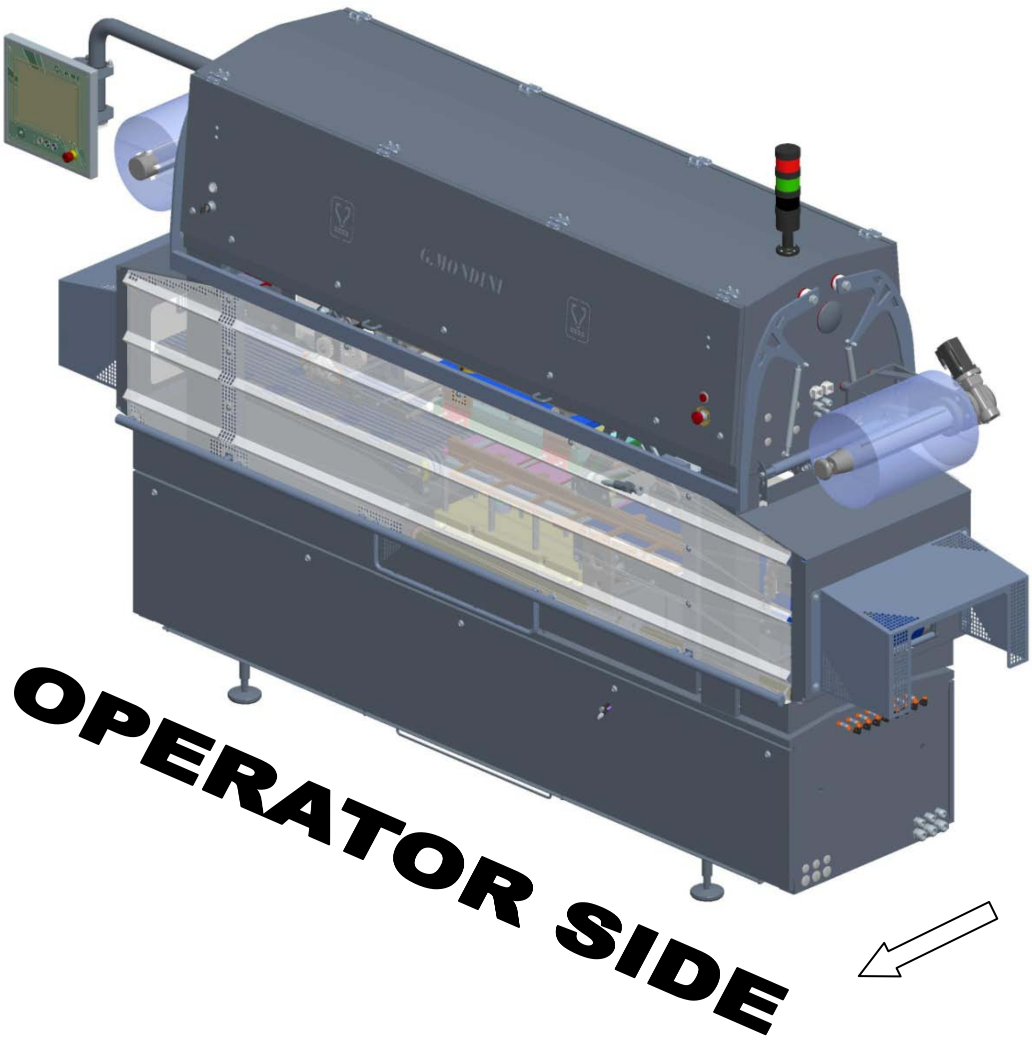
They are to be restored to normal functioning as soon as the reason for their temporary removal has been solved, and in any case before starting up the line.

3.2 DESCRIPTION OF THE WORK STATION

G.MONDINI S.p.A. has always designed and manufactured their machines with explicit focus on the safety of the operator, who shall perform his tasks safely and with full control at every stage of the operating cycle.

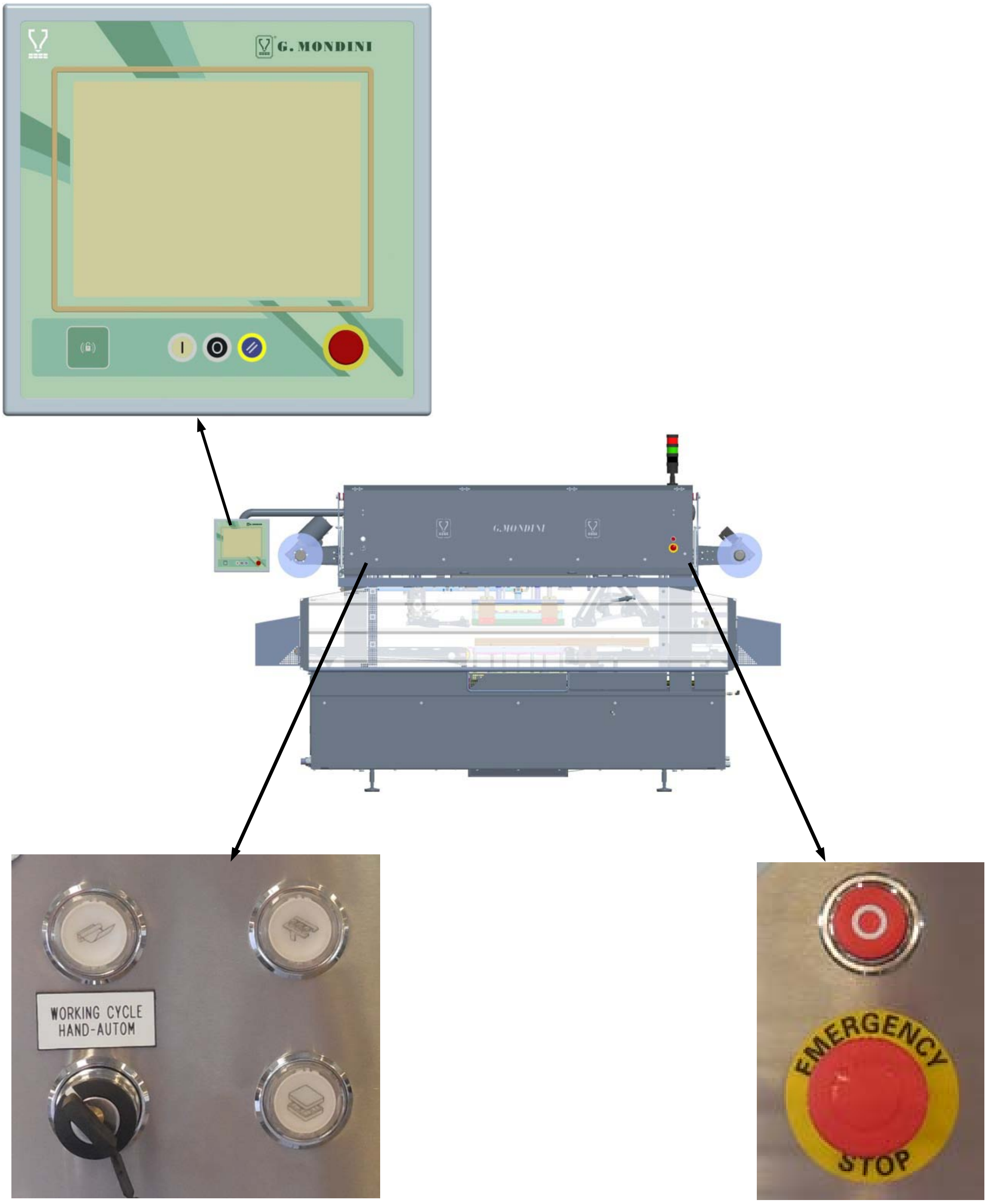
The machine was designed and built by identifying an "*operator side*" that is agreed jointly with the customer, where the following main operations can be carried out:

- installation and removal of the closing tool;
- replacement of sealing film coils;
- removal of sealing trimmings.



Nevertheless, the operator can move along the machine perimeter and have access to all controls during operation.

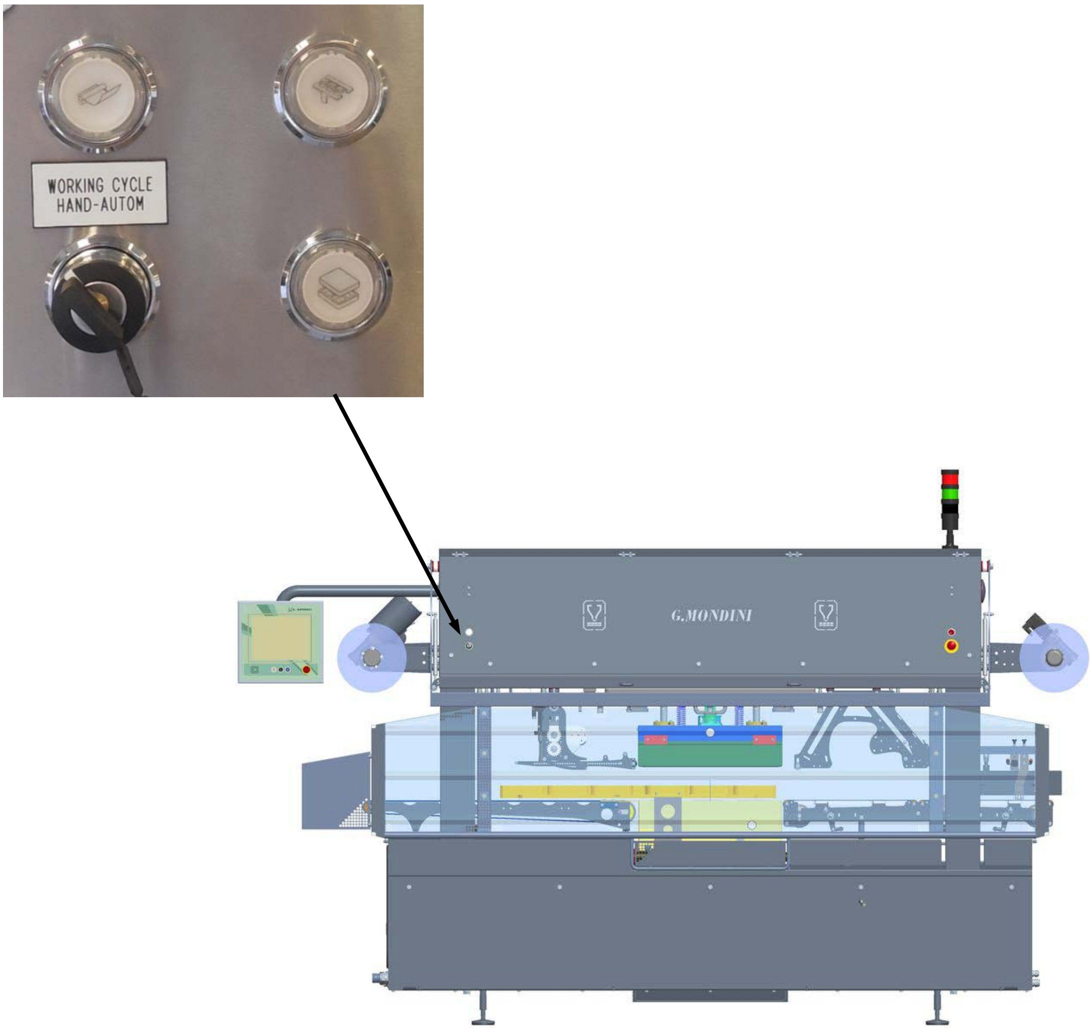
All the control devices required for machine operation are accessible from the *“operator side”*. All the commands are clearly visible and identifiable with plates with a description explaining their use. The controls are located at a safety distance from hazardous areas, including the emergency stop buttons that cause the immediate halt of the machine in case of hazard or as required.



The machine cannot be operated by more than one person at a time, despite the operator can, from the operating position, ensure at any time that there is no one else operating the rest of the installation. A sound and/or visual warning activates on turning the machine to allow anyone present to leave the hazardous area or to prevent the machine from starting up.

The machine can be started and restarted after a stop only by the operator in charge of operation using the appropriate commands.

The machine can be run either in "automatic" or "manual" mode, which can be selected by the selector switch knob situated on the "operator side". The selection of one operating mode excludes the other.



3.3 FORESEEN, UNFORESEEN AND IMPROPER USE

The line described herein has been designed to execute packing operations on certain foodstuff products.

> ⚠ DANGER
>
> The use of the line for purposes and processes that are not described in this handbook is **IMPROPER USE**. G. MONDINI S.p.A. shall not be held in any way liable for damages caused to goods and / or persons and shall consider any type of guarantee of the line as null and void.

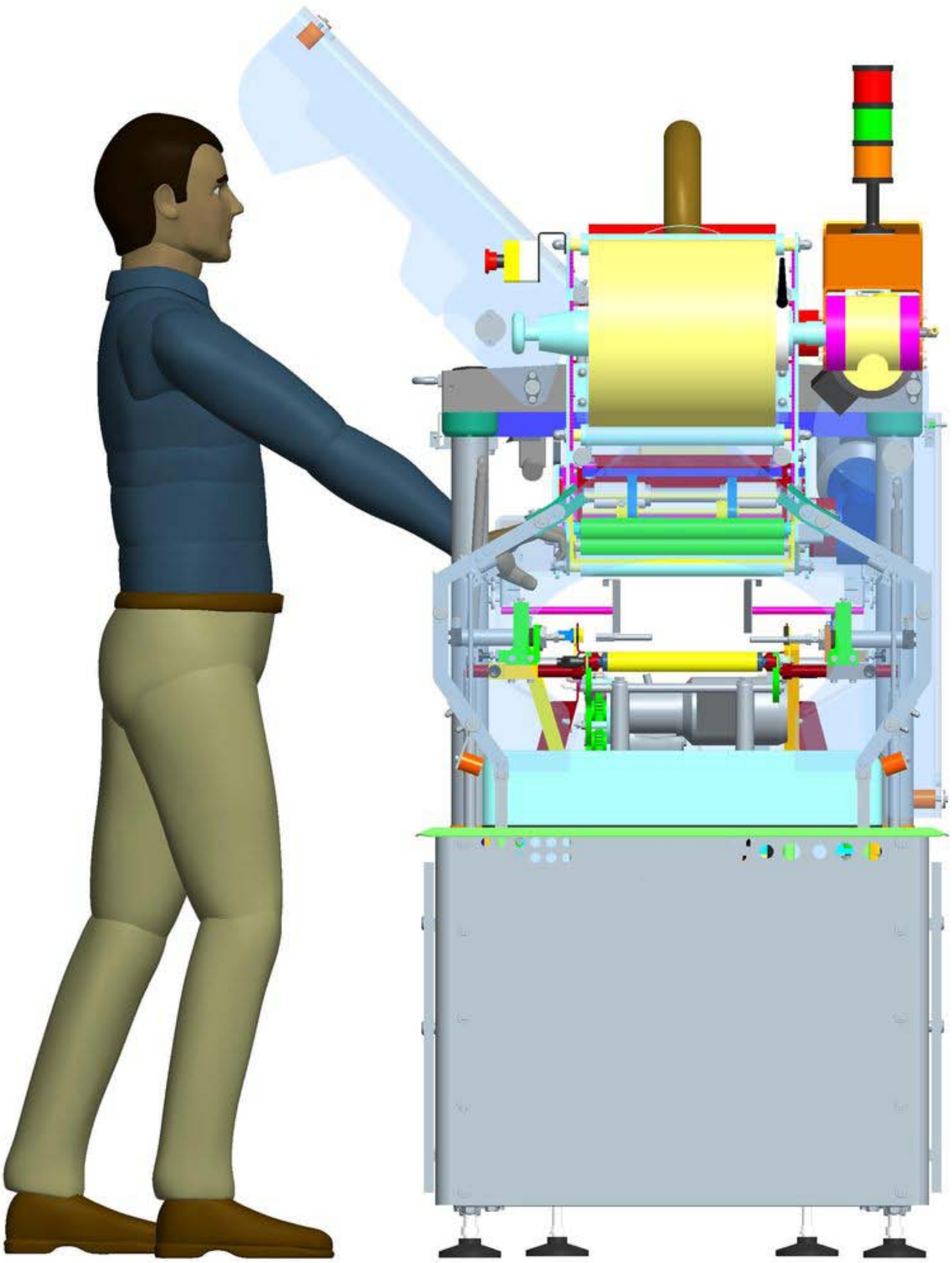
It is forbidden to use the line with a number of workers over the number foreseen in this instructions handbook.

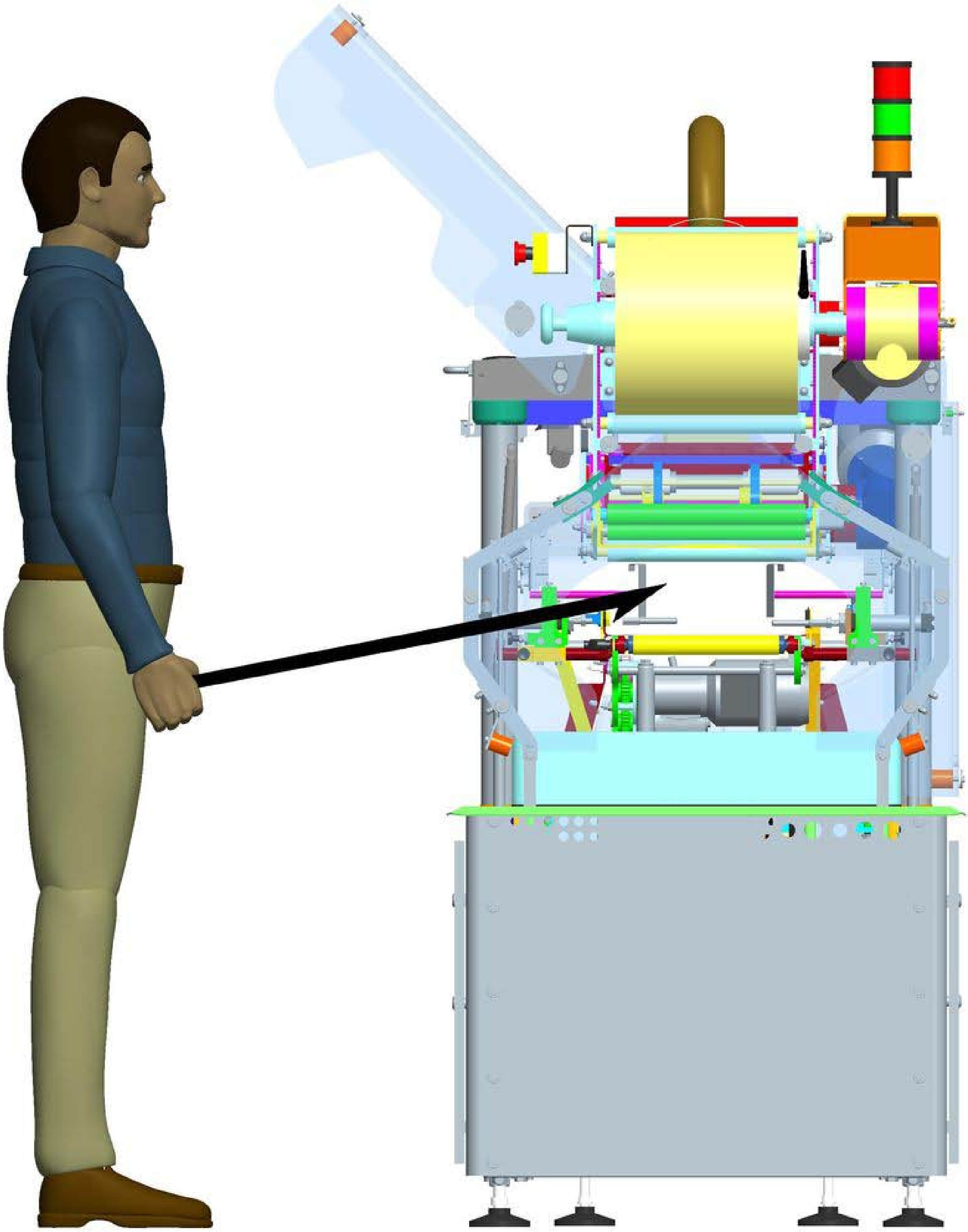
As a function to the use of the fixture, the presence of others in its vicinity when working is not foreseen.

It is the responsibility of the user to install the line so it is not adjacent to passageways; otherwise the user is to provide appropriate barriers to prevent others accidentally entering the work zone.

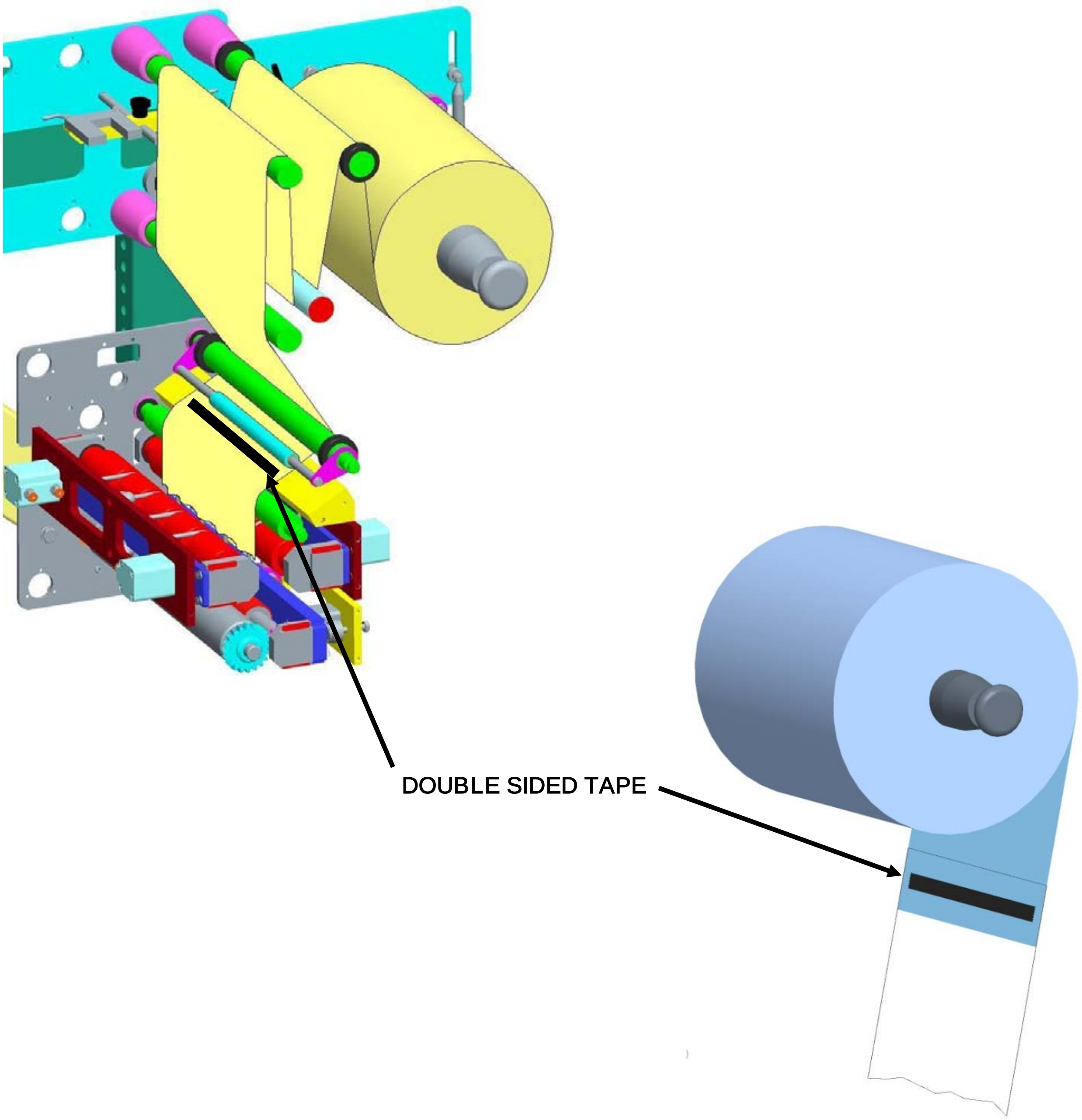
The following is not allowed:

1. Use of materials that are not specifically foreseen by the manufacturer.
2. Handling, installation, inspection, power-on, preparation, use, shut-down, maintenance methods that are not those indicated.
3. Use as a means of transport or to move in the space with the line moving parts running .
4. Allow unskilled persons to use the line.
5. Run the line without protections or with only partial protections.
6. Install them in a way that is not as indicated in the Installation and the Commissioning chapters (in the case of re-installation).
7. Connect the power supplies to the line inadequately or in manner that does not comply to the instructions.
8. Serious neglect in maintenance .
9. Make unauthorised changes or interventions (especially on safety devices).
10. Use of spare parts that are not originals or not for that model.
11. Total or partial non observance of the instructions.
12. Use of the fixture in a manner that does not comply with the methods described in the handbook.
13. Modification of system programs without serious assessment and authorisation from the manufacturer.
14. Put your hands into any section of the system, even when it is disconnected from the electricity. Both standard and reactive machine cleaning, maintenance and the sealing film bobbin change operations must be performed using appropriate tools, to ensure complete and overall operator safety.
15. Before resuming production after any type of maintenance, always check that the nuts and bolts have been tightened correctly. Every six months it is advisable to check all the nuts and bolts on the machine, especially those subjected to vibration like the whole sealing tool group.
16. Automatic solenoid valve operating mode.
17. More than one operator running the machine at the same time.





G.Mondini S.p.A. recommends for the sealing film bobbin substitution, to fix with double sided tape the exhausted film border the new bobbin film.



3.4 RESIDUAL RISKS

3.4.1 General information

During the design, all the zones and parts that incur risks have been assessed and as a result, all the necessary precautions have been taken to avoid risks to persons and damage to **Closing Machine** components.

> **!**
>
> WARNING!
>
> Periodically check that all safety devices function correctly. Never remove the fixed or movable guards from the system. Do not leave objects or tools in the working area.

3.4.2 Residual risks and precautions

After carefully studying all the possible risks on the line, all the necessary solutions have been applied to remove the risks and limit the dangers for exposed persons.

On the line, although fitted with these safety systems, some risks remain that can be eliminated or reduced by the relevant precautions.

Table 3.1 summarises the list of residual risks that remain on the line.

Table 3.1 - Summary of residual risks

Type of risk	Description of the risk	Precautions
	The safety circuits could be subject to a lowering of the safety level.	Make periodical checks (once a month) on the serviceability of the safety circuits.
	Connections between safety circuits and control relays could be subject to failures that are not immediately detected .	Make periodical checks (once a month) on the preservation and efficiency of the control relays.
	Crushing risk following access to moving parts caused by the removal of fixed parts, or bypassing safety devices when an automatic or manual cycle is running.	The exposure of persons to moving parts of the machine can cause seriously hazardous situations. The machine is not to be started without all the fixed and movable protections correctly installed.
	Accident risk caused by deliberate tampering with safety devices or machine parts.	It is forbidden to make any type of mechanical, electrical or fluidic modification, to avoid creating additional hazards and consequent risks that have not been foreseen.
	Electrical hazards caused by unauthorised access to electric cabinets.	The keys giving access to electrical equipment are not to be given to unauthorised operators.

Type of risk	Description of the risk	Precautions
⚠	Breaking or damage risks, with possible lowering of the safety level, of electrical equipment components following a short circuit .	Before making the connection check that the d.c. current at the installation point is not higher than that indicated on the protection switches on the electric panel; otherwise the user is to provide the appropriate limiting devices.
⚠	Accident risk caused by poor lighting in the area where the machine is installed.	The user customer is to provide lighting in the area surrounding the machine, especially at the operator work stations, to comply to the relevant standards in force in the user country and according to the EU Directives.
⚠	Accident risk caused by unexpected movements or contact with live parts.	Before starting maintenance activities on the machine that require the replacement of assemblies or components, cut-out and padlock the energy sources using the specific cut-out devices.

3.4.3 Grease families NLGI 00

Composition, information on the components

Components in concentrations equal to or over 0.1% of the weight (classified as toxic or very toxic) or 1% (classified as harmful, irritating or corrosive).

DANGEROUS COMPONENTS	DANGEROUS CONCENTRATION
Not classified	

Type of risk

Product consisting of basic oils at high degree of refining and additives. The product is characterised by low oral and skin toxicity values and under normal use conditions should not present health hazards. Scrupulously observe the recommendations for use.

First Aid

INHALING

In the case of over exposure to mist, vapours and fumes, take the injured person immediately into the open and, if necessary give artificial respiration.

Call a doctor immediately.

SKIN CONTACT

Carefully wash with abundant water, and if available use neutral soap.

Remove shoes and contaminated clothing. If the irritation persists, go to the doctor. If the material, due to improper use of the high pressure greaser, has been injected under the skin, call a doctor immediately.

EYE CONTACT

Wash immediately with abundant water until the irritation ceases. If the irritation continues go to the doctor.

SWALLOWING

If swallowed, do NOT cause vomiting. Keep the injured person resting.

Call a doctor immediately.

Fire fighting measures

Extinguishing agents: foam, chemical powder, carbon dioxide.

FIRE AND EXPLOSION HAZARDS

Low risk combustible material. The product may form inflammable mixtures and burn only is heated to temperatures over its flash point. However, the presence of small quantities of more volatile hydrocarbons could increase the risk.

SPECIAL FIRE FIGHTING MEASURES

Use nebulized water to cool the surfaces exposed to the fire and to protect the personnel putting out the fire. Fire fighters exposed to fumes and vapours are to wear adequate protection (respirators and masks).

DANGEROUS COMBUSTIBLE PRODUCTS

In the case of incomplete combustion, smoke, sulphur oxides and carbon monoxide could develop.

Measures to control accidental spillage

PRECAUTIONS FOR THE PERSONNEL

See EXPOSURE CONTROL AND PERSONAL PROTECTION

SPILLAGE ON THE GROUND

Collect up the residual material; spread absorbing substances on the area then collect up in containers. If necessary dispose of the absorbed residue. Inform the competent Authorities if the product has reached water ways or sewers or if it has contaminated the ground or vegetation. Take measures to minimise the effects on waterbeds.

SPILLAGE ON WATER

Not available.

Handling and Storage

Keep the product in cool ventilated areas, away from heat sources. Use suitable equipment for lifting and transporting the bins and heavy packs under safe conditions. The electrical equipment used ids to conform to local ruling concerning fire prevention for these types of materials:

Loading/unloading temperature	°C : ambient
Storage temperature	°C : ambient

SPECIAL PRECAUTIONS

Keep the containers closed.

Exposure control and personal protection

EXPOSURE LIMITS

Oil mists: 5mg/m^3 - limit ACGHIH TLV-TWA, 8 h

Personal protection

In cases of potential contact, use safety goggles, clothing and gloves resistant to chemical agents. If only accidental contact is probable, wear safety goggles with side protection. Other special protection is not necessary if contact with the skin and eyes has been safeguarded. Should the concentration of the product in the air exceed the exposure limits and if the systems, operating methods and other means to reduce the exposure of the workers are not adequate, protection for the respiratory organs is necessary

Stability and reactivity

Stable.

CONDITIONS TO BE AVOIDED

Keep away from heat sources, naked flames and all other sources of ignition.

MATERIALS TO BE AVOIDED

Avoid contact with oxidisers such as liquid chlorine and concentrated oxygen.

DANGEROUS DECOMPOSITION PRODUCTS

The product is stable at ambient temperature.

Toxicity information

OVEREXPOSURE EFFECTS

Inhaling:

Negligible risk at ambient temperature or with normal handling. At high temperatures it may form high concentrations of vapours and mist that could irritate the eyes and respiratory organs. Do not breathe in the vapours and mist.

Skin contact:

Low level of acute toxicity. Frequent or prolonged contact could cause irritation or dermatitis: in the case of wounds caused by jets of grease at high pressure and should this become injected into the skin or other part of the body, since in time it could cause serious damage to the soft tissues, surgical treatment is necessary immediately.

Eye contact:

Slightly irritating: the optical tissues are not damaged.

Swallowing:

Low level of acute/systematic toxicity.

Chronic effects

The basic oils contained in this product have not shown cancerous activity in tests on guinea pigs (long term skin tests).

TOXICITY DATA

Acute toxicity:

Experimental data on the finished product is not available. The heath risks indicated derive from the current knowledge on the toxicity of the basic mineral oils and additives used, taking into consideration the concentration in the finished product. The general effects of this type of mineral oils are well known and indicated in many scientific papers and in the CONCAWE report “Health Aspect of Lubricants” - May 1987.

Chronic toxicity

Not having specific data on all the basic oils used in the product and by comparison with basic oils of similar composition and refining potential cancerous effects are not expected. The basic oils assessed have undergone biological tests according to the Exxon standard protocol. These basic oils have not shown cancerous effects.

Ecological information

Not having specific environmental data concerning this product, it has been assessed on information relevant to the hydrocarbons contained in the lubricating oils of mineral origin. When spilt they remain mainly on the soil or water surface. On the basis of informative literature, this product should not present particular hazards for the earth and/or water habitat. The product is poorly biodegradable and persists in the atmosphere. It may contain additives for which environmental data is not available. Therefore this assessment only concerns the basic mineral oils contained.

Remarks concerning disposal

Collect and dispose of the refuse products in authorised dumps according to the current national and EEC ruling.

Information concerning transport

CONTAINERS FOR TRANSPORT

Bins, buckets, cartridges.

Transport/storage temperature °C: ambient

Other information

TYPE/USE OF PRODUCT

Multipurpose lubricating grease of high quality for industrial and marine use.

Sources of information

The information contained herein refers only to the specific product and may not be valid should the product be used in combination with other products or in other types of processing. The information has been updated to the best of current knowledge at the date of the last revision. However we give no guarantee regarding the accuracy, reliability or completeness of this information. In fact, it is the responsibility of the user to ascertain the suitability and completeness of the information given, according to the particular use to be made.

3.5 SAFETY DEVICES

The closing machine described herein is fitted with the following safety devices/solutions:

Type of device/solution	Function	Description in paragraph
Main switch	Cuts out the machine electrical power	3.5.1
Air cut-out valve	Cuts out the pneumatic supply	3.5.2
Fixed and movable protections	Confine the machine hazardous zones	3.5.3
Emergency stop	Immediate stopping of the line.	3.5.4
Safety limit switches with interlock on protection doors	Controls the closing condition of the access doors inside the machine and locks them when the machine is running.	3.5.5
Automatic circuit breakers	Cuts out electric supply in the case of overload or short circuit	3.5.6
Individual protection devices	Protection for operators when carrying out the job	3.5.7

3.5.1 Power supply sectioning and padlocking

Function:

Cuts out power supply sources to the machine.

Method:

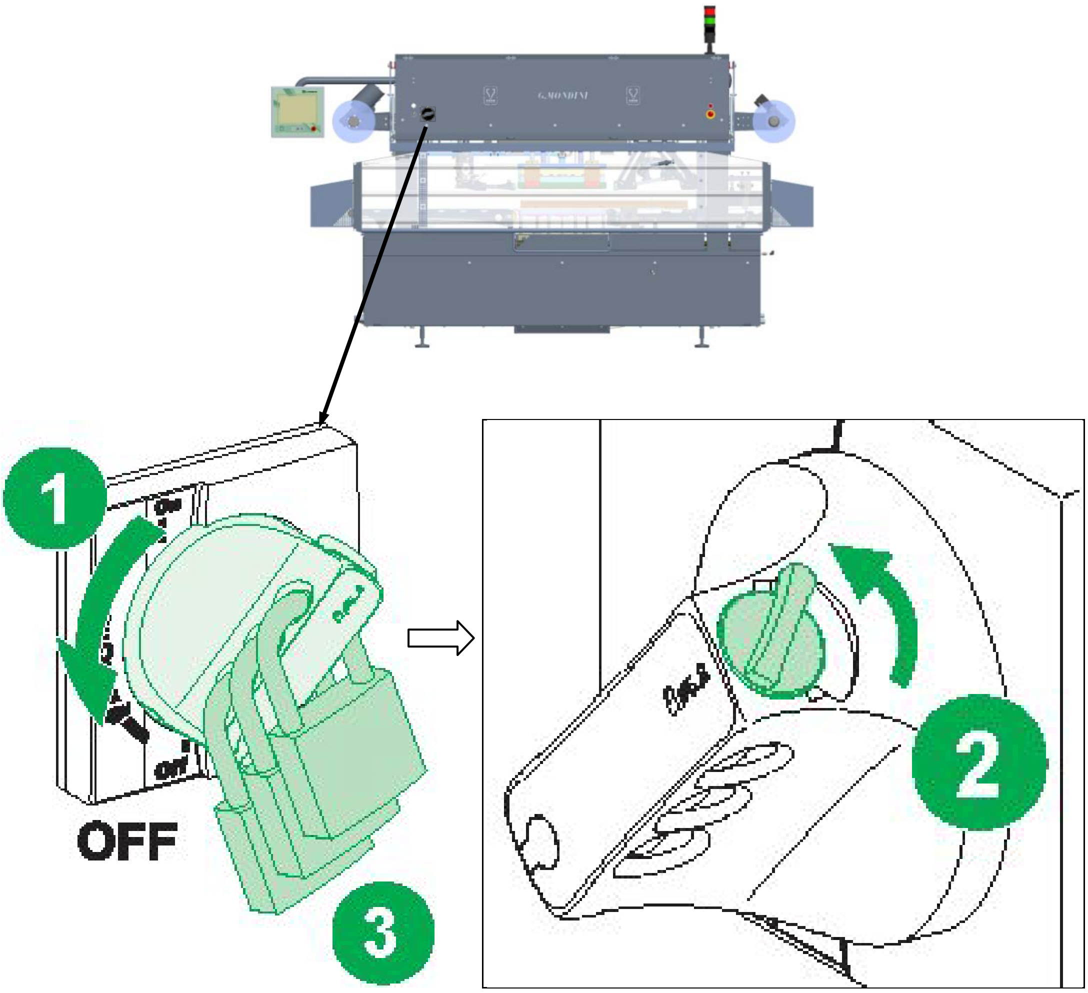
Before carrying out any type of maintenance operation on the machine the energy sources must be cut out and any accumulated energy discharged.

The electric power is cut out through the main switch on the bench.

The master switch on the machine can be padlocked to prevent the risk of injury due to unexpected start-up or contact with live parts.

For a description of risks and precautions, please refer to table 3.1 in this chapter.

G. Mondini machines are normally not supplied with a padlock. This accessory must be requested specifically on finalisation of the contract.



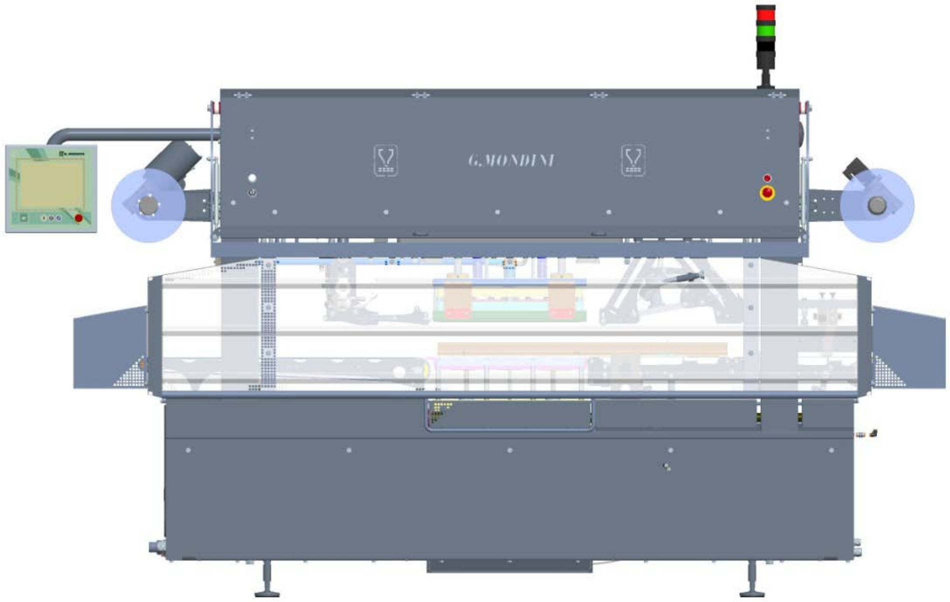
3.5.2 Air cut-out valve

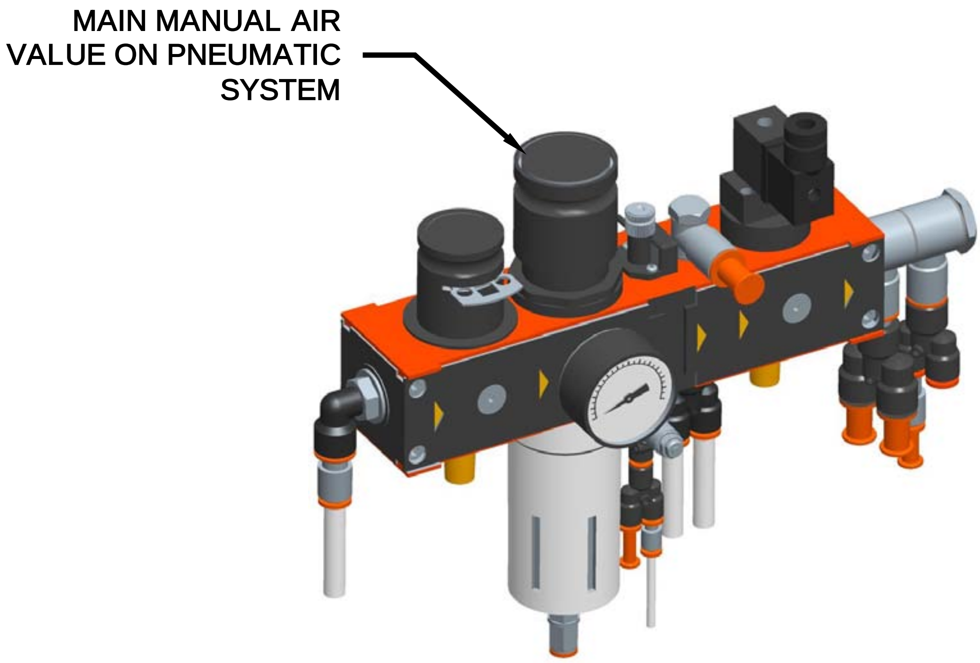
Function:

Cuts out the air supply

Method:

The pneumatic supply is to be cut out through the manual valve on the air treatment assembly.





Manual pneumatic valves are circuit selector valves and their activation inhibits the system air supply.

They shall be used only in maintenance phases, while the machine is not in operation.

Their aim is to increase safety during maintenance operations and they shall be padlocked to prevent involuntary activation.

Pressure restoration through these valves shall occur in full safety, with the guards closed, since it restores only air supply ignoring the state of the solenoid valves that control pistons; the machine shall not be in operation.

These devices are not intended to be emergency devices (they are not connected with any safety unit) and, therefore, cannot be used during the normal production cycle.

Their correct use includes the following steps:

- Guard closing
- Circuit breaker activation
- Disabling of the restoration control with a padlock (the key must be kept by the maintenance person)
- Interruption of power supplies
- Guard opening
- Maintenance operations
- Restoration of safety conditions (guard closing and fitting)
- Restoration of power supplies
- Removal of the padlock from the breaker
- Restoration of the network pressure

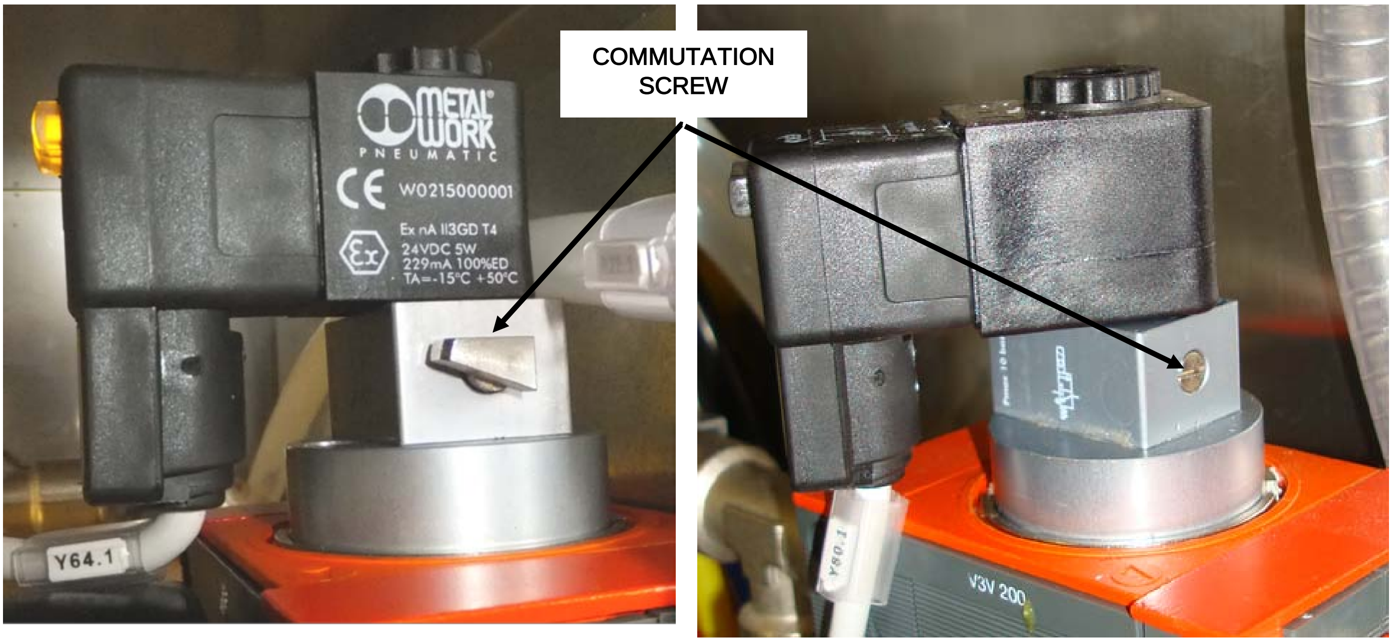
The solenoid valves can work on automatic or manual, depending on the position of the commutation screw on the coil solenoid.

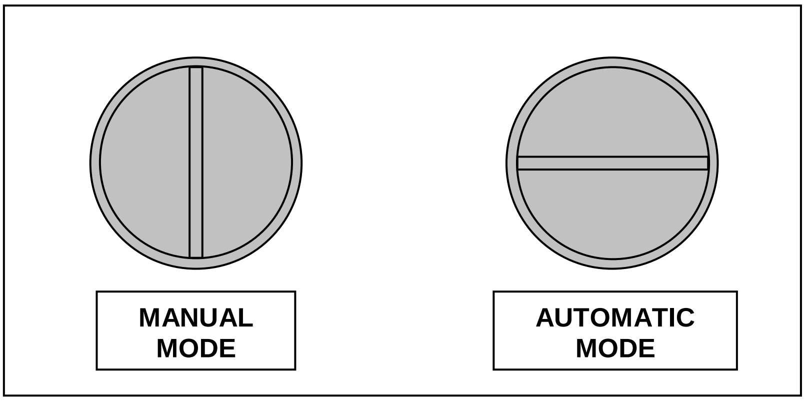
IT IS MANDATORY FOR THE SOLENOID VALVES TO WORK IN AUTOMATIC MODE.

To make sure the solenoid valves are on automatic, check that the commutation screw is in the horizontal position, as shown below. Before switching on the machine, make sure all the solenoid valves are in the automatic position.

Switching to manual mode (screw slot vertical), bypasses the safety circuit and is hence considered as tampering with the system. If this is done, G.Mondini S.p.A. disclaims all liability for damage or injury and any warranties will be invalidated.

NEVER SWITCH SOLENOID VALVES FROM AUTOMATIC TO MANUAL MODE.





3.5.3 Fixed and movable protections

Function:

Confinement of the line hazardous zones of the closing machine **TRAVE 350 VG**.

Method:

Fixed guards

The fixed guards surround the work area to prevent contact with the moving components.

They consist of plexiglass and stainless steel structures that are fixed onto the machine and stainless steel structures at the base of the machine.

The fixed guards are not controlled and must only be removed for maintenance operations, which must only be performed with the power supplies cut out.

After maintenance work, always remember to lock the hinged guards and the drop-down ones at the base of the machine.

Movable guards

The area where the sealing machine belt runs is protected by a gate fitted with an interlock microswitch that stops the machine when the gate is opened.

3.5.4 Emergency stop

Function:

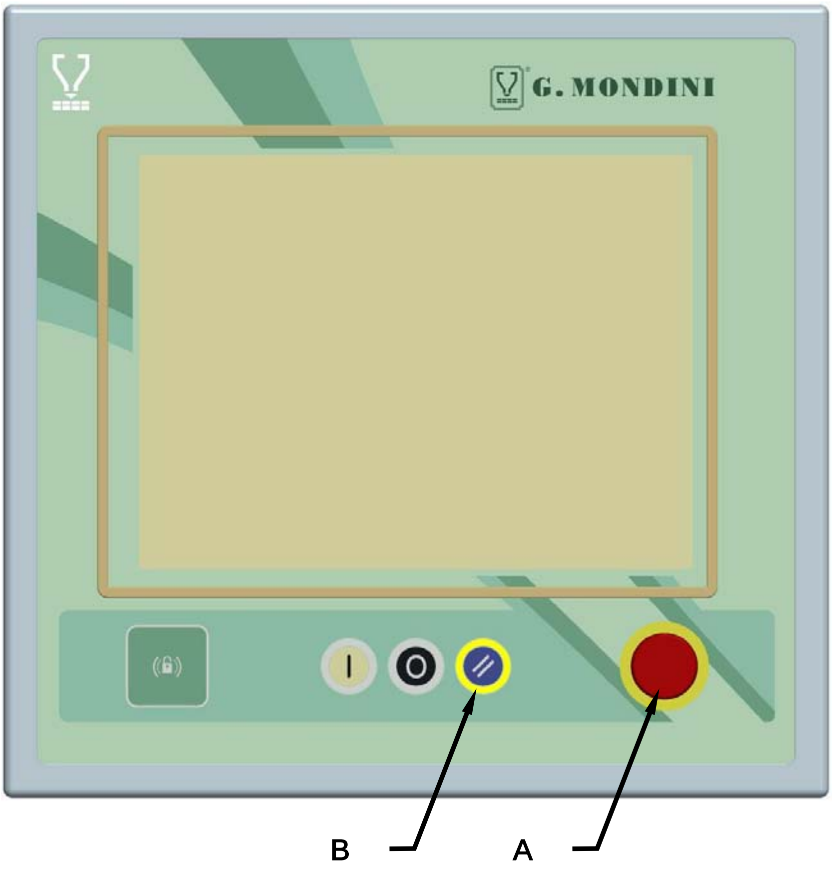
Immediate stop of the closing machine **TRAVE 350 VG** processing cycle

Method:

If one of the red “mushroom” buttons (**A**) on the operator panels is pressed, the line processing cycle is stopped, because the power to the control PLC outputs of all the line actuators is deactivated.

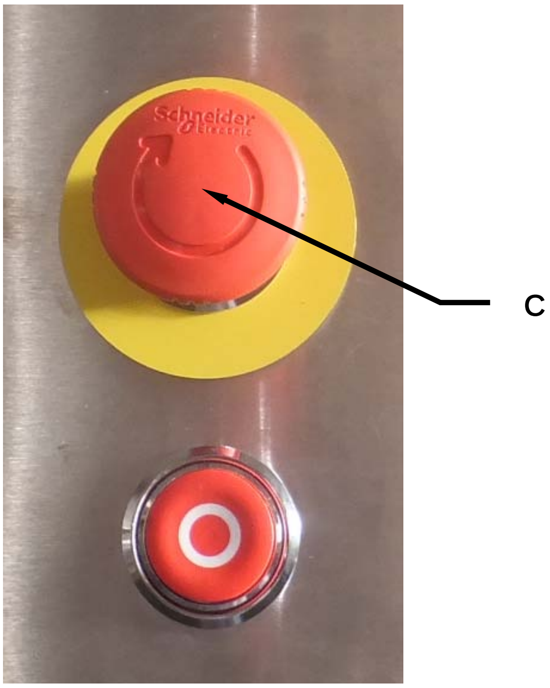
If pressed, the button is lowered and remains down in this position. To rearm, turn the button clockwise.

To restore the power is necessary press the EMERGENCY RESET pushbutton (**B**) on the panel.



If one of the red “mushroom” buttons (**C**) on the operator panels is pressed, the line processing cycle is stopped, because the power to the control PLC outputs of all the line actuators is deactivated.

To restore the power is necessary press the EMERGENCY RESET pushbutton (**B**) on the panel.



DANGER

Before servicing the machine or performing any other intervention, always press the emergency button associated with the opening of the moving guards to disconnect the machine from the power supply. Check operation of this safety device regularly.

The emergency circuits are to be checked periodically to ensure the correct functioning.

3.5.5 Safety limit switch with interlock on access door

Function:

Belt sliding protection door closure control.

Method:

The closure of the access door is controlled by safety limit switch.

The limit switch has a mechanical lock to prevent the door opening during the operating cycle.

⚠ DANGER

Before servicing the machine or performing any other intervention, always press the emergency button associated with the opening of the moving guards to disconnect the machine from the power supply. Check operation of this safety device regularly.

The safety device described above is to be checked periodically to ensure the correct functioning.

3.5.6 Automatic circuit breakers

Function:

Control the electric circuits and cut out the supply before dangerous overloads or short circuits are generated

Method:

If there is an overload and/or a short circuit, the switches cut out the supply to the circuits, causing the machine to stop immediately.

These switches are in the main electric cabinet.

3.5.7 Individual protection devices

Function:

To protect the operator when working.

Method:

The operators who are qualified and authorised to operate on the line are to use the individual protection devices to reduce risks and keep them unharmed.

> WARNING!
>
> The clothing of those who run the line or carry out maintenance on it is to conform the safety requirements specified in the EU directive 89/656/EEC and the laws in force in the country where the line is installed.

> DANGER!
>
> To avoid mechanical risks, such as dragging, trapping and others, do not wear bracelets, wristwatches, rings or chains during operations in processing cycles and maintenance.

4 TRANSPORT AND INSTALLATION

CONTENTS

4.1 TRANSPORT AND MARKING

The operations described in the following pages are only to be carried out by authorised persons.

The persons assigned to all the installation operations described in this chapter regarding packing, handling, transport, unloading, positioning, connections, checks and controls of the Closing Machine, are to be well trained and have thorough knowledge of accident prevention standards.

Unauthorised persons must remain outside the area of these operations.

The accident prevention precautions and the operations to be carried out that are contained in this chapter are to be always strictly observed during the various operations, to avoid accidents to persons and damage to the fixtures.

☞ NOTE:
For information regarding the handling and lifting of commercial components that are installed on the line, see the specific use and maintenance handbooks.

4.2 PACKING

The size and structure of the line does not permit the shipping of the same fully assembled; therefore the structure and the enclosures have to be dismantled.

All machined surfaces are to be covered with a film of lubricant.

Before starting operations, it is best to define with certainty all the lifting points on the line components, so that this stage is carried out in safety.

If the line is installed for preliminary tests in the manufacturer's plant, the packing and re-assembly in the Customer's plant will be carried out by G. MONDINI S.p.A.

The lifting of the line assemblies can be by means of lift truck or with a crane/hoist.

In the first case, position the assembly on an appropriate wooden pallet. The entry points for the trolley forks should be properly identified so that the assembly is lifted in a stable and safe manner (see figure 4-1).

In the second case, sling the assembly with the ropes, taking care to protect parts that could become damaged by the rubbing of the cords with rags and wooden blocks and balance the part before lifting it (see figure 4-2).

!

WARNING!

When the load has been lifted to a height of more than 30 cm, the operators assigned to the lifting must remain at a safety distance of more than 2 metres from the perimeter of the load. A breakdown on the lifting equipment or an uncontrolled movement of the load is a serious hazard for the safety of the persons.

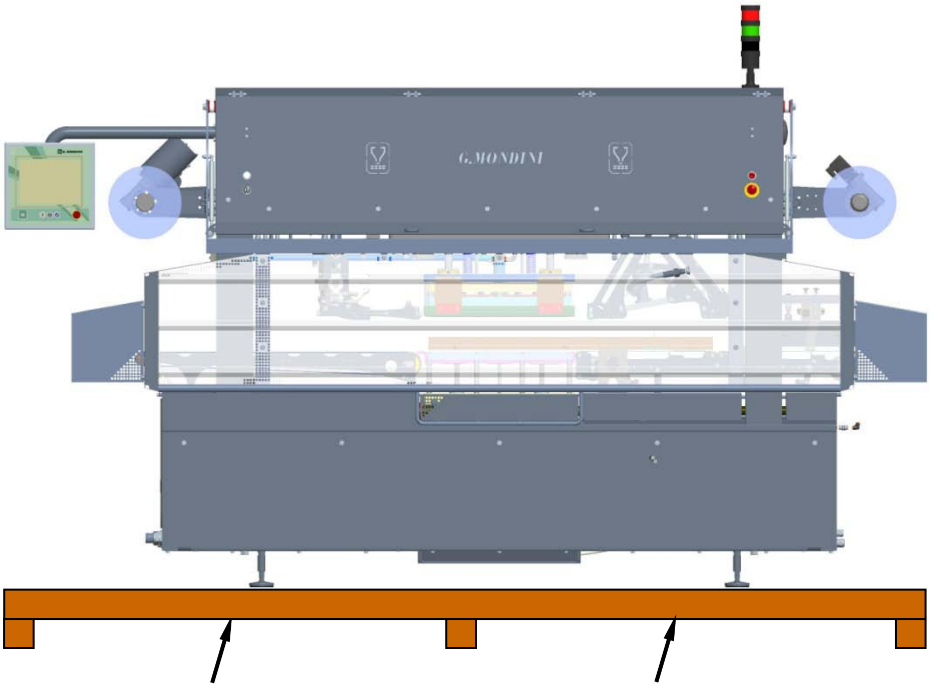


Figure 4-1: Lifting using a forklift truck.

> NOTE:
> When transporting the machine on a pallet, make sure that the forks are properly inserted at the points shown by the arrows.

Take care to lift the machine keeping the centre of the machine as a reference and make sure it is properly balanced.

If the machine is lifted without a pallet, make sure that the forks are placed at machine's bearing points and they do not damage any parts of the machine that protrude or collapse.

Do not place any wooden inserts, shims or cloths between the forks and the machine to prevent it from slipping.

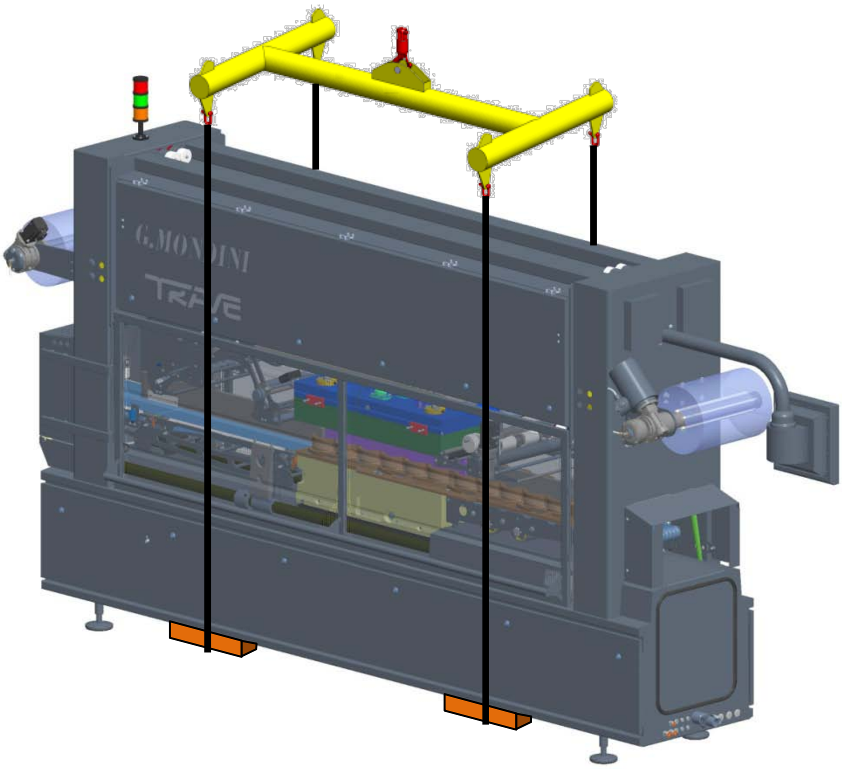
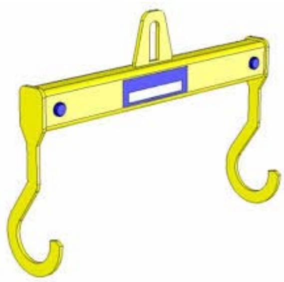
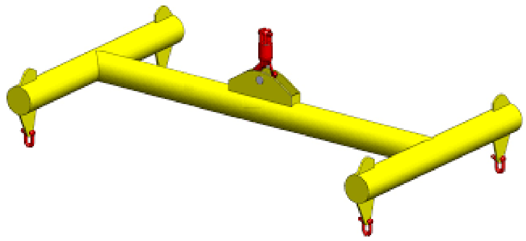


Figure 4-2: Lifting using ropes, cables or belts (crane/hoist)

NOTE:
We recommend using a lifting sling based on one of the systems shown in the pictures below so as to keep the ropes detached from the external cover of the machine.

NOTE:
It is recommended to place wooden inserts between the machine and the ropes in order to prevent sheet metal from being crushed as a result of the stress generated by the ropes.





4.3 TRANSPORT

☞ **NOTE:**
G. MONDINI S.p.A. shall not be held liable for any damage to persons or goods caused by wrong handling of the loads, executed by unskilled persons and/or using inadequate lifting equipment.

The line parts are shipped, according to the destination, in these ways:

BY SEA ⇒ the assembly is packed in a flat bottom crate and anchored by tie-bars. The crate is lined with tarred paper and has a flap for Customs inspections; it contains bags with drying salts to protect against humidity and salt.

BY AIR ⇒ the assembly is packed in a flat bottom crate and anchored by tie-bars. The crate is lined with tarred paper and has a flap for Customs inspections; it contains bags with drying salts to protect against humidity and other atmospheric agents.

BY LAND ⇒ Land transport can be divided into two types:

LONG DISTANCE TRANSPORT ⇒ the assembly is covered with protective sheets, enclosed in a flat bottom wooden cage and clamped with tie-rods to the loading bed of the semi-trailer.

To lift the crate, scrupulously follow the instructions stamped on the outside of the packing. The packaging can be kept for further use; it is therefore a good practice to keep it in a protected area so that it does not become damaged and hence not reliable. If it is to be disposed of, it will be the responsibility of the **Customer** to dispose of it in compliance with the laws in force in the country.

MEDIUM AND SHORT DISTANCE TRANSPORT⇒ each separate component of the line is fastened to a bed and covered with protective sheets.

4.4 CHARACTERISTICS OF THE INSTALLATION AREA

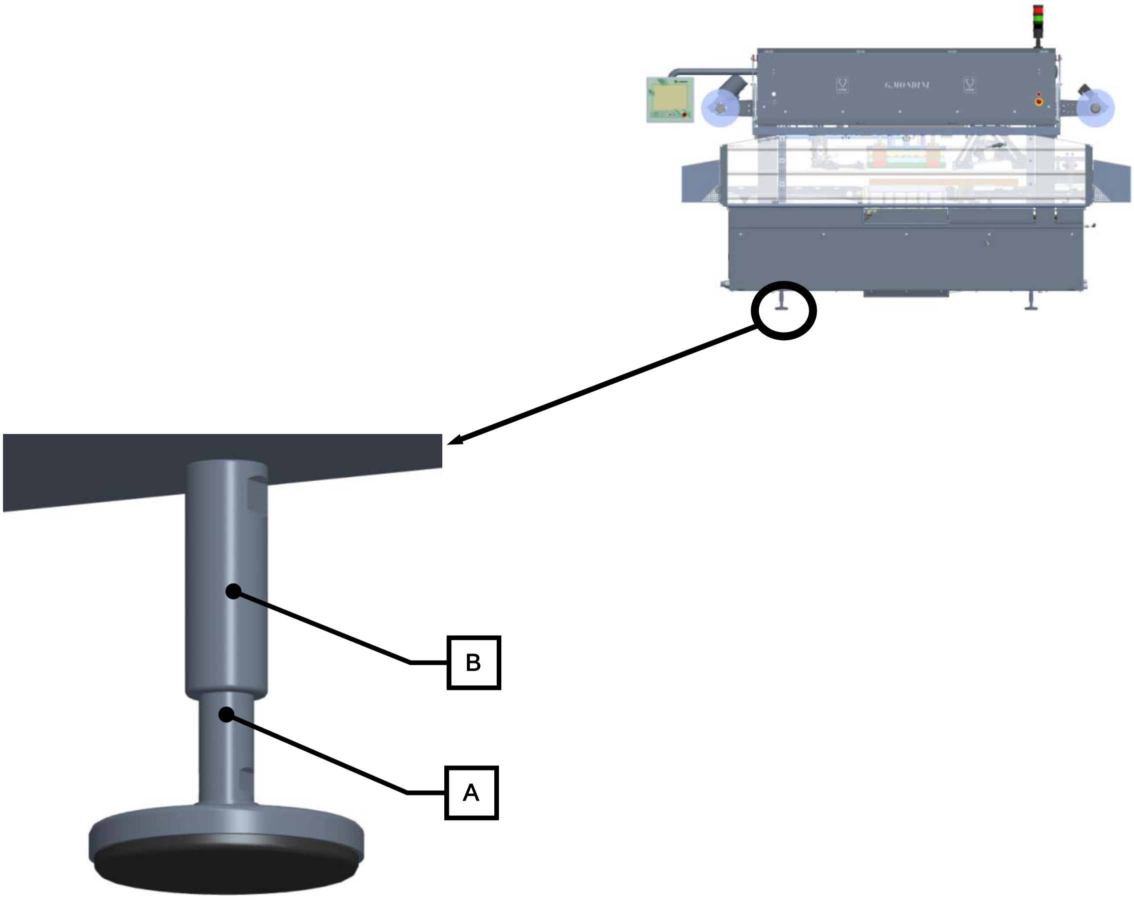
Check that the area selected for the installation is compatible with the overall dimensions on the lay-out and that there are the required environmental conditions.

There must be sufficient space around the line to permit operations and especially maintenance operations.

The floor must be level and flat, dimensioned to sustain the weight of the line to be installed and without ruts or steps.

4.5 POSITIONING

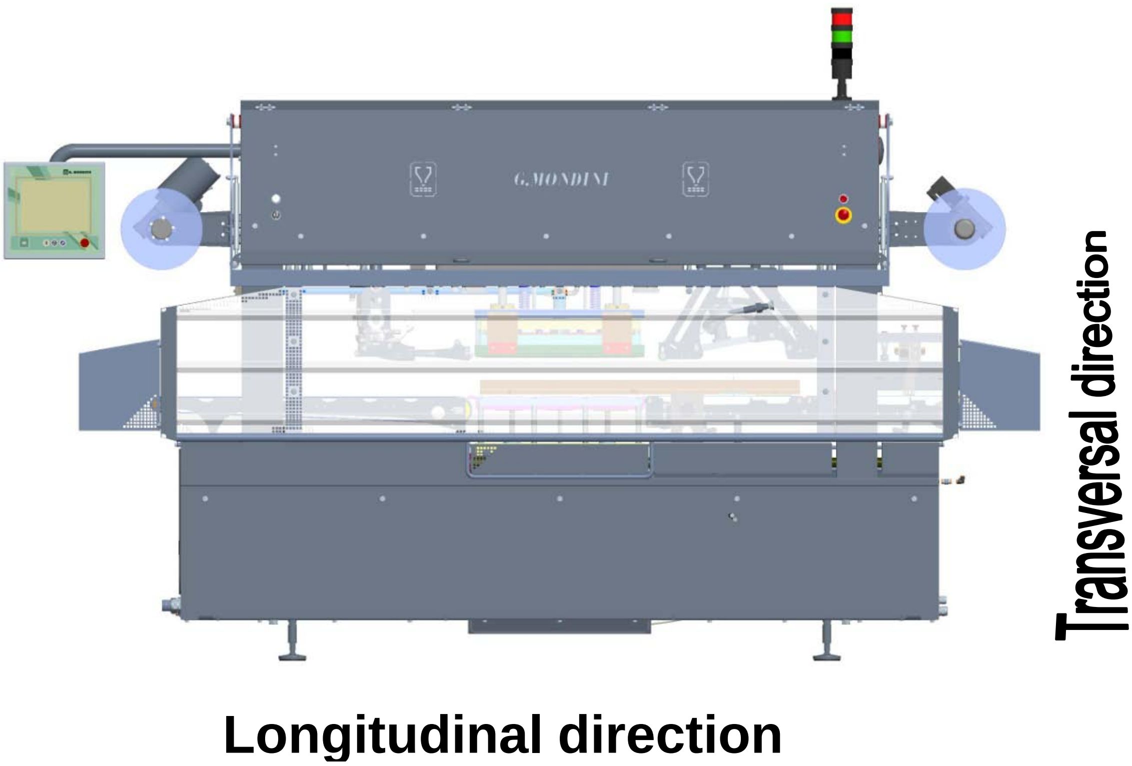
After the site has been decided, the line assemblies are to be positioned, placing them on their feet, to which the rubber vibration dampers are added.



Once the line is in position, the planarity adjustment operations have to be performed.

Make a first alignment with the level adjustment screws (**A**) on the feet (screw to lower and unscrew to raise).

Place a level (of adequate precision) on a line bearing surface and with the alignment screws adjust the planarity, both lengthwise and crosswise. After adjustment has been made, lock all the fastening feet nuts (**B**).



4.6 CONNECTION TO EXTERNAL ENERGY SOURCES

4.6.1 Electrical circuit connection

!

WARNING!

The machine power supply connection operations are only to be carried out by skilled personnel who are to use all the necessary personal protection devices.

The electrical power supply connection is to be carried out in compliance with the safety standards of the country where the line is to be used.

!

WARNING!

Before starting the electrical connection, check that the power supply cut-out switch is locked in open position.

Check that the power line is dimensioned to the system total load.

Connect the system to earth before making any other connection to the mains.

Check the correct voltage value indicated on the line ID plate and the type of connection on the power supply transformer on the electric panel.

It is advised to install an automatic cut-out device at the beginning of the supply line for easier maintenance operations.

Upstream of the main electric panel, use cut-out devices with an adequate cut-out capacity for the switches on the panel.

Before powering the system, make the checks indicated in the “ELECTRICAL CONNECTION CHECK” paragraph.

The point of connection is located inside the main console.

This position is safe as there is no interference with the moving parts and does not affect the insulation of the console and all the parts connected to it.

The cables are to be connected directly on to the terminals of the main switch. The connection of the earth conductor is to be fastened on the earth terminal as specified on the electrical drawings.

After all connections have been made, refit the terminal covers.

4.6.2 Pneumatic connection

⚠ **WARNING!**

Connection to the machine supplies are only to be carried out by skilled personnel and they are to use all the necessary individual protection devices.

⚠ **WARNING!**

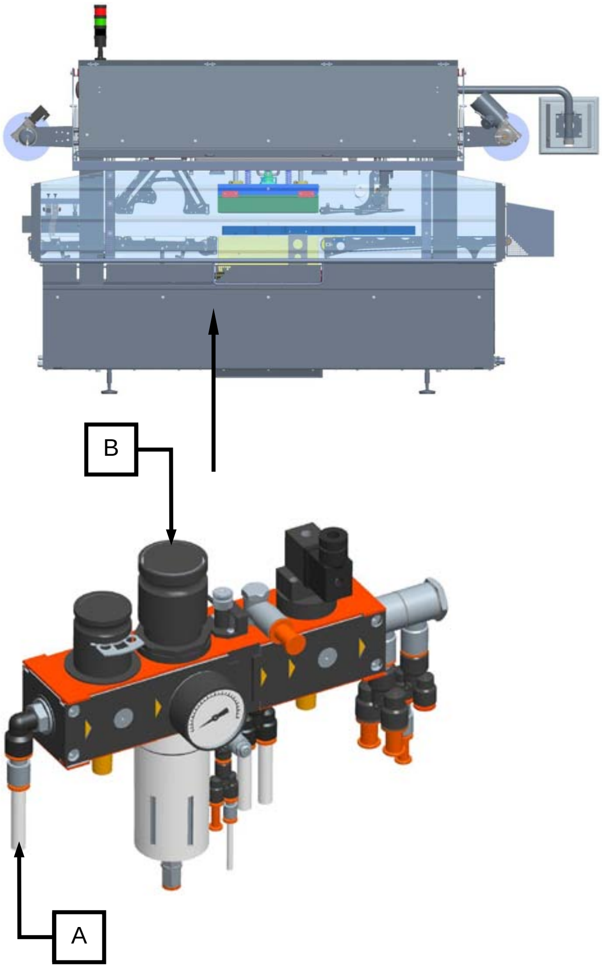
Make the machine pneumatic connections correctly, checking the correct pressure of the pneumatic system.

Make sure that the manual cut-off valve is closed.

Before switching on the system pneumatic supply, make the checks indicated in the “PNEUMATIC CONNECTION CHECK” paragraph.

When the pneumatic supply is activated, there could be uncontrolled movements of actuators. Therefore when carrying out the operations be sure there is nobody in the hazardous working zone.

The connection (**a**) of the compressed air tube is to be made on the main air treatment unit of the actuators supply branch, upstream of the manual cut-out valve (**b**).



Use fittings and tubes that have a higher resistance than the maximum compressed air network pressure.

4.6.3 Vacuum circuit connection

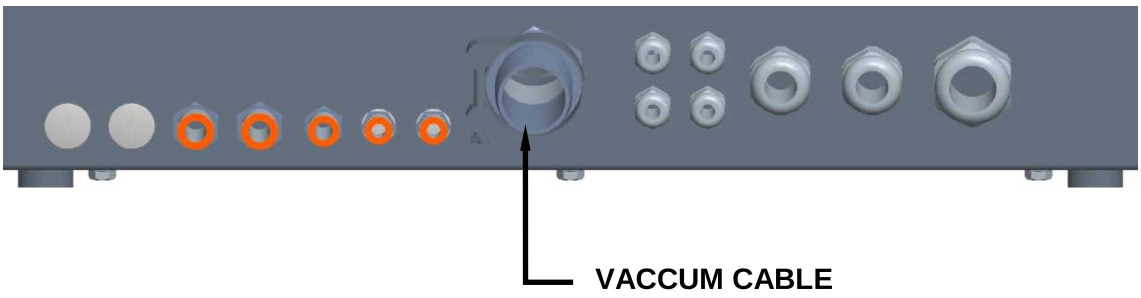
WARNING!

Connection to the machine supplies are only to be carried out by skilled personnel and they are to use all the necessary individual protection devices.

WARNING!

Make the machine vacuum connections correctly, checking the correct pressure of the vacuum system.

The connection point of the closing unit is inside the vacuum panel (behind the electric panel). To reach it , use the hole in the pedestal that sustains the vacuum panel, as shown in the next figure.



4.6.4 Gas circuit connection

WARNING!

Connection to the machine supplies are only to be carried out by skilled personnel and they are to use all the necessary individual protection devices.

WARNING!

Make the machine gas connections correctly, checking the correct pressure of the gas system. MAX 8 bar!

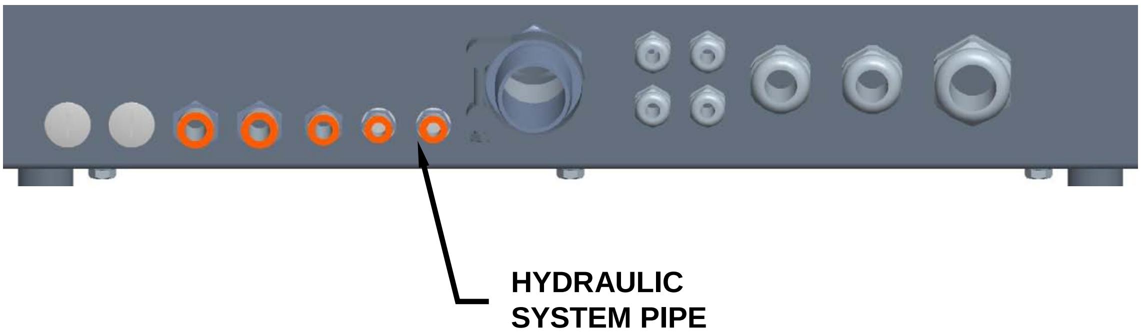
To connect to the gas circuit, the tube coming from the plant mains line is to be inserted in the machine mask, as shown in the next figure.

4.6.5 Hydraulic system connection (if provided)

> **WARNING!**
>
> The connection operations of the various machine supplies must be performed solely by appropriately specialised and qualified personnel and are subject to compliance with personal protection equipment requirements.

> **WARNING!**
>
> Proceed with correct hook-up of the machine's hydraulic system connections by checking to ensure that the system pressure levels are appropriate. **MAX 4 bar!**

For hook-up to the hydraulic system, insert the pipe leading from the factory supply circuit into the machine mask, as illustrated in the figure hereunder.



4.7 FIRST START-UP

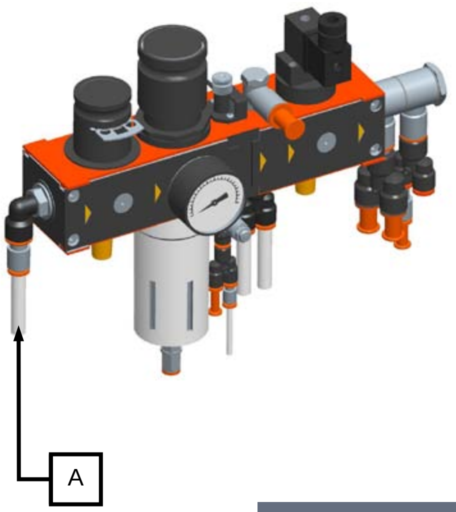
4.7.1 Preparation for the first start-up

Before starting up the line, check the serviceability of all the connections made and described in the previous paragraphs, and make the checks indicated below.

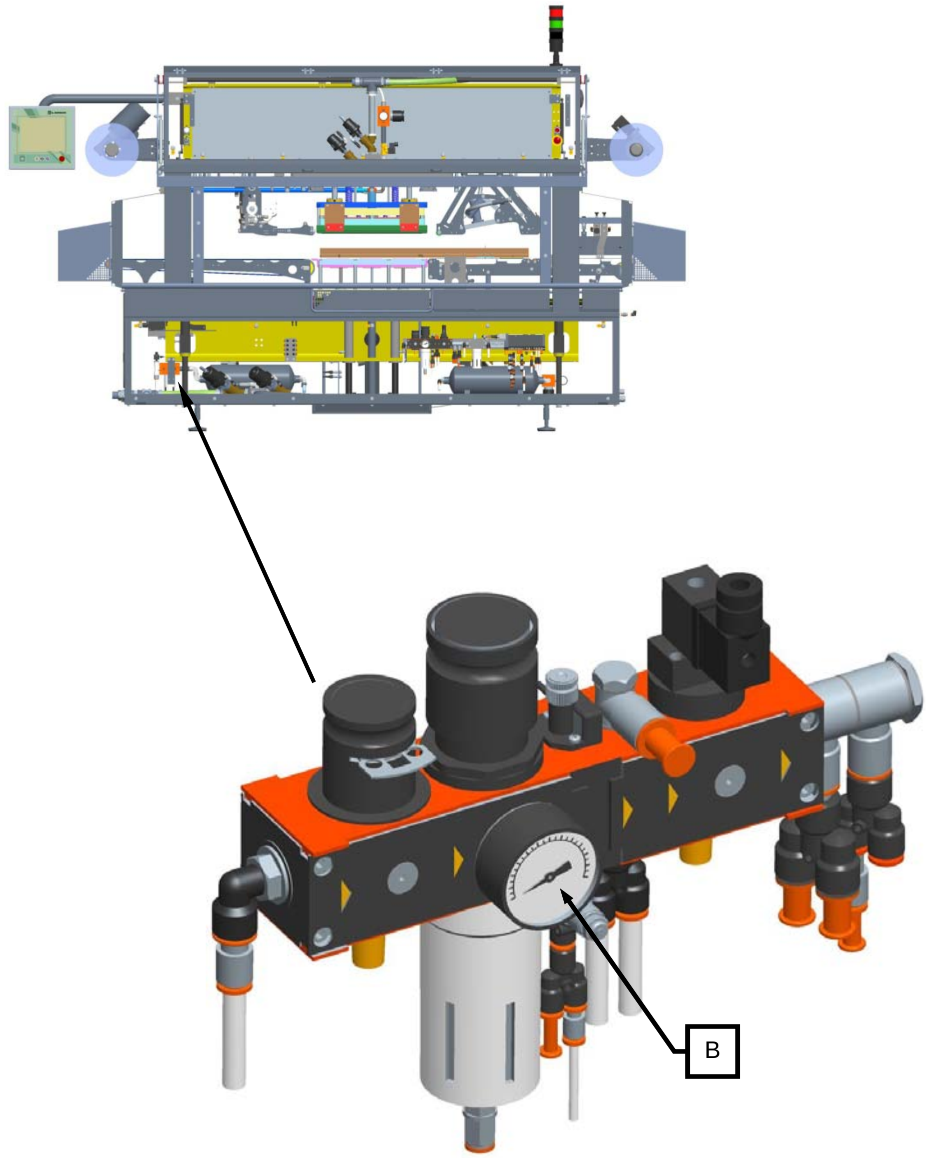
4.7.2 Pneumatic connection check

Proceed as follows:

1. Check that the mains air tube (**A**) has been connected correctly.



2. Check that the pressure value indicated on the air treatment unit gauge (**b**) is not lower than 6 bar.

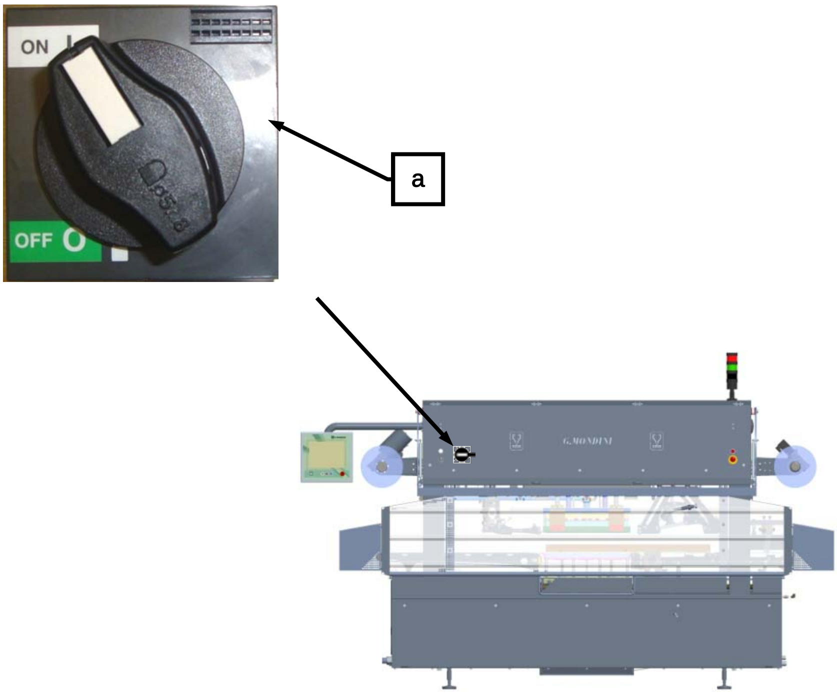


3. Check that the air supply tube fittings to the components of the pneumatic system have been tightened correctly

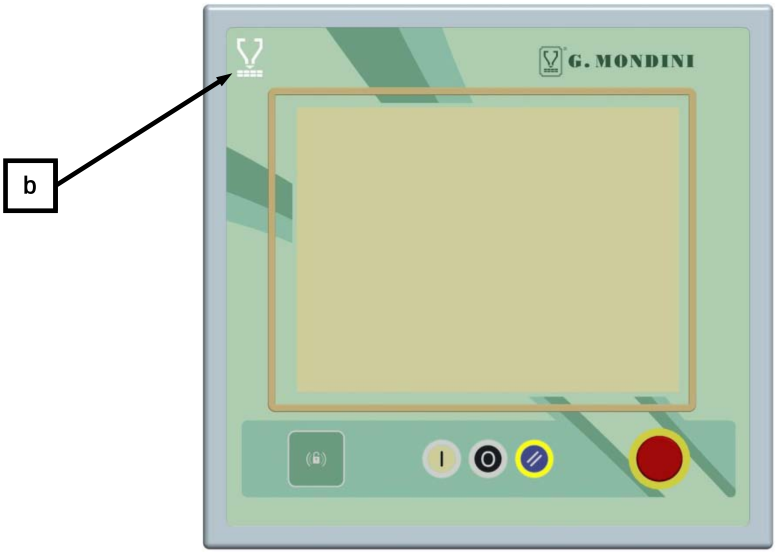
4.7.3 Electrical connection check

Proceed as follows:

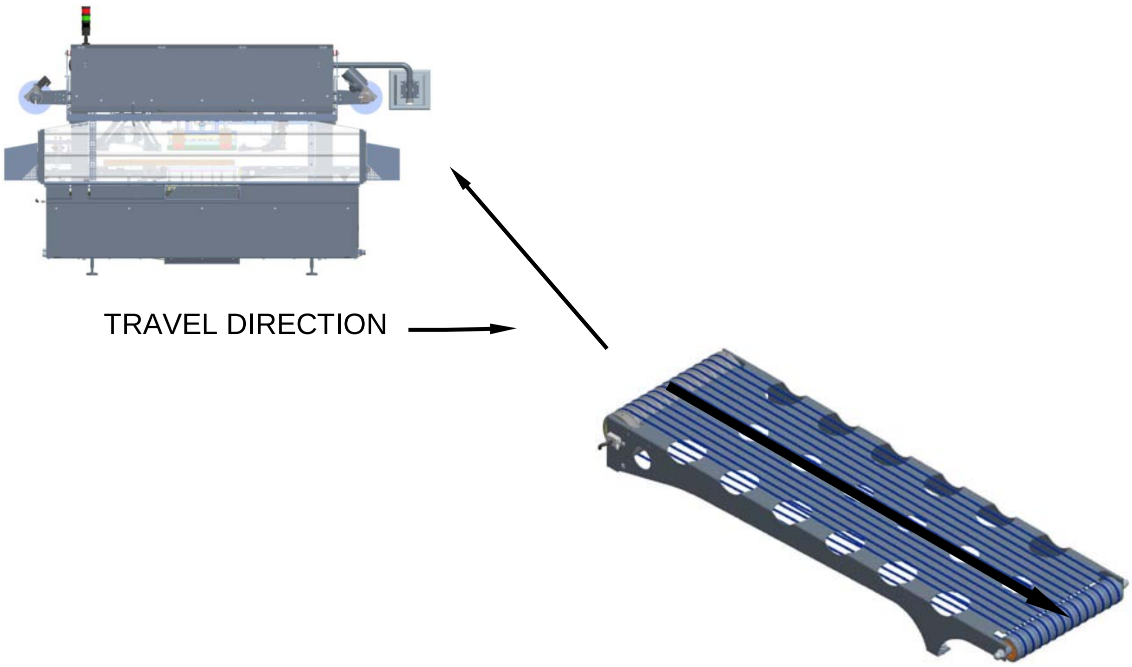
1. Supply the line by positioning the selector switch (**a**) on the electric power panel to 1.



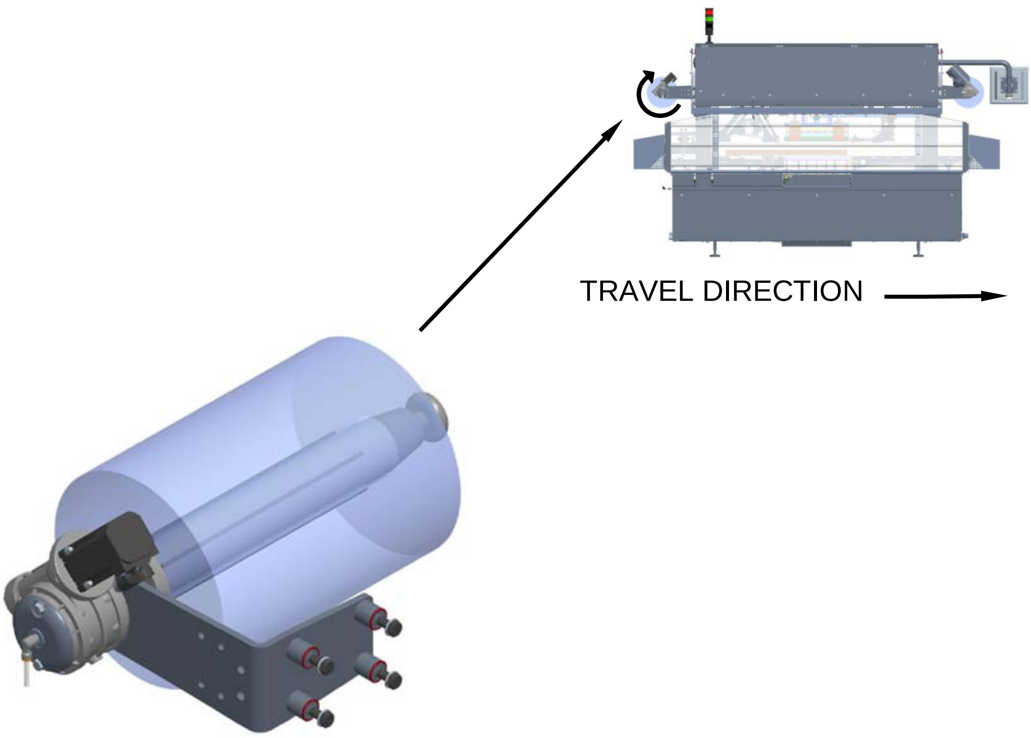
2. Check that on the control panel the “CURRENT INSERTED” (**b**) warning light switches on.



3. Check the closing machine exit belts move in the direction indicated by the arrow in the next figure.



A further check can be made on the direction of the bobbins rotation, which is to be as indicated on the shaft itself.



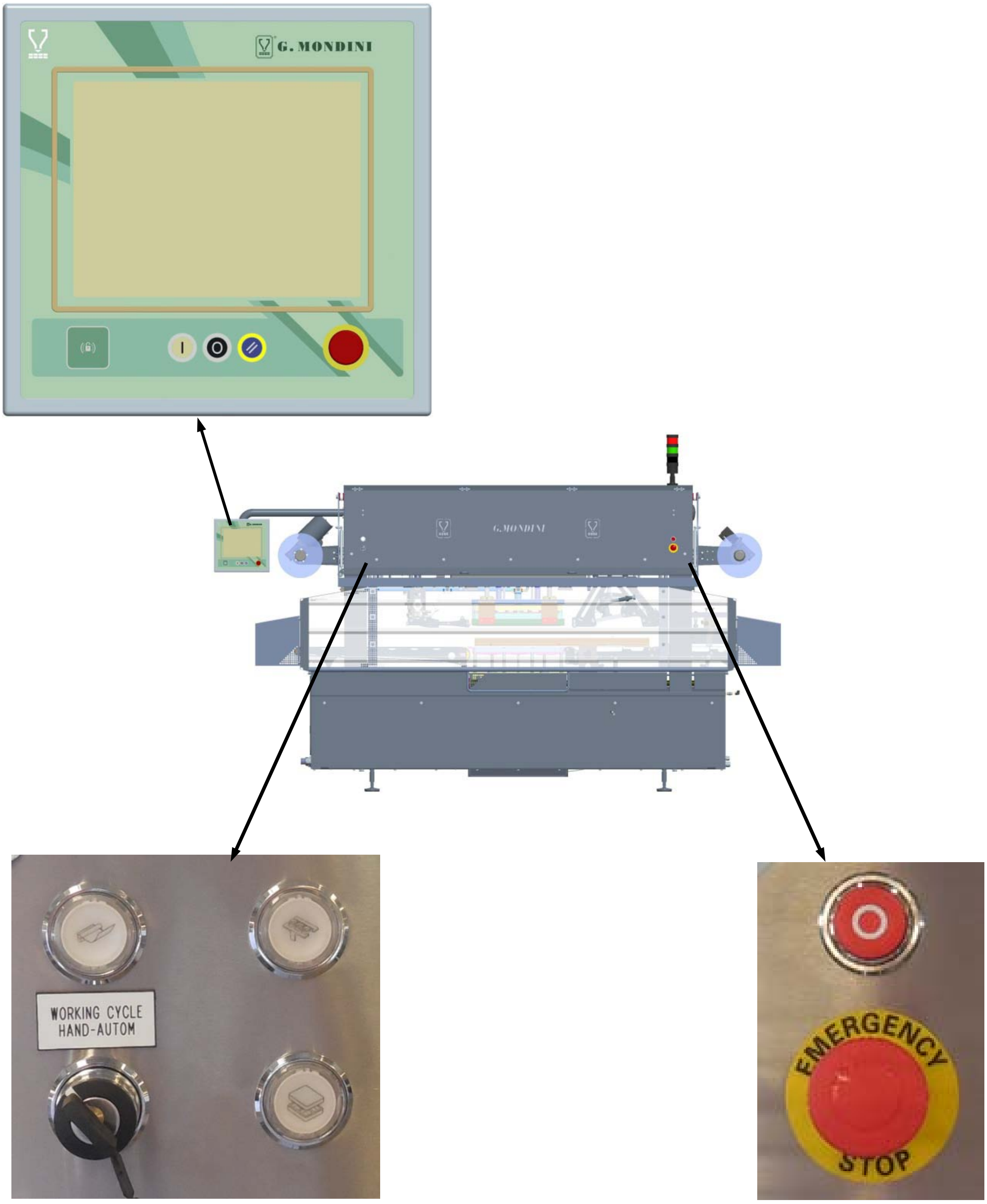
If one of these components should rotate in the wrong direction, shut off the machine immediately and reverse the cables on the terminals of the main switch.

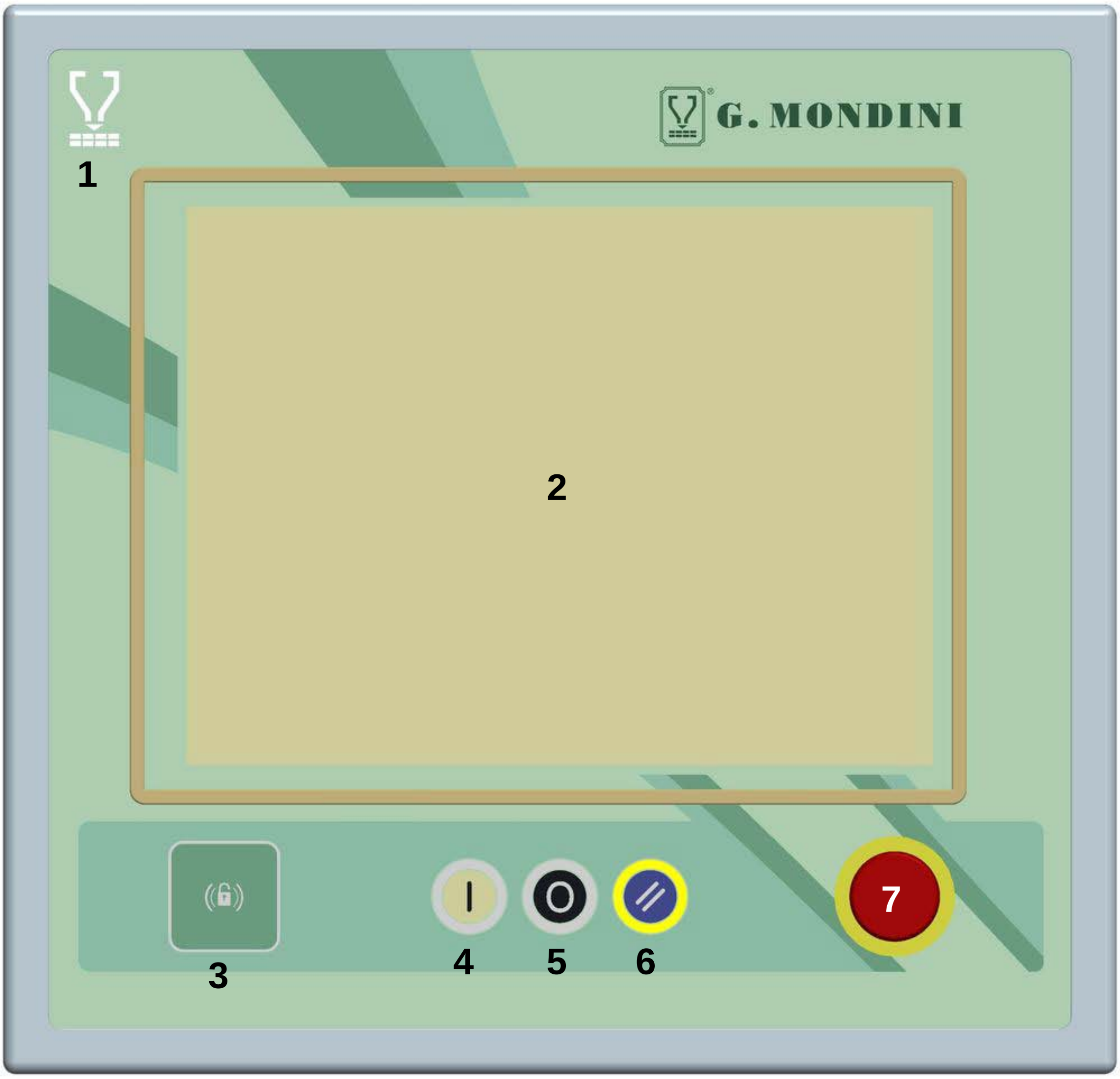
5 CONTROL SYSTEM

INDEX

5.1 GENERAL DATA

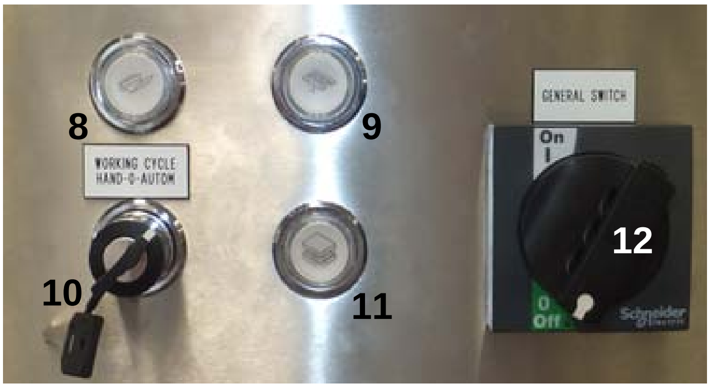
The Sealing Machine's Control System is constituted by a control panel comprising all the commands needed to guarantee control and operation of the machine. Assembled onto a swivel arm, it is fixed onto the machine's top panel.





1	When it lights up, the white lamp signals that current is inserted to the machine, i.e. that electric power supply is provided.
2	This operator panel is connected up to the PLC module through which all necessary production parameters are set and entered.
3	This device allows to effect the "Login" to a determined level of password. For this machine this device is optional.

4	This button has the function to activate the machine when ready for the production. When this button is pressed with the machine in condition of alarm the machine it doesn't work.
5	The stop cycle pushbutton is used for immediate interruption of machine operations. The peculiarity of this pushbutton is that it provides for “focused” intervention, i.e. the intervention takes place only at the end of the cycle underway. This allows the machine, prior to stopping, to position all the elements subject to cyclic operation, back into their cycle start positions. As mentioned, the stop command does not involve all the machine elements, but only those elements that operate according to subsequent closed cycle chains. An example thereof is the pusher unit assembly motion. Elements subject to continuous operations such as the sealing or snap-on lid outfeed belt or the film waste winding device motor, remain active. For the machine to resume operations after this pushbutton has been pressed, simply press on the “Start” pushbutton again.
6	Button of zero resetting of the alarms in progress. This button is used before using the button of march to take back the production.
7	Pressing this pushbutton will instantly stop all operations and movement of the sealing machine’s in-line elements, regardless of the position they are in.



8	The white "manual cycle lid unrolling/decoiler pushbutton" will activate the machine's relative unrolling cycle. If this pushbutton is set accordingly, it will work both in the "Automatic" cycle and in the "Manual" cycle modes.
9	The white "manual cycle pusher arms" pushbutton will only work in the "Manual" cycle mode. Pressing this pushbutton continuously will execute a complete cycle by the pusher arms, that will come to a stop in cycle-start position. If this pushbutton is instead pushed in intermittently, the same cycle is executed but in a fragmented manner. This pushbutton, furthermore, also has a function when the machine is set in the "Automatic" cycle mode. This option is implemented when, at the end of production, the very last containers sealed by the machine needs to be removed from the sealing area. As there are no other containers being fed in along the line, the machine stops in standby mode. Pressing this pushbutton will provide for one complete cycle enabling the sealing area to be totally emptied.
10	The position of this switch determines the type of machine operation mode required: When this selector is in "Manual" position, the machine will run only if the relative commands are active, i.e. only if the manual operation pushbuttons have been activated. When this selector is in "Automatic" position, once the start pushbutton has been pressed, the machine will operate completely independently and automatically. When this selector is in "0" position, the machine goes into standby mode. To reset the machine operating conditions, simply switch this selector either to the "Manual" or to the "Automatic" position and then proceed as if to start up operations.

11	The white “manual cycle tool closure” pushbutton will only work in the “Manual” cycle mode. Holding this pushbutton pressed in continuously will execute a complete cycle by the sealing tool, that will come to a stop in cycle-start position. If this pushbutton is instead pushed in intermittently, the same cycle is executed but in a fragmented manner. This pushbutton does not have an active function during machine operations and it is only used for check-ups and verification of the tool and of the various tool movements.
12	This selector activates power supply to the machine.
13	The stop cycle pushbutton is used for interruption of machine operations at the end of cycle.
14	Pressing this pushbutton will instantly stop all operations and movement of the sealing machine’s in-line elements, regardless of the position they are in.

ATTENTION! While the panel is starting don't touch the screen up to the project loading bar (**A**) in the lower part of the page disappears.

It is possible touch the screen only when the following page is visualized:

5.2 NAVIGATION KEYS

Each page on the operator panel displays a bar of navigation keys.

The arrow keys at the ends of the bar can be used to scroll through the various icons.

Touching any icons displays related sub-pages containing parameters or a description of the associated functions.

 MACHINE SYNOPTIC PAGE (HOME PAGE)

 SAFETY PAGE (SAFETY DEVICES)

 PLU PAGE (WHERE AVAILABLE)

 RECIPE PAGE

 LOCKING PAGE

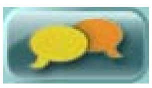
 COMMANDS PAGE

 MAINTENANCE PAGE

 TOOL HEATING PAGE

 MOST USED PAGE

 FAST ICONS

 CHAT FOR TECHNICAL ASSISTANCE WITH G.MONDINI FIRM

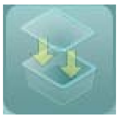

LINE PAGE (WHERE AVAILABLE)


TRAYS DENESTER PAGE (WHERE AVAILABLE)


DOSER PAGE (WHERE AVAILABLE)


DEVIATOR PAGE (WHERE AVAILABLE)


TRAY TURNER PAGE (WHERE AVAILABLE)


SNAP ON LID MACHINE PAGE (WHERE AVAILABLE)

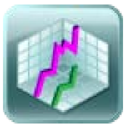

MOBILE HEAD PAGE (WHERE AVAILABLE)

ENERGY PAGE

OMAC PAGE

CAM VIEW PAGE

PRODUCTION DATA PAGE



TRENDS PAGE

QUALITY PAGE

NOT USED KEY


FILE CONTROL PAGE (USB)


MULTI-LANGUAGE TEXT PAGE


USER AND PASSWORD PAGE

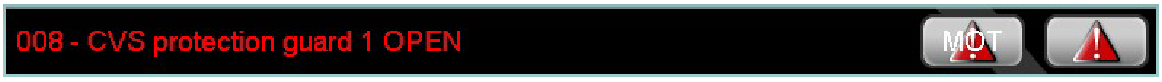

WORKSHIFT PAGE


NOT USED KEY

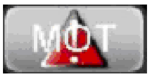

PANEL CLEANING PAGE


HELP PAGE


WINDOWS EXIT (ONLY WITH "MONDINI" USER AUTHORISATION)




ACTIVE ALARMS PAGE



MOTION ACTIVE ALARMS PAGE (if available)

5.3 PAGE HEADING

Access to machine mode setting (maintenance or pause) and shift display.

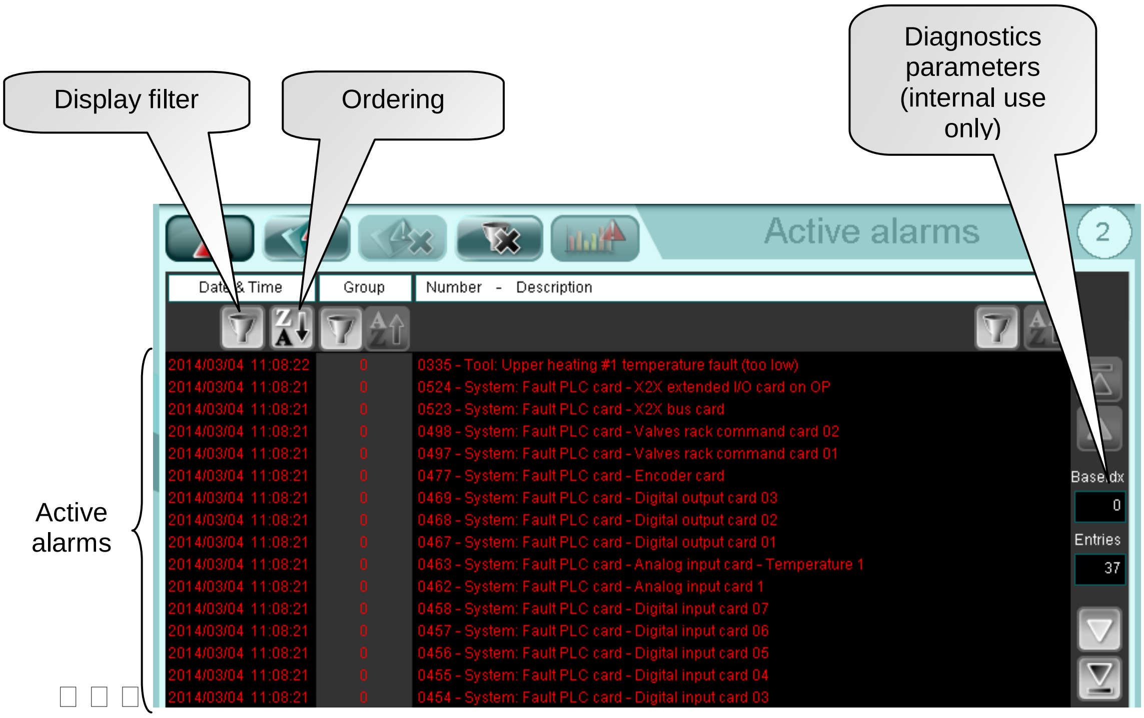
Date and time setting and display

User login

Language selection

5.4 ALARMS

This page displays a list of active alarms occurring during production due to system malfunctions.




ACTIVE ALARMS PAGE


ALARM HISTORY PAGE

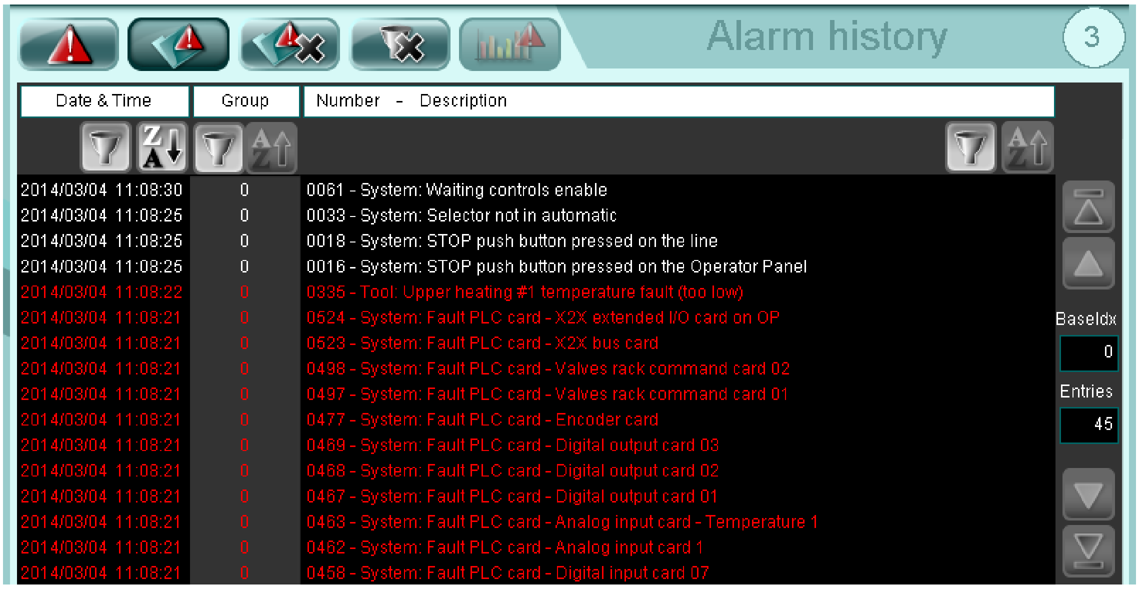

SEARCH FILTER REMOVAL


ALARM LIST ORDER


ALARM LIST FILTER

5.5 ALARM HISTORY

This page displays a list of the alarms that have occurred during production since the last reset.




ACTIVE ALARMS PAGE


ALARM HISTORY PAGE


ERASE ALARM HISTORY


DISABLE SEARCH FILTER


ORDER ALARM LIST


ALARM LIST SEARCH FILTER

5.6 INVERTER AND AXES ALARMS

From the main menù bar press the 

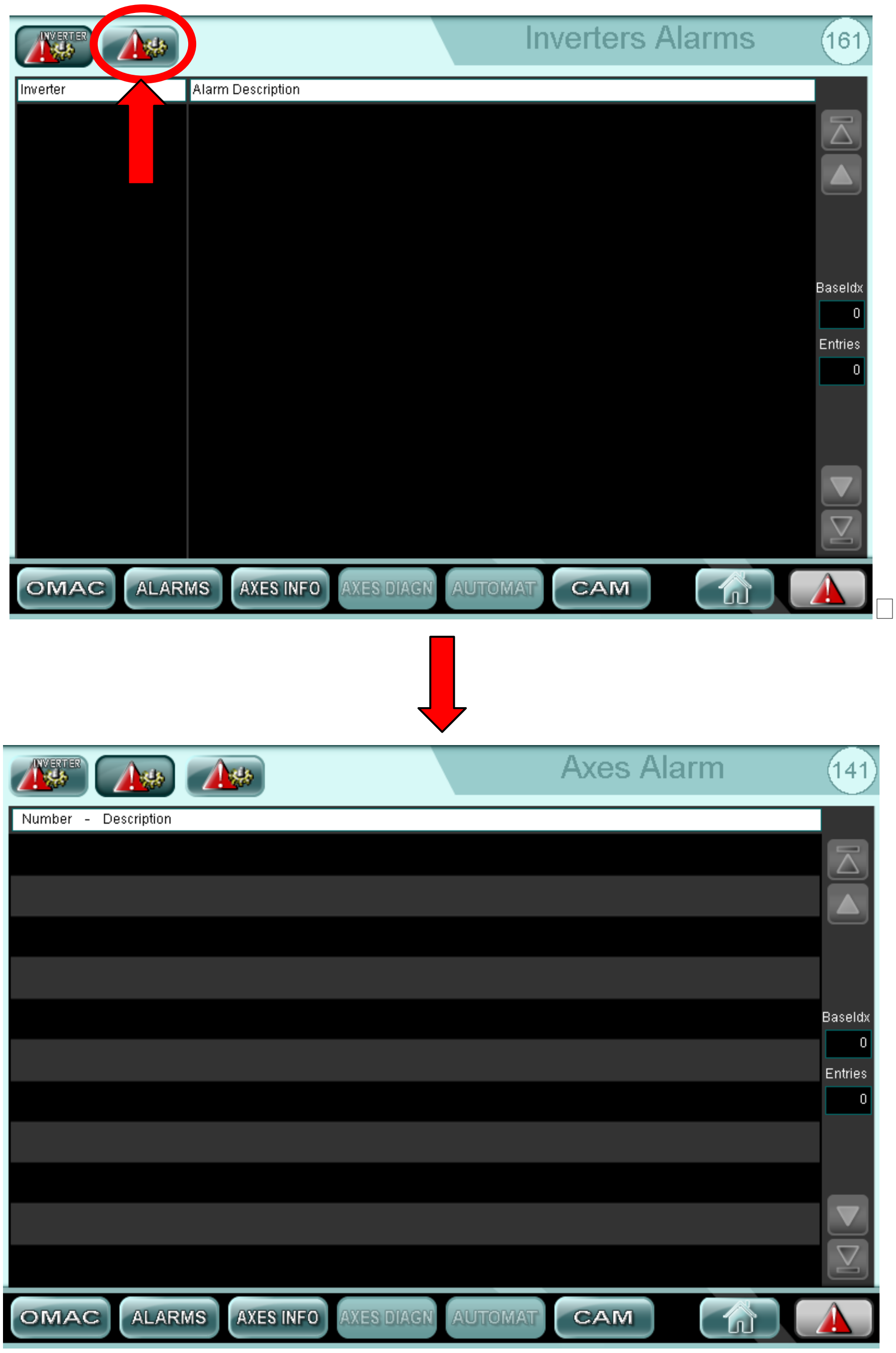
key to access the following page.

INVERTER ALARMS PAGE

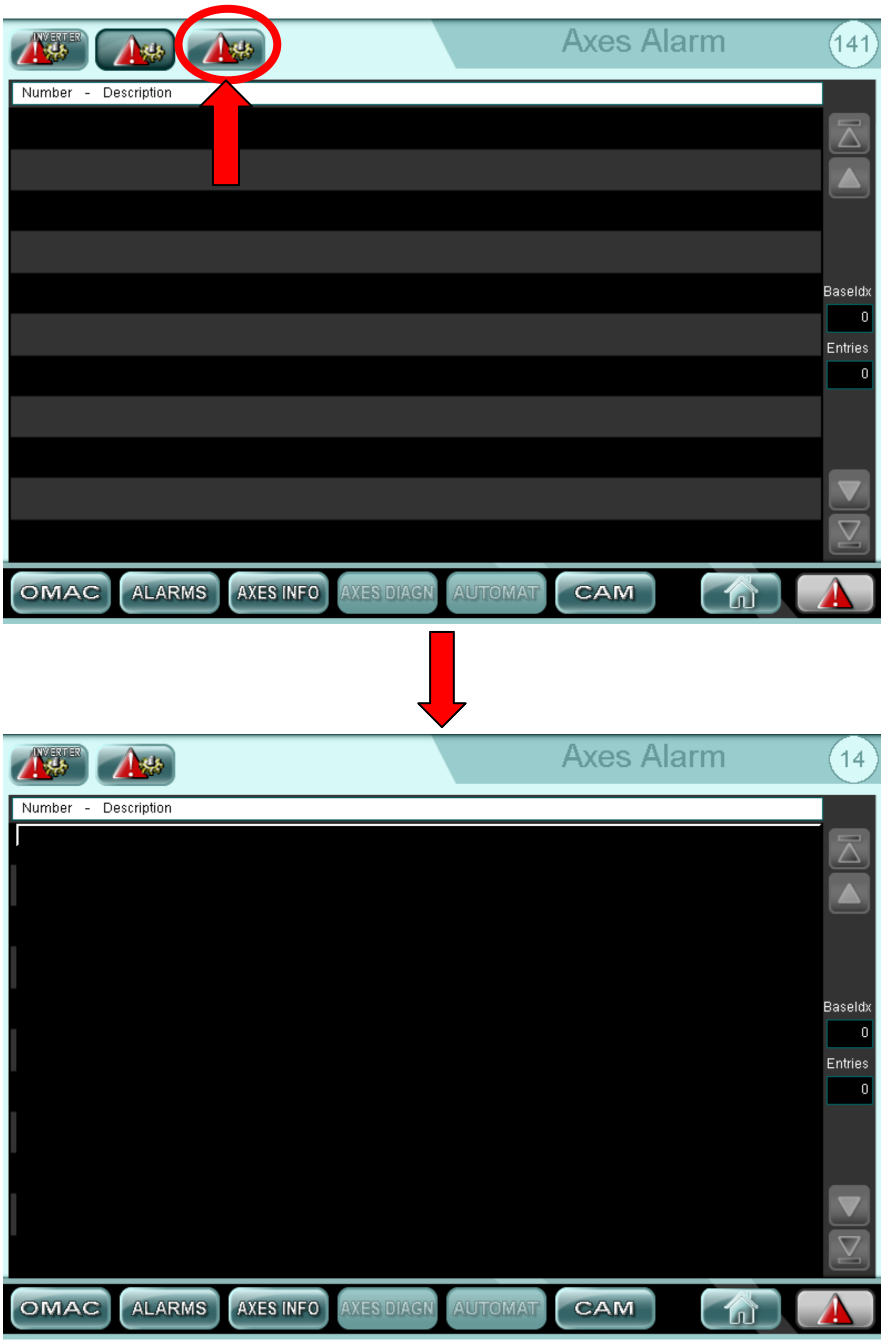
ACTIVE ALARMS PAGE

From this page you can see the list of alarms concerning the inverter installed on the machine.

By using the key  you enter the page which lists the alarms concerning the axes.



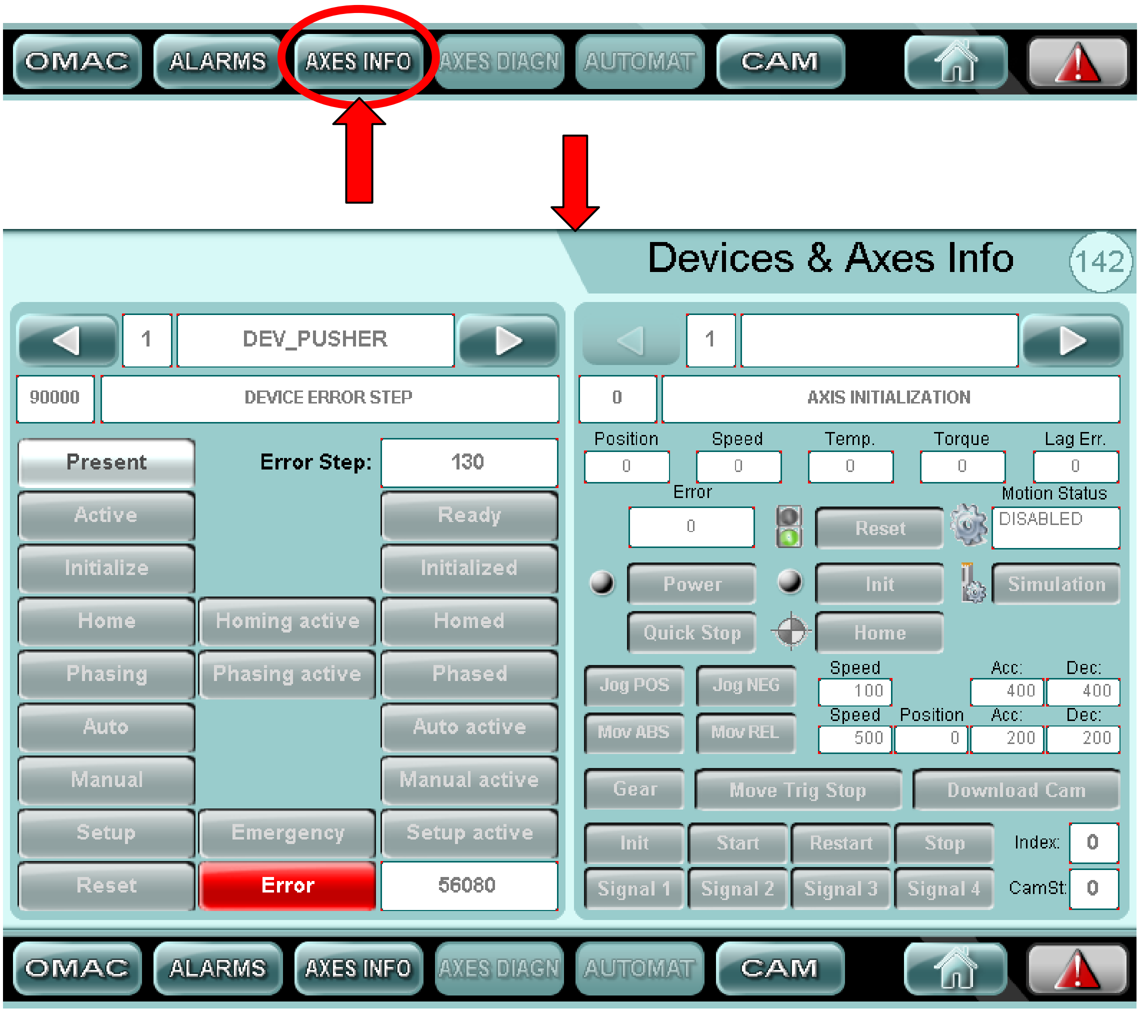
By using the key  you enter the page which lists the advanced level alarms concerning the axes.

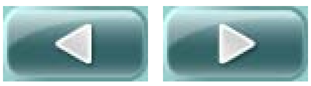


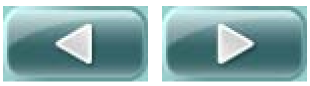
5.6.1 Devices & Axes Info

From the inverter and axes alarm page by using the 

key is possible to access the devices and axes info page. From this page you can see the status and the correct operation of the devices and the axes.



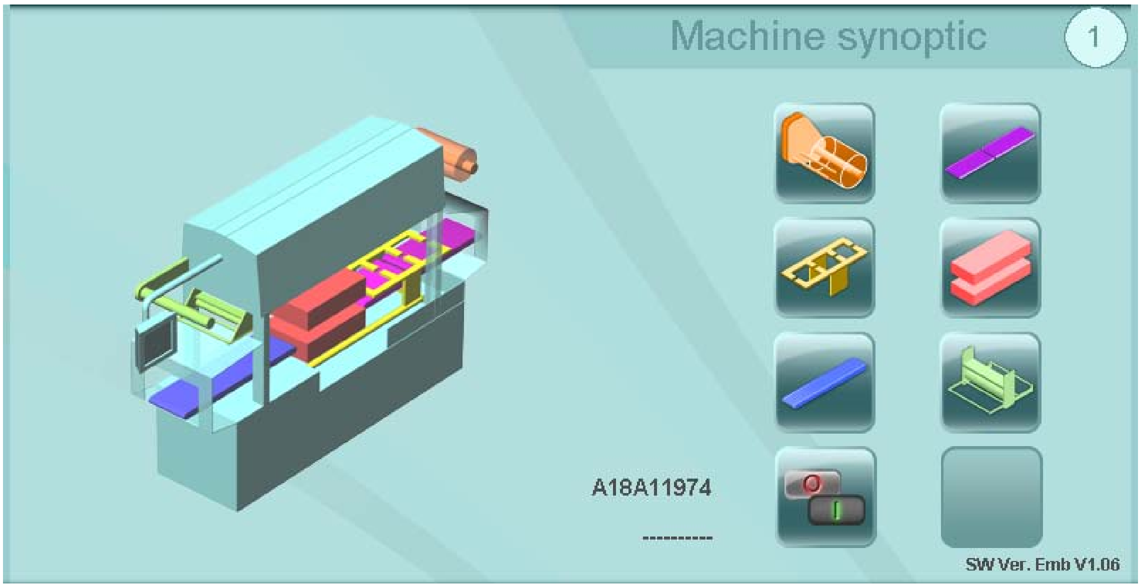
In left side of this page by using the  keys it's possible to scroll the devices installed on the machine

In right side of this page by using the  keys it's possible to scroll the axes of the machine

In case there were any malfunctions you can scroll these pages and check for alarm red.

This page allows the customer to provide the assistance service a detailed state of the state of the axes of the machine.

5.7 HOME PAGE – MACHINE SYNOPTIC



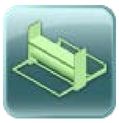
FILM UNWINDER PAGE

SPACER BELT, FEED BELT AND SYNCRHONIZER (WHERE AVAILABLE) PAGE

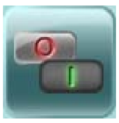
PUSHER PAGE

TOOL PAGE

DELIVERY BELT PAGE

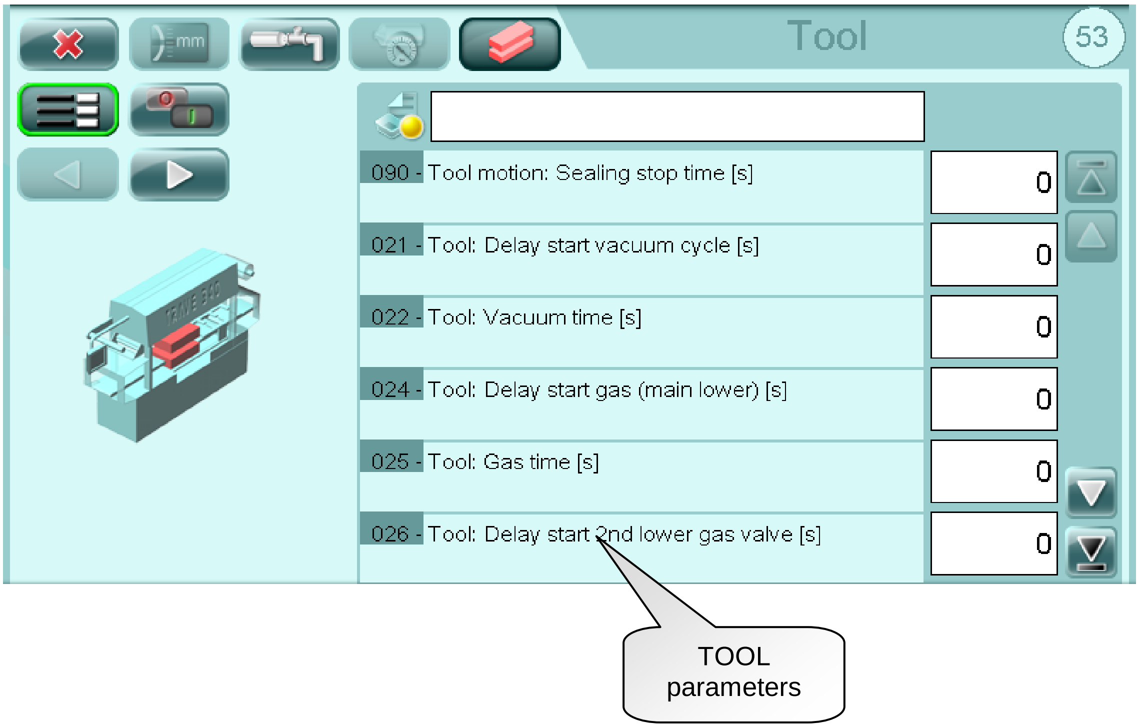


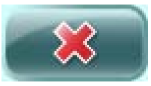
FILM UNWINDER

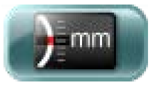


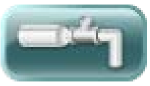
SYSTEM ENABLE

5.7.1 Tool



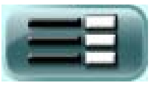

RETURN TO MACHINE SYNOPTIC (HOME PAGE)


TOOL AXIS PAGE


PNEUMATIC SYSTEM PAGE


GAS MIXER PAGE (WHERE AVAILABLE)

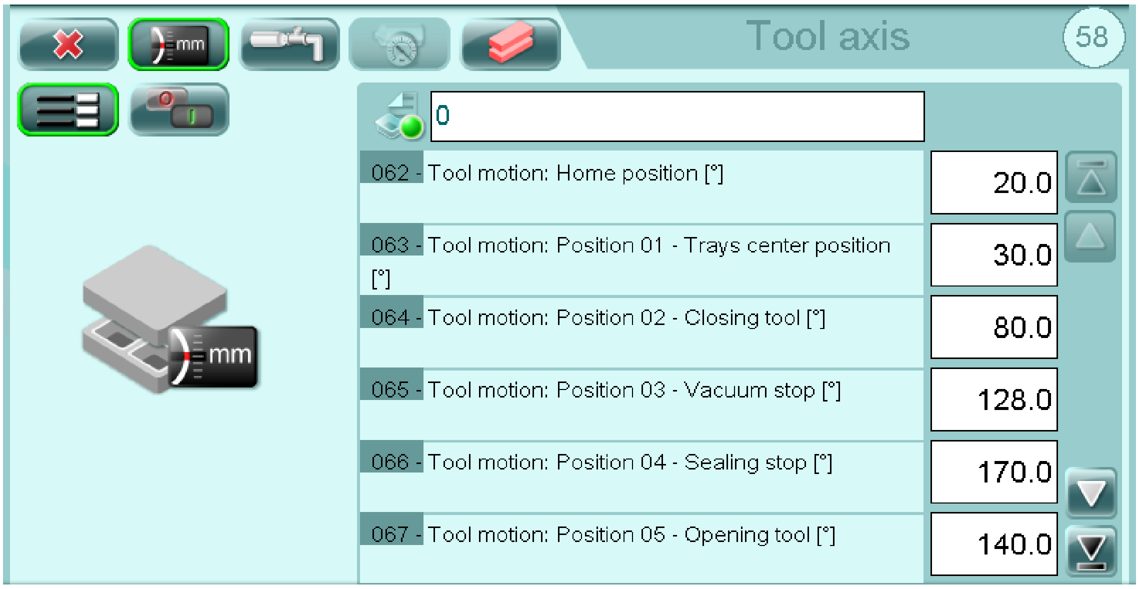

RETURN TO TOOL PAGE

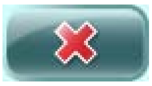

PARAMETERS TOOL PAGE


ENABLES TOOL PAGE

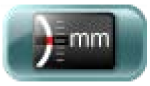

TOOL PARAMETERS TYPE CHANGE

5.7.1.1 Tool axis





RETURN TO MACHINE SYNOPTIC (HOME PAGE)



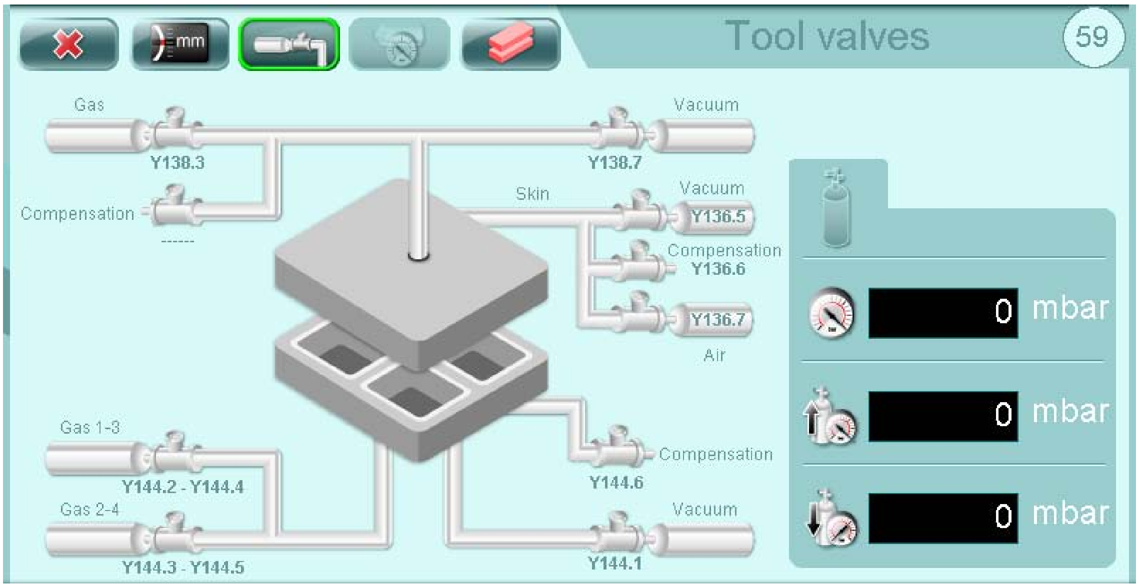
TOOL AXIS PAGE

PNEUMATIC SYSTEM PAGE

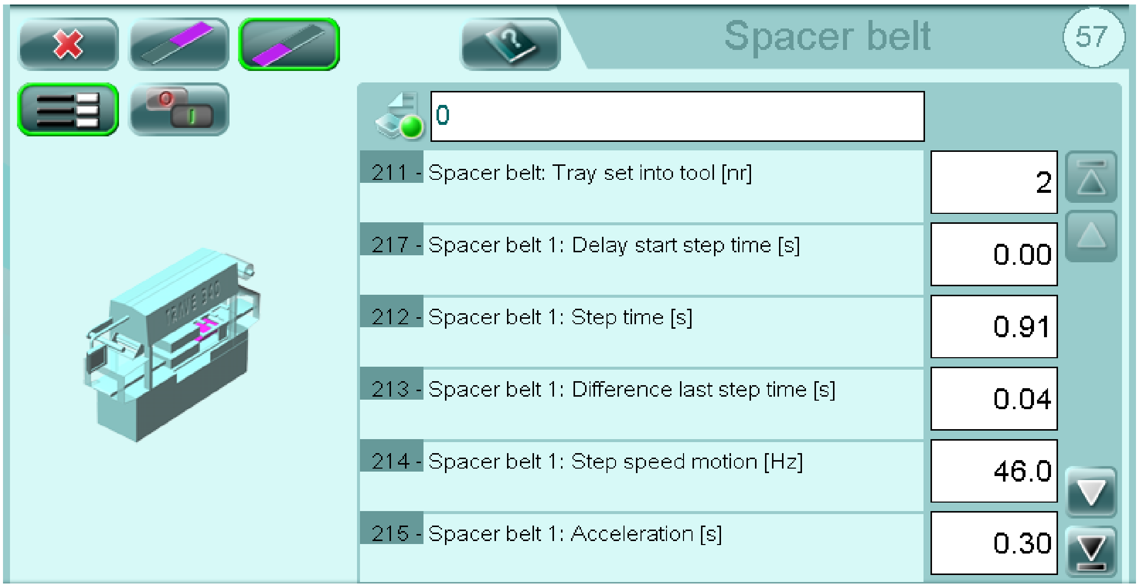
GAS MIXER PAGE (WHERE AVAILABLE)

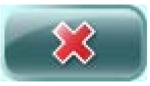
RETURN TO TOOL PAGE

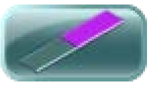
5.7.1.2 Valve layout

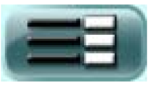


5.7.2 Spacer belt



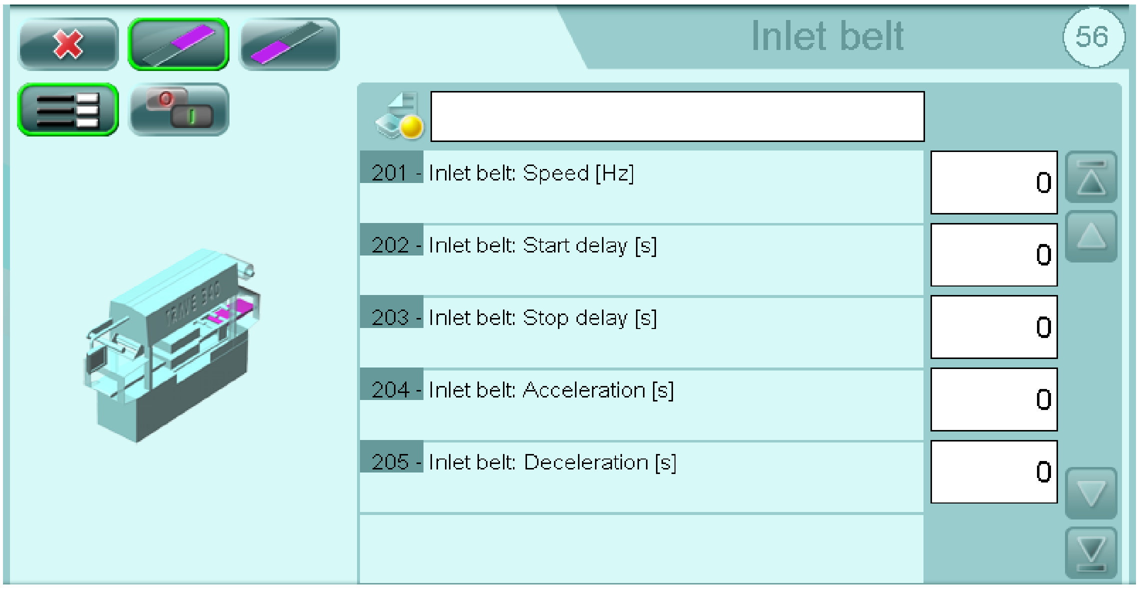

RETURN TO MACHINE SYNOPTIC (HOME PAGE)

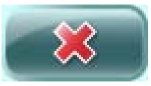

FEED BELT PAGE (WHERE AVAILABLE)

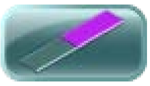

PARAMETERS SPACER BELT PAGE


ENABLES SPACER BELT PAGE

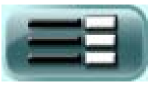
5.7.3 Inlet belt



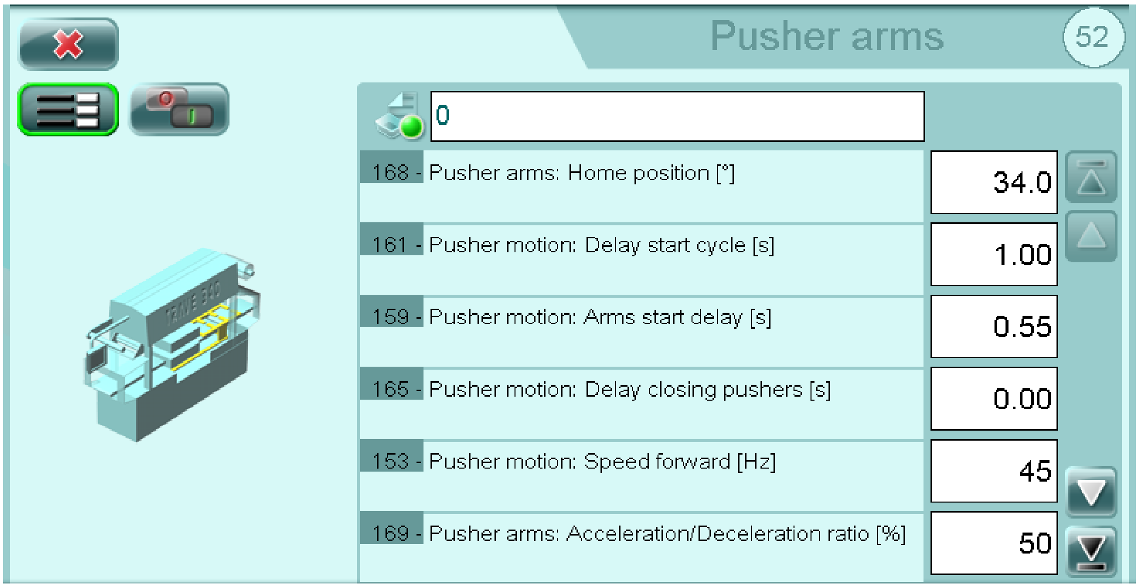

RETURN TO MACHINE SYNOPTIC (HOME PAGE)

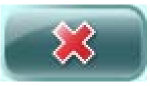

SPACER BELT PAGE

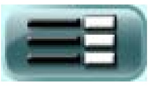

ENABLES INLET BELT PAGE


PARAMETERS INLET BELT PAGE

5.7.4 Pusher arms

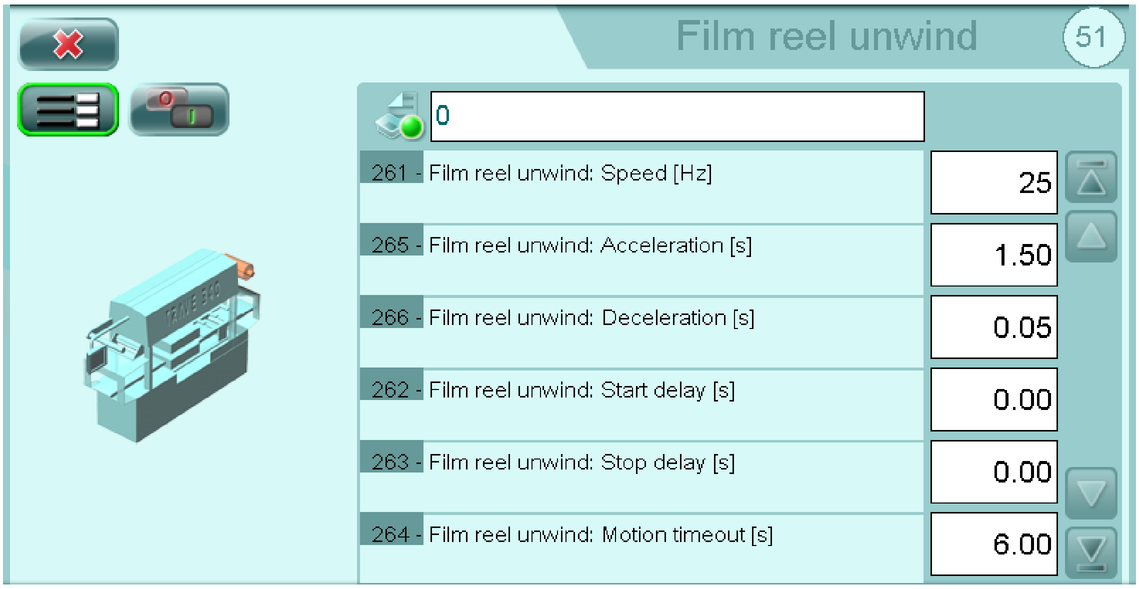


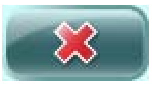

RETURN TO MACHINE SYNOPTIC (HOME PAGE)


PARAMETERS PUSHER ARMS PAGE

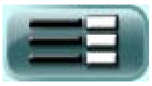

ENABLES PUSHER ARMS PAGE

5.7.5 Film reel unwind





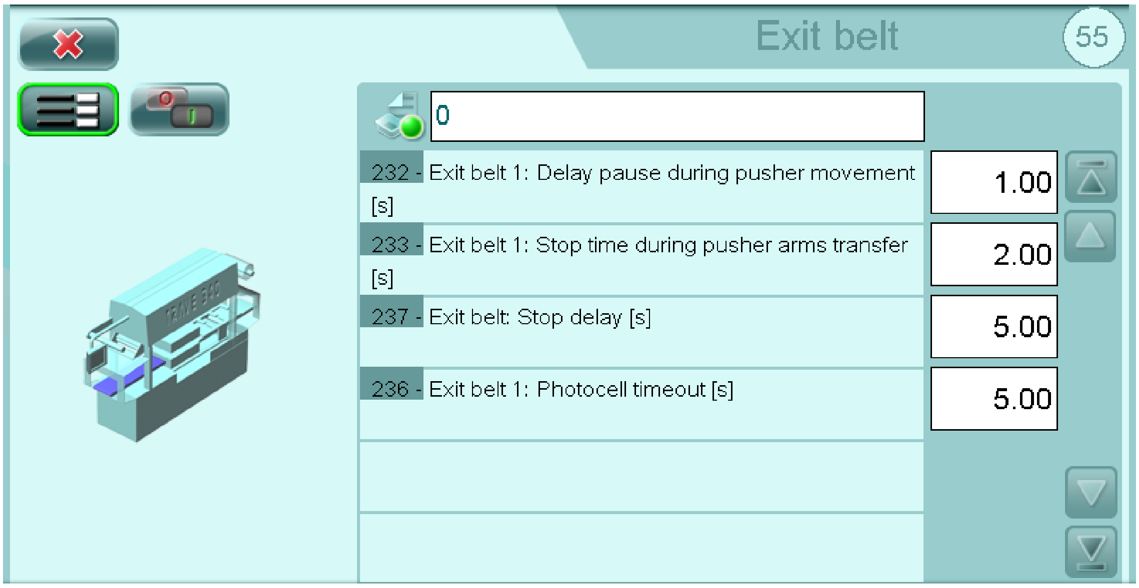
RETURN TO MACHINE SYNOPTIC (HOME PAGE)

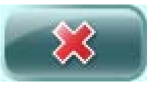


PARAMETERS FILM REEL UNWIND PAGE

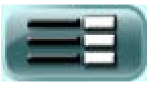
ENABLES FILM REEL UNWIND PAGE

5.7.6 Exit belt





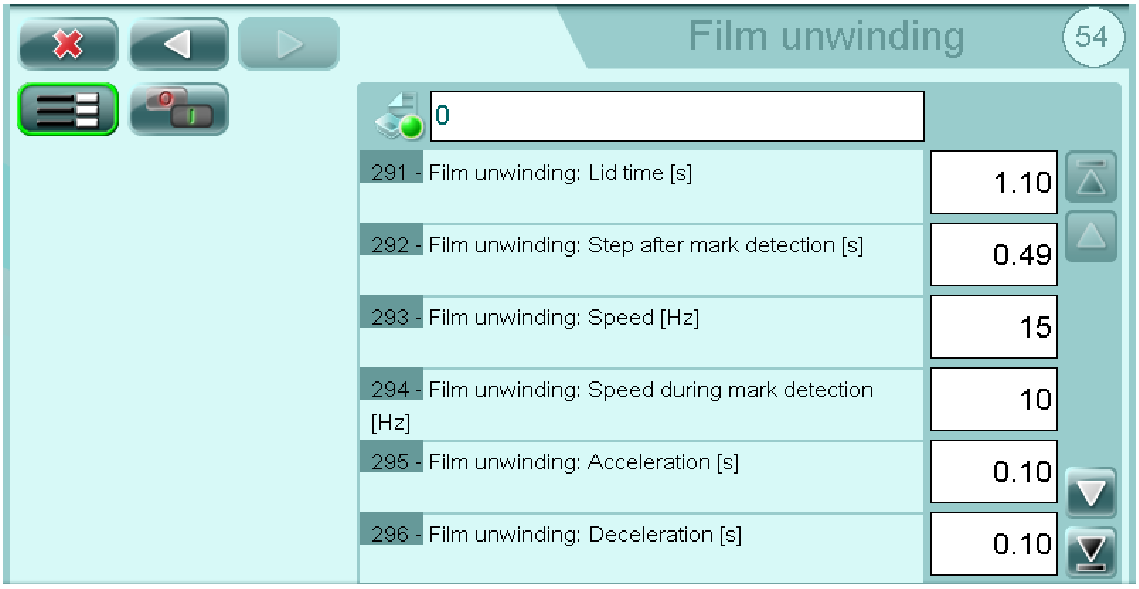
RETURN TO MACHINE SYNOPTIC (HOME PAGE)

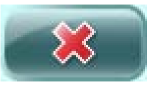


PARAMETERS EXIT BELT PAGE

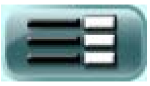
ENABLES EXIT BELT PAGE

5.7.7 Film unwinding





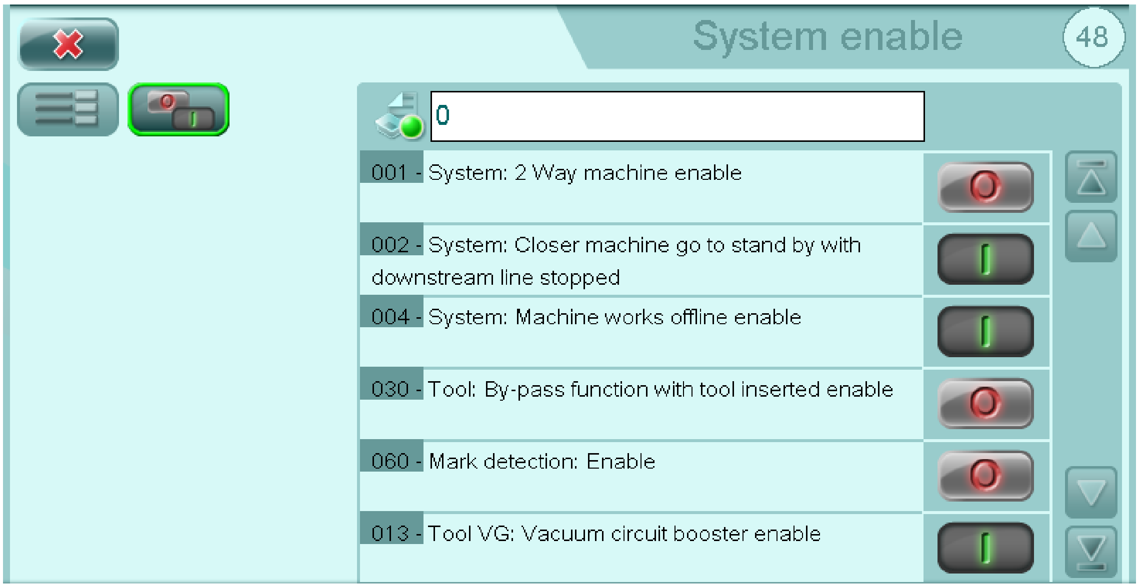
RETURN TO MACHINE SYNOPTIC (HOME PAGE)

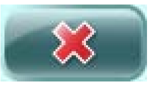


PARAMETERS FILM UNWINDING PAGE

ENABLES FILM UNWINDING PAGE

5.7.8 System enable





RETURN TO MACHINE SYNOPTIC (HOME PAGE)

5.8 RECIPE MANAGEMENT

IMPORTANT! The “000_default.csv” recipe cannot be deleted or modified as it is the default recipe used at the test stage at the Mondini plant.

Scroll through the recipes to display images of them.
Selection is automatic and the parameters are loaded with a slight delay.

After changing a parameter, it is necessary to save the recipe (for an OP level change) or load the data in a recipe and save it (for a PLC level change).

Data overwritten in the PLC cannot be recovered, so it is advisable to create a new recipe when you wish to change more than one parameter.

Before putting a recipe into production, it is very important for the machine to be stopped and all the mechanical parts ready to receive a new setting.

When the recipe is sent to the PLC, the current one is overwritten. So, when the current recipe parameters have not been saved in a recipe, it is advisable to save them before loading a new recipe.


RECIPE MANAGEMENT PAGE

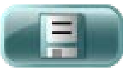

RECIPE PARAMETER PAGE


RECIPE ENABLE PAGE


DOWNLOAD RECIPE FROM PLC (PLC -> OP) Download PLC parameters to the OP parameters of the current recipe


UPLOAD RECIPE TO PLC (OP -> PLC) Copy OP parameters to PLC parameters

Parameter limit SETTING and consents

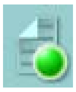

SAVE SELECTED RECIPE DATA (OP parameters)


NEW RECIPE (a copy of the “000_default” recipe is created and must be renamed)

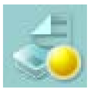

COPY RECIPE (a copy of the selected recipe is created and must be renamed)


RENAME SELECTED RECIPE

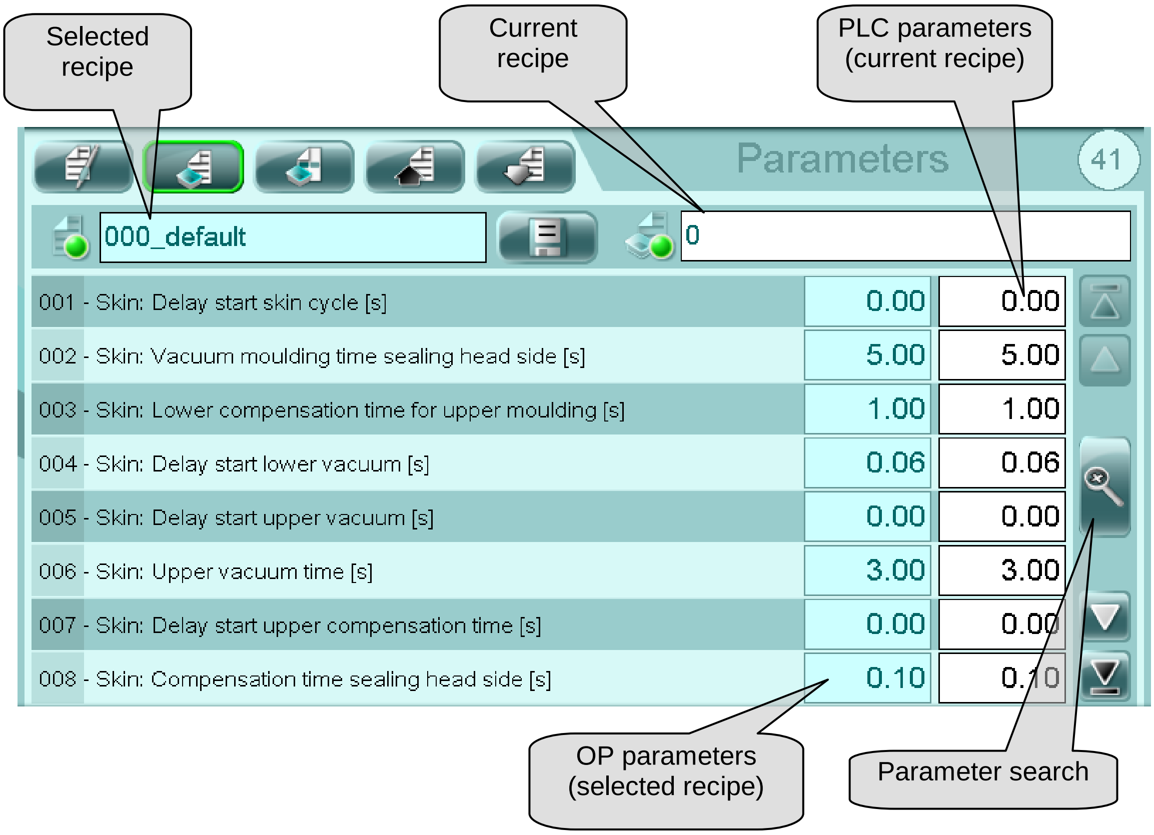

DELETE SELECTED RECIPE


RECIPE STATUS (MODIFIED OR SAVED)


RECIPE REQUESTED BY THE SUPERVISOR


RECIPE DISPLAYED DIFFERENT FROM THOSE IN THE PLC

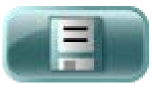
5.8.1 Recipe parameters



RECIPE MANAGEMENT PAGE

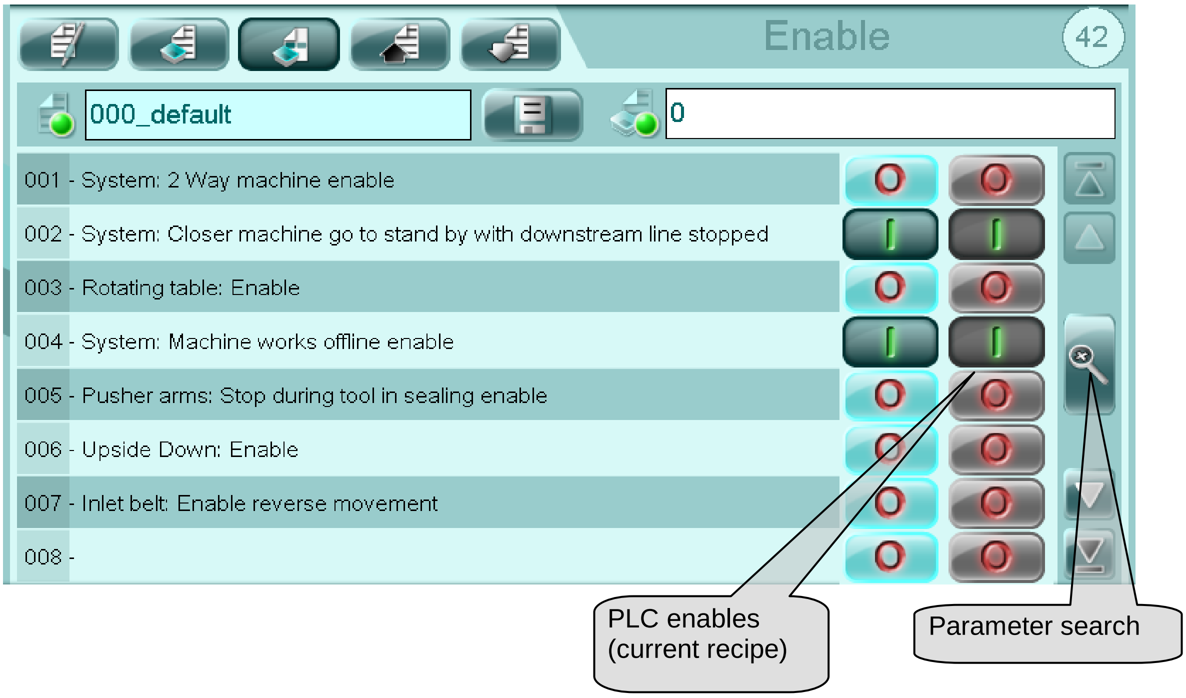
RECIPE PARAMETER PAGE

RECIPE ENABLE PAGE



SAVE SELECTED RECIPE DATA (OP parameters)

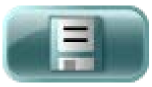
5.8.2 Enables recipe




RECIPE MANAGEMENT PAGE


RECIPE PARAMETER PAGE

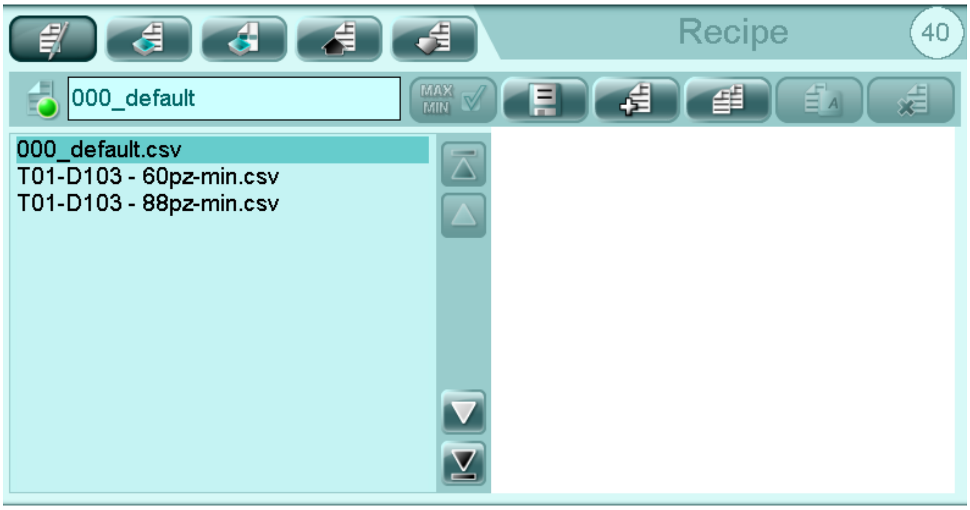

ENABLES RECIPE PAGE


SAVE SELECTED RECIPE DATA

5.8.3 Creating a new recipe

Proceed as follows to create a new recipe.

Touch  from the main menu bar to display the main recipe page.

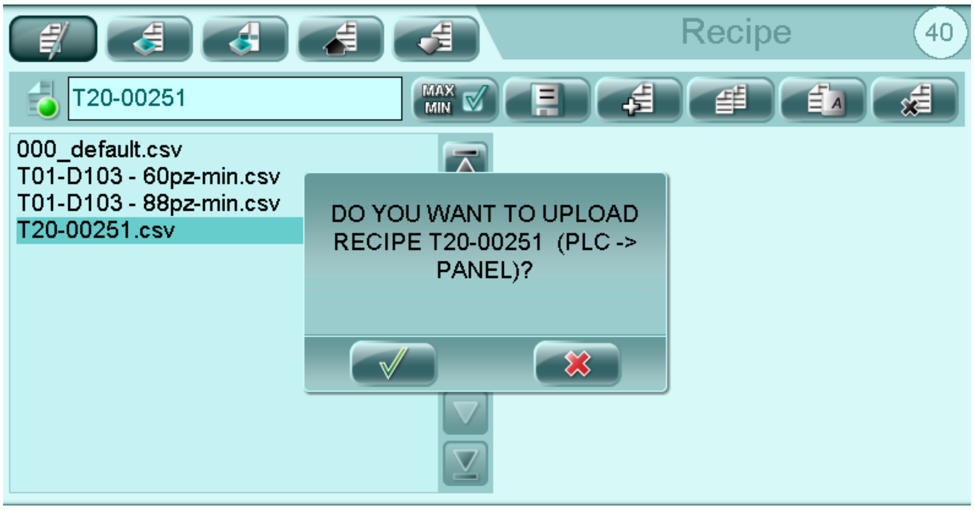


Select  to open the new Recipe window.

Touch  to confirm the name of the recipe.

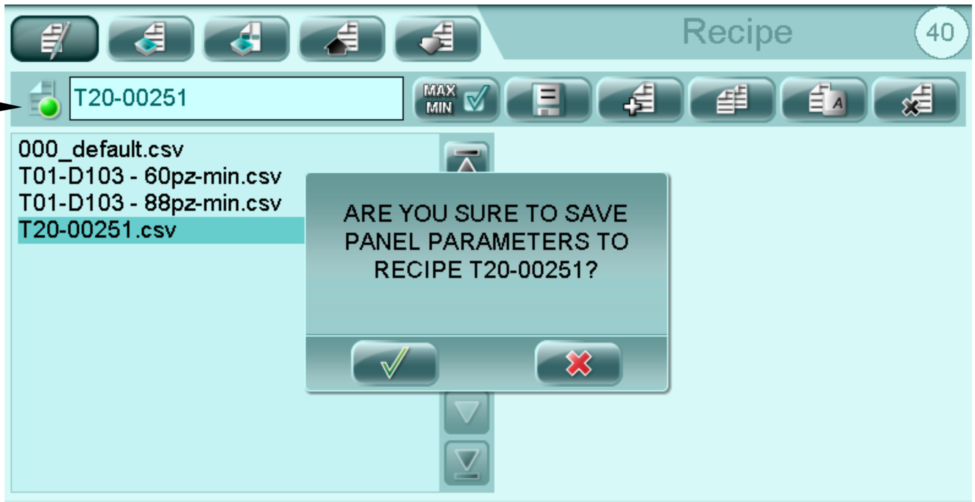
After the new recipe has been created, select it from the list of available recipes on the right-hand side of the panel display and press  to copy the current recipe data to the newly-created recipe.

Touch  to confirm.



Press  to save.

A

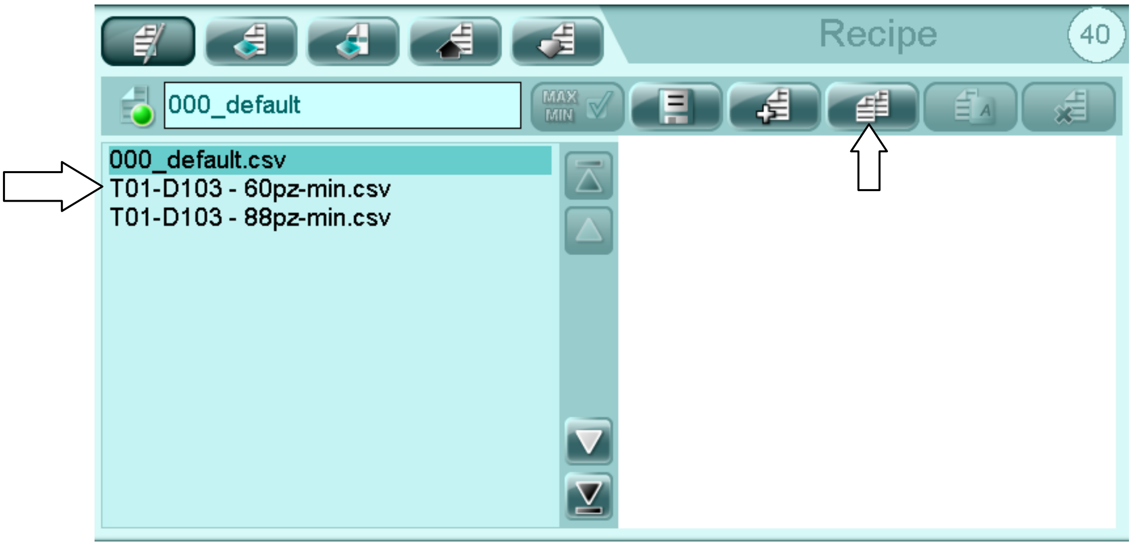


Touch  to confirm.
In order to verify whether the recipe has been correctly saved, you need to check that icon (**A**) light next to the recipe name is green and not orange.

5.8.4 Creating a new recipe from an existing one

Proceed as follows to create a copy of an existing recipe.

Select the recipe you wish to copy and touch .



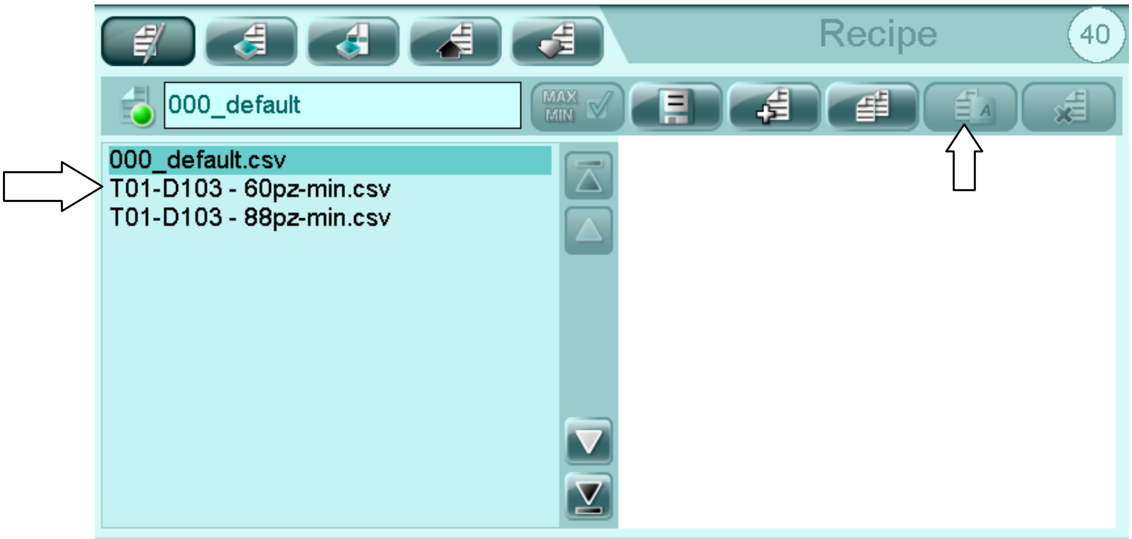
Insert the name of the new recipe using the alphanumeric keypad.

Touch  to confirm.

5.8.5 Renaming an existing recipe

Proceed as follows to rename an existing recipe.

Select the recipe that you wish to rename and touch .



Insert the new name of the selected recipe using the alphanumeric keypad.

Touch  to confirm.

5.8.6 Recipe image

If you want insert a image into recipe:

- Copy the image into a USB stick (supplied with machine) into a “root”.
- Insert USB stick on the rear of the panel.
- Login with right password level.
- Scroll through the recipes and click for select recipe.
- Click on the image list saved on USB stick.
- Select the correct image and confirm.

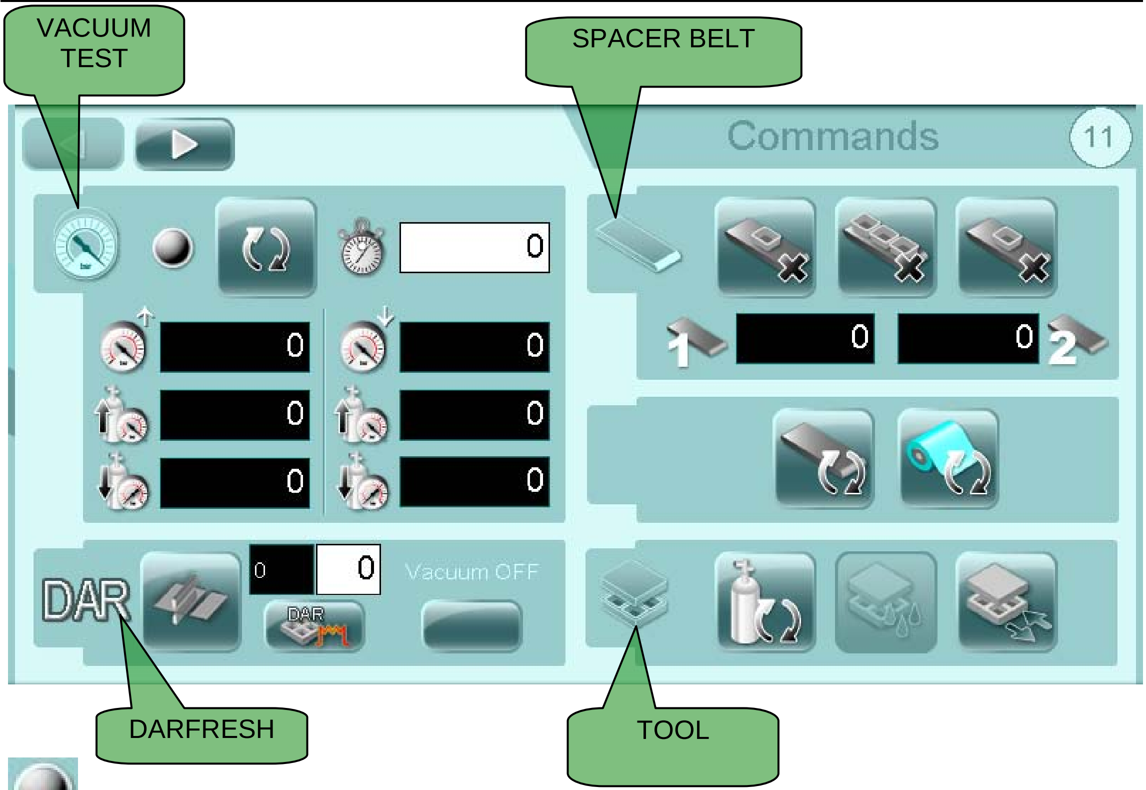
IMPORTANT!

Image properties:

- Format: JPG (for TRAVE 1xxx, EVO 367-384) or PNG (for TRAVE 3xx)
- Dimension: max 100-150 Kb
- Size recommended: 370 x 290 pixel.

Default recipe image is not changeable!

5.9 COMMANDS



VACUUM TEST ON

START VACUUM TEST

UPPER INSTALLATION CURRENT PRESSURE

LOWER INSTALLATION CURRENT PRESSURE

GAS VALUE REACHED

VACUUM VALUE REACHED

VACUUM TEST TIME

TRAYS ON SPACER BELT 1

TRAYS ON SPACER BELT 2

REMOVE ONE TRAY FROM SPACER BELT

REMOVE ALL TRAYS FROM SPACER BELT

START BELT TEST

START FILM TEST

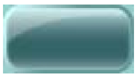
WASHING WITH GAS

TOOL WASHING CYCLE (TOOL IN VACUUM/GAS POSITION)

TOOL IN TOOL-CHANGE POSITION

CUT FILM CYCLE FOR DARFRESH APPLICATION

DARFRESH SHUTTLE IN CHANGE POSITION



VACUUM OFF IN DARFRESH PLATES

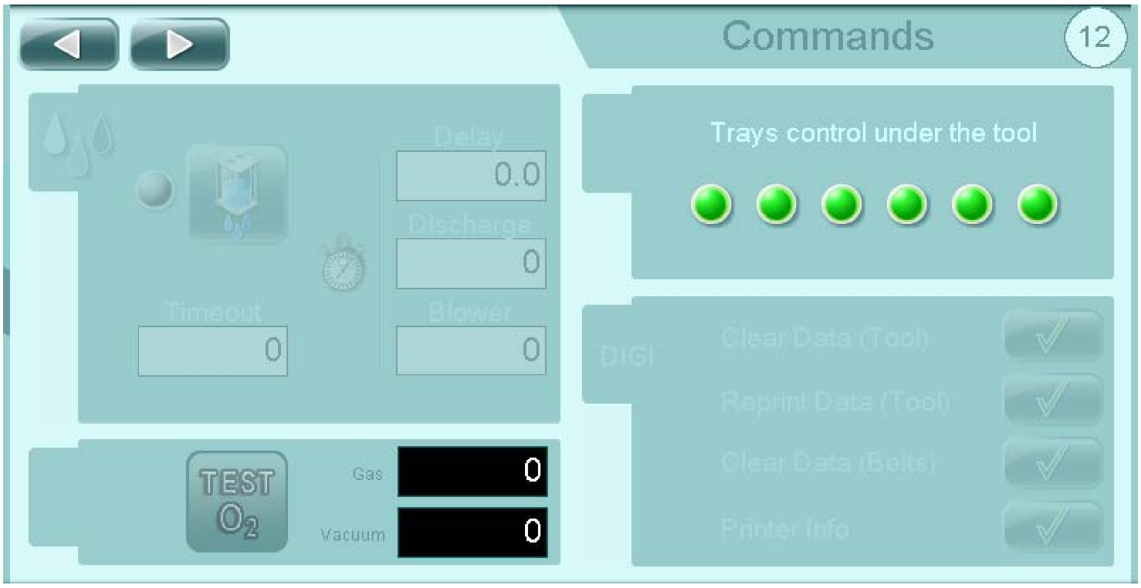
DARFRESH SHUTTLE UNDER THE SEALING TOOL

12

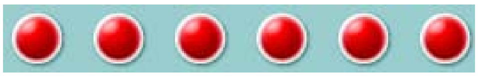
SHUTTLE POSITIONING TIME UNDER THE SEALING TOOL (IN MINUTES)

0

TIME REMAINING FOR THE POSITIONING OF THE SHUTTLE UNDER THE SEALING TOOL (IN SECONDS)



LIQUID SEPARATOR MANUAL START (WHERE AVAILABLE)

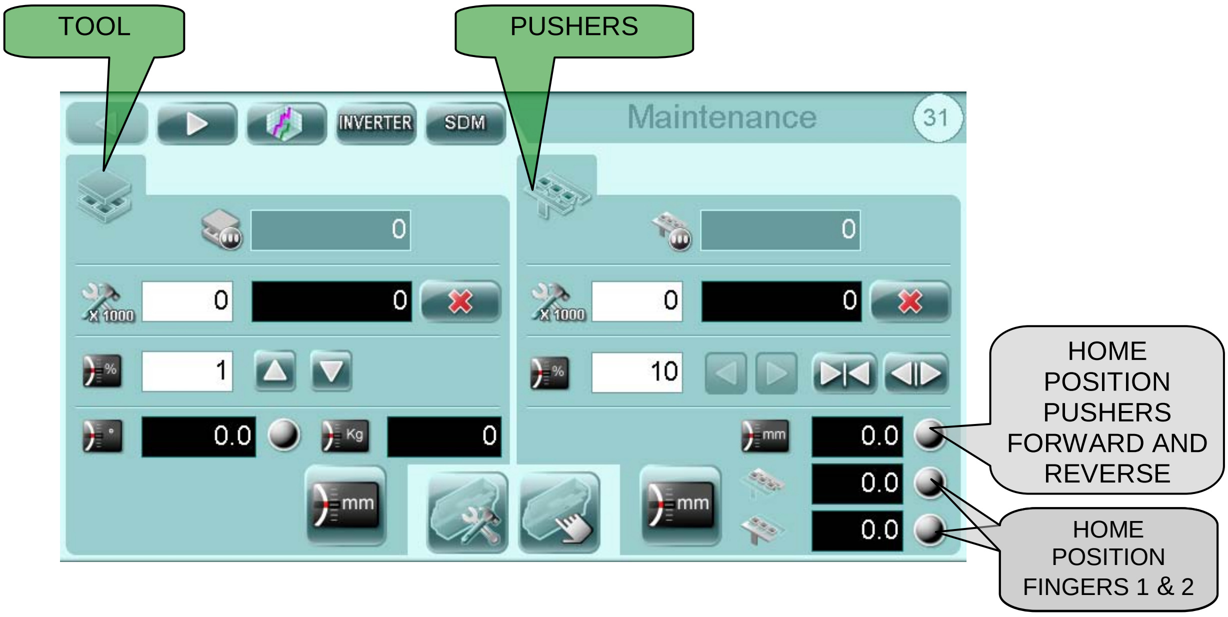


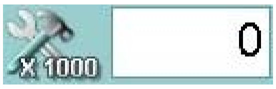
TRAYS CONTROL UNDER THE TOOL

START OXYGEN TEST

LINE EMPTYING

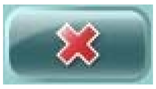
5.10 MAINTENANCE

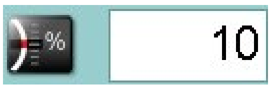




FIXED PROGRAMMED MAINTENANCE

TOOL CYCLES COUNTER

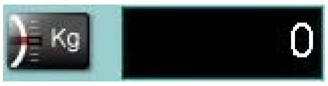

RESET PARTIAL COUNTER



% SPEED MANUAL MOVEMENT

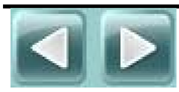

UP AND DOWN MANUAL COMMAND

CURRENT TOOL POSITION

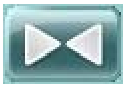


PRESSIURE

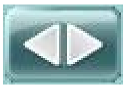
PUSHERS CYCLES COUNTER



FORWARD AND REVERSE MANUAL COMMAND



CLOSE FINGERS



OPEN FINGERS

INVERTER AMPER

ACTUAL PUSHERS POSITION

FINGER 1 POSITION

FINGER 2 POSITION

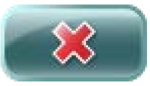
SET TOOL CHANGE POSITION

ENABLE TOOL INVERSION (WHERE AVAILABLE)

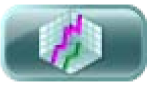
HOME KEYS (TOOL AND PUSHERS)

ENABLE MAINTENANCE (JOG MODE)

ENABLE MANUAL CYCLES

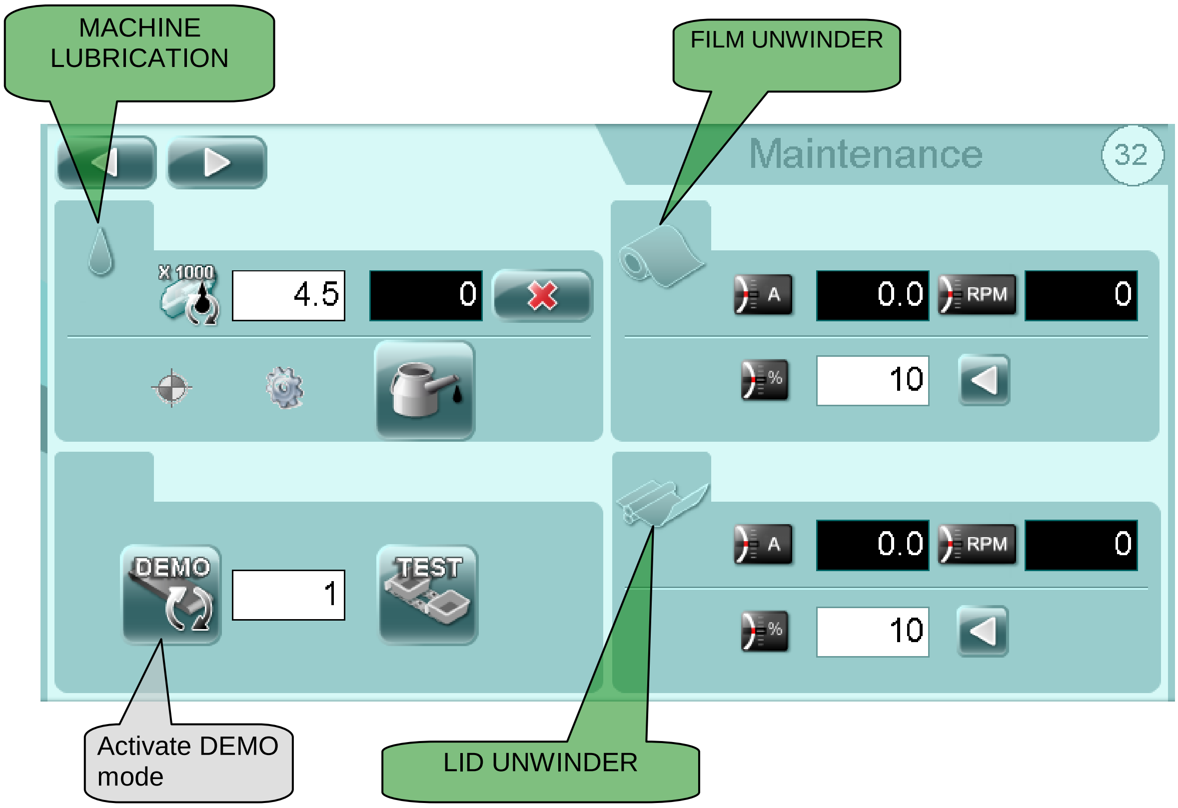


RESET TOOL MAINTENANCE CYCLES



TRENDS PAGE

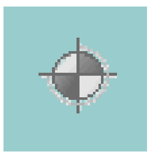
INVERTER PARAMETERS PAGE



LUBRICATION THRESHOLD

LUBRICATION CYCLES

RESET LUBRICATION CYCLES



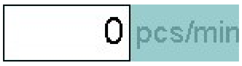
SENSOR OPERATION LED

INDICATOR OF THE MOTOR ACTIVE

MANUAL LUBRICATION

FEED BELT DEMO MODE (IF AVAILABLE). Used to simulate tray feed at a specified rate.

TEST (IF AVAILABLE): Allows to enable the transport chain test.

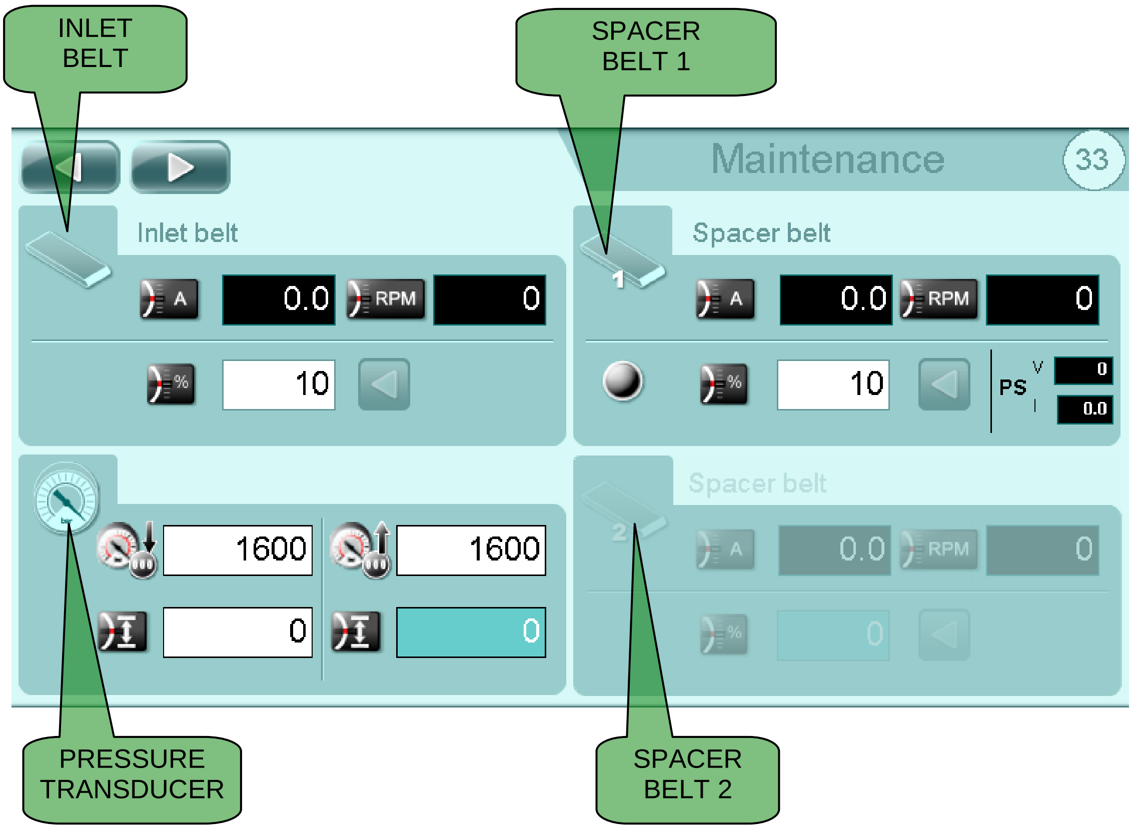


ENTER NUMBER OF TRAYS TO SIMULATE

INVERTER AMPER

INVERTER SPEED

MANUAL SPEED AND FORWARD COMMAND



INVERTER AMPER

INVERTER SPEED

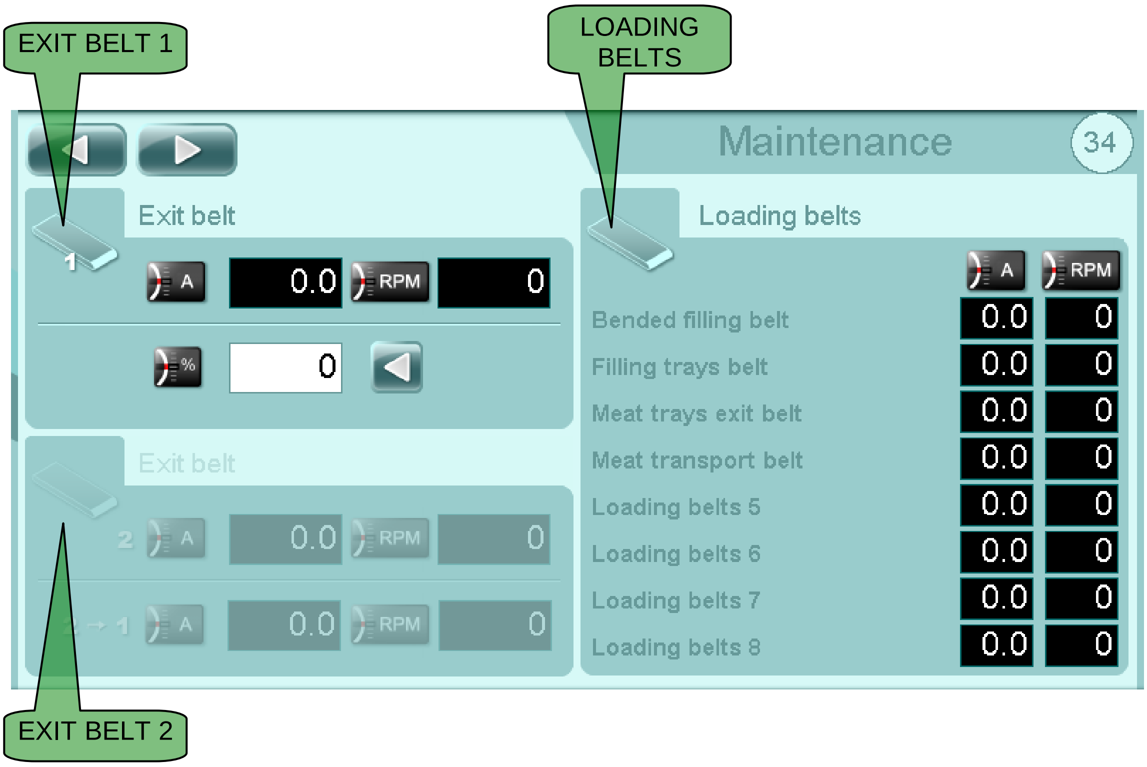
MANUAL SPEED AND FORWARD COMMAND

UPPER PRESSURE TRANSDUCER COEFFICIENT

LOWER PRESSURE TRANSDUCER COEFFICIENT

UPPER PRESSURE TRANSDUCER OFFSET

LOWER PRESSURE TRANSDUCER OFFSET

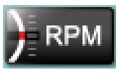


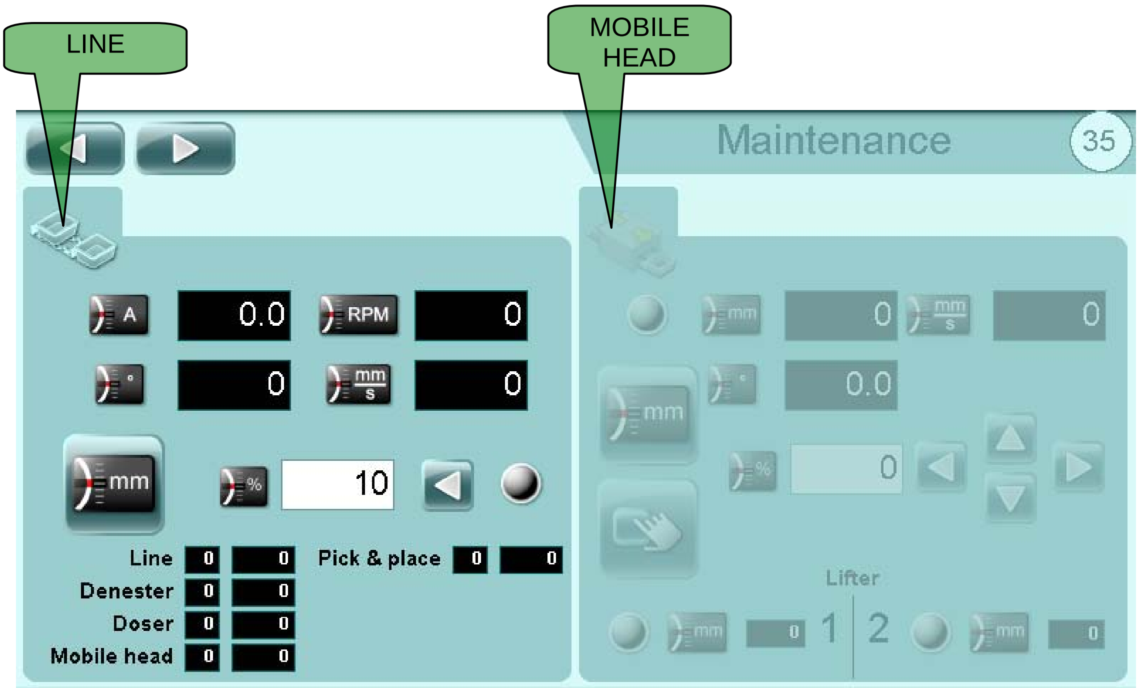

INVERTER AMPER


INVERTER SPEED


MANUAL SPEED AND FORWARD COMMAND

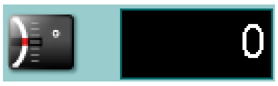

DISPLAY DEVICE AMPERE CONSUMPTION


DISPLAY DEVICE REVOLUTIONS PER MINUTE

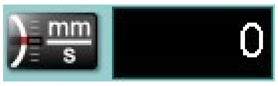


INVERTER AMPER

INVERTER SPEED

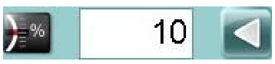


CURRENT LINE POSITION

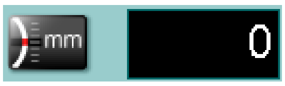


CURRENT SPEED

LINE HOME

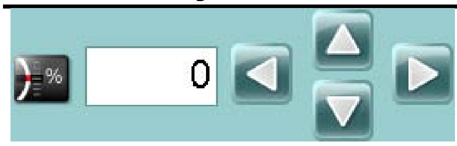


MANUAL SPEED AND FORWARD COMMAND



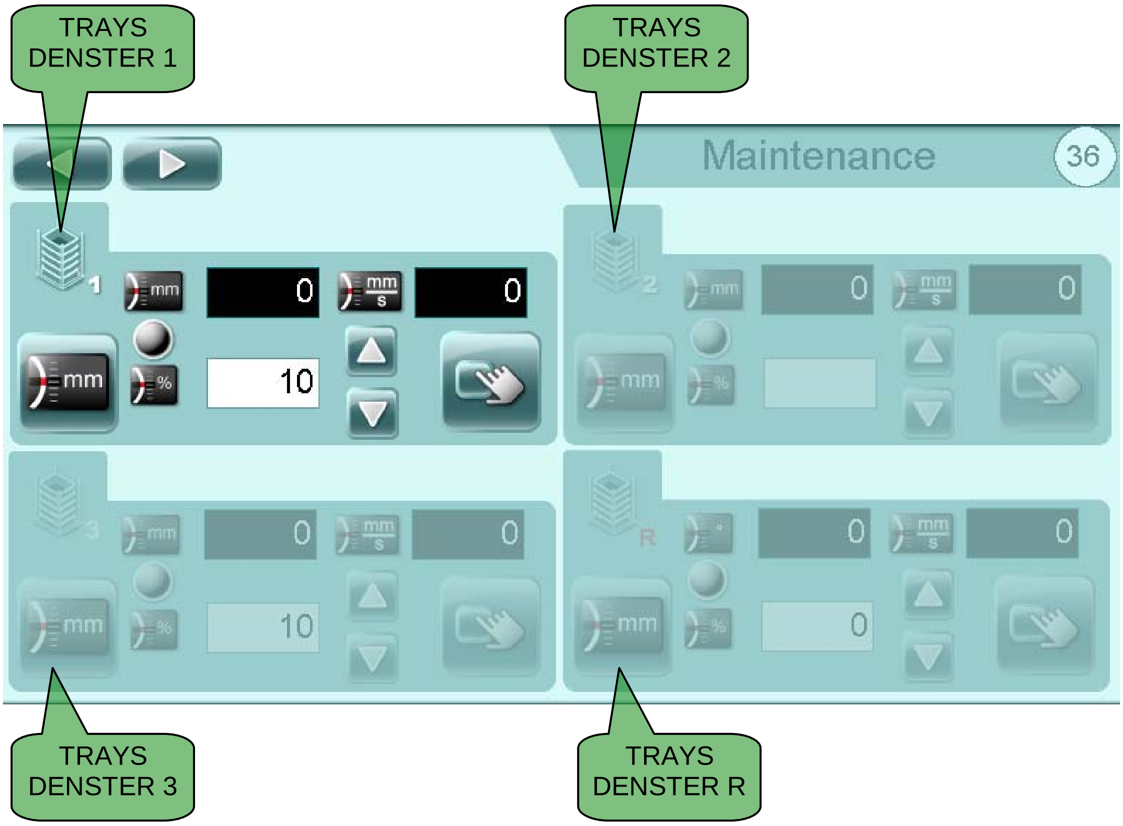
CURRENT MOBILE HEAD POSITION

MOBILE HEAD HOME



MANUAL SPEED AND COMMAND

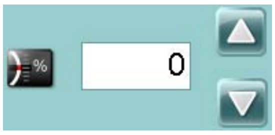
MANUAL CYCLE COMMAND (WHERE AVAILABLE)



CURRENT POSITION

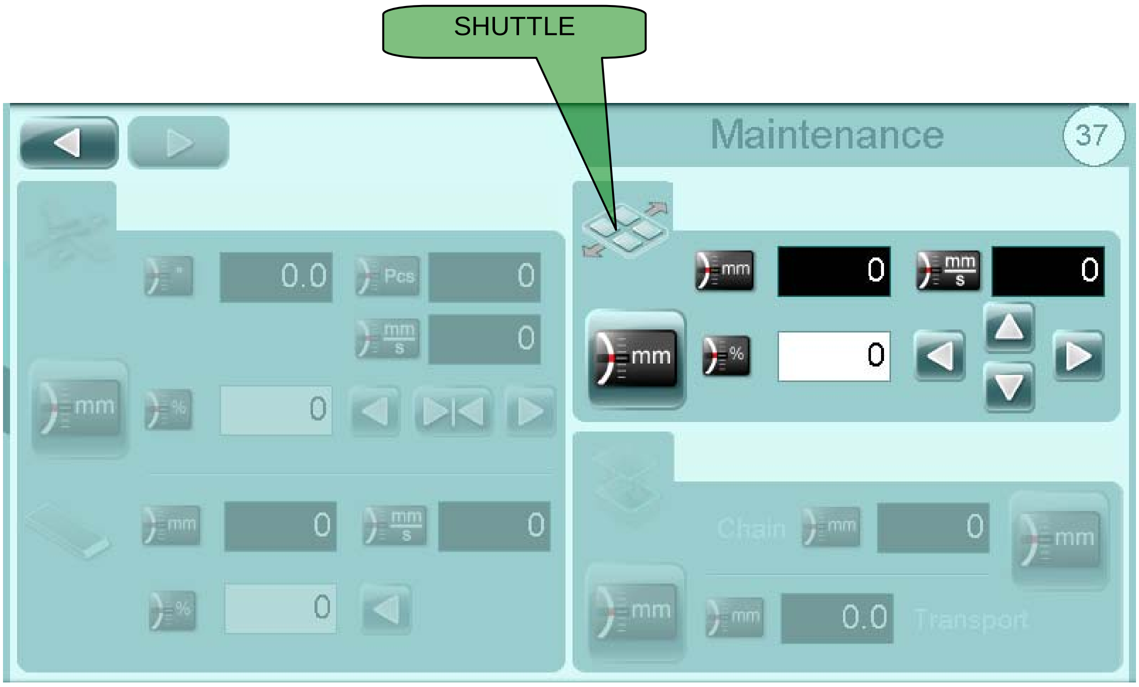
CURRENT SPEED

TRAYS DENESTER HOME



MANUAL SPEED AND COMMAND

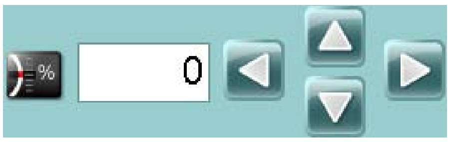

MANUAL CYCLE COMMAND (WHERE AVAILABLE)



CURRENT POSITION

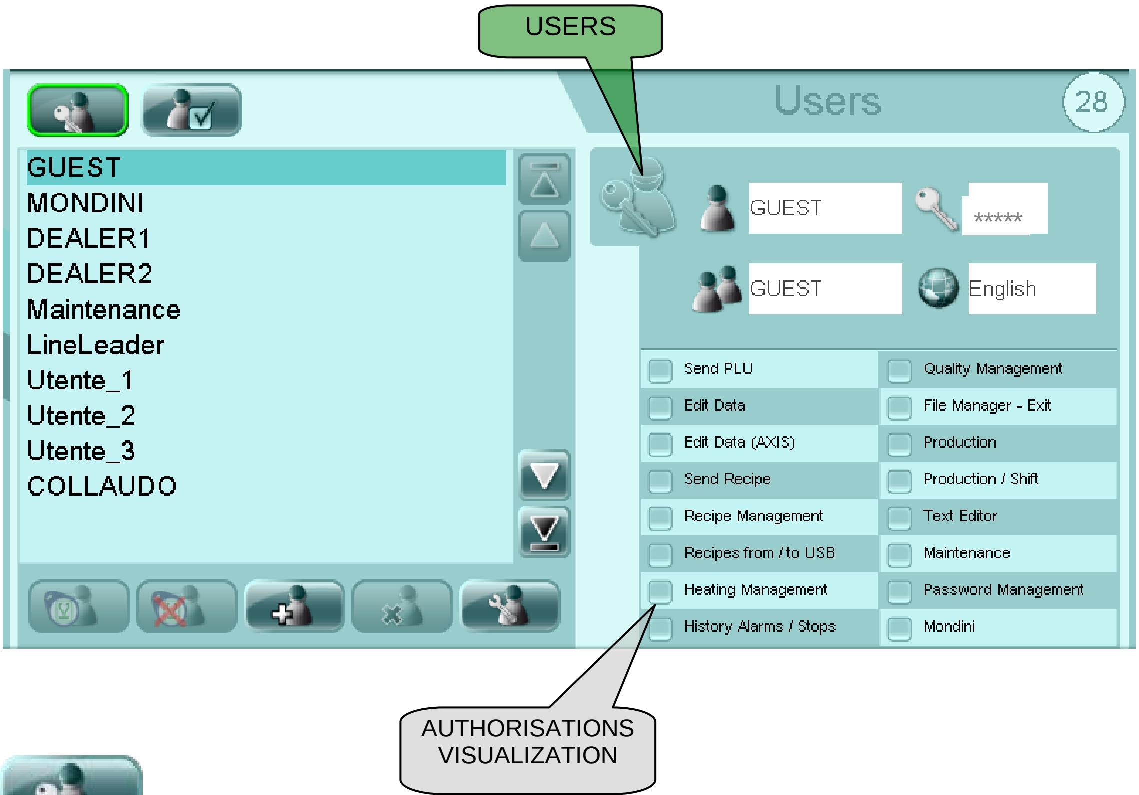
CURRENT SPEED

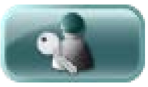
SHUTTLE HOME



MANUAL SPEED AND COMMAND

5.11 USER AND PASSWORD MANAGEMENT



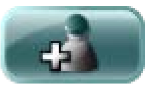


USER AND PASSWORD MANAGEMENT PAGE

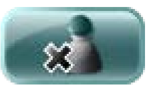
CONSENT PAGE

IT ASSOCIATES A TAG TO THE USER

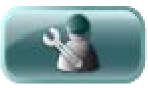
IT CANCELS ASSOCIATION USER TO A TAG



NEW USER



DELETE USER



MODIFY USER (VIEW PASSWORD)

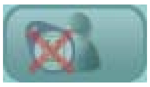
USER

ENCRIPTED PASSWORD (NEARBY THE SYMBOL MEANS IF THE LOGGED USER HAS A TAG ASSOCIATED)

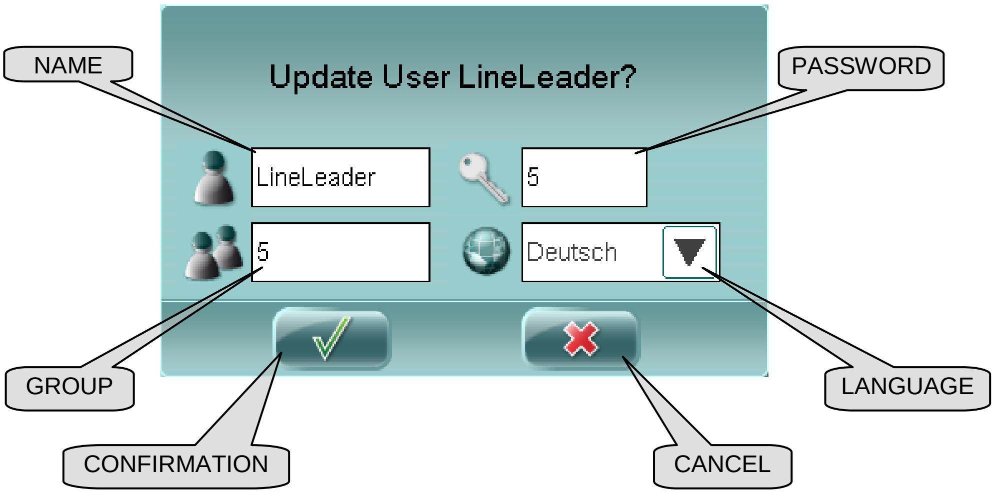
GROUP

LANGUAGE

IT ASSOCIATES THE TAG TO THE USER



IT CANCELS THE ASSOCIATION OF THE TAG WITH THE USER



IMPORTANT! GUEST and MONDINI users cannot be modified or deleted.

To log in a user, touch the icon at the top of the screen .

Users in possession of higher level passwords are advised to log out (GUEST) on completion of the modification.

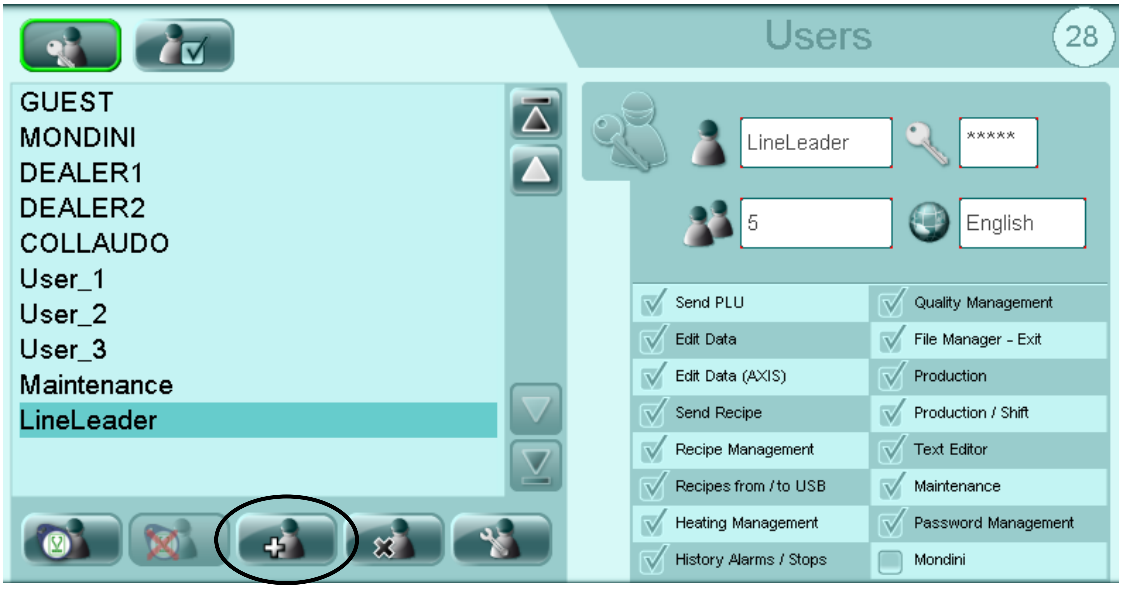
If the panel is not used for a set time, the active password expires (GUEST).

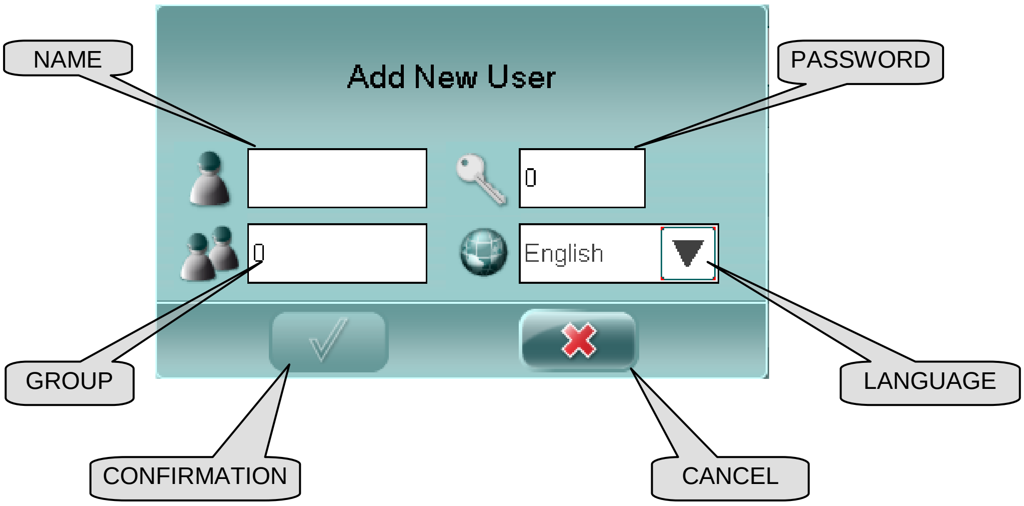
Each password gives access to different modification and processing levels. It is there advisable to keep access passwords secret and only communicate them to authorised personnel.

Authorisations are linked to a group and the user belongs to a group. So first create a group with specific authorisations and then create the users belonging to this group (same logic as Windows).

5.11.1 Creation new user

To create a new user touch the button  and compile the fields in the window.

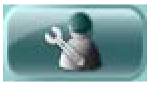


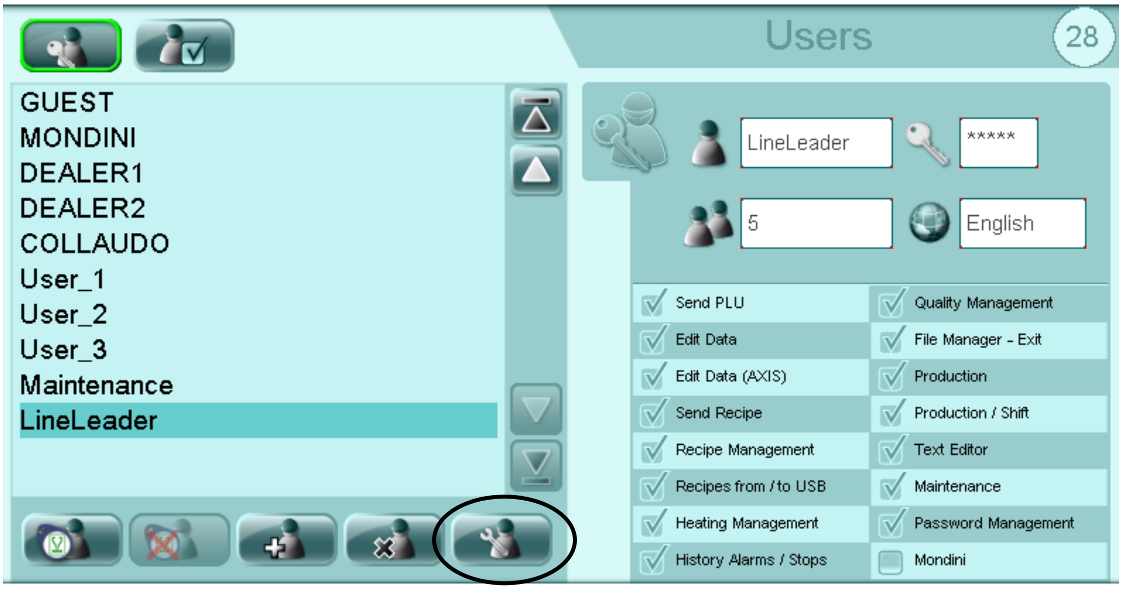


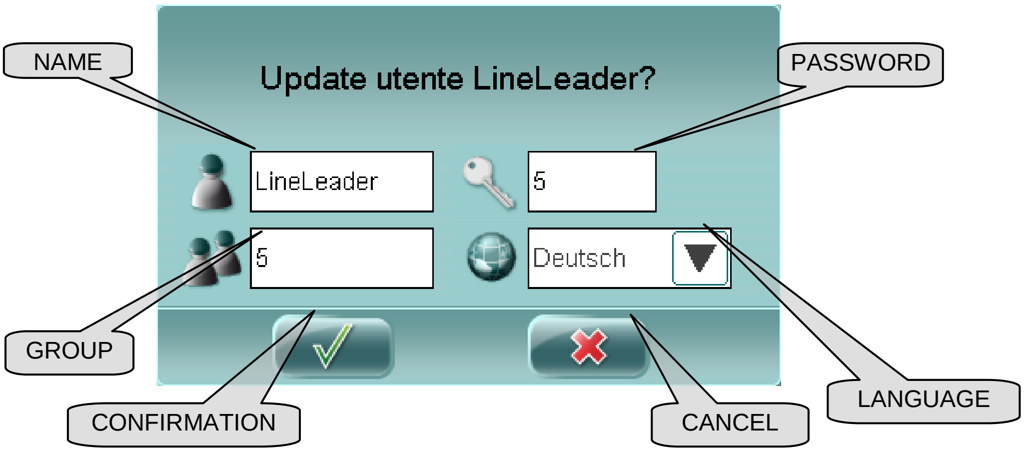
ATTENTION!

The authorizations are connected to the group and the user is join to a group. Then first a group is created (if it is not already existing) with some specific authorizations and then the users are created that have to join to this group (same philosophy of windows).

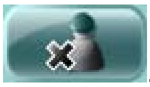
5.11.2 Modification existing user

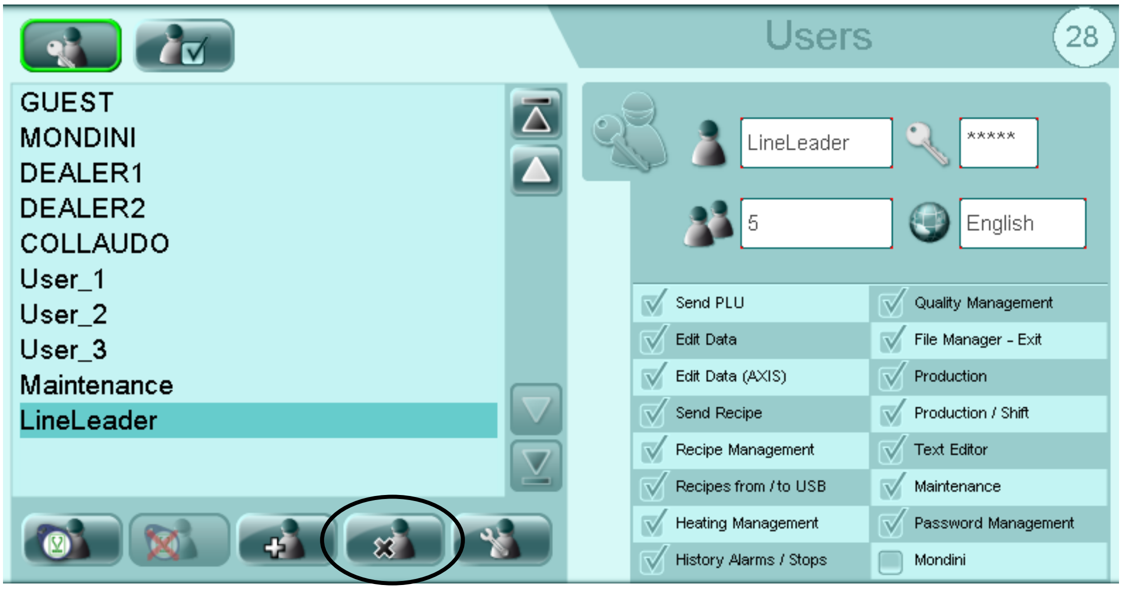
To modify an existing user touch the user to modify from the list of the users, touch the button  and to modify the data desired in the window t.





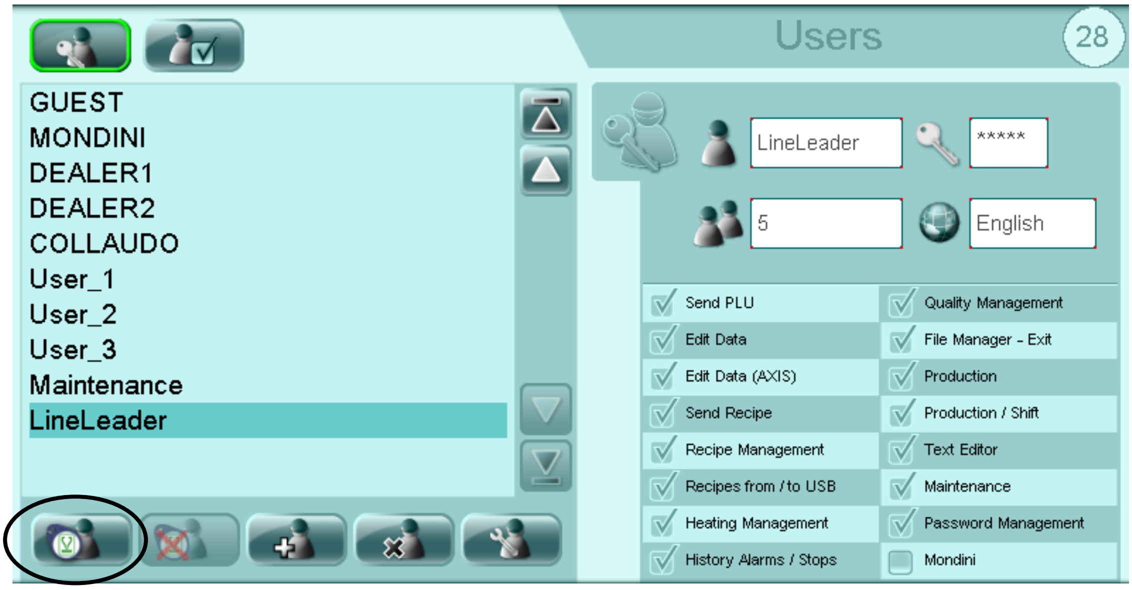
5.11.3 Eliminate existing user

To eliminate an existing user touch the user to eliminate from the list of the users and touch the button .

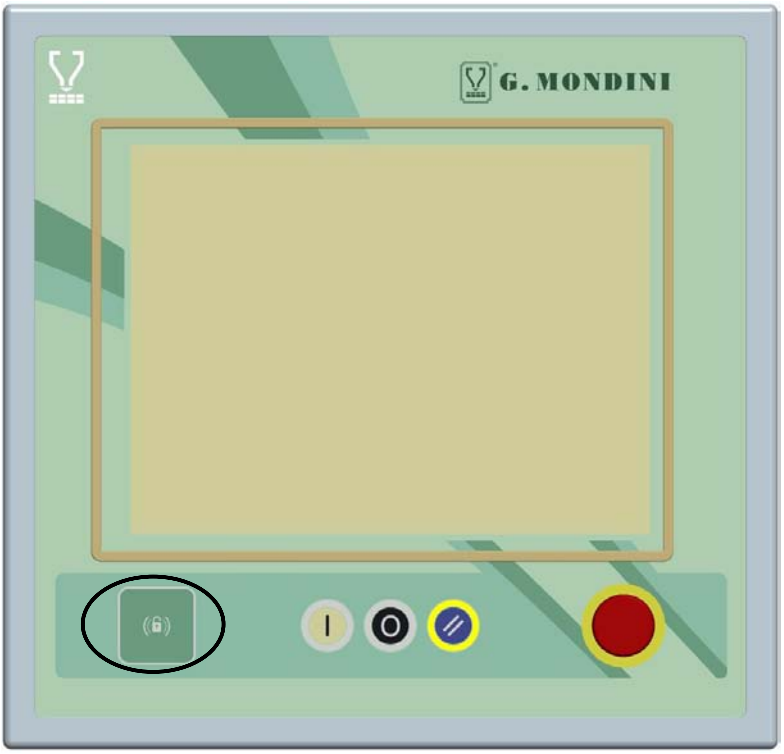


5.11.4 Associate user to TAG

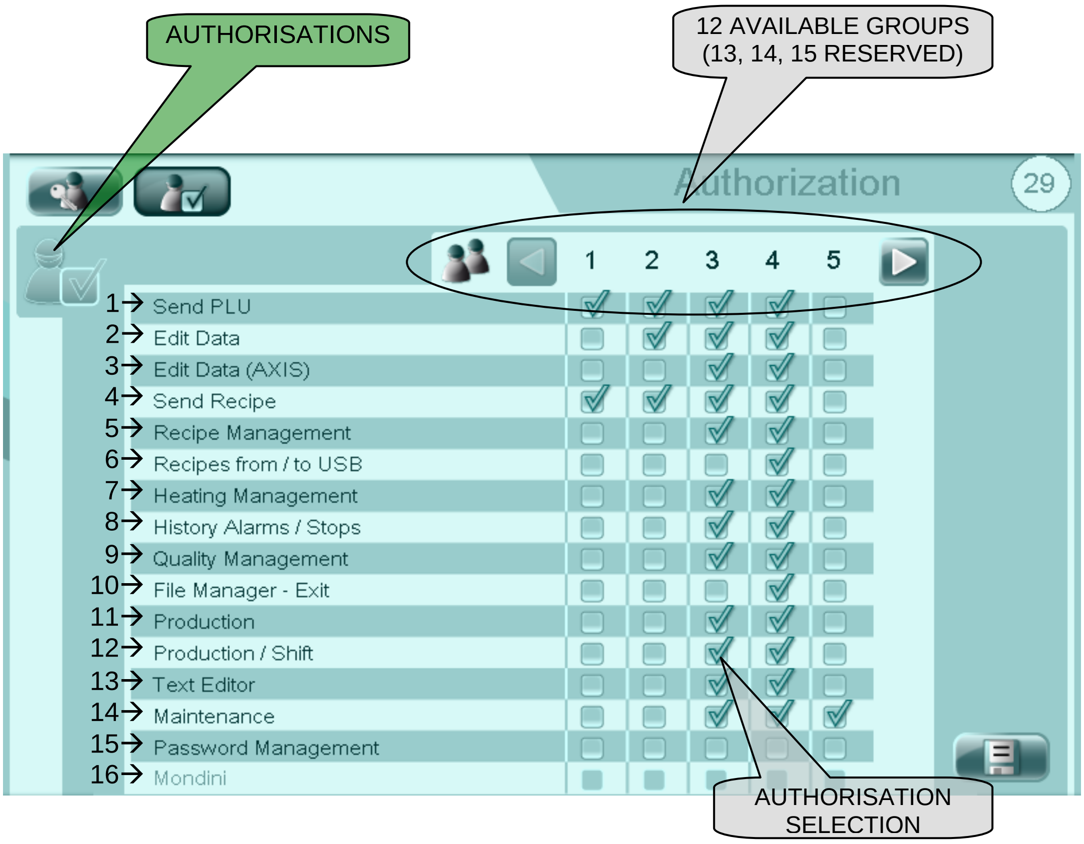
To associate an existing user to a TAG touch the user to associate from the list of the users and touch the button .



Position the TAG in correspondence of the device on the left bottom side of the operator panel  and confirm on the screen of the operator panel.



5.11.5 Authorisations



To train an authorization in a group is enough to touch the corresponding box so that the symbol ☑ appears.

SAVE CHANGES TO AUTHORISATIONS

1 Send PLU: basic authorization for the use of the page "most used";

2 Edit Data: it allows to modify all the "not critical parameters" of the machine;

3 Edit Data (AXIS): it allows to modify the advanced parameters;

4 Send Recipe: it allows to send the recipe from the panel to the PLC;

5 Recipe Management: it allows to create, to cancel and to effect changes to the recipes;

6 Recipes from/to USB: it allows to effect the transfer of the recipes from/to USB;

7 Heating Management: it allows to effect changes to the heating of the sealing tool;

8 History Alarms: it allows to visualize the "alarm history" page;

9 Quality Management: it allows to visualize the "quality" page;

10 File Manager - Exit: it allows to effect the management of the files for the export and the importation through USB;

11 Production: it allows the visualization of the production data of the machine;

12 Production/Shift: it allows the production shifts management;

13 Edit texts: it allows the texts management in the different languages;

14 Maintenance: it allows to access the maintenance page;

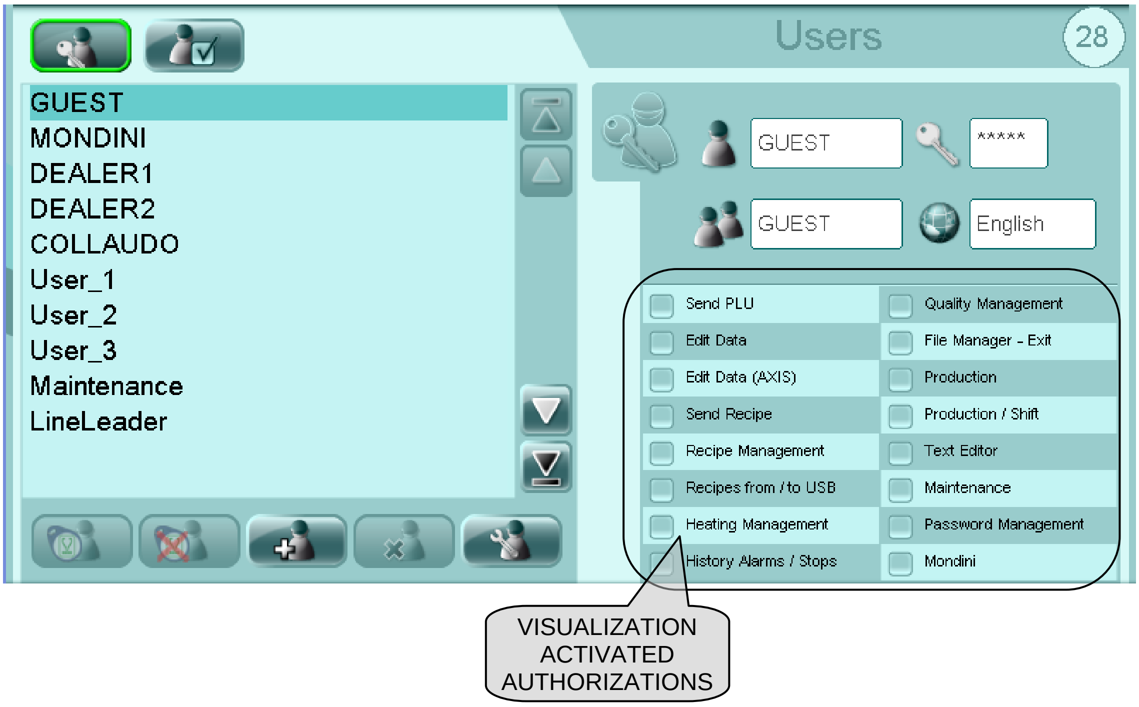
15 Password management: it allows the creation, the elimination and the change of the users and of the user groups;

16 Mondini: this authorization is accessible only to the "G.Mondini" technicians.

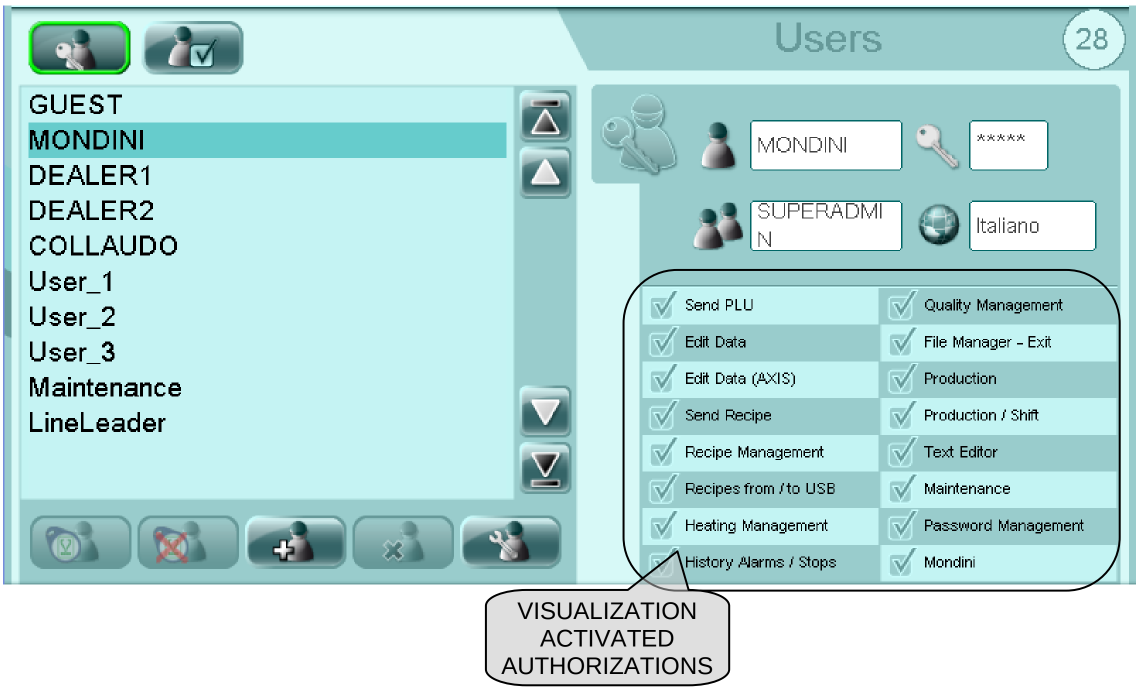
5.11.6 Standard “G.Mondini” users

In this paragraph the users and the relative activated authorizations are described as they have been thought and realized by the "G.Mondini" firm.

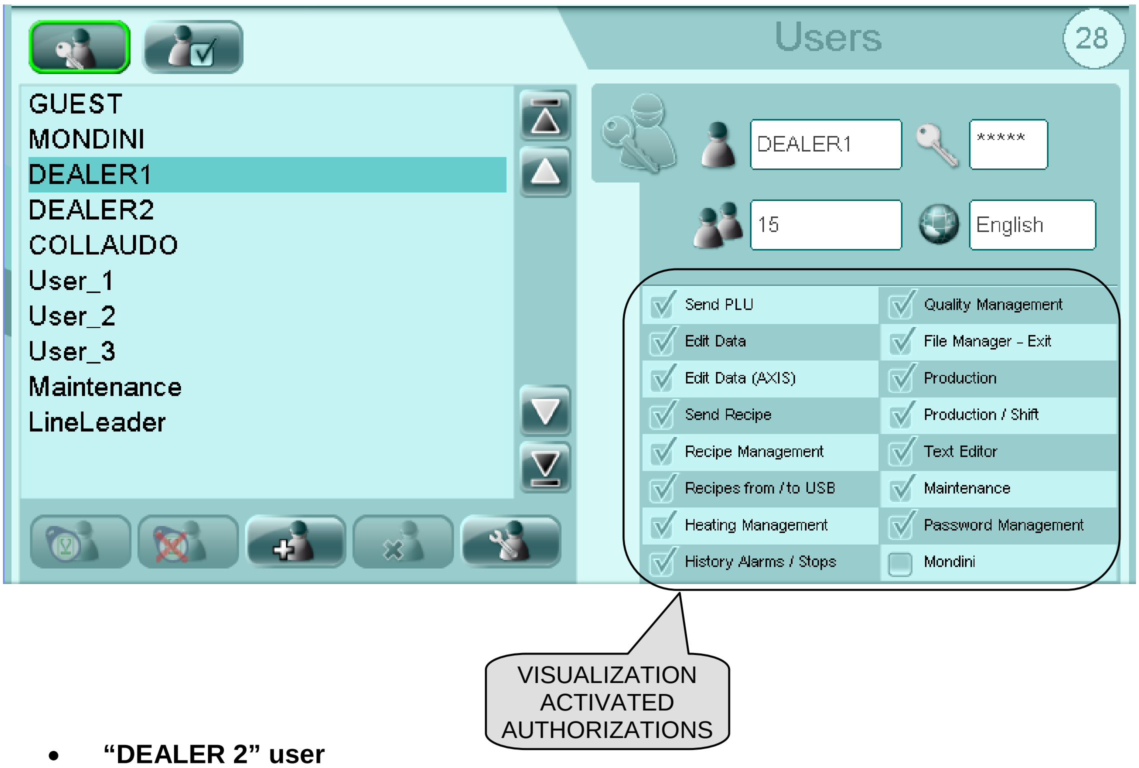
- **“GUEST” user**



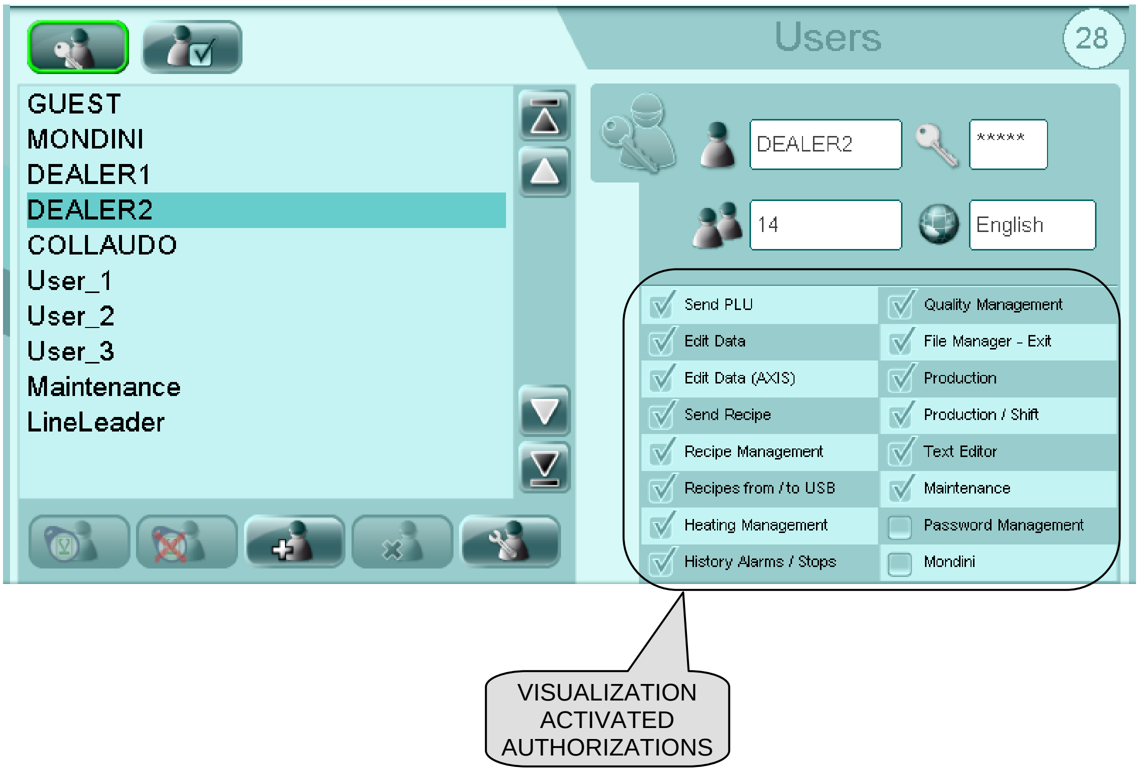
- **“MONDINI” user**



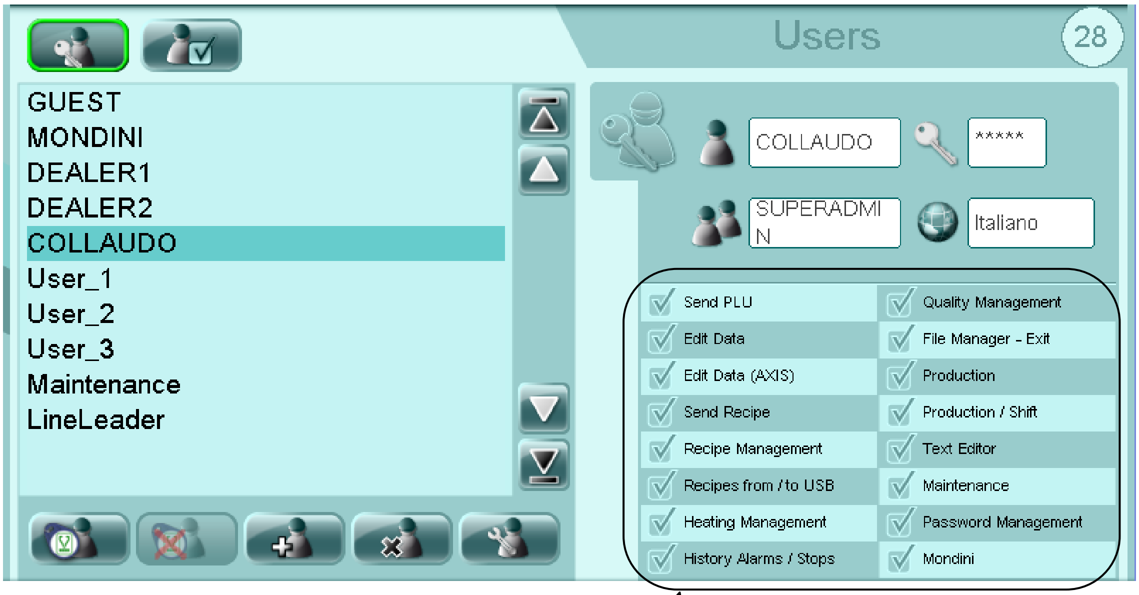
- **"DEALER 1" user**



- **"DEALER 2" user**

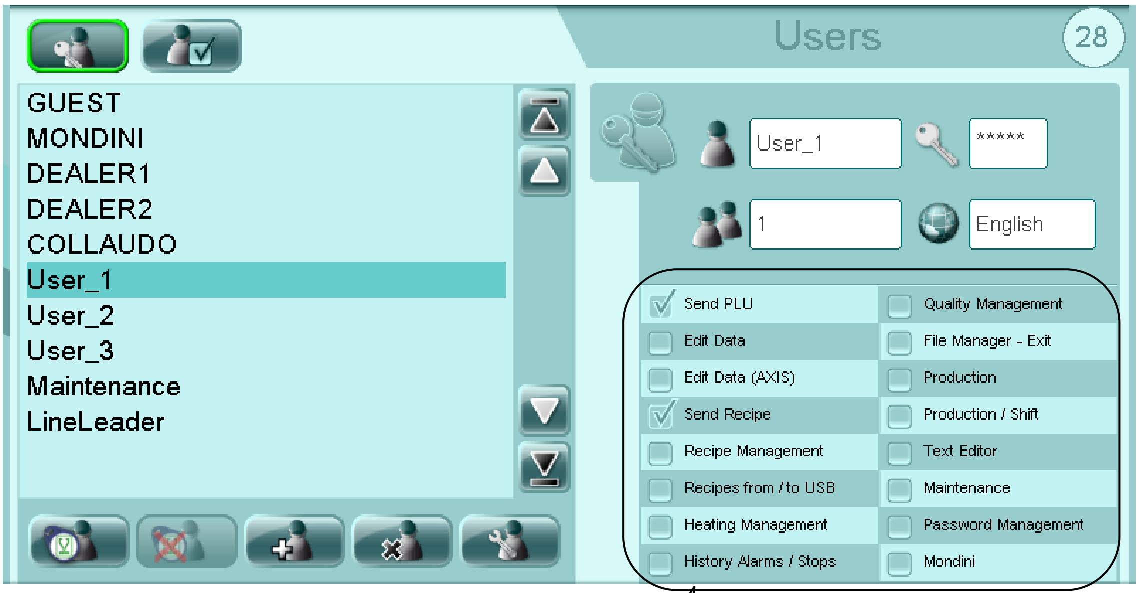


- **“COLLAUDO” user**



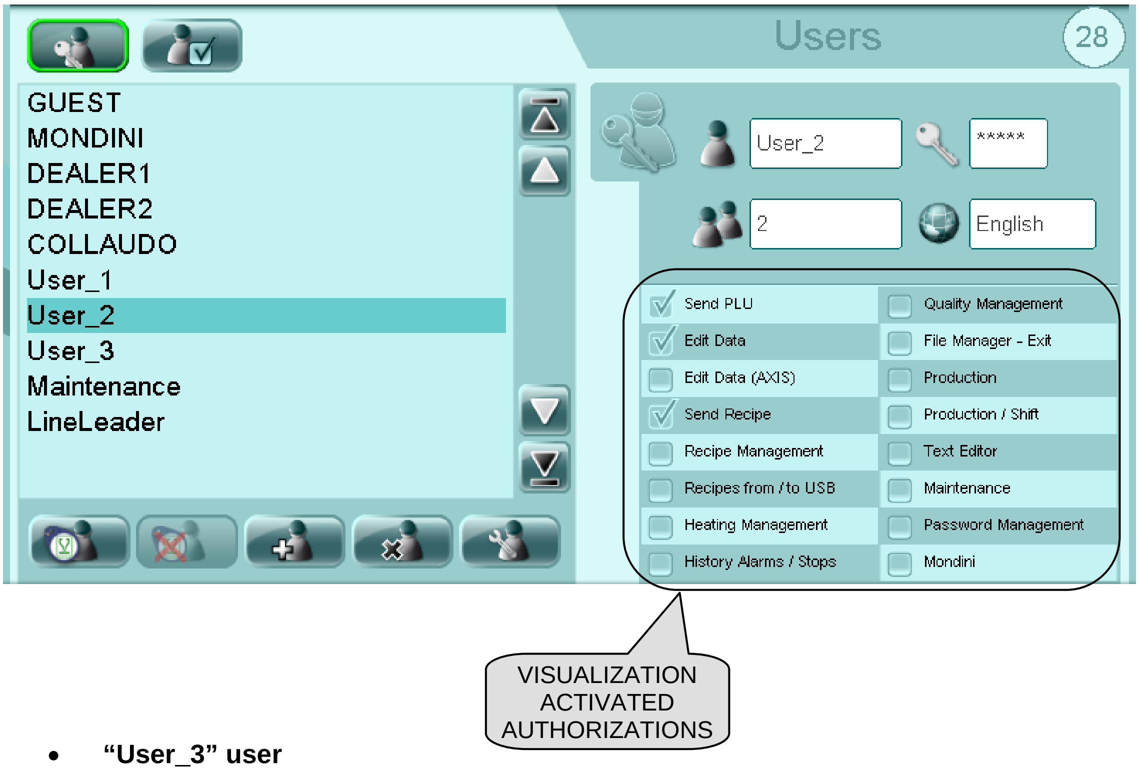
VISUALIZATION
ACTIVATED
AUTHORIZATIONS

- **“User_1” user**

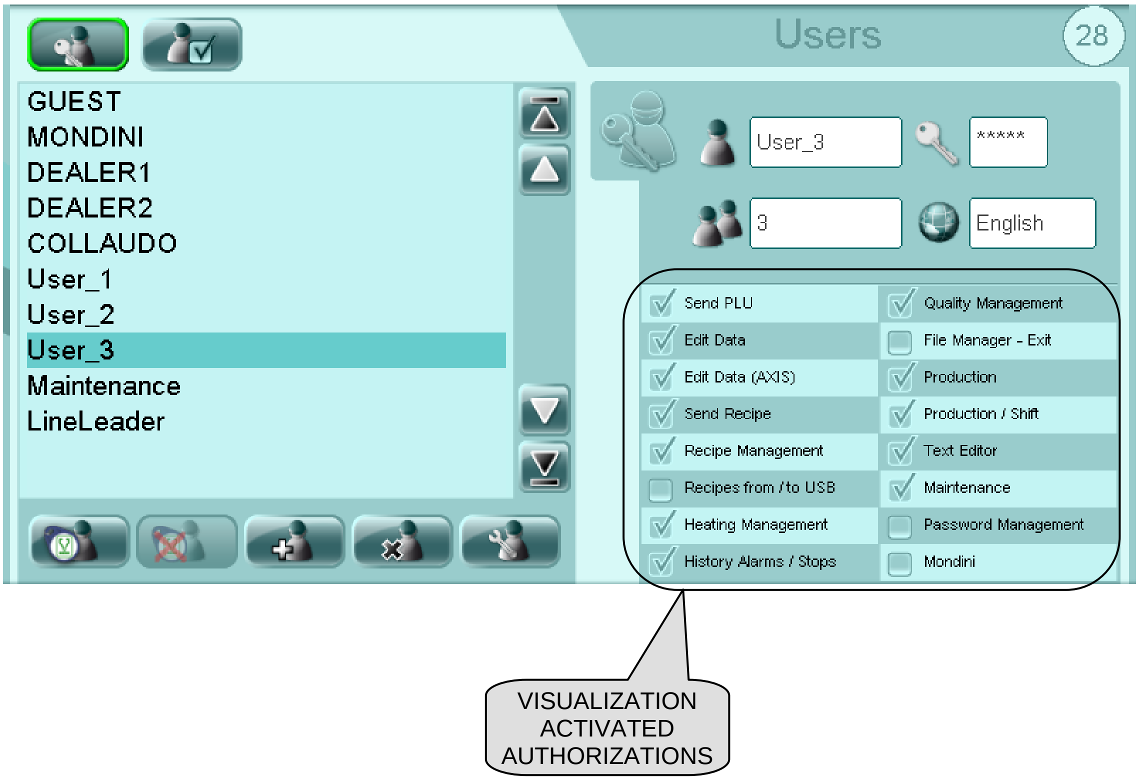


VISUALIZATION
ACTIVATED
AUTHORIZATIONS

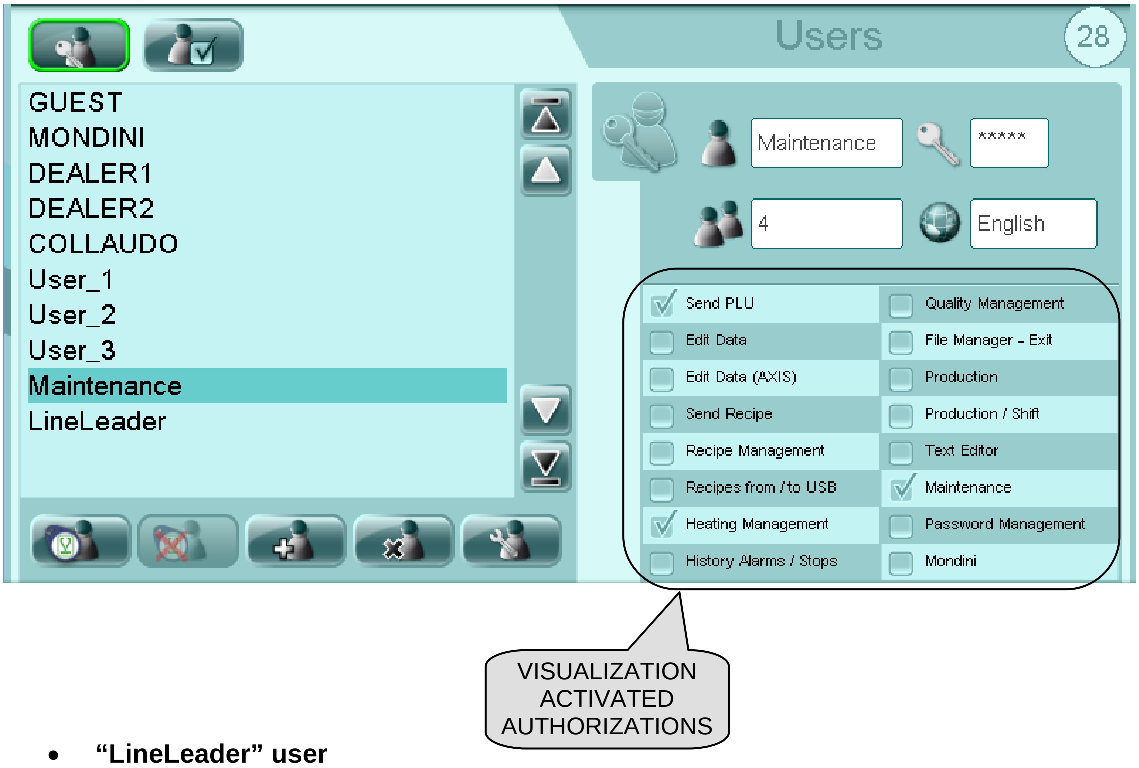
- **"User_2" user**



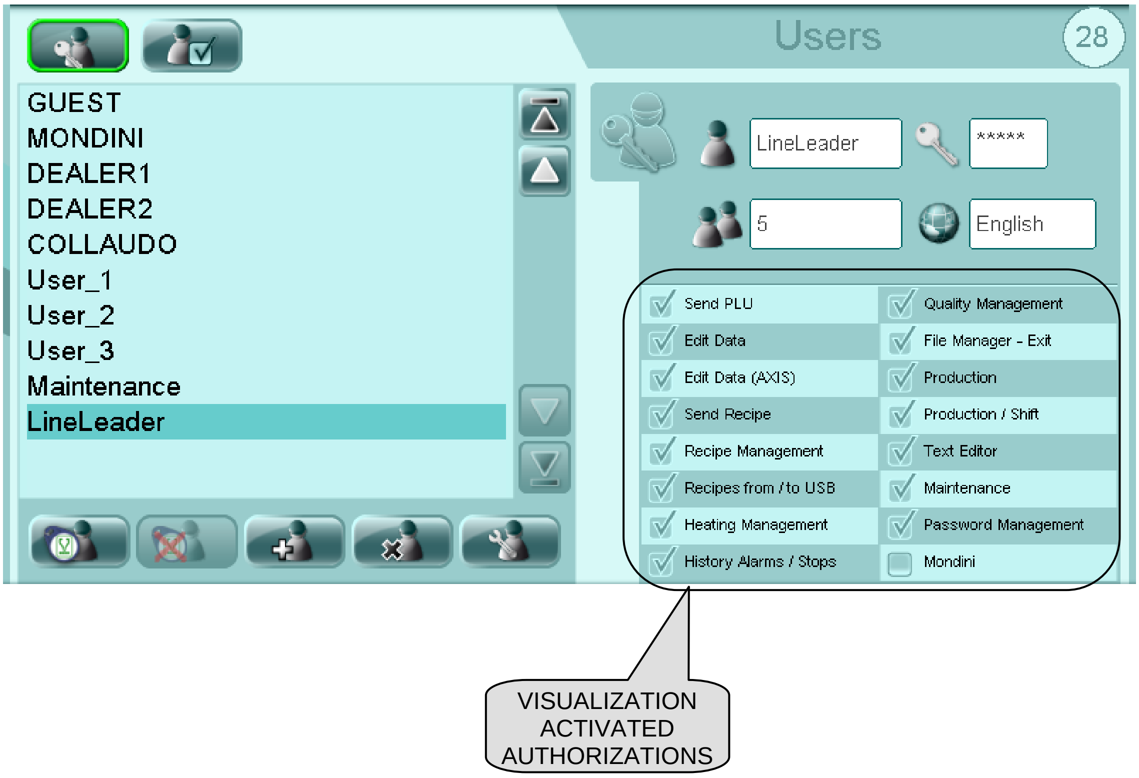
- **"User_3" user**



- **“Maintenance” user**

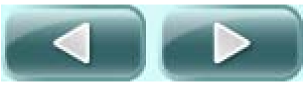


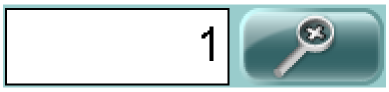
- **“LineLeader” user**



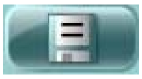
5.12 MULTILANGUAGE TEXT MANAGEMENT

ATTENTION! This page is accessible only if the user is logged with the authorization "Edit texts."

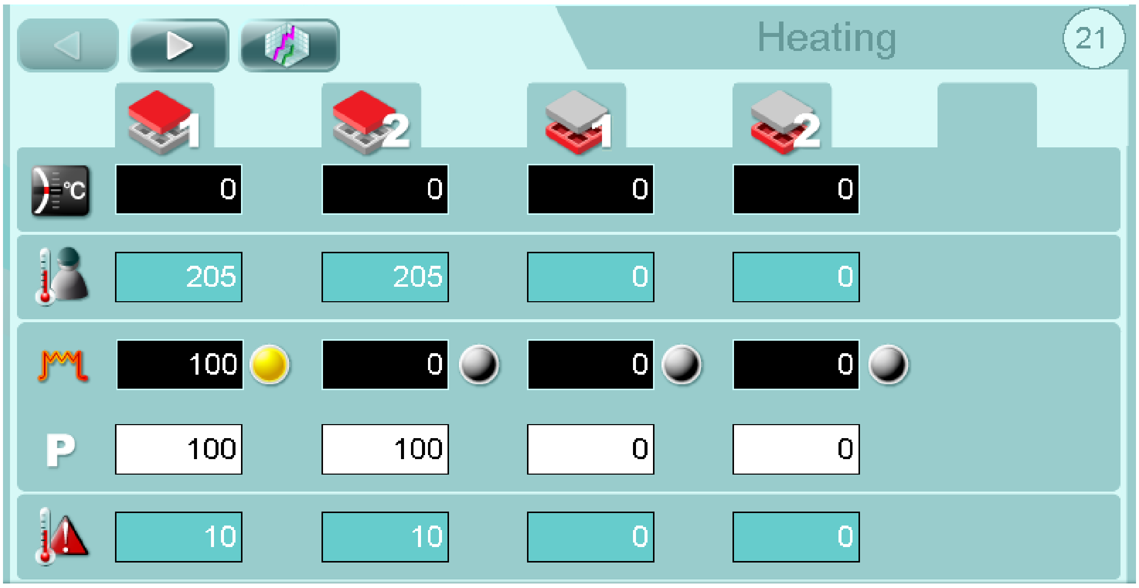
Scroll the kind of text (parameter, enables, alarm and stops) with  pushbuttons.

Search the text number toching the search text  area and insert the text number or scroll the text with  pushbuttons.

Write the new text.

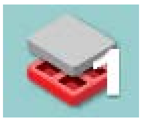
Save the changes with  pushbutton.

5.13 TOOL HEATING



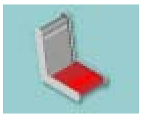
UPPER TOOL 1 HEATING

UPPER TOOL 2 HEATING

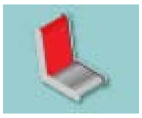


LOWER TOOL 1 HEATING

LOWER TOOL 2 HEATING



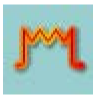
SKIN CYCLE HORIZONTAL PREHEATING



SKIN CYCLE VERTICAL PREHEATING

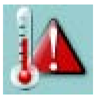
CURRENT TEMPERATURE

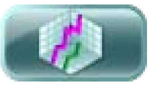
SET POINT TEMPERATURE (RECIPE DATA)

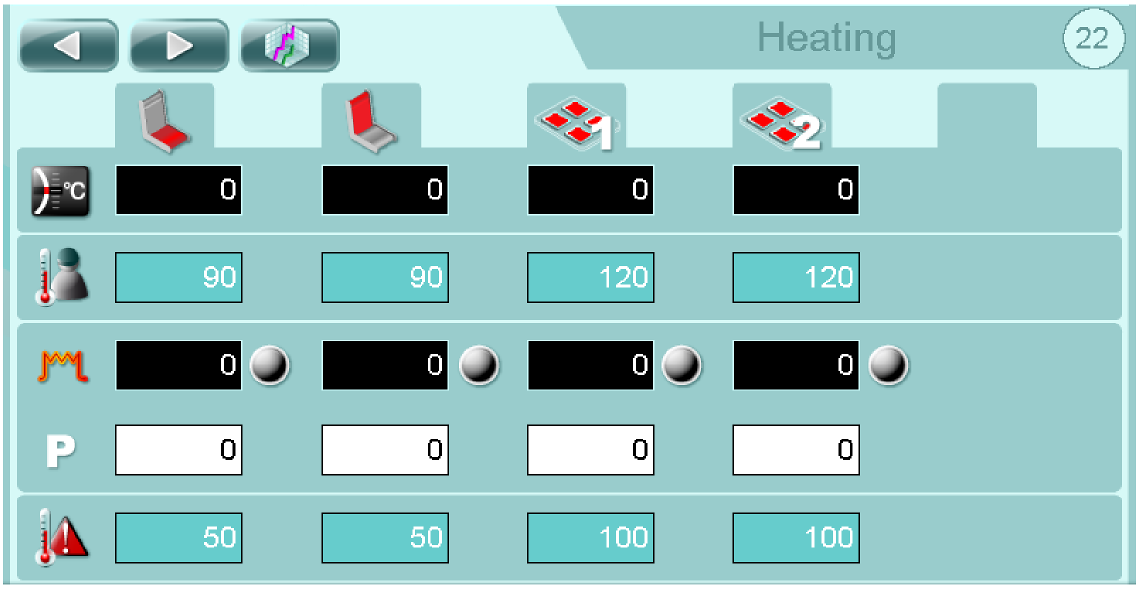

RESISTANCE HEATING POWER (%)

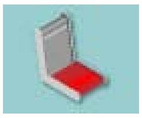

LIGHT SHOWS WHEN HEATING ON


PROPORTIONAL PARAMETER FOR HEATING SYSTEM

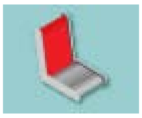

TEMPERATURE ALARM (RECIPE DATA)


TRENDS PAGE

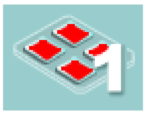




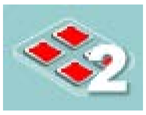
SKIN CYCLE HORIZONTAL PREHEATING



SKIN CYCLE VERTICAL PREHEATING



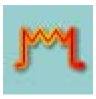
SKIN RESISTANCE 1 HEATING



SKIN RESISTANCE 2 HEATING

CURRENT TEMPERATURE

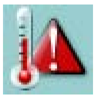
SET POINT TEMPERATURE (RECIPE DATA)

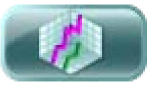


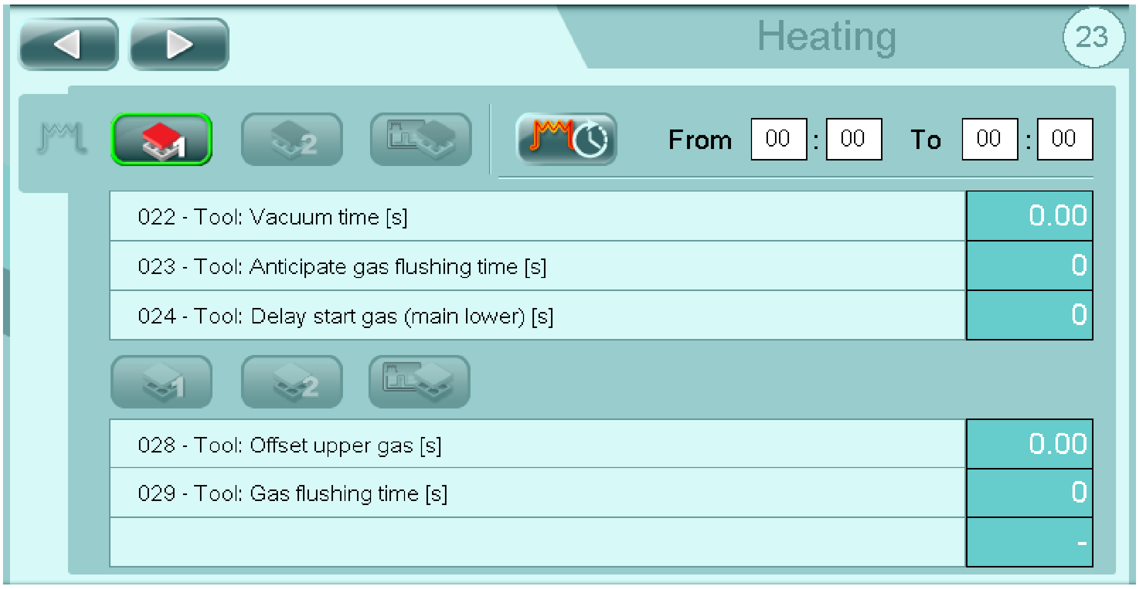
RESISTANCE HEATING POWER (%)


LIGHT SHOWS WHEN HEATING ON

PROPORTIONAL PARAMETER FOR HEATING SYSTEM


TEMPERATURE ALARM (RECIPE DATA)


TRENDS PAGE



ENABLE UPPER TOOL 1 HEATING (RECIPE DATA)

ENABLE UPPER TOOL 2 HEATING (RECIPE DATA)

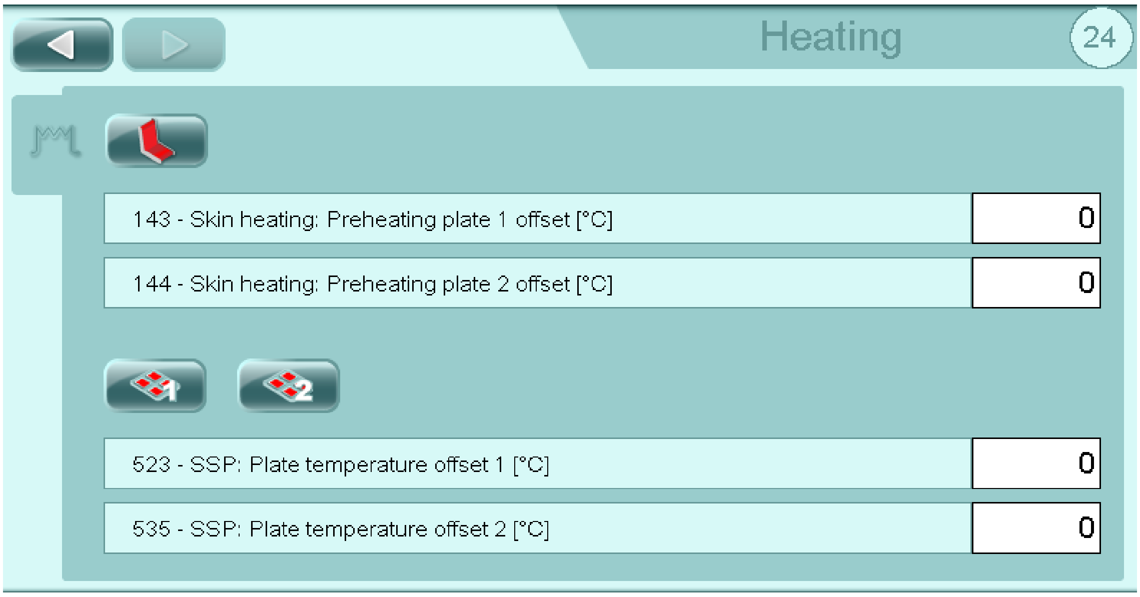
ENABLE ONE SHOT HEATING (RECIPE DATA)

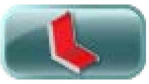
ENABLE PROGRAMMED HEATING CYCLE

NOTE: To change the time for switching on and off of the heating cycle programmed tool simply enter the time of ignition in the "From 00:00" and the turn-off time in the "To 00:00".

ENABLE LOWER TOOL HEATING (RECIPE DATA)

ENABLE LOWER TOOL ONE SHOT HEATING (RECIPE DATA)



 ENABLE VERTICAL AND HORIZONTAL PREHEATING FOR SKIN CYCLE (IF AVAILABLE)

 ENABLE THE PLATE PREHEATING MOUNTED ON THE SHUTTLE (IF AVAILABLE)

5.14 LOCKING

REEL 1

REEL 2

DRIVE ROLLER (WHEN AUTOMATIC REEL CHANGE IS AVAILABLE)

NOT USED

UPPER TOOL

LOWER TOOL

LID UNWINDER

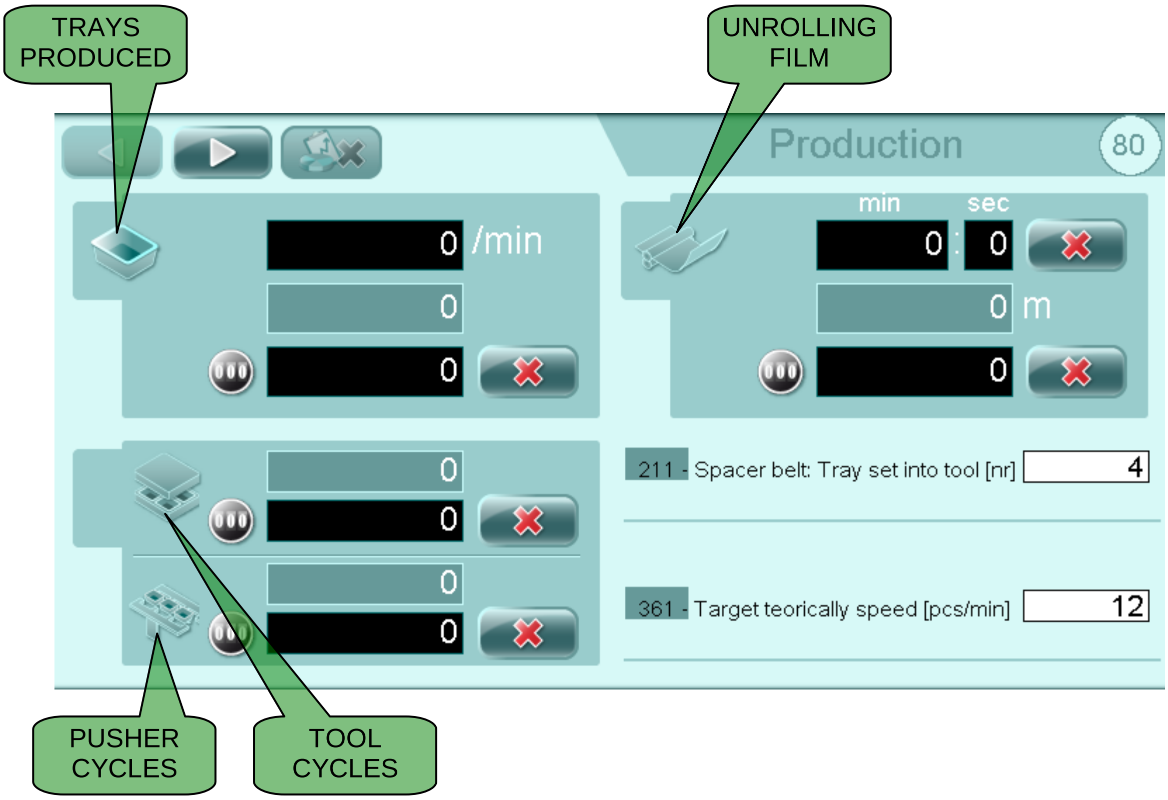
DISABLE THE COMPRESSED AIR CIRCUIT

DARFRESH TOOL

WASTE WINDER

WASTE WINDER WASTE WINDER CYCLE

5.15 PRODUCTION DATA



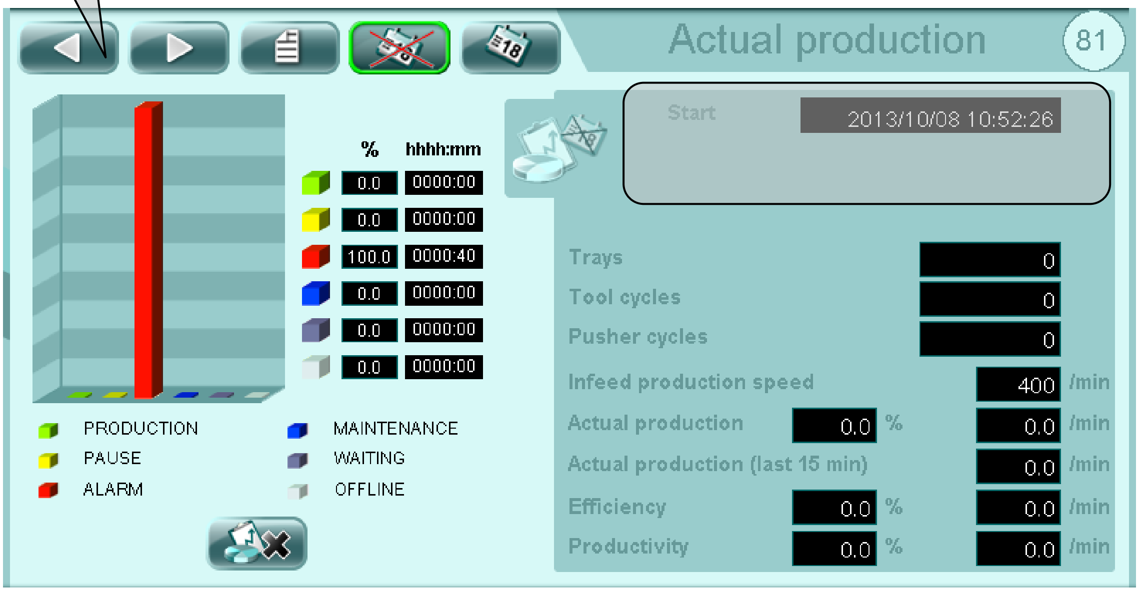
CURRENT PRODUCTION

TOTAL MACHINE OUTPUT

PARTIAL MACHINE OUTPUT

RESET PIECE COUNTER

MACHINE PRODUCTION GRAPH



PRODUCTION RESET (NOT BY SHIFT)

PRODUCTION DATA (NOT BY SHIFT)

PRODUCTION DATA SEARCH BY SHIFT

Trays: number of items produced by the machine.

Tool cycles: number of cycles effected by the tool.

Pusher cycles: number of cycles effected by the pusher arms.

Actual Performance %: items produced in consideration to the items to produce.

Efficiency/min: Items / time OPERATION.

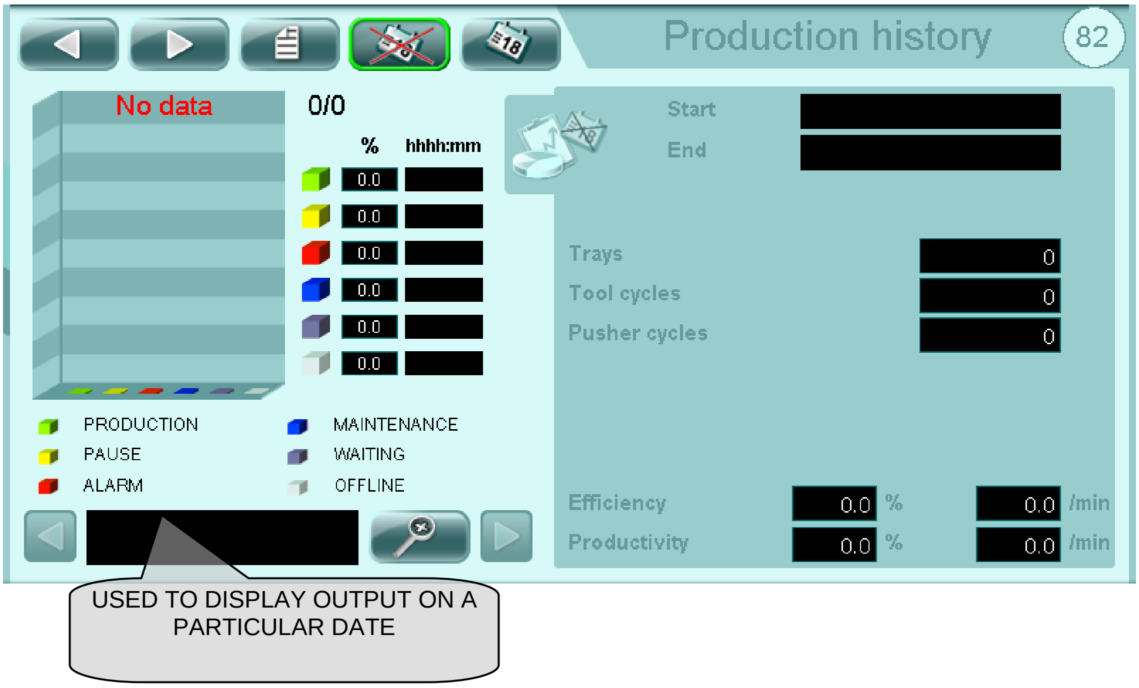
Productivity/min: Items / RUN time.

Performance/min: Items / time

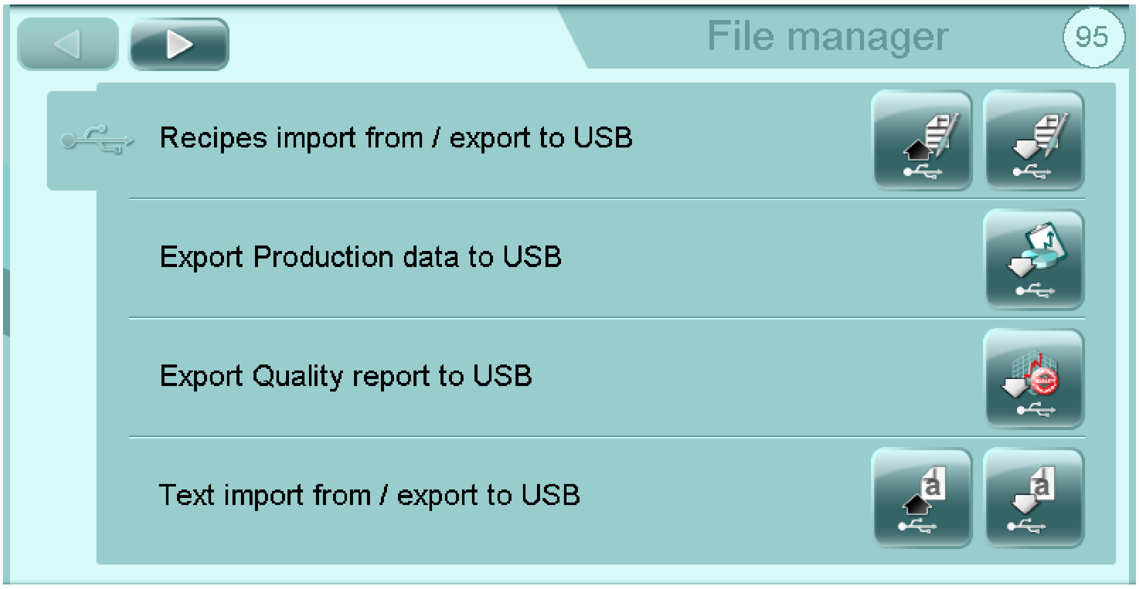
Efficiency %: (run time x theoretical production x 100) / ietms.

Productivity %: (function time x theoretical production x 100) / items.

Performance %: (time x theoretical production x 100) / items.


Production history
82
No data
0/0
%
hhhh:mm
Start
End
Trays
Tool cycles
Pusher cycles
PRODUCTION
PAUSE
ALARM
MAINTENANCE
WAITING
OFFLINE
Efficiency
Productivity
/min
USED TO DISPLAY OUTPUT ON A PARTICULAR DATE

5.16 FILE MANAGER



Imports from USB the .csv file for a recipe

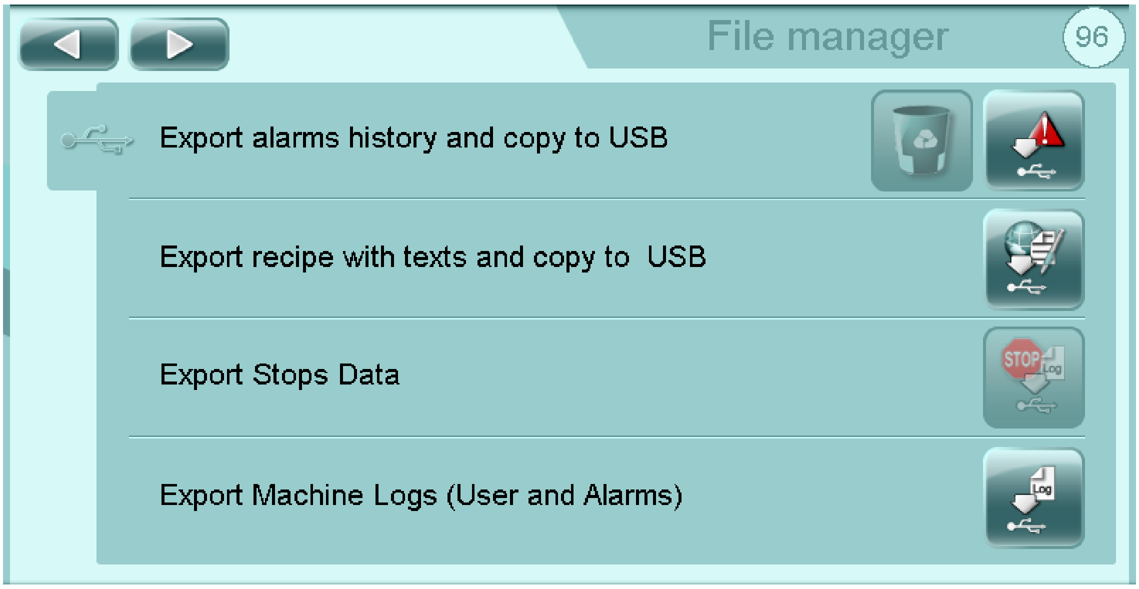
It allows to export on USB or to care from USB on her or from the folder "RECIPE" the file ". csv" for every recipe and a folder with the same name of the recipe containing the image.

It allows to export on USB a file ". csv" in the folder "PRODUCTION" where the data of the production are copied separated in the modality "with shifts" and in the modality "without shifts".

It allows to export on USB a file ". csv" in the folder "QUALITY" for every effected recording.

Imports from USB, parameter, enable, alarm and stoppage texts

It allows to export a file ". csv" in the folder "TEXTS" of the USB with parameters, enable, alarms and locks.



Exports to USB a .csv file in the EXP_AL_STORY folder with alarms in the selected language for a given time period.

It allows to export on USB a file ". csv" with the parameters of the recipe, the texts in language and the image related to the recipe.

It allows to export on USB the LOG of the users in a file ". csv" in the folder "LOGGER."

Imports from the USERS folder in USB the list of users with passwords and groups.

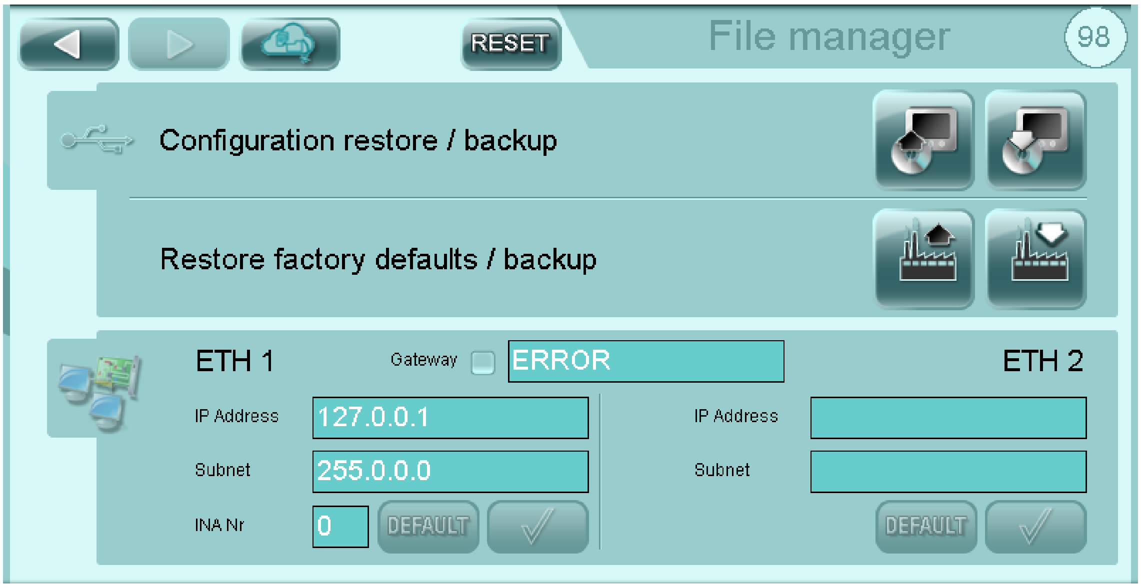
Exports on USB a file ". csv" in the folder "USERS" the users list with the relative passwords and the groups.

Imports from the AL_CONF folder in USB the alarm configuration for the alarm group management

Exports on USB a file ". csv" in the folder "AL_CONF" with the configuration of the alarms for the management of the groups alarm.

Imports from the USB the inverters parameters file.

Exports in the USB the inverters parameters file.



Enable the option to request a recipe by the Supervisor.

Complete system restart, every temporary data will be lost!

Restores the files of the last back up saved.

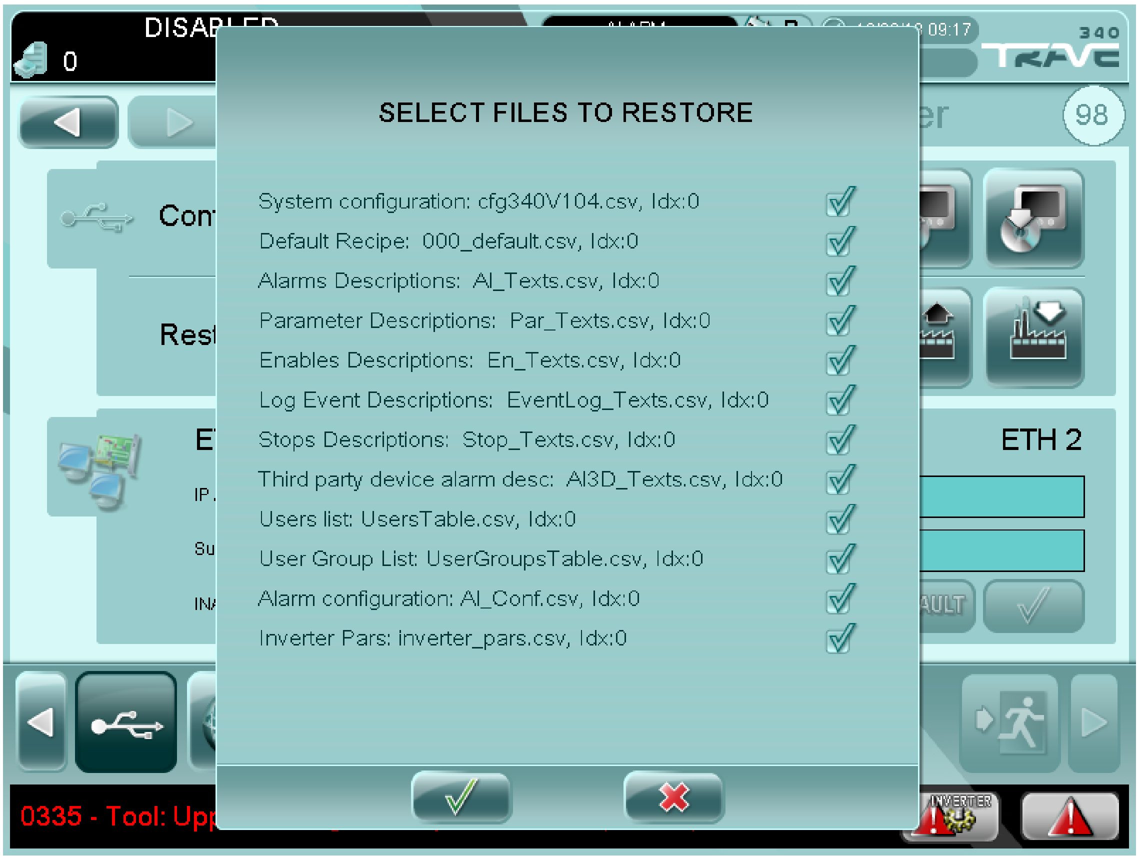
Exports on USB a back up copy of the whole files used byu the application with the exclusion of manuals and video.

It allows the selective restoration of the factory settings.

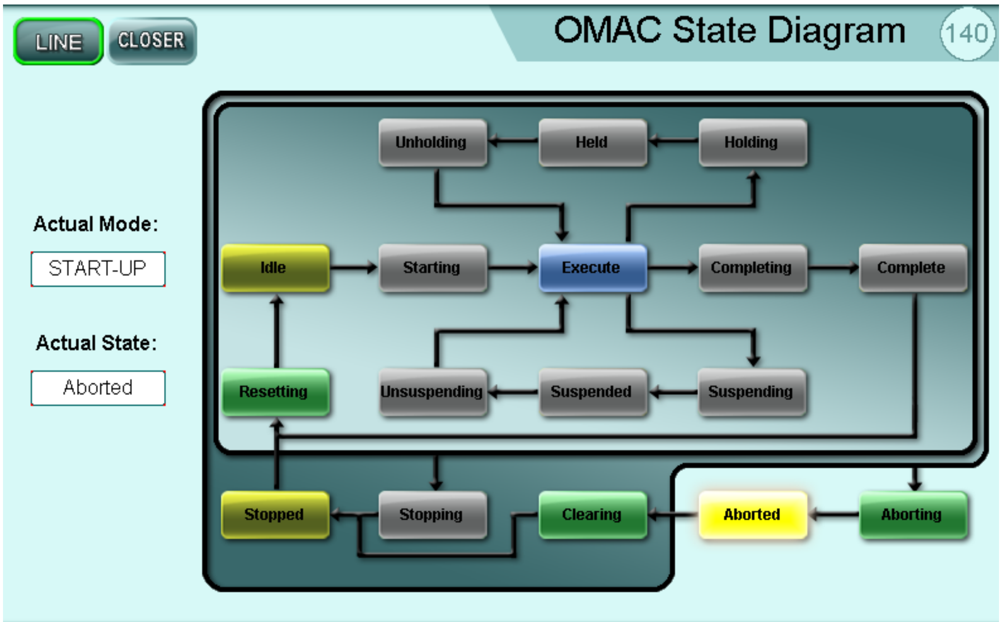
It allows to create the factory settings (this operation must be executes only after having contacted the "G.Mondini" technicians).

In the lower part of the page there are the exit ethernet data of the panel.

5.16.1 Restore factory defaults



5.17 OMAC



This machine has been built according to the standard OMAC. This page shows the state of job of the machine. Using the buttons in this page is possible to effect a visual diagnostic to make more effective the technical support with the client.

OMAC PAGE

AXIS ALARMS PAGE

AXIS INFORMATIONS PAGE

AXIS DIAGNOSTIC PAGE

AXIS AUTOMAT PAGE

CAM PAGE

HOME PAGE

ACTIVE ALARMS PAGE

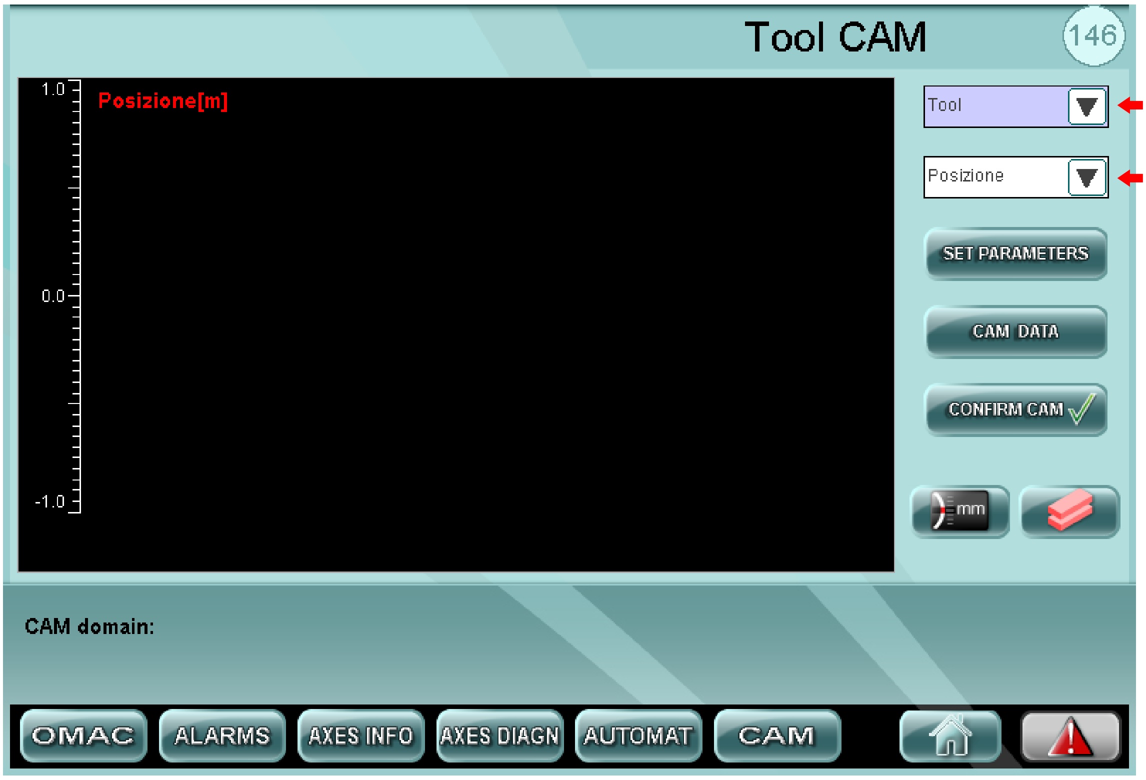
5.18 CAM VIEW PAGE

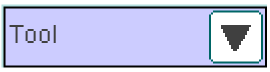
5.18.1 Tool

IMPORTANT! This page can only be accessed if the logged-in user has “Edit Data 3” authorisation.

From the CAM VIEW  page, you can control the cams of the tool and denester (if present). Cam selection is via drop-down menu A.
Use drop-down menu B to select the physical size to display on the graph (position, speed, acceleration).

The cam parameters can only be entered on these pages. On the tool axis pages, these data are only displayed with a decimal point.



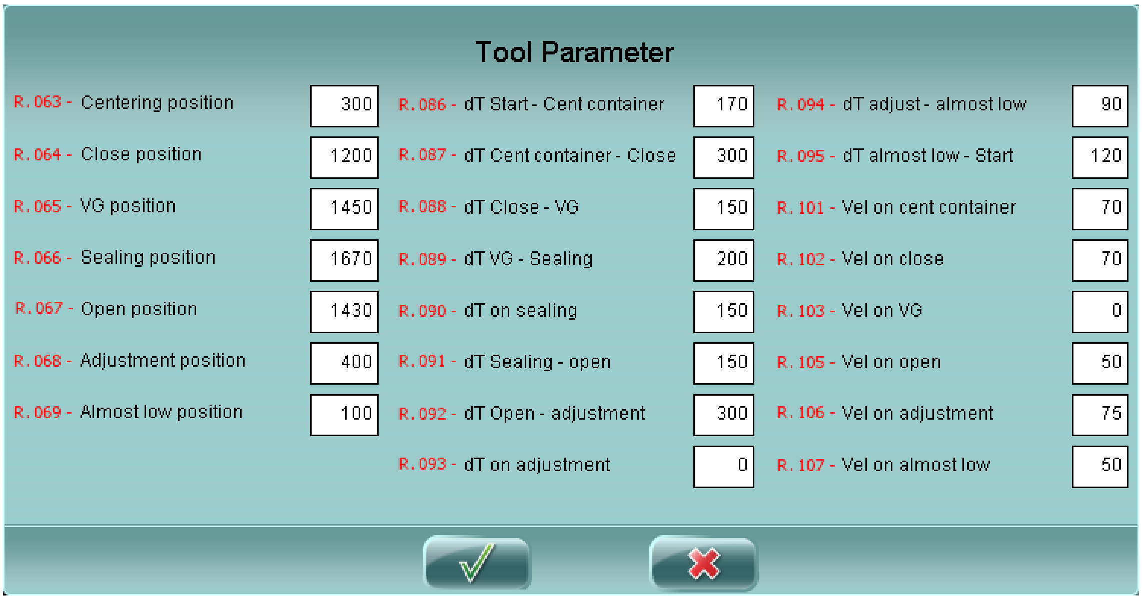


: tool or denester cam selection (if envisaged)

: graph size selection

Click 

to enter the cam parameters.



N.B. These parameters must only be modified by fully qualified and authorised personnel as they involve the entire machine cycle.

Positions are expressed in tenths of a degree (with connecting rod/crank movement) or in tenths of a millimetre (with screw movement).

Time intervals for moving from one position to another are expressed in milliseconds. They must all be different from 0, except for R093, which can be 0 if there is not stop for tray settling.

After  has been clicked to confirm the data, the software runs calculations to generate the cam.

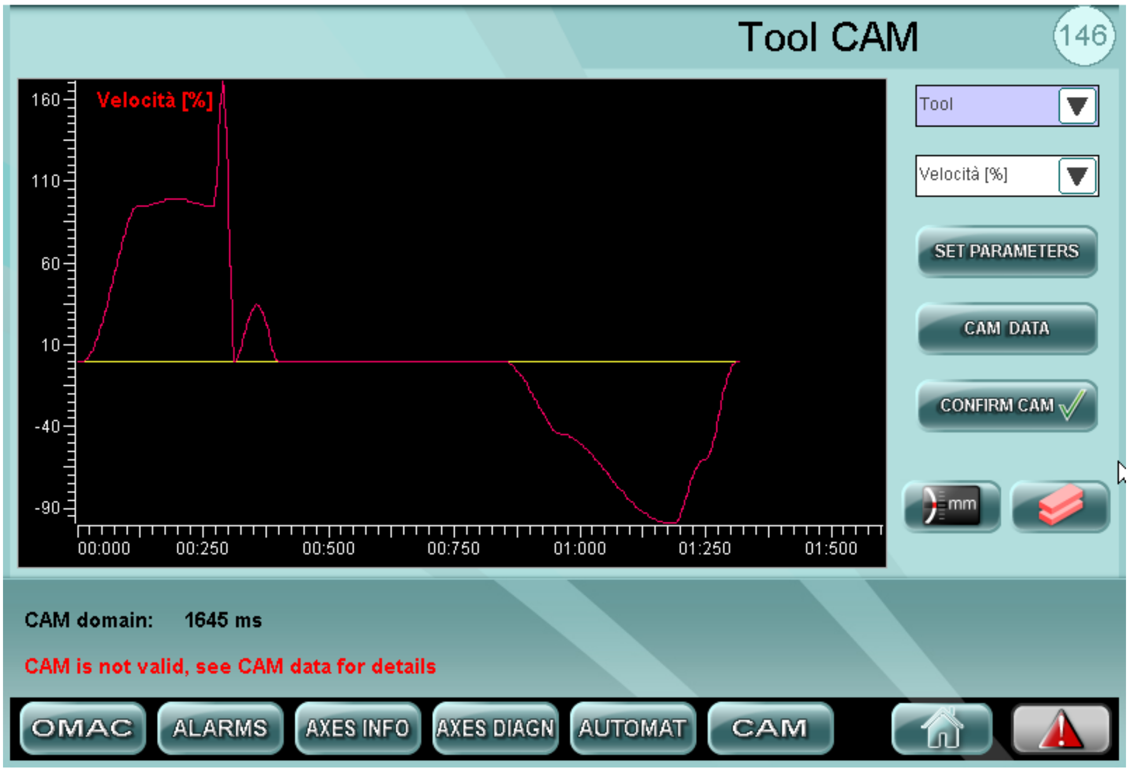
If the correct parameters have been entered, the message "CAM respects limit, you can download it" is displayed in green at the bottom of the screen. You can now download the cam by clicking 

.

If the parameters entered generate speeds exceeding 100%, the message "CAM is not valid, see CAM data for details" is displayed in red at the bottom of the screen. In this case the cam cannot be downloaded.

Click 

to view the time intervals that generated the malfunction.

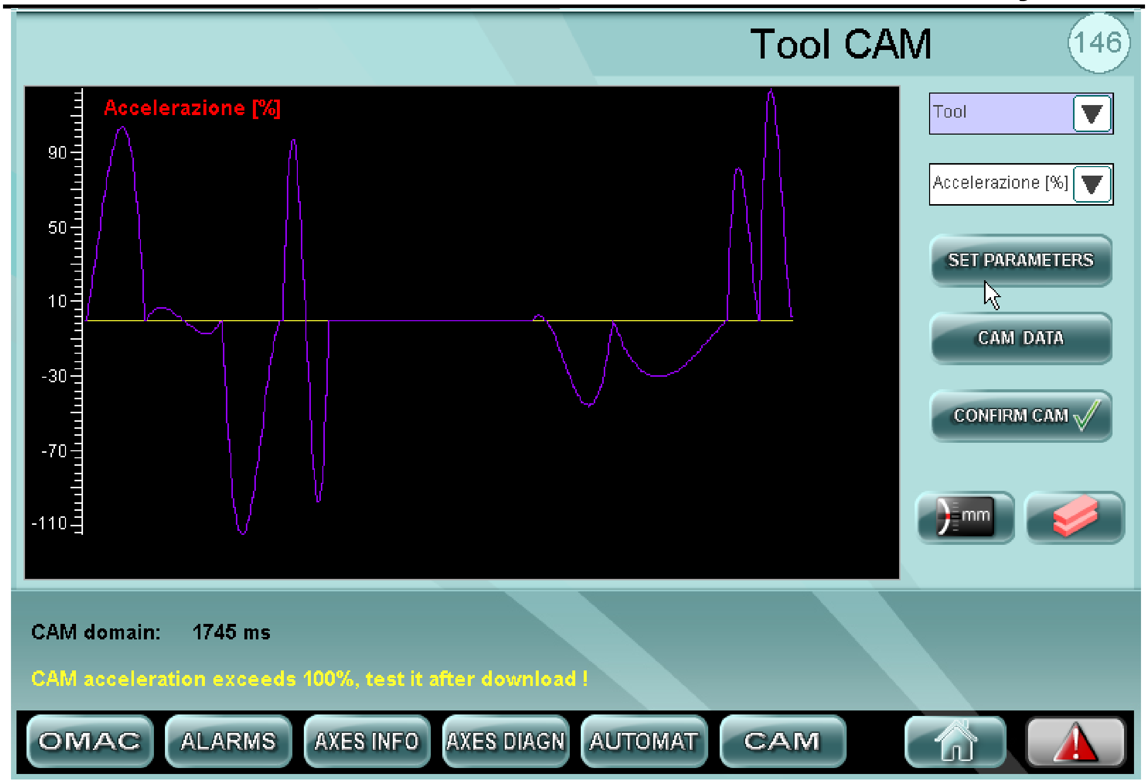


If the parameters entered generate accelerations exceeding 100%, the message "CAM acceleration exceeds 100%, test it after download!" is displayed in yellow at the bottom of the screen. The cam can be loaded by clicking 

, but it is advisable to check by how much the calculated accelerations exceed 100% (see graph below).

To verify the incidence of accelerations, select them from drop-down menu B to access the data graph. The best cam presents the lowest number of acceleration and lowest intensity peaks.

The same rule that "the lower the peaks, the better the cam" also applies to the speeds.


Tool CAM
146
Accelerazione [%]
90
50
10
-30
-70
-110
Tool
Accelerazione [%]
SET PARAMETERS
CAM DATA
CONFIRM CAM
mm
CAM domain: 1745 ms
CAM acceleration exceeds 100%, test it after download !
OMAC
ALARMS
AXES INFO
AXES DIAGN
AUTOMAT
CAM

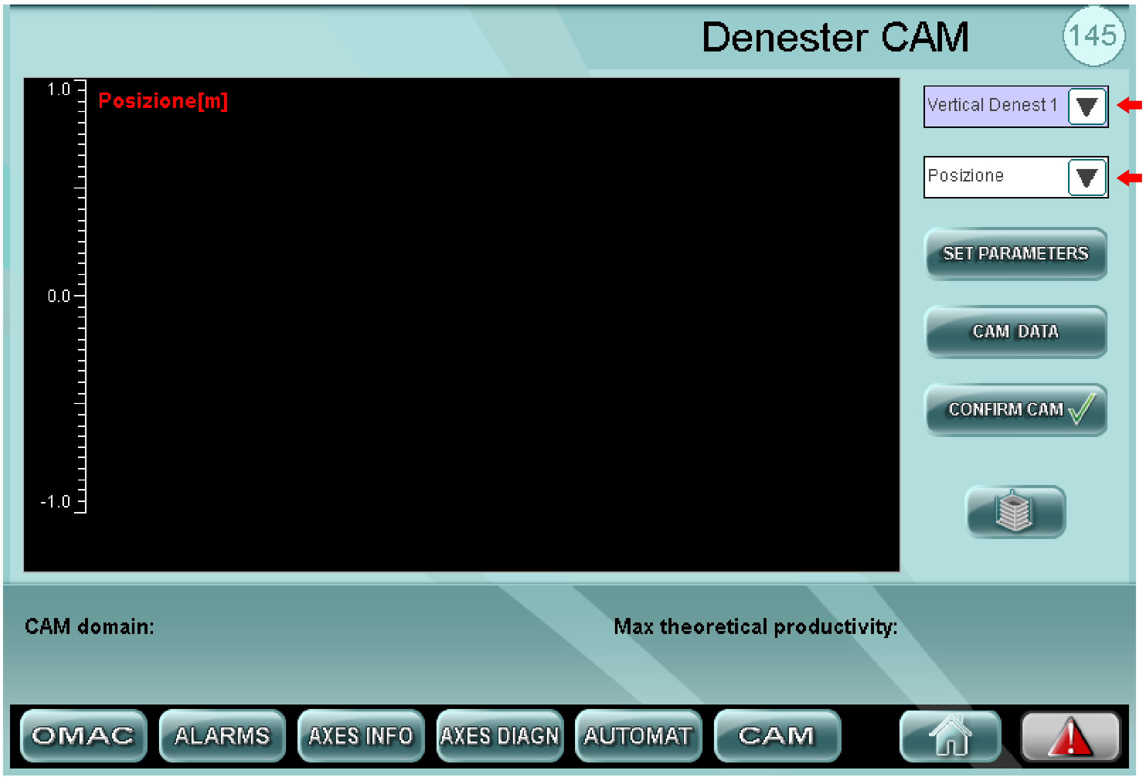
5.18.2 Denester

IMPORTANT! This page can only be accessed if the logged-in user has "Edit Data 3" authorisation.

From the CAM VIEW 

page, you can control the cams of the tool and denester (if present). Cam selection is via drop-down menu A. Use drop-down menu B to select the physical size to display on the graph (position, speed, acceleration).

The cam parameters can only be entered on these pages. On the denester pages, these data are only displayed with a decimal point.

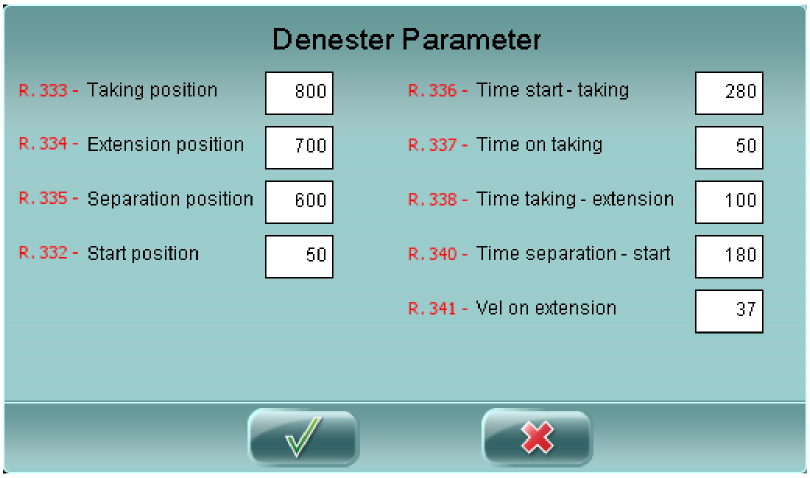


: tool or denester cam selection (if envisaged)

: graph size selection

Click 

to enter the cam parameters.



N.B. These parameters must only be modified by fully qualified and authorised personnel as they involve the entire machine cycle.

Positions are expressed in tenths of a millimetre. Time intervals for moving from one position to another are expressed in milliseconds. They must all be different from 0.

After  has been clicked to confirm the data, the software runs calculations to generate the cam.

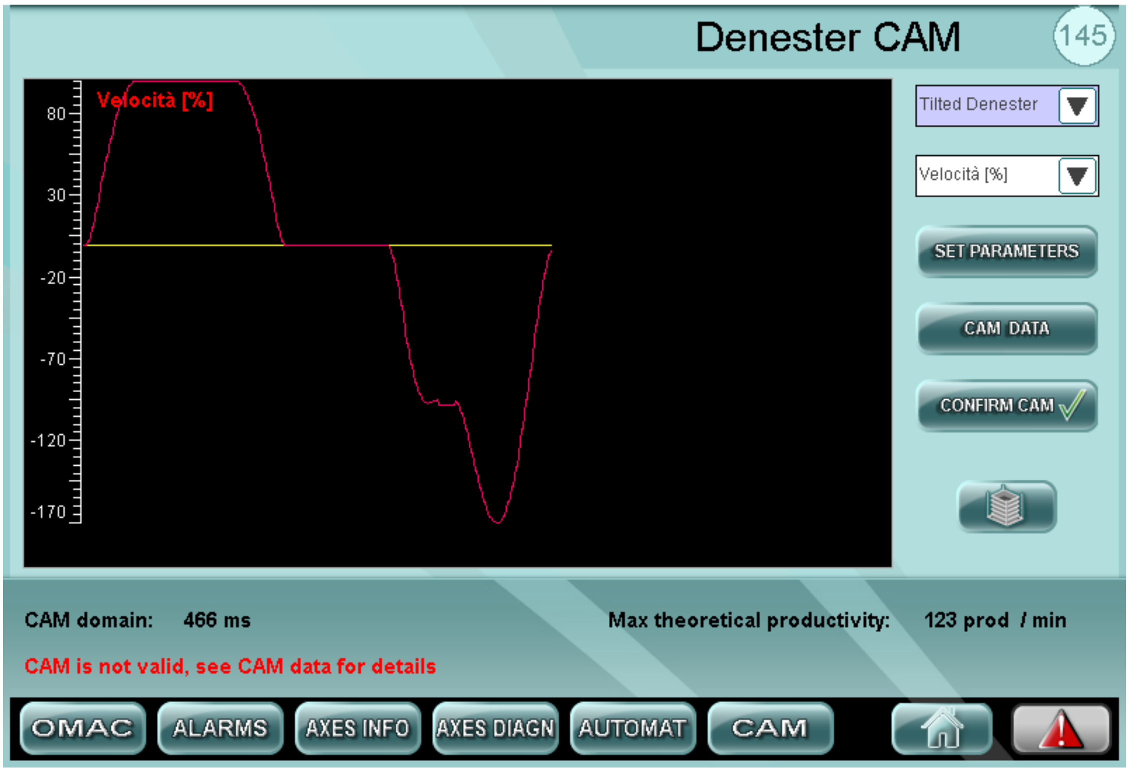
If the correct parameters have been entered, the message “CAM respects limit, you can download it” is displayed in green at the bottom of the screen. You can now download the cam by clicking 

.

If the parameters entered generate speeds exceeding 100%, the message “CAM is not valid, see CAM data for details” is displayed in red at the bottom of the screen. In this case the cam cannot be downloaded.

Click 

to view the time intervals that generated the malfunction.

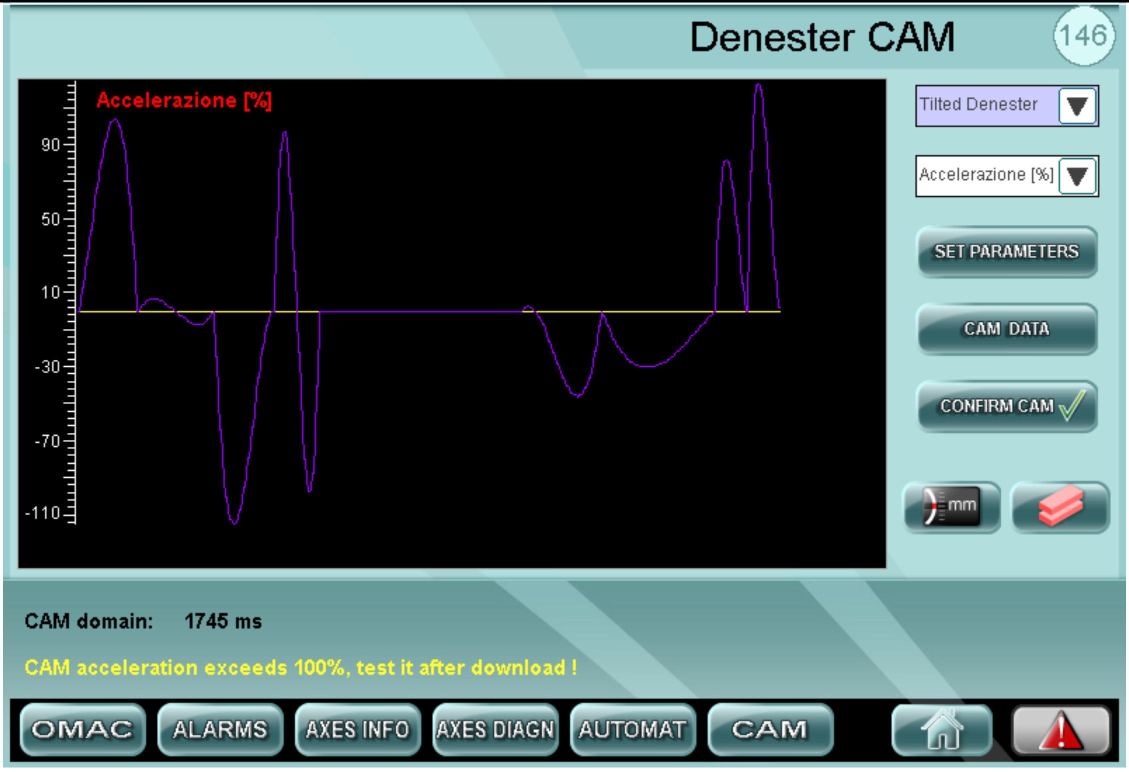


If the parameters entered generate accelerations exceeding 100%, the message “CAM acceleration exceeds 100%, test it after download!” is displayed in yellow at the bottom of the screen. The cam can be loaded by clicking 

, but it is advisable to check by how much the calculated accelerations exceed 100% (see graph below).

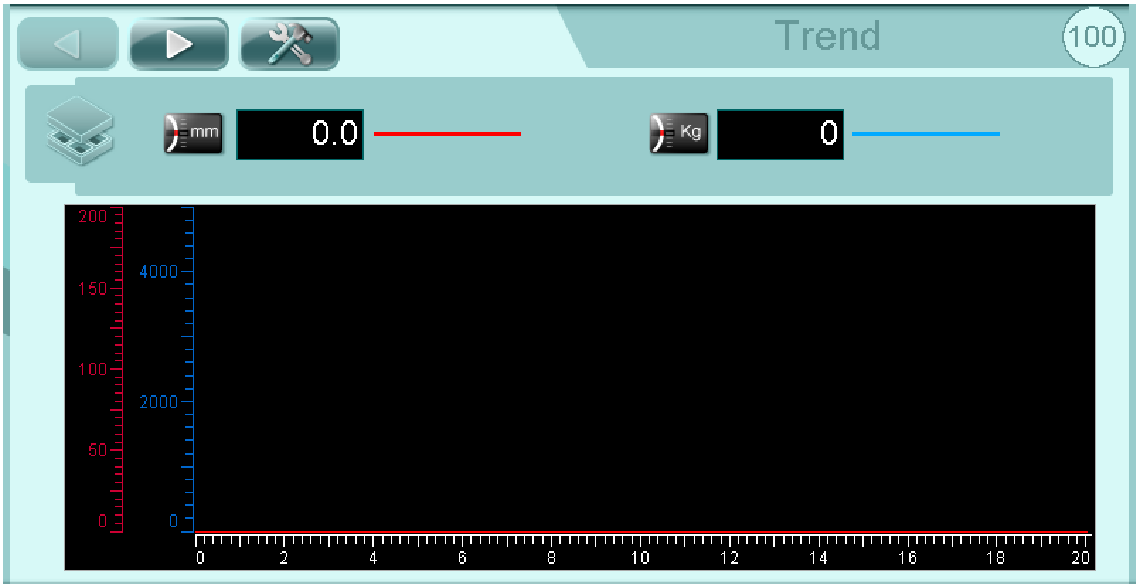
To verify the incidence of accelerations, select them from drop-down menu B to access the data graph. The best cam presents the lowest number of acceleration and lowest intensity peaks.

The same rule that “the lower the peaks, the better the cam” also applies to the speeds.
I


Denester CAM
146
Accelerazione [%]
90
50
10
-30
-70
-110
Tilted Denester
Accelerazione [%]
SET PARAMETERS
CAM DATA
CONFIRM CAM
mm
CAM domain: 1745 ms
CAM acceleration exceeds 100%, test it after download !
OMAC
ALARMS
AXES INFO
AXES DIAGN
AUTOMAT
CAM

5.19 TRENDS

These pages display graphs representing tool data trends during the operating cycle.

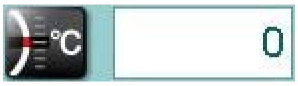


ACCESS TO THE MANAGEMENT PAGE

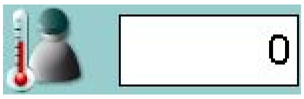
TOOL POSITION IN MILLIMETRES

PRESSURE CLOSING TOOL

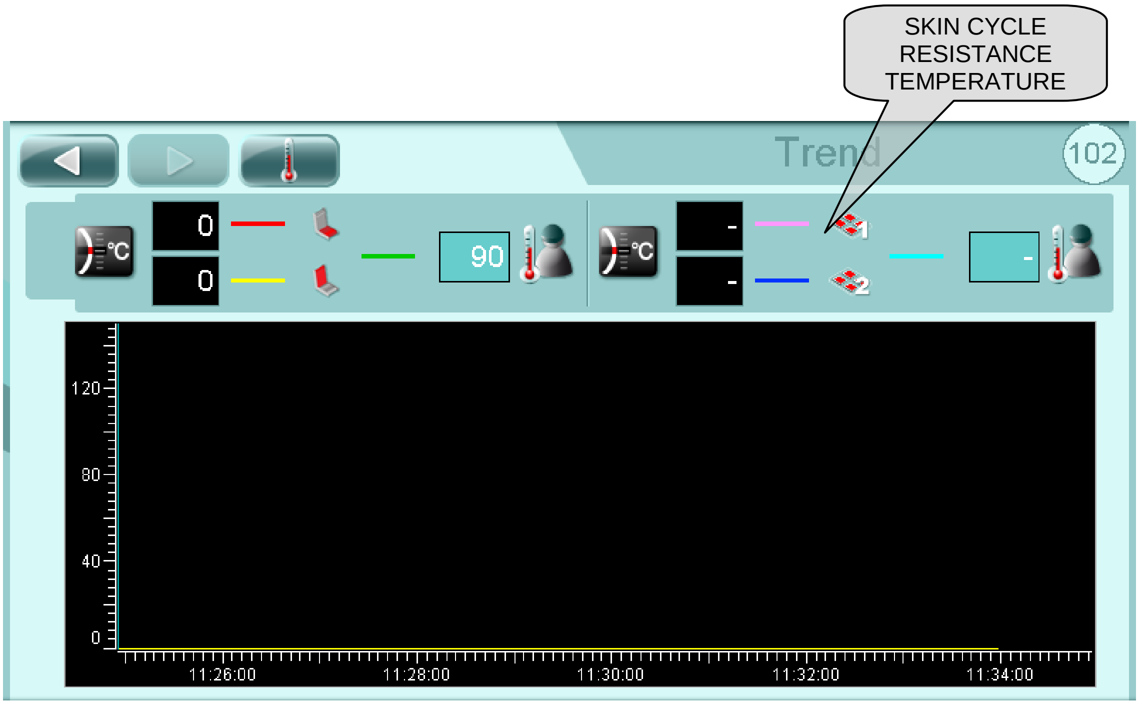
ACCESS TO TEMPERATURE PAGE

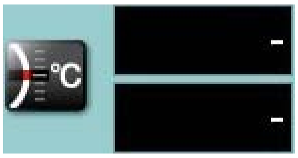


(CURRENT) TOOL TEMPERATURE

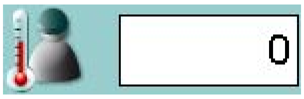


SET POINT TEMPERATURE



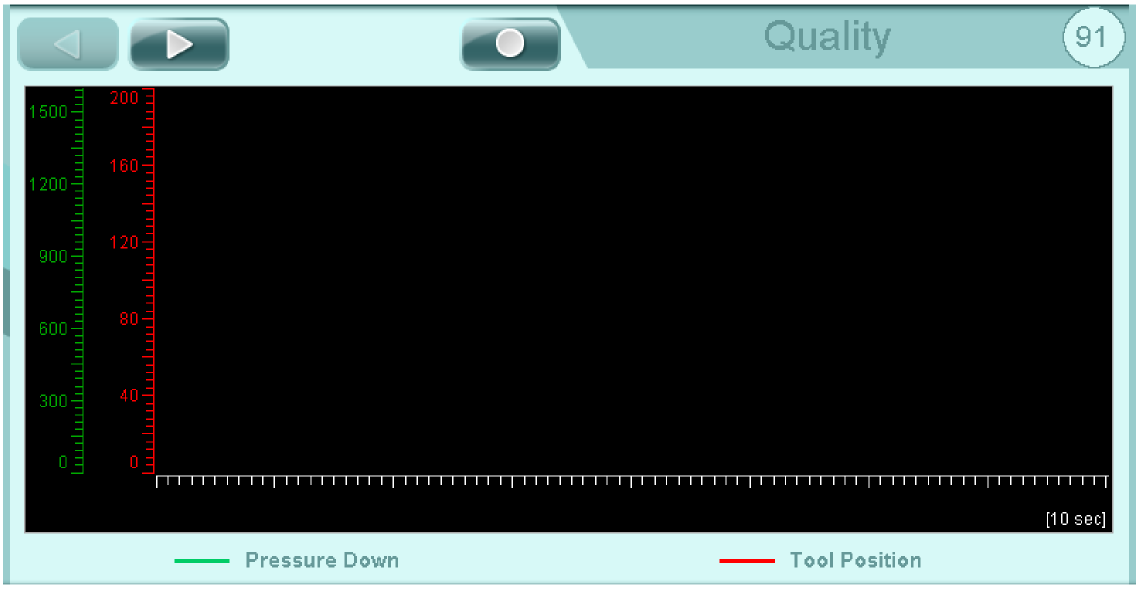


SKIN CYCLE RESISTANCE TEMPERATURE

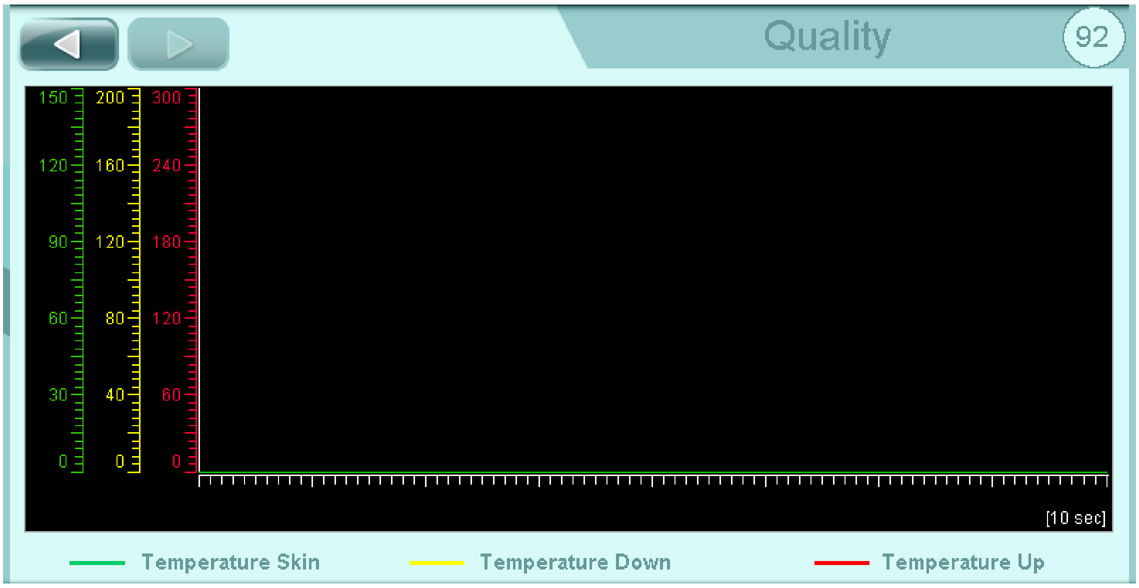


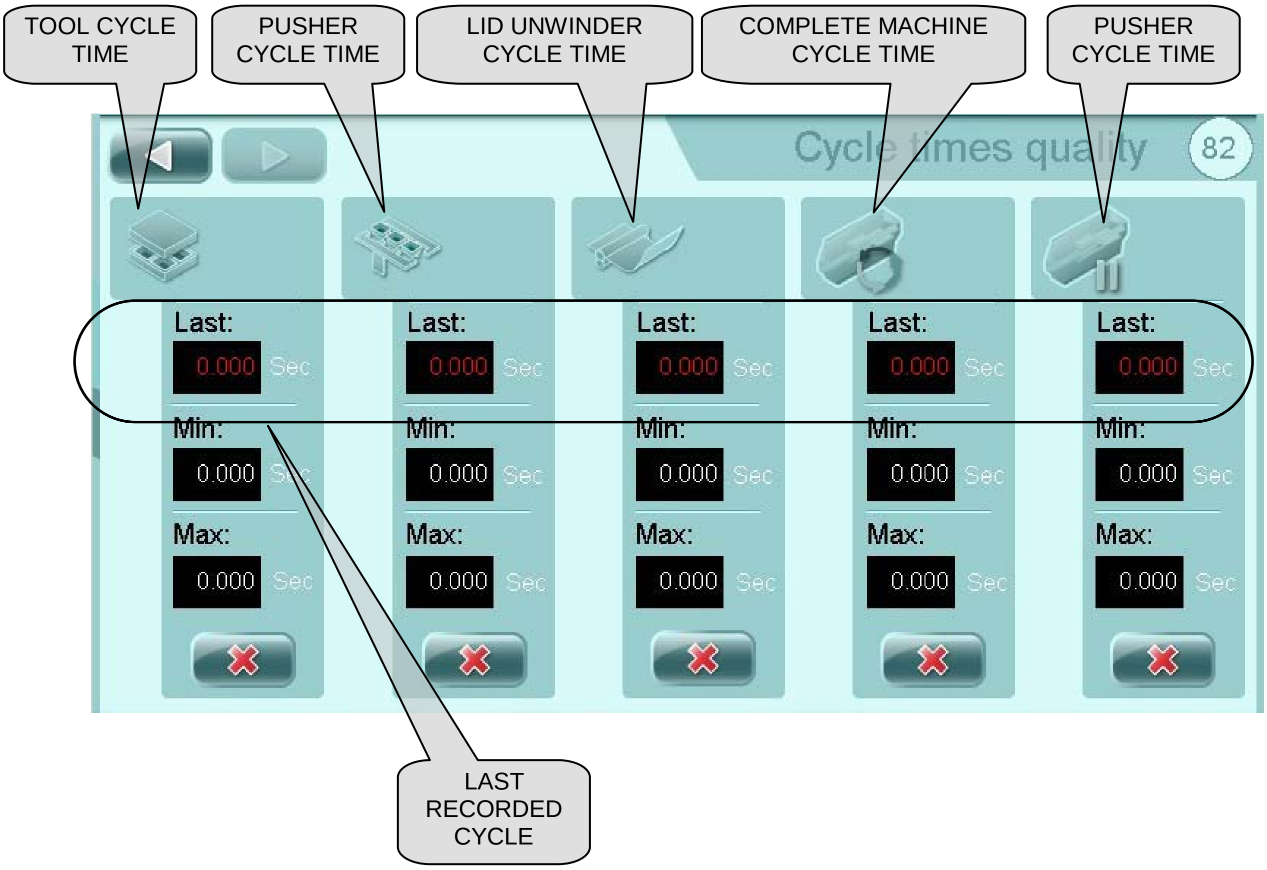
SKIN CYCLE RESISTANCE SET POINT TEMPERATURE

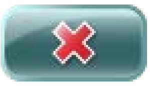
5.20 QUALITY



START QUALITY DATA RECORDING (1 CYCLE). When this cycle is activated, the operator panel starts to record data as from the first available cycle for a time of 10 seconds. The panel records the operating cycle situation every 10ms and saves the data in an Excel file. The data recorded by the operator panel refer to the sealing tool, and in particular to the position, force, upper and lower pressure, upper and lower temperature, and skin temperature.

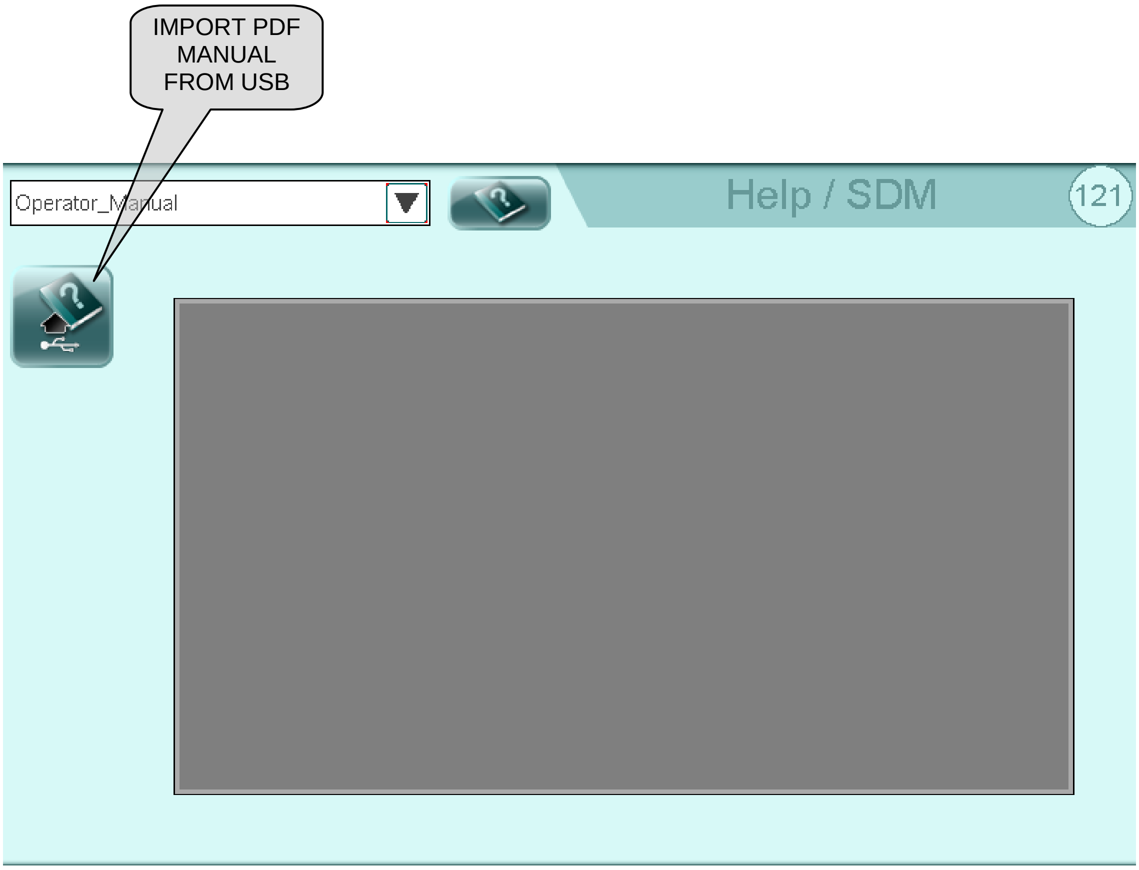

Quality
92
150
120
90
60
30
0
200
160
120
80
40
0
300
240
180
120
60
0
[10 sec]
Temperature Skin
Temperature Down
Temperature Up



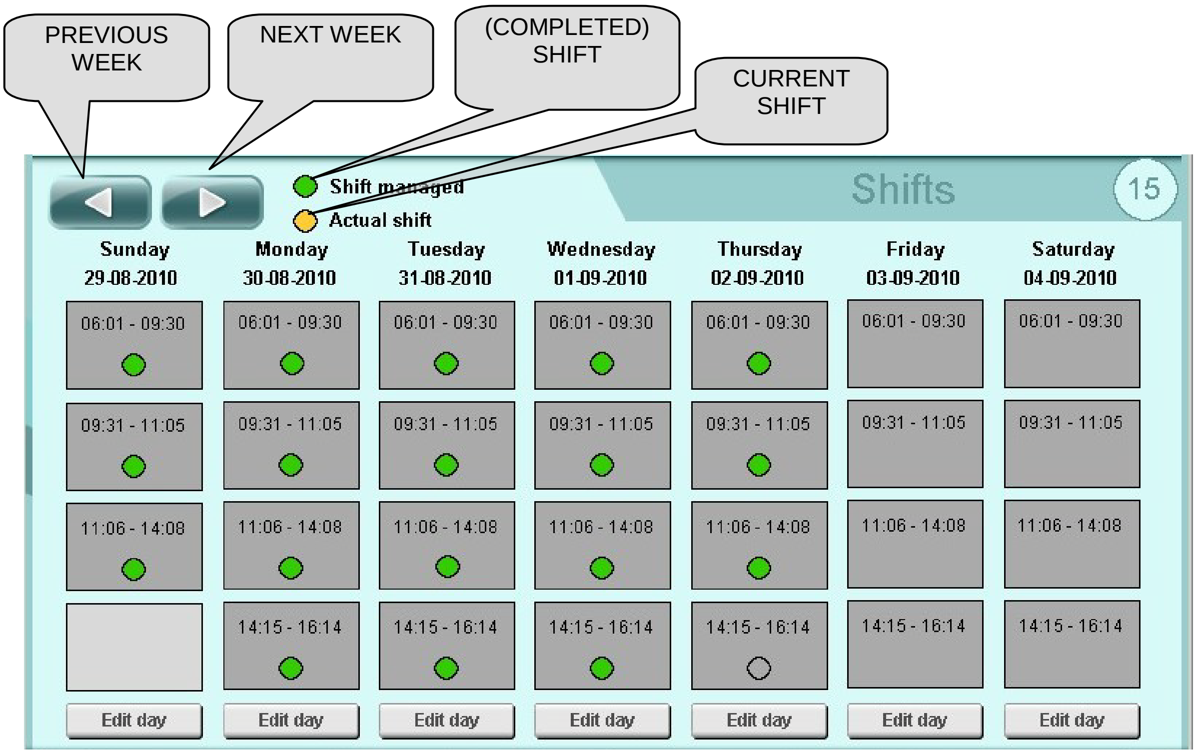


Reset cycle times

5.21 HELP



5.22 WORK SHIFTS



5.23 USB ENTRY

To access to the USB port is necessary remove the cover placed on the back side of the operator panel.

5.24 CF-CARD RELPACEMENT

For CF-CARD replacement, proceed as follows:

1. Save the recipes to have a backup copy.
2. Switch off the machine.
3. Unscrew the TORX - T 20 screws and remove panel screen.

4. Remove the old CF-CARD.

5. Insert the new CF-CARD.
6. Reassemble the screen panel.
7. Turn on the machine.

5.25 REPLACE B&R PANEL

For replace CF-CARD is necessary:

1. Save recipes for back up.
2. Switch OFF the machine.
3. Unscrew 16 TORX - T 20, open the frontal display, unplug the connectors and E-stop; note the right position and numbers.

4. Remove CF-CARD.

5. Remove the rear of panel from arm.

6. Assemble then new rear of panel to the arm.

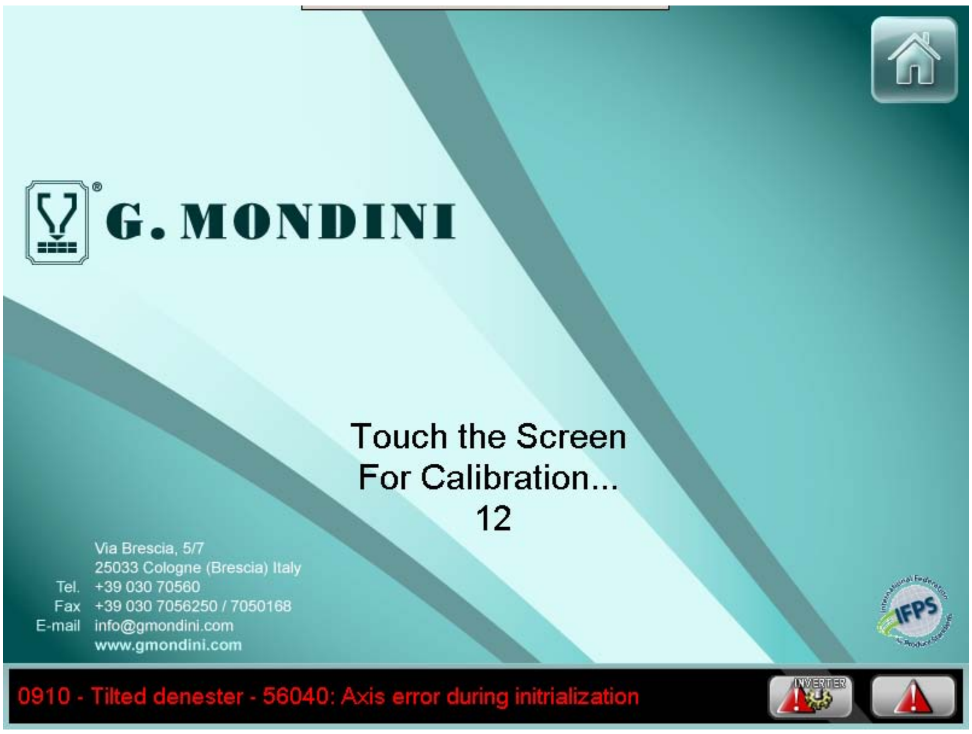
7. Insert the CF-CARD into new panel (or insert the new CF-CARD if available).

8. Plug the connectors and E-stop.

9. Switch ON the machine.

5.26 MANUAL CALIBRATION OF B&R PANEL

For manual calibration of B&R panel is necessary loggin with “COLLAUDO” and touch the logo machine in the right upper corner.
After this, press the screen for calibration.



5.27 LIST AND DESCRIPTION OF RECIPE PARAMETERS

The various recipe parameters are described in the following pages.

0001 - Skin: Delay start skin cycle [s]	The delay required to start the vacuum cycle to form the film in the tool during a skin cycle.
0002 - Skin: Vacuum moulding time sealing head side [s]	The vacuum time required for film formation in the tool during a skin cycle.
0003 - Skin: Lower compensation time for upper moulding [s]	The opening time of the valve regulating air supply to the lower tool during a skin cycle.
0004 - Skin: Delay start lower vacuum [s]	The delay time regulating opening of the lower vacuum valve during a skin cycle.
0005 - Skin: Delay start upper vacuum [s]	The delay time regulating opening of the upper vacuum valve during a skin cycle.
0006 - Skin: Upper vacuum time [s]	The time required to create a vacuum in the tool during a skin cycle.
0007 - Skin: Delay start upper compensation time [s]	The opening time of the sealer compensation valve during a skin cycle.
0008 - Skin: Lower vacuum delay after upper compensation [s]	The time required to fill the tray with gas.
0009 - Skin: Upper compensation time before sealing [s]	The advance time regulating closing of lower gas valve 2.
0010 - Skin: Delay start compensation time sealing head side [s]	The advance time regulating closing of the upper gas valve.
0011 - Skin super: Delay start upper compensation time [s]	The delay time regulating opening of the upper tool compensation valve.
0012 - Skin: Start upper compensation pressure threshold [mbar]	The pressure to reach in the upper tool for the compensation cycle start.
0013 - Skin: Stop upper compensation pressure threshold [mbar]	The pressure to reach in the upper tool for the compensation cycle stop.
0014 - Skin: Stop sealing compensation pressure threshold [mbar]	The pressure to reach in the sealing head for the compensation cycle stop.
0015 - Skin: Stop upper tool vacuum pressure threshold [mbar]	The pressure to reach for the upper tool vacuum cycle stop.
0016 - Film reel unwind: Photocell-rollers distance for reel autochange [mm]	The distance between the film sensor and the closing rollers.
0017 - Film reel unwind: Working rollers distance for reel autochange [mm]	The amount of film to unwind with the sealing rollers closed to ensure perfect joining of the ends of the bi-adhesive film.
0018 - Film reel unwind: Tool-rollers distance for reel autochange [mm]	The distance between the film sensor and the sealing tool.
0019 - Skin: Delay start compensation time sealing head side 2 [s]	The delay time regulating opening of the sealing head compensation valve in the side 2 of the tool.

0020 - SP: Lower vacuum time [s]	The time required to create a vacuum in the lower tool during a super protruding cycle.
0021 - Tool: Delay start vacuum cycle [s]	The delay time regulating the start of the vacuum cycle.
0022 - Tool: Vacuum time [s]	The vacuum time required to remove air from the tray before sealing.
0023 - Tool: Anticipate gas flushing time [s]	The anticipate time for the gas flushing cycle.
0024 - Tool: Delay start gas (main lower) [s]	The delay time regulating opening of the lower main gas valve.
0025 - Tool: Gas time [s]	The time required to fill the tray with gas.
0026 - Tool: Delay start 2nd lower gas valve [s]	The delay time regulating opening of the lower gas valve 2.
0027 - Tool: Delay start upper gas [s]	The delay time regulating opening of the upper gas valve.
0028 - Tool: Offset upper gas [s]	It's the gas correction to add inside the sealing tool.
0029 - Tool: Gas flushing time [s]	The gas flushing cycle.
0030 - Tool: Delay start tool motion after gas time [s]	The delay time regulating the tool notion start after gas cycle start.
0031 - Tool: Delay start Slice Pack [s]	The delay time regulating Slice Pack cycle start.
0032 - Tool: Slice Pack time [s]	The Slice Pack time cycle.
0033 - Tool: Delay stop vacuum Slice Pack after pusher movement start [s]	The delay time regulating closing vacuum valve after the pusher arms movement start during the Slice Pack cycle.
0034 - Gas mixer: Set point Oxygen (O2) [%]	The oxygen percentage in the tray during the sealing cycle.
0035 - Gas mixer: Set point Carbon Dioxide (CO2) [%]	The carbon dioxide percentage in the tray during the sealing cycle.
0036 - Gas mixer: Set point Nitrogen (N2) [%]	The nitrogen percentage in the tray during the sealing cycle.
0037 - Tool: G Close closing horizontal piston time [s]	The closing time for the horizontal piston of the tool during the G closing cycle.
0038 - Tool: G Close sensor timeout [s]	The maximum reading time of the sensor detecting the G closing cycle.
0039 - Tool: Delay start lower vacuum [s]	The delay time regulating the start of the vacuum cycle in the lower tool.
0040 - Tool: G Close vertical piston time [s]	The closing time for the vertical piston of the tool during the G closing cycle.
0041 - Tool pressure: Lower tool vacuum threshold [mbar]	The pressure to reach in the lower tool during the vacuum cycle.
0042 - Tool pressure: Lower tool gas threshold [mbar]	The pressure to reach in the lower tool during the gas cycle.
0043 - Tool pressure: Lower tool gas 2 threshold (anticipate respect gas 1) [mbar]	The pressure to reach in the lower tool during the gas 2 cycle.
0044 - Tool pressure: Upper tool air threshold [mbar]	The pressure to reach in the upper tool during the gas cycle.
0045 - Tool pressure: Electronic vacuum cycle timeout [s]	The maximum reading time of the sensor detecting the electronic vacuum cycle.
0046 - Tool pressure: Electronic gas cycle timeout [s]	The maximum reading time of the sensor detecting the electronic gas cycle.

0047 - Tool pressure: Offset gas to vacuum overlap [mbar]	The pressure inside the tool for the opening of the gas valves for the washing cycle.
0048 - Tool: G Close delay lower contr. to cut film [s]	The time of delay for the lower control for the film cutting during the G closing cycle.
0049 - DOT: Air sealing head delay [s]	The delay time regulating air sealing head cycle start.
0050 - Film Stretch: Pre-moulding time [s]	The time for the film pre-moulding cycle operation during the film stretch cycle.
0051 - Film Stretch: Blowing time [s]	The time for the film blowing cycle during the film stretch cycle.
0052 - Film Stretch: Delay tool lifting in film stretch cycle [s]	The delay time for the tool lifting during the film stretch cycle.
0053 - Film Stretch: Delay start compensation time [s]	The delay time for opening of the compensation valves.
0054 - SP: Delay start lower vacuum after sealing [s]	The delay time for the opening of the vacuum valves in the lower tool during a super protruding cycle.
0055 - Tool: Vacuum circuit cleaning cycle running	The number of vacuum circuit cleaning cycle
0056 - DOT: Sealing head air time [s]	The duration time of the sealing head air.
0057 - Skin: Delay start second skin valve [s]	The delay time for the opening of the second skin valve.
0058 - Tool: Gas purge time during recipe change [s]	The purge time of the gas cycle duting the recipe change.
0059 - Tool: Vacuum circuit cleaning final time [s]	The time of the air for the tool cleaning after the last cycle
0060 - Film Stretch: Second pre-moulding valve delay [s]	The delay time for the opening of the second pre-moulding valve.
0061 - Tool: Sealing position threshold [mm]	The sealing position threshold.
0062 - Tool motion: Home position [°]	The start-of-cycle position of sealing tool.
0063 - Tool motion: Position 01 - Trays center position [°]	The trays centring position.
0064 - Tool motion: Position 02 - Closing tool [°]	The sealing tool closing position.
0065 - Tool motion: Position 03 - Vacuum stop [°]	The sealing tool position to run a vacuum/gas cycle.
0066 - Tool motion: Position 04 - Sealing stop [°]	The delay time regulating opening of the lower gas valve 2.
0067 - Tool motion: Position 05 - Opening tool [°]	The sealing tool opening position.
0068 - Tool motion: Position 06 - Trays settings [°]	The trays setting position.
0069 - Tool motion: Position 07 - Trays based [°]	The lower tool position.
0070 - Tool motion: Offset start vacuum position [s]	The sealing tool advance time to run a vacuum cycle.
0071 - Tool motion: Film unwinding start enable anticipated [°]	The tool position for the lid unrolling cycle start.
0072 - Tool motion: Internal guide timeout [s]	The maximum reading time of the tray transport chain photocell beyond which the machine signals an alarm.

0073 - Tool motion: Start trays of position control windows	The start for the trays control position window during the tool motion cycle.
0074 - Tool motion: End trays of position control windows	The end for the trays control position window during the tool motion cycle.
0075 - Tool motion: Trays out of position control time [s]	The time for the trays control position window during the tool motion cycle.
0076 - Tool motion: Stop position for special tool	The stop position for the special tool cycle.
0077 - Tool motion: Stop position during lifting	The stop position of the tool during the lifting movement.
0078 - Tool motion: Stop position moulding Hybric [°]	The stop position for moulding Hybric.
0079 - Tool motion: Stop position speed moulding Hybric [%]	The speed to stop position moulding Hybric [%]
0080 - Tool motion: Stop position Acc/Dec moulding Hybric [%]	The Acc/Dec to stop position moulding Hybric [%]
0081 - Tool motion: Forming (Skin) vacuum pause advance on vacuum position [mm]	The advance position of the tool for the vacuum pause for the film forming during the skin cycle.
0082 - Tool: Lower heads piston riactivation quote	The lower heads piston riactivation quote.
0083	
0084	
0085	
0086 - Tool motion: Time 01 from start to trays center [ms]	It's the time from the sealing cycle beginning to the trays centring position.
0087 - Tool motion: Time 02 from trays center to closing tool [ms]	It's the time from the trays centring position to the closing tool position.
0088 - Tool motion: Time 03 from closing tool to vacuum stop [ms]	It's the time from the closing tool position to the vacuum stop position.
0089 - Tool motion: Time 04 from vacuum stop to sealing stop [ms]	It's the time from the vacuum stop position to the sealing stop position.
0090 - Tool motion: Sealing time / DOT perforation time [s]	The time of the tool for the sealing/perforation cycle.
0091 - Tool motion: Time 05 from sealing stop to opening tool [ms]	It's the time from the sealing stop position to the opening tool position.
0092 - Tool motion: Time 06 from opening tool to trays settings [ms]	It's the time from the opening tool position to the trays setting position.
0093 - Tool motion: Pause trays settings time [ms]	The sealing tool pause time for the trays setting.
0094 - Tool motion: Time 07 from trays settings to trays in base [ms]	It's the time from the trays setting position to the trays in base position.
0095 - Tool motion: Time 08 from trays in base to end tool position [ms]	It's the time from the trays in base position to the base to end tool position.
0096 - Tool motion: Timeout [s]	The time for the alarm relating to the movement of the tool.
0097	
0098	
0099	
0100	
0101 - Tool motion: Speed 01 - Crossing trays center position [%]	It's the speed to the trays center position crossing.

0102 - Tool motion: Speed 02 - Crossing closing tool position [%]	It's the speed to the closing tool position crossing.
0103 - Tool motion: Speed 03 - Crossing vacuum stop position [%]	It's the speed to the vacuum stop position crossing.
0104 - Tool motion: Speed 04 - Crossing sealing stop position [%]	It's the speed to the sealing stop position crossing.
0105 - Tool motion: Speed 05 - Crossing opening tool position [%]	It's the speed to the opening stop position crossing.
0106 - Tool motion: Speed 06 - Crossing trays settings position [%]	It's the speed to the trays setting position crossing.
0107 - Tool motion: Speed 07 - Crossing trays in base position [%]	It's the speed to the trays in base position crossing.
0108 - Tool motion: Speed 08 - Going to end position [%]	It's the tool speed going to the end cycle position.
0109 - Tool: Catch film vacuum delay [s]	The delay time regulating the falling lower heads.
0110 - Tool: Falling lower heads delay [s]	The delay time regulating the start of the vacuum to the catch film cycle.
0111 - Inlet belts: Buffer full time [s]	This parameter refers to the maximum reading time of the photocell for the belt full.
0112 - Shuttle: Fixed clamp lift delay [s]	The delay time for the lift of the fixed clamp.
0113	
0114	
0115	
0116 - Tilted denester: Blow Time [s]	The blow duration time.
0117 - Tool: Vacuum pause time [s]	The pause time of the vacuum cycle.
0118 - Tool: Vacuum time 2 [s]	The time of the vacuum cycle 2.
0119 - Tool: Speed [%]	The operating speed of the sealing tool.
0120 - Tool: Sealing torque set point [Kg]	The operating torque of the sealing tool.
0121 - Mirabella: Temperature set point [°C]	The set-point temperature during the "Mirabella" sealing cycle.
0122 - Mirabella: Alarm threshold [°C]	The temperature threshold during the "Mirabella" sealing cycle.
0123 - Mirabella: Cavity 1 detector temperature offset [°C]	The offset temperature for the first tool cavity during the "Mirabella" sealing cycle.
0124 - Mirabella: Cavity 2 detector temperature offset [°C]	The offset temperature for the second tool cavity during the "Mirabella" sealing cycle.
0125 - Mirabella: Cavity 3 detector temperature offset [°C]	The offset temperature for the third tool cavity during the "Mirabella" sealing cycle.
0126 - Mirabella: Cavity 4 detector temperature offset [°C]	The offset temperature for the fourth tool cavity during the "Mirabella" sealing cycle.
0127 - Mirabella: Cavity 5 detector temperature offset [°C]	The offset temperature for the fifth tool cavity during the "Mirabella" sealing cycle.
0128 - Mirabella: Cavity 6 detector temperature offset [°C]	The offset temperature for the sixth tool cavity during the "Mirabella" sealing cycle.
0129 - Mirabella: Cavity 7 detector temperature offset [°C]	The offset temperature for the seventh tool cavity during the "Mirabella" sealing cycle.

0130 - Mirabella: Cavity 8 detector temperature offset [°C]	The offset temperature for the eighth tool cavity during the "Mirabella" sealing cycle.
0131 - Tool heating: Upper temperature set point [°C]	The operating temperature of the sealing tool.
0132 - Tool heating: Upper tool temperature alarm threshold [°C]	The maximum upper and lower temperature deviation from the upper tool set point temperature beyond which a machine alarm is generated.
0133 - Tool heating: Upper tool one shot heating time [s]	The duration of the upper one-shot heating cycle.
0134 - Tool heating: Upper heating plate 1 offset [°C]	The temperature difference between the set point and the value measured by the sensor of upper tool 1.
0135 - Tool heating: Upper heating plate 2 offset [°C]	The temperature difference between the set point and the value measured by the sensor of upper tool 2.
0136 - Tool heating: Lower heating plate 2 offset [°C]	The temperature difference between the set point and the value measured by the sensor of lower tool 2.
0137 - Tool heating: Lower temperature set point [°C]	The operating temperature of the lower tool.
0138 - Tool heating: Lower tool temperature alarm threshold [°C]	The maximum upper and lower temperature deviation from the lower tool set point temperature beyond which a machine alarm is generated.
0139 - Tool heating: Lower heating plate 1 offset [°C]	The duration of the lower one-shot heating cycle.
0140 - Tool heating: Lower tool one shot heating time [s]	The temperature difference between the set point and the value measured by the sensor of lower tool 1.
0141 - Skin heating: Film preheating temperature set point [°C]	The operating temperature of the film heating.
0142 - Skin heating: Film preheating alarm threshold [°C]	The maximum film preheating temperature deviation from the upper tool set point temperature beyond which a machine alarm is generated.
0143 - Skin heating: Preheating plate 1 offset [°C]	The temperature difference between the set point and the value measured by the sensor of preheating plate 1.
0144 - Skin heating: Preheating plate 2 offset [°C]	The temperature difference between the set point and the value measured by the sensor of preheating plate 2.
0145 - Mirabella: Cavity 9 detector temperature offset [°C]	The offset temperature for the ninth tool cavity during the "Mirabella" sealing cycle.
0146 - Mirabella: Cavity 10 detector temperature offset [°C]	The offset temperature for the tenth tool cavity during the "Mirabella" sealing cycle.
0147 - Vibrator: Frequency [Hz]	This parameter refers to the frequency for the vibrator device speed.
0148 - Line: Open discharge hopper phase [°]	The hopper opening duration.
0149 - Vibrator: Frequency [%]	The vibration frequency of the vibrator device.

0150 - Tilted denester: Chain position for start phase of blower cycle [°]	The position of the chain for the blower start.
0151 - Pusher motion: Forward low speed time [s]	The run-time for the forward movement in low speed of the pusher arms.
0152 - Pusher motion: Stroke [mm]	The stroke of the pusher motion
0153 - Pusher motion: Speed forward [Hz]	The pusher arms speed during the forward movement.
0154 - Pusher motion: Acceleration [s]	The pusher arms acceleration during the forward movement.
0155 - Pusher motion: Deceleration [s]	The pusher arms deceleration during the forward movement.
0156 - Pusher motion: Speed backward [Hz]	The pusher arms speed during the backward movement.
0157 - Pusher motion: Backward high speed time [s]	The run-time for the backward movement in high speed of the pusher arms.
0158 - Pusher motion: Backward low speed time [s]	The run-time for the backward movement in low speed of the pusher arms.
0159 - Pusher motion: Arms start delay [s]	The delay time for the pusher arms advancement start.
0160 - Pusher motion: Arm delay start backward [s]	The delay time for the pusher arms backward movement start.
0161 - Pusher motion: Delay start cycle [s]	The delay time for the pusher arms cycle start.
0162 - Pusher motion: Arm pliers timeout [s]	The maximum reading time of the sensor detecting arm pliers movements.
0163 - Pusher motion: High speed forward time [s]	The run-time for the forward movement in high speed of the pusher arms.
0164 - Pusher motion: Stop position searching speed [Hz]	The run-speed during the stop position searching of the pusher arms.
0165 - Pusher motion: Delay closing pushers [s]	The delay time for the closing of the pusher arms.
0166 - Pusher arms: Home position [mm]	The delay time for the opening of the pusher arms.
0167 - Pusher motion: Timeout movement [s]	The maximum reading time of the sensor detecting pusher arms movements.
0168 - Pusher arms: Home position [°]	The start position for the pusher arms cycle.
0169 - Pusher arms: Acceleration/Deceleration ratio [%]	The ratio between the acceleration and the deceleration of the pusher arms working cycle.
0170 - Pusher arms: Opening advance [°]	The advance position for the pusher arms opening.
0171 - Pusher arms: Acceleration foreward movement [%]	The pusher arms acceleration during the forward movement.
0172 - Pusher arms: Deceleration foreward movement [%]	The pusher arms deceleration during the forward movement.
0173 - Pusher arms: Acceleration backward movement [%]	The pusher arms acceleration during the backward movement.
0174 - Pusher arms: Deceleration backward movement [%]	The pusher arms deceleration during the backward movement.
0175 - Pusher arms: Advance start time [s]	The advance time for the pusher arms start.

0176 - Pusher: Speed forward [%]	The pusher motion speed during the forward movement.
0177 - Pusher arms: Speed backward [%]	The pusher motion speed during the backward movement.
0178 - Pushe: Arm pliers opening advance threshold [mm]	The advance position for the pushers opening.
0179 - Pusher motion: Chain phase for start cycle [°]	The chain phase for the pushers motion start.
0180 - Pusher motion: Open delay [s]	The delay time for the opening of pusher motion.
0181 - Tool - Lower vacuum Proportional EV: Opening percentage [%]	The percentage for the opening of the lower vacuum valve.
0182 - Line: Load belt delay stop [s]	The delay time for the load belt stop.
0183 - Line: Inlet pacer start phase [°]	It’s the phase for the start of the inlet pacer.
0184 - Line: Inlet pacer piston 1 close delay [s]	The delay time for the closing of inlet pacer piston.
0185 - Line: Inlet pacer piston 2 open delay [s]	The delay time for the opening of inlet pacer piston.
0186 - Line: Load belt deceleration [s]	This parameter refers to the load belt deceleration.
0187 - Line: Load belt acceleration [s]	This parameter refers to the load belt acceleration.
0188	
0189	
0190 - Spacer belt: Start phase [°]	It’s the phase for the start of the distancer belt in relation to the moment when the trays arrive from the line.
0191 - Inlet belt: Speed [%]	This parameter refers to the inlet belt 1 speed.
0192 - Inlet belt: Acceleration [%]	This parameter refers to the inlet belt 1 acceleration.
0193 - Inlet belt: Deceleration [%]	This parameter refers to the inlet belt 1 deceleration.
0194 - Belt 1: Running time [s]	The running time of the belt 1.
0195 - Belt 1: Pause time [s]	The delay time for the belt 1 start.
0196 - Belt 1: Speed [Hz]	This parameter refers to the belt 1 speed.
0197 - Belt 1: Start delay [s]	This parameter refers to the delay time for the belt 1 start.
0198 - Belt 1: Acceleration [s]	This parameter refers to the belt 1 acceleration.
0199 - Belt 1: Deceleration [s]	This parameter refers to the belt 1 deceleration.
0200 - Inlet belt phaser: Timeout photocells [s]	The maximum reading time of the inlet belt phaser start photocell beyond which the machine signals an alarm.
0201 - Inlet belt: Speed [Hz]	The inlet belt operating speed in automatic mode.
0202 - Inlet belt: Start delay [s]	The inlet belt start delay.
0203 - Inlet belt: Stop delay [s]	The inlet belt stop delay.
0204 - Inlet belt: Acceleration [s]	The inlet belt operating acceleration in automatic mode.
0205 - Inlet belt: Deceleration [s]	The inlet belt operating deceleration in automatic mode.

0206 - Inlet belt phaser: Type [1-2]	It allows to select the operation mode of the inlet belt.
0207 - Inlet belt phaser: Delay open piston 1 [s]	The delay time for the opening of piston 1.
0208 - Inlet belt phaser: Delay close piston 1 [s]	The delay time for the closing of piston 1.
0209 - Inlet belt phaser: Delay open piston 2 [s]	The delay time for the opening of piston 2.
0210 - Inlet belt phaser: Delay close piston 2 [s]	The delay time for the closing of piston 2.
0211 - Spacer belt: Tray set into tool [nr]	The number of tool cavities (number of trays sealed at each cycle).
0212 - Spacer belt 1: Step time [s]	The function time for the execution of one step of the spacer belt.
0213 - Spacer belt 1: Difference last step time [s]	The difference time from the step time for the last step time execution.
0214 - Spacer belt 1: Step speed motion [Hz]	Spacer belt operating speed.
0215 - Spacer belt 1: Acceleration [s]	Acceleration of spacer belt 1 during operation.
0216 - Spacer belt 1: Deceleration [s]	Deceleration of spacer belt 1 during operation.
0217 - Spacer belt 1: Delay start step time [s]	The delay time for the spacer belt step start.
0218 - Spacer belt: Tray length verification time [s]	It's the sensor reading time for the tray position verification.
0219 - Inlet belt phaser: Delay open piston 1 after pusher start [s]	The delay time for the opening of piston 1 after the pusher start.
0220 - Spacer belt: Photocell timeout [s]	The maximum reading time of the spacer belt start photocell beyond which the machine signals an alarm.
0221 - Spacer belt: Tray length [mm]	It's the tray length.
0222 - Spacer belt: Minimum take over time for pusher arms [s]	The minimum time for the trays taking by the pusher arms.
0223 - Spacer belt: Acceleration for stopping phase positioning [%]	The acceleration of the pusher time for the during the stop position movement.
0224 - Spacer belt: Start photocell distance to belt [mm]	It’s the distance in which is located the start photocell
0225 - Spacer belt: Tray to tray gap [mm]	It’s the distance that must exist between a container and the other
0226 - Spacer belt: Last step length [mm]	It's the moving by the belt during the last step
0227 - Spacer belt: Tray length verify [mm]	It’s the length used for the containers control
0228 - Spacer belt: Ratio speed	It’s the ratio speed to the spacer belt
0229 - Inlet belt: Reverse mouvement position [mm]	The position in which it starts the reversal of the belt movement
0230 - Exit belt 1: Stop position (Refer o mouvement pusherarms) [mm]	The stop position of the exit belt.
0231 - Exit belt 1: Speed [Hz]	Exit belt speed in automatic mode.
0232 - Exit belt 1: Delay pause during pusher movement [s]	It's the delay time for the exit belt pause during the pusher arms movement.

0233 - Exit belt 1: Stop time during pusher arms transfer [s]	It's the exit belt pause time during the pusher arms trays transfer.
0234 - Exit belt 1: Acceleration [s]	Exit belt acceleration in automatic mode.
0235 - Exit belt 1: Deceleration [s]	Exit belt deceleration in automatic mode.
0236 - Exit belt 1: Photocell timeout [s]	The maximum reading time of the exit belt start photocell beyond which the machine signals an alarm.
0237 - Exit belt: Stop delay [s]	The delay time for the exit belt stop.
0238 - Exit belt 1-2: Delay start 2nd belt after trays crossing [s]	The delay time for the second belt start after the trays crossing.
0239 - Exit belt 1: Close piston delay [s]	The delay time for the closing of piston.
0240 - Spacer belt: Photocell distance [step]	Number of steps for the distance from the distancer belt.
0241 - Exit belt 2: Speed [Hz]	Exit belt speed in automatic mode.
0242 - Doser 2: Lifter go down phase [°]	It is the position in which the mobile head lowers for drawing near to the container.
0243 - Doser 2: Lifter go up phase [°]	It is the position in which, effected the dosing, the mobile head gets up and move away from the container.
0244 - Exit belt 2: Acceleration [s]	Exit belt acceleration in automatic mode.
0245 - Exit belt 2: Deceleration [s]	Exit belt deceleration in automatic mode.
0246 - Exit belt 2: Photocell timeout [s]	The maximum reading time of the exit belt start photocell beyond which the machine signals an alarm.
0247 - Exit belt 3 (2>1): Speed [Hz]	Exit belt speed in automatic mode.
0248 - Exit belt 3 (2>1): Acceleration [s]	Exit belt acceleration in automatic mode.
0249 - Exit belt 3 (2>1): Deceleration [s]	Exit belt deceleration in automatic mode.
0250 - Exit belt 3 (2>1): Photocell timeout [s]	The maximum reading time of the exit belt start photocell beyond which the machine signals an alarm.
0251 - Scrap rewind: speed [%]	Scrap rewind operating speed.
0252 - Scrap rewind: acceleration [%]	Acceleration of scrap rewind during operation.
0253 - Scrap rewind: deceleration [%]	Deceleration of scrap rewind during operation.
0254 - Tool: tray in tool timeout [s]	it is the maximum reading time of the sensor that notices the trays in the tool over which the machine enters alarm.
0255 - Spacer belt 2: Tray length [mm]	It's the tray length.
0256 - Spacer belt 2: Start photocell distance to belt [mm]	It’s the distance in which is located the start photocell
0257 - Spacer belt 2: Tray to tray gap [mm]	It’s the distance that must exist between a container and the other
0258 - Spacer belt 2: Last step length [mm]	It's the moving by the belt during the last step
0259 - Spacer belt 2: Tray length verify [mm]	It’s the length used for the containers control
0260 - Doser 1: Product falling time [s]	the time during which the valve remains open to perform the unloading of the product in the container.

0261 - Film reel unwind: Speed [Hz]	The reel unwinder speed in automatic mode.
0262 - Film reel unwind: Start delay [s]	The delay time regulating start of the reel unwinder.
0263 - Film reel unwind: Stop delay [s]	The delay time regulating stop of the reel unwinder.
0264 - Film reel unwind: Motion timeout [s]	The maximum reading time of the film uncoil photocell beyond which the machine signals an alarm.
0265 - Film reel unwind: Acceleration [s]	The reel unwinder acceleration in automatic mode.
0266 - Film reel unwind: Deceleration [s]	The reel unwinder deceleration in automatic mode.
0267 - Film reel unwind: Speed [%]	The reel unwinder speed in automatic mode.
0268 - Film reel unwind: Acceleration [%]	The reel unwinder acceleration in automatic mode.
0269 - Film reel unwind: Deceleration [%]	The reel unwinder deceleration in automatic mode.
0270 - Lid preformed: Home position [mm]	The start-of-cycle position of lid preformed shuttle.
0271 - Lid preformed: 1st pick position [mm]	The shuttle position for the first lid taking.
0272 - Lid preformed: 2nd pick position [mm]	The shuttle position for the second lid taking.
0273 - Lid preformed: Waiting position [mm]	The shuttle position for the trays waiting.
0274 - Lid preformed: Release position [mm]	The shuttle position for the lid release in the closing tool.
0275 - Lid preformed: Threshold lifting pick-up/release piston [mm]	It's the position where the shuttle start the lifting pick-up/release movement.
0276 - Lid preformed: Speed going to tool [%]	The shuttle speed during the movement under the tool.
0277 - Lid preformed: Speed returning to lid stock [%]	The shuttle speed during the returning to the lid stock.
0278 - Lid preformed: Acceleration going to tool [%]	The shuttle acceleration during the movement under the tool.
0279 - Lid preformed: Deceleration going to tool [%]	The shuttle deceleration during the movement under the tool.
0280 - Lid preformed: Acceleration returning to lid stock [%]	The shuttle acceleration during the returning to the lid stock.
0281 - Lid preformed: Deceleration returning to lid stock [%]	The shuttle deceleration during the returning to the lid stock.
0282 - Lid preformed: Number of pick-up positions [n]	Number of pick-up positions during the shuttle cycle.
0283 - Lid preformed: Pick-up time [s]	Time of the shuttle in the lid pick-up position.
0284 - Lid preformed: Release time [s]	Time of the shuttle in the lid release position.
0285 - Lid preformed: Timeout Lifting/lowering trolley [s]	The maximum reading time of the shuttle start photocell during lifting/lowering movement beyond which the machine signals an alarm.
0286 - Lid preformed: Suction cups cleaning time [s]	The blowing time in the suction cups for the cleaning.
0287 - System: machine closer empty time [s]	The emptying time the trays from the closing machine.

0288 - Lid preformed: Start tool delay after preformed cycle [s]	The delay time regulating start of the tool after preformed cycle.
0289 - Lid preformed: Timeout Forward/backward trolley [s]	
0290 - Scrap uwinding: Stop delay [s]	The delay time regulating stop of the scrap uwinding.
0291 - Film unwinding: Lid time [s]	The amount of film to unwind for each sealing cycle.
0292 - Film unwinding: Step after mark detection [s]	The distance between one reference mark on the sealing film and the next detected by the corresponding sensors.
0293 - Film unwinding: Speed [Hz]	The operating speed of the lid unwinder in automatic mode.
0294 - Film unwinding: Speed during mark detection [Hz]	The speed of the sensor reading the reference mark on the sealing film.
0295 - Film unwinding: Acceleration [s]	Lid unwinder acceleration.
0296 - Film unwinding: Deceleration [s]	Lid unwinder deceleration.
0297 - Film unwinding: Working mode: 1=Normal, 2=Last tray, 3=Open tool, 4=Following [x]	Used to select the lid unwinder operating mode.
0298 - Film unwinding: Mark detector length timeout [s]	The maximum reading time of the film mark detector photocell beyond which the machine signals an alarm.
0299 - Film unwinding: Mark detector timeout [s]	The maximum reading time of the film mark search photocell beyond which the machine signals an alarm.
0300 - Film unwinding: Start shaping delay [s]	It's the stop following action position.
0301 - Film unwinding: Shaping time [s]	The time for the shaping of the sealing film.
0302 - Film unwinding: Pusher arm position for ending gear [mm]	The pusher arms position for the ending cycle of the film unwinding device.
0303 - Film unwinding: Speed ratio with chain	It’s the speed ratio between the film unwinding device and the chain line.
0304 - Lid unroll: Mark photocell selection	The mark photocell selection.
0305 - Doser 1: Lifter/stopper timeout [s]	it is the maximum reading time of the sensor that notices the correct operation of the lifter over which the machine enters alarm.
0306 - Chain: Line acceleration/deceleration ratio [%]	The ratio between the acceleration and the deceleration of the line.
0307 - Chain: Offset back motion [°]	This parameter refers to the exit acceleration belt speed.
0308 - Chain: Speed [steps/min]	This parameter refers to the delay time for the exit acceleration belt start.
0309 - Chain: Acceleration [%]	This parameter refers to the transport chain acceleration.
0310 - Chain: Deceleration [%]	This parameter refers to the transport chain deceleration.
0311 - Chain: Home position [°]	The start-of-cycle position of the tray transport chain.
0312 - Chain: Step number [steps]	This is the chain steps number.

0313 - Chain: Speed [Hz]	The operating speed of the chain in automatic mode.
0314 - Chain: Acceleration [s]	The operating acceleration of the chain in automatic mode.
0315 - Chain: Deceleration [s]	The operating deceleration of the chain in automatic mode.
0316 - Chain: Step pause time [s]	This is the pause time of the chain.
0317 - Chain: Delay start [s]	The delay time regulating start of the tray transport chain operating cycle.
0318 - Chain: Photocell timeout [s]	The maximum reading time of the tray transport chain photocell beyond which the machine signals an alarm.
0319 - Chain: Tilted denester delay start [s]	The delay time for the tilted denester cycle start.
0320 - Chain: Stopper piston start phase [°]	The position for stopper piston start under the doser.
0321 - Chain: Stopper piston working time [s]	The working time for the stopper piston under the doser.
0322 - Line: Tray presence for pusher arms phase	the presence of the trays for the pusher arms phase start.
0323 - Feed belt 1: Speed [%]	The feed belt 1 speed during the working cycle.
0324 - Feed belt 1: Acceleration [%]	The acceleration feed belt 1 speed during the working cycle.
0325 - Feed belt 1: Deceleration [%]	The deceleration feed belt 1 speed during the working cycle.
0326 - Line: Start pusher arms phase	The line phase for the pusher arms cycle start.
0327 - Line ejector: Delay start ejection [s]	The delay time for the line ejector start.
0328 - Line ejector: Ejection time [s]	The ejection time for the line ejector.
0329 -	
0330 - Doser 1: Steps frequency start dosing [nr]	It is the number of steps which adjusts the dosing frequency.
0331 - Tilted denester: Home position [mm]	The start-of-cycle position of the trays denester.
0332 - Tilted denester: Lower position [mm]	It's the lower position of the trays denester.
0333 - Tilted denester: Taking position [mm]	It's the taking position of the trays denester.
0334 - Tilted denester: Extension position [mm]	It's the extension position of the trays denester.
0335 - Tilted denester: Separation position [mm]	It's the separation position of the trays denester.
0336 - Tilted denester: Time from start to taking position [ms]	It's the time from the starting position to the taking position of the trays denester.
0337 - Tilted denester: Taking time [ms]	It's the taking time of the trays denester.
0338 - Tilted denester: Time from taking to extension position [ms]	It's the time from the taking position to the extension position of the trays denester.
0339 - Tilted denester: Time from extension to separation position [ms]	It's the time from the extension position to the separation position of the trays denester.

0340 - Tilted denester: Time from separation to start position [ms]	It's the time from the separation position to the start position of the trays denester.
0341 - Tilted denester: Speed on extension [%]	The operating speed of the trays denester in automatic mode.
0342 - Tilted denester: Out impediment (clearance) time [ms]	It's the delay time for the trays denester cycle start.
0343 - Tilted denester: Tray sliding time [ms]	It's the necessary time for the tray sliding.
0344 - Tilted denester: Delay stop vacuum sucker [ms]	It's the delay time for vacuum sucker cycle stop.
0345 - Tilted denester: Delay switch rotation after cycle start [ms]	It's the delay time for the switch rotation after the trays denester cycle start.
0346 - Tilted denester: Delay return switch rotation after taking [ms]	It's the delay time for the return switch rotation after the trays denester cycle start.
0347 - Tilted denester: Missing tray in tray stock - advise [s]	The maximum reading time of the trays presence in the tray stock photocell beyond which the machine signals an advise.
0348 - Tilted denester: Missing tray in tray stock - alarm [s]	The maximum reading time of the trays presence in the tray stock photocell beyond which the machine signals an alarm.
0349 - Tilted denester: Delay start cycle [s]	The delay time for the tilted denester start cycle.
0350 - Tilted denester: Pneumatical piston motion timeout [s]	The maximum reading time for the correct operation of the pneumatic piston of the tilted denester.
0351 - Tilted denester: Start phase	The tilted denester start phase.
0352 - Lid preformed: Suction cups cleaning delay [s]	The delay time for the suction cup cleaning start.
0353 - Customer doser: Delay start dosing [s]	The delay time for the customer doser start cycle.
0354 - Customer doser: Dosing time [s]	The dosing time of the customer doser.
0355 - Customer doser: Delay start lifter [s]	The delay time for the lifter cycle start.
0356 - Customer doser: Lifter working time [s]	The working time of the lifter device.
0357 - Doser 1: Stopper lift phase [°]	It is the position in which, effected the dosing, the stopper gets up.
0358 - Doser 1: Stopper lowering phase [°]	It is the position in which the stopper lowers.
0359 - Doser 1: Stopper working time [s]	The working time for the stopper piston.
0360 - Vertical denester 1: Trays into denester [Trays for chain steps]	The number of containers to be extracted for each step of the chain.
0361 - Target teorically speed [pcs/min]	The theoretical speed of production.
0362 - System: Pulse time length (for DIGI) [s]	The time for the DIGI pulse.
0363 - Doser 1: Timeout for dosing [s]	The maximum reading time for the correct operation of the doser 1.
0364 - Doser 2: Timeout for dosing [s]	The maximum reading time for the correct operation of the doser 2.

0365 - Doser 3: Timeout for dosing [s]	The maximum reading time for the correct operation of the doser 3.
0366 - Doser 3: Position [steps]	this parameter refers to the number of chain steps where the doser 1 is positioned.
0367 - Doser 3: Start dosing phase [°]	this parameter refers to the start phase of the dosing cycle for the doser 1.
0368 - Line ejector: Delay start ejection 2 [s]	The delay time for the line ejector start.
0369 - Line ejector: Ejection time 2 [s]	The ejection time for the line ejector.
0370 - Doser 1: Position [steps]	this parameter refers to the number of chain steps where the doser 1 is positioned.
0371 - Doser 1: Start dosing phase [°]	this parameter refers to the start phase of the dosing cycle for the doser 1.
0372 - Doser 2: Position [steps]	this parameter refers to the number of chain steps where the doser 2 is positioned.
0373 - Doser 2: Start dosing phase [°]	this parameter refers to the start phase of the dosing cycle for the doser 2.
0374 - Doser 1: Delay start doser [s]	This parameter refers to the delay time for the doser 1 start.
0375 - Doser 2: Delay start doser [s]	This parameter refers to the delay time for the doser 2 start.
0376 - Doser 3: Delay start doser [s]	This parameter refers to the delay time for the doser 3 start.
0377 - Line: Hopper discharge start phase [°]	this parameter refers to the start phase of the hopper discharge.
0378 - Line: Hopper discharge time [s]	The duration time of the hopper discharge cycle.
0379 - Doser 2: Lifter working time [s]	The working time of the lifter on the mobile head 2.
0380 - Mobile head doser 1: Step position [steps]	this parameter refers to the number of chain steps where the mobile head is positioned.
0381 - Mobile head doser 1: Start dosing phase [°]	this parameter refers to the start phase of the dosing cycle for the mobile head.
0382 - Doser 1: Lifter go down phase [°]	It is the position in which the mobile head lowers for drawing near to the container.
0383 - Doser 1: Lifter go up phase [°]	It is the position in which, effected the dosing, the mobile head gets up and move away from the container.
0384 - Mobile head doser 1: Cam opening for dosing verification [°]	The cam opening for the dosing verification.
0385 - Mobile head: Module cycle steps [steps]	The chain steps necessary to execute a mobile head cycle.
0386 - Mobile head: Start sincronized position [mm]	The mobile head position where it's synchronized with the line.
0387 - Mobile head: Acceleration space [mm]	The mobile head acceleration space used to reach the line speed.
0388 - Mobile head: Following sincronization space [mm]	The mobile head space for the line following.
0389 - Mobile head: Deceleration space [mm]	The mobile head deceleration space.
0390 - Mobile head: Gear ratio MH / chain	The mobile head/chain speed gear ratio.
0391 - Mobile head: Home offset [°]	The offset for the mobile head home position.

0392 - Mobile head doser 2: Step position [steps]	The mobile head doser 2 position on the chain line.
0393 - Mobile head doser 2: Start dosing phase [°]	The mobile head doser 2 position for the start dosing.
0394 - Mobile head doser 2: Lifter go down phase [°]	The mobile head doser 2 position for the start lifter go down.
0395 - Mobile head doser 2: Lifter go up phase [°]	The mobile head doser 2 position for the start lifter go up.
0396 - Mobile head doser 2: Cam opening for dosing verification [°]	The cam opening for the dosing verification.
0397 - Doser 1: Delay start lifter [s]	The delay time for the lifter start on the mobile head 1.
0398 - Doser 1: Lifter working time [s]	The working time for the lifter on the mobile head 1.
0399 - Doser 2: Delay start lifter [s]	The delay time for the lifter start on the mobile head 2.
0400 - Deviator: Belt speed [Hz]	The speed of the deviator belt.
0401 - Deviator: Belt acceleration [s]	The acceleration of the deviator belt.
0402 - Deviator: Belt deceleration [s]	The deceleration of the deviator belt.
0403 - Deviator: Photocell timeout [s]	The maximum reading time of the deviator belt photocell beyond which the machine signals an alarm.
0404 - Doser 2: Lifter/stopper timeout [s]	it is the maximum reading time of the sensor that notices the correct operation of the lifter over which the machine enters alarm.
0405 - Deviator: Pieces to move per way [pcs]	Number of trays to move per way.
0406 - Deviator: Number of exit way [nr]	Number of exit way.
0407 - Deviator: Rest position [°]	The deviator belt rest position during the working cycle.
0408 - Deviator: 1st way position [°]	The deviator belt first way position during the working cycle.
0409 - Deviator: 2nd way position [°]	The deviator belt second way position during the working cycle.
0410 - Deviator: 3rd way position [°]	The deviator belt third way position during the working cycle.
0411 - Deviator: 4th way position [°]	The deviator belt fourth way position during the working cycle.
0412 - Deviator: Reject way position [°]	The deviator belt fourth way position during the working cycle.
0413 - Deviator: Motion delay [s]	The delay time for deviator belt motion start.
0414 - Deviator: change way speed [%]	The deviator belt speed during the change way.
0415 - Deviator: Turn tray foreward speed [%]	The turn tray speed during the foreward motion.
0416 - Deviator: Turn tray backward speed [%]	The turn tray speed during the backward motion.
0417 - Deviator: Change way acceleration [%]	The deviator belt acceleration during the change way.
0418 - Deviator: Change way decceleration [%]	The deviator belt deceleration during the change way.

0419 - Deviator: Turn tray foreward acc/dec [%]	The turn tray acceleration/deceleration during the foreward motion.
0420 - Deviator: Turn tray backward acc/dec [%]	The turn tray acceleration/deceleration during the backward motion.
0421 - Deviator: Home position [°]	The deviator belt home position.
0422 - Deviator: Tray detection filter (tray length) [mm]	The photocell reading time for the trays detection in the deviator belt.
0423 - Deviator: turn tray stroke [°]	The stroke of the turn tray piston.
0424 - Deviator: turn tray filter for tray lenght [mm]	The filter time for the tray length oh the deviator belt.
0425 - Line: Delay foreward dosing translation [s]	The delay time for the forward dosing translation motion.
0426 - Line: Delay backward dosing translation [s]	The delay time for the backward dosing translation motion.
0427 - Doser 1: Pulse dosing time [s]	The time for the dosing cycle of the doser 1.
0428 - Doser 2: Pulse dosing time [s]	The time for the dosing cycle of the doser 2.
0429 - Doser 3: Pulse dosing time [s]	The time for the dosing cycle of the doser 3.
0430 - Platformer: Start phase [°]	It’s the phase for the start of the platformer machine.
0431 - Feed belt 2: Acceleration [%]	The acceleration feed belt 2 speed during the working cycle.
0432 - Feed belt 2: Deceleration [%]	The deceleration feed belt 2 speed during the working cycle.
0433 - Mobile head: Start phase respect line position [°]	This parameter refers to the mobile head start phase respect line position.
0434 - Mobile head: Following accerleration [%]	The acceleration of the mobile head for the following of the line.
0435 - Mobile head: Following deceleration [%]	The deceleration of the mobile head for the following of the line.
0436 - Accelerator belt: Speed [Hz]	This parameter refers to the accelerator belt speed.
0437	
0438 - Exit belt 1: Open piston delay [s]	It’s the delay time for opening of the piston on the exit belt 1
0439 - Accelerator belt: Start delay [s]	This parameter refers to the accelerator belt acceleration.
0440 - Accelerator belt: Phaser start delay [s]	The delay time for the phaser device cycle start.
0441 - Accelerator belt: Phaser photocell timeout [s]	The maximum reading time of the sensor for the correct operation of the phaser device cycle.
0442 - Exit guide 1: speed [%]	The exit guide 1 operating speed.
0443 - Loading belt: Speed [Hz]	This parameter refers to the accelerator belt speed.
0444 - Loading belt: Acceleration [s]	This parameter refers to the accelerator belt acceleration.
0445 - Loading belt: Deceleration [s]	This parameter refers to the accelerator belt deceleration.
0446 - Loading belt: Start delay [s]	The delay time for the loading belt cycle start.

0447 - Loading belt: Photocell timeout [s]	The maximum reading time of the sensor for the correct operation of the loading belt cycle.
0448 - Loading belt: Stop delay [s]	The delay time for the loading belt cycle stop.
0449 - Exit guide 2: speed [%]	The exit guide 2 operating speed.
0450 - Line: Ejector 2 position (after doser) [steps]	The second ejector position (step) sfter the doser.
0451 - Vertical denester 1: Home position [mm]	The vertical denester 1 home position.
0452 - Vertical denester 1: Lower position [mm]	The release trays position of the vertical denester 1.
0453 - Vertical denester 1: Taking position [mm]	The taking trays position of the vertical denester 1.
0454 - Vertical denester 1: Extension position [mm]	The suction cups extension position of the vertical denester 1.
0455 - Vertical denester 1: Separation position [mm]	The trays separation from the tray stock position of the vertical denester 1.
0456 - Vertical denester 1: Time from start to taking position [ms]	Time from start position to taking position of the vertical denester 1.
0457 - Vertical denester 1: Taking time [ms]	Time in taking position of the vertical denester 1.
0458 - Vertical denester 1: Time from taking to extension position [ms]	Time from taking position to the suction cups extension position of the vertical denester 1.
0459 - Vertical denester 1: Time from extension to separation position [ms]	Time from the suction cups extension position to the trays separation position of the vertical denester 1.
0460 - Vertical denester 1: Time from separation to start position [ms]	Time from the trays separation position to the start position of the vertical denester 1.
0461 - Vertical denester 1: Speed on extension [%]	The extension motion speed of the vertical denester 1.
0462 - Vertical denester 1: Out impediment (clearance) time [ms]	The delay time for the restart vertical denester 1 cycle.
0463 - Vertical denester 1: Tray transfering time to line [ms]	The transferring time of the trays from the vertical denester 1 to the line.
0464 - Vertical denester 1: Delay stop vacuum sucker [ms]	The delay time for the suction cup vacuum stop.
0465 - Vertical denester 1: Delay switch rotation after cycle start [ms]	The delay time for the switch rotation after the vertical denester 1 cycle start.
0466 - Vertical denester 1: Delay return switch rotation after taking [ms]	The delay time for the switch return rotation after the trays taking cycle start.
0467 - Vertical denester 1: Missing tray in tray stock - advise [s]	The maximum reading time of the trays presence in the tray stock photocell beyond which the machine signals an advise.
0468 - Vertical denester 1: Missing tray in tray stock - alarm [s]	The maximum reading time of the trays presence in the tray stock photocell beyond which the machine signals an alarm.
0469 - Vertical denester 1: Tray release time to line [s]	Time in tray release position of the vertical denester 1 to the line.
0470 - Vertical denester 1: Transfer piston timeout [s]	The maximum reading of the transfer piston photocell beyond which the machine signals the alarm.

0471 - Vertical denester 1: Trays into denester (chain steps for cycle)	The chain steps number for every vertical denester 1 cycle.
0472 - Vertical denester 1: Lift pack working time [s]	The working time of the lift pack device.
0473 - Vertical denester 1: Delay start lift tray pack [s]	The delay time for the lift pack device start cycle.
0474 - Vertical denester 1: Delay start cycle [s]	The delay time for the vertical denester start cycle.
0475 - Vertical denester 1: Taking time [ms]	The time for the taking trays cycle on the vertical denester.
0476 - Vertical denester 1: Delay stop vacuum sucker [ms]	The delay time for the closing of the vacuum suckers on the vertical denester.
0477 - Vertical denester 1: Out impediment (clearance) time [ms]	The clearance time of the trays for start the next cycle of the vertical denester.
0478 - Vertical denester 1: Pneumatical piston motion timeout [s]	The maximum reading time for the correct operation of the pneumatic piston of the vertical denester.
0479 - Vertical denester 1: Open selector delay time [s]	The delay time for the opening of the selectors on the vertical denester.
0480 - Vertical denester 1: Closing selector delay time [s]	The delay time for the closing of the selectors on the vertical denester.
0481 - Vertical denester 2: Home position [mm]	The vertical denester 2 home position.
0482 - Vertical denester 2: Lower position [mm]	The release trays position of the vertical denester 2.
0483 - Vertical denester 2: Taking position [mm]	The taking trays position of the vertical denester 2.
0484 - Vertical denester 2: Extension position [mm]	The suction cups extension position of the vertical denester 2.
0485 - Vertical denester 2: Separation position [mm]	The trays separation from the tray stock position of the vertical denester 2.
0486 - Vertical denester 2: Time from start to taking position [ms]	Time from start position to taking position of the vertical denester 2.
0487 - Vertical denester 2: Taking time [ms]	Time in taking position of the vertical denester 2.
0488 - Vertical denester 2: Time from taking to extension position [ms]	Time from taking position to the suction cups extension position of the vertical denester 2.
0489 - Vertical denester 2: Time from extension to separation position [ms]	Time from the suction cups extension position to the trays separation position of the vertical denester 2.
0490 - Vertical denester 2: Time from separation to start position [ms]	Time from the trays separation position to the start position of the vertical denester 2.
0491 - Vertical denester 2: Speed on extension [%]	The extension motion speed of the vertical denester 2.
0492 - Vertical denester 2: Out impediment (clearance) time [ms]	The delay time for the restart vertical denester 2 cycle.
0493 - Vertical denester 2: Tray transfering time to line [ms]	The transferring time of the trays from the vertical denester 2 to the line.
0494 - Vertical denester 2: Delay stop vacuum sucker [ms]	The delay time for the suction cup vacuum stop.
0495 - Vertical denester 2: Delay switch rotation after cycle start [ms]	The delay time for the switch rotation after the vertical denester 2 cycle start.

0496 - Vertical denester 2: Delay return switch rotation after taking [ms]	The delay time for the switch return rotation after the trays taking cycle start.
0497 - Vertical denester 2: Missing tray in tray stock - advise [s]	The maximum reading time of the trays presence in the tray stock photocell beyond which the machine signals an advise.
0498 - Vertical denester 2: Missing tray in tray stock - alarm [s]	The maximum reading time of the trays presence in the tray stock photocell beyond which the machine signals an alarm.
0499 - Vertical denester 2: Tray release time to line [s]	Time in tray release position of the vertical denester 2 to the line.
0500 - Vertical denester 2: Transfer piston timeout [s]	The maximum reading of the transfer piston photocell beyond which the machine signals the alarm.
0501 - Vertical denester 2: Trays into denester (chain steps for cycle)	The chain steps number for every vertical denester 2 cycle.
0502 - Vertical denester 2: Open selector delay time [s]	The delay time for the opening of the selectors on the vertical denester.
0503 - Vertical denester 1: Closing stopper delay time [s]	The delay time for the closing of the stopper on the vertical denester.
0504 - Vertical denester 1: Open stopper delay time [s]	The delay time for the opening of the stopper on the vertical denester.
0505	
0506 -	
0507 -	
0508 - Deviator: Delay start belt [s]	The delay time for the deviator belt cycle start.
0509 - Deviator: Delay stop belt [s]	The delay time for the deviator belt cycle stop.
0510 - Downstream line stopped timeout [s]	it is the maximum reading time of the sensor that notices the trays at the downstream line over which the machine enters alarm.
0511 - SSP: Home position [mm]	The Super protruding cycle home position.
0512 - SSP: Rest position [mm]	The Super protruding rest position during the working cycle.
0513 - SSP: Heating position [mm]	The Super protruding heating position.
0514 - SSP: Threshold lifting pick-up/release piston [mm]	It's the position where the piston start the lifting pick-up/release movement.
0515 - SSP: Speed going to tool [%]	The motion speed going to tool during the Super protruding cycle.
0516 - SSP: Speed back to rest position [%]	The motion speed back to rest position during the Super protruding cycle.
0517 - SSP: Acceleration going to tool [%]	The motion acceleration going to tool during the Super protruding cycle.
0518 - SSP: Deceleration going to tool [%]	The motion deceleration going to tool during the Super protruding cycle.
0519 - SSP: Acceleration back to rest position [%]	The motion acceleration back to rest position during the Super protruding cycle.
0520 - SSP: Deceleration back to rest position [%]	The motion deceleration back to rest position during the Super protruding cycle.
0521 - SSP: Film heating time [s]	Time in the film heating position during the Super protruding cycle.
0522 - SSP: Plate temperature set point [°C]	The working temperature of the plate during the Super protruding cycle.
0523 - SSP: Plate temperature offset 1 [°C]	The offset temperature of the plate during the Super protruding cycle.

0524 - SSP: Plate temperature alarm threshold [°C]	The threshold temperature of the plate during the Super protruding cycle for the temperature alarm.
0525 - SSP: Timeout Lifting/lowering heating plate [s]	The maximum reading of the lifting/lowering piston photocell beyond which the machine signals the alarm.
0526 - SSP: Blowing time after heating [s]	The time for the blowing device cycle after the heating cycle.
0527 - SSP: Delay start dome forming [s]	The delay time for the film dome forming cycle start.
0528 - SSP: Dome forming time [s]	The time for the film dome forming cycle.
0529 - SSP: Main vacuum time [s]	The time for the opening of the main vacuum valve.
0530 - SSP: Sealing head compensation time [s]	The time for the opening of the head compensation valve.
0531 - SSP: Upper compensation time [s]	The time for the opening of the upper tool compensation valve.
0532 - Tool: Film cutting time [s]	The time for the sealing film cutting cycle.
0533 - Tool: Delay start film cutting [s]	The delay time for the sealing film cutting start.
0534 - SSP: Upper vacuum pulse during moulding time [s]	The time for the upper vacuum pulse during the moulding cycle.
0535 - SSP: Plate temperature offset 2 [°C]	The temperature offset of the heating plate 2.
0536 - SSP: Shuttle timeout [s]	The maximum reading time for the correct operation of the shuttle for the super protruding cycle.
0537 - SSP: Lowering upper tool delay [s]	The delay time for the lowering of the upper tool during the super protruding/skin cycle.
0538 - SSP: External upper vacuum delay [s]	The delay time for the upper vacuum delay during the super protruding/skin cycle.
0539 - Tool: Delay start lift tool after forming time [s]	The delay time for the start of the tool lift after the forming time.
0540 - SSP: Dome forming extra time [s]	The extra time for the closing of the vacuum valves in the dome forming during the super protruding/skin cycle.
0541 - SSP: Start lower compensation delay [s]	The delay time for the compensation of pressure in the lower tool during the super protruding/skin cycle.
0542 - SSP: Lift upper tool delay [s]	The delay time for the lift motion of the upper tool during the super protruding/skin cycle.
0543 - SSP: Delay start lower vacuum [s]	The delay time for the lower vacuum cycle start.
0544 - SSP: Delay start upper vacuum [s]	The delay time for the upper vacuum cycle start.
0545 - Doser 4: Position [steps]	this parameter refers to the number of chain steps where the doser 4 is positioned.
0546 - Doser 4: Start dosing phase [°]	this parameter refers to the start phase of the dosing cycle for the doser 4.
0547 - Doser 4: Delay start doser [s]	This parameter refers to the delay time for the doser 4 start.
0548 - Doser 4: Timeout for dosing [s]	The maximum reading time for the correct operation of the doser 4.

0549 - Doser 4: Pulse dosing time [s]	The time for the dosing cycle of the doser 4.
0550 - MicVac dispenser: Label stop [mm]	This parameter refers to the label stop position.
0551 - MicVac dispenser: High speed [mm/s]	It's the carrying out speed of the film.
0552 - MicVac dispenser: Acceleration [%]	It's the carrying out acceleration of the film.
0553 - Film unwinding: Lid length [mm]	The length of the film for the lid unwinding.
0554 - Film unwinding: Step after mark detection [mm]	The film unwinding after the film mark detection.
0555 - Film unwinding: Speed [%]	The film unwinding working cycle speed.
0556 - Film unwinding: Speed during mark detection [%]	The film unwinding working cycle speed during the film mark detection.
0557 - Film unwinding: Acceleration [%]	The film unwinding working cycle acceleration.
0558 - Film unwinding: Deceleration [%]	The film unwinding working cycle deceleration.
0559 - SSP: Delay start upper compensation time [s]	The delay time for the opening of the upper compensation valve.
0560 - MicVac labeller: Label start [mm]	It's the departure position of the MicVac labeller.
0561 - MicVac labeller: Label space [mm]	it is the distance among the labels applied on the film.
0562 - MicVac labeller: Start speed [mm/s]	It's the start speed of the MicVac labeller.
0563 - MicVac labeller: Application speed [mm/s]	It's the MicVac labeller speed during the labels application movement.
0564 - MicVac labeller: Return speed [mm/s]	It's the MicVac labeller speed during the return movement.
0565 - MicVac labeller: Acceleration [%]	It's the MicVac labeller acceleration during the speeds change.
0566 - MicVac labeller: Knife ON time [ms]	It's the operation time of the film cutter pistons.
0567 - MicVac labeller: Knife OFF time [ms]	It's the pause time of the film cutter pistons.
0568 - MicVac labeller: Number of labels	It's the labels number applied on the film to every operation cycle.
0569 - DOT: Delay start dome forming [s]	The delay time for the forming film cycle.
0570 - Rotating belt 1: Speed [Hz]	The speed of the rotating belt 1.
0571 - Rotating belt 1: Acceleration [s]	The acceleration of the rotating belt 1.
0572 - Rotating belt 1: Deceleration [s]	The deceleration of the rotating belt 1.
0573 - Rotating belts: Start delay [s]	The delay time for the rotating belts start.
0574 - Rotating belts: Stop delay [s]	The delay time for the rotating belts stop.
0575 - Rotating belt 2: Speed [Hz]	The speed of the rotating belt 1.
0576 - Rotating belt 2: Acceleration [s]	The acceleration of the rotating belt 1.
0577 - Rotating belt 2: Deceleration [s]	The deceleration of the rotating belt 1.

0578 - Rotating belts: Phaser close delay [s]	The delay time for the closing of the phaser on the rotating belts.
0579 - Rotating belts: Phaser close time [s]	The closing time of the phaser on the rotating belts.
0580 - Rotating belts: Phaser photocell timeout [s]	The maximum reading time of the sensor detecting the correct operation of the phaser on the rotating belts.
0581 -	
0582 - Doser 2: Stopper lift phase [°]	It is the position in which, effected the dosing, the stopper gets up.
0583 - Doser 2: Stopper lowering phase [°]	It is the position in which the stopper lowers.
0584 - Doser 2: Stopper working time [s]	The working time for the stopper piston.
0585 -	
0586 - Tilted denester: Delay start blow [s]	This parameter refers to the delay time for the blow start.
0587 - Vertical denester 1: Delay start stopper [ms]	This parameter refers to the delay time for the stopper start.
0588 - Vertical denester 2: Delay start stopper [ms]	This parameter refers to the delay time for the stopper start.
0589 - Film unwinding: After detect mark speed [%]	The speed of the film unwinding after detect mark.
0590 - Load denester belt: Acceleration [s]	The acceleration of the load denester belt.
0591 - Load denester belt: Deceleration [s]	The deceleration of the load denester belt.
0592 - Load denester belt: Speed [Hz]	The speed of the load denester belt.
0593 - Load denester belt: foreward stop belt delay [s]	The delay time for the stop of load denester during forward movement.
0594 - Load denester belt: Backward stop belt delay [s]	The delay time for the stop of load denester during forward movement.
0595 - Load denester belt: Stop photocell timeout [s]	The maximum reading time of the stop sensor detecting the correct operation on the load denester belt.
0596	
0597	
0598 - Load denester belt: Delay close separation gate [s]	The delay time for the separation gate closing.
0599 - Doser 2: Steps frequency start dosing [nr]	It is the number of steps which adjusts the dosing frequency.
0600	
0601	
0602 - Mobile head: wait time before back movement [s]	It's the pause time of the mobile head before the backward movement.
0603 - Mobile head: back movement speed [%]	It's the mobile head speed during the backward movement.
0604 - Mobile head: back movement acceleration [%]	It's the mobile head acceleration during the backward movement.
0605 - Mobile head: back movement deceleration [%]	It's the mobile head deceleration during the backward movement.

0606 - Mobile head: Ahead acceleration time [s]	It's the acceleration time of the mobile head during the ahead movement.
0607 - Mobile head: Pause time [s]	It's the pause time of the mobile head.
0608 - Doser 3: Lifter go down phase [°]	It is the position in which the mobile head lowers for drawing near to the container.
0609 - Doser 3: Lifter go up phase [°]	It is the position in which, effected the dosing, the mobile head gets up and move away from the container.
0610 - Trays bended feeding belt: Speed [Hz]	This parameter refers to the trays bended feeding belt speed.
0611 - Trays bended feeding belt: Delay start [s]	This parameter refers to the delay time for the trays bended feeding belt start.
0612	
0613 - Trays bended feeding belt: Acceleration [s]	This parameter refers to the trays bended feeding belt acceleration.
0614 - Trays bended feeding belt: Deceleration [s]	This parameter refers to the trays bended feeding belt deceleration.
0615 - Filling trays belt: Speed [Hz]	This parameter refers to the filling trays belt speed.
0616 - Filling trays belt: Delay start [s]	This parameter refers to the delay time for the filling trays belt start.
0617 -	
0618 - Filling trays belt: Acceleration [s]	This parameter refers to the filling trays belt acceleration.
0619 - Filling trays belt: Deceleration [s]	This parameter refers to the filling trays belt deceleration.
0620 - Meat trays exit belt: Speed [Hz]	This parameter refers to the meat trays exit belt speed.
0621 - Meat trays exit belt: Delay start [s]	This parameter refers to the delay time for the meat trays exit belt start.
0622 - Meat trays exit belt: Photocell timeout [s]	This parameter refers to the maximum reading time of the photocell for the correct operation of the belt.
0623 - Meat trays exit belt: Acceleration [s]	This parameter refers to the meat trays exit belt acceleration.
0624 - Meat trays exit belt: Deceleration [s]	This parameter refers to the meat trays exit belt deceleration.
0625 - Meat transport belt: Speed [Hz]	This parameter refers to the meat trays exit belt speed.
0626 - Meat transport belt: Delay start [s]	This parameter refers to the delay time for the meat trays exit belt start.
0627 -	
0628 - Meat transport belt: Acceleration [s]	This parameter refers to the meat trays exit belt acceleration.
0629 - Meat transport belt: Deceleration [s]	This parameter refers to the meat trays exit belt deceleration.
0630 - Meat transport belt: Belt full [s]	This parameter refers to the maximum reading time of the photocell for the belt full.
0631 - Filling trays belt: Open phaser delay [s]	This parameter refers to the phaser device opening.
0632 - Filling trays belt: Closer phaser delay [s]	This parameter refers to the phaser device closing.

0633 - Filling trays belt: Photocell timeout [s]	This parameter refers to the maximum reading time of the photocell for the correct operation of the belt.
0634 - Meat transport belt: Photocell timeout [s]	This parameter refers to the maximum reading time of the photocell for the correct operation of the belt.
0635 - Meat transport belt: Maximum meat cadence [piece/min]	This parameter refers to the maximum product meat cadence for the correct operation of the belt.
0636 - Infeed meat belt: Speed [Hz]	This parameter refers to the infeed meat belt speed.
0637 - Infeed meat belt: Delay start [s]	This parameter refers to the delay time for the infeed meat belt start.
0638 - Infeed meat belt: Acceleration [s]	This parameter refers to the infeed meat belt acceleration.
0639 - Infeed meat belt: Deceleration [s]	This parameter refers to the infeed meat belt deceleration.
0640 - Line follower: Home position [mm]	It’s the beginning position of the line follower device.
0641 - Line follower: Acceleration foreward movement [%]	This parameter refers to the line follower acceleration during foreward movement.
0642 - Line follower: Deceleration foreward movement [%]	This parameter refers to the combiner belt deceleration during foreward movement.
0643 - Line follower: Acceleration Backward movement [%]	This parameter refers to the line follower acceleration during backward movement.
0644 - Line follower: Deceleration Backward movement [%]	This parameter refers to the combiner belt deceleration during backward movement.
0645 - SSP: Air skin delay respect main compensation skin valve [s]	The air skin delay respect main compensation skin valve.
0646 - Line follower: Speed backward [%]	This parameter refers to the line follower speed during backward movement.
0647 - Line follower: Start phase [°]	It’s the phase for the start of the line follower device.
0648 - Line follower: Follower stroke [mm]	It’s the trays follower stroke.
0649 - Lifter 1: Working position for stoppers [mm]	It’s the lifter working position for stoppers.
0650 - Lifter 1: Home position [mm]	It’s the beginning position of the lifter 1 device.
0651 - Lifter 1: Acceleration foreward movement [%]	It’s the acceleration of the lifter 1 device during the forward movement.
0652 - Lifter 1: Deceleration foreward movement [%]	It’s the deceleration of the lifter 1 device during the forward movement.
0653 - Lifter 1: Acceleration Backward movement [%]	It’s the acceleration of the lifter 1 device during the backward movement.
0654 - Lifter 1: Deceleration Backward movement [%]	It’s the deceleration of the lifter 1 device during the backward movement.
0655 - Lifter 1: Speed forward [%]	It’s the speed of the lifter 1 device during the forward movement.
0656 - Lifter 1: Speed backward [%]	It’s the speed of the lifter 1 device during the backward movement.
0657 - Lifter 1: Working time [s]	It’s the working time of the lifter 1 device. È il tempo di lavoro del dispositivo alzatore1.

0658 - Lifter 1: Working position [mm]	It's the working position of the lifter 1 device.
0659 - Lifter 1: Line phase for start ahead movement [°]	It's the line position for the lowering movement of the lifter 1.
0660 - Lifter 2: Home position [mm]	It's the beginning position of the lifter 2 device.
0661 - Lifter 2: Acceleration foreward movement [%]	It's the acceleration of the lifter 2 device during the forward movement.
0662 - Lifter 2: Deceleration foreward movement [%]	It's the deceleration of the lifter 2 device during the forward movement.
0663 - Lifter 2: Acceleration Backward movement [%]	It's the acceleration of the lifter 2 device during the backward movement.
0664 - Lifter 2: Deceleration Backward movement [%]	It's the deceleration of the lifter 2 device during the backward movement.
0665 - Lifter 2: Speed forward [%]	It's the speed of the lifter 2 device during the forward movement.
0666 - Lifter 2: Speed backward [%]	It's the speed of the lifter 2 device during the backward movement.
0667 - Lifter 2: Working time [s]	It's the working time of the lifter 2 device. È il tempo di lavoro del dispositivo alzatore1.
0668 - Lifter 2: Working position [mm]	It's the working position of the lifter 2 device.
0669 - Lifter 2: Line phase for start lowering movement [°]	It's the line position for the lowering movement of the lifter 2.
0670 - Rotating denester: Home position [°]	It's the beginning cycle position of the rotating denester.
0671 - Rotating denester: Rotating speed [%]	It's the speed for the rotating denester.
0672 - Rotating denester: Start delay time [s]	It's the delay time for the rotating denester start.
0673 - Rotating denester: Tray production speed [pz/min]	It's the production speed of the rotating denester.
0674 - Rotating denester: Rotating acceleration [%]	It's the acceleration of the rotating denester.
0675 - Rotating denester: Rotating deceleration [%]	It's the deceleration of the rotating denester.
0676 - Rotating denester: Denester position for start phase of stopper cycle [°]	It's the denester position for start phase of stopper cycle.
0677 - Rotating denester: Missing tray in tray stock advise [s]	It's the time for the advise of missing tray in tray stock.
0678 - Rotating denester: Missing tray in tray stock alarm [s]	It's the time for the alarm of missing tray in tray stock.
0679 - Rotating denester: Delay restart stopper cycle after cycle stop [nr]	It's the delay for the restart stopper cycle after cycle stop.
0680 - Rotating denester: Pause time between stopper cycles [s]	It's the pause time between the stopper cycle.
0681 - Rotating denester - Low stopper: Open delay time [s]	It's the delay time for the low stopper device opening.
0682 - Rotating denester - Low stopper: Working time [s]	It's the working time for the low stopper device.
0683 - Rotating denester - Upper stopper: Open delay time [s]	It's the delay time for the upper stopper device opening.

0684 - Rotating denester: Denester position for start phase of blower cycle [°]	The denester position for the blower cycle start.
0685 - Rotating denester: Blolow working time [s]	The working time of the blow cycle.
0686 - Line: Stop middle position 1 [°]	Stop middle position 1 of the line
0687 - Line: Stop middle position 2 [°]	Stop middle position 2 of the line
0688 - Line: Stop middle position 3 [°]	Stop middle position 3 of the line
0689 - Line: Middle stop time 1 [s]	The time to stop in the middle position 1.
0690 - Line: Middle stop time 2 [s]	The time to stop in the middle position 2.
0691 - Line: Middle stop time 3 [s]	The time to stop in the middle position 3.
0692 - Line: Middle moving speed [%]	The operation speed between the middle pauses
0693 - Rotating denester: Discharge area free filter time [s]	The reading time of the sensor for the trays discharge area free.
0694 - Rotating denester: Discharge area busy filter time [s]	The reading time of the sensor for the trays discharge area busy.
0695 - Rotating denester: Selectors cyclinders movement time [s]	The movement time of the selectors cylinders.
0696 - Rotating denester: Photocell timeout below denester [s]	The maximum reading time of the photocell below denester for the correct operation of the device.
0697 - SSP: Delay start compensation time sealing head [s]	The delay time for the sealing head compensation valve opening.
0698 - SSP: Delay start compensation time sealing head side 2 [s]	The delay time for the compensation valve opening.
0699 - SSP: Delay start second skin valve [s]	The delay time for the second skin valve opening.
0700 - DOT: Vacuum time [s]	The time for the vacuum cycle during the "DARFRESH" cycle.
0701 - DOT: Start compensation delay [s]	The delay time for the compensation cycle start during the "DARFRESH" cycle.
0702 - DOT: Lower pre-vacuum time [s]	The time for the pre-vacuum cycle.
0703 - DOT: Delay stop lower vacuum during tool lowering motion [s]	The delay time for the closing of the lower vacuum valves during the "DARFRESH" cycle.
0704 - DOT: Compensation time [s]	The delay time for the opening of the compensation valves during the "DARFRESH" cycle.
0705 - DOT: Vacuum time [s]	The time for the vacuum cycle during the "DARFRESH" cycle.
0706 - DOT: Stop vacuum delay [s]	The delay time for the closing of the vacuum valves during the "DARFRESH" cycle.
0707 - DOT: Cooling plates time [s]	The time for the cooling of the plates during the "DARFRESH" cycle.
0708 - DOT: Release air delay time [s]	The delay time for the film release air blowing cycle start during the "DARFRESH" cycle.
0709 - DOT: Release air blowing time [s]	The blowing time for the film release during the "DARFRESH" cycle.

0710 - Shuttle: Start position [mm]	The start position of the "DARFRESH" cycle.
0711 - Shuttle: Stop position [mm]	The stop position of the "DARFRESH" cycle.
0712 - Shuttle: Ahead speed [%]	The forward motion speed during the "DARFRESH" sealing cycle.
0713 - Shuttle: Back speed [%]	The backward motion speed during the "DARFRESH" sealing cycle.
0714 - Shuttle: Ahead Acceleration [%]	The forward motion acceleration during the "DARFRESH" sealing cycle.
0715 - Shuttle: Ahead Deceleration [%]	The forward motion deceleration during the "DARFRESH" sealing cycle.
0716 - Shuttle: Shuttle position for anticipate tool rising motion [mm]	The shuttle position for the anticipate tool rising motion.
0717 - Shuttle: Back acceleration [%]	The backward motion acceleration during the "DARFRESH" sealing cycle.
0718 - Shuttle: Back deceleration [%]	The backward motion deceleration during the "DARFRESH" sealing cycle.
0719 - Shuttle: Home position [mm]	The home position for the "DARFRESH" sealing cycle.
0720 - Shuttle: Shuttle position for anticipate tool lowering motion [mm]	The shuttle position for the anticipate tool lowering motion.
0721 - Shuttle: Wait position [mm]	The wait position for the "DARFRESH" sealing cycle.
0722 - DOT: Pick up time [s]	The pick up time during the "DARFRESH" cycle.
0723 - DOT: Release time [S]	The release time during the "DARFRESH" cycle.
0724 - Shuttle: Tool position for shuttle starting [mm]	The sealing tool position for the shuttle plate start.
0725 - DOT: Delay start lowering tool for pickup film [s]	The delay time for the lower tool pickup up of the film.
0726 - DOT: Delay start compensation 2 [s]	The delay time for the compensation valves opening during the "DARFRESH" cycle.
0727 - DOT: Pick-up film vacuum pause during cutting phase [s]	The pick-up pause time for the film cutting phase during the "DARFRESH" cycle.
0728 - DOT: Delay start vacuum sealing head plate [s]	The delay time for the vacuum on the suckers of the sealing head plate during the "DARFRESH" cycle.
0729 - DOT: Delay start compensation sealing head plate [s]	The delay time for the compensation on the suckers of the sealing head plate during the "DARFRESH" cycle.
0730 - DOT: Pre-compensation pressure [mbar]	The pressure for the pre-compensation cycle during the "DARFRESH" cycle.
0731 - DOT: Delay start sealing head final compensation [s]	The delay time for the final compensation cycle of the sealing head during the "DARFRESH" cycle.
0732 - DOT: Vacuum pause during compensation [s]	The pause time of the vacuum during the compensation cycle during the "DARFRESH" cycle.
0733 - DOT: Vacuum plate release delay [s]	The delay time for the closing of plates vacuum valves.
0734 - DOT: Delay start second skin valve [s]	The delay time regulating start of the second valve during the "DARFRESH" cycle.

0735 - Tool: Forming (Skin) vacuum pause time while opening lower vacuum[s]	The pause time of the forming vacuum while opening the lower vacuum.
0736 - DOT: Backward stroke to pause position [mm]	The carriage back moving to the pause position during the DOT cycle.
0737 - DOT: Backward stroke speed [%]	The carriage movement speed back to the pause position during the DOT cycle.
0738 - DOT: Tool down time when in pause position [s]	The time in which the tool remains in pause position during the DOT cycle.
0739 - DOT: Cutting time [s]	is the working time of the blades for cutting the film.
0740 - SP: Skin air delay [s]	The delay time for the skin air cycle.
0741 - SP: Skin air time [s]	The skin air cycle time.
0742 - SP: Lift tool start delay [s]	The delay time for the lift tool movement.
0743 - SP: Dome forming time [s]	The time for the film forming cycle.
0744 - SP: Upper vacuum pulse during moulding time [s]	The time for the vacuum pulse during film moulding cycle.
0745 - SP: Vacuum moulding time sealing head side [s]	The vacuum moulding time cycle.
0746 - SP: Delay start cycle [s]	The delay time for the “Skin Protruding” cycle.
0747 - SP: Lower compensation time for upper moulding [s]	The lower compensation time for the upper moulding.
0748 - SP: Delay start lower vacuum [s]	The delay time for the lower vacuum cycle start.
0749 - SP: Delay start upper vacuum [s]	The delay time for the upper vacuum cycle start.
0750 - Snap on lid: Chain acceleration [%]	The acceleration of the chain on the snap on lid machine.
0751 - Snap on lid: Chain deceleration [%]	The deceleration of the chain on the snap on lid machine.
0752 - Snap on lid: Chain speed [Hz]	The speed of the chain on the snap on lid machine.
0753 - Snap on lid: Chain home position [°]	The position for the beginning cycle of the chain on the snap on lid machine.
0754 - Snap on lid: Chain position cam for tool stopper [°]	The position of the cam on the chain for the tool stopper start.
0755 - Snap on lid: Chain position cam for load piston [°]	The position of the cam on the chain for the load piston start.
0756 - Snap on lid: Chain position cam for tool start [°]	The position of the cam on the chain for the tool start.
0757 - Snap on lid: Chain stopper working time [s]	The time for the chain stopper cycle.
0758 - Snap on lid: Chain step [steps]	The number of steps for a chain cycle.
0759 - Snap on lid: Chain step pause time [s]	The time for the step pause of the chain.
0760 - Snap on lid: Tool step position [steps]	The number of steps of the chain for the tool position.
0761 - Snap on lid: Tool cavity [nr]	The number of cavity in the tool of the snap on lid machine.
0762 - Snap on lid: Lid pick-up time [s]	The time in the pick-up position of the lids.

0763 - Snap on lid: Lid release time [s]	The time in the release position of the lids.
0764 - Snap on lid: Lid shuttle lift timeout [s]	The maximum reading time for the sensor that notice the correct operation of the lid shuttle lift.
0765 - Snap on lid: Delay start lid presence in tool [s]	The delay time for the check-in of the presence of the lids in the tool.
0766 - Snap on lid: Tool closing time [s]	The closing time of the tool.
0767 - Snap on lid: Line loading piston timeout [s]	The maximum reading time for the sensor that notice the correct operation of the line loading piston.
0768 - Snap on lid: Timeout closing tool [s]	The maximum reading time for the sensor that notice the correct operation of the closing tool piston.
0769 - Snap on lid: Timeout moving shuttle [s]	The maximum reading time for the sensor that notice the correct operation of the shuttle.
0770 - Snap on lid: Timeout chain tray photocell [s]	The maximum reading time for the sensor that notice the correct operation of the chain of the snap on lid machine.
0771 - Snap on lid: Delay start shuttle transfer [s]	The delay time for the shuttle transfer start.
0772 - Snap on lid: Delay open tray stopper [s]	The delay time for the tray stopper open.
0773 - Line follower: Ahead pause time (before come back) [s]	This parameter refers to the line follower pause after ahead movement. (before come back)
0774 - SP: Delay start compensation time sealing head [s]	The delay time for the sealing head compensation valve opening.
0775 - SP: Delay start second skin valve [s]	The delay time for the second skin valve opening.
0776 - SP: Delay start compensation time sealing head side 2 [s]	The delay time for the compensation valve opening.
0777 - SP: Delay start upper compensation time [s]	The delay time for the upper compensation opening.
0778 - SP: Lower vacuum time [s]	The lower vacuum time.
0779 - SP: Upper vacuum time [s]	The upper vacuum time.
0780 - Pliers: Open speed [%]	It’s the open speed for the pliers movement.
0781 - Pliers: Open acceleration [%]	It’s the open acceleration for the pliers movement.
0782 - Pliers: Open deceleration [%]	It’s the open deceleration for the pliers movement.
0783 - Pliers: Open position plier 1 [mm]	It’s the open position for the plier 1.
0784 - Pliers: Open position plier 2 [mm]	It’s the open position for the plier 2.
0785 - Pliers: Close speed [%]	It’s the close speed for the pliers movement.
0786 - Pliers: Close acceleration [%]	It’s the close acceleration for the pliers movement.
0787 - Pliers: Close deceleration [%]	It’s the close deceleration for the pliers movement.
0788 - Pliers: Close position plier 1 [mm]	It’s the close position for the plier 1.

0789 - Pliers: Close position plier 2 [mm]	It' the close position for the plier 2.
0790 - Pliers: Home position plier 1 [mm]	It's the home position of the plier 1.
0791 - Pliers: Home position plier 2 [mm]	It's the home position of the plier 2.
0792 - DOT: Film cutting delay [s]	is the delay time which controls the starting of the working cycle of the blades for cutting the film.
0793	
0794	
0795	
0796	
0797 - Line follower: Speed ratio with chain	It's the ratio speed to the spacer belt (among chain).
0798 - Pacer Line: Close pacer phase [°]	The line position where the pacer closing.
0799 - Pacer Line: Pacer working time [s]	The working time of the pacer device.
0800 - Line: Flatten piston phase start [°]	The position for the flatten piston start cycle.
0801 - Line: Flatten piston working time [s]	The working time of the flatten piston.
0802 - Vertical denester 1: Discharge area free filter time [s]	The reading time of the sensor for the trays discharge area free.
0803 - Vertical denester 1: Discharge area busy filter time [s]	The reading time of the sensor for the trays discharge area busy.
0804 - Vertical denester 1: Extractor delay [s]	The delay time for the extractor cycle start.
0805 - Vertical denester 1: Extractor working time [s]	The working time of the extractor cycle.
0806 - Vertical denester 1: Selectors cyclinders movement time [s]	The working time of the selectors cylinders movement.
0807 - Vertical denester 1 - Low stopper: Open delay time [s]	The delay time for the low stopper opening.
0808 - Vertical denester 1 - Low stopper: Working time [s]	The working time of the low stopper cycle.
0809 - Vertical denester 1 - Upper stopper: Open delay time [s]	The delay time for the upper stopper opening.
0810 - Vertical denester 1: Upper stoppers system cycle restart delay [s]	The delay time for the upper stoppers system cycle restart.
0811 - Vertical denester 1: Extractor system cycle restart delay [s]	The delay time for the extractor system cycle restart.
0812 - Vertical denester 1: Photocell timeout below denester [s]	The maximum reading time of the stop sensor detecting the correct operation below the denester.
0813 - Line: Flatten piston position [steps]	the position in which is mounted the flatten (among step chain)
0814 - Doser: Delay start product call [s]	The delay time for the start of the call product device inside the doser hopper.
0815 - Doser: Delay stop product call [s]	The delay time for the stop of the call product device inside the doser hopper

0816 - Doser: Alarm product call [s]	The maximum reading time for the call product device alarm
0817 - Tool cavities heating: Upper temperature set point [°C]	The operating temperature of upper heating plate in the tool area.
0818 - Tool cavities heating: Upper tool temperature alarm threshold [°C]	The maximum time it takes the sensor to detect a upper heating plate temperature, beyond which a machine alarm is generated.
0819 - Tool cavities heating: Upper heating plate 1 offset [°C]	The temperature difference between the set point and the value measured by the sensor of upper heating plate.
0820 - Tool cavities heating: Upper heating plate 2 offset [°C]	The temperature difference between the set point and the value measured by the sensor of upper heating plate.
0821 - Tool cavities heating: Maximum temperature [°C]	The maximum temperature of the tool cavities.
0822	
0823 - Line: Cleaning blow time [s]	The cleaning blow duration time.
0824 - Line: Cleaning blow start phase [°]	The cleaning blow start phase.
0825 - DOT heating plates: Start delay [s]	The delay time for the DOT heating plates.

6 OPERATION

INDEX

6.1 DAILY START UP OPERATIONS

Every day, prior to starting production, the operator must run a series of procedures, and specifically:

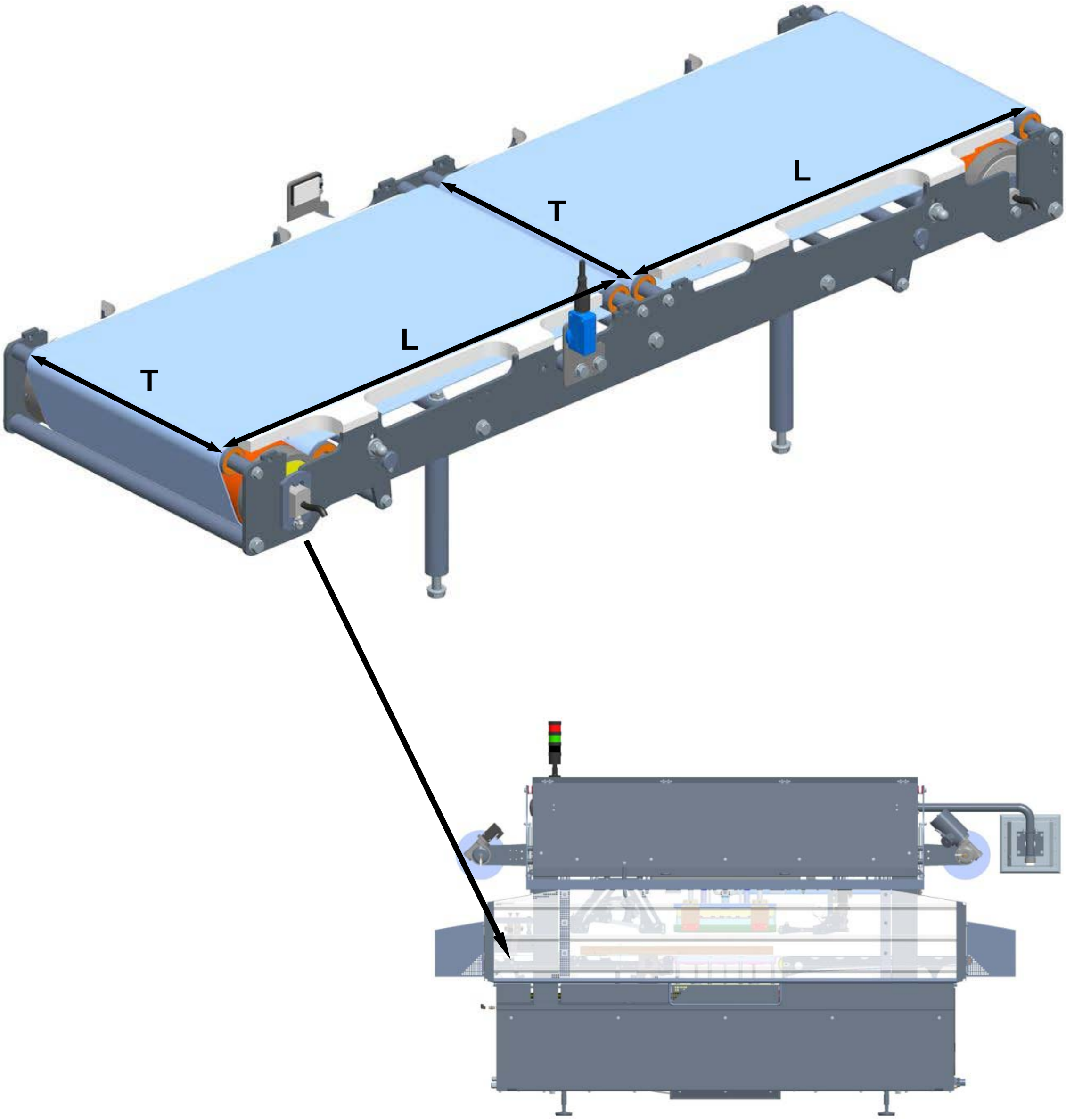
1. Inspection of the spacer belt.
2. Inspection of the tray detection photocell.
3. Sealing film assembly operations.
4. Check up on the work cycle recipe to be run.
5. Check run of the tool assembled on the machine;
6. Checking to ensure that the machine is clean. A wash cycle must also be run.
7. Inspection of safety guards and doors to see that they are closed.

A description on how to proceed with each one of the above points is provided herein as follows:

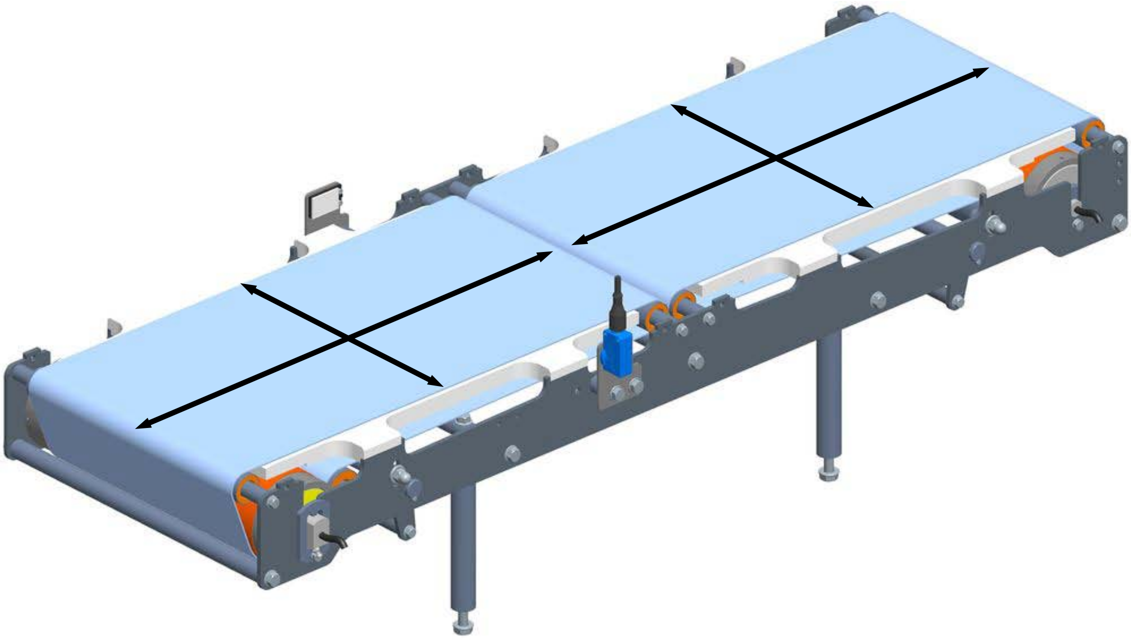
DANGER
Before servicing the machine or performing any other intervention, always press the emergency button associated with the opening of the moving guards to disconnect the machine from the power supply. Check operation of this safety device regularly.

6.1.1 Inspection of the spacer belt

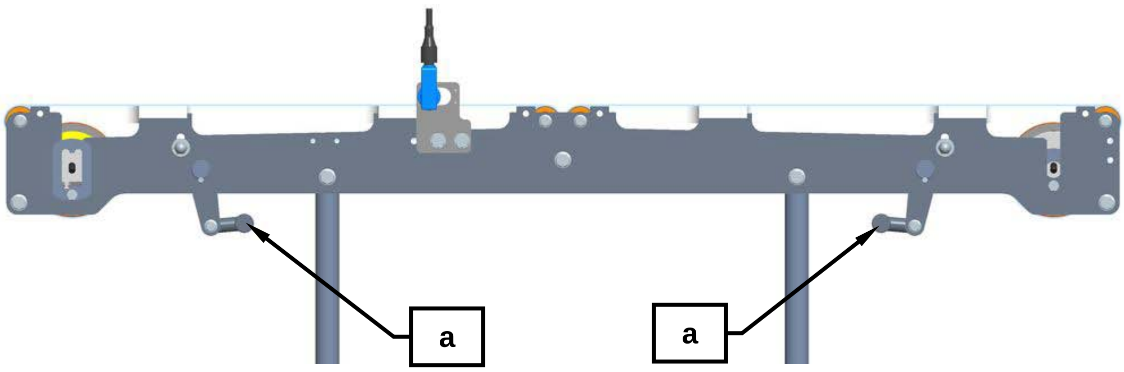
1. Check the spacer belt is assembled and centred correctly and that the tray travels both in longitudinal (**L**) and in transversal (**T**) directions.



If necessary, the operator must see that they are accurately aligned and centred by hand.



2. Use the appropriate levers (**a**) to lock the short infeed belts.

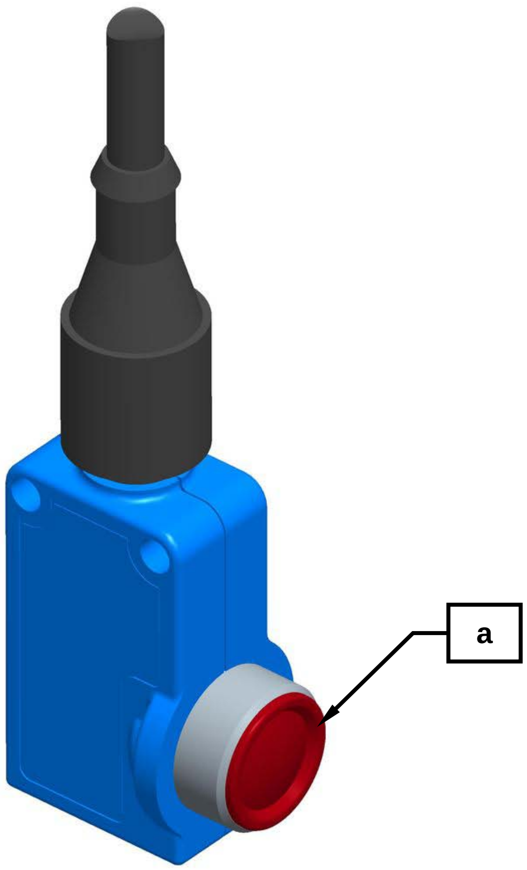


6.1.2 Inspection of the tray detection photocell

1. Remove the safety and protection covering from the photocell (**a**) checking to ensure that it is neither dirty nor wet (as a consequence of the washing cycle). If necessary, wipe it clean using a soft, dry cloth.

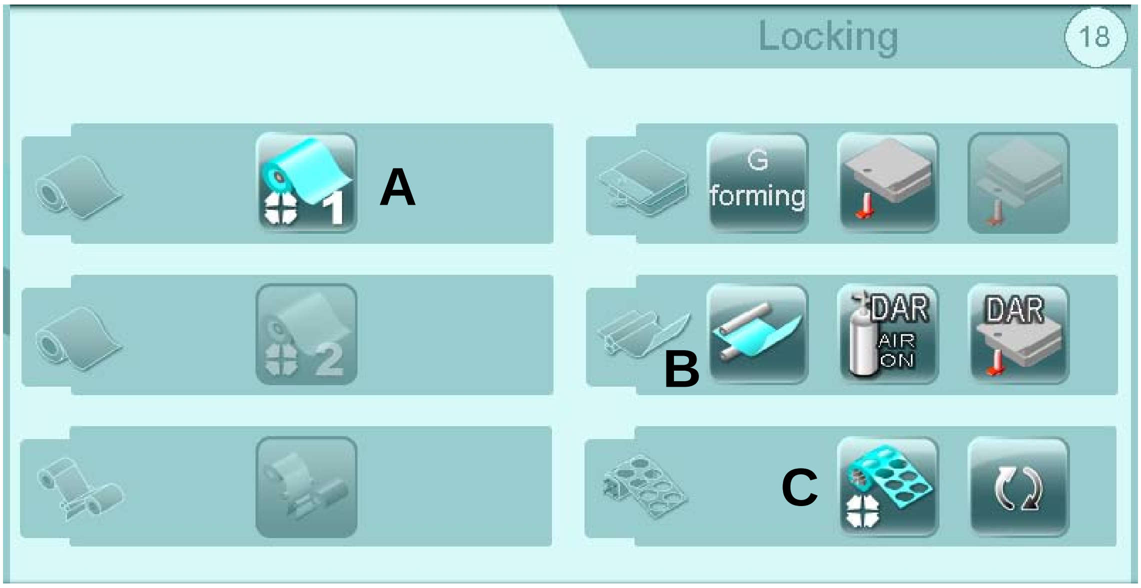
Important!

- NEVER wash the photocells, the reflectors and uncovered motors using high-pressure water jets.
- NEVER wash the photocells, the reflectors and uncovered motors using solvents or thinners. This because solvents or thinners contain chemical substances that will damage the plastic photocell eyes thereby compromising their operation.
- NEVER wash the photocells, the reflectors and uncovered motors using hot water as this will damage the electronic components. This is particularly important to observe after having powered off the line because the photocells are still hot.

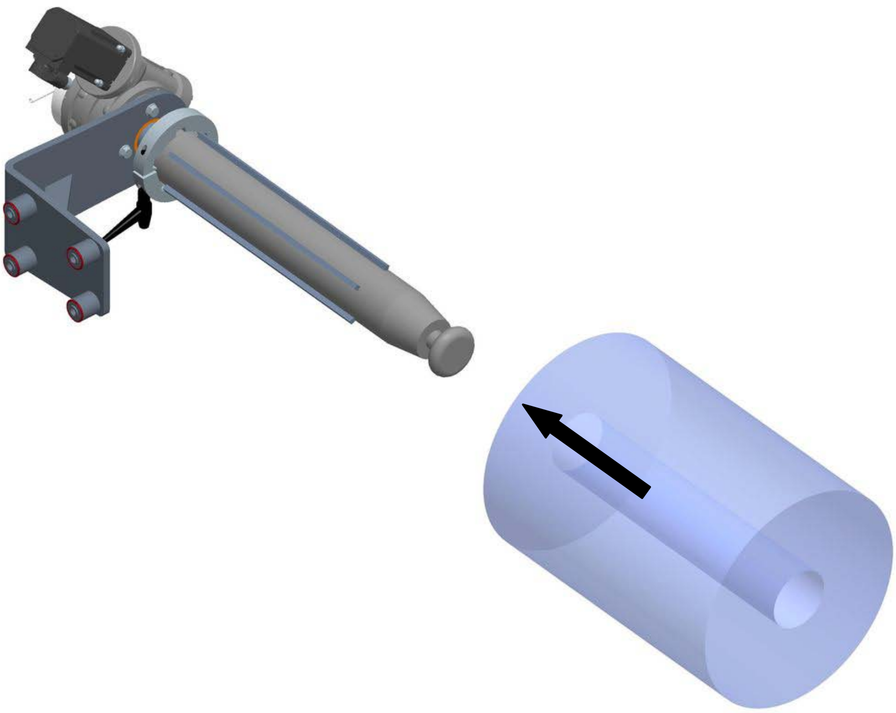


6.1.3 Sealing film assembly operations

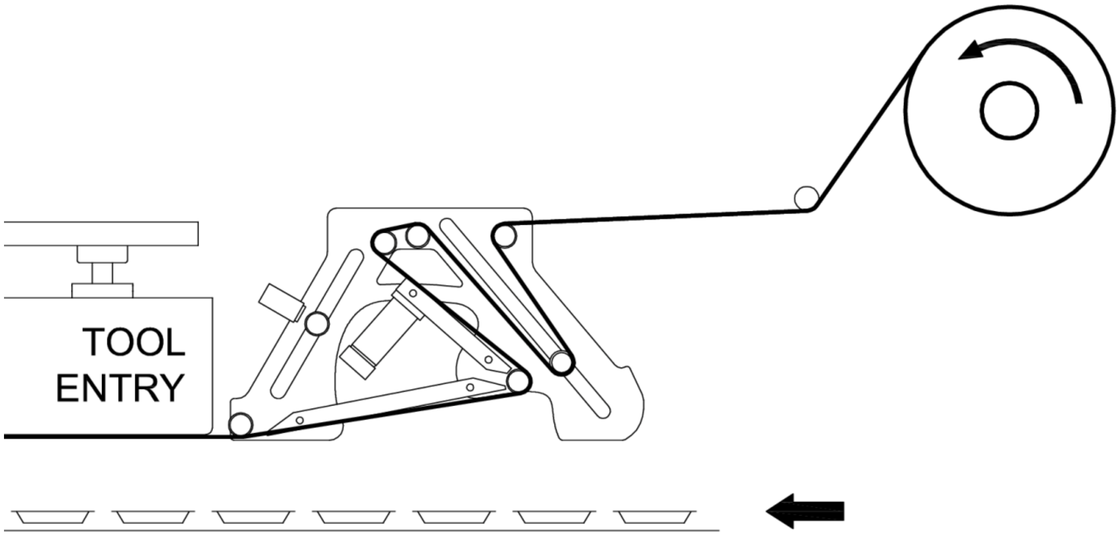
1. Check that the system locking functions on the Lid Unrolling, on the Waste Winding and on the Film Reel are all deactivated by acting on the relative selector keys (**A**), (**B**) and (**C**).

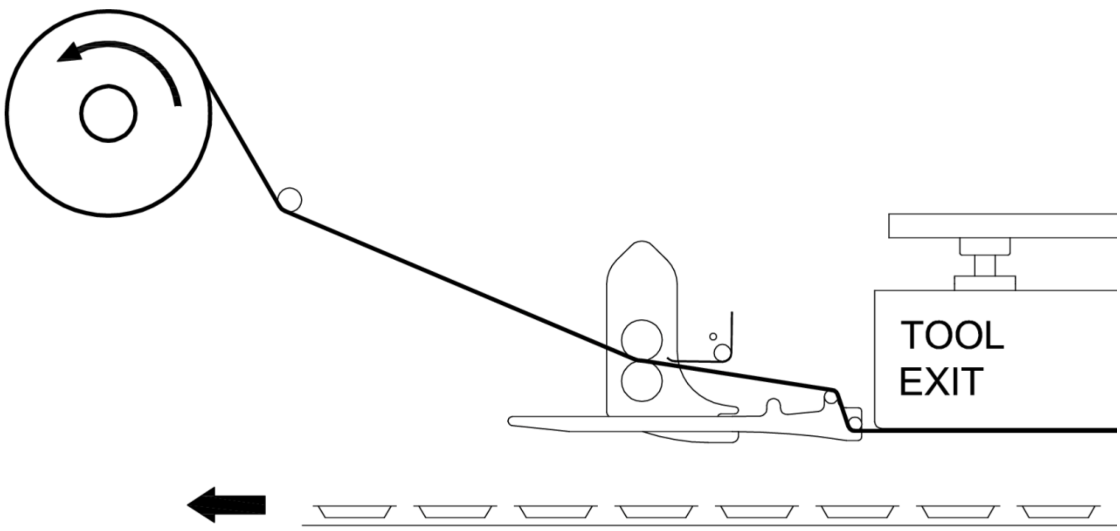


2. Fit on the sealing film reel.

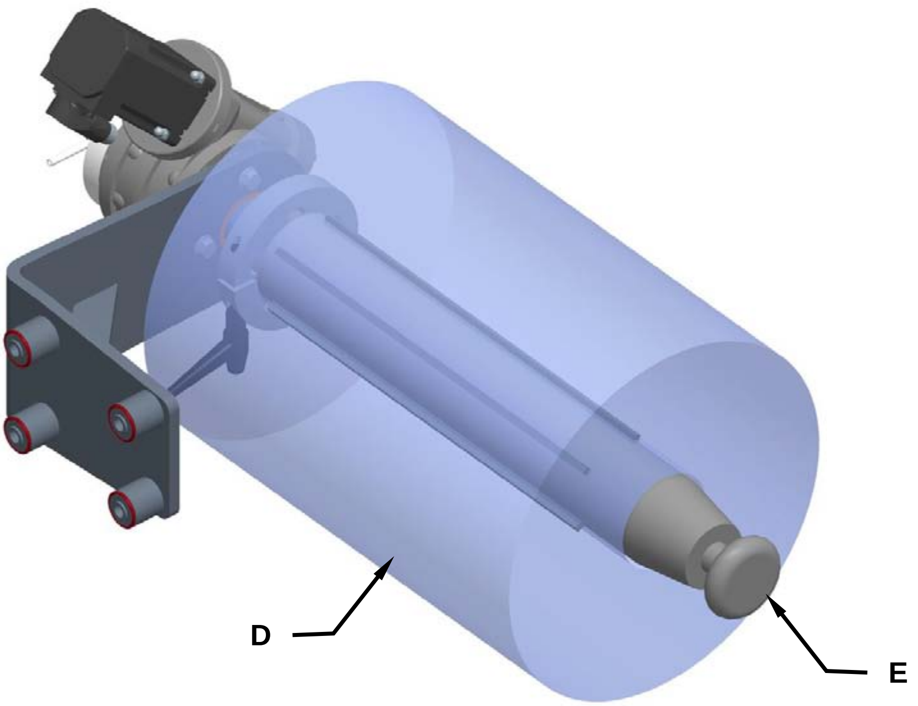


3. Thread the film in through the bobbin unwinder rollers as per the directions indicated by the arrows provided near each one of the rollers.

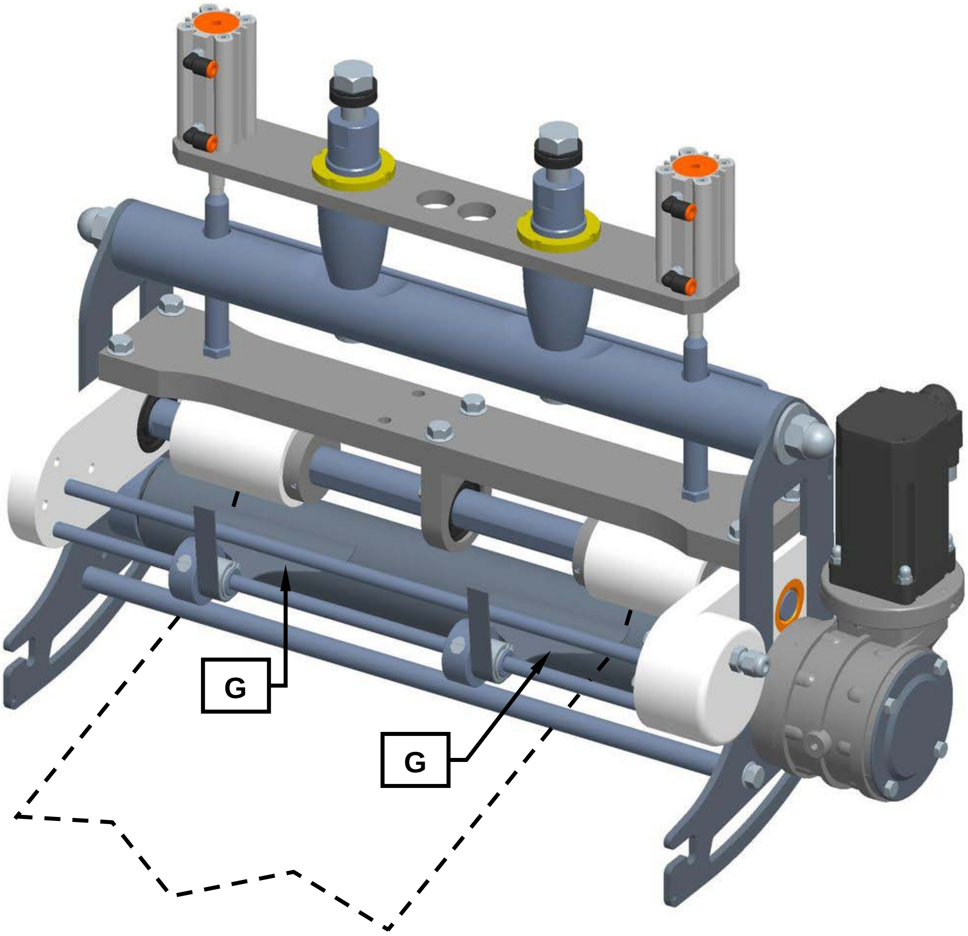




4. Check to ensure that the bobbin (**D**) is correctly centred on the film unwinder system shaft (**E**) and lock it in place using the pushbutton (**F**).



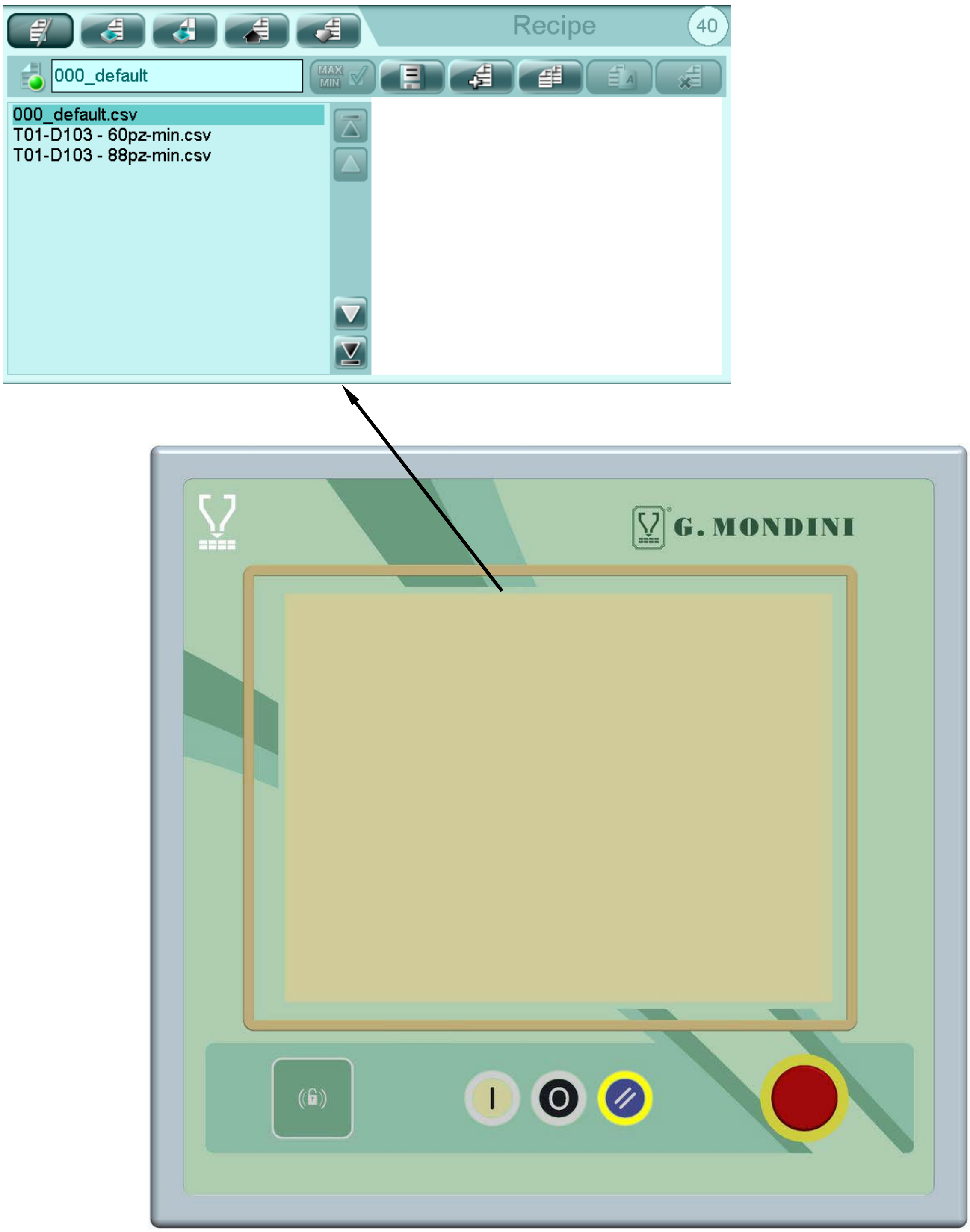
5. Check to see that the sealing film waste blade plates (**G**) are positioned on the relative outer edges. If not, move them so that they are positioned correctly as shown.



6. Push the button (**H**) to block the film unrolling pullers and wind the waste onto the waste winding shaft. To lock the waste push the button (**L**).

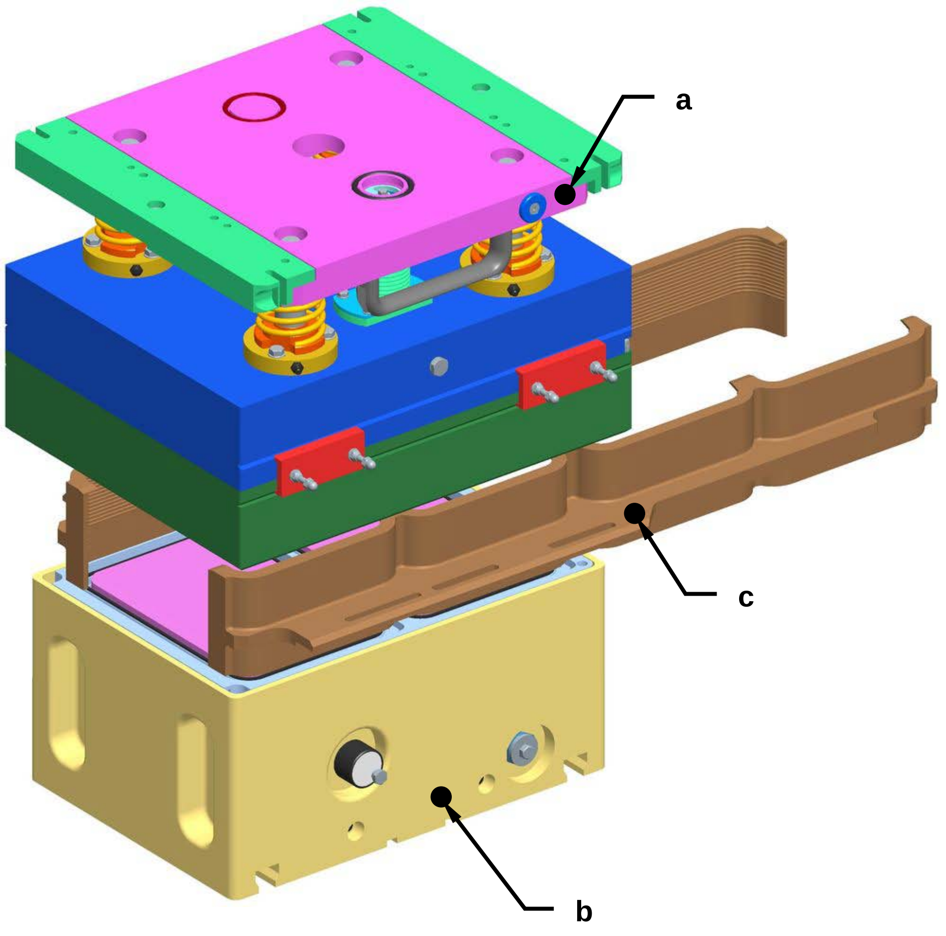
6.1.4 Check up on the work cycle recipe to be run

1. Check the system to ensure that the correct work cycle recipe has been selected for machine operation. To do so, it is necessary to consult the relative “SCREEN PAGE” section in Chapter 5 in this manual.



6.1.5 Check run of the tool assembled on the machine

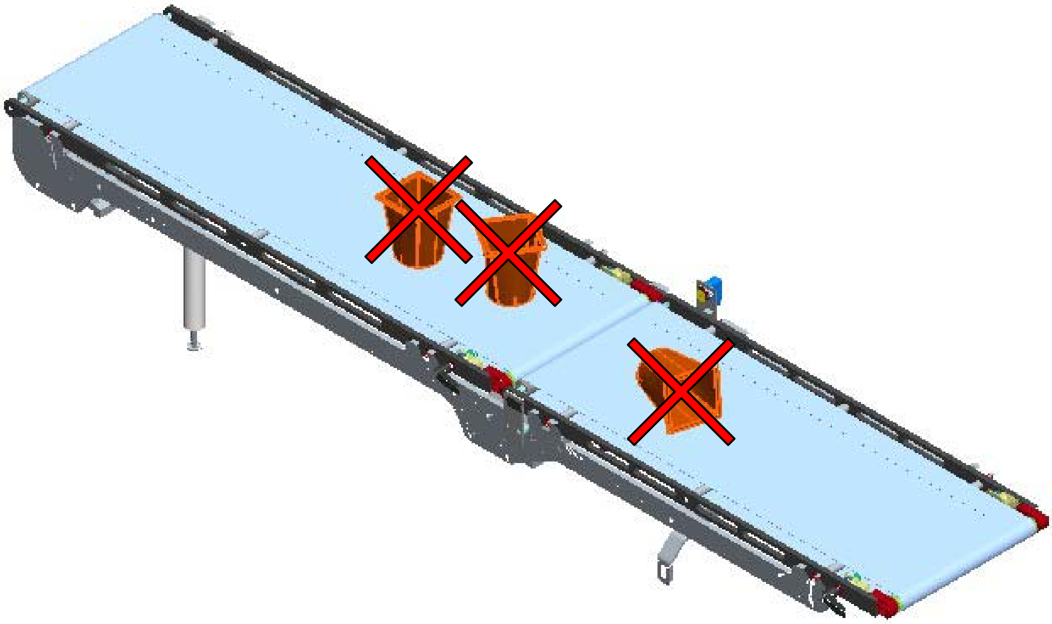
1. Check to ensure that all the tool components belong with one another (i.e. with matching coloured dots and/or numbers).



a. Top tool half
b. Bottom tool half
c. Right and left pusher units

6.1.6 Checking to ensure that the machine is clean - running a wash cycle

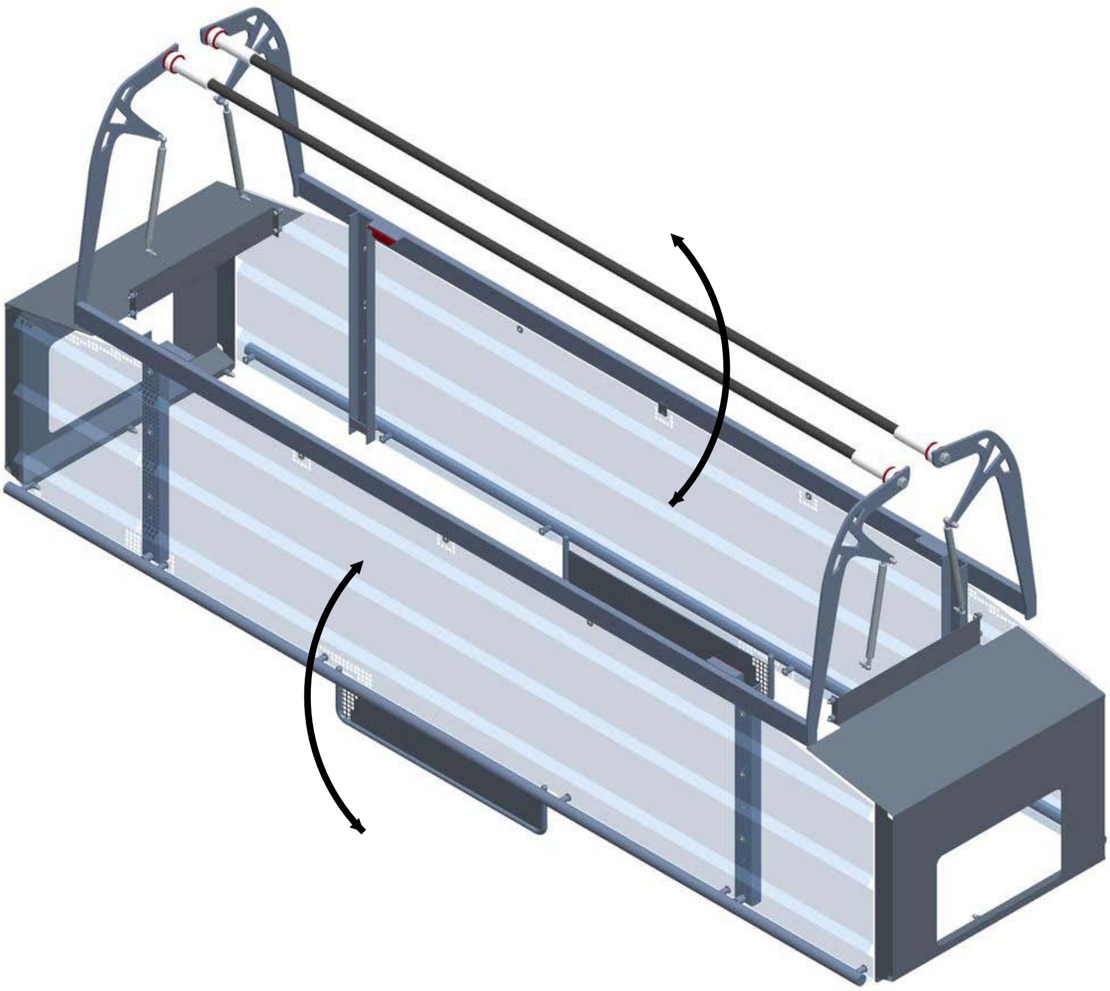
1. Check to ensure that there is nothing lying around on the machine, on the belts or in the vicinity of the tool.



2. Disengage the wash cycle (if selected) by acting on the relative selector key.

6.1.7 Inspection of safety guards and doors to see that they are closed

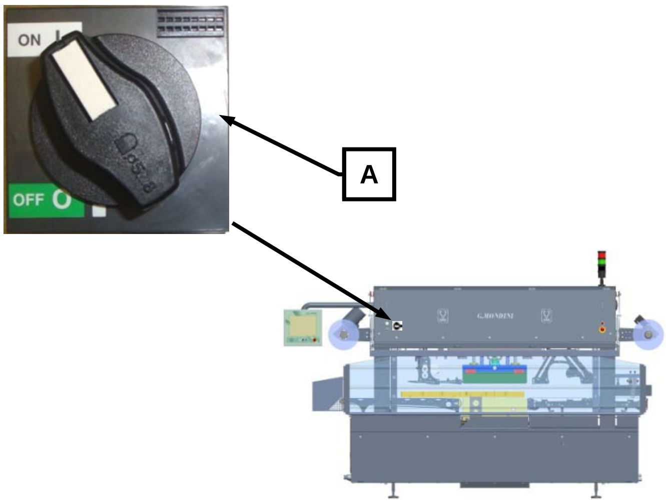
1. Check to make sure that all the sealing machine cover guards and doors are perfectly closed.



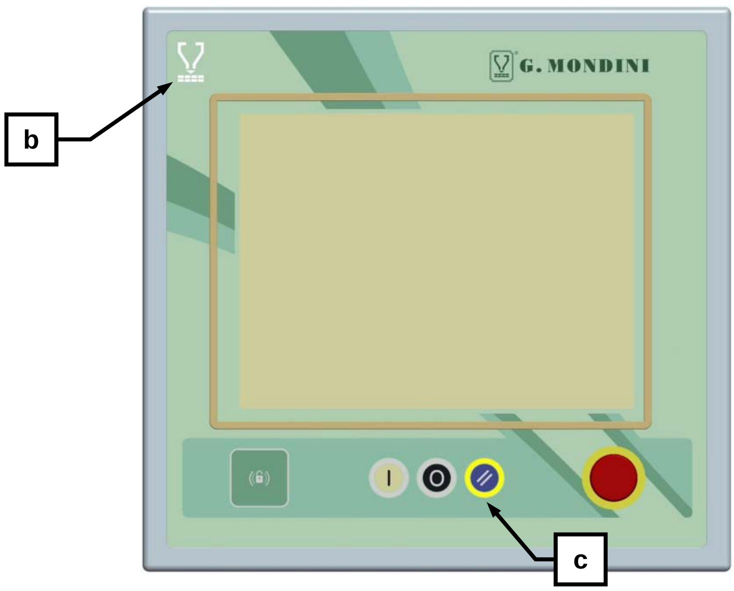
6.2 FIRST-TIME MACHINE START-UP

After having run all the inspection operations as specified and described in the previous pages, proceed as follows for machine start-up.

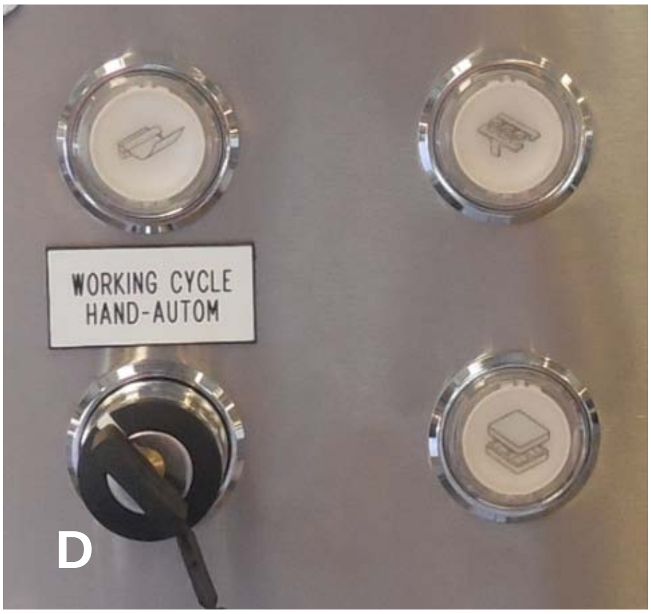
1. Power up the machine's power supply by turning main switch (**A**) into position 1.

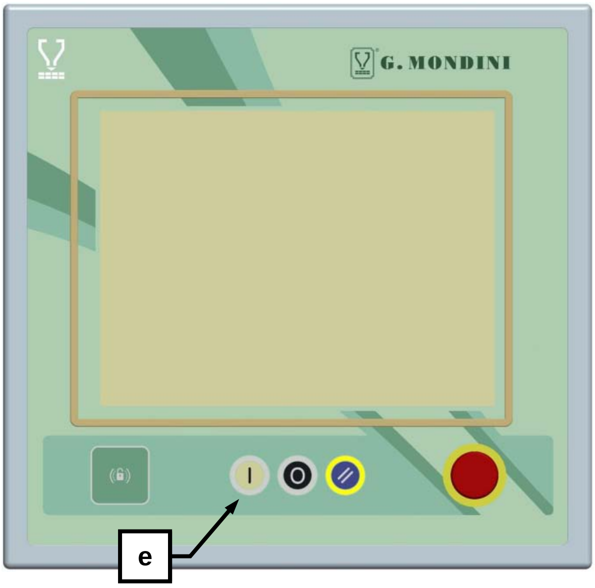


2. To check whether the machine has been powered up, check the electric controls panel to see that the white signal light (**B**) is on, meaning that POWER IS CONNECTED.
3. After having checked the machine according to the alarm messages displayed on screen, press the pushbutton (**C**) for ALARM RESET.



4. Switch selector (**D**) to the required machine operation mode, i.e. either to MANUAL or to AUTOMATIC.
5. Press the START pushbutton (**E**) to activate machine operations.
    - Every time this pushbutton is pressed, with selector (**D**) in MANUAL position, the machine runs one work cycle.
    - With selector (**D**) in the AUTOMATIC position and pressing the green START pushbutton, the machine work cycles are controlled based on the signals sent by the photocells and based on the parameters set into the recipe selected for operation.





6.3 END OF PRODUCTION (DAILY)

Every day, at the end of production, before departing from the machine the operator must perform a set of routine procedures and specifically:

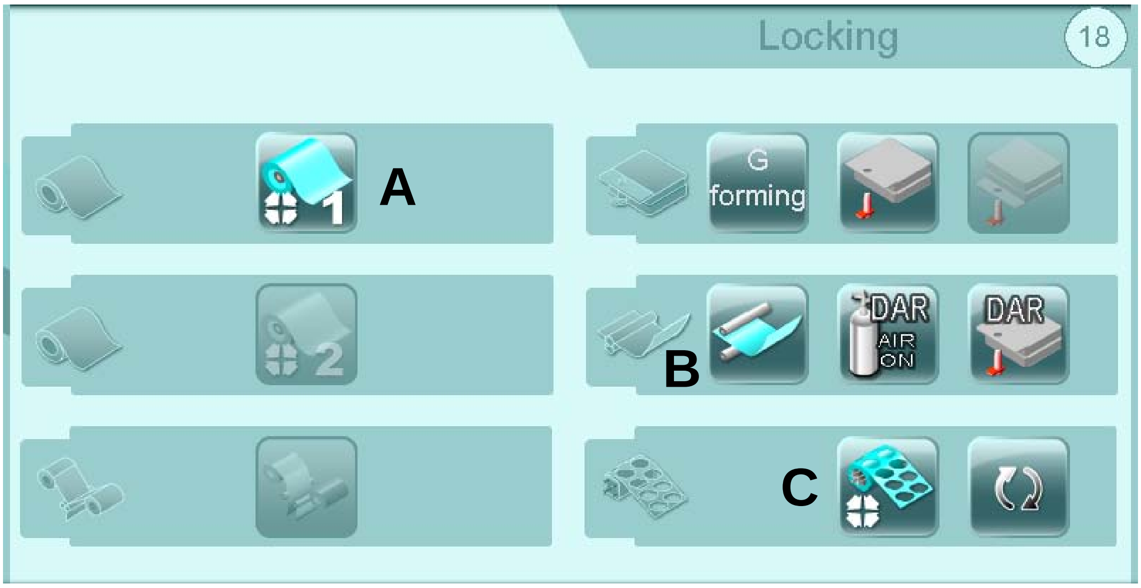
1. Sealing film removal operations
2. Wash cycle activation and covering up of the container detection photocell
3. Unlocking of the short infeed belts
4. Removal of the short infeed belts (only in the event of sterilisation).

A description on how to proceed with each one of the above points is provided herein as follows:

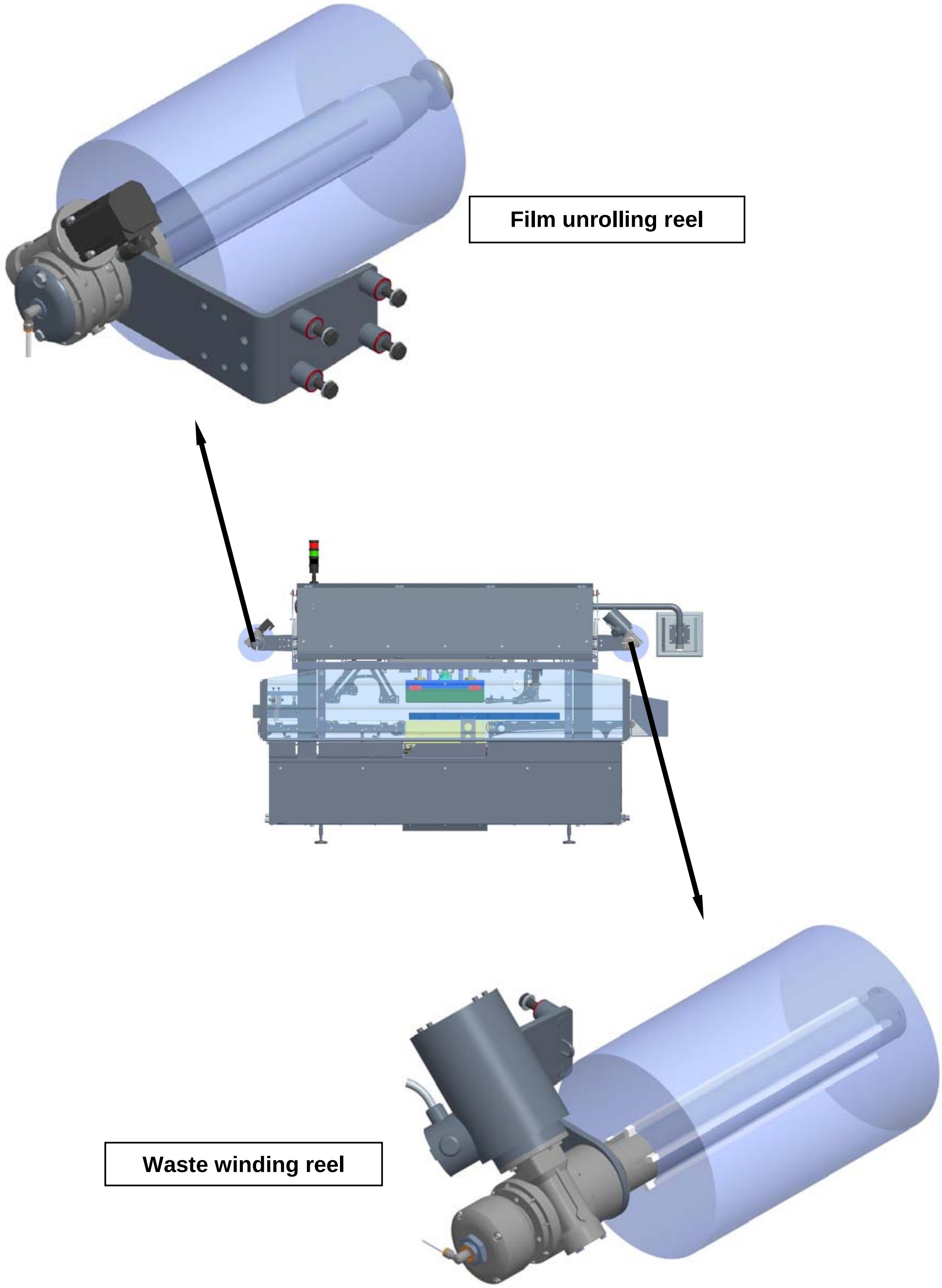
DANGER
Before servicing the machine or performing any other intervention, always press the emergency button associated with the opening of the moving guards to disconnect the machine from the power supply. Check operation of this safety device regularly.

6.3.1 Sealing film removal operations

1. Check that the system locking functions on the Lid Unrolling, on the Film Unrolling and on the Waste Winding are all deactivated by acting on the relative selector keys (**A**), (**B**) and (**C**).



2. Break off the sealing film and remove the film reels as well as on the waste winding reels.

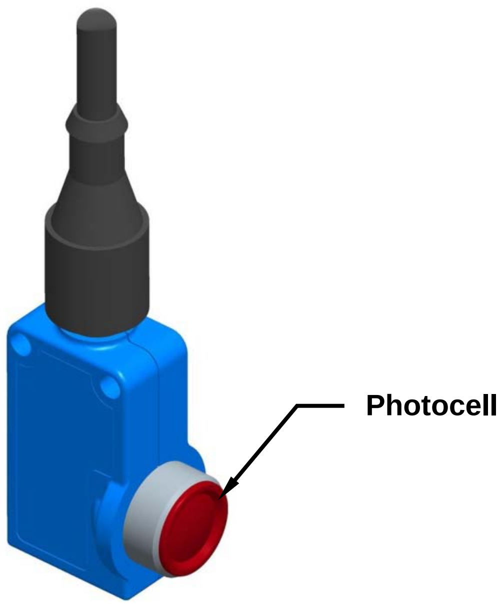


6.3.2 Wash cycle activation + cover-up of the tray detection photocell

1. Activate the wash cycle and cover up all photocells, reflectors and uncovered motors installed in the system (in-line, sealing machines, doser units, etc…) to prevent them from being damaged during the cleaning phases. All the photocells are provided with an **IP 65** or higher protection class to ensure overall safety against contacts with energised machine parts as well as protection against dust and/or water jet penetration hazards. We recommend that, prior to beginning machine washing operations, the above mentioned parts are covered using plastic bags. Should it nevertheless occur that the photocells get wet, we recommend drying them off using a soft, dry cloth so as not to damage or scratch the photocell eyes.

WARNING!

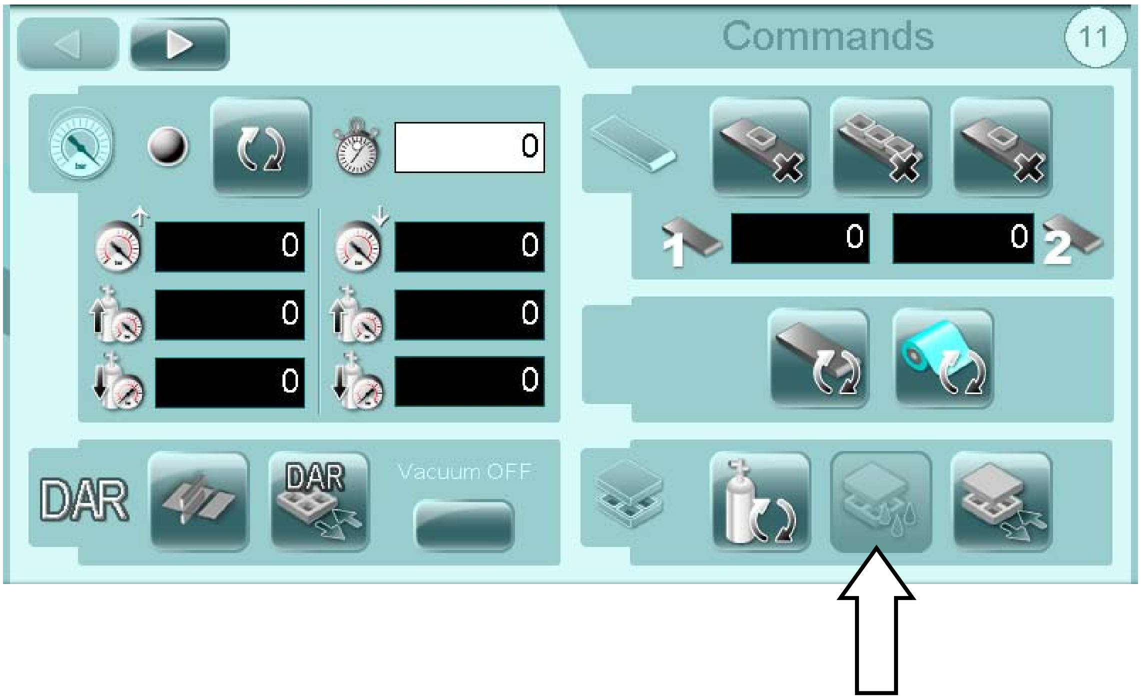
- It is strictly prohibited to wash the photocells, the reflectors and uncovered motors using high-pressure water jets.
- It is strictly prohibited to wash the photocells, the reflectors and uncovered motors using solvents or thinners. The chemical composition of said products could damage the plastic photocell eyes thereby compromising their operation.
- To prevent damages to the relative electronic systems, it is strongly recommended NOT to wash the photocells using hot water. This is very important especially after the machine has been powered off because the photocell is still hot.



2. Check to make sure that all the machine's safety coverings are firmly closed.

3. On the operator panel's "Main Menu" page, access the "Commands" page.

4. From the "Commands" page, activate the ""wash cycle" function by switching the relative selector to "enable".



5. After having checked again that all containers and trays as well as the film bobbins have been correctly removed, press on the “OK” pushbutton key.

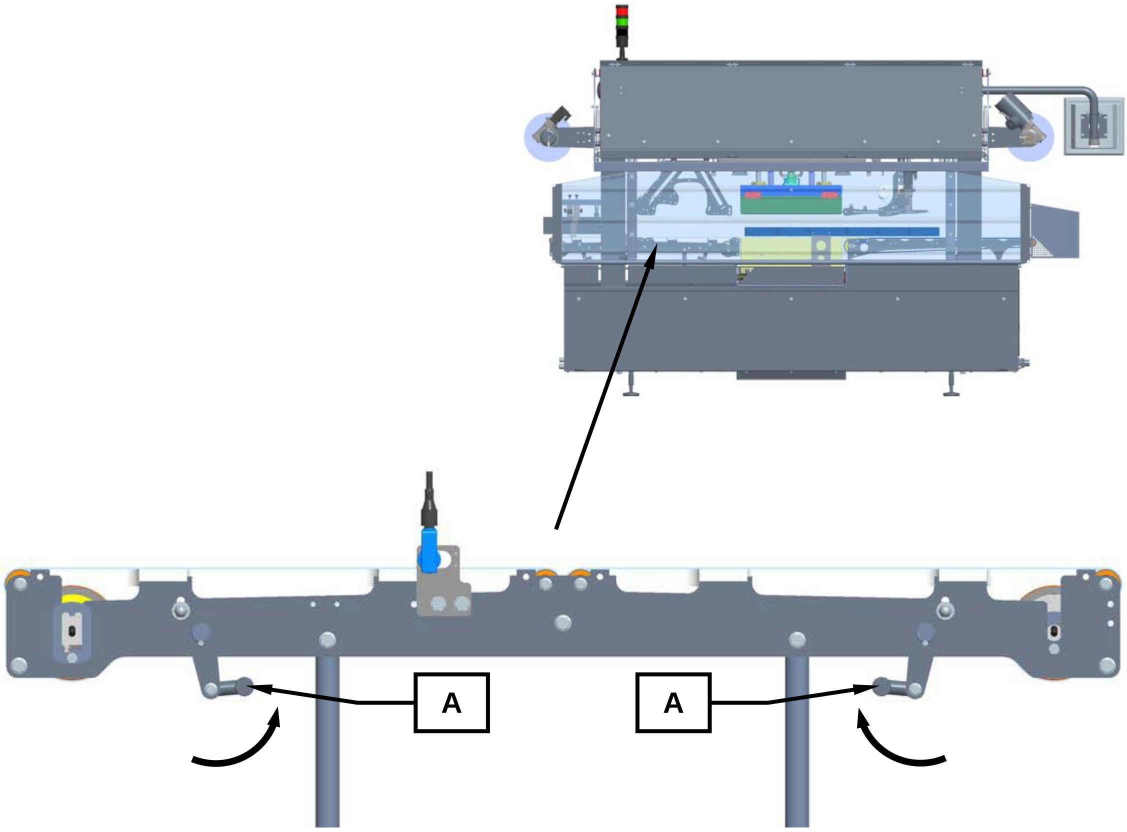
6. Once the “washing cycle” has been enabled and activated, the tool will close up to prevent water and detergents possibly entering the bell frames.

7. Open up the coverings and start the machine washing operations.

WARNING! We strongly recommend using neutral Ph detergents and foams, that combine disinfectant and sanitising properties with grease-removal properties.

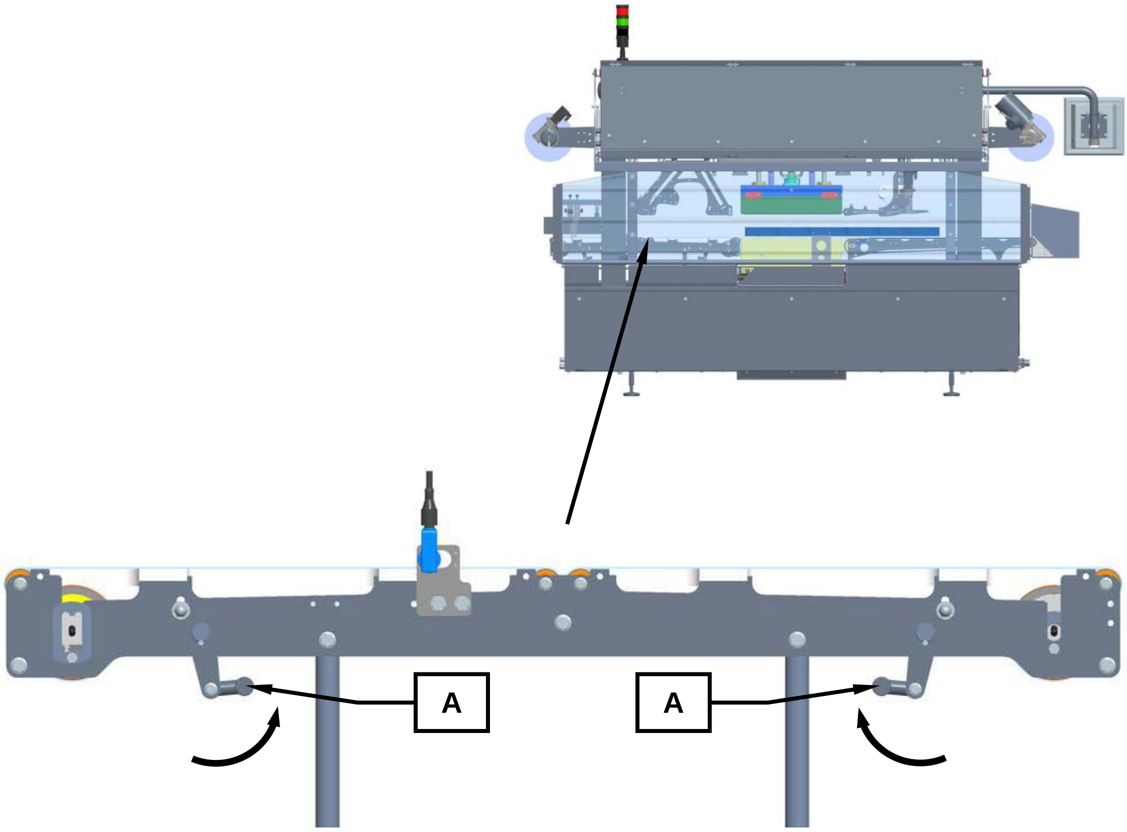
6.3.3 Unlocking the short infeed belts

1. Use the appropriate lever (**A**) to unlock the short infeed belts.

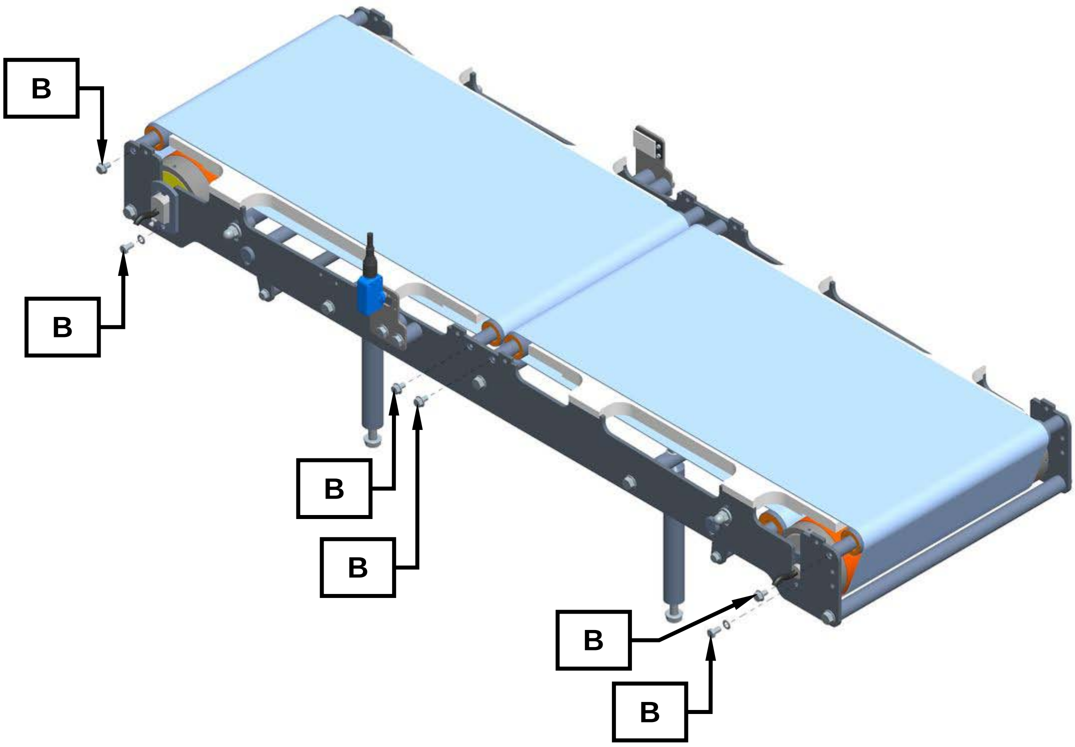


6.3.4 Removal of the spacer belt (only in the event of sterilisation)

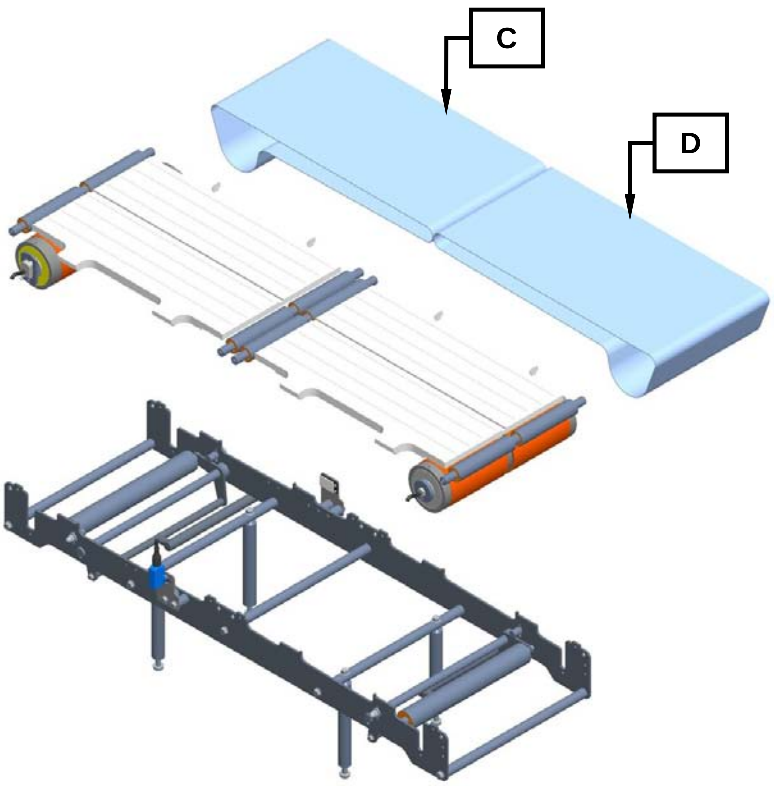
1. Use the appropriate lever (**A**) to unlock the spacer belt.



2. Unscrew the screws (**B**) an remove the belts groups.

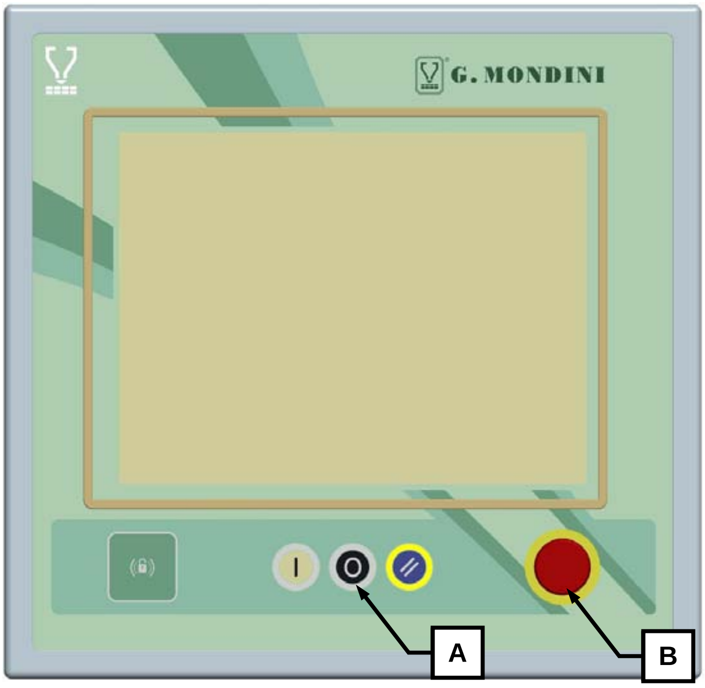


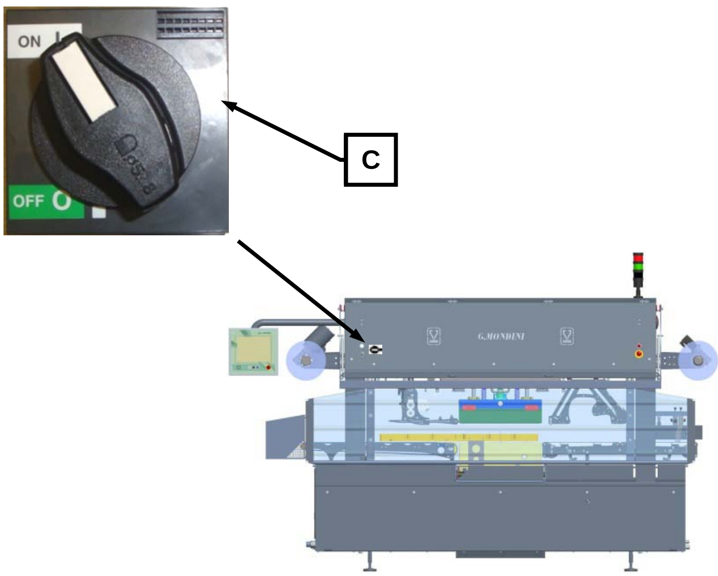
3. Remove the belt (**C**) and (**D**).



6.4 MACHINE CYCLE STOPS

1. To stop the machine work cycle at the end of production it is necessary to press the red “STOP PROGRAM CYCLE” pushbutton (**A**), thereby allowing all the machine elements to travel back into cycle start position.
2. Carry out all the operations involved with “production program end” then switch off the machine by first pressing the red EMERGENCY pushbutton (**B**) then proceed with cutting off the machine’s power supply by positioning the main switch (**C**) into position 0.



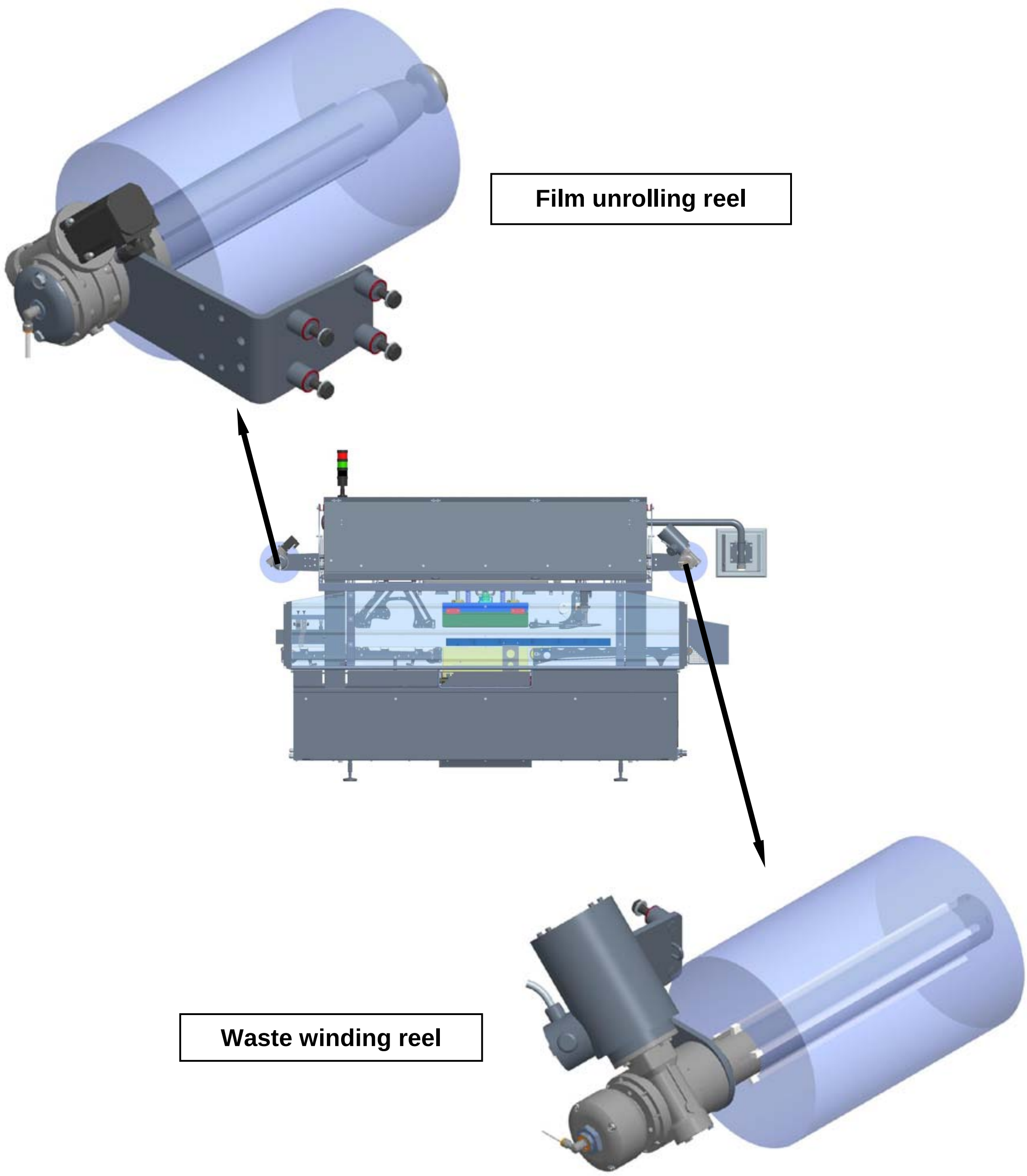


6.5 TRAY FORMAT CHANGES

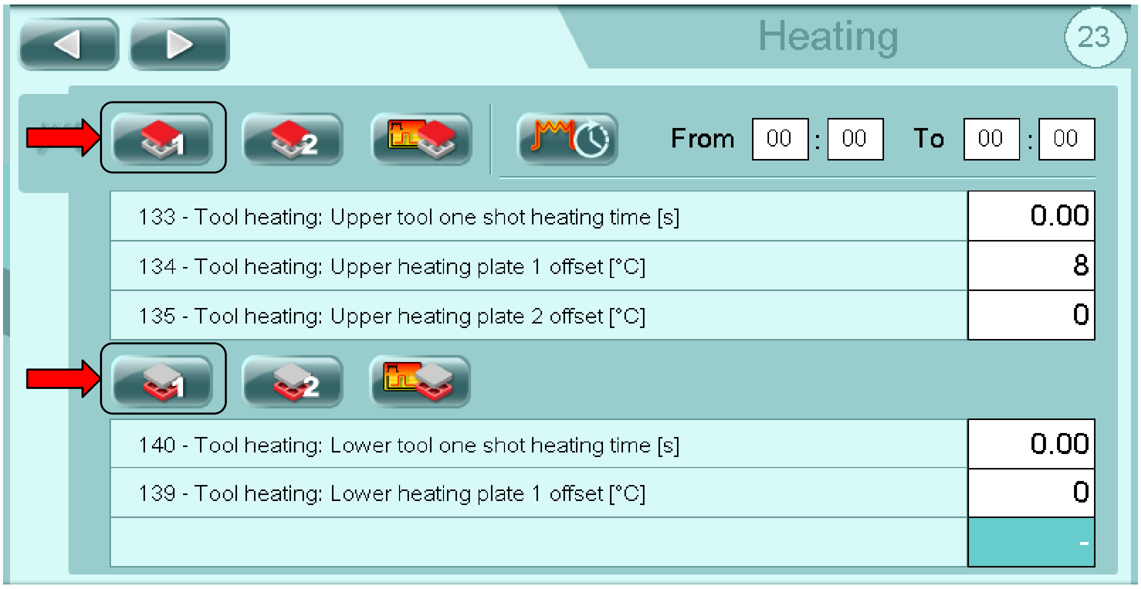
To carry out tray format changes, proceed as follows:

6.5.1 Extraction/removal of the tool

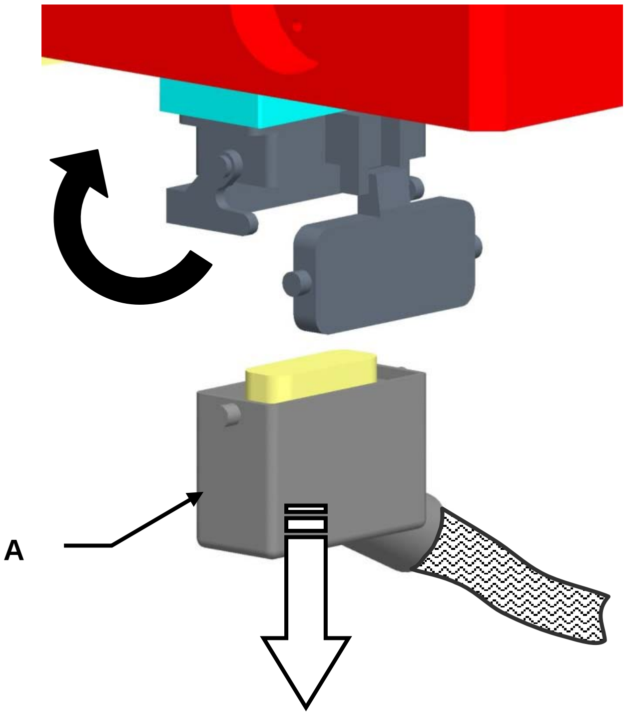
1. Tear off the film at tool level and gather one end onto the film unrolling reel and one end onto the film waste collector shaft. If necessary, extract to remove the two reels.



2. Disable operation of the resistance of the tool from page 23 of the operator panel.

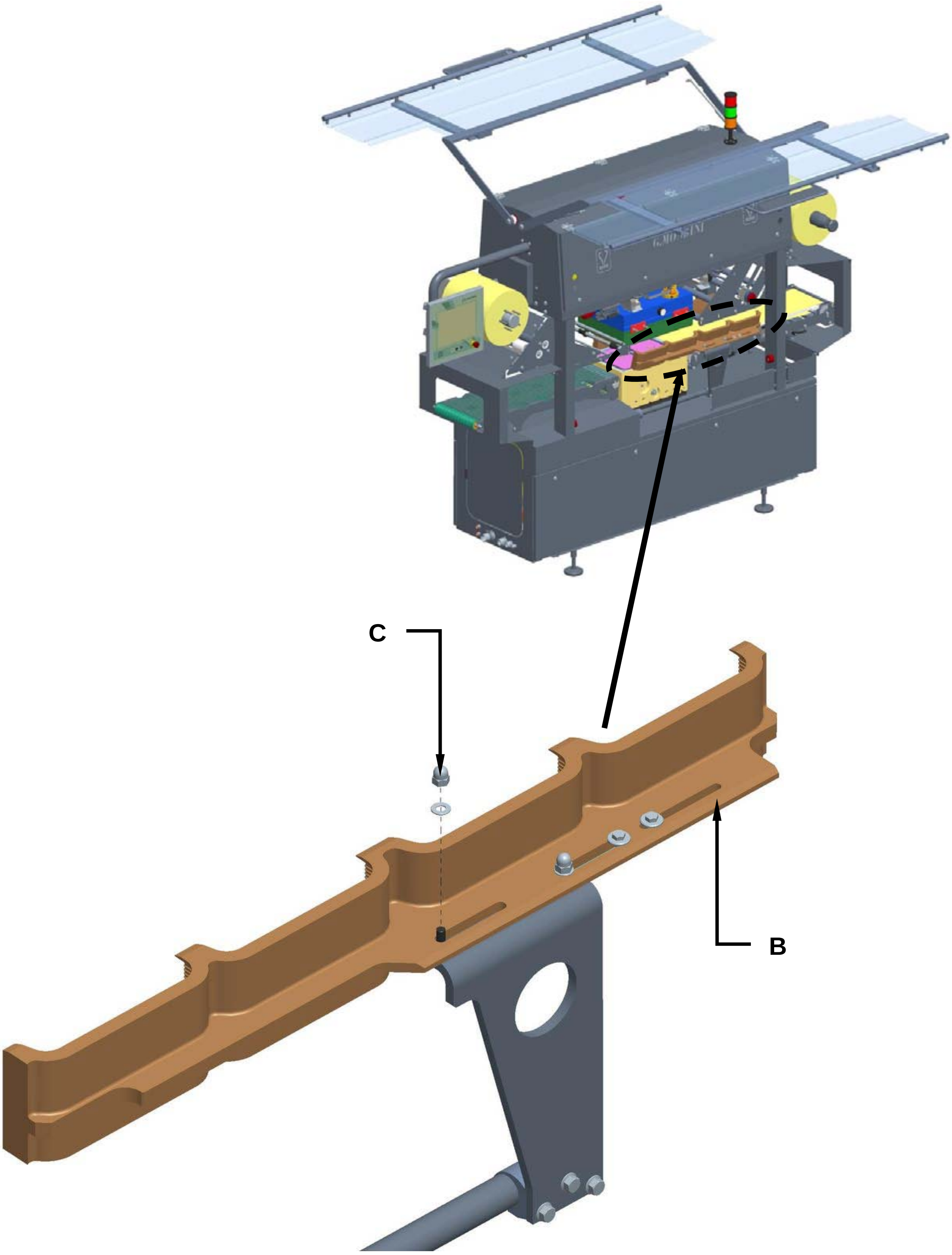


3. Disconnect the tool's electrical connections by disconnecting the HARTING socket (**A**) as illustrated.

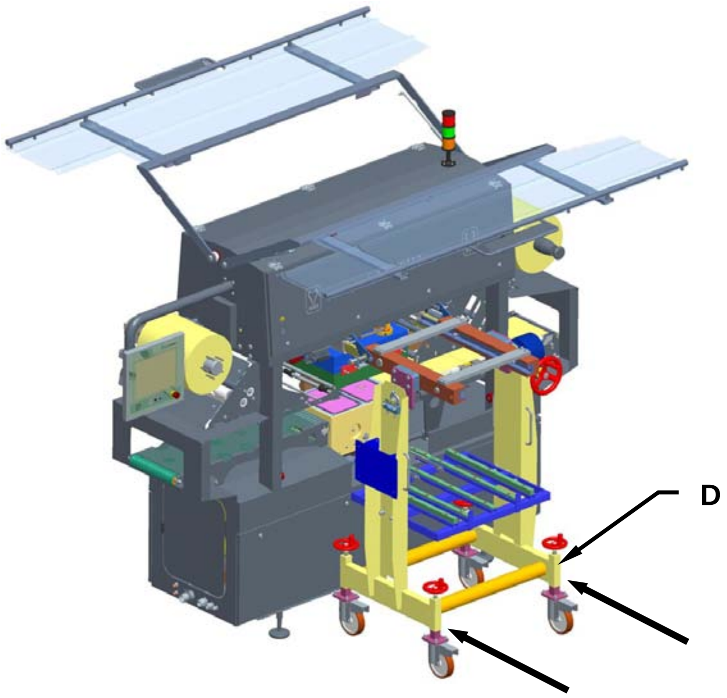


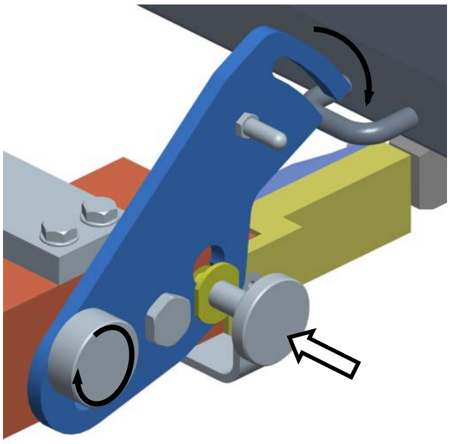
⚠ WARNING!	Before disconnecting the plug HARTING you must disable the operation of the tool resistances!

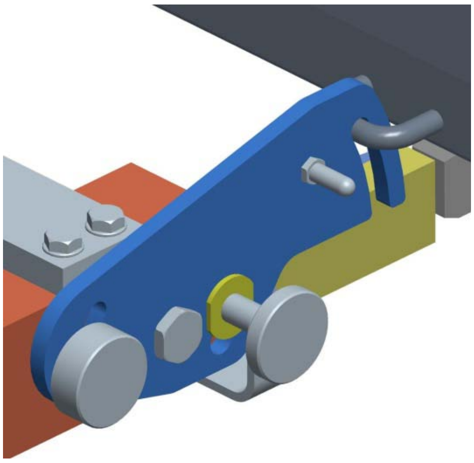
4. Remove the two left and right pusher units (**B**), by acting on the two lock nuts (**C**)



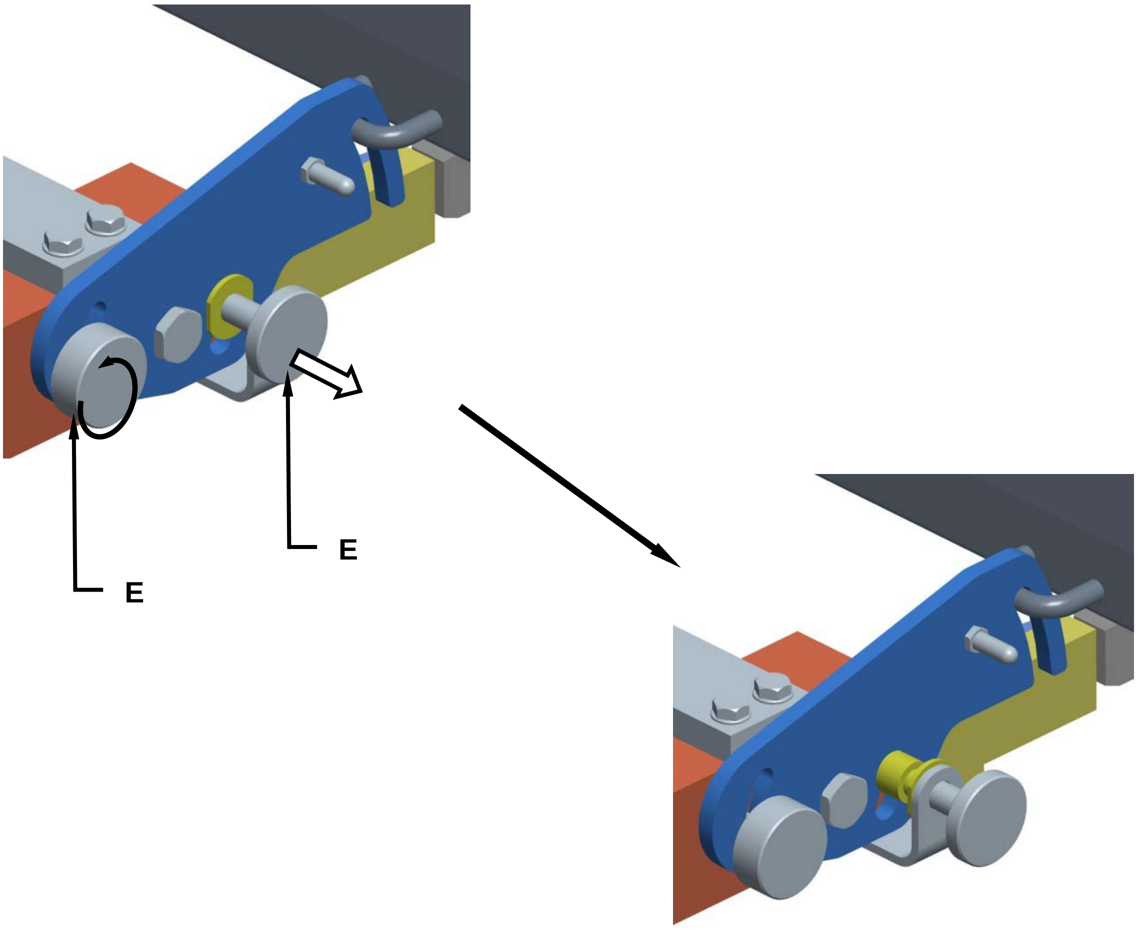
5. Draw the tool removal trolley (**D**) up alongside the machine and hook it up to the machine.





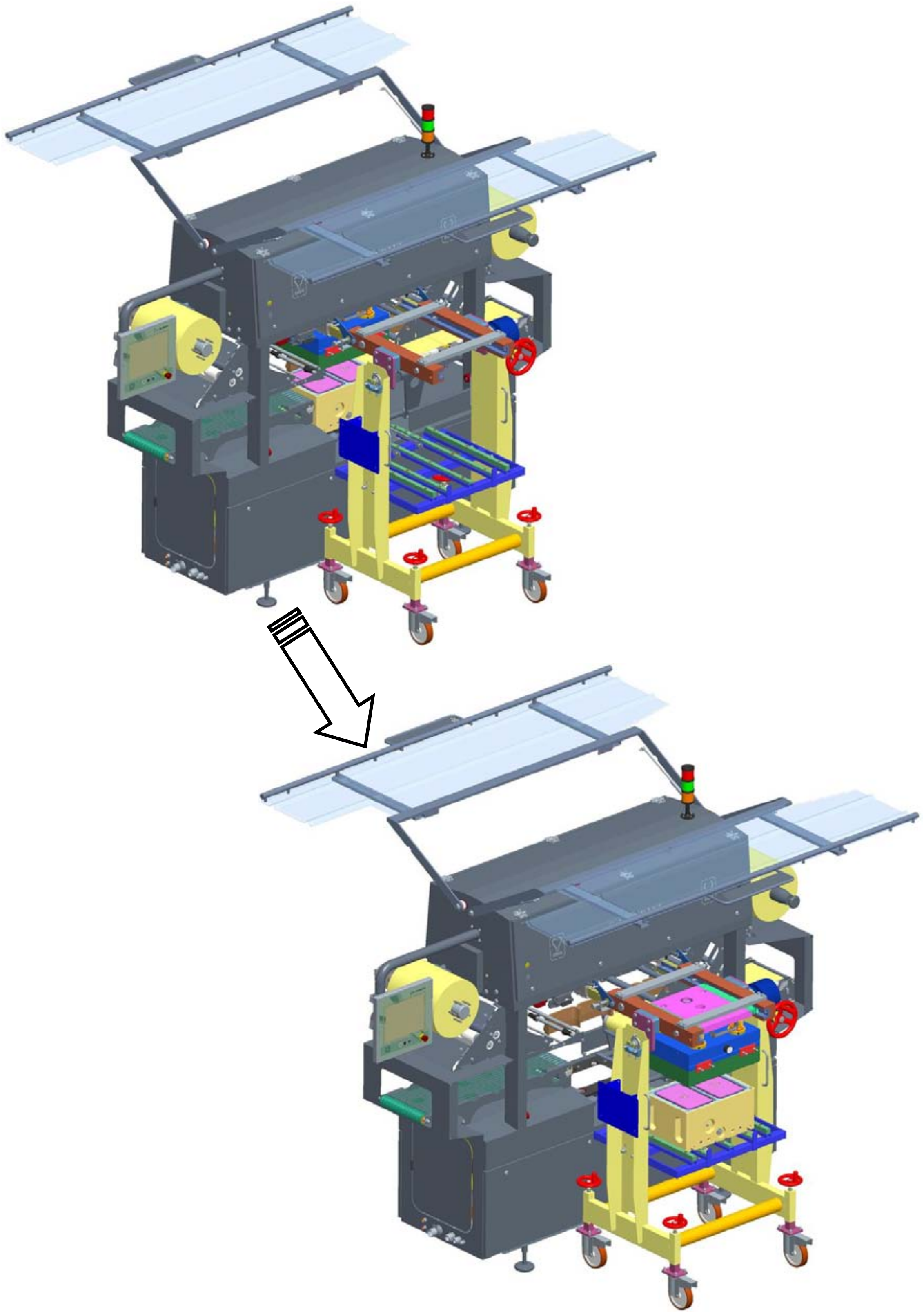


6. Turn out the two lock screws (**E**).

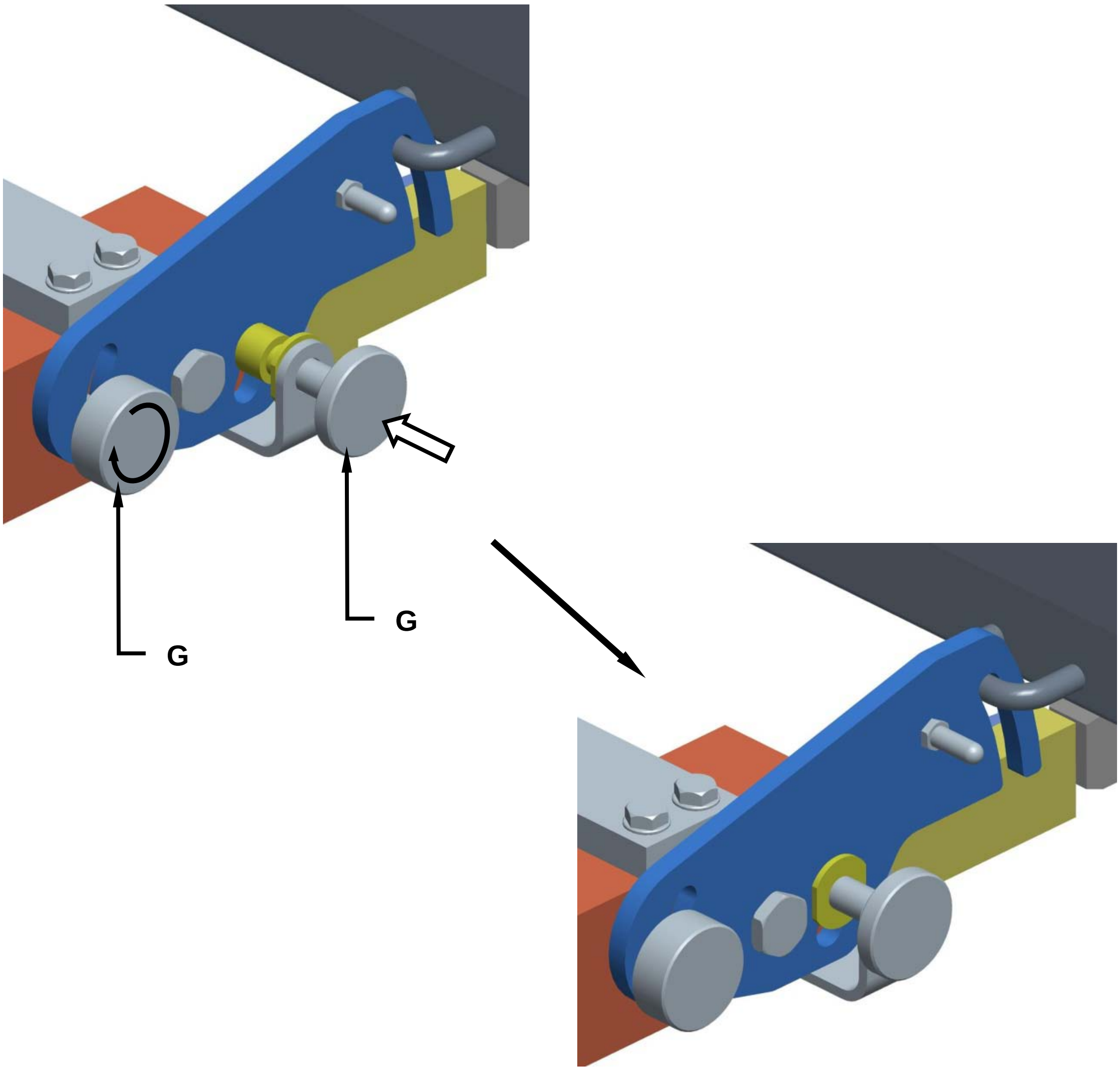


7. Unlock the tool by acting on the “TOOL LOCK” selector switch (**F**).

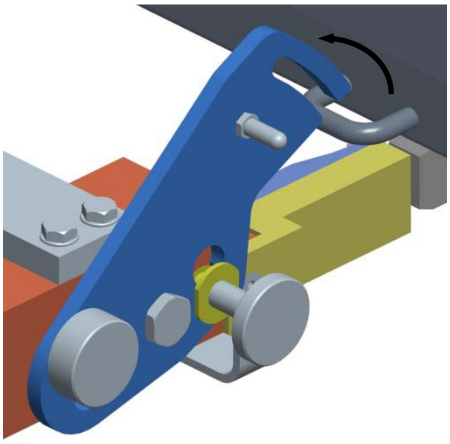
8. Extract the two tool ends using the appropriate handles (both the bottom and top tool edges).

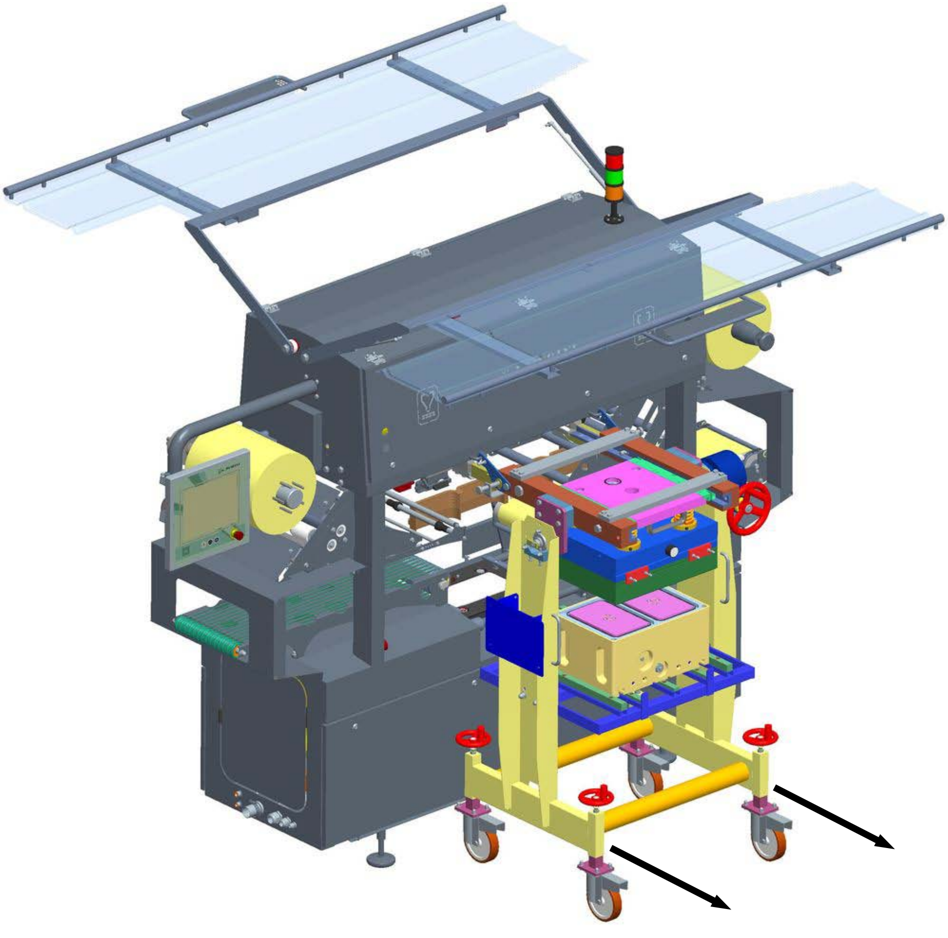


9. Screw in the two lock screws (**G**).



10. Unhook the tool removal trolley from the machine and move it away whilst checking to ensure that the trolley lock pins are locked in correctly.

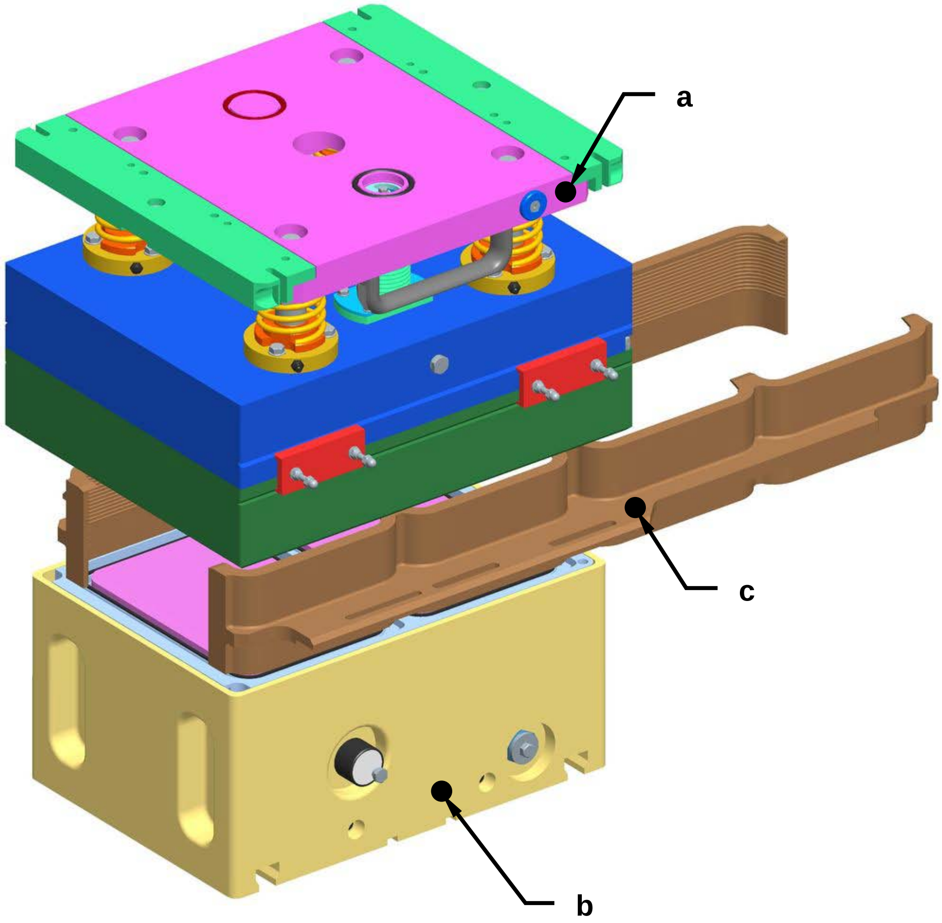




6.5.2 Tool insertion

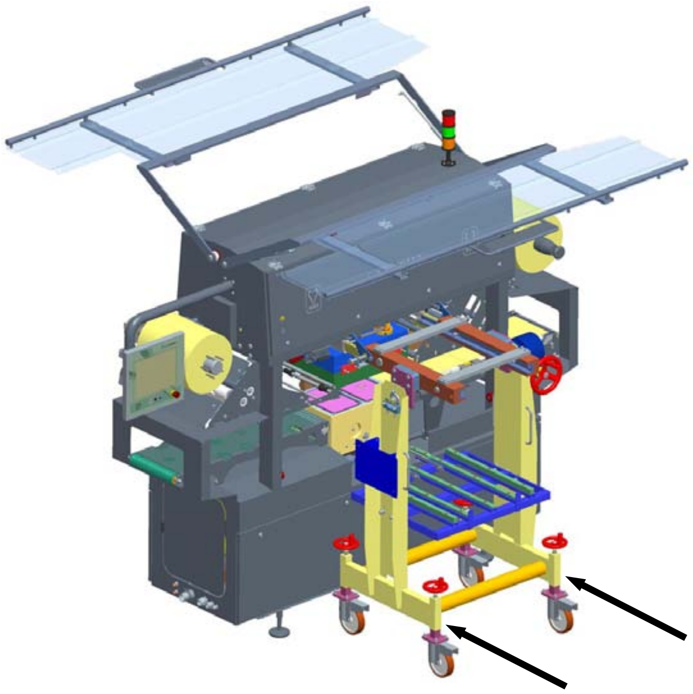
For tool insertion into the sealing machine proceed as follows:

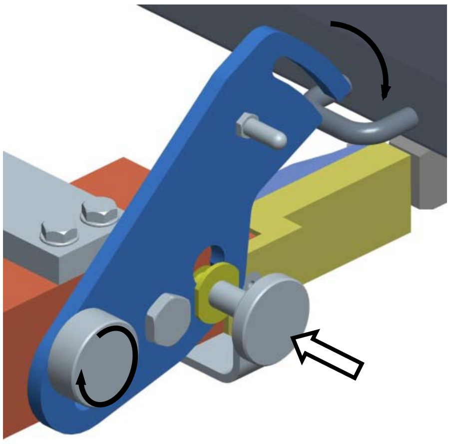
1. Check to see that all tool details and components are of the same type (i.e. the same colour or the same number).

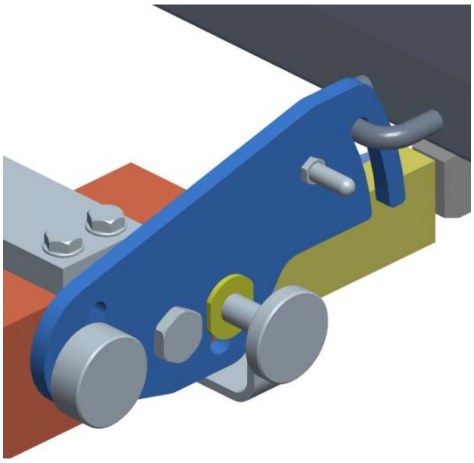


a. Top tool half
b. Bottom tool half
c. Right and left pusher units

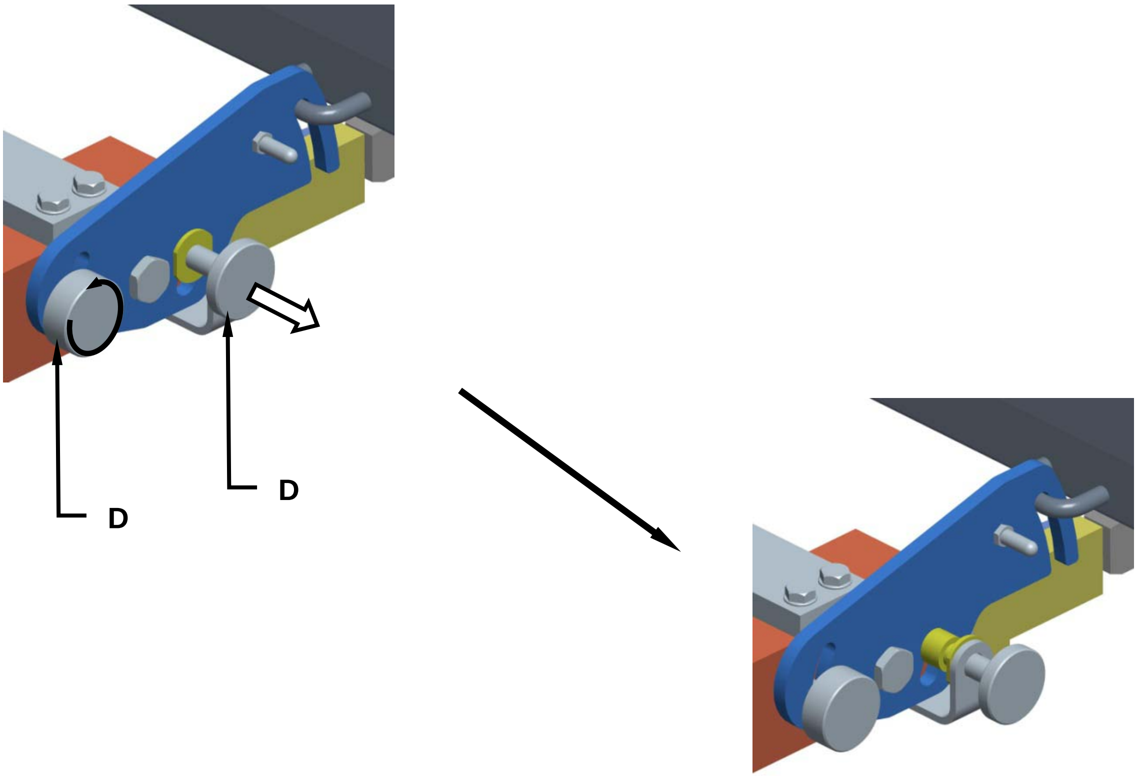
2. Position the tool removal trolley up next to the sealing machine and hook it on.



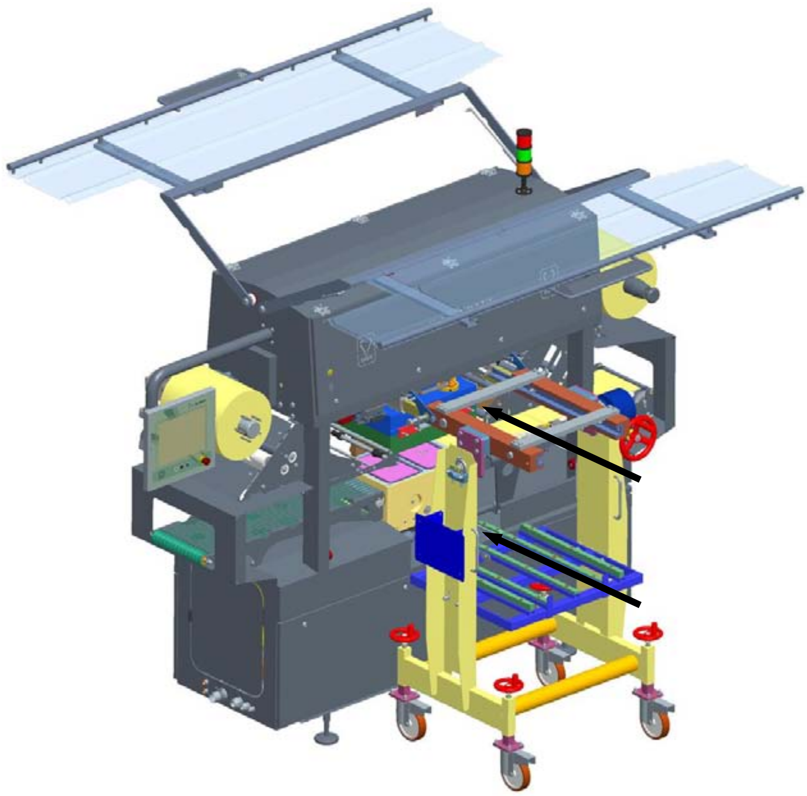




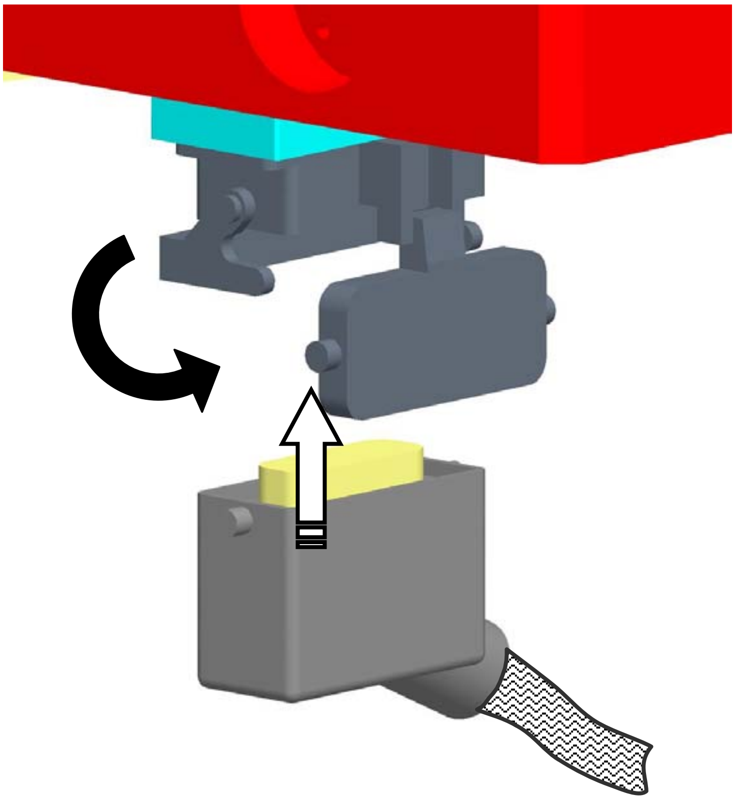
3. Remove the two lock screws (**D**).



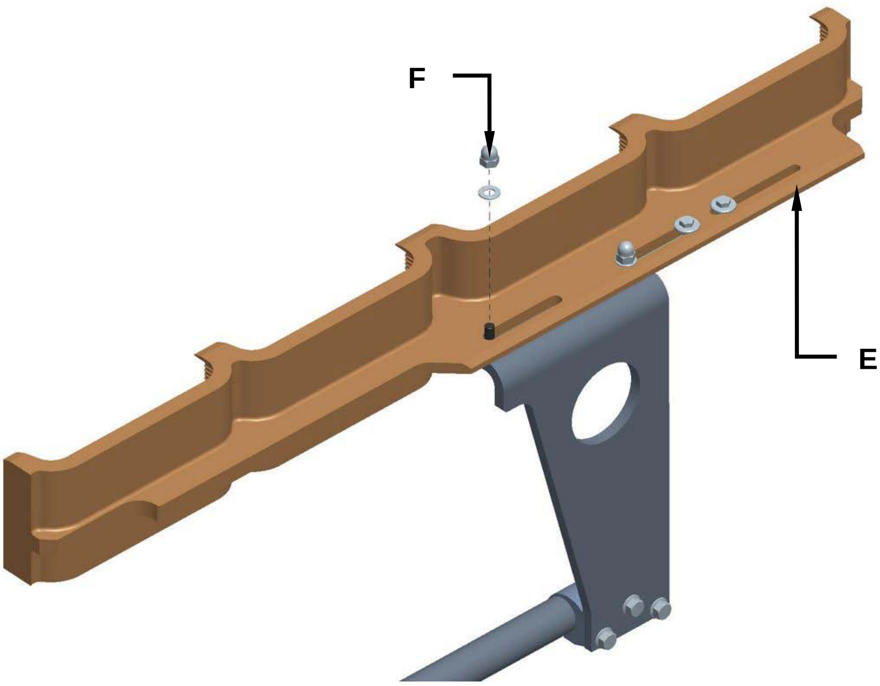
4. Insert the two half tools using the appropriate handles (both the top and the bottom halves) and ensure they are accurately housed.



5. Connect the tool HARTING socket up to the connector on the machine.

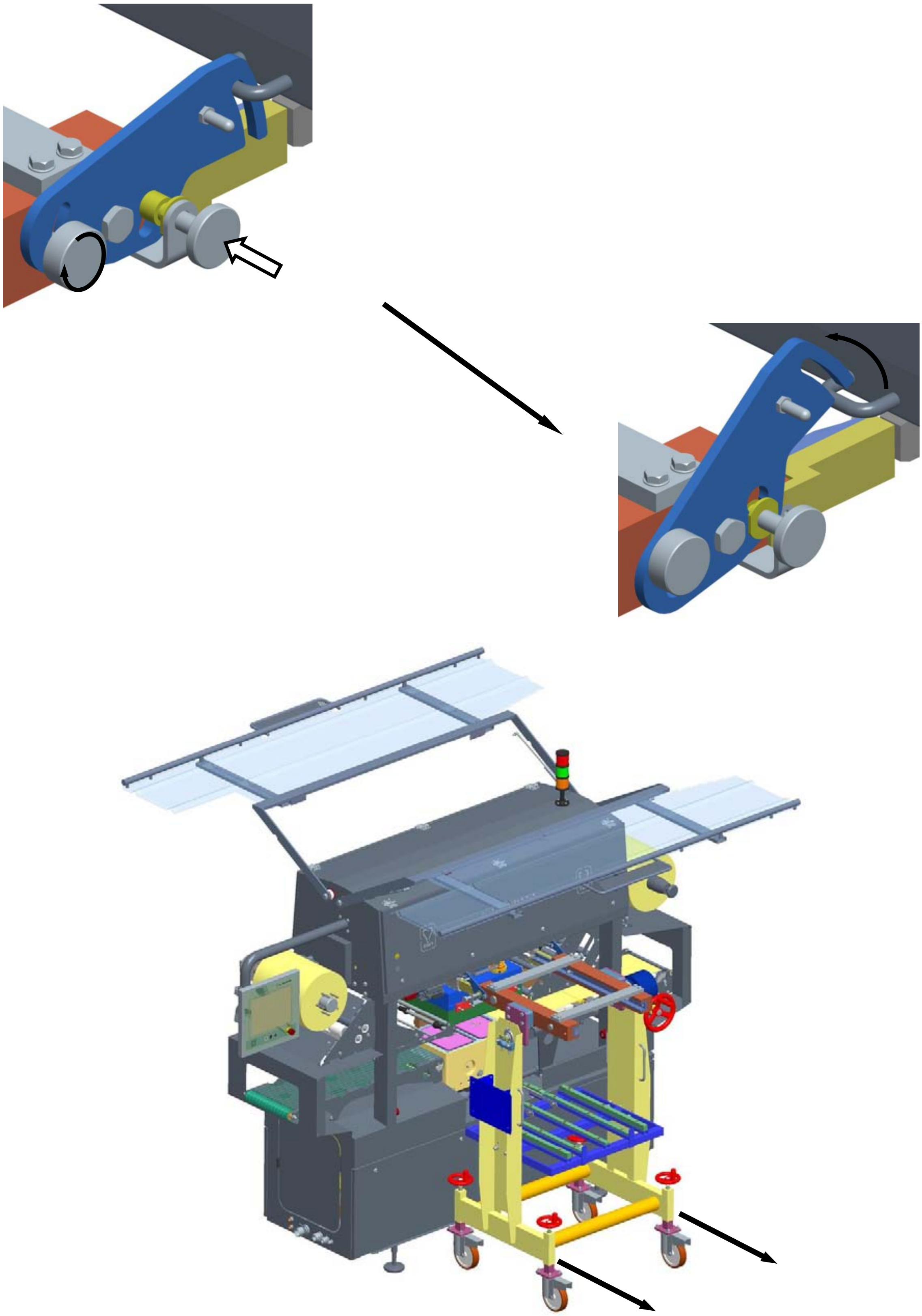


6. Assemble on both the right and the left pusher arms (**E**) and, after ensuring that they are perfectly housed in position, lock them on using the relative lock nuts (**F**).

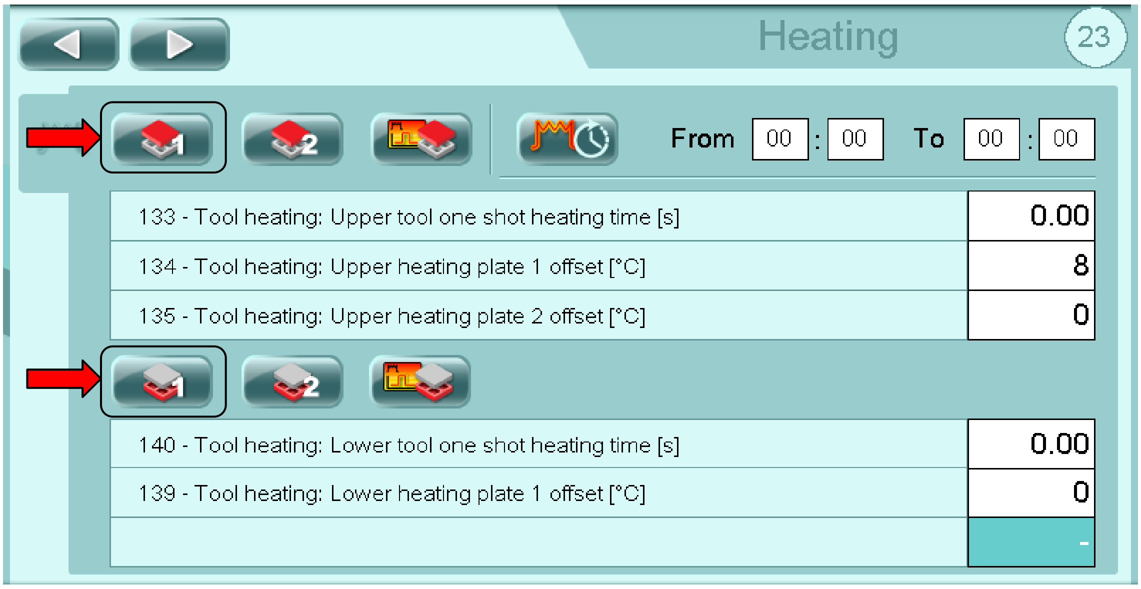


7. Unlock the tool by acting on the "TOOL LOCK" selector switch (**G**).

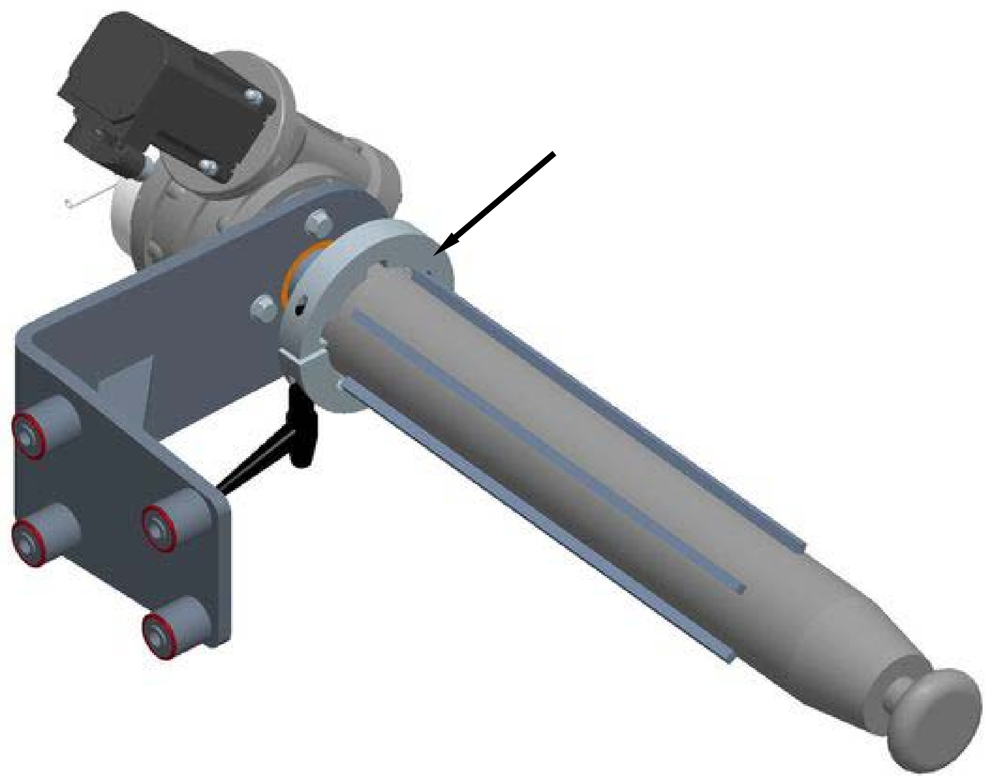
8. Unhook the tool removal trolley from the sealing machine and remove it.



9. Enable the operation of the resistance of the tool from page 23 of the operator panel.

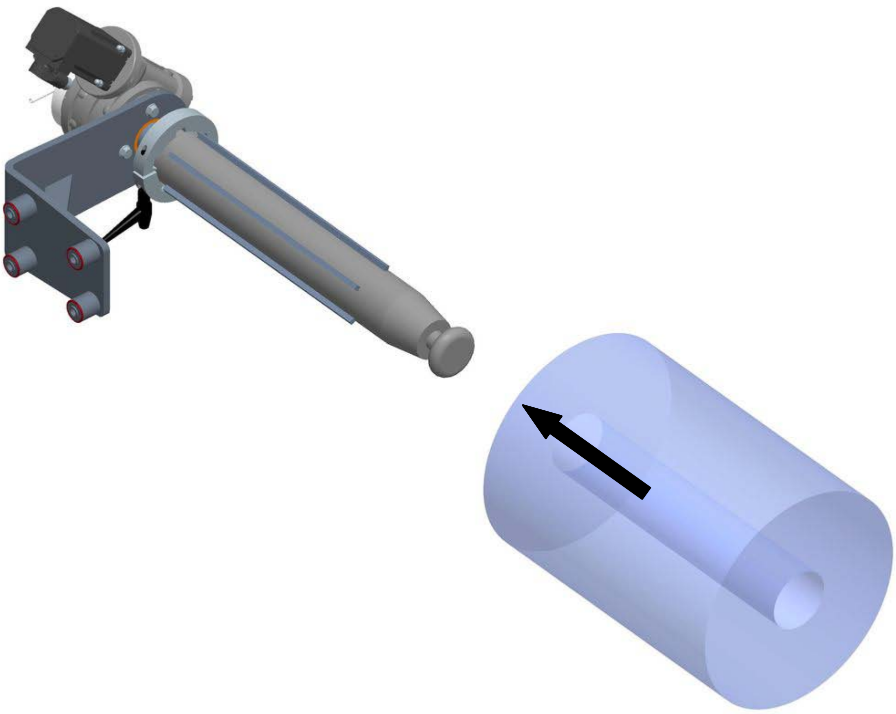


10.Insert spacers on the film unwinder bobbin, as the case may be.

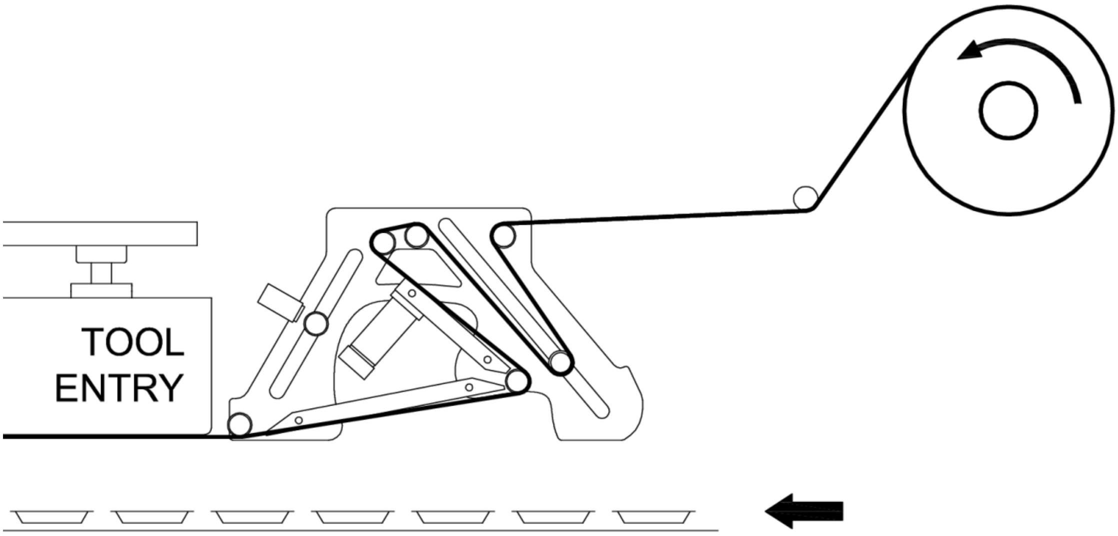


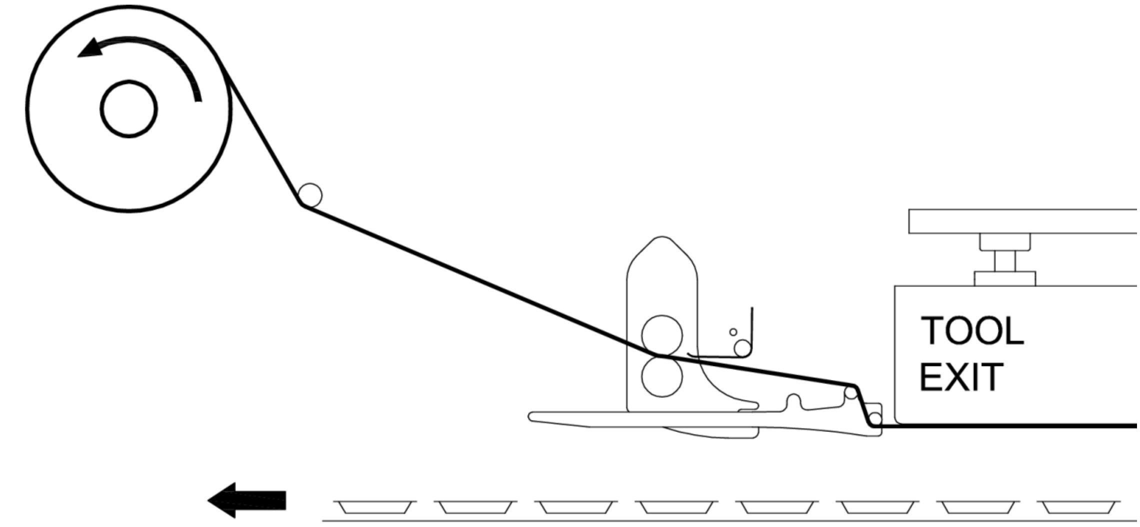
11.Check that the system locking functions on the Lid Unrolling, on the Waste Winding and on the Film Reel are all deactivated by acting on the relative selector keys (**H**), (**L**) and (**M**).

12.Fit on the sealing film reel.

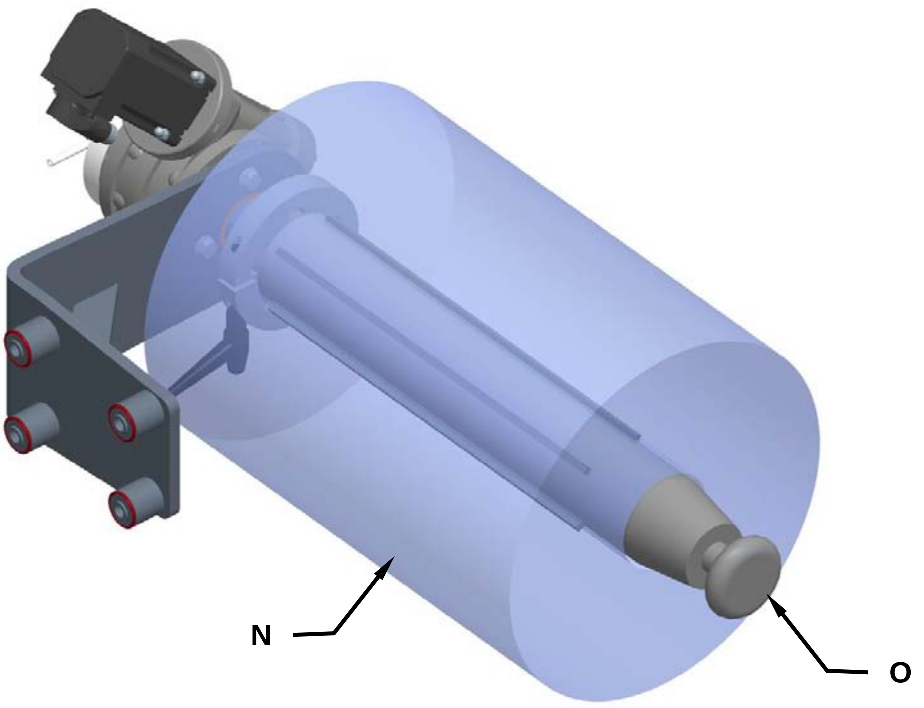


13. Thread the film in through the bobbin unwinder rollers as per the directions indicated by the arrows provided near each one of the rollers.

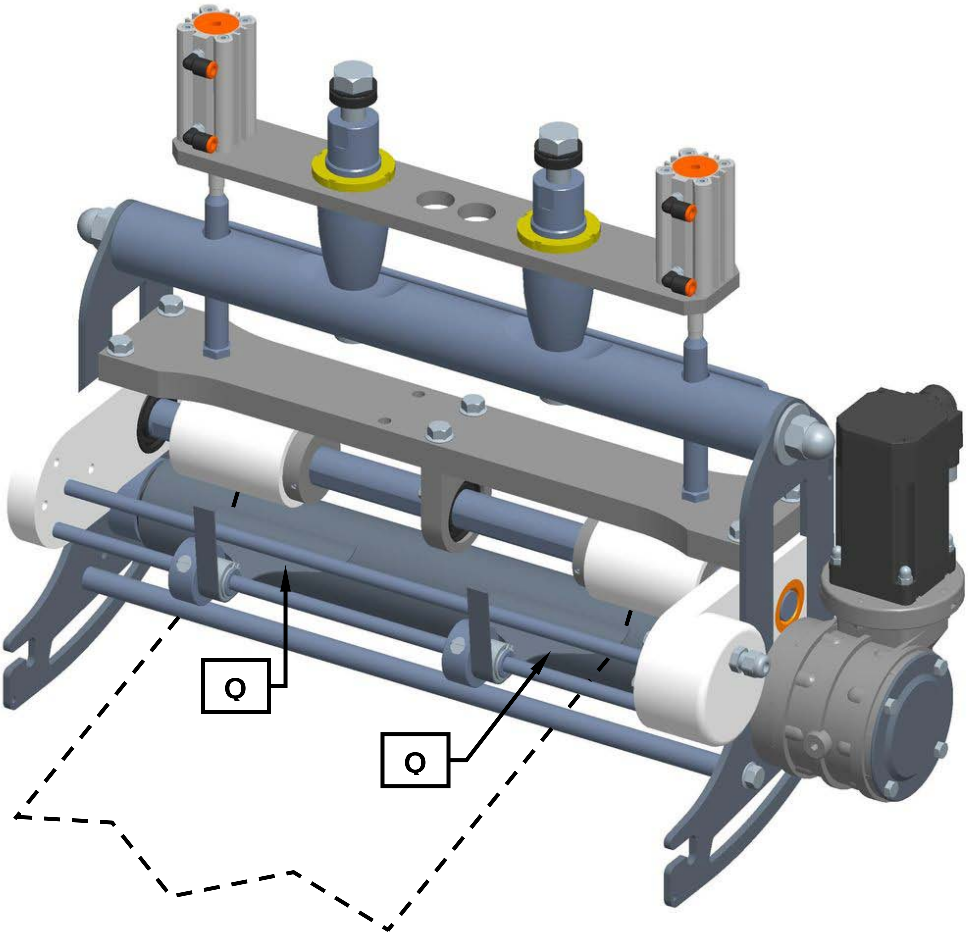




14. Check to ensure that the bobbin (**N**) is correctly centred on the film unwinder system shaft (**O**) and lock it in place using the pushbutton (**P**).

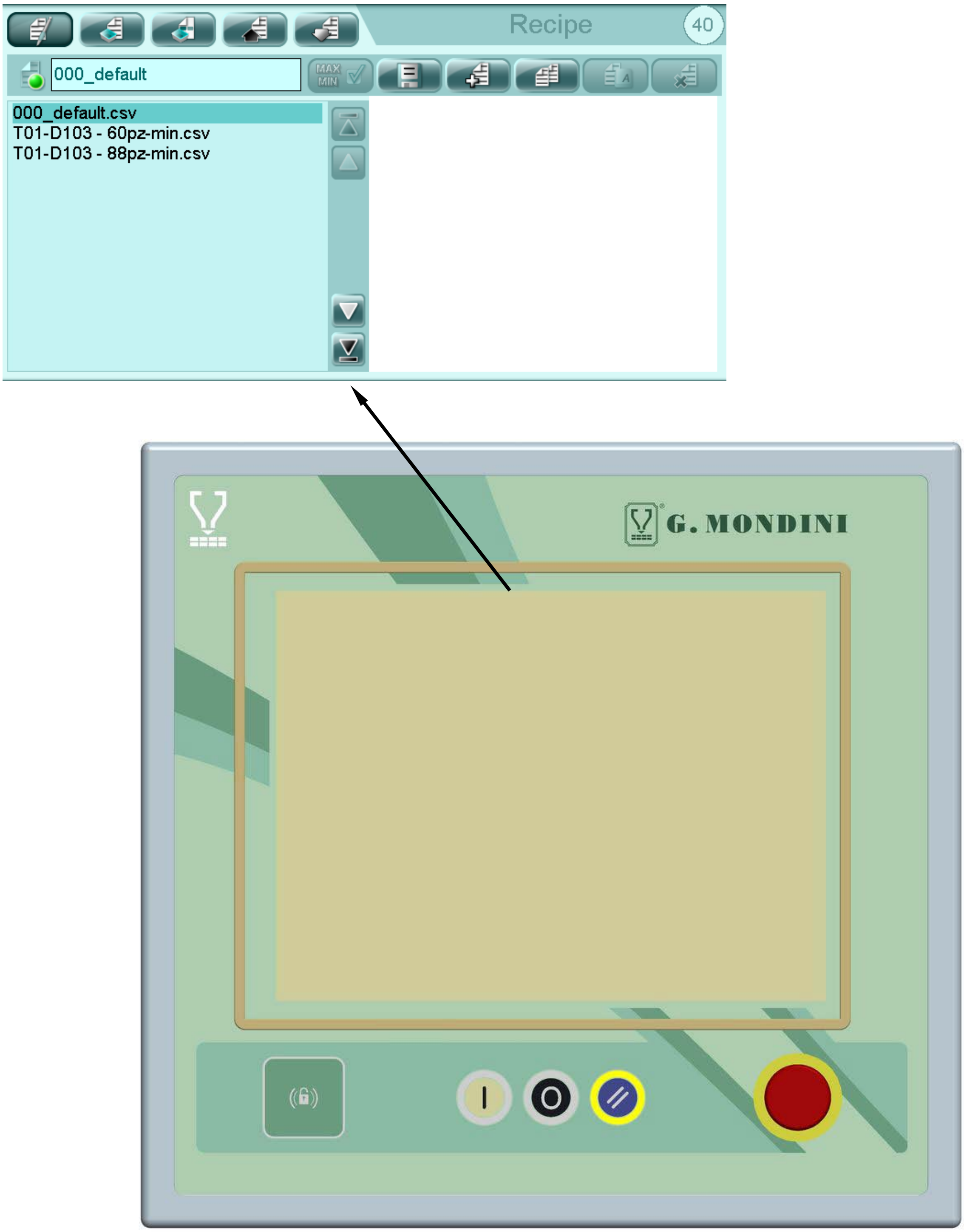


15. Check to see that the sealing film waste blade plates (**Q**) are positioned on the relative outer edges. If not, move them so that they are positioned correctly as shown.



16. Push the button (**R**) to block the film unrolling pullers and wind the waste onto the waste winding shaft. To lock the waste push the button (**S**).

17.Enter the recipe relative to the tool you have just inserted.



6.6 TOOL CENTERING

TIME REQUIRED:	60/90 min.
EQUIPMENTS:	Workshop equipment.

WARNING!

The clothing of those who run the line or carry out maintenance on it is to conform the safety requirements specified in the EU directive 89/656/EEC and the laws in force in the country where the line is installed.

DANGER!

To avoid mechanical risks, such as dragging, trapping and others, do not wear bracelets, wristwatches, rings or chains during operations in processing cycles and maintenance.

It's forbidden more than one operator running on the machine at the same time.

It's obligatory to operate in the presence of a supervisor, who monitors the work phases and intervenes in danger case.

WARNING! Danger of cutting, abrasion and burns.

It's obligatory the use of appropriate personal protective equipment, such as gloves for high temperatures and resistant to cuts. These must be confortable to be able to wear them during all work phases.

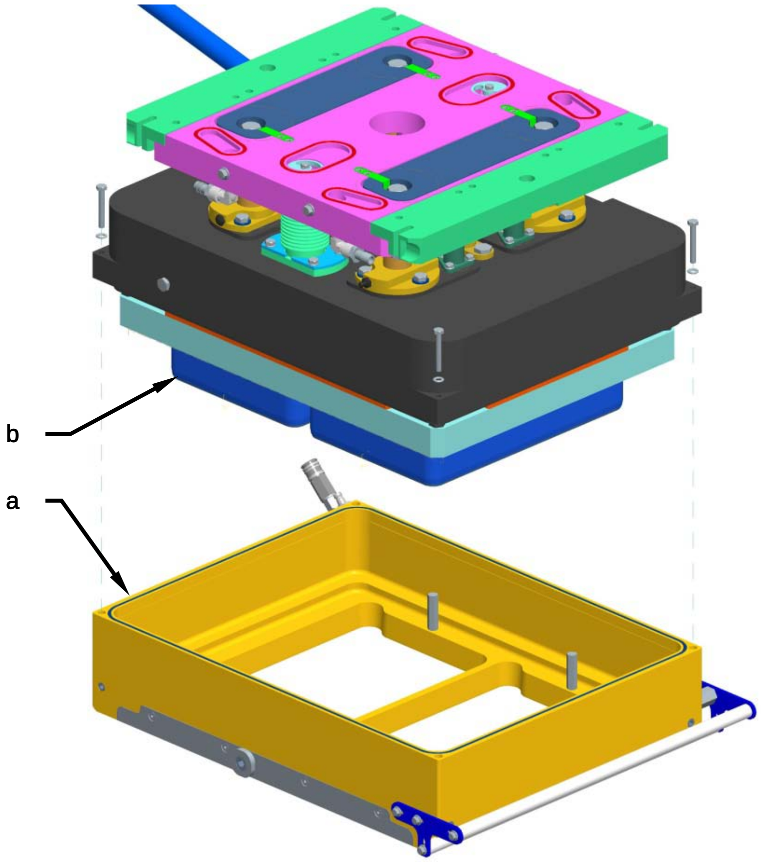
WARNING! The maintainer, before performing these operations must be trained and informed, he have to also make sure of any "off-center" between the bells and note these differences, they are to be considered with the measures to note down for the centering.

WARNING!

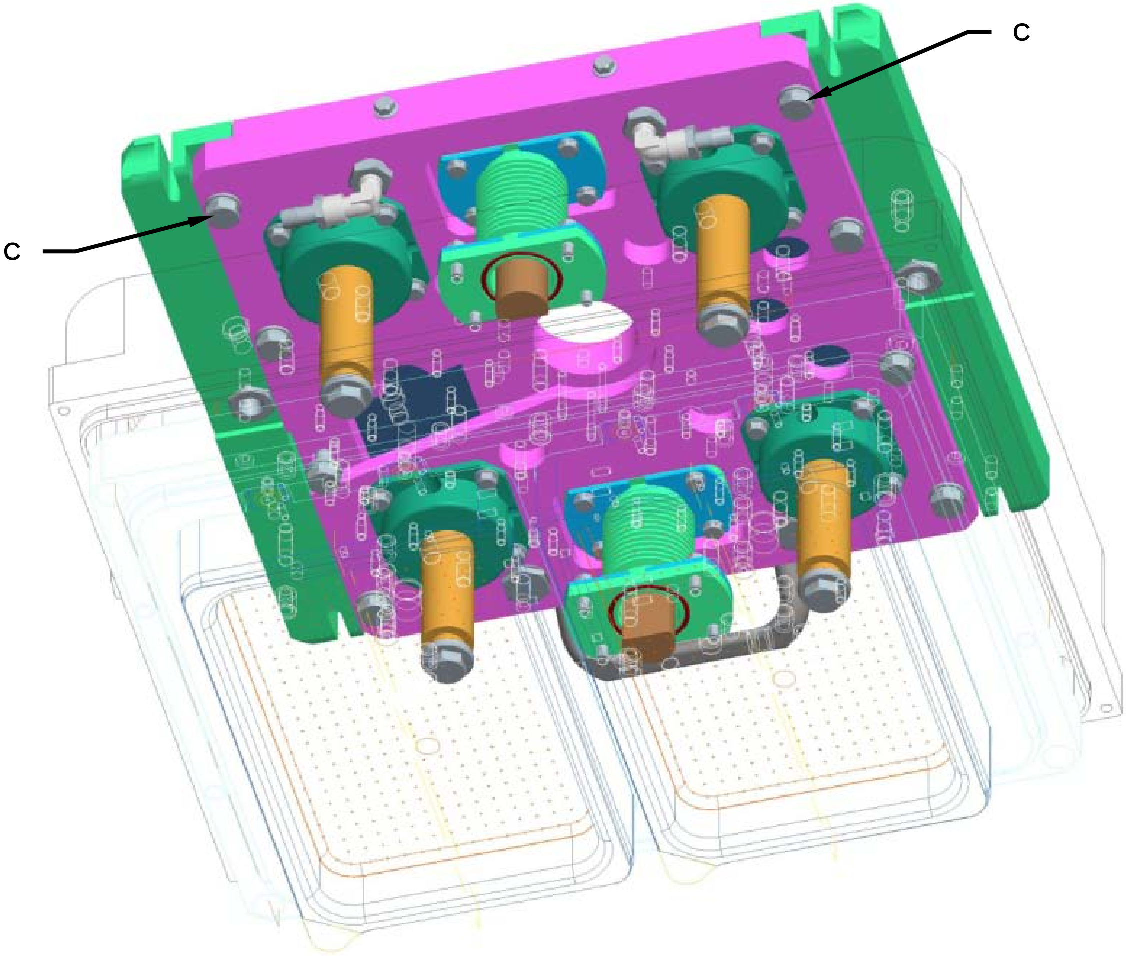
G. Mondini S.p.A. is not responsible for any damage caused by untrained and uninformed maintenance personnel.

For a correct centering operation scrupulously perform the following operations:

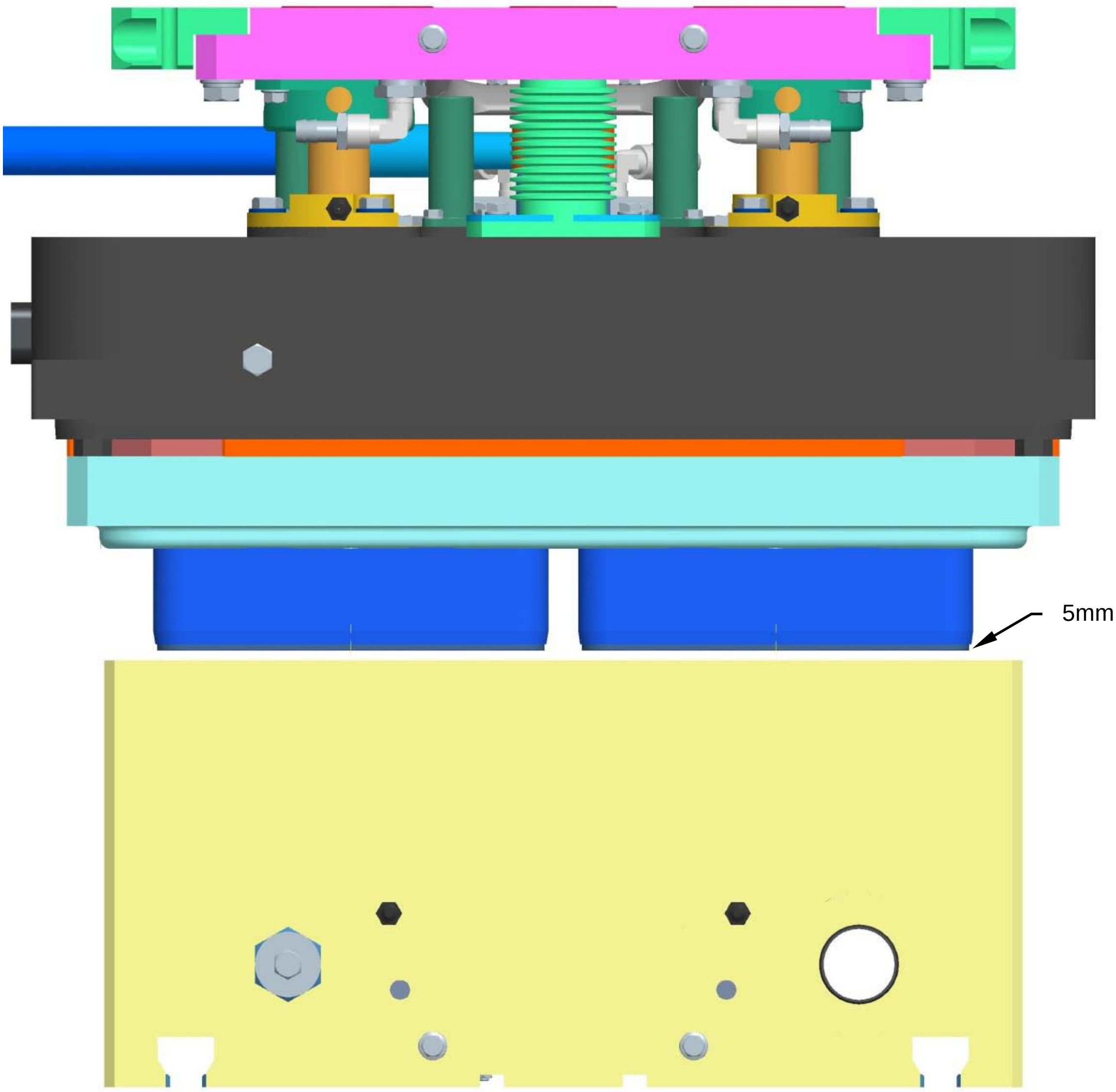
1. Lock the tool and heat up the soldering resistance to the operating temperature.
2. Remove the middle bell (**A**) and the blades (**B**) from the upper tool.



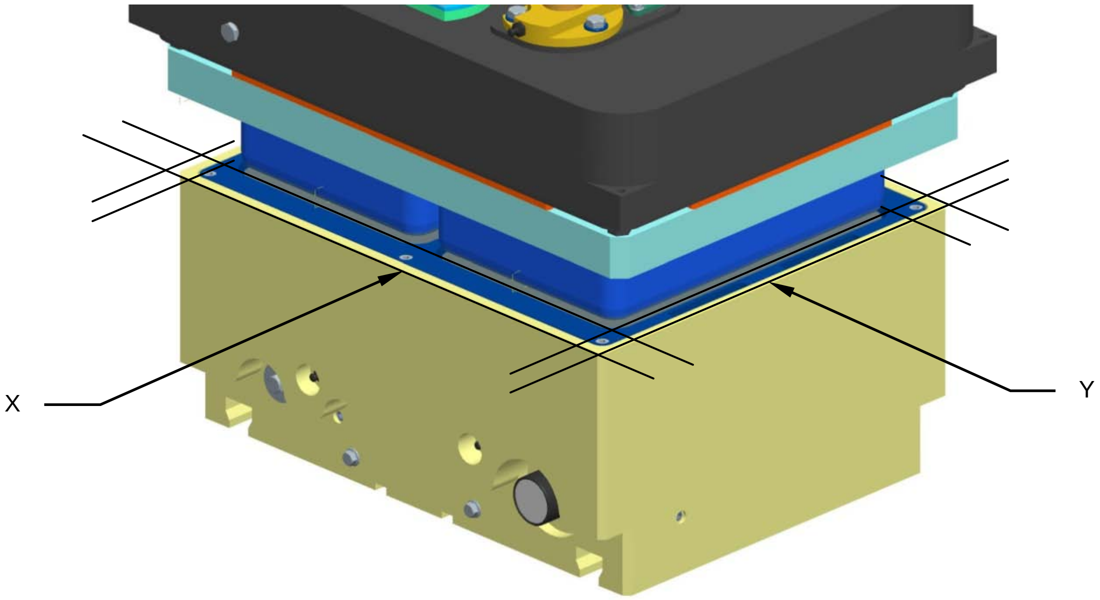
3. Loosen the screws (**C**) of the brackets.



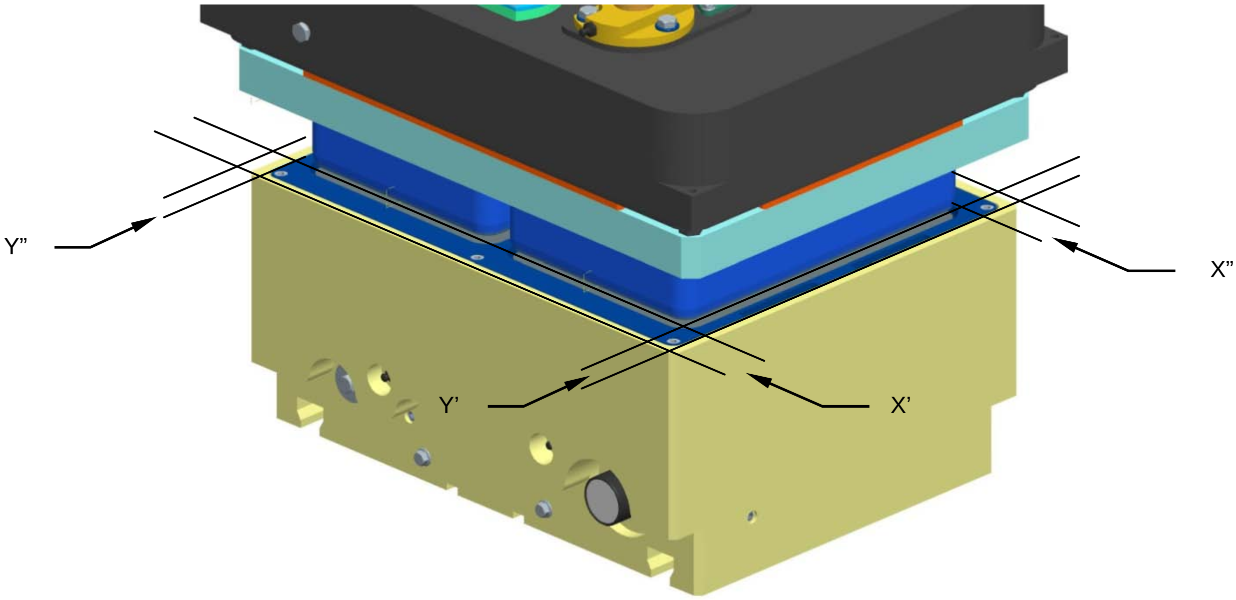
4. In "manual cycle", move the lower tool very slowly, up to a distance of about 5 mm.



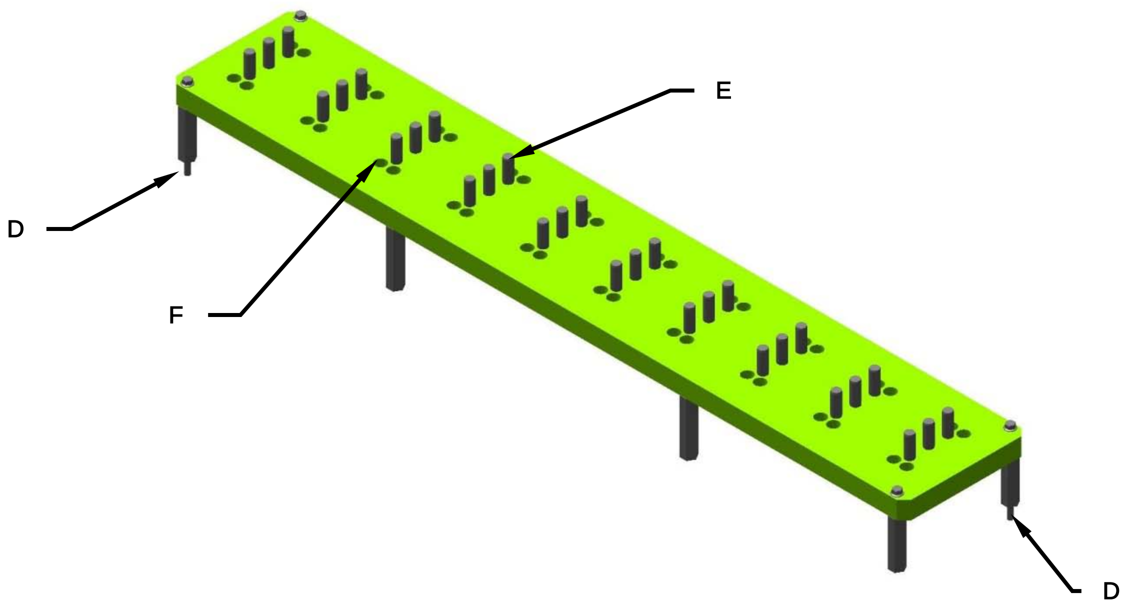
5. Measure with a caliber the distance between the outside of the lower bell and the plate of blades in the directions (**X**) and (**Y**).



6. Move the tool manually until the same dimensions are detected in all directions (**X'=X"; Y'=Y"**).



7. Lock the screws (**C**) of the brackets.
8. Lower the lower tool.
9. Loosen the fixing screws of the solders.
10. Insert the centering plate, on the counter-tool, centering with the appropriate reference pins (**D**).



11. Heat up the soldering resistance to the operating temperature.
12. Move the lower tool very slowly, until the centering pins (**E**) come into contact with the welding elements, centering each other.
13. Through the appropriate holes (**F**), tighten the fixing screws of the welding elements.
14. Retract with the lower tool and remove the centering plate.
15. Reposition the blades (**B**) and the middle bell (**A**).
16. Restore the "automatic cycle".

WARNING! The operations described above must be observed scrupulously, operating with the utmost care and complying with all the safety regulations.

7 DIAGNOSTICS

Malfunctions may occur the sealing cycle that prevent correct operation of the machine.

These malfunctions, referred to as ALARMS, are displayed promptly by the system on synoptic tables showing alarm messages and possible solutions.

ALARM	SOLUTION
0001 - Safety_Emergency: Main safety unit to be rearmed	- Check the safety control unit.
0002 - Safety_Emergency: E-STOP pressed on Operator Panel	- Release the emergency button by turning clockwise. - Reset the alarm by pressing the memory reset button. - Check operation of the safety relay inside the main control panel.
0003 - Safety_Emergency: E-STOP pressed on film reel unwind	- Release the emergency button by turning clockwise. - Reset the alarm by pressing the memory reset button. - Check operation of the safety relay inside the main control panel.
0004 - Safety_Emergency: E-STOP pressed on line (begin)	- Release the emergency button by turning clockwise. - Reset the alarm by pressing the memory reset button. - Check operation of the safety relay inside the main control panel.
0005 - Safety_Emergency: E-STOP pressed on line (middle)	- Release the emergency button by turning clockwise. - Reset the alarm by pressing the memory reset button. - Check operation of the safety relay inside the main control panel.
0006 - Safety_Emergency: E-STOP pressed on line (end)	- Release the emergency button by turning clockwise. - Reset the alarm by pressing the memory reset button. - Check operation of the safety relay inside the main control panel.
0007 - Safety_Emergency: E-STOP pressed from Customer	- Release the emergency button by turning clockwise. - Reset the alarm by pressing the memory reset button. - Check operation of the safety relay inside the main control panel.
0008 - Safety_Guard: CVS protection guard inlet not operator OPEN	- Check that the guards are closed and the microswitches are active. - Reset the alarm by pressing the memory reset button. - Check operation of the safety relay inside the control panel.

0009 - Safety_Guard: CVS protection guard main not operator OPEN	- Check that the guards are closed and the microswitches are active. - Reset the alarm by pressing the memory reset button. - Check operation of the safety relay inside the control panel.
0010 - Safety_Guard: CVS protection guard exit not operator OPEN	- Check that the guards are closed and the microswitches are active. - Reset the alarm by pressing the memory reset button. - Check operation of the safety relay inside the control panel.
0011 - Safety_Guard: CVS protection guard exit operator OPEN	- Check that the guards are closed and the microswitches are active. - Reset the alarm by pressing the memory reset button. - Check operation of the safety relay inside the control panel.
0012 - Safety_Guard: CVS protection guard main operator OPEN	- Check that the guards are closed and the microswitches are active. - Reset the alarm by pressing the memory reset button. - Check operation of the safety relay inside the control panel.
0013 - Safety_Guard: CVS protection guard inlet operator OPEN	- Check that the guards are closed and the microswitches are active. - Reset the alarm by pressing the memory reset button. - Check operation of the safety relay inside the control panel.
0014 - Safety_Guard: CVS protection guard inlet tunnel OPEN	- Check that the guards are closed and the microswitches are active. - Reset the alarm by pressing the memory reset button. - Check operation of the safety relay inside the control panel.
0015 - Safety_Emergency: General Safe Emergency circuit fault	- Check the safety control unit.
0016 - System: STOP push button pressed on the Operator Panel	- Reset the alarm by pressing the memory reset button. - Check operation of the safety relay inside the main control panel.
0017 - System: STOP push button pressed at the film reel unwind	- Reset the alarm by pressing the memory reset button. - Check operation of the safety relay inside the main control panel.
0018 - System: STOP push button pressed on the line	- Reset the alarm by pressing the memory reset button. - Check operation of the safety relay inside the main control panel.

0019 - System: Metal detector fault	- Check coherence cams. - Restart the machine. - Call the assistance.
0020 - Safety_Emergency: E-STOP pressed on film reel unwind (back)	- Release the emergency button by turning clockwise. - Reset the alarm by pressing the memory reset button. - Check operation of the safety relay inside the main control panel.
0021 - Safety_Emergency: E-STOP pressed on line doser 1	- Release the emergency button by turning clockwise. - Reset the alarm by pressing the memory reset button. - Check operation of the safety relay inside the main control panel.
0022 - Safety_Emergency: E-STOP pressed on line doser 2	- Release the emergency button by turning clockwise. - Reset the alarm by pressing the memory reset button. - Check operation of the safety relay inside the main control panel.
0023 - Safety_Emergency: E-STOP pressed on intralox inlet belt	- Release the emergency button by turning clockwise. - Reset the alarm by pressing the memory reset button. - Check operation of the safety relay inside the main control panel.
0024 - Safety_Emergency: E-STOP pressed on exit belt 2	- Release the emergency button by turning clockwise. - Reset the alarm by pressing the memory reset button. - Check operation of the safety relay inside the main control panel.
0025 - System: Closer machine initialize in running	- - -
0026 - Safety: CVS safety circuit fault	- Restore the safety circuit of the CVS (emergency and guards).
0027 - Safety: Line safety circuit fault	- Restore the safety circuit of the line (emergency and guards).
0028 - Safety: Thermal protection CPU	- Reset the automatic circuit breaker. - Check the current absorption.
0029 - System: Thermal protection opearator panel	- Reset the automatic circuit breaker. - Check the current absorption.
0030 - Safety: Safety CPU is NOT running	- Check the CPU state (wirings, power supply and powerLink net). - Check the led diagnoses on the CPU.
0031 - System: Line machine initialize in running	- - -
0032 - Safety: CVS unit to be rearmed	- Check the safety control unit in the absence of other guard alarms.

0033 - System: Selector not in automatic	- Rotate the selector in automatic position. - Reset the alarm by pressing the memory reset button.
0034 - Safety_Guard: Line protection guard OPEN (inlet phaser)	- Check that the guards are closed and the microswitches are active. - Reset the alarm by pressing the memory reset button. - Check operation of the safety relay inside the control panel.
0035 - System: Software fatal error - pointer null	- Check parameters setting. - Restart the system. - Call the assistance.
0036 - System: Software fatal error master speed null	- Check parameters setting. - Restart the system. - Call the assistance.
0037 - Safety: K01-K02 fault - press and release emergency pushbutton	- Press and release emergency pushbutton.
0038 - Safety: K03-K04 fault - open and close mobile head doser 1 safety guard	- Press and release emergency pushbutton.
0039 - System: Auxiliary cooling fan protection fault	- Verify the good conditions of the cooling fan system of the electrical panel.
0040 - System: Machine closer not inizialized	- Starter message.
0041 - Lubrication: Grease pump in rest	- WARNING: attend the end of the rest cycle.
0042 - Lubrication: Pump power protection FAULT	- Reset the automatic circuit breaker. - Check the current absorption.
0043 - Lubrication: Grease tank low level	- Check the level of grease in the automatic greaser tank. - Reset the alarm via the relevant button on the operator panel.
0044 - System: Machine line not inizialized	- Starter message.
0045 - Lubrication: Timeout grease cycle on dispenser 1	- Check the good conditions of the control sensor. - Check the grease level in the tank and that the grease comes to the distributor.
0046 - Safety_Emergency: E-STOP pressed on line (middle 2)	- Release the emergency button by turning clockwise. - Reset the alarm by pressing the memory reset button. - Check operation of the safety relay inside the main control panel.
0047 - Lubrication: Timeout grease cycle on dispenser 2	- Check the good conditions of the control sensor. - Check the grease level in the tank and that the grease comes to the distributor.

0048 - System: Compressed air supply pressure FAULT	- Check the air supply pressure, which must not be below 5.5 bar. - Check the pressure is sufficient to meet the machine’s requirements. - Check the air supply pipes.
0049 - System: Gas supply pressure FAULT	- Check the gas supply pressure
0050 - Liquid separator: Timeout discharge system	- Check the timeout setting. - Check the control sensor.
0051 - System: Gas mixer fault	- Check the gas mixer setting and wiring.
0052 - System: Downstream line stopped - CVS in pause	- Restart the downstream line. - To operate the tray sealer with the downstream line stopped, switch the machine to autonomous operation via the selector on the operator panel.
0053 - Printer: Not ready	- Switch ON the printer.
0054 - Printer: Film low level	- Check the film level.
0055 - System: ACOPOS MICRO X2X comunication not active	- Check the correct wiring. - Call the service.
0056 - System: Downstream line fault - closer arrested	- Restart the downstream line. - To operate the tray sealer with the downstream line stopped, switch the machine to autonomous operation via the selector on the operator panel.
0057 - OMAC: Closer in homing	- Message home cycle in progress.
0058 - OMAC: Line in homing	- Message home cycle in progress.
0059 - System: ACOPOS MICRO power supply fault	- Check the correct wiring. - Call the service.
0060 - System: CVS cycle delayed	- Check that the tray sealer cycle time is shorter than the tray entry time and reduce the sealing- vacuum- gas time. - Check the feed line speed. - Check the distance between trays.
0061 - System: Waiting controls enable	- Release the emergency button by turning clockwise. - Reset the alarm by pressing the memory reset button. - Check operation of the safety relay inside the main control panel.
0062 - Safety_Emergency: E-STOP pressed on line (middle 3)	- Release the emergency button by turning clockwise. - Reset the alarm by pressing the memory reset button. - Check operation of the safety relay inside the main control panel.

0063 - Safety_Emergency: E-STOP pressed on line (middle 4)	- Release the emergency button by turning clockwise. - Reset the alarm by pressing the memory reset button. - Check operation of the safety relay inside the main control panel.
0064 - Safety_Guard: CVS protection guard open on lid preformed stock	- Check that the guards are closed and the microswitches are active. - Reset the alarm by pressing the memory reset button. - Check operation of the safety relay inside the control panel.
0065 - Safety: Mobile head 1 safety unit to be rearmed	- Check the safety control unit in the absence of other guard alarms.
0066 - Safety_Guard: Mobile head 1 protection guard 1 OPEN	- Check that the guards are closed and the microswitches are active. - Reset the alarm by pressing the memory reset button. - Check operation of the safety relay inside the control panel.
0067 - Safety_Guard: Mobile head 1 protection guard 2 OPEN	- Check that the guards are closed and the microswitches are active. - Reset the alarm by pressing the memory reset button. - Check operation of the safety relay inside the control panel.
0068 - Safety_Guard: Mobile head 1 protection guard 3 OPEN	- Check that the guards are closed and the microswitches are active. - Reset the alarm by pressing the memory reset button. - Check operation of the safety relay inside the control panel.
0069 - Safety_Guard: Mobile head 1 protection guard 4 OPEN	- Check that the guards are closed and the microswitches are active. - Reset the alarm by pressing the memory reset button. - Check operation of the safety relay inside the control panel.
0070 - Safety_Emergency: E-STOP pressed on exit belt 3	- Release the emergency button by turning clockwise. - Reset the alarm by pressing the memory reset button. - Check operation of the safety relay inside the main control panel.
0071 - Safety_Guard: Line protection guard 1 OPEN (CVS inlet tunnel)	- Check that the guards are closed and the microswitches are active. - Reset the alarm by pressing the memory reset button. - Check operation of the safety relay inside the control panel.

0072 - Safety_Guard: Line protection guard OPEN (tunnel to exit linr)	- Check that the guards are closed and the microswitches are active. - Reset the alarm by pressing the memory reset button. - Check operation of the safety relay inside the control panel.
0073 - Safety_Guard: Lifter protection guard 1 OPEN	- Check that the guards are closed and the microswitches are active. - Reset the alarm by pressing the memory reset button. - Check operation of the safety relay inside the control panel.
0074 -	
0075 -	
0076 -	
0077 - Safety_Emergency: E-STOP pressed on tilted denester	- Release the emergency button by turning clockwise. - Reset the alarm by pressing the memory reset button. - Check operation of the safety relay inside the main control panel.
0078 - Safety_Emergency: E-STOP pressed on vertical denester	- Release the emergency button by turning clockwise. - Reset the alarm by pressing the memory reset button. - Check operation of the safety relay inside the main control panel.
0079 - System: STOP push button 2 pressed on the line	- Reset the alarm by pressing the memory reset button. - Check operation of the safety relay inside the main control panel.
0080 - System: Labeller fault	- Restart the machine. - Call the assistance.
0081 - Safety_Emergency: E-STOP pressed on meat transport belt	- Release the emergency button by turning clockwise. - Reset the alarm by pressing the memory reset button. - Check operation of the safety relay inside the main control panel.
0082 - Safety_Emergency: E-STOP pressed on meat transport belt operator panel	- Release the emergency button by turning clockwise. - Reset the alarm by pressing the memory reset button. - Check operation of the safety relay inside the main control panel.

0083 - Safety_Guard: Dancer protection guard OPEN	- Check that the guards are closed and the microswitches are active. - Reset the alarm by pressing the memory reset button. - Check operation of the safety relay inside the control panel.
0084 - Safety: Dancer safety unit to be rearmed	- Release the emergency button by turning clockwise. - Reset the alarm by pressing the memory reset button. - Check operation of the safety relay inside the main control panel.
0085 - System: Thermal protection fault film marker	- Reset the motor circuit breaker. - Check motor current absorption.
0086 - Line: Compressed air supply pressure FAULT	- Check the air supply pressure, which must not be below 5.5 bar. - Check the pressure is sufficient to meet the machine's requirements. - Check the air supply pipes.
0087 - Line: Missing trays	- Put in the missing containers.
0088 - System: Metal detector fault	- Restart the machine. - Call the assistance.
0089 - Safety_Emergency: Line safety unit to be rearmed	- Release the emergency button by turning clockwise. - Reset the alarm by pressing the memory reset button. - Check operation of the safety relay inside the main control panel.
0090 - Line: Missing trays after doser	- Check the doser function.
0091 - Liquid separator: Discharge cycle in running	- - -
0092 - Tool: Motor fan overload	- Reset the motor circuit breaker. - Check motor current absorption.
0093 - Belt 1: Waiting inverter enable	- Message of inverter enable in progress.
0094 - Belt 1: Motor Temperature fault	- Check motor current absorption. - Check there are no obstacles to motor operation. - Check the cables linking the motor and servo (power and encoder). - Check the motor temperature.
0095 - Belt 1: Inverter not ready	- Reset the motor circuit breaker. - Check motor current absorption.
0096 - Belt 1: Thermal protection	- Reset the motor circuit breaker. - Check motor current absorption.
0097 - Inlet belt: Thermal protection	- Reset the motor circuit breaker. - Check motor current absorption.

0098 - Inlet belt: Motor temperature fault	- Check motor current absorption. - Check there are no obstacles to motor operation. - Check the cables linking the motor and servo (power and encoder). - Check the motor temperature.
0099 - Inlet belt: Waiting inverter enable	- Message of inverter enable in progress.
0100 - Inlet belt: Inverter not ready	- Reset the motor circuit breaker. - Check motor current absorption.
0101 - Accelerator belt: Waiting inverter enable	- Message of inverter enable in progress.
0102 - Accelerator belt: Motor temperature fault	- Check motor current absorption. - Check there are no obstacles to motor operation. - Check the cables linking the motor and servo (power and encoder). - Check the motor temperature.
0103 - Accelerator belt: Inverter not ready	- Reset the motor circuit breaker. - Check motor current absorption.
0104 - Accelerator belt: Thermal protection fault	- Reset the motor circuit breaker. - Check motor current absorption.
0105 - Line: Photocell timeout	- Remove the tray in front of the photocell. - Check that the set alarm time is compatible with the current tray type. - Check operation on the photocell and retro- reflector.
0106 - Line: Home missing	- Perform a home position search cycle.
0107 - Line: Repositioning	- Message repositioning in progress.
0108 - Line: General logic error	- Check the alarms on the operator panel. - Restart the system.
0109 - System: Alarm leaking vacuum valve tool side	- Check the correct tightness of the vacuum circuit.
0110 - System: Alarm leaking vacuum valve pump side	- Check the correct tightness of the vacuum circuit.
0111 - Line: Axis fault	- Check there are no obstacles to motor operation. - Check the timeout setting. - Check any sensors.
0112 - Line: Motion not possible for home missing	- Execute an home position cycle. - Check the correct operation of the control sensors.
0113 -	
0114 - Inlet belt: Waiting distancer belt empty	- - - -

0115 - Line: Tray set incomplete going to closer	- Check the machine entrance. - Check the correct operation of the control sensors. - Remove the trays and repeat the cycle.
0116 - Safety_Emergency: E-STOP pulled on life line	- Release the emergency button by turning clockwise. - Reset the alarm by pressing the memory reset button. - Check operation of the safety relay inside the main control panel.
0117 - Line: Photocell to closer side 1 timeout	-Remove the tray in front of the photocell. -Check that the alarm time set into the system is adequate enough for the type of tray being used. -Check to ensure that the photocell and the reflector are working correctly.
0118 - Line: Photocell to closer side 2 timeout	-Remove the tray in front of the photocell. -Check that the alarm time set into the system is adequate enough for the type of tray being used. -Check to ensure that the photocell and the reflector are working correctly.
0119 - Line: Tray couple missing at the closer (inlet side 1)	-Put in the missing containers.
0120 - Line: Tray couple missing at the closer (inlet side 2)	-Put in the missing containers.
0121 - Line: Tray moving to spacer belt during home procedure	Remove the container on the deviator belt and reset the counter.
0122 - Inlet belt: Home missing	- Perform a home position search cycle.
0123 - Line: Waiting inverter enable	- Message of inverter enable in progress.
0124 - Loading belt: Waiting inverter enable	- Message of inverter enable in progress.
0125 - Loading belt: Motor temperature fault	- Check motor current absorption. - Check there are no obstacles to motor operation. - Check the cables linking the motor and servo (power and encoder). - Check the motor temperature.
0126 - Loading belt: Inverter not ready	- Reset the motor circuit breaker. - Check motor current absorption.
0127 - Loading belt: Thermal protection fault	- Reset the motor circuit breaker. - Check motor current absorption.
0128 - Spacer belt 1: Inverter not ready	- Reset the motor circuit breaker. - Check motor current absorption.

0129 - Spacer belt 1: Motor overtemperature protection	- Check motor current absorption. - Check there are no obstacles to motor operation. - Check the cables linking the motor and servo (power and encoder). - Check the motor temperature.
0130 - Spacer belt 1: Waiting inverter enable	- Message of inverter enable in progress.
0131 - Spacer belt 1: Home missing	- Perform a home position search cycle.
0132 - System: Closer in delay	- Check the operation time of the sealing machine.
0133 - Spacer belt 1: Axis fault	- Check there are no obstacles to motor operation. - Check the timeout setting. - Check any sensors.
0134 -	
0135 - Spacer belt 1: Axis error - see axis error data table	- Check the type of alarm in the related charts on the operator panel.
0136 -	
0137 - Spacer belt 1: Photocel fault	- Remove the tray stopped in front of the photocell. - Verify that the set alarm time is correct for the tray type used. - Check aligner photocell & reflector working.
0138 - Spacer belt 1: Trays positioning error	- Check tray position on the spacer belt. - Check the control sensors.
0139 - Spacer belt 1: Remove tray in spacer belt photocell area	- Remove the trays in spacer belt photocell area.
0140 - 2->1 inlet belt: Photocell timeout	- Remove the tray in front of the photocell. - Check that the set alarm time is compatible with the current tray type. - Check operation on the photocell and retro- reflector.
0141 - Tool: Lower - exit side not locked or cleaning bypass inserted	- Lock the tool. - Remove the by-pass. - Switch-off the cleaning cycle.
0142 - Inlet belt: Axis error - see axis error data table	- Check the type of alarm in the related charts on the operator panel.
0143 - System: Labeller thermal protection fault	- Reset the motor circuit breaker. - Check motor current absorption.
0144 - Spacer belt 1: Trays inlet over belt too fast	- Check the flow of the incoming containers on the spacer belt.
0145 - Spacer belt 1: Starting photocell timeout	- Remove the tray stopped in front of the photocell. - Verify that the set alarm time is correct for the tray type used. - Check aligner photocell & reflector working.

0146 - Accelerator belt: Movement phaser timeout	- Check the piston position. - Check there are no obstacles to the piston movement. - Check the correct operation of the control sensors.
0147 - Accelerator belt: Thermal protection phaser	- Reset the motor circuit breaker. - Check motor current absorption.
0148 - Loading belt: Photocell 1 timeout	-Remove the tray in front of the photocell. -Check that the alarm time set into the system is adequate enough for the type of tray being used. -Check to ensure that the photocell and the reflector are working correctly.
0149 - Loading belt: Photocell 2 timeout	-Remove the tray in front of the photocell. -Check that the alarm time set into the system is adequate enough for the type of tray being used. -Check to ensure that the photocell and the reflector are working correctly.
0150 - Loading belt: Phaser photocell timeout	-Remove the tray in front of the photocell. -Check that the alarm time set into the system is adequate enough for the type of tray being used. -Check to ensure that the photocell and the reflector are working correctly.
0151 - Line: Motor wiring change running	- - -
0152 - Line: Speed wiring change not posssible (inverter power on)	- Message of inverter enable in progress.
0153 - Line: Thermal protection fault for remote line cabinet	- Reset the automatic circuit breaker. - Check the current absorption.
0154 - Line: Missing tray couple - side 1	-Put in the missing containers.
0155 - Line: Missing tray couple - side 2	-Put in the missing containers.
0156 - System: Metal detector thermal protection fault	- Restart the machine. - Call the assistance.
0157 - Safety_Guard: Line protection guard OPEN (mobile head to remove)	- Check that the guards are closed and the microswitches are active. - Reset the alarm by pressing the memory reset button. - Check operation of the safety relay inside the control panel.
0158 - Line: Wait missing couple reset pushbutton	- - - -
0159 - Pusher arms: HOME missing	- Execute an home position cycle.

0160 - Pusher arms plier 2: Timeout closing	- Check the piston position. - Check there are no obstacles to the piston movement. - Check the correct operation of the control sensors.
0161 - Pusher arms plier 2: Timeout opening	- Check the piston position. - Check there are no obstacles to the piston movement. - Check the correct operation of the control sensors.
0162 - Pusher arms: Open sensor fault	- Check the correct operation of the control sensors.
0163 - Pusher arms transfer: Rest position sensor fault	- Check the correct operation of the control sensors.
0164 - Pusher arms: Axis fault	- Check there are no obstacles to motor operation. - Check the timeout setting. - Check any sensors.
0165 - Pusher arms transfer: Ahead position sensor signal missing	- Check the correct operation of the control sensors.
0166 - Pusher arms: Timeout starting forward motion	- Check the correct operation of the control sensors.
0167 - Pusher arms: Timeout searching forward stop position	- Check the correct operation of the control sensors.
0168 - Pusher arms: Timeout starting backward motion	- Check the correct operation of the control sensors.
0169 - Pusher arms: Timeout searching backward stop position	- Check the correct operation of the control sensors.
0170 - Pusher arms: Inverter not ready	- Reset the motor circuit breaker. - Check motor current absorption.
0171 -	
0172 - Pusher arms: Disable start pusher arms for tool not low (sensor and position)	- Check the tool position sensors. - Check the machine cycle.
0173 - Pusher arms: Repositioning	- Message repositioning in progress.
0174 -	
0175 -	
0176 - Pusher arms: Thermal protection fan	- Check the correct operation of the fan and that there are no obstacles that prevent its movement. - Reset the motor circuit breaker. - Check motor current absorption.
0177 - Pusher arms: Waiting inverter enable	- Check the condition of the encoder ring. - Check the correct wiring.
0178 - Tool: Reading encoder fault	- Check the condition of the encoder ring. - Check the correct wiring.
0179 - Pusher arms: Reading encoder fault	- Reposition the pusher arms axis.

0180 - Pusher arms clamp 1: Rest position sensor signal missing	- Check the correct operation of the control sensors.
0181 - Pusher arms clamp 2: Rest position sensor signal missing	- Check the correct operation of the control sensors.
0182 - Pusher arms plier 1: Timeout closing	- Check the piston position. - Check there are no obstacles to the piston movement. - Check the correct operation of the control sensors.
0183 - Pusher arms plier 1: Timeout opening	- Check the piston position. - Check there are no obstacles to the piston movement. - Check the correct operation of the control sensors.
0184 - Pusher arms: Motor overtemperature protection	- Check motor current absorption. - Check there are no obstacles to motor operation. - Check the cables linking the motor and servo (power and encoder). - Check the motor temperature.
0185 - Pusher arms: Central guide still open	- Check the pusher arms parameters setting.
0186 - Pusher arms: Exit full	- Remove the tray stopped in front of the photocell. - Verify that the set alarm time is correct for the tray type used. - Check aligner photocell & reflector working.
0187 - Pusher arms: Servodrive thermal protection	- Reset the motor circuit breaker. - Check motor current absorption.
0188 - Pusher arms: Home running	- - -
0189 - Pusher arms: Cycle in delay	- Check the pusher arms parameters setting.
0190 - Pusher arms: Maintenance threshold #2 required	- Execute the maintenance and reset the counter.
0191 - Pusher arms: Maintenance threshold #1 required	- Execute the maintenance and reset the counter.
0192 - Pusher arms: Brake thermal protection fault	- Reset the motor circuit breaker. - Check motor current absorption. - Check there are no obstacles to motor operation. - Check the power supply at the motor terminals.
0193 - Pusher arms: Transfering out of position	- Check there are no obstacles to motor operation. - Check the timeout setting. - Check any sensors.

0194 - Pusher arms: Clamp not in position	- Check the parameters of the clamp system. - Check for sensors.
0195 - Pusher arms: Servodrive thermal protection fault	- Reset the servomotor circuit breaker. - Check servomotor current absorption.
0196 - Pusher arms: Movement not possible for trays presence in snap on lid chain	- Remove the trays.
0197 - Safety_Guard: CVS protection guard 2 open on lid preformed stock	- Check that the guards are closed and the microswitches are active. - Reset the alarm by pressing the memory reset button. - Check operation of the safety relay inside the control panel.
0198 - Safety_Guard: Mobile head 1 protection guard 5 OPEN	- Check that the guards are closed and the microswitches are active. - Reset the alarm by pressing the memory reset button. - Check operation of the safety relay inside the control panel.
0199 - Safety_Guard: Mobile head 1 protection guard 6 OPEN	- Check that the guards are closed and the microswitches are active. - Reset the alarm by pressing the memory reset button. - Check operation of the safety relay inside the control panel.
0200 - ACOPOS MICRO Power supply: Output fault	- Check the correct wiring. - Call the service.
0201 - ACOPOS MICRO Power supply: Overload fault	- Check the correct wiring. - Call the service.
0202 - ACOPOS MICRO Power supply: Phase detection fault	- Check the correct wiring. - Call the service.
0203 - ACOPOS MICRO Power supply: Overtemperature	- Check the correct wiring. - Call the service.
0204 - ACOPOS MICRO Power supply: Overvoltage	- Check the correct wiring. - Call the service.
0205 - Inlet belt phaser: Timeout photocell phaser side 1	- Remove the tray stopped in front of the photocell. - Verify that the set alarm time is correct for the tray type used. - Check aligner photocell & reflector working.
0206 - Inlet belt phaser: Timeout photocell phaser side 2	- Remove the tray stopped in front of the photocell. - Verify that the set alarm time is correct for the tray type used. - Check aligner photocell & reflector working.

0207 - Inlet phaser: Fault during trays loading	- Remove the tray stopped in front of the photocell. - Verify that the set alarm time is correct for the tray type used. - Check aligner photocell & reflector working.
0208 - Line: Trays tall fault	- Check the trays. - Check the trays tall parameters setting.
0209 - Line: Timeout photocell tray tall	- Remove the tray stopped in front of the photocell. - Verify that the set alarm time is correct for the tray type used. - Check aligner photocell & reflector working.
0210 - System: Thermal protection fault film marker 2	- Reset the automatic circuit breaker. - Check the current absorption.
0211 - Printer 2: Not ready	- Switch ON the printer.
0212 - Printer: Warning	- Check the film level.
0213 - Safety_Emergency: E-STOP pressed on rotating denester	- Release the emergency button by turning clockwise. - Reset the alarm by pressing the memory reset button. - Check operation of the safety relay inside the main control panel.
0214 - Rotating denester: Safety guard unit to be rearmed	- Check the safety control unit.
0215 - Denester: Compressed air supply pressure FAULT	- Check the air supply pressure, which must not be below 5.5 bar. - Check the pressure is sufficient to meet the machine’s requirements. - Check the air supply pipes.
0216 - Meat load: Compressed air supply pressure FAULT	- Check the air supply pressure, which must not be below 5.5 bar. - Check the pressure is sufficient to meet the machine’s requirements. - Check the air supply pipes.
0217 - Meat load: STOP push button pressed	- Release the emergency button by turning clockwise. - Reset the alarm by pressing the memory reset button. - Check operation of the safety relay inside the main control panel.
0218 - System: Gas analizer thermal protection fault	- Restart the machine. - Call the assistance.
0219 - System: Gas analizer fault	- Restart the machine. - Call the assistance.

0220 - System: STOP push button 2 pressed	- Release the emergency button by turning clockwise. - Reset the alarm by pressing the memory reset button. - Check operation of the safety relay inside the main control panel.
0221 - System: Downstream line fault 1	- Restart the downstream line. - To operate the tray sealer with the downstream line stopped, switch the machine to autonomous operation via the selector on the operator panel.
0222 - System: Downstream line fault 2	- Restart the downstream line. - To operate the tray sealer with the downstream line stopped, switch the machine to autonomous operation via the selector on the operator panel.
0223 - System: Downstream line fault 3	- Restart the downstream line. - To operate the tray sealer with the downstream line stopped, switch the machine to autonomous operation via the selector on the operator panel.
0224 - Belt 1: Thermal protection fan	- Reset the motor circuit breaker. - Check motor current absorption.
0225 - Line: Pause time too long	- Decrease the pause time.
0226 - Deviator: Tray cadence too high	- Check the line speed parameters setting.
0227 - Line: Photocell timeout side 1	- Remove the tray stopped in front of the photocell. - Verify that the set alarm time is correct for the tray type used. - Check aligner photocell & reflector working.
0228 - Line: Photocell timeout side 1	- Remove the tray stopped in front of the photocell. - Verify that the set alarm time is correct for the tray type used. - Check aligner photocell & reflector working.
0229 - Line: Thermal protection fan	- Check the correct operation of the fan and that there are no obstacles that prevent its movement. - Reset the motor circuit breaker. - Check motor current absorption.
0230 - Line: Doser shift register photocell timeout side 1	- Remove the tray stopped in front of the photocell. - Verify that the set alarm time is correct for the tray type used. - Check aligner photocell & reflector working.

0231 - Line: Doser shift register photocell timeout side 2	- Remove the tray stopped in front of the photocell. - Verify that the set alarm time is correct for the tray type used. - Check aligner photocell & reflector working.
0232 - Line: Change running mode requested but not allowed	- Check the line parameters setting.
0233 - Line: Inverter not ready	- Reset the motor circuit breaker. - Check motor current absorption.
0234 - Safety_Guard: Mobile head 2 protection guard 1 OPEN	- Check that the guards are closed and the microswitches are active. - Reset the alarm by pressing the memory reset button. - Check operation of the safety relay inside the control panel.
0235 - Safety_Guard: Mobile head 2 protection guard 2 OPEN	- Check that the guards are closed and the microswitches are active. - Reset the alarm by pressing the memory reset button. - Check operation of the safety relay inside the control panel.
0236 - Safety_Guard: Mobile head 2 protection guard 3 OPEN	- Check that the guards are closed and the microswitches are active. - Reset the alarm by pressing the memory reset button. - Check operation of the safety relay inside the control panel.
0237 - Safety_Guard: Mobile head 2 protection guard 4 OPEN	- Check that the guards are closed and the microswitches are active. - Reset the alarm by pressing the memory reset button. - Check operation of the safety relay inside the control panel.
0238 - Exit acceleration belt: Motor temperature fault	- Check motor current absorption. - Check there are no obstacles to motor operation. - Check the cables linking the motor and servo (power and encoder). - Check the motor temperature.
0239 - Exit acceleration belt: Inverter not ready	- Reset the motor circuit breaker. - Check motor current absorption.
0240 - Rotating table: Motor overtemperature protection	- Check motor current absorption. - Check there are no obstacles to motor operation. - Check the cables linking the motor and servo (power and encoder). - Check the motor temperature.

0241 - Rotating table: Jamm detected	- Remove the jamming and check the control sensor.
0242 - Rotating table: Thermal protection	- Reset the motor circuit breaker. - Check motor current absorption.
0243 - Exit acceleration belt: Thermal protection fault	- Reset the motor circuit breaker. - Check motor current absorption.
0244 - Exit acceleration belt: Waiting inverter enable	- Message of inverter enable in progress.
0245 - Mirabella_Cavity_1: Low temperature	- Link the connector. - Check the good operation of the PT 100.
0246 - Mirabella_Cavity_1: High temperature	- Link the connector. - Check the good operation of the PT 100.
0247 - Mirabella_Cavity_1: Pt 100 unplugged	- Link the connector. - Check the good operation of the PT 100.
0248 - Mirabella_Cavity_2: Low temperature	- Link the connector. - Check the good operation of the PT 100.
0249 - Mirabella_Cavity_2: High temperature	- Link the connector. - Check the good operation of the PT 100.
0250 - Mirabella_Cavity_2: Pt 100 unplugged	- Link the connector. - Check the good operation of the PT 100.
0251 - Mirabella_Cavity_3: Low temperature	- Link the connector. - Check the good operation of the PT 100.
0252 - Mirabella_Cavity_3: High temperature	- Link the connector. - Check the good operation of the PT 100.
0253 - Mirabella_Cavity_3: Pt 100 unplugged	- Link the connector. - Check the good operation of the PT 100.
0254 - Mirabella_Cavity_4: Low temperature	- Link the connector. - Check the good operation of the PT 100.
0255 - Mirabella_Cavity_4: High temperature	- Link the connector. - Check the good operation of the PT 100.
0256 - Mirabella_Cavity_4: Pt 100 unplugged	- Link the connector. - Check the good operation of the PT 100.
0257 - Line: Remove mobile head gearing	- Remove mobile head gearing.
0258 - Line: Thermal protection of servomotor	- Reset the motor circuit breaker. - Check motor current absorption.

0259 - Distancer Belt: Servodrive thermal protection	- Reset the motor circuit breaker. - Check motor current absorption.
0260 - Tool: Tool cooling water low flow	- Check that the inlet and exit valve are open and water arrives to the plant.
0261 -	
0262 - Tool: Lower heating #1 resistor power line protection fault	- Check the thermal protection. - Check the actual temperature. - Check the SCR good conditions. - Check the tool resistance good conditions and the wiring.
0263 - Tool: Lower heating #1 SCR FAULT	- Reset the alarm by pressing the memory reset button. - Check actuator power supply.
0264 - Tool: Lower heating #1 temperature fault (too high)	-Check the control sensor. - Check there isn't water in the connector. - Check temperature proportional parameter.
0265 - Tool: Lower heating #1 temperature fault (too low)	- Check the control sensor. - Check there isn't water in the connector. - Check temperature proportional parameter.
0266 - Tool: Lower heating #1 PT100 NOT plugged	- Connect the connector. - Check the PT 100 good conditions.
0267 - Tool: Central vacuum fault	- Check the correct operation of the control sensor. - Check the parameters setting.
0268 - System: Gas analizer enabled:Disconnect pipes	- Disconnect pipes.
0269 - Tool: Cooling water can't open while tool is unlocked	- Lock the tool.
0270 - Tool: Home running	- - - -
0271 - Tool: Axis not Ready	- Check the transmission organs. - Repeat the home cycles.
0272 - Tool: Axis fault	- Check there are no obstacles to motor operation. - Check the timeout setting. - Check any sensors.
0273 - Tool: Shuttle not in position	- Check the system stop position. - Check control sensor operation and position.
0274 - Tool: Not moving for Super Protruding shuttle not in position	- Check the system stop position. - Check control sensor operation and position.
0275 - Film reel unwind: Waiting inverter enable	- Message of inverter enable in progress.
0276 - Film unwind: Waiting inverter enable	- Message of inverter enable in progress.
0277 - Film reel unwind: Inverter not ready	- Reset the motor circuit breaker. - Check motor current absorption.

0278 - Film reel unwind: Motor overload	- Reset the motor circuit breaker. - Check motor current absorption.
0279 - Film reel unwind 1: Motor temperature FAULT	- Check motor current absorption. - Check there are no obstacles to motor operation. - Check the cables linking the motor and servo (power and encoder). - Check the motor temperature.
0280 - Film reel unwind 2: Motor temperature FAULT	- Check motor current absorption. - Check there are no obstacles to motor operation. - Check the cables linking the motor and servo (power and encoder). - Check the motor temperature.
0281 - Film reel unwind 1: Film dancer high	- Check the film dancer position.
0282 - Film reel unwind: Motion not possible for home missing	- Execute an home position cycle. - Check the correct operation of the control sensors.
0283 - Tool: VG position too high	- Check the parameters setting.
0284 - Tool: G close proximity fault	- Check the correct operation of the control sensors.
0285 - Doser 3: Thermal protection fault	- Reset the motor circuit breaker. - Check motor current absorption.
0286 - Doser 1: Thermal protection fault	- Reset the motor circuit breaker. - Check motor current absorption.
0287 - Doser 2: Thermal protection fault	- Reset the motor circuit breaker. - Check motor current absorption.
0288 - Film: Scrap broken	- Verify position of film scrap. - Check breakage sensor site after sealing area. - Check die-cutters working (uncorrected trim is possible).
0289 - Film unwinding: Short distance value setted for stop film after mark detection	- Correct the parameter search mark speed, ending position after search and deceleration parameters.
0290 - Film unwinding: Lid cycle slow	- Increase lid unrolling speed or decrease pusher arms speed.
0291 - Film unwinding: Cycle not done	- Execute a manual cycle.
0292 - Film unwinding: Roller not closed	- Activate the relevant pneumatic selector.
0293 - Scrap winder: Rollers not locked	- Activate the relevant pneumatic selector.
0294 - Film reel unwind: Film coil 1 not locked	- Activate the relevant pneumatic selector.
0295 - Film unwinding: Inverter not ready	- Reset the motor circuit breaker. - Check motor current absorption.

0296 - Film reel unwind: Film coil 2 not locked	- Activate the relevant pneumatic selector.
0297 -	
0298 - Film reel unwind: Motion timeout	- Check the timeout parameter. - Check the coil unrolling speed parameter. - Check if the bobbin is finished and the roller movement.
0299 - Film unwinding: Mark detection sensor timeout	- Check that the set alarm time is compatible with the current film type. - Check operation on the photocell and retro- reflector.
0300 - Film unwinding: Timeout during mark detection	- Check that the set alarm time is compatible with the current film type. - Check operation on the photocell and retro- reflector.
0301 - Film: Film end or breaked on uncoil system	- Reel of film finished, replace with a new one. - Check the control sensors.
0302 - Film unwinding: Command with tool not in low position	- Lower completely the tool before activating the film cycle.
0303 - Film reel unwind: Home missing	- Perform a home position search cycle.
0304 - Film unwinding: Home missing	- Perform a home position search cycle.
0305 - Film unwinding: Motor overtemperature protection	- Check motor current absorption. - Check there are no obstacles to motor operation. - Check the cables linking the motor and servo (power and encoder). - Check the motor temperature.
0306 - Film unwinding: Servodrive thermal protection	- Reset the motor circuit breaker. - Check motor current absorption.
0307 - Safety: K emergency feedback fault	- Check the correct operation of K01/K02 contactors.
0308 - Film unwinding: Axis fault	- Check there are no obstacles to motor operation. - Check the timeout setting. - Check any sensors.
0309 -	
0310 -	
0311 -	
0312 - Film unwinding: Start disable for tool out of position	- Lower completely the tool before activating the film cycle.
0313 - Lid preformed: Home missing	- Perform a home position search cycle.
0314 - Lid preformed: General logic error	- Check the alarms on the operator panel. - Restart the system.

0315 - Safety_Guard: Reel auto change protection guard OPEN	- Check that the guards are closed and the microswitches are active. - Reset the alarm by pressing the memory reset button. - Check operation of the safety relay inside the control panel.
0316 -	
0317 - Lid preformed: Axis fault	- Check there are no obstacles to motor operation. - Check the timeout setting. - Check any sensors.
0318 -	
0319 - Lid preformed: Axis error - see axis error data table	- Check the type of alarm in the related charts on the operator panel.
0320 - Tool: Vacuum gas sensor reading - Sensor fault	- Check the correct operation of the control sensor.
0321 - Tool: Sealing sensor reading - Sensor fault	- Check the correct operation of the control sensor.
0322 - Tool: Timeout moving to vacuum position	- Check the correct operation of the control sensor.
0323 - Tool: Timeout moving to sealing position	- Check the correct operation of the control sensor.
0324 - Tool: Timeout moving to tray setting position	- Check the correct operation of the control sensor.
0325 - Tool: Low sensor missing - Tool out position	- Check the correct operation of the control sensor.
0326 - Tool: Timeout moving to rest position	- Check the correct operation of the control sensor.
0327 - Tool: Motor overload protection	- Reset the motor circuit breaker. - Check motor current absorption.
0328 - Scrap unwinding: Thermal protection fault	- Reset the motor circuit breaker. - Check motor current absorption.
0329 - Film unwinding: Sealing not possible for reel autochange in run	- Wait.
0330 - Tool: Trays into tool out of position	- Remove the tray stopped in front of the photocell. - Verify that the set alarm time is correct for the tray type used. - Check aligner photocell & reflector working.
0331 - Tool: Tool pause at VG position selected and speed setted not 0 !!!	- Check the parameters setting.
0332 - Tool: Upper heating #1 resistor power line protection fault	- Check the thermal protection. - Check the actual temperature. - Check the SCR good conditions. - Check the tool resistance good conditions and the wiring.

0333 - Tool: Upper heating #1 SCR FAULT	- Reset the alarm by pressing the memory reset button. - Check actuator power supply.
0334 - Tool: Upper heating #1 temperature fault (too high)	-Check the control sensor. - Check there isn't water in the connector. - Check temperature proportional parameter.
0335 - Tool: Upper heating #1 temperature fault (too low)	- Check the control sensor. - Check there isn't water in the connector. - Check temperature proportional parameter.
0336 - Tool: Upper heating #1 PT100 NOT plugged	- Connect the connector. - Check the PT 100 good conditions.
0337 - Film unwinding: Reel autochange running	- Wait.
0338 - Tool: Pause at tray setting position and speed setted not compatible !!!	- Check the parameters setting.
0339 - Tool: Tool by-pass belt not in right position - lower tool inlet side proximity	- Check the parameters setting.
0340 - Tool: Inlet central guide out of position - closing command	- Check the control sensor and the sliding guides.
0341 - Tool: Inlet central guide out of position - opening command	- Check the control sensor and the sliding guides.
0342 - Tool: Outlet tool central guide out of position - closing command	- Check the control sensor and the sliding guides.
0343 - Tool: Outlet tool central guide out of position - opening command	- Check the control sensor and the sliding guides.
0344 - Scrap unwinding: Motor temperature fault	- Check motor current absorption. - Check there are no obstacles to motor operation. - Check the cables linking the motor and servo (power and encoder). - Check the motor temperature.
0345 - Tool can not unlocked wile cooling water flowing	- Close the cooling circuit.
0346 - Tool: Vacuum position and sealing position not conpatible	- Check the correct position of the vacuum/gas control sensors.
0347 -	
0348 -	
0349 - Tool: Repositioning	- Message repositioning in progress.
0350 - Tool: Home sensor missing - Tool out of position	- Check the correct operation of the control sensor.
0351 - Tool: Sealing position and setting position not compatible	- Check the correct position of the sealing cycle control sensors.

0352 - Tool: Upper - exit side not locked	- Check the tool position - turn the pneumatic selector on the control panel to lock it. - Check control sensor operation and position.
0353 - Tool: Upper - inlet side not locked	- Check the tool position - turn the pneumatic selector on the control panel to lock it. - Check control sensor operation and position.
0354 - Tool: Lower - exit side not locked	- Check the tool position - turn the pneumatic selector on the control panel to lock it. - Check control sensor operation and position.
0355 - Tool: Lower - inlet side not locked	- Check the tool position - turn the pneumatic selector on the control panel to lock it. - Check control sensor operation and position.
0356 - Tool: Home missing	- Execute an home position cycle.
0357 - Tool: Axis wrong position	- Check the transmission organs. - Repeat the home cycles.
0358 - Tool: Tool not low	- Check the correct operation of the control sensors.
0359 - Tool: Waiting enable from other machine vacuum cycle completed	- Wait the vacuum cycle of the other machines is finished or check the feedback signals from the other machines.
0360 - Tool: Vacuum pump motor 1 protection FAULT-Overload	- Check the motor absorption, the wiring and the thermal protection setting.
0361 -	
0362 - Tool: Vacuum pump motor 2 protection FAULT-Overload	- Check the motor absorption, the wiring and the thermal protection setting.
0363 - Tool: Booster motor protection FAULT-Overload	- Check the motor absorption, the wiring and the thermal protection setting.
0364 - Tool: Washing cycle in progress	- - - -
0365 - Tool: Vacuum pressure fault during tool cycle	- Check vacuum cycle parameter settings. - Check vacuum circuit seal. - Check the pressure transducer. - Check the pressure is enough. - Check the valves opening.
0366 - Tool: Gas pressure fault during tool cycle	- Check gas cycle parameter settings. - Check gas circuit seal. - Check the pressure transducer. - Check the pressure is enough. - Check the valves opening.

0367 - Tool: Electronic vacuum cycle timeout	- Check vacuum cycle parameter settings. - Check vacuum circuit seal. - Check the pressure transducer. - Check the pressure is enough. - Check the valves opening.
0368 - Tool: Electronic gas cycle timeout	- Check gas cycle parameter settings. - Check gas circuit seal. - Check the pressure transducer. - Check the pressure is enough. - Check the valves opening.
0369 - Tool: Timeout opening upper compensation	- Check the parameters setting. - Check the valve opening.
0370 - Tool: Timeout closing upper compensation	- Check the parameters setting. - Check the valve opening.
0371 - Tool: Maintenance threshold #2 required	- Execute the maintenance and reset the counter.
0372 - Tool: Maintenance threshold #1 required	- Execute the maintenance and reset the counter.
0373 - Tool: Timeout closing seal skin compensation	- Check the parameters setting. - Check the valve opening.
0374 - Tool: Timeout upper vacuum for skin	- Check the parameters setting. - Check the valve opening.
0375 - Tool: By-pass belt thermal protection	- Reset the motor circuit breaker. - Check motor current absorption. - Check there are no obstacles to motor operation. - Check the power supply at the motor terminals.
0376 - Tool: By-pass belt connector not linked	- Link the connector.
0377 - Tool: By-pass belt motor temperature fault	- Check motor current absorption. - Check there are no obstacles to motor operation. - Check the cables linking the motor and servo (power and encoder). - Check the motor temperature.
0378 - System: Machine need grease	- Perform a lubrication cycle the machine. - Reset the counter on the operator panel.
0379 - Tool: Tool motion not possible for bypass function selected	- Remove the by-pass.
0380 - Skin heating: Plate #1 power line protection fault	- Check the motor absorption, the wiring and the thermal protection setting.
0381 - Skin heating: Temperature fault plate 1 (too high)	- Check the alarm setting. - Check the thermoregulator is on. - Check operation of the thermal resistance and the heating resistances.

0382 - Skin heating: Temperature fault plate 1 (too low)	- Check the alarm setting. - Check the thermoregulator is on. - Check operation of the thermal resistance and the heating resistances.
0383 - Skin heating: PT100 plate #1 NOT plugged	- Connect the connector. - Check the PT 100 good conditions.
0384 - Skin heating: Plate #2 power line protection fault	- Check the motor absorption, the wiring and the thermal protection setting.
0385 - Skin heating: Temperature fault plate 2 (too high)	- Check the alarm setting. - Check the thermoregulator is on. - Check operation of the thermal resistance and the heating resistances.
0386 - Skin heating: Temperature fault plate 2 (too low)	- Check the alarm setting. - Check the thermoregulator is on. - Check operation of the thermal resistance and the heating resistances.
0387 - Skin heating: PT100 plate #2 NOT plugged	- Connect the connector. - Check the PT 100 good conditions.
0388 - Tool: Size changed - Reposition required	- Perform repositioning.
0389 - Tilted denester: Rising piston timeout	- Check the piston position. - Check there are no obstacles to the piston movement. - Check the correct operation of the control sensors.
0390 - Tilted denester: Falling piston timout	- Check the piston position. - Check there are no obstacles to the piston movement. - Check the correct operation of the control sensors.
0391 - Denester: Release position changed	- Check the parameters setting.
0392 - Tilted denester: Cycle in delay	- Check the parameters setting.
0393 - Denester: Need phasing	- Perform phasing.
0394 - Tilted denester: Line setted for half step	- Check the parameters setting.
0395 -	
0396 -	
0397 - Tilted denester: Axis fault	- Check there are no obstacles to motor operation. - Check the timeout setting. - Check any sensors.
0398 -	
0399 - Tilted denester: Axis error - see axis error data table	- Check the type of alarm in the related charts on the operator panel.
0400 - Tilted denester: Too slow	- Check the tray denester parameters.
0401 - Tilted denester: Trays missing (warning)	- Load the trays in the denester.

0402 - Tilted denester: Trays missing FAULT	- Load the trays in the denester.
0403 - Tilted denester: Vacuum pump motor protection FAULT-Overload	- Check the motor absorption, the wiring and the thermal protection setting.
0404 - Tilted denester: Extra stroke high position	- Check the piston stroke and the control sensor.
0405 - Tilted denester: Home missing	- Perform a home position search cycle.
0406 - Tilted denester: Repositioning	- Message repositioning in progress.
0407 -	
0408 - Doser 3: Lifting lifter timeout	- Check the piston position. - Check there are no obstacles to the piston movement. - Check the correct operation of the control sensors.
0409 - Doser 3: Lowering lifter timeout	- Check the piston position. - Check there are no obstacles to the piston movement. - Check the correct operation of the control sensors.
0410 - Doser: Not ready #1	- Switch ON the doser.
0411 - Doser: Delay #1	- Check doser parameters setting.
0412 - Doser: Not ready #2	- Switch ON the doser.
0413 - Doser: Delay #2	- Check doser parameters setting.
0414 - Doser: Waiting for dosing signal 1 done	- Check the doser cycle.
0415 - Doser: Waiting for dosing signal 2 done	- Check the doser cycle.
0416 - Exit belt 1: Motor overload	- Control alarm code on inverter display. - Check automatic switch on inverter feeding line. - Verify motor current absorption. - Check there are no obstacles preventing the motor from running. - Check the inverter ' feeding line.'
0417 - Doser: Not ready #3	- Switch ON the doser.
0418 - Exit belt 1: Motor temperature FAULT	- Check motor current absorption. - Check there are no obstacles to motor operation. - Check the cables linking the motor and servo (power and encoder). - Check the motor temperature.
0419 - Doser 1: Lifting stopper timeout	- Check the piston position. - Check there are no obstacles to the piston movement. - Check the correct operation of the control sensors.
0420 - Doser: Delay #3	- Check doser parameters setting.

0421 - Doser: Alarm product call	- Check doser parameters setting. - Check the correct operation of the control sensors.
0422 - Doser1: Lowering stopper timeout	- Check the piston position. - Check there are no obstacles to the piston movement. - Check the correct operation of the control sensors.
0423 - Doser: Waiting for dosing signal 3 done	- Check the doser cycle.
0424 - Film reel unwind 2: Home missing	- Perform a home position search cycle.
0425 - System: Fault PLC card - Tool Encoder card	- Check the correct operation of the PLC card installed form in the machine. - Check the state led.
0426 - Mirabella_Cavity_1: Thermal protection fault	- Reset the automatic circuit breaker. - Check the current absorption.
0427 - Mirabella_Cavity_2: Thermal protection fault	- Reset the automatic circuit breaker. - Check the current absorption.
0428 - Exit belt 1: Waiting inverter enable	- - - -
0429 - Tilted denester: Thermal protection fault	- Reset the motor circuit breaker. - Check motor current absorption.
0430 -	
0431 - Dosers: Dosers stop By button on line	- Release the emergency button by turning clockwise. - Reset the alarm by pressing the memory reset button. - Check operation of the safety relay inside the main control panel.
0432 - Exit belt 1: Jamm detected	- Remove the jamming and check the control sensor.
0433 -	
0434 -	
0435 - Mobile head: External condition do not permits axis to start	- Check the cycle state in execution from the OMAC pages of the operator panel.
0436 - Mobile head: Axis error - see axis error data table	- Check the type of alarm in the related charts on the operator panel.
0437 -	
0438 - Mobile head: Device error	- Check the correct operation of the device.
0439 -	
0440 - Mobile head: Doser 1 not ready	- Switch ON the mobile head.
0441 - Mobile head: Doser 1 delay	- Check dozer cycle time and reduce it.
0442 - Mobile head: Home missing	- Perform an home cycle positioning.
0443 - Mobile head: Repositioning	- Message repositioning in progress.
0444 - Mobile head: Extra stroke ahead position	- Check the mobile head stroke and the control sensor.

0445 - Mobile head: Extra stroke back position (home side)	- Check the mobile head stroke and the control sensor.
0446 - Tool: G close sensor timeout	- Check the transmission organs. - Repeat the home cycles.
0447 - Exit belt 2: Home missing	- Perform an home cycle positioning.
0448 - System: PC battery low voltage - please replace it	- Replace the PC battery.
0449 - Alarm not defined	
0450 - System: Fault PLC card - X20 BC0083 - Bus coupler Powerlink	- Check the correct operation of the PLC card installed form in the machine. - Check the state led.
0451 - System: Fault PLC card - X20 PS9400 - Power supply (for bus coupler)	- Check the correct operation of the PLC card installed form in the machine. - Check the state led.
0452 - System: Fault PLC card - Digital input card 01	- Check the correct operation of the PLC card installed form in the machine. - Check the state led.
0453 - System: Fault PLC card - Digital input card 02	- Check the correct operation of the PLC card installed form in the machine. - Check the state led.
0454 - System: Fault PLC card - Digital input card 03	- Check the correct operation of the PLC card installed form in the machine. - Check the state led.
0455 - System: Fault PLC card - Digital input card 04	- Check the correct operation of the PLC card installed form in the machine. - Check the state led.
0456 - System: Fault PLC card - Digital input card 05	- Check the correct operation of the PLC card installed form in the machine. - Check the state led.
0457 - System: Fault PLC card - Digital input card 06	- Check the correct operation of the PLC card installed form in the machine. - Check the state led.
0458 - System: Fault PLC card - Digital input card 07	- Check the correct operation of the PLC card installed form in the machine. - Check the state led.
0459 - System: Fault PLC card - Digital input card 08	- Check the correct operation of the PLC card installed form in the machine. - Check the state led.
0460 - System: Fault PLC card - Digital input card 09	- Check the correct operation of the PLC card installed form in the machine. - Check the state led.
0461 - System: Fault PLC card - Digital input card 10	- Check the correct operation of the PLC card installed form in the machine. - Check the state led.
0462 - System: Fault PLC card - Analog input card 1	- Check the correct operation of the PLC card installed form in the machine. - Check the state led.

0463 - System: Fault PLC card - Analog input card - Temperature 1	- Check the correct operation of the PLC card installed form in the machine. - Check the state led.
0464 - System: Fault PLC card - Analog input card - Temperature 2	- Check the correct operation of the PLC card installed form in the machine. - Check the state led.
0465 - System: Fault PLC card - Analog input card 4	- Check the correct operation of the PLC card installed form in the machine. - Check the state led.
0466 - System: Fault PLC card - Analog input card 5	- Check the correct operation of the PLC card installed form in the machine. - Check the state led.
0467 - System: Fault PLC card - Digital output card 01	- Check the correct operation of the PLC card installed form in the machine. - Check the state led.
0468 - System: Fault PLC card - Digital output card 02	- Check the correct operation of the PLC card installed form in the machine. - Check the state led.
0469 - System: Fault PLC card - Digital output card 03	- Check the correct operation of the PLC card installed form in the machine. - Check the state led.
0470 - System: Fault PLC card - Digital output card 04	- Check the correct operation of the PLC card installed form in the machine. - Check the state led.
0471 - System: Fault PLC card - Digital output card 05	- Check the correct operation of the PLC card installed form in the machine. - Check the state led.
0472 - System: Fault PLC card - Digital output card 06	- Check the correct operation of the PLC card installed form in the machine. - Check the state led.
0473 - System: Fault PLC card - Digital output card 07	- Check the correct operation of the PLC card installed form in the machine. - Check the state led.
0474 - System: Fault PLC card - Digital output card 08	- Check the correct operation of the PLC card installed form in the machine. - Check the state led.
0475 - System: Fault PLC card - Digital output card 09	- Check the correct operation of the PLC card installed form in the machine. - Check the state led.
0476 - System: Fault PLC card - Digital output card 10	- Check the correct operation of the PLC card installed form in the machine. - Check the state led.
0477 - System: Fault PLC card - Encoder card	- Check the correct operation of the PLC card installed form in the machine. - Check the state led.
0478 - System: Fault PLC card - Safe digital input card 02	- Check the correct operation of the PLC card installed form in the machine. - Check the state led.

0479 - System: Fault PLC card - Safe digital input card 03	- Check the correct operation of the PLC card installed form in the machine. - Check the state led.
0480 - System: Fault PLC card - Safe digital input card 04	- Check the correct operation of the PLC card installed form in the machine. - Check the state led.
0481 - System: Fault PLC card - Safe digital input card 05	- Check the correct operation of the PLC card installed form in the machine. - Check the state led.
0482 - System: Fault PLC card - Safe digital input card 06	- Check the correct operation of the PLC card installed form in the machine. - Check the state led.
0483 - System: Fault PLC card - X20 PS3300 - Power supply (for heating card)	- Check the correct operation of the PLC card installed form in the machine. - Check the state led.
0484 - System: Fault PLC card - Digital output card 01 (for heating)	- Check the correct operation of the PLC card installed form in the machine. - Check the state led.
0485 - System: Fault PLC card - Digital output card 02 (for heating)	- Check the correct operation of the PLC card installed form in the machine. - Check the state led.
0486 - System: Fault PLC card - DOT_1 Heating card - Low current	- Check the correct operation of the PLC card installed form in the machine. - Check the state led.
0487 - System: Fault PLC card - Safe digital input/output card 01	- Check the correct operation of the PLC card installed form in the machine. - Check the state led.
0488 - System: Fault PLC card - Safe digital input/output card 02	- Check the correct operation of the PLC card installed form in the machine. - Check the state led.
0489 - System: Fault PLC card - DOT_2 Heating card - Low current	- Check the correct operation of the PLC card installed form in the machine. - Check the state led.
0490 - System: Fault PLC card - Safe digital output card 01	- Check the correct operation of the PLC card installed form in the machine. - Check the state led.
0491 - System: Fault PLC card - Safe digital output card 02	- Check the correct operation of the PLC card installed form in the machine. - Check the state led.
0492 - System: Fault PLC card - Safe digital output card 03	- Check the correct operation of the PLC card installed form in the machine. - Check the state led.
0493 - System: Fault PLC card - Safe digital output card 04	- Check the correct operation of the PLC card installed form in the machine. - Check the state led.
0494 - System: Fault PLC card - Digital output card 03 (for heating)	- Check the correct operation of the PLC card installed form in the machine. - Check the state led.

0495 - System: Fault PLC card - Digital output card 04 (for heating)	- Check the correct operation of the PLC card installed form in the machine. - Check the state led.
0496 - Fault CPU safe PLC	- Check the correct operation of the PLC card installed form in the machine. - Check the state led.
0497 - System: Fault PLC card - Valves rack command card 01	- Check the correct operation of the PLC card installed form in the machine. - Check the state led.
0498 - System: Fault PLC card - Valves rack command card 02	- Check the correct operation of the PLC card installed form in the machine. - Check the state led.
0499 -	
0500 - Exit belt 1: Trays not present	- Check the closing machine operation.
0501 - System: Fault PLC card - Analog output card 02	- Check the correct operation of the PLC card installed form in the machine. - Check the state led.
0502 - Exit belt 3 (2>1): Motor temperature FAULT	- Check motor current absorption. - Check there are no obstacles to motor operation. - Check the cables linking the motor and servo (power and encoder). - Check the motor temperature.
0503 - Exit belt 3 (2>1): Jamm detected	- Remove the jamming and check the control sensor.
0504 - Exit belt 3 (2>1): Motor overload	- Check motor current absorption. - Check there are no obstacles to motor operation. - Check the cables linking the motor and servo (power and encoder). - Check the motor temperature.
0505 - Film reel unwind 1: Axis error - see axis error data table	- Check the type of alarm in the related charts on the operator panel.
0506 - Load belt phaser: Timeout photocell phaser side 1	- Remove the tray in front of the photocell. - Check that the set alarm time is compatible with the current tray type. - Check operation on the photocell and retro- reflector.
0507 - Load belt phaser: Timeout photocell phaser side 2	- Remove the tray in front of the photocell. - Check that the set alarm time is compatible with the current tray type. - Check operation on the photocell and retro- reflector.
0508 - Film reel unwind 2: Axis error - see axis error data table	- Check the type of alarm in the related charts on the operator panel.

0509 -Tool: Home not possible for preformed not in safe position	- Place the plate in a safe position. - Check the correct operation of the control sensors.
0510 - System: Fault PLC card in remote node (line) - Bus coupler Powerlink for external ethernet	- Check the type of alarm in the related charts on the operator panel.
0511 - System: Fault PLC card - Analog output card 01	- Check the transmission organs. - Repeat the home cycles.
0512 - System: Fault PLC card in remote node (line) - Safe digital input card 01	- Check the correct operation of the PLC card installed form in the machine. - Check the state led.
0513 - System: Fault PLC card in remote node (line) - Safe digital input card 02	- Check the correct operation of the PLC card installed form in the machine. - Check the state led.
0514 - System: Fault PLC card in remote node (line) - Safe digital input card 03	- Check the correct operation of the PLC card installed form in the machine. - Check the state led.
0515 - System: Fault PLC card in remote node (line) - Safe digital input card 04	- Check the correct operation of the PLC card installed form in the machine. - Check the state led.
0516 - System: Fault PLC card in remote node (line) - Safe digital input card 05	- Check the correct operation of the PLC card installed form in the machine. - Check the state led.
0517 - System: Fault PLC card in remote node (line) - Safe digital input/output card 01	- Check the correct operation of the PLC card installed form in the machine. - Check the state led.
0518 - System: Fault PLC card in remote node (line) - Safe digital input/output card 02	- Check the correct operation of the PLC card installed form in the machine. - Check the state led.
0519 - System: Fault PLC card in remote node (line) - Safe digital input/output card 03	- Check the correct operation of the PLC card installed form in the machine. - Check the state led.
0520 - System: Fault PLC card - Safe digital output card 08	- Check the correct operation of the PLC card installed form in the machine. - Check the state led.
0521 - System: Fault PLC card - Safe digital output card 09	- Check the correct operation of the PLC card installed form in the machine. - Check the state led.
0522 - System: Fault PLC card - X20 HB8815 - Bus coupler Powerlink for external ethernet	- Check the correct operation of the PLC card installed form in the machine. - Check the state led.
0523 - System: Fault PLC card - X2X bus card	- Check the correct operation of the PLC card installed form in the machine. - Check the state led.
0524 - System: Fault PLC card - X2X extended I/O card on OP	- Check the correct operation of the PLC card installed form in the machine. - Check the state led.
0525 - System: Fault PLC card in remote node (line) - X20 BC0083 - Bus coupler Powerlink	- Check the correct operation of the PLC card installed form in the machine. - Check the state led.

0526 - System: Fault PLC card in remote node (line) - X20 PS9400 - Power supply	- Check the correct operation of the PLC card installed form in the machine. - Check the state led.
0527 - System: Fault PLC card in remote node (line) - X20 PS9400 - Power supply (for output card)	- Check the correct operation of the PLC card installed form in the machine. - Check the state led.
0528 - System: Fault PLC card in remote node (line) - Digital input card 01	- Check the correct operation of the PLC card installed form in the machine. - Check the state led.
0529 - System: Fault PLC card in remote node (line) - Digital input card 02	- Check the correct operation of the PLC card installed form in the machine. - Check the state led.
0530 - System: Fault PLC card in remote node (line) - Digital input card 03	- Check the correct operation of the PLC card installed form in the machine. - Check the state led.
0531 - System: Fault PLC card in remote node (line) - Digital input card 04	- Check the correct operation of the PLC card installed form in the machine. - Check the state led.
0532 - System: Fault PLC card in remote node (line) - Digital input card 05	- Check the correct operation of the PLC card installed form in the machine. - Check the state led.
0533 - System: Fault PLC card in remote node (line) - Digital input card 06	- Check the correct operation of the PLC card installed form in the machine. - Check the state led.
0534 - System: Fault PLC card - Digital input card 11	- Check the correct operation of the PLC card installed form in the machine. - Check the state led.
0535 - System: Fault PLC card in remote node (line) - Digital output card 01	- Check the correct operation of the PLC card installed form in the machine. - Check the state led.
0536 - System: Fault PLC card in remote node (line) - Digital output card 02	- Check the correct operation of the PLC card installed form in the machine. - Check the state led.
0537 - System: Fault PLC card in remote node (line) - Digital output card 03	- Check the correct operation of the PLC card installed form in the machine. - Check the state led.
0538 - System: Fault PLC card in remote node (line) - Digital output card 04	- Check the correct operation of the PLC card installed form in the machine. - Check the state led.
0539 - System: Fault PLC card in remote node (line) - Digital output card 05	- Check the correct operation of the PLC card installed form in the machine. - Check the state led.
0540 - Deviator: Motor temperature FAULT	- Check motor current absorption. - Check there are no obstacles to motor operation. - Check the cables linking the motor and servo (power and encoder). - Check the motor temperature.

0541 - Deviator: Jamm detected	- Remove the jamming and check the control sensor.
0542 - Deviator: Home missing	- Perform an home cycle positioning.
0543 -	
0544 -	
0545 -	
0546 - Deviator: Axis fault	- Reset the alarm pressing the zero resetting memory pushbutton. - Check the feeding on the driving.
0547 -	
0548 - Deviator: Axis error - see axis error data table	- Reset the alarm pressing the zero resetting memory pushbutton. - Check the feeding on the driving.
0549 -	
0550 - Deviator belt: Motor temperature FAULT	- Check motor current absorption. - Check there are no obstacles to motor operation. - Check the cables linking the motor and servo (power and encoder). - Check the motor temperature.
0551 - Deviator belt: Jamm detected	- Remove the jamming and check the control sensor.
0552 - Deviator belt: Home missing	- Perform an home cycle positioning.
0553 - Deviator belt: General logic error	- Check the alarms on the operator panel. - Restart the system.
0554 -	
0555 - Doser 2: Safety guard unit to be riarmed	- Check the safety control unit in the absence of other guard alarms.
0556 - Doser 2: Safety guard 1 open	- Release the emergency button by turning clockwise. - Reset the alarm by pressing the memory reset button. - Check operation of the safety relay inside the main control panel.
0557 - Doser 2: Safety guard 2 open	- Release the emergency button by turning clockwise. - Reset the alarm by pressing the memory reset button. - Check operation of the safety relay inside the main control panel.
0558 - Doser 2: Safety guard 3 open	- Release the emergency button by turning clockwise. - Reset the alarm by pressing the memory reset button. - Check operation of the safety relay inside the main control panel.

0559 - Doser 2: Safety guard 4 open	- Release the emergency button by turning clockwise. - Reset the alarm by pressing the memory reset button. - Check operation of the safety relay inside the main control panel.
0560 - Mobile head: Doser 2 not ready	- Switch ON the mobile head.
0561 - Mobile head: Doser 2 delay	- Switch ON the mobile head.
0562 - Safety: Mobile head 2 safety unit to be rearmed	- Release the emergency button by turning clockwise. - Reset the alarm by pressing the memory reset button. - Check operation of the safety relay inside the main control panel.
0563 - Safety: Mobile head 3 safety unit to be rearmed	- Release the emergency button by turning clockwise. - Reset the alarm by pressing the memory reset button. - Check operation of the safety relay inside the main control panel.
0564 - Safety: Mobile head 4 safety unit to be rearmed	- Release the emergency button by turning clockwise. - Reset the alarm by pressing the memory reset button. - Check operation of the safety relay inside the main control panel.
0565 - Mobile head 1: Timeout lifter up	- Check the piston position. - Check there are no obstacles to the piston movement. - Check the correct operation of the control sensors.
0566 - Mobile head 1: Timeout lifter down	- Check the piston position. - Check there are no obstacles to the piston movement. - Check the correct operation of the control sensors.
0567 - Mobile head 2: Timeout lifter up	- Check the piston position. - Check there are no obstacles to the piston movement. - Check the correct operation of the control sensors.
0568 - Mobile head 2: Timeout lifter down	- Check the piston position. - Check there are no obstacles to the piston movement. - Check the correct operation of the control sensors.
0569 - Line: Chain in homing	- - - -
0570 -	
0571 -	

0572 - Lid preformed: Ahead position extra stroke	- Check the control sensor. - Check the parameters setting and the mechanical parts.
0573 - Lid preformed: Lid stock empty	- Load the lids in the denester.
0574 - Lid preformed: Trolley repositioning	- Message repositioning in progress.
0575 - Lid preformed: Timeout during lifting pick-up trolley in position 1	- Check the piston position. - Check there are no obstacles to the piston movement. - Check the correct operation of the control sensors.
0576 - Lid preformed: Timeout during lowering pick-up trolley in position 1	- Check the piston position. - Check there are no obstacles to the piston movement. - Check the correct operation of the control sensors.
0577 - Lid preformed: Timeout during lifting pick-up trolley in position 2	- Check the piston position. - Check there are no obstacles to the piston movement. - Check the correct operation of the control sensors.
0578 - Lid preformed: Timeout during lowering pick-up trolley in position 2	- Check the piston position. - Check there are no obstacles to the piston movement. - Check the correct operation of the control sensors.
0579 - Lid preformed: Timeout during lifting trolley in release position	- Check the piston position. - Check there are no obstacles to the piston movement. - Check the correct operation of the control sensors.
0580 - Lid preformed: Timeout during lowering trolley in release position	- Check the piston position. - Check there are no obstacles to the piston movement. - Check the correct operation of the control sensors.
0581 - Lid preformed: Missing lids under the tool	- Check the vacuum circuit. - Check the parameters setting.
0582 - Lid preformed: Vacuum pompe thermal protection	- Check the thermal protection. - Check the actual temperature. - Check the SCR good conditions. - Check the tool resistance good conditions and the wiring.
0583 - Lid preformed: Lid transport trolley not low	- Check the piston movement. - Control the control sensor.
0584 - Lid preformed: Command with tool out of position	- Reposition the tool.
0585 - Lid preformed: Cycle in delay	- Check the parameters setting.

0586 - Lid preformed: Command not possible for By-pass selected	- Disable the by-pass device.
0587 - Lid preformed: Pick-up 1 or 2 positions setted wrong value	- Check the pick-up parameters setting.
0588 - Lid preformed: Pick-up 2 or waiting positions setted wrong value	- Check the pick-up parameters setting.
0589 - Lid preformed: Waiting or release positions setted wrong value	- Check the pick-up parameters setting.
0590 - Tool: Alarm ok closing g sensor	- Check the correct operation of the control sensors.
0591 - Safety_Emergency: E-STOP pressed on rotating table	- Release the emergency button by turning clockwise. - Reset the alarm by pressing the memory reset button. - Check operation of the safety relay inside the main control panel.
0592 - Doser 1: Safety guard 1 open	- Check that the guards are closed and the microswitches are active. - Reset the alarm by pressing the memory reset button. - Check operation of the safety relay inside the control panel.
0593 - Doser 1: Safety guard 2 open	- Check that the guards are closed and the microswitches are active. - Reset the alarm by pressing the memory reset button. - Check operation of the safety relay inside the control panel.
0594 - Doser 1: Safety guard 3 open	- Check that the guards are closed and the microswitches are active. - Reset the alarm by pressing the memory reset button. - Check operation of the safety relay inside the control panel.
0595 - Doser 1: Safety guard 4 open	- Check that the guards are closed and the microswitches are active. - Reset the alarm by pressing the memory reset button. - Check operation of the safety relay inside the control panel.
0596 - Doser 1: Head 2 in delay	- Check dozer cycle time and reduce it.
0597 - Lid preformed: Missing lids in pick-up position 1	- Check the vacuum circuit. - Check the parameters setting.
0598 - Vertical denester 1: Thermal protection of servomotor	- Reset the motor circuit breaker. - Check motor current absorption.
0599 - Vertical denester 1: Safety guard unit to be rearmed	- Check the safety control unit in the absence of other guard alarms.
0600 -	

0601 -	
0602 - Vertical denester 1: Axis fault	- Check there are no obstacles to motor operation. - Check the timeout setting. - Check any sensors.
0603 -	- Execute an home position cycle. - Check the correct operation of the control sensors.
0604 - Vertical denester 1: Axis error - see axis error data table	- Check the type of alarm in the related charts on the operator panel.
0605 -	
0606 - Vertical denester 1: Trays missing (warning)	- Load the trays in the denester.
0607 - Vertical denester 1: Trays missing FAULT	- Load the trays in the denester.
0608 - Vertical denester 1: Vacuum pump motor protection FAULT-Overload	- Check the motor absorption, the wiring and the thermal protection setting.
0609 - Vertical denester 1: Extra stroke high position	- Check the piston stroke and the control sensor.
0610 - Vertical denester 1: Home missing	- Perform an home cycle positioning.
0611 - Vertical denester 1: Repositioning	- Message repositioning in progress.
0612 -	
0613 -	
0614 -	
0615 - Vertical denester 1: Timeout transfer piston moving to release position	- Check the correct movement of the piston. - Check the pneumatic low regulators. - Check the timeout parameter setting. - check the trays are correctly lined up.
0616 - Vertical denester 1: Timeout transfer piston moving to pick position	- Check the correct movement of the piston. - Check the pneumatic low regulators. - Check the timeout parameter setting. - check the trays are correctly lined up.
0617 - Vertical denester 1: Line not setted for half step	- Check the line parameters setting.
0618 - Vertical denester 1: Safety guard 1 open	- Check that the guards are closed and the microswitches are active. - Reset the alarm by pressing the memory reset button. - Check operation of the safety relay inside the control panel.

0619 - Vertical denester 1: Safety guard 2 open	- Check that the guards are closed and the microswitches are active. - Reset the alarm by pressing the memory reset button. - Check operation of the safety relay inside the control panel.
0620 -	
0621 -	
0622 - Vertical denester 2: Axis fault	- Check there are no obstacles to motor operation. - Check the timeout setting. - Check any sensors.
0623 -	
0624 - Vertical denester 2: Axis error - see axis error data table	- Check the type of alarm in the related charts on the operator panel.
0625 -	
0626 - Vertical denester 2: Trays missing (warning)	- Load the trays in the denester.
0627 - Vertical denester 2: Trays missing FAULT	- Load the trays in the denester.
0628 - Vertical denester 2: Vacuum pump motor protection FAULT-Overload	- Check the motor absorption, the wiring and the thermal protection setting.
0629 - Vertical denester 2: Extra stroke high position	- Check the piston stroke and the control sensor.
0630 - Vertical denester 2: Home missing	- Perform a home position search cycle.
0631 - Vertical denester 2: Repositioning	- Message repositioning in progress.
0632 -	
0633 -	
0634 -	
0635 - Vertical denester 2: Timeout transfer piston moving to release position	- Check the correct movement of the piston. - Check the pneumatic low regulators. - Check the timeout parameter setting. - check the trays are correctly lined up.
0636 - Vertical denester 2: Timeout transfer piston moving to pick position	- Check the correct movement of the piston. - Check the pneumatic low regulators. - Check the timeout parameter setting. - check the trays are correctly lined up.
0637 - Vertical denester 2: Line not setted for half step	- Check the line parameters setting.
0638 - Doser 1: Waiting dosing done for head 2	- - - -
0639 - Doser 1: Safety guard unit to be riarmed	- Check the safety control unit in the absence of other guard alarms.

0640 - Lid preformed: Forward mouvement timeout	- Check the piston position. - Check there are no obstacles to the piston movement. - Check the correct operation of the control sensors.
0641 - Lid preformed: Backward mouvement timeout	- Check the piston position. - Check there are no obstacles to the piston movement. - Check the correct operation of the control sensors.
0642 - Vertical denester 1: Safery guard 3 open	- Check that the guards are closed and the microswitches are active. - Reset the alarm by pressing the memory reset button. - Check operation of the safety relay inside the control panel.
0643 - Vertical denester 1: Safery guard 4 open	- Check that the guards are closed and the microswitches are active. - Reset the alarm by pressing the memory reset button. - Check operation of the safety relay inside the control panel.
0644 - Line: Waiting denester	- - - -
0645 - Line: Waiting doser	- - - -
0646 - Vertical denester 1: Need phasing	- Perform phasing.
0647 - Vertical denester 2: Need phasing	- Perform phasing.
0648 - Vertical denester 1: Falling piston timout	- Check the piston position. - Check there are no obstacles to the piston movement. - Check the correct operation of the control sensors.
0649 - Vertical denester 1: Rising piston timeout	- Check the piston position. - Check there are no obstacles to the piston movement. - Check the correct operation of the control sensors.
0650 - Vertical denester 1: Safery guard 5 open	- Check that the guards are closed and the microswitches are active. - Reset the alarm by pressing the memory reset button. - Check operation of the safety relay inside the control panel.
0651 - Vertical denester 1: Safery guard 6 open	- Check that the guards are closed and the microswitches are active. - Reset the alarm by pressing the memory reset button. - Check operation of the safety relay inside the control panel.

0652 -	
0653 - SSP: Axis fault	- Reset the alarm pressing the zero resetting memory pushbutton. - Check the feeding on the driving.
0654 -	
0655 - SSP: Axis error - see axis error data table	- Reset the alarm pressing the zero resetting memory pushbutton. - Check the feeding on the driving.
0656 -	
0657 - SSP: Home missing	- Execute an home position cycle. - Check the piston stroke and the control sensor operation.
0658 - SSP: Start disablo for tool out of position	- Reposition the tool.
0659 - SSP: Ahead position extra stroke	- Check the control sensor. - Check the respective parameters and the mechanical organs.
0660 - SSP: Command with tool out of position	- Reposition the tool.
0661 - SSP: Cycle in delay	- Check the working cycle of the machine. - Check the respective parameters.
0662 - SSP: Repositioning	- Message of repositioning in progress.
0663 - SSP: Timeout during lifting shuttle in tool position	- Check the piston position. - Check there are no obstacles to the piston movement. - Check the correct operation of the control sensors.
0664 - SSP: Timeout during lowering shuttle in tool position	- Check the piston position. - Check there are no obstacles to the piston movement. - Check the correct operation of the control sensors.
0665 - SSP: Shuttle not in low position	- Check the piston position. - Check there are no obstacles to the piston movement. - Check the correct operation of the control sensors.
0666 - SSP: Cooling water can't open wile tool is unlocked	- Lock the tool.
0667 - SSP: Preheating plate low temperature	- Reset the automatic interrupter. - Check the absorption of the heating resistances.
0668 - SSP: Preheating plate high temperature	- Reset the automatic interrupter. - Check the absorption of the heating resistances.
0669 - SSP: Heating resistor thermal protection	- Reset the automatic switch. - Verify motor current absorption. - Check there are no obstacles preventing the motor from running. - Check motor terminals are powered.

0670 - SSP: Heating resistor PT100 not plugged	- Link the connector. - Check the good operation of the PT 100.
0671 - SSP: Tool cooling water low flow	- Check that the inlet and exit valve are open and water arrives to the plant.
0672 - SSP: Timeout during shuttle move to tool position	- Check the correct operation of the control sensor.
0673 - SSP: Timeout during shuttle move to rest position	- Check the correct operation of the control sensor.
0674 - SSP: SKIN plate not in SAFE position	- Position the SKIN plate in SAFE position. - Check the SKIN plate position. - Check the correct operation of the control sensor.
0675 - Line: Waiting distancer belt empty	- - - -
0676 - Line: Waiting reset tray set incomplete	- - - -
0677 - Exit pacer: Photocell timeout	- Remove the tray in front of the photocell. - Check that the set alarm time is compatible with the current tray type. - Check operation on the photocell and retro- reflector.
0678 - Lifter 1: Home missing	- Perform a home position search cycle.
0679 - Lifter 2: Home missing	- Perform a home position search cycle.
0680 - System: Air press depurate fault	- Check the air press.
0681 - System: Gas mixer thermal protection fault	- Reset the automatic switch. - Verify current absorption.
0682 - Line: Semi Auto mode not possible for pause time = 0	- Check parameters setting.
0683 - Line: Mobile head plugged but carter not plugged	- Check the carter.
0684 - Line: Mobile head not present and carter not present	- Check the mobile head and carter.
0685 - Doser: Not ready #4	- Check doser parameters setting.
0686 - Doser: Delay #4	- Check doser parameters setting.
0687 - Doser: Waiting for dosing signal 4 done	- - - -
0688 - Doser 1: Thermal protection fault	- Reset the motor circuit breaker. - Check motor current absorption.
0689 - Doser 1: Head 3 in delay	- Check parameters setting. - Check the correct operation of the control sensor.
0690 - Tool: Sealing position greater than maximum extra stroke	- Reset the automatic switch. - Verify current absorption.

0691 - Exit belt 2: Motor temperature FAULT	- Check motor current absorption. - Check there are no obstacles to motor operation. - Check the cables linking the motor and servo (power and encoder). - Check the motor temperature.
0692 - Exit belt 2: Jamm detected	- Remove the jamming and check the control sensor.
0693 - Exit belt 2: Motor overload	- Control alarm code on inverter display. - Check automatic switch on inverter feeding line. - Verify motor current absorption. - Check there are no obstacles preventing the motor from running. - Check the inverter ' feeding line.'
0694 - Exit belt 2: Waiting inverter enable	- Message of inverter enable in progress.
0695 - Exit belt 2 inverter: General fault - See axis error data table	- To check the type of alarm in the related charts on the operator panel.
0696 - Lifter 1: Axis error - see axis error data table	- To check the type of alarm in the related charts on the operator panel.
0697 - Lifter 2: Axis error - see axis error data table	- To check the type of alarm in the related charts on the operator panel.
0698 - Lifter 1: Device error	- Check the correct operation of the device.
0699 - Lifter 2: Device error	- Check the correct operation of the device.
0700 - Pusher arms: Fault X2X communication to inverter	- Check the communication LEDS status inside the inverter and on the transmitter. - Check the wiring of the net X2X. - Check the alarm code on the display inverter (if the display shows the alarm CFF perform the procedure described in paragraph 7.1).
0701 - Distancer belt: Fault X2X communication to inverter	- Check the communication LEDS status inside the inverter and on the transmitter. - Check the wiring of the net X2X. - Check the alarm code on the display inverter (if the display shows the alarm CFF perform the procedure described in paragraph 7.1).
0702 - Inlet belt: Fault X2X communication to inverter	- Check the communication LEDS status inside the inverter and on the transmitter. - Check the wiring of the net X2X. - Check the alarm code on the display inverter (if the display shows the alarm CFF perform the procedure described in paragraph 7.1).

0703 - Exit belt: Fault X2X communication to inverter	- Check the communication LEDS status inside the inverter and on the transmitter. - Check the wiring of the net X2X. - Check the alarm code on the display inverter (if the display shows the alarm CFF perform the procedure described in paragraph 7.1).
0704 - Film unwinding: Fault X2X communication to inverter	- Check the communication LEDS status inside the inverter and on the transmitter. - Check the wiring of the net X2X. - Check the alarm code on the display inverter (if the display shows the alarm CFF perform the procedure described in paragraph 7.1).
0705 - Film reel unwind: Fault X2X communication to inverter	- Check the communication LEDS status inside the inverter and on the transmitter. - Check the wiring of the net X2X. - Check the alarm code on the display inverter (if the display shows the alarm CFF perform the procedure described in paragraph 7.1).
0706 - Feed belt 1: Fault X2X communication to inverter	- Check the communication LEDS status inside the inverter and on the transmitter. - Check the wiring of the net X2X. - Check the alarm code on the display inverter (if the display shows the alarm CFF perform the procedure described in paragraph 7.1).
0707 - Accelerator belt: Fault X2X communication to inverter	- Check the communication LEDS status inside the inverter and on the transmitter. - Check the wiring of the net X2X. - Check the alarm code on the display inverter (if the display shows the alarm CFF perform the procedure described in paragraph 7.1).
0708 - Loading belt: Fault X2X communication to inverter	- Check the communication LEDS status inside the inverter and on the transmitter. - Check the wiring of the net X2X. - Check the alarm code on the display inverter (if the display shows the alarm CFF perform the procedure described in paragraph 7.1).
0709 - Accelerator exit belt: Fault X2X communication to inverter	- Check the communication LEDS status inside the inverter and on the transmitter. - Check the wiring of the net X2X. - Check the alarm code on the display inverter (if the display shows the alarm CFF perform the procedure described in paragraph 7.1).

0710 - MicVac labeller: Linear unit not ready	- Turn on the labeller.
0711 - MicVac labeller: Timeout	- Check the correct operation of the control sensors.
0712 - MicVac labeller: Labels roll empty	- Replace the labels roller. - Check the correct operation of the control sensors.
0713 - MicVac: Thermal protection fault	- Reset the automatic switch. - Verify motor current absorption.
0714 - Line: Fault X2X communication to inverter	- Check the communication LEDS status inside the inverter and on the transmitter. - Check the wiring of the net X2X. - Check the alarm code on the display inverter (if the display shows the alarm CFF perform the procedure described in paragraph 7.1).
0715 - Trays bended feeding belt: Fault X2X communication to inverter	- Check the communication LEDS status inside the inverter and on the transmitter. - Check the wiring of the net X2X. - Check the alarm code on the display inverter (if the display shows the alarm CFF perform the procedure described in paragraph 7.1).
0716 - Filling trays belt: Fault X2X communication to inverter	- Check the communication LEDS status inside the inverter and on the transmitter. - Check the wiring of the net X2X. - Check the alarm code on the display inverter (if the display shows the alarm CFF perform the procedure described in paragraph 7.1).
0717 - Meat trays exit belt: Fault X2X communication to inverter	- Check the communication LEDS status inside the inverter and on the transmitter. - Check the wiring of the net X2X. - Check the alarm code on the display inverter (if the display shows the alarm CFF perform the procedure described in paragraph 7.1).
0718 - Meat transport belt: Fault X2X communication to inverter	- Check the communication LEDS status inside the inverter and on the transmitter. - Check the wiring of the net X2X. - Check the alarm code on the display inverter (if the display shows the alarm CFF perform the procedure described in paragraph 7.1).

0719 - Infeed meat belt: Fault X2X communication to inverter	- Check the communication LEDS status inside the inverter and on the transmitter. - Check the wiring of the net X2X. - Check the alarm code on the display inverter (if the display shows the alarm CFF perform the procedure described in paragraph 7.1).
0720 - Pusher arm inverter: General fault - See axis error data table	- Reset the alarm pressing the zero resetting memory pushbutton. - Check the feeding on the driving.
0721 - Spacer belt inverter: General fault - See axis error data table	- Reset the alarm pressing the zero resetting memory pushbutton. - Check the feeding on the driving.
0722 - Inlet belt inverter: General fault - See axis error data table	- Reset the alarm pressing the zero resetting memory pushbutton. - Check the feeding on the driving.
0723 - Lid inverter: General fault - See axis error data table	- Reset the alarm pressing the zero resetting memory pushbutton. - Check the feeding on the driving.
0724 - Reel inverter: General fault - See axis error data table	- Reset the alarm pressing the zero resetting memory pushbutton. - Check the feeding on the driving.
0725 - Feedbelt inverter: General fault - See axis error data table	- Reset the alarm pressing the zero resetting memory pushbutton. - Check the feeding on the driving.
0726 - Line inverter: General fault - See axis error data table	- Reset the alarm pressing the zero resetting memory pushbutton. - Check the feeding on the driving.
0727 - Accelerator belt inverter: General fault - See axis error data table	- Reset the alarm pressing the zero resetting memory pushbutton. - Check the feeding on the driving.
0728 - Loading belt inverter: General fault - See axis error data table	- Reset the alarm pressing the zero resetting memory pushbutton. - Check the feeding on the driving.
0729 - Exit belt inverter: General fault - See axis error data table	- Reset the alarm pressing the zero resetting memory pushbutton. - Check the feeding on the driving.
0730 - Accelerator exit belt inverter: General fault - See axis error data table	- Reset the alarm pressing the zero resetting memory pushbutton. - Check the feeding on the driving.
0731 - Trays bended feeding belt: General fault - See axis error data table	- Reset the alarm pressing the zero resetting memory pushbutton. - Check the feeding on the driving.
0732 - Filling trays belt: General fault - See axis error data table	- Reset the alarm pressing the zero resetting memory pushbutton. - Check the feeding on the driving.
0733 - Meat trays exit belt: General fault - See axis error data table	- Reset the alarm pressing the zero resetting memory pushbutton. - Check the feeding on the driving.

0734 - Meat transport belt: General fault - See axis error data table	- Reset the alarm pressing the zero resetting memory pushbutton. - Check the feeding on the driving.
0735 - Infeed meat belt: General fault - See axis error data table	- Reset the alarm pressing the zero resetting memory pushbutton. - Check the feeding on the driving.
0736 - Combiner belt inverter: General anomaly - See table	- Reset the alarm pressing the zero resetting memory pushbutton. - Check the feeding on the driving.
0737 - Belt 1_1 inverter: General anomaly - See table	- Reset the alarm pressing the zero resetting memory pushbutton. - Check the feeding on the driving.
0738 - Belt 1_2 inverter: General anomaly - See table	- Reset the alarm pressing the zero resetting memory pushbutton. - Check the feeding on the driving.
0739 - Belt 2_1 inverter: General anomaly - See table	- Reset the alarm pressing the zero resetting memory pushbutton. - Check the feeding on the driving.
0740 - Belt 2_2 inverter: General anomaly - See table	- Reset the alarm pressing the zero resetting memory pushbutton. - Check the feeding on the driving.
0741 - Vibrator inverter: General fault - See axis error data table	- To check the type of alarm in the related charts on the operator panel.
0742 - Rotating belt 1: Fault X2X communication to inverter	- Check the communication LEDS status inside the inverter and on the transmitter. - Check the wiring of the net X2X. - Check the alarm code on the display inverter (if the display shows the alarm CFF perform the procedure described in paragraph 7.1).
0743 - Rotating belt 2: Fault X2X communication to inverter	- Check the communication LEDS status inside the inverter and on the transmitter. - Check the wiring of the net X2X. - Check the alarm code on the display inverter (if the display shows the alarm CFF perform the procedure described in paragraph 7.1).
0744 - Vibrator inverter: Fault X2X communication to inverter	- Check the communication LEDS status inside the inverter and on the transmitter. - Check the wiring of the net X2X. - Check the alarm code on the display inverter (if the display shows the alarm CFF perform the procedure described in paragraph 7.1).

0745 - Snap on lid chain: Fault X2X communication to inverter	- Check the communication LEDS status inside the inverter and on the transmitter. - Check the wiring of the net X2X. - Check the alarm code on the display inverter (if the display shows the alarm CFF perform the procedure described in paragraph 7.1).
0746 - Feed belt 1_1: Fault X2X communication to inverter	- Check the communication LEDS status inside the inverter and on the transmitter. - Check the wiring of the net X2X. - Check the alarm code on the display inverter (if the display shows the alarm CFF perform the procedure described in paragraph 7.1).
0747 - Feed belt 1_2: Fault X2X communication to inverter	- Check the communication LEDS status inside the inverter and on the transmitter. - Check the wiring of the net X2X. - Check the alarm code on the display inverter (if the display shows the alarm CFF perform the procedure described in paragraph 7.1).
0748 - Feed belt 2_1: Fault X2X communication to inverter	- Check the communication LEDS status inside the inverter and on the transmitter. - Check the wiring of the net X2X. - Check the alarm code on the display inverter (if the display shows the alarm CFF perform the procedure described in paragraph 7.1).
0749 - Feed belt 2_2: Fault X2X communication to inverter	- Check the communication LEDS status inside the inverter and on the transmitter. - Check the wiring of the net X2X. - Check the alarm code on the display inverter (if the display shows the alarm CFF perform the procedure described in paragraph 7.1).
0750 - Trays bended feeding belt: Inverter not ready	- Reset the motor circuit breaker. - Check motor current absorption.
0751 - Filling trays belt: Inverter not ready	- Reset the motor circuit breaker. - Check motor current absorption.
0752 - Meat trays exit belt: Inverter not ready	- Reset the motor circuit breaker. - Check motor current absorption.
0753 - Meat transport belt: Inverter not ready	- Reset the motor circuit breaker. - Check motor current absorption.
0754 - Trays bended feeding belt: Waiting inverter enable	- Message of inverter enable in progress.

0755 - Filling trays belt: Waiting inverter enable	- Message of inverter enable in progress.
0756 - Meat trays exit belt: Waiting inverter enable	- Message of inverter enable in progress.
0757 - Meat transport belt: Waiting inverter enable	- Message of inverter enable in progress.
0758 - Trays bended feeding belt: Motor temperature fault	- Check motor current absorption. - Check there are no obstacles to motor operation. - Check the cables linking the motor and servo (power and encoder). - Check the motor temperature.
0759 - Filling trays belt: Motor temperature fault	- Check motor current absorption. - Check there are no obstacles to motor operation. - Check the cables linking the motor and servo (power and encoder). - Check the motor temperature.
0760 - Meat trays exit belt: Motor temperature fault	- Check motor current absorption. - Check there are no obstacles to motor operation. - Check the cables linking the motor and servo (power and encoder). - Check the motor temperature.
0761 - Meat transport belt: Motor temperature fault	- Check motor current absorption. - Check there are no obstacles to motor operation. - Check the cables linking the motor and servo (power and encoder). - Check the motor temperature.
0762 - Trays bended feeding belt: Thermal protection fault	- Reset the motor circuit breaker. - Check motor current absorption. - Check there are no obstacles to motor operation. - Check the power supply at the motor terminals.
0763 - Filling trays belt: Thermal protection fault	- Reset the motor circuit breaker. - Check motor current absorption. - Check there are no obstacles to motor operation. - Check the power supply at the motor terminals.
0764 - Meat trays exit belt: Thermal protection fault	- Reset the motor circuit breaker. - Check motor current absorption. - Check there are no obstacles to motor operation. - Check the power supply at the motor terminals.

0765 - Meat transport belt: Thermal protection fault	- Reset the motor circuit breaker. - Check motor current absorption. - Check there are no obstacles to motor operation. - Check the power supply at the motor terminals.
0766 - Safety: Meat belts safety unit to be rearmed	- Check the safety control unit in the absence of other guard alarms.
0767 - Safety_Guard: Protection guard meat belts OPEN	- Check that the guards are closed and the microswitches are active. - Reset the alarm by pressing the memory reset button. - Check operation of the safety relay inside the control panel.
0768 - Filling trays belt: Photocell timeout	- Remove the tray in front of the photocell. - Check that the set alarm time is compatible with the current tray type. - Check operation on the photocell and retro- reflector.
0769 - Filling trays belt: Tray missing	- Put in the missing containers. - Check the correct operation of the control sensors.
0770 - Meat transport belt: Filling photocell fault	- Remove the tray stopped in front of the photocell. - Verify that the set alarm time is correct for the tray type used. - Check aligner photocell & reflector working.
0771 - Meat transport belt: Prefilling photocell fault	- Remove the tray stopped in front of the photocell. - Verify that the set alarm time is correct for the tray type used. - Check aligner photocell & reflector working.
0772 - Meat transport belt: Meat cadence too quickly	- Check the parameters settings of the meat transport belt.
0773 - Infeed meat belt: Inverter not ready	- Reset the motor circuit breaker. - Check motor current absorption.
0774 - Infeed meat belt: Waiting inverter enable	- Message of inverter enable in progress.
0775 - Infeed meat belt: Motor temperature fault	- Check motor current absorption. - Check there are no obstacles to motor operation. - Check the cables linking the motor and servo (power and encoder). - Check the motor temperature.

0776 - Infeed meat belt: Thermal protection fault	- Reset the motor circuit breaker. - Check motor current absorption. - Check there are no obstacles to motor operation. - Check the power supply at the motor terminals.
0777 - Vibrator: Servo drive thermal protection fault	- Reset the motor circuit breaker. - Check motor current absorption.
0778 - Vibrator: Waiting inverter enable	- Message of inverter enable in progress.
0779 - Vibrator: Inverter not ready	- Reset the motor circuit breaker. - Check motor current absorption.
0780 - Exit 1 super belt: Fault X2X communication to inverter	- Check the communication LEDS status inside the inverter and on the transmitter. - Check the wiring of the net X2X. - Check the alarm code on the display inverter (if the display shows the alarm CFF perform the procedure described in paragraph 7.1).
0781 - Lifter 1: Repositioning	- Message repositioning in progress.
0782 - Lifter 2: Repositioning	- Message repositioning in progress.
0783 - Rotating denester: Home missing	- Perform a home position search cycle.
0784 - Rotating denester: wrong speed parameter setted	- Correct the rotating denester parameter.
0785 - Rotating denester: Home position changed	- Message of home position changed.
0786 - Vertical denester 1: Release position changed	- Check the piston position. - Check there are no obstacles to the piston movement. - Check the correct operation of the control sensors.
0787 - Vertical denester 2: Release position changed	- Check the piston position. - Check there are no obstacles to the piston movement. - Check the correct operation of the control sensors.
0788 - Vertical denester 1: Cycle in delay	- Check the vertical denester parameters.
0789 - Vertical denester 2: Cycle in delay	- Check the vertical denester parameters.
0790 - Mirabella_Detector: Thermal protection fault	- Reset the motor circuit breaker. - Check motor current absorption. - Check there are no obstacles to motor operation. - Check the power supply at the motor terminals.
0791 - Mirabella_Detector: Not ready	- Switch ON the Mirabella detector.

0792 - Mirabella_Detector: Insert film	- Check the film insertion.
0793 - Mirabella_Detector: Film fault	- Check the film level. - Check the film is not break.
0794 - Rotating denester: Trays missing (warning)	- Load the trays in the denester.
0795 - Rotating denester: Trays missing FAULT	- Load the trays in the denester.
0796 - Rotating denester: Thermal protection fault	- Reset the motor circuit breaker. - Check motor current absorption.
0797 - Tool: limit software position reached	
0798 - Exit belt 3 (2>1): Fault X2X communication to inverter	- Check the communication LEDS status inside the inverter and on the transmitter. - Check the wiring of the net X2X. - Check the alarm code on the display inverter (if the display shows the alarm CFF perform the procedure described in paragraph 7.1).
0799 - Exit belt 3 (2>1): Inverter communication by X2X not active	- Check the communication LEDS status inside the inverter and on the transmitter. - Check the wiring of the net X2X. - Check the alarm code on the display inverter (if the display shows the alarm CFF perform the procedure described in paragraph 7.1).
0800 - Snap on lid: Safety guard to be rearmed	- Check the safety control unit in the absence of other guard alarms.
0801 - Snap on lid: Safety guard 1 open	- Check that the guards are closed and the microswitches are active. - Reset the alarm by pressing the memory reset button. - Check operation of the safety relay inside the control panel.
0802 - Snap on lid: Safety guard 2 open	- Check that the guards are closed and the microswitches are active. - Reset the alarm by pressing the memory reset button. - Check operation of the safety relay inside the control panel.
0803 - Snap on lid: Safety guard 3 open	- Check that the guards are closed and the microswitches are active. - Reset the alarm by pressing the memory reset button. - Check operation of the safety relay inside the control panel.

0804 - Snap on lid: Safety guard 4 open	- Check that the guards are closed and the microswitches are active. - Reset the alarm by pressing the memory reset button. - Check operation of the safety relay inside the control panel.
0805 - Snap on lid: Emergency pushbutton	- Release the emergency button by turning clockwise. - Reset the alarm by pressing the memory reset button. - Check operation of the safety relay inside the main control panel.
0806 - Snap on lid: Compressed air low pressure	- Check the air supply pressure, which must not be below 5.5 bar. - Check the pressure is sufficient to meet the machine's requirements. - Check the air supply pipes.
0807 - Snap on lid: Chain motor overload	- Reset the motor circuit breaker. - Check motor current absorption.
0808 - Snap on lid: Vacuum pompe overload	- Check motor current absorption. - Check there are no obstacles to motor operation. - Check the cables linking the motor and servo (power and encoder). - Check the motor temperature.
0809 - Snap on lid: Chain encoder reading fault	- Check the correct operation of the control sensors.
0810 - Snap on lid: Chain is waiting tool in right position	- - - -
0811 - Snap on lid: Chain is waiting tool stopper in right position	- - - -
0812 - Snap on lid: Chain is waiting loading piston in right position	- - - -
0813 - Snap on lid: Chain home missing	- Perform a home position search cycle.
0814 - Snap on lid: Chain in homing	- - - -
0815 - Snap on lid: Tool wait lid shuttle	- - - -
0816 - Snap on lid: Lid stok empty	- Load the lids. - Check the correct operation of the control sensors.
0817 - Snap on lid: Stop pushbutton pressed	- Reset the alarm by pressing the memory reset button. - Check operation of the safety relay inside the main control panel.
0818 - Snap on lid: Shattle manual movement not possible for tool not low	- Lower the tool. - Check the correct operation of the control sensors.

0819 - Snap on lid: Timeout lifting shuttle in pick-up position 1	- Check the piston position. - Check there are no obstacles to the piston movement. - Check the correct operation of the control sensors.
0820 - Snap on lid: Timeout lowering shuttle in pick-up position 1	- Check the piston position. - Check there are no obstacles to the piston movement. - Check the correct operation of the control sensors.
0821 - Snap on lid: Timeout lifting shuttle in pick-up position 2	- Check the piston position. - Check there are no obstacles to the piston movement. - Check the correct operation of the control sensors.
0822 - Snap on lid: Timeout lowering shuttle in pick-up position 2	- Check the piston position. - Check there are no obstacles to the piston movement. - Check the correct operation of the control sensors.
0823 - Snap on lid: Timeout lifting shuttle in release position	- Check the piston position. - Check there are no obstacles to the piston movement. - Check the correct operation of the control sensors.
0824 - Snap on lid: Timeout lowering shuttle in release position	- Check the piston position. - Check there are no obstacles to the piston movement. - Check the correct operation of the control sensors.
0825 - Snap on lid: Missing lid in tool	- Check the operation of the Shuttle. - Check the vacuum system.
0826 - Snap on lid: Shuttle not in low position	- Check parameters setting. - Check the correct operation of the control sensors.
0827 - Snap on lid: Chain loading piston lowering timeout	- Check the piston position. - Check there are no obstacles to the piston movement. - Check the correct operation of the control sensors.
0828 - Snap on lid: Chain loading piston lifting timeout	- Check the piston position. - Check there are no obstacles to the piston movement. - Check the correct operation of the control sensors.
0829 - Snap on lid: Timeout lifting tool	- Check the piston position. - Check there are no obstacles to the piston movement. - Check the correct operation of the control sensors.

0830 - Snap on lid: Timeout lowering tool	- Check the piston position. - Check there are no obstacles to the piston movement. - Check the correct operation of the control sensors.
0831 - Snap on lid: Stopper under tool piston lowering timeout	- Check the piston position. - Check there are no obstacles to the piston movement. - Check the correct operation of the control sensors.
0832 - Snap on lid: Stopper under tool piston lifting timeout	- Check the piston position. - Check there are no obstacles to the piston movement. - Check the correct operation of the control sensors.
0833 - Snap on lid: Timeout movement shuttle going to pick-up position 1	- Check the piston position. - Check there are no obstacles to the piston movement. - Check the correct operation of the control sensors.
0834 - Snap on lid: Timeout movement shuttle going to pick-up position 2	- Check the piston position. - Check there are no obstacles to the piston movement. - Check the correct operation of the control sensors.
0835 - Snap on lid: Timeout movement shuttle going to release position - tool	- Check the piston position. - Check there are no obstacles to the piston movement. - Check the correct operation of the control sensors.
0836 - Snap on lid: Timeout chain photocell	- Remove the tray stopped in front of the photocell. - Verify that the set alarm time is correct for the tray type used. - Check aligner photocell & reflector working.
0837 - Tilted denester: Sliding plate not in position	- Check the position of the sliding plate. - Check the correct operation of the control sensors.
0838 - Spacer belt 2: Servodrive thermal protection	- Reset the motor circuit breaker. - Check motor current absorption.
0839 - Spacer belt 2: Axis error - see axis error data table	- To check the type of alarm in the related charts on the operator panel.
0840 - Spacer belt 2: Home missing	- Perform a home position search cycle.
0841 - Spacer belt 2: Axis fault	- Check there are no obstacles to motor operation. - Check the timeout setting. - Check any sensors.

0842 - Spacer belt 2: Starting photocell timeout	- Remove the tray stopped in front of the photocell. - Verify that the set alarm time is correct for the tray type used. - Check aligner photocell & reflector working.
0843 - Spacer belt 2: Trays positioning error	- Check tray position on the spacer belt. - Check the control sensors.
0844 - Spacer belt 2: Trays inlet over belt too fast	- Check the flow of the incoming containers on the spacer belt.
0845 - Spacer belt 2: Photocel fault	- Remove the tray stopped in front of the photocell. - Verify that the set alarm time is correct for the tray type used. - Check aligner photocell & reflector working.
0846 - Exit belt 3 (2>1): Inverter not ready	- Reset the motor circuit breaker. - Check motor current absorption.
0847 - Exit belt 3 (2>1): Waiting inverter enable	- Message of inverter enable in progress.
0848 - Exit belt 1 super inverter: General fault - See axis error data table	- To check the type of alarm in the related charts on the operator panel.
0849 - Exit belt 1 super: Inverter communication by X2X not active	- Check the communication LEDS status inside the inverter and on the transmitter. - Check the wiring of the net X2X. - Check the alarm code on the display inverter (if the display shows the alarm CFF perform the procedure described in paragraph 7.1).
0850 - Pusher arms: Inverter communication by X2X not active	- Check the communication LEDS status inside the inverter and on the transmitter. - Check the wiring of the net X2X. - Check the alarm code on the display inverter (if the display shows the alarm CFF perform the procedure described in paragraph 7.1).
0851 - Spacer belt: Inverter communication by X2X not active	- Check the communication LEDS status inside the inverter and on the transmitter. - Check the wiring of the net X2X. - Check the alarm code on the display inverter (if the display shows the alarm CFF perform the procedure described in paragraph 7.1).

0852 - Inlet belt: Inverter communication by X2X not active	- Check the communication LEDS status inside the inverter and on the transmitter. - Check the wiring of the net X2X. - Check the alarm code on the display inverter (if the display shows the alarm CFF perform the procedure described in paragraph 7.1).
0853 - Lid unroll: Inverter communication by X2X not active	- Check the communication LEDS status inside the inverter and on the transmitter. - Check the wiring of the net X2X. - Check the alarm code on the display inverter (if the display shows the alarm CFF perform the procedure described in paragraph 7.1).
0854 - Reel: Inverter communication by X2X not active	- Check the communication LEDS status inside the inverter and on the transmitter. - Check the wiring of the net X2X. - Check the alarm code on the display inverter (if the display shows the alarm CFF perform the procedure described in paragraph 7.1).
0855 - Line: Inverter communication by X2X not active	- Check the communication LEDS status inside the inverter and on the transmitter. - Check the wiring of the net X2X. - Check the alarm code on the display inverter (if the display shows the alarm CFF perform the procedure described in paragraph 7.1).
0856 - Exit belt: Inverter communication by X2X not active	- Check the communication LEDS status inside the inverter and on the transmitter. - Check the wiring of the net X2X. - Check the alarm code on the display inverter (if the display shows the alarm CFF perform the procedure described in paragraph 7.1).
0857 - Inlet feed line: Inverter communication by X2X not active	- Check the communication LEDS status inside the inverter and on the transmitter. - Check the wiring of the net X2X. - Check the alarm code on the display inverter (if the display shows the alarm CFF perform the procedure described in paragraph 7.1).
0858 - Acceleration belt: Inverter communication by X2X not active	- Check the communication LEDS status inside the inverter and on the transmitter. - Check the wiring of the net X2X. - Check the alarm code on the display inverter (if the display shows the alarm CFF perform the procedure described in paragraph 7.1).

0859 - Loading belt: Inverter communication by X2X not active	- Check the communication LEDS status inside the inverter and on the transmitter. - Check the wiring of the net X2X. - Check the alarm code on the display inverter (if the display shows the alarm CFF perform the procedure described in paragraph 7.1).
0860 - Exit acceleration belt: Inverter communication by X2X not active	- Check the communication LEDS status inside the inverter and on the transmitter. - Check the wiring of the net X2X. - Check the alarm code on the display inverter (if the display shows the alarm CFF perform the procedure described in paragraph 7.1).
0861 - Bended belt: Inverter communication by X2X not active	- Check the communication LEDS status inside the inverter and on the transmitter. - Check the wiring of the net X2X. - Check the alarm code on the display inverter (if the display shows the alarm CFF perform the procedure described in paragraph 7.1).
0862 - Meat tray load belt: Inverter communication by X2X not active	- Check the communication LEDS status inside the inverter and on the transmitter. - Check the wiring of the net X2X. - Check the alarm code on the display inverter (if the display shows the alarm CFF perform the procedure described in paragraph 7.1).
0863 - Exit meat belt: Inverter communication by X2X not active	- Check the communication LEDS status inside the inverter and on the transmitter. - Check the wiring of the net X2X. - Check the alarm code on the display inverter (if the display shows the alarm CFF perform the procedure described in paragraph 7.1).
0864 - Meat belt: Inverter communication by X2X not active	- Check the communication LEDS status inside the inverter and on the transmitter. - Check the wiring of the net X2X. - Check the alarm code on the display inverter (if the display shows the alarm CFF perform the procedure described in paragraph 7.1).
0865 - Meat load belt: Inverter communication by X2X not active	- Check the communication LEDS status inside the inverter and on the transmitter. - Check the wiring of the net X2X. - Check the alarm code on the display inverter (if the display shows the alarm CFF perform the procedure described in paragraph 7.1).

0866 - Snap on lid chain: Inverter communication by X2X not active	- Check the communication LEDS status inside the inverter and on the transmitter. - Check the wiring of the net X2X. - Check the alarm code on the display inverter (if the display shows the alarm CFF perform the procedure described in paragraph 7.1).
0867 - Available: Inverter communication by X2X not active	- Check the communication LEDS status inside the inverter and on the transmitter. - Check the wiring of the net X2X. - Check the alarm code on the display inverter (if the display shows the alarm CFF perform the procedure described in paragraph 7.1).
0868 - Rotating belt 1: Inverter communication by X2X not active	- Check the communication LEDS status inside the inverter and on the transmitter. - Check the wiring of the net X2X. - Check the alarm code on the display inverter (if the display shows the alarm CFF perform the procedure described in paragraph 7.1).
0869 - Rotating belt 2: Inverter communication by X2X not active	- Check the communication LEDS status inside the inverter and on the transmitter. - Check the wiring of the net X2X. - Check the alarm code on the display inverter (if the display shows the alarm CFF perform the procedure described in paragraph 7.1).
0870 - System: Inverter warm-up	- - - -
0871 - Tool: Vacuum pump motor 3 protection FAULT-Overload	- Check the motor absorption, the wiring and the thermal protection setting.
0872 - Denester: Tray stock photocell fault	- Remove the tray stopped in front of the photocell. - Verify that the set alarm time is correct for the tray type used. - Check aligner photocell & reflector working.
0873 - Rotating denester: Photocell timeout under discharge area	- Remove the tray stopped in front of the photocell. - Verify that the set alarm time is correct for the tray type used. - Check aligner photocell & reflector working.
0874 - Doser 1: Too many trays not dosed	- Check the doser functions.
0875 -	
0876 -	

0877 - System: STOP push button pressed on the remote control	- Reset the alarm by pressing the memory reset button. - Check operation of the safety relay inside the main control panel.
0878 - Safety_Guard: CVS protection guard main 2 not operator OPEN	- Check that the guards are closed and the microswitches are active. - Reset the alarm by pressing the memory reset button. - Check operation of the safety relay inside the control panel.
0879 - Safety_Guard: CVS protection guard main operator 2 OPEN	- Check that the guards are closed and the microswitches are active. - Reset the alarm by pressing the memory reset button. - Check operation of the safety relay inside the control panel.
0880 - System: Fault PLC card - X20 PS2100 - Power supply (for v/g valves)	- Check the correct operation of the PLC card installed form in the machine. - Check the state led.
0881 - System: Fault PLC card - X20 PS2100 - Power supply (x2x bus coupler)	- Check the correct operation of the PLC card installed form in the machine. - Check the state led.
0882 - Doser 1 motor 2: Fault X2X communication to inverter	- Check the communication LEDS status inside the inverter and on the transmitter. - Check the wiring of the net X2X. - Check the alarm code on the display inverter (if the display shows the alarm CFF perform the procedure described in paragraph 7.1).
0883 - Doser 1 motor 3: Fault X2X communication to inverter	- Check the communication LEDS status inside the inverter and on the transmitter. - Check the wiring of the net X2X. - Check the alarm code on the display inverter (if the display shows the alarm CFF perform the procedure described in paragraph 7.1).
0884 - Doser 1 motor 4: Fault X2X communication to inverter	- Check the communication LEDS status inside the inverter and on the transmitter. - Check the wiring of the net X2X. - Check the alarm code on the display inverter (if the display shows the alarm CFF perform the procedure described in paragraph 7.1).
0885 - Servo Distancer belt - 56120: Axis error during initrialization	- Check the type of alarm in the related charts on the operator panel.
0886 - Servo Distancer belt - 56121: Axis error during homing	- Check the type of alarm in the related charts on the operator panel.
0887 - Servo Distancer belt - 56122: Axis error during phasing	- Check the type of alarm in the related charts on the operator panel.

0888 - Servo Distancer belt - 56124: Axis error during start setup	- Check the type of alarm in the related charts on the operator panel.
0889 - Servo Distancer belt - 56125: Axis error during automatic starting	- Check the type of alarm in the related charts on the operator panel.
0890 - Servo Distancer belt - 56126: Axis error during manual starting	- Check the type of alarm in the related charts on the operator panel.
0891 - Servo Distancer belt - 56127: Fault during quickstop	- Check the type of alarm in the related charts on the operator panel.
0892 - Servo Distancer belt - 56128: Axis error - see axis error data table	- Check the type of alarm in the related charts on the operator panel.
0893 - Doser 3: Safety guard unit to be riarmed	- Check the safety control unit in the absence of other guard alarms.
0894 - Doser 3: Safety guard 1 open	- Release the emergency button by turning clockwise. - Reset the alarm by pressing the memory reset button. - Check operation of the safety relay inside the main control panel.
0895 - Doser 3: Safety guard 2 open	- Release the emergency button by turning clockwise. - Reset the alarm by pressing the memory reset button. - Check operation of the safety relay inside the main control panel.
0896 - Doser 3: Safety guard 3 open	- Release the emergency button by turning clockwise. - Reset the alarm by pressing the memory reset button. - Check operation of the safety relay inside the main control panel.
0897 - Doser 3: Safety guard 4 open	- Release the emergency button by turning clockwise. - Reset the alarm by pressing the memory reset button. - Check operation of the safety relay inside the main control panel.
0898 - Line load belt: Fault X2X communication to inverter	- Check the communication LEDS status inside the inverter and on the transmitter. - Check the wiring of the net X2X. - Check the alarm code on the display inverter (if the display shows the alarm CFF perform the procedure described in paragraph 7.1).
0899 - Doser 1 motor 1: Fault X2X communication to inverter	- Check the communication LEDS status inside the inverter and on the transmitter. - Check the wiring of the net X2X. - Check the alarm code on the display inverter (if the display shows the alarm CFF perform the procedure described in paragraph 7.1).

0900 - Line - 56020: Axis error during initrialization	- Check the type of alarm in the related charts on the operator panel.
0901 - Line - 56021: Axis error during homing	- Check the type of alarm in the related charts on the operator panel.
0902 - Line - 56022: Axis error during phasing	- Check the type of alarm in the related charts on the operator panel.
0903 - Line - 56023: Axis error during automatic starting	- Check the type of alarm in the related charts on the operator panel.
0904 - Line - 56024: Cam calculation error	- Check the type of alarm in the related charts on the operator panel.
0905 - Line - 56025: Fault during quickstop	- Check the type of alarm in the related charts on the operator panel.
0906 - Line - 56026: Axis error - see axis error data table	- Check the type of alarm in the related charts on the operator panel.
0907 - Line load belt: Inverter communication by X2X not active	- Check the communication LEDS status inside the inverter and on the transmitter. - Check the wiring of the net X2X. - Check the alarm code on the display inverter (if the display shows the alarm CFF perform the procedure described in paragraph 7.1).
0908 - Doser 1 motor 1: Inverter communication by X2X not active	- Check the communication LEDS status inside the inverter and on the transmitter. - Check the wiring of the net X2X. - Check the alarm code on the display inverter (if the display shows the alarm CFF perform the procedure described in paragraph 7.1).
0909 - Doser 1 motor 2: Inverter communication by X2X not active	- Check the communication LEDS status inside the inverter and on the transmitter. - Check the wiring of the net X2X. - Check the alarm code on the display inverter (if the display shows the alarm CFF perform the procedure described in paragraph 7.1).
0910 - Tilted denester - 56040: Axis error during initrialization	- Check the type of alarm in the related charts on the operator panel.
0911 - Tilted denester - 56041: Axis error during homing	- Check the type of alarm in the related charts on the operator panel.
0912 - Tilted denester - 56042: Axis error during phasing	- Check the type of alarm in the related charts on the operator panel.
0913 - Tilted denester - 56045: Axis error during automatic starting	- Check the type of alarm in the related charts on the operator panel.
0914 - Tilted denester - 56043: Cam calculation error	- Check the type of alarm in the related charts on the operator panel.
0915 - Tilted denester - 56047: Fault during quickstop	- Check the type of alarm in the related charts on the operator panel.
0916 - Tilted denester - 56046: Axis error during manual starting	- Check the type of alarm in the related charts on the operator panel.

0917 - Tilted denester - 56044: Axis error during setup	- Check the type of alarm in the related charts on the operator panel.
0918 - Tilted denester - 56048: Axis error - see axis error data table	- Check the type of alarm in the related charts on the operator panel.
0919 - Doser 1 motor 3: Inverter communication by X2X not active	- Check the communication LEDS status inside the inverter and on the transmitter. - Check the wiring of the net X2X. - Check the alarm code on the display inverter (if the display shows the alarm CFF perform the procedure described in paragraph 7.1).
0920 - Vertical denester 1 - 56040: Axis error during initrialization	- Check the type of alarm in the related charts on the operator panel.
0921 - Vertical denester 1 - 56041: Axis error during homing	- Check the type of alarm in the related charts on the operator panel.
0922 - Vertical denester 1 - 56042: Axis error during phasing	- Check the type of alarm in the related charts on the operator panel.
0923 - Vertical denester 1 - 56045: Axis error during automatic starting	- Check the type of alarm in the related charts on the operator panel.
0924 - Vertical denester 1 - 56043: Cam calculation error	- Check the type of alarm in the related charts on the operator panel.
0925 - Vertical denester 1 - 56047: Fault during quickstop	- Check the type of alarm in the related charts on the operator panel.
0926 - Vertical denester 1 - 56046: Axis error during manual starting	- Check the type of alarm in the related charts on the operator panel.
0927 - Vertical denester 1 - 56044: Axis error during setup	- Check the type of alarm in the related charts on the operator panel.
0928 - Vertical denester 1 - 56048: Axis error - see axis error data table	- Check the type of alarm in the related charts on the operator panel.
0929 - Doser 1 motor 4: Inverter communication by X2X not active	- Check the communication LEDS status inside the inverter and on the transmitter. - Check the wiring of the net X2X. - Check the alarm code on the display inverter (if the display shows the alarm CFF perform the procedure described in paragraph 7.1).
0930 - DOT: Home missing	- Perform a home position search cycle.
0931 - DOT: Film cutter not high	- Check the piston position. - Check there are no obstacles to the piston movement. - Check the correct operation of the control sensors.
0932 - DOT: Film cutter not low	- Check the piston position. - Check there are no obstacles to the piston movement. - Check the correct operation of the control sensors.

0933 - DOT: Film plate not packed	- Check the piston position. - Check there are no obstacles to the piston movement. - Check the correct operation of the control sensors.
0934 - DOT: Film plate not un-packed	- Check the piston position. - Check there are no obstacles to the piston movement. - Check the correct operation of the control sensors.
0935 - DOT: Tool not in high position	- Check the piston position. - Check there are no obstacles to the piston movement. - Check the correct operation of the control sensors.
0936 - DOT: Film transport shuttle not in pick-up position	- Check the piston position. - Check there are no obstacles to the piston movement. - Check the correct operation of the control sensors.
0937 - DOT: Film transport shuttle not in release position	- Check the piston position. - Check there are no obstacles to the piston movement. - Check the correct operation of the control sensors.
0938 - DOT: Film brake or finished	- Check the level of the film bobbin. - Check the good conditions of the film. - Check the correct operation of the control sensors.
0939 - DOT: Film not inizialized	- Check the correct insertion of the film. - Check the correct operation of the control sensors.
0940 - DOT: Film inizializing	- - - - -
0941 - DOT: Initialize cycle is waiting cutting pushbutton	- - - - -
0942 - DOT: Tool not open	- - - - -
0943 - DOT: Change tool cycle running	- - - - -
0944 - DOT: Servo drive thermal protection fault	- Reset the motor circuit breaker. - Check motor current absorption.
0945 - DOT: Lid missing in tool	- Check the correct insertion of the film. - Check the correct operation of the control sensors.
0946 - DOT: Shuttle back position fault	- Check there are no obstacles to motor operation. - Check the power supply at the motor terminals. - Check the correct operation of the control sensors.
0947 - DOT: Homing	- - - - -

0948 - DOT: Thermal protection fault vacuum pompe	- Reset the motor circuit breaker. - Check motor current absorption. - Check there are no obstacles to motor operation. - Check the power supply at the motor terminals.
0949 - DOT: Extra stroke ahead	- Check the piston stroke and the control sensor.
0950 - DOT: Extra stroke back	- Check the piston stroke and the control sensor.
0951 - DOT: Repositioning	- - - - -
0952 - DOT: Vacuum and tool data not compatible	- Check the pusher arms parameters setting.
0953 - DOT: Fault device - See axis error table	- Reset the alarm pressing the zero resetting memory pushbutton. - Check the feeding on the driving.
0954 - DOT: Manual mode required	- Enable manual mode.
0955 - DOT/SHUTTLE SKIN: Inlet hook not in safe position	- Place the safety hook in the correct position.
0956 - DOT/SHUTTLE SKIN: Exit hook not in safe position	- Place the safety hook in the correct position.
0957 - DOT: High tool timeout	- Check the correct operation of the control sensor.
0958 -	
0959 -	
0960 - Vertical denester 2 - 56040: Axis error during initrialization	- Check the type of alarm in the related charts on the operator panel.
0961 - Vertical denester 2 - 56041: Axis error during homing	- Check the type of alarm in the related charts on the operator panel.
0962 - Vertical denester 2 - 56042: Axis error during phasing	- Check the type of alarm in the related charts on the operator panel.
0963 - Vertical denester 2 - 56045: Axis error during automatic starting	- Check the type of alarm in the related charts on the operator panel.
0964 - Vertical denester 2 - 56043: Cam calculation error	- Check the type of alarm in the related charts on the operator panel.
0965 - Vertical denester 2 - 56047: Fault during quickstop	- Check the type of alarm in the related charts on the operator panel.
0966 - Vertical denester 2 - 56046: Axis error during manual starting	- Check the type of alarm in the related charts on the operator panel.
0967 - Vertical denester 2 - 56044: Axis error during setup	- Check the type of alarm in the related charts on the operator panel.
0968 - Vertical denester 2 - 56048: Axis error - see axis error data table	- Check the type of alarm in the related charts on the operator panel.
0969 -	

0970 - Rotating belt 1: Thermal protection fault	- Reset the motor circuit breaker. - Check motor current absorption. - Check there are no obstacles to motor operation. - Check the power supply at the motor terminals.
0971 - Rotating belt 2: Thermal protection fault	- Reset the motor circuit breaker. - Check motor current absorption. - Check there are no obstacles to motor operation. - Check the power supply at the motor terminals.
0972 - Rotating belt 1: Inverter not ready	- Reset the motor circuit breaker. - Check motor current absorption.
0973 - Rotating belt 1: Waiting inverter enable	- Message of inverter enable in progress.
0974 - Rotating belt 1: Motor temperature fault	- Check motor current absorption. - Check there are no obstacles to motor operation. - Check the cables linking the motor and servo (power and encoder). - Check the motor temperature.
0975 - Rotating belt 2: Inverter not ready	- Check motor current absorption.
0976 - Rotating belt 2: Waiting inverter enable	- Message of inverter enable in progress.
0977 - Rotating belt 2: Motor temperature fault	- Check motor current absorption. - Check there are no obstacles to motor operation. - Check the cables linking the motor and servo (power and encoder). - Check the motor temperature.
0978 - Rotating belt 1 inverter: General anomaly - See table	- Reset the alarm pressing the zero resetting memory pushbutton. - Check the feeding on the driving.
0979 - Rotating belt 2 inverter: General anomaly - See table	- Reset the alarm pressing the zero resetting memory pushbutton. - Check the feeding on the driving.
0980 - Rotating belts: Phaser photocell timeout	- Remove the tray in front of the photocell. - Check that the set alarm time is compatible with the current tray type. - Check operation on the photocell and retro- reflector.

0981 - Load chain belt: Thermal protection fault	- Reset the motor circuit breaker. - Check motor current absorption. - Check there are no obstacles to motor operation. - Check the power supply at the motor terminals.
0982 - Load chain belt: Inverter not ready	- Reset the motor circuit breaker. - Check motor current absorption.
0983 - Load chain belt: Waiting inverter enable	- Message of inverter enable in progress.
0984 - Load chain belt: Motor temperature fault	- Check motor current absorption. - Check there are no obstacles to motor operation. - Check the cables linking the motor and servo (power and encoder). - Check the motor temperature.
0985 - Load denester belt: Thermal protection fault	- Reset the motor circuit breaker. - Check motor current absorption. - Check there are no obstacles to motor operation. - Check the power supply at the motor terminals.
0986 - Load denester belt: Motor temperature fault	- Check motor current absorption. - Check there are no obstacles to motor operation. - Check the cables linking the motor and servo (power and encoder). - Check the motor temperature.
0987 - Load denester belt: Inverter not ready	- Check motor current absorption.
0988 - Load denester belt: Waiting inverter enable	- Message of inverter enable in progress.
0989 - Load denester belt: Sliding gate open timeout	- Check the piston position. - Check there are no obstacles to the piston movement. - Check the correct operation of the control sensors.
0990 - Load denester belt: Sliding gate close timeout	- Check the piston position. - Check there are no obstacles to the piston movement. - Check the correct operation of the control sensors.
0991 - Load denester belt: Stop belt photocell timeout	- Remove the tray in front of the photocell. - Check that the set alarm time is compatible with the current tray type. - Check operation on the photocell and retro- reflector.

0992 - Load denester belt: General anomaly - See table	- Reset the alarm pressing the zero resetting memory pushbutton. - Check the feeding on the driving.
0993 - Safety_Emergency: E-STOP pressed on load denester belt	- Release the emergency button by turning clockwise. - Reset the alarm by pressing the memory reset button. - Check operation of the safety relay inside the main control panel.
0994 - Safety_Guard: Tray load denester belt protection guard 1 OPEN	- Check that the guards are closed and the microswitches are active. - Reset the alarm by pressing the memory reset button. - Check operation of the safety relay inside the control panel.
0995 - Load denester belt: Fault X2X communication to inverter	- Check the communication LEDS status inside the inverter and on the transmitter. - Check the wiring of the net X2X. - Check the alarm code on the display inverter (if the display shows the alarm CFF perform the procedure described in paragraph 7.1).
0996 - Load denester belt: Inverter communication by X2X not active	- Check the communication LEDS status inside the inverter and on the transmitter. - Check the wiring of the net X2X. - Check the alarm code on the display inverter (if the display shows the alarm CFF perform the procedure described in paragraph 7.1).
0997 - Load denester belt: Lift compact piston timeout	- Check the piston position. - Check there are no obstacles to the piston movement. - Check the correct operation of the control sensors.
0998 - Load denester belt: Lower compact piston timeout	- Check the piston position. - Check there are no obstacles to the piston movement. - Check the correct operation of the control sensors.
0999 - Load chain belt: General anomaly - See table	- Reset the alarm pressing the zero resetting memory pushbutton. - Check the feeding on the driving.
1000 - Combiner belt: Motor temperature fault	- Check motor current absorption. - Check there are no obstacles to motor operation. - Check the cables linking the motor and servo (power and encoder). - Check the motor temperature.
1001 - Combiner belt: Inverter not ready	- Check motor current absorption.

1002 - Combiner belt: Thermal protection fault	- Reset the motor circuit breaker. - Check motor current absorption. - Check there are no obstacles to motor operation. - Check the power supply at the motor terminals.
1003 - Combiner belt: Waiting inverter enable	- Message of inverter enable in progress.
1004 - Combiner belt: Photocell A side 1 timeout	- Remove the tray in front of the photocell. - Check that the set alarm time is compatible with the current tray type. - Check operation on the photocell and retro- reflector.
1005 - Combiner belt: Photocell A side 2 timeout	- Remove the tray in front of the photocell. - Check that the set alarm time is compatible with the current tray type. - Check operation on the photocell and retro- reflector.
1006 - Combiner belt: Photocell B side 1 timeout	- Remove the tray in front of the photocell. - Check that the set alarm time is compatible with the current tray type. - Check operation on the photocell and retro- reflector.
1007 - Combiner belt: Photocell B side 2 timeout	- Remove the tray in front of the photocell. - Check that the set alarm time is compatible with the current tray type. - Check operation on the photocell and retro- reflector.
1008 - Combiner belt: Obstruction	- Remove the tray in front of the photocell. - Check that the set alarm time is compatible with the current tray type. - Check operation on the photocell and retro- reflector.
1009 -	
1010 - Belt 1_1: Motor temperature fault	- Check motor current absorption. - Check there are no obstacles to motor operation. - Check the cables linking the motor and servo (power and encoder). - Check the motor temperature.
1011 - Belt 1_1: Inverter not ready	- Check motor current absorption.

1012 - Belt 1_1: Thermal protection fault	- Reset the motor circuit breaker. - Check motor current absorption. - Check there are no obstacles to motor operation. - Check the power supply at the motor terminals.
1013 - Belt 1_1: Waiting inverter enable	- Message of inverter enable in progress.
1014 - Safety_Guard: CVS protection guard inlet not operator 1 OPEN (MicVac)	- Check that the guards are closed and the microswitches are active. - Reset the alarm by pressing the memory reset button. - Check operation of the safety relay inside the control panel.
1015 - Safety_Guard: CVS protection guard inlet not operator 2 OPEN (MicVac)	- Check that the guards are closed and the microswitches are active. - Reset the alarm by pressing the memory reset button. - Check operation of the safety relay inside the control panel.
1016 - Safety_Guard: CVS protection guard inlet operator 1 OPEN (MicVac)	- Check that the guards are closed and the microswitches are active. - Reset the alarm by pressing the memory reset button. - Check operation of the safety relay inside the control panel.
1017 - Safety_Guard: CVS protection guard inlet operator 2 OPEN (MicVac)	- Check that the guards are closed and the microswitches are active. - Reset the alarm by pressing the memory reset button. - Check operation of the safety relay inside the control panel.
1018 - MicVac: Safety guard unit to be rearmed	- Check the safety control unit in the absence of other guard alarms.
1019 - MicVac: Home missing	- Perform a home position search cycle.
1020 - Belt 1_2: Motor temperature fault	- Check motor current absorption. - Check there are no obstacles to motor operation. - Check the cables linking the motor and servo (power and encoder). - Check the motor temperature.
1021 - Belt 1_2: Inverter not ready	- Check motor current absorption.
1022 - Belt 1_2: Thermal protection fault	- Reset the motor circuit breaker. - Check motor current absorption. - Check there are no obstacles to motor operation. - Check the power supply at the motor terminals.

1023 - Belt 1_2: Waiting inverter enable	- Message of inverter enable in progress.
1024 - MicVac dispenser: Not ready	- Turn on the MicVac - Dispenser.
1025 - MicVac dispenser: Missing two labels	- Replace the labels roller. - Check the correct operation of the control sensors.
1026 - MicVac dispenser: Timeout	- Check the correct operation of the control sensors.
1027 - MicVac: Timeout during data save command	- Check the MicVac cables. - Restart the MicVac.
1028 - MicVac: Comunication timeout	- Check the MicVac cables. - Restart the MicVac.
1029 - MicVac: Saving in progress	- Wait.
1030 - Belt 2_1: Motor temperature fault	- Check motor current absorption. - Check there are no obstacles to motor operation. - Check the cables linking the motor and servo (power and encoder). - Check the motor temperature.
1031 - Belt 2_1: Inverter not ready	- Check motor current absorption.
1032 - Belt 2_1: Thermal protection fault	- Reset the motor circuit breaker. - Check motor current absorption. - Check there are no obstacles to motor operation. - Check the power supply at the motor terminals.
1033 - Belt 2_1: Waiting inverter enable	- Message of inverter enable in progress.
1034 -	
1035 -	
1036 -	
1037 -	
1038 -	
1039 -	
1040 - Belt 2_2: Motor temperature fault	- Check motor current absorption. - Check there are no obstacles to motor operation. - Check the cables linking the motor and servo (power and encoder). - Check the motor temperature.
1041 - Belt 2_2: Inverter not ready	- Check motor current absorption.
1042 - Belt 2_2: Thermal protection fault	- Reset the motor circuit breaker. - Check motor current absorption. - Check there are no obstacles to motor operation. - Check the power supply at the motor terminals.
1043 - Belt 2_2: Waiting inverter enable	- Message of inverter enable in progress.

1044 -	
1045 -	
1046 -	
1047 -	
1048 -	
1049 -	
1050 - Plier 1 - 56280: Axis error during initrialization	- Check the type of alarm in the related charts on the operator panel.
1051 - Plier 1 - 56281: Axis error during homing	- Check the type of alarm in the related charts on the operator panel.
1052 - Plier 1 - 56282: Axis error during phasing	- Check the type of alarm in the related charts on the operator panel.
1053 - Plier 1 - 56285: Axis error during automatic starting	- Check the type of alarm in the related charts on the operator panel.
1054 - Plier 1 - 56283: Cam calculation error	- Check the type of alarm in the related charts on the operator panel.
1055 - Plier 1 - 56287: Fault during quickstop	- Check the type of alarm in the related charts on the operator panel.
1056 - Plier 1 - 56286: Axis error during manual starting	- Check the type of alarm in the related charts on the operator panel.
1057 - Plier 1 - 56284: Axis error during setup	- Check the type of alarm in the related charts on the operator panel.
1058 - Plier 1 - 56288: Axis error - see axis error data table	- Check the type of alarm in the related charts on the operator panel.
1059 - Plier 1: Device error	
1060 - Plier 1: Repositioning	
1061 - Plier 1: Home missing	
1062 - Plier 1: Axis error - See axis error data table	
1063 - Plier 1: Positioning error	
1064 -	
1065 -	
1066 -	
1067 -	
1068 -	
1069 -	
1070 - Plier 2 - 56290: Axis error during initrialization	- Check the type of alarm in the related charts on the operator panel.
1071 - Plier 2 - 56291: Axis error during homing	- Check the type of alarm in the related charts on the operator panel.
1072 - Plier 2 - 56292: Axis error during phasing	- Check the type of alarm in the related charts on the operator panel.
1073 - Plier 2 - 56295: Axis error during automatic starting	- Check the type of alarm in the related charts on the operator panel.
1074 - Plier 2 - 56293: Cam calculation error	- Check the type of alarm in the related charts on the operator panel.
1075 - Plier 2 - 56297: Fault during quickstop	- Check the type of alarm in the related charts on the operator panel.

1076 - Plier 2 - 56296: Axis error during manual starting	- Check the type of alarm in the related charts on the operator panel.
1077 - Plier 2 - 56294: Axis error during setup	- Check the type of alarm in the related charts on the operator panel.
1078 - Plier 2 - 56298: Axis error - see axis error data table	- Check the type of alarm in the related charts on the operator panel.
1079 - Plier 2: Device error	
1080 - Plier 2: Repositioning	
1081 - Plier 2: Home missing	
1082 - Plier 2: Axis error - See axis error data table	
1083 - Plier 2: Positioning error	
1084 -	
1085 -	
1086 - Tool_Cavity_1: Maximum temperature reached	- Check the parameters setting.
1087 - Tool_Cavity_2: Maximum temperature reached	- Check the parameters setting.
1088 - Tool_Cavity_2: Thermal protection fault	- Check the thermal protection. - Check the actual temperature. - Check the SCR good conditions. - Check the tool resistance good conditions and the wiring.
1089 - Tool_Cavity_2: SCR Fault	- Reset the alarm by pressing the memory reset button. - Check actuator power supply.
1090 - Tool_Cavity_2: High temperature	- Reset the automatic interrupter. - Check the absorption of the heating resistances.
1091 - Tool_Cavity_2: Low temperature	- Reset the automatic interrupter. - Check the absorption of the heating resistances.
1092 - Tool_Cavity_2: Pt 100 for alarm plugged	- Link the connector. - Check the good operation of the PT 100.
1093 - Tool_Cavity_2: Pt 100 plugged	- Link the connector. - Check the good operation of the PT 100.
1094 - Tool_Cavity_1: Thermal protection fault	- Check the thermal protection. - Check the actual temperature. - Check the SCR good conditions. - Check the tool resistance good conditions and the wiring.
1095 - Tool_Cavity_1: SCR Fault	- Reset the alarm by pressing the memory reset button. - Check actuator power supply.
1096 - Tool_Cavity_1: High temperature	- Reset the automatic interrupter. - Check the absorption of the heating resistances.

1097 - Tool_Cavity_1: Low temperature	- Reset the automatic interrupter. - Check the absorption of the heating resistances.
1098 - Tool_Cavity_1: Pt 100 for alarm plugged	- Link the connector. - Check the good operation of the PT 100.
1099 - Tool_Cavity_1: Pt 100 plugged	- Link the connector. - Check the good operation of the PT 100.
1100 - Valve safety: Test required	- Perform the test.
1101 - Valve safety: Test running	- Message of test in progress.
1102 - Valve safety: Timeout of reset pushbutton	- Check operation of the reset pushbutton.
1103 - Valve safety: Timeout of safety rele	- Check operation of the safety rele.
1104 - Valve safety: Timeout of test rele	- Check operation of the test rele.
1105 - Valve safety - Lower vacuum valve: Valve not closed	- Check the wiring of the valve. - Check operation of the sensor. - Check operation of the valve.
1106 - Valve safety - Lower vacuum valve: Valve not open	- Check the wiring of the valve. - Check operation of the sensor. - Check operation of the valve.
1107 - Valve safety - Lower vacuum valve: Plc output fault	- Check the Plc output.
1108 - Valve safety - Lower vacuum valve: Contemporaneity open to gas valve	- Check the wiring of the valve. - Check operation of the sensor. - Check operation of the valve.
1109 - Valve safety - Upper vacuum valve: Valve not closed	- Check the wiring of the valve. - Check operation of the sensor. - Check operation of the valve.
1110 - Valve safety - Upper vacuum valve: Valve not open	- Check the wiring of the valve. - Check operation of the sensor. - Check operation of the valve.
1111 - Valve safety - Upper vacuum valve: Plc output fault	- Check the Plc output.
1112 - Valve safety - Upper vacuum valve: Contemporaneity open to gas valve	- Check the wiring of the valve. - Check operation of the sensor. - Check operation of the valve.
1113 - Valve safety - Skin vacuum valve: Valve not closed	- Check the wiring of the valve. - Check operation of the sensor. - Check operation of the valve.
1114 - Valve safety - Skin vacuum valve: Valve not open	- Check the wiring of the valve. - Check operation of the sensor. - Check operation of the valve.
1115 - Valve safety - Skin vacuum valve: Plc output fault	- Check the Plc output.
1116 - Valve safety - Skin vacuum valve: Contemporaneity open to gas valve	- Check the wiring of the valve. - Check operation of the sensor. - Check operation of the valve.

1117 - Valve safety - Lower gas 1 valve: Valve not closed	- Check the wiring of the valve. - Check operation of the sensor. - Check operation of the valve.
1118 - Valve safety - Lower gas 1 valve: Valve not open	- Check the wiring of the valve. - Check operation of the sensor. - Check operation of the valve.
1119 - Valve safety - Lower gas 1 valve: Plc output fault	- Check the Plc output.
1120 - Valve safety - Lower gas 1 valve: Contemporaneity open to vacuum valve	- Check the wiring of the valve. - Check operation of the sensor. - Check operation of the valve.
1121 - Valve safety - Lower gas 2 valve: Valve not closed	- Check the wiring of the valve. - Check operation of the sensor. - Check operation of the valve.
1122 - Valve safety - Lower gas 2 valve: Valve not open	- Check the wiring of the valve. - Check operation of the sensor. - Check operation of the valve.
1123 - Valve safety - Lower gas 2 valve: Plc output fault	- Check the Plc output.
1124 - Valve safety - Lower gas 2 valve: Contemporaneity open to vacuum valve	- Check the wiring of the valve. - Check operation of the sensor. - Check operation of the valve.
1125 - Valve safety - Lower gas 3 valve: Valve not closed	- Check the wiring of the valve. - Check operation of the sensor. - Check operation of the valve.
1126 - Valve safety - Lower gas 3 valve: Valve not open	- Check the wiring of the valve. - Check operation of the sensor. - Check operation of the valve.
1127 - Valve safety - Lower gas 3 valve: Plc output fault	- Check the Plc output.
1128 - Valve safety - Lower gas 3 valve: Contemporaneity open to vacuum valve	- Check the wiring of the valve. - Check operation of the sensor. - Check operation of the valve.
1129 - Valve safety - Lower gas 4 valve: Valve not closed	- Check the wiring of the valve. - Check operation of the sensor. - Check operation of the valve.
1130 - Valve safety - Lower gas 4 valve: Valve not open	- Check the wiring of the valve. - Check operation of the sensor. - Check operation of the valve.
1131 - Valve safety - Lower gas 4 valve: Plc output fault	- Check the Plc output.
1132 - Valve safety - Lower gas 4 valve: Contemporaneity open to vacuum valve	- Check the wiring of the valve. - Check operation of the sensor. - Check operation of the valve.
1133 - Valve safety - Upper gas valve: Valve not closed	- Check the wiring of the valve. - Check operation of the sensor. - Check operation of the valve.

1134 - Valve safety - Upper gas valve: Valve not open	- Check the wiring of the valve. - Check operation of the sensor. - Check operation of the valve.
1135 - Valve safety - Upper gas valve: Plc output fault	- Check the Plc output.
1136 - Valve safety - Upper gas valve: Contemporaneity open to vacuum valve	- Check the wiring of the valve. - Check operation of the sensor. - Check operation of the valve.
1137 - Valve safety - Stretch vacuum valve: Valve not closed	- Check the wiring of the valve. - Check operation of the sensor. - Check operation of the valve.
1138 - Valve safety - Stretch vacuum valve: Valve not open	- Check the wiring of the valve. - Check operation of the sensor. - Check operation of the valve.
1139 - Valve safety - Stretch vacuum valve: Plc output fault	- Check the Plc output.
1140 - Valve safety - Stretch vacuum valve: Contemporaneity open to gas valve	- Check the wiring of the valve. - Check operation of the sensor. - Check operation of the valve.
1141 - Valve safety - Stretch blowing valve: Valve not closed	- Check the wiring of the valve. - Check operation of the sensor. - Check operation of the valve.
1142 - Valve safety - Stretch blowing valve: Valve not open	- Check the wiring of the valve. - Check operation of the sensor. - Check operation of the valve.
1143 - Valve safety - Stretch blowing valve: Plc output fault	- Check the Plc output.
1144 - Valve safety - Stretch blowing valve: Contemporaneity open to gas valve	- Check the wiring of the valve. - Check operation of the sensor. - Check operation of the valve.
1145 - Valve safety - Skin vacuum valve 2: Valve not closed	- Check the wiring of the valve. - Check operation of the sensor. - Check operation of the valve.
1146 - Valve safety - Skin vacuum valve 2: Valve not open	- Check the wiring of the valve. - Check operation of the sensor. - Check operation of the valve.
1147 - Valve safety - Skin vacuum valve 2: Plc output fault	- Check the Plc output.
1148 - Valve safety - Skin vacuum valve 2: Contemporaneity open to gas valve	- Check the wiring of the valve. - Check operation of the sensor. - Check operation of the valve.
1149 - Valve safety: General vacuum valve fault	- Check the wiring of the valve. - Check operation of the sensor. - Check operation of the valve.
1150 - Valve safety: Vacuum circuit fault	- Check the wiring of the valve. - Check operation of the sensor. - Check operation of the valve.

1151 - Valve safety: General gas valve fault	- Check the wiring of the valve. - Check operation of the sensor. - Check operation of the valve.
1152 - Valve safety: Gas circuit fault	- Check the wiring of the valve. - Check operation of the sensor. - Check operation of the valve.
1153 - Valve safety: Control cycle in run	- - - - -
1154 - Valve safety: General vacuum fault	- Check the wiring of the valve. - Check operation of the sensor. - Check operation of the valve.
1155 - Valve safety: Gas low pressure	- Check the wiring of the valve. - Check operation of the sensor. - Check operation of the valve.
1156 -	
1157 -	
1158 -	
1159 -	
1160 - Rotating denester - 56040: Axis error during initrialization	- Check the type of alarm in the related charts on the operator panel.
1161 - Rotating denester - 56041: Axis error during homing	- Check the type of alarm in the related charts on the operator panel.
1162 - Rotating denester - 56042: Axis error during phasing	- Check the type of alarm in the related charts on the operator panel.
1163 - Rotating denester - 56045: Axis error during automatic starting	- Check the type of alarm in the related charts on the operator panel.
1164 - Rotating denester - 56043: Cam calculation error	- Check the type of alarm in the related charts on the operator panel.
1165 - Rotating denester - 56047: Fault during quickstop	- Check the type of alarm in the related charts on the operator panel.
1166 - Rotating denester - 56046: Axis error during manual starting	- Check the type of alarm in the related charts on the operator panel.
1167 - Rotating denester - 56044: Axis error during setup	- Check the type of alarm in the related charts on the operator panel.
1168 - Rotating denester - 56048: Axis error - see axis error data table	- Check the type of alarm in the related charts on the operator panel.
1169 - Rotating denester: Need phasing	
1170 -	
1171 -	
1172 -	
1173 -	
1174 -	
1175 - Tool: Gas change requested	- Make the gas change.
1176 - Tool: It's not allowed to select two gas at the same time	- Select only one type of gas.
1177 - Tool: Gas selection requested	- Select a gas.
1178 -	
1179 -	

1180 - Darfresh heating plate 1: Resistor power line protection fault	- Check the thermal protection. - Check the actual temperature. - Check the SCR good conditions. - Check the heating plate resistance good conditions and the wiring.
1181 -Darfresh heating plate 1: SCR FAULT	- Reset the alarm by pressing the memory reset button. - Check actuator power supply.
1182 -Darfresh heating plate 1: Temperature fault (too high)	- Reset the automatic interrupter. - Check the absorption of the heating resistances.
1183 -Darfresh heating plate 1:Temperature fault (too low)	- Reset the automatic interrupter. - Check the absorption of the heating resistances.
1184 -Darfresh heating plate 1: PT100 NOT plugged	- Link the connector. - Check the good operation of the PT 100.
1185 - Darfresh heating plate 2: Resistor power line protection fault	- Check the thermal protection. - Check the actual temperature. - Check the SCR good conditions. - Check the heating plate resistance good conditions and the wiring.
1186 -Darfresh heating plate 2: SCR FAULT	- Reset the alarm by pressing the memory reset button. - Check actuator power supply.
1187 -Darfresh heating plate 2: Temperature fault (too high)	- Reset the automatic interrupter. - Check the absorption of the heating resistances.
1188 -Darfresh heating plate 2:Temperature fault (too low)	- Reset the automatic interrupter. - Check the absorption of the heating resistances.
1189 -Darfresh heating plate 2: PT100 NOT plugged	- Link the connector. - Check the good operation of the PT 100.
1190 - DOT heating plate 3: Resistor power line protection fault	- Check the thermal protection. - Check the actual temperature. - Check the SCR good conditions. - Check the heating plate resistance good conditions and the wiring.
1191 - DOT heating plate 3: SCR FAULT	- Reset the alarm by pressing the memory reset button. - Check actuator power supply.
1192 - DOT heating plate 3: Temperature fault (too high)	- Reset the automatic interrupter. - Check the absorption of the heating resistances.
1193 - DOT heating plate 3:Temperature fault (too low)	- Reset the automatic interrupter. - Check the absorption of the heating resistances.

1194 - DOT heating plate 3: PT100 NOT plugged	- Link the connector. - Check the good operation of the PT 100.
1195 - DOT heating plate 4: Resistor power line protection fault	- Check the thermal protection. - Check the actual temperature. - Check the SCR good conditions. - Check the heating plate resistance good conditions and the wiring.
1196 - DOT heating plate 4: SCR FAULT	- Reset the alarm by pressing the memory reset button. - Check actuator power supply.
1197 - DOT heating plate 4: Temperature fault (too high)	- Reset the automatic interrupter. - Check the absorption of the heating resistances.
1198 - DOT heating plate 4:Temperature fault (too low)	- Reset the automatic interrupter. - Check the absorption of the heating resistances.
1199 - DOT heating plate 4: PT100 NOT plugged	- Link the connector. - Check the good operation of the PT 100.

2034 - Panel: Texts too long	- Shorten the texts.
2035 - Panel:	
2036 - Panel:	
2037 - Panel:	
2038 - Panel:	
2039 - Panel: recipe sync error doser (ishida)	- Missed alignment with third parts. - Check the connection, the working and the good state of the third parts.
2040 - Panel: recipe sync error printer (domino)	- Missed alignment with third parts. - Check the connection, the working and the good state of the third parts.
2041 - Panel: recipe sync error weigher (digi)	- Missed alignment with third parts. - Check the connection, the working and the good state of the third parts.
2042 - Panel: recipe sync error x-ray (loma)	- Missed alignment with third parts. - Check the connection, the working and the good state of the third parts.
2043 - Panel: recipe sync error stacker (ulma)	- Missed alignment with third parts. - Check the connection, the working and the good state of the third parts.
2044 - Panel: recipe fry date not accepted	- Check Ethernet connection between PLC LENZE and PC B&R.
2045 - Panel: max number of recipes reached	- Cancel the recipes not used for recovering free spaces.
2046 - Panel: max alarm story log almost reached	- Cancel the alarm story list.
2047 - Panel: CPU ip cannot be reached check connection	- Check wiring connections.

7.1 RESET INVERTER DEFAULT DATA

With inverter alarm “CFF”, showed on inverter display, need inverter reset and reload default data.

Procedure:

1. Push pushbutton “enter” (weel selector).
2. Display show menu “FCs”
3. Press pushbutton “enter” and open menu.
4. Turn enter pushbutton up to display “InI”
5. Press and hold for 1 sec. minimum enter pushbutton.
6. If operation is successfully, display shows “No”.
7. Switch off machine (complete power-off).
8. Few seconds after complete display off, is possible to power-on machine.
9. After start-up, inverter display must show “rdY” and then “nSt”

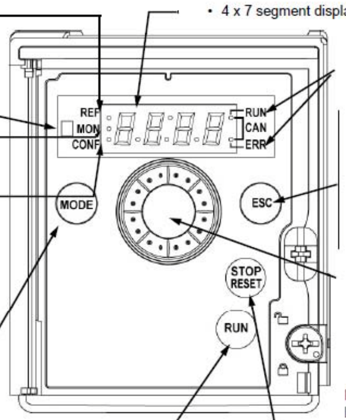
Menu “FCs” is present also in other data group as “drC”,”I_O”, “CtL”, “FUn”.

FCS	❑ **[Restore config.]**	(1)	[No] (nO)
2 s	**⚠ DANGER** **UNINTENDED EQUIPMENT OPERATION** Check that the changes made to the current configuration are compatible with the wiring diagram used. **Failure to follow these instructions will result in death or serious injury.**		
nO	☐ [No] (nO): Function inactive		
rEC1	☐ [Internal] (rEC1): The current configuration becomes identical to the backup configuration previously saved by [Saving config.] (SCS) = [Config 1] (Str1). [Internal] (rEC1) is only visible if the backup has been carried out. [Restore config.] (FCS) automatically switches to [No] (nO) as soon as this action has been performed.		
InI	☐ [Factory Set.] (InI): The current configuration is replaced by the configuration selected by the [Macro configuration] (CFG) parameter (2). [Restore config.] (FCS) automatically switches to [No] (nO) as soon as this action has been performed. **Note**: If nAd appears on the display briefly before the parameter switches to [No] (nO), this means that the configuration transfer is not possible and has not been performed (different drive ratings, for example). If ntr appears on the display briefly before the parameter switches to [No] (nO), this means that an invalid configuration transfer has occurred and that the factory settings will need to be restored using [Factory Set.] (InI). In both cases, check the configuration to be transfered before trying again.		

Description of the HMI

Functions of the display and the keys

- REF LED, illuminated if [SPEED REFERENCE] (rEF-) menu is active
- Load LED
- MON LED, illuminated if [MONITORING] (SUP-) menu is active.
- CONF LED, illuminated if the [SETTINGS] (SEt-), [MOTOR CONTROL] (drC-),[INPUTS / OUTPUTS CFG] (I_O-), [COMMAND] (CtL-), [APPLICATION FUNCT] (FUn-), [FAULT MANAGEMENT] (FLt-) or [COMMUNICATION] (COM-) menus are active.
- 4 x 7 segment display



- Status LEDs of integrated communication interface
- Used to quit a menu or parameter or to clear the value displayed in order to revert to the value in the memory.
- In LOCAL configuration, 2s press on ESC button switches between the control/ programming modes
- Jog dial - can be used for navigation by turning it clockwise or counter-clockwise - pressing the jog dial enables the user to make a selection or confirm information. = ENT

Functions as a potentiometer in LOCAL configuration and in REMOTE configuration if [Ref.1 channel] (Fr1-) in the [COMMAND] (CtL-) menu is set to [Image input AIV1] (AIV1).

- MODE button (1): 3s press on MODE button switches between the REMOTE/LOCAL configurations. If [SPEED REFERENCE] (rEF-) is displayed, this will take you to the [SETTINGS] (SEt-) menu. If not, it will take you to the [SPEED REFERENCE] (rEF-) menu.
- RUN button: Controls powering up of the motor for forward running in LOCAL configuration and in REMOTE configuration if the [2/3 wire control] (tCC) parameter in the [INPUTS / OUTPUTS CFG] (I_O-) menu is set to [Local] (LOC), page 44. (could be hidden by door if function disabled)

STOP/RESET button
- Enables detected fault to be reset
- Can be used to control motor stopping
  - If [2/3 wire control] (tCC) is not set to [Local] (LOC), freewheel stop
  - If [2/3 wire control] (tCC) is set to [Local] (LOC), stop on ramp or freewheel stop during DC injection braking.

Note 1: In LOCAL configuration, the three LEDs REF, MON and CONF are blinking simultaneously in programming mode and are working as a LED chaser in control mode.

PROGRAMMING

Tips:

- Before beginning programming, complete the customer setting tables, page 106.
- Use the [Restore config.] (FCS) parameter, page 43, to return to the factory settings at any time.
- To locate the description of a function quickly, use the index of functions on page 105.
- Before configuring a function, read carefully the "Function compatibility" section on pages 18 and 19.
- Note:
  The following operations must be performed for optimum drive performance in terms of accuracy and response time:
  - Enter the values indicated on the (motor) rating plate in the [MOTOR CONTROL] (drC-) menu, page 38.
  - Perform auto-tuning with the motor cold and connected using the [Auto-tuning] (tun) parameter, page 40.
  - Adjust the [FreqLoopGain] (FLG) parameter, page 30 and the [Fr.Loop.Stab] (StA) parameter, page 31.

2. Apply input power to the drive, but do not give a run command.

3. Configure:

- ☐ The nominal frequency of the motor [Standard mot. freq] (bFr) page 38 if this is not 50 Hz,
- ☐ The motor parameters in the [MOTOR CONTROL] (drC-) menu, page 38, only if the factory configuration of the drive is not suitable,
- ☐ The application functions in the [INPUTS / OUTPUTS CFG] (I_O-) menu, page 44, the [COMMAND] (CtL-) menu, page 47, and the [APPLICATION FUNCT.] (FUn-) menu, page 59, only if the factory configuration of the drive is not suitable.

4. In the [SETTINGS] (SEt-) menu, adjust the following parameters:

- ☐ [Acceleration] (ACC), page 29 and [Deceleration], (dEC) page 29,
- ☐ [Low speed] (LSP), page 30 and [High speed] (HSP), page 30,
- ☐ [Mot. therm. current] (ItH), page 30.

5. Start the drive.

8 MAINTENANCE

INDEX

8.1 SAFETY PRECAUTIONS

8.1.1 Introduction

The personnel in charge of the machine operation and maintenance must be well trained and qualified and placed into positions that the customer requires, with an extensive knowledge of the safety and accident prevention standards and regulations in force.

Non-authorised personnel must strictly stay out of the work area during machine operations.

The safety and accident prevention warnings and precautions specified in this section must be accurately observed at all times during machine operation and maintenance, in order to avoid and prevent personal injuries as well as machine damages.

Said warnings and precautions are consistently referred to and further detailed throughout the Manual whenever there is a procedure involving injury or damage hazards, by way of **WARNING** and **DANGER** notes.

1. The **WARNING** notes normally precede an operation that, if not correctly performed, may incur in machine damages.
2. The **DANGER** notes normally precede an operation that, if not correctly performed, may lead to personal injuries.

DANGER
Before servicing the machine or performing any other intervention, always press the emergency button associated with the opening of the moving guards to disconnect the machine from the power supply. Check operation of this safety device regularly.

8.2 DANGER NOTES

- High voltage can cause death by electrocution. Operate at all times with maximum carefulness and in total compliance with the accident prevention standards and regulations in force in the country the machine is installed in.
- Prior to carrying out any operation at all on the machine, always ensure that all the main and secondary power and feed-in supplies have been totally cut-off. Hang specific warning signs such as “**MACHINE MAINTENANCE OPERATIONS UNDERWAY – DO NOT SWITCH ON THE POWER MAINS**” in the vicinity of the mains switch.
- During operations, moving machine parts all constitute a hazard and may lead to serious personal injuries. Strictly avoid coming into contact with said moving parts. Prior to starting any intervention at all on the machine, always ensure that there is no way that it can be started up by any one of the other components attached to it.
- Never remove the safety guards or cut off the protection devices installed on the machine.
- Failure to provide machine grounding may lead to serious personal injuries. Always make sure that the machine is accurately grounded and that all grounding operations are compliant with regulations in force.
- Avoid use of hazardous solvent products.
- Always ensure, prior to starting the machine up, that the personnel in charge of maintenance is at a safe distance from the machine and that no tools or materials have been left in the vicinity of the machine itself.
- Always use accident prevention and safety equipment during machine maintenance operations.
- Machine installation must always be updated and maintained according to all the accident prevention provisions in force. All machine moving parts and drives must be covered to prevent accidental contacts.
- Always check to make sure that all the machine safety guards and protection devices have been correctly closed prior to starting machine up.

- Extended overloading or failures may lead to electric motor and electrical equipment overheating, with the subsequent issue of hazardous fumes; in these cases, immediately power the machine down and do not approach the smoking areas until the smoke has been dispersed via adequate ventilation. Avoid inhaling the fumes forming internally to the machine devices during repair and maintenance interventions.
- Should the machine ever catch fire, never use water jets. Cut off all the power and feed-in supplies and use CO_2 fire extinguishers.
- Be very careful not to mix air and hydraulic oil when the system is under pressure. This to avoid explosive atmosphere hazards.
- Always ensure that the systems are all totally depressurized, prior to proceeding with any maintenance operation on the relative components.
- Keep well away from any discharge tap or hole during all system depressurising operations.
- Accurately inspect all the fittings and make sure there is no dust, oil, dirt or wear and tear on the threads, prior to proceeding with connecting them up.
- Make sure that all fittings and joints are accurately closed and tightened prior to pressurising the plant systems, after maintenance operations.
- Do not handle hydraulic fluid in the presence of electric sparks and/or naked flames.
- Always keep well away from any component that can be activated or moved by pressure, if the pressure contents have not been totally discharged from the plant systems. Do not wear objects that can tangle in with the machine and relative devices and act as conductors (chain necklaces, bracelets, etc.)
- **Ensure that all tools and equipment for use are always in perfect working conditions** and, where mandatory, that they are provided with **insulating handles and grips**. Check to ensure that the insulation on test equipment cables and conductors does not show the minimum sign of wear and tear, and/or damage.

8.3 WARNING NOTES

– Prior to putting the machine back into operation after a breakdown, it must be accurately inspected and controlled for evidence of any possible further damages.

– When using a ground detector to check any electrical device insulation, be sure to check that all the electronic control equipment is disconnected, to avoid relative component damages.

– Always use perfectly dry air during air pressure cleaning operations, with a pressure that does not go above 2 kg/cm².

– Always use tools and equipment that is in perfectly kept conditions and appropriate for the operations that need to be performed; the use of unsuitable tools and equipment may cause some serious damages.

– Any repair operations that may be possibly required, must be conducted in clean working environments, as free of dust as possible. Protect all the connection lights with plastic cover caps and keep dismantled pieces with machined surfaces accurately covered, until they are reassembled back onto the machine.

– Make sure that the lubrication system is working and distributing correctly at all times. Lubrication failures may seriously damage the sealing machine components.

– Never act on the adjustments and positioning of the overstroke or limit micro switches, unless expressly required to eliminate an anomaly. Tampering with them can lead to serious machine damages.

– Prior to proceeding with the reassembly of any assembly unit, always spread a thin layer of oil on the internal parts and coupling surfaces. Replace all gaskets with original spares prior to component reassembly.

✍ = **Note:**
Some of the machine components weigh over 20 kg. It is therefore necessary to use appropriate handling equipment for disassembly and reassembly operations thereof. Never work on said parts by yourself. G.MONDINI S.p.A. cannot be held in any way liable for damages incurring to persons and/or objects due to failure to observe this rule and/or any other safety-related rule and provision.

8.4 QUALIFICATION OF PERSONNEL ASSIGNED TO MACHINE MAINTENANCE OPERATIONS

8.4.1 General Duties

In order to meet the ever-rising need for qualification in the field of entirely automatic operated manufacturing system maintenance, personnel assigned to maintenance duties is required:

- to be knowledgeable of the standards and regulations in force on accident prevention during the performance of operations on motor-driven machines and to be capable of implementing them;
- to have read and understood the section on “Accident Prevention Safety” in the FOREWORD chapter;
- to be knowledgeable of the fundamental construction and operation principles of special component manufacturing systems
- to be capable of using and consulting the manufacturing documentations as well as the automatic sealing machine documents;
- to being responsible for independent decision-making, in as far as interventions on entirely automatic manufacturing systems are concerned;
- to be willing to adapt to any technological change made to the machine;
- to acknowledge production process irregularities and/or anomalies and, whenever necessary, to go ahead with any measure, as required.

8.4.2 Qualified Personnel duties and tasks

The make-up and qualification of the personnel teams specified in the maintenance plans, are as recommended by G.MONDINI S.p.A.

The different operations can, if necessary, be also performed by personnel with equal or higher qualifications, provided they have attended the relative training courses.

The professional job descriptions for personnel required to operate on the machine are:

8.4.2.1 Technician in charge of machine

Typical tasks:

Control and upkeep of production quality on entirely automatic manufacturing systems, and specifically:

- implementation and assessment of the user's guide and diagnostics system results
- use of the machine under standard operating conditions (in automatic work mode) and restoration of operations after emergency stop switch interventions
- the provision (upon request, as the case may be) of the pieces to be processed and of the assessment tools
- where necessary, quality control and implementation of provisions necessary for the maintenance of the nominal operation quality.
- cleaning of certain machine parts (rest-on and lock-on elements and parts)

cooperation for the performance of the following activities:

- maintenance activities
- troubleshooting and repairs

Performance of regular inspections/verifications, particularly:

- inspection of tubing and pipe tightness
- inspection to check lubrication efficiency
- inspection of the protection devices for any wear and tear
- inspection of the flexible pipe and tube systems, for any wear and tear
- inspection for any visible oil leaks around machine moving parts
- inspection for the absence of any foreign objects or foreign matter in the machine operation radius.
- inspection to see that all indicator and signal lamps are working
- work pressure and delivery level inspections involving the hydraulic, pneumatic and lubrication systems.

Technical knowledge required:

- knowledge on the use and operation of the **Closing machine TRAVE 350 VG**
- knowledge of the lubricants used in the machine and the hazards connected with use thereof
- machine breakdown logic research methods and assessment of the results
- organisation skills for the management and control of the measures necessary to restore the machine back into its functional production status.
- professional experience on automatic manufacturing and operation systems (automatic handling systems, element handling systems, etc.)
- ground knowledge of pneumatic, hydraulic and electric control and adjustment techniques.

Qualifications required:

- Complete industrial mechanic education and training, with specific skills in the technical automated system sectors
- Instruction and training on the Sealing machine are warranted by G.MONDINI S.p.A.

8.4.2.2 Technician in charge of lubrication

Typical tasks:

Lubricant tank filling and emptying operations, to be performed at regular intervals on entirely automatic manufacturing systems:

- inspection of the remaining lubricant levels
- lubricant tank cleaning and replacement of the lubricant tank contents
- top-up of the lubricant reserves depleted by grease pumps, oiling devices, oil canisters, etc.
- replacement of lubricants that are too old or used.

Technical knowledge required:

- Knowledge on lubricant qualities and the greases required for the various operations
- independent working skills according to pre-established lubrication plans
- knowledge of the correct spent lubricant scrapping methods as per provisions on environmental protection.

Qualifications required:

These operations can be performed by qualified personnel, that has undergone a sufficiently long training period on the machine.

8.4.2.3 Mechanical Maintenance Technician

Typical tasks:

Performance of preventive maintenance operations, revision and, where necessary, repairs on mechanical assembly units pertaining to entirely automatic manufacturing systems and specifically:

- drive belt replacement
- inspection of machine moving parts
- repairs on mechanical unit assemblies

Technical knowledge required:

- good knowledge of mechanical, pneumatic and hydraulic installations;
- knowledge on numeric control systems implemented in the devices
- knowledge on the fundamental principles of electric control and adjustment techniques
- inspection results assessment skills and decision making skills on necessary action
- knowledge on issuing revision reports
- knowledge on measurement and testing methods for the determination of the effective machine/handling status.

Qualifications required:

- Complete industrial mechanic education and training, with specific skills in the technical automated system sectors.
- Experience in automated handling system maintenance operations. Instruction and training on the machine are warranted by G.MONDINI S.p.A.

8.4.2.4 Electric/electronic Maintenance Technician

Typical tasks:

Performance of preventive maintenance operations, revision and, where necessary, repairs on electric and electronic assembly units pertaining to entirely automatic manufacturing systems and specifically:

- backup battery replacement operations
- analysis of microprocessor controlled device failures
- converter replacement and adaptation for three-phase servo motors.

Technical knowledge required:

- knowledge on three phase servo motors and relative adaptation to converter
- knowledge on research and repair methods of control system failures, performed via diagnostics systems, of computerised control systems and/or similar equipment.

Qualifications required:

- Complete industrial electronic technician education and training, with specific skills in the technical apparatus sectors
- Experience in machine tools maintenance operations. Instruction and training on the machine are warranted by G.MONDINI S.p.A.

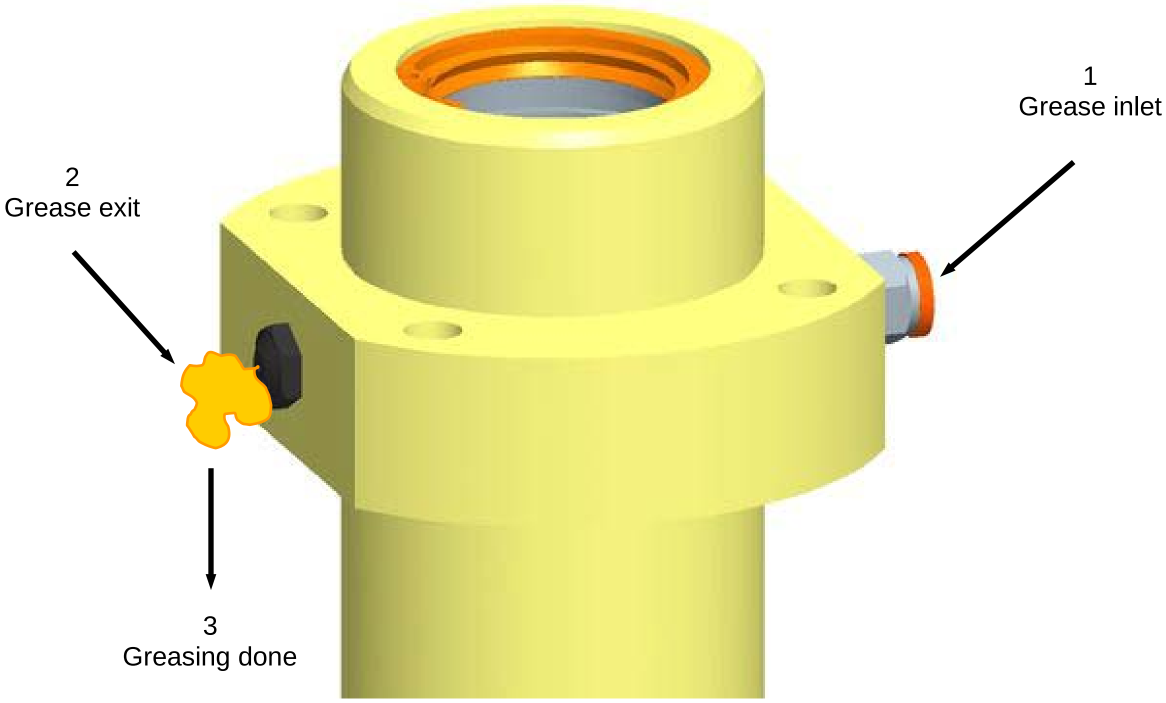
8.5 LUBRICATION SYSTEM

DANGER
Prior to carrying out any operation at all on the machine, always ensure that all the main and secondary power and feed-in supplies have been totally cut-off. Hang specific warning signs such as "MACHINE MAINTENANCE OPERATIONS UNDERWAY – DO NOT SWITCH ON THE POWER MAINS" in the vicinity of the mains switch.

The machine must be greased manually, using a pump grease.

To ensure proper lubrication of the machine is recommended to carry out the greasing operations when corresponding message appears on the operator panel.

To ensure that the greasing operations are performed correctly it is necessary to check that there is grease coming out of the bleed valves assembled on the part being greased (see the figure). This confirms the fact that greasing has been performed successfully.



During the assembly and manufacturer factory testing runs, our corporate selection is for a non-toxic food grease that is appropriate for the packaging of food products. Said grease is:

- **“NILS FOOD 00”** edible grease;

The general features of this grease are comparable to other greases available from a number of different companies:

COMPANY	NAME	GENERAL FEATURES
NILS	FOOD 00	
KLÜBER	Paraliq GA 3400	White medicinal oil with a paraffin base, and a complex aluminium NLGI 00 thickening agent
MOBIL	Mobilgrease FM 003	White medicinal oil with a paraffin base, and a complex aluminium NLGI 00 thickening agent

For what concerns lubricating oil of the clutch blocks our Company uses the following oil in the assembly phases:

- **“TOTAL Fluide ATX”** oil.

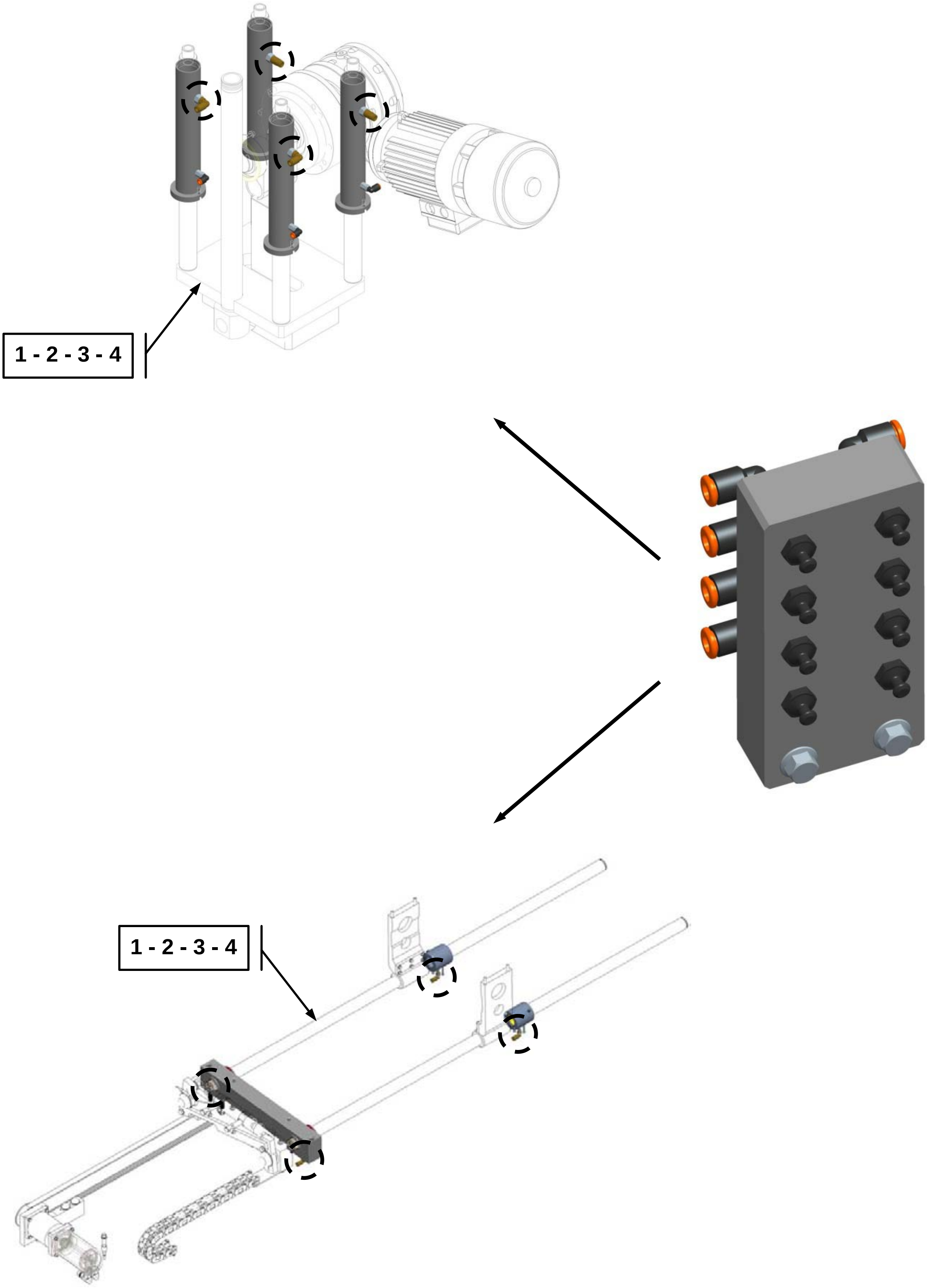
This product is comparable to a set of different oils available on the market, offering the same technical characteristics (the products listed as follows may have a different name on overseas markets)

COMPANY	NAME	GENERAL FEATURES
TOTAL	FLUIDE ATX	Dexron II D, Ford Mercon, MB 236.6, CAT TO 2
AGIP	ATF II D	
ELF	MAGIC G 2	
ESSO	A.T.F. D	
Q8	AUTO 14	
SHELL	DONAX TA	
TAMOIL	ATF II D	

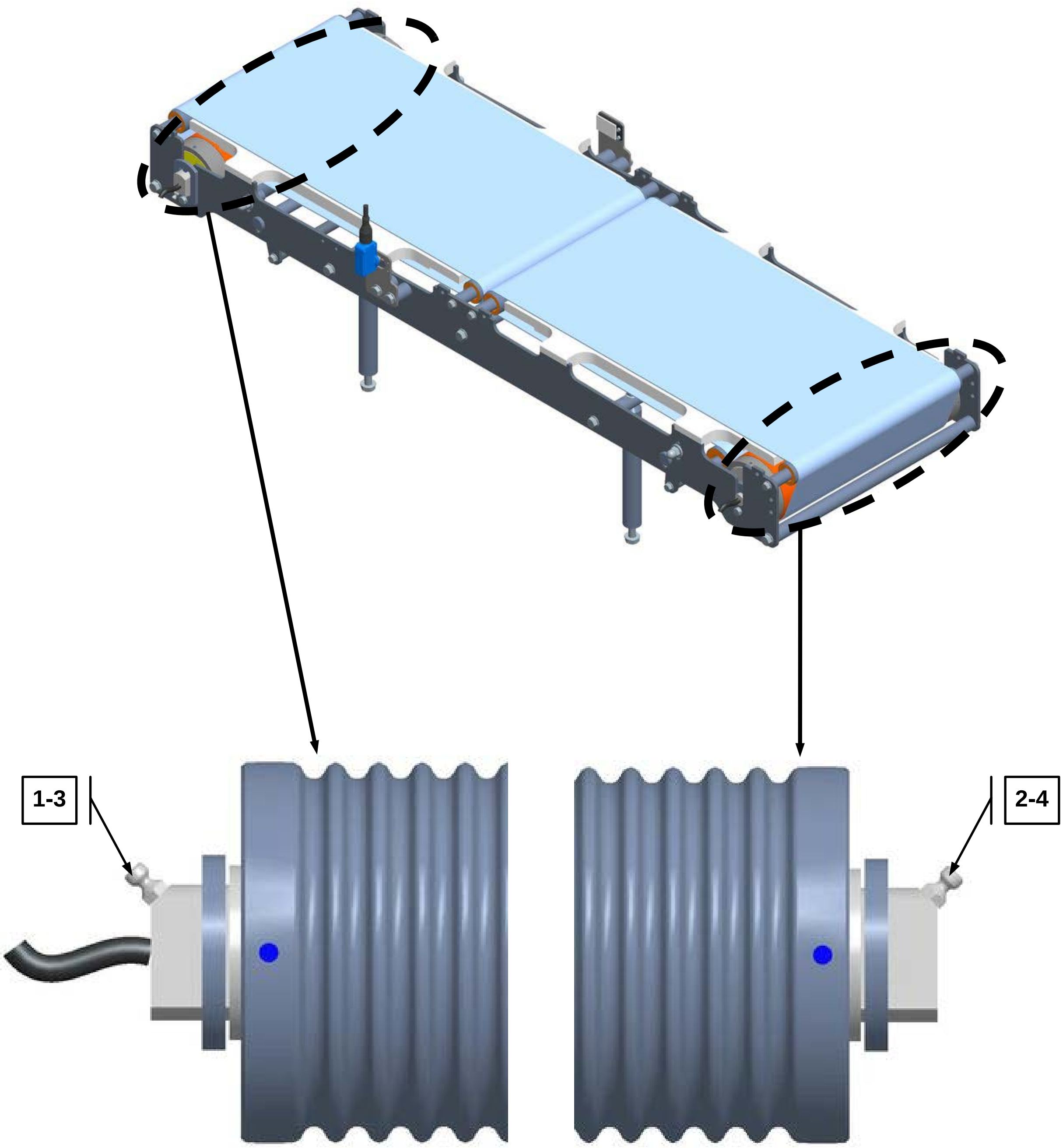
The groups that need a manual greasing are:

- Tool crank-handle (vertical tool movement) **whit manual distributor**;
- Pusher unit assembly (sliding group) **whit manual distributor**;
- Inlet belt Motor drums;
- Exit belt Motor drum;
- Sealing tool.

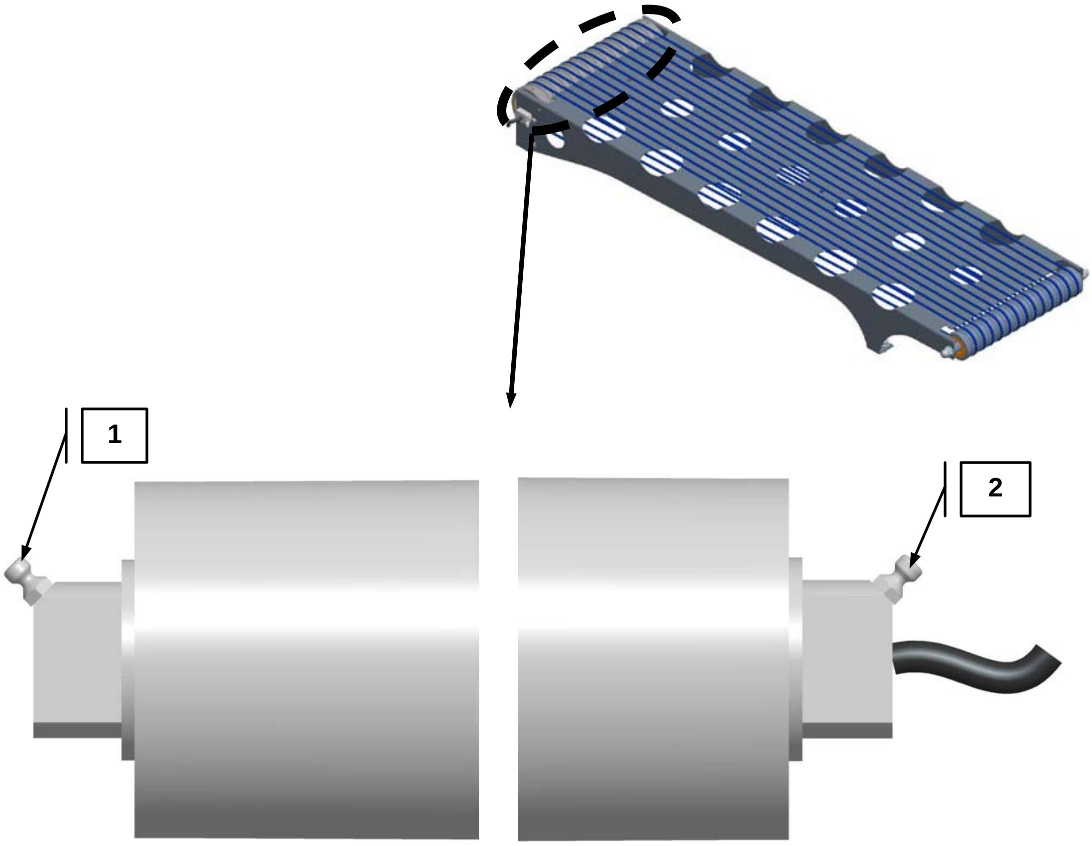
1. The tool crank-handle cam area is provided with **4** points that are lubricated with the use of the manual distributor.
2. The pusher unit areas are provided with **4** points that are lubricated with the use of the manual distributor.



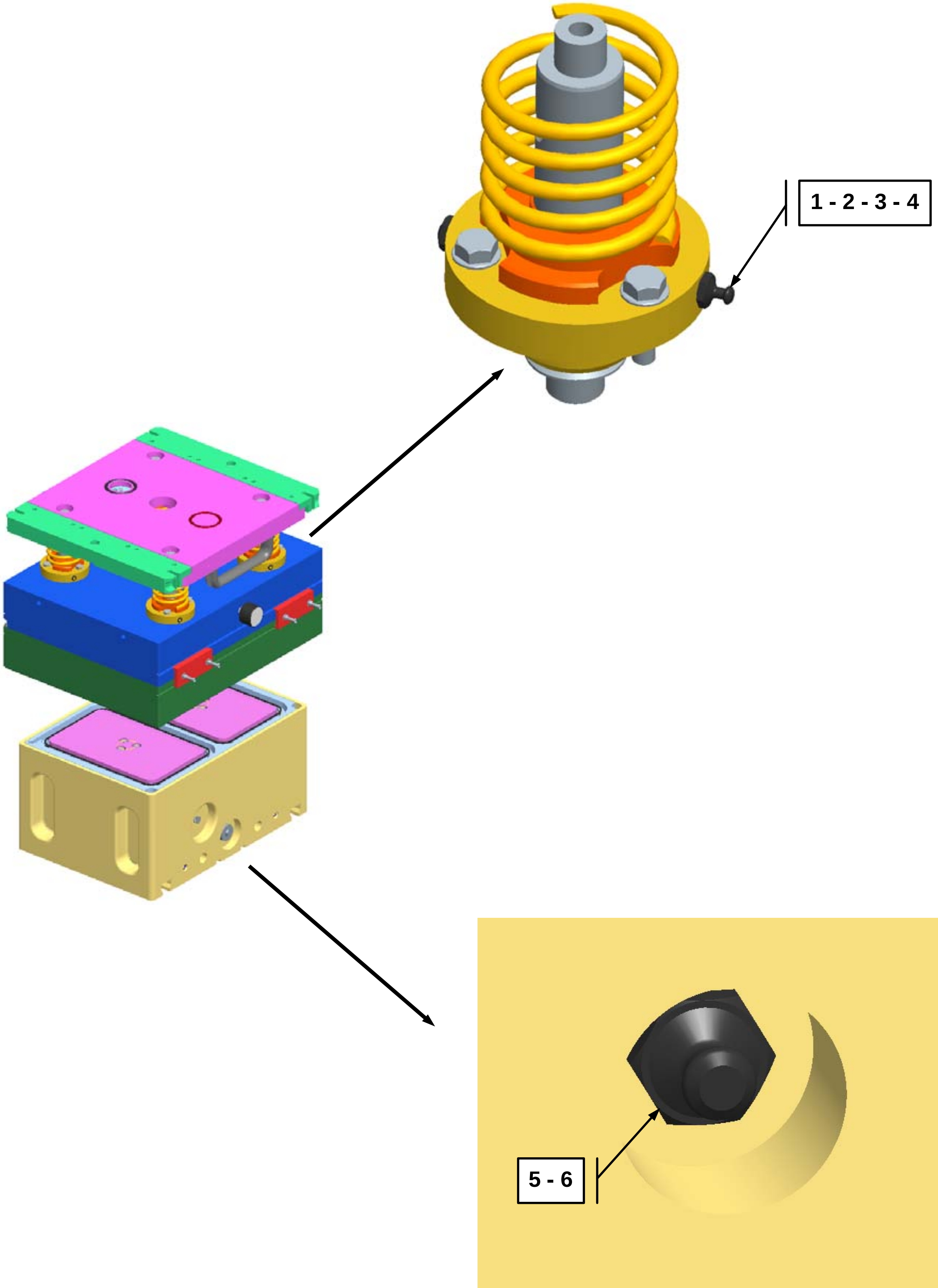
3. The inlet belt is provided with 2 or 4 greasing points.



4. The exit belt area, that has been provided with 2 greasing points that require manual lubrication.



5. The sealing tool area, that has been provided with **6** greasing points that require manual lubrication.



8.6 MACHINE CLEANING

As is the case for most of the machines in the G.MONDINI S.p.A. production range, the **Closing machine TRAVE 350 VG** has been designed and engineered to package foodstuff and derivatives, in any condition.

Said processes are thus all subject to very strict hygienic standards and regulations, to aptly avoid all the inconveniencies and hazards that could arise in the event that said standards and regulations are not observed.

Total machine operation cleanliness and hygiene is therefore of fundamental importance.

It is very important to proceed with machine cleaning operations straight after the machine has been switched off for the day or whenever it has terminated a production process cycle to avoid the formation of bacteria occurring in product residues left on the machine.

Prior to proceeding with machine cleaning, it is important to ensure that it is totally safe to do so. This to avoid incurring dangerous situations and hazards.

Machine cleaning operations can be taken care of by using high-pressure jets (with a maximum pressure of 40 bar and a temperature not higher than 50°C) on most of the machine surfaces. Nevertheless, it is necessary to be very careful not to direct the high pressure jets on any machine element that is particularly delicate, or that is not provided with a safety covering.

To avoid water filtering internally to the machine and thus causing damages to machine mechanical elements, motors and electrical plants, it is important that all the relative tightness safety seals are accurately checked at regular intervals.

WARNING! Do not point high-pressure water jets directly onto machine mechanical elements, motors and electrical system elements. Check to make sure that all the safety seals are accurately closed and be sure to also check the relative tightness safety seals at regular intervals.

As already underlined at the beginning of this paragraph, it is necessary to conduct machine cleaning operations not only for removal of dust accumulation, of substances that are extraneous to machine operations and any product residues but also in order to disinfect and sanitise any machine part that may come into contact with the products being packaged.

There are a number of non-dangerous detergent products on the market, non-harmful to persons, to the environment and to the machine itself, that can be removed with a simple rinsing operation.

It is the machine user's responsibility to further provide the machine cleaning personnel with all the products, equipment and tools necessary for their personal safety and protection and to duly inform said personnel on all the risks and hazards involved with the use and implementation of said cleaning products, equipment and tools.

The concentration and temperature levels of the detergent/disinfectant products selected for use must fully comply with all the relative product safety and technical specifications and must also be compatible with the machine construction materials.

Use of neutral Ph detergents and foams that combine disinfectant and sanitising properties with grease-removal properties is thus hereby recommended.

8.6.1 Table of recommended products

Most of the packaging systems designed and manufactured by G. MONDINI S.p.A. consist of light alloys, such as aluminium, and painted surfaces that may be etched by the chemical products used. It is therefore advisable to use as much as possible safe cleaning solutions. The use of aggressive chemicals for an effective cleaning affects the integrity of the materials used on G. MONDINI S.p.A. systems.

The table below provides a list of different recommended chemicals that have been tested on G. MONDINI S.p.A. equipment for all types of application in order to prevent all potential contamination under all possible weather conditions.

All cleaning products must comply with local requirements. Locally approved products may differ from this list.

If any of the recommended products is not available on the market, please feel free to contact G. MONDINI S.p.A. and jointly identify any other equally compatible products.

Application	Type of product	Trade name
Descaling	Acid SMS	Aluwash VA3
External foam cleaning	Alkaline SMS	EnduroFree VE14
	Alkaline Chlorinated SMS	Diverfoam SMS Chlor VF18
	Acid SMS	Acifoam P Free
Manual cleaning	Mid alkaline	Fatsolve VF21
	Daily cleaning and disinfection	Delladet VS2
	Neutral	Shureclean Plus VK9
Manual disinfection by immersion	Surfactant	Suredis VT1
	Surfactant	Divosan QC VT50
Manual disinfection by spraying	Alcohol-based solution	Divodes FG VT29
	Alcohol-based solution	Alcosan VT10
Manual disinfection by paper wipes	Alcohol-based solution	Divodes FG Wipes VT75
Hands wash	Sanitizing soap	Softcare Bac H4
Hands disinfection	Alcoholic solution	Softcare Med H5

8.6.2 Cleaning protocol

Cleaning equipment	
Products	**Concentr.**
Foam	Follow the label directions
General cleaner	Follow the label directions
Descaling agent	Follow the label directions
Disinfectant	Follow the label directions
Test kits required	
Water temperature	max. 50°C
Water pressure	max. 40 bar
Washing frequency	Daily
Washing methods	Manual and foaming
Personal protection equipment	
Eye protection	✓
Safety gloves	✓
Safety boots	✓
Protective suit	✓
Hard cap	✓
Tools and equipment	
Safety precautions	Always wear personal protective equipment. Never add water to chemicals. Always add chemicals to water. Immediately report any accidents!

Cleaning Protocol

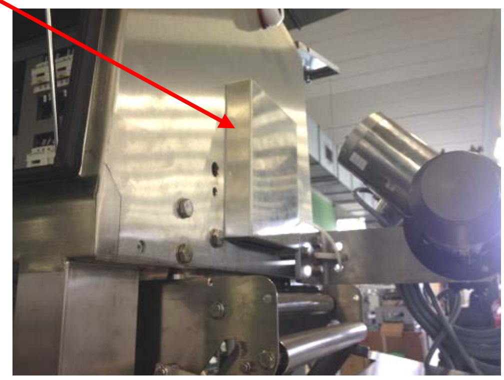
Immediately after production:

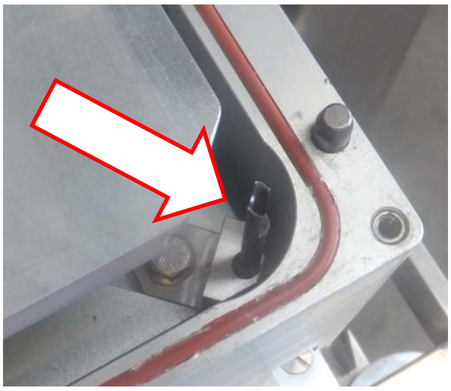
1. Activate all the stops required (disconnect the power and compressed air supply) and lock all the component parts before cleaning. Put in place the appropriate warning signs indicating that cleaning is in progress.
2. Clear the area involved of any packaging and processing materials.
3. Pre-clean the area with a shovel, brush and suitable non-abrasive tools. Place food waste in bags/containers that are suitable for disposal.
   Notes. Do not use compressed air to remove any dust from the equipment. This will cause dissemination of dust and microorganisms all around and on the equipment to be cleaned. If the use of compressed air is required, air must be filtered and applied at the lowest possible pressure.

During disinfection:

4. Make sure that the procedures under points 1 to 3 above have been properly performed.
5. Clean the switchboards and water-sensitive components by hand, and cover everything with disposable plastic bags.
6. Rinse the equipment that is sensitive to high pressures, the supporting structures and the surfaces in contact with the floor, using plain water at a medium-low pressure (not exceeding 40 bar). Do not rinse at a high pressure. Take great care and prevent water and chemicals from getting into contact with sensitive surfaces.
7. Rinse the floor, collect any contaminated food and put them into bins for disposal. Do not dispose of waste fluids in the drain system. They must be gathered and stored for final disposal.
8. Remove the equipment, such as belts and conveyors, moulding tools, valves and metering pipes (see details further on in this protocol) and position them in designated areas or on storage carts, and clean them separately (do not place them on the floor).
9. Manually clean the areas of the moulding tools in contact with the machine. Cover any vacuum and air ducts that have remained open due to the removal of equipment, using disposable plastic bags to prevent water from entering the pneumatic circuits.
10. Clear the machine of any deposits of dirt. Thoroughly rinse the area and remove contaminated food from the collecting bins.
11. Inspect the area and rinse again, if necessary. Wash up the floor and collection bins again.
12. Manually clean the equipment and sensitive surfaces that have not been washed in step 5 above.
13. Apply the cleaning solution or foaming cleanser on all washable surfaces of the component parts and floors, both under and around the system. Leave the solution to activate for the time needed (see the datasheet for details). Rub your hands vigorously and wash them thoroughly, if necessary.
14. Rinse with fresh water. Gather and remove all traces of cleaning solutions or contaminants.
15. Carefully check all the surfaces and make sure they are properly clean. If it is not so, apply the cleaning solution again and, if necessary, clean by hand.
16. If cleaning solutions have been re-applied, rinse thoroughly to remove any traces of contaminants and solution.
17. Repeat steps 14 and 15 if necessary.
18. Apply one of the recommended disinfectants or a similar one. When using a disinfectant other than the one recommended, it may be necessary to rinse the surfaces thoroughly.
19. Clean and dry the photoelectric cells using a soft non-abrasive cloth and a suitable disinfectant.
20. Remove and dispose of the plastic bags used to protect the electrical panels, the water-sensitive surfaces and the open vacuum and air ducts.
21. Remount the equipment as specified in the maintenance plan, taking care to keep it clean and disinfected.

See below for further details on cleaning.

Details	**Photographs**
Clean the control panels and water-sensitive components manually and protect them with a plastic bag. Do not spray water, cleaning solutions or disinfectants directly on panels or other electrical components – they must be protected against water. Example: Operator panel, electric motors and electrical connectors.	
Do not spray water, cleaning solutions or disinfectants directly on the detection points of the machine. The photoelectric cells must be cleaned using a non-abrasive soft cloth and be disinfected with specific alcohol-based products. Example: sensors, photoelectric cells Switchboard ventilation	 
Remove all additional equipment as specified in the maintenance plan (see also the details shown in this protocol) and position it on dedicated surfaces or storage carts for cleaning (do not place on the floor). Examples: Conveyor belts	

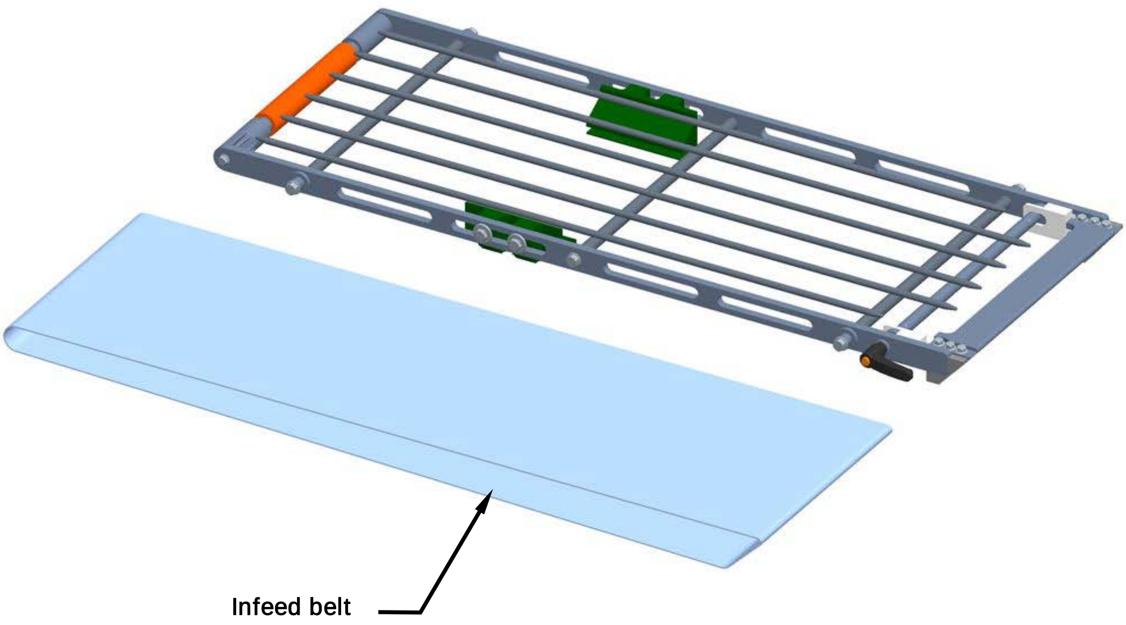
Cleaning the tool: The moulding tool must be removed from the machine, cleaned and disinfected manually, using recommended chemicals. Use a soft cloth and non-abrasive brushes, taking care of water-sensitive component parts. It must be disinfected using one of the recommended alcohol-based products.	
Disinfecting the Darfresh® tool during production: The lower Darfresh® tool consists of needles that perforate the tray and can therefore become contaminated with the product. Cleaning and disinfecting the needles during production would be difficult. We therefore suggest to disinfect them using one of the alcohol-based sprays.	 

8.6.3 Washing the infeed belts

The infeed belts are components of the sealing machine that require special care in terms of cleaning and disinfection. Here the trays are not yet sealed and there is the risk for the food coming into contact with the belt.

It is advisable to wash and disinfect the belt on a daily basis, using the most suitable solution among those recommended to prevent the onset of bacteria proliferation in the organic product exposed to the air for long, which may affect the hygiene of the machine.

Daily cleaning of the infeed belts must be effected with the belts removed from the machine. For extraordinary cleaning, the infeed belts must be removed completely, following the procedure described in subsection 8.11.2 of this handbook.



8.6.4 Cleaning the sealing tool

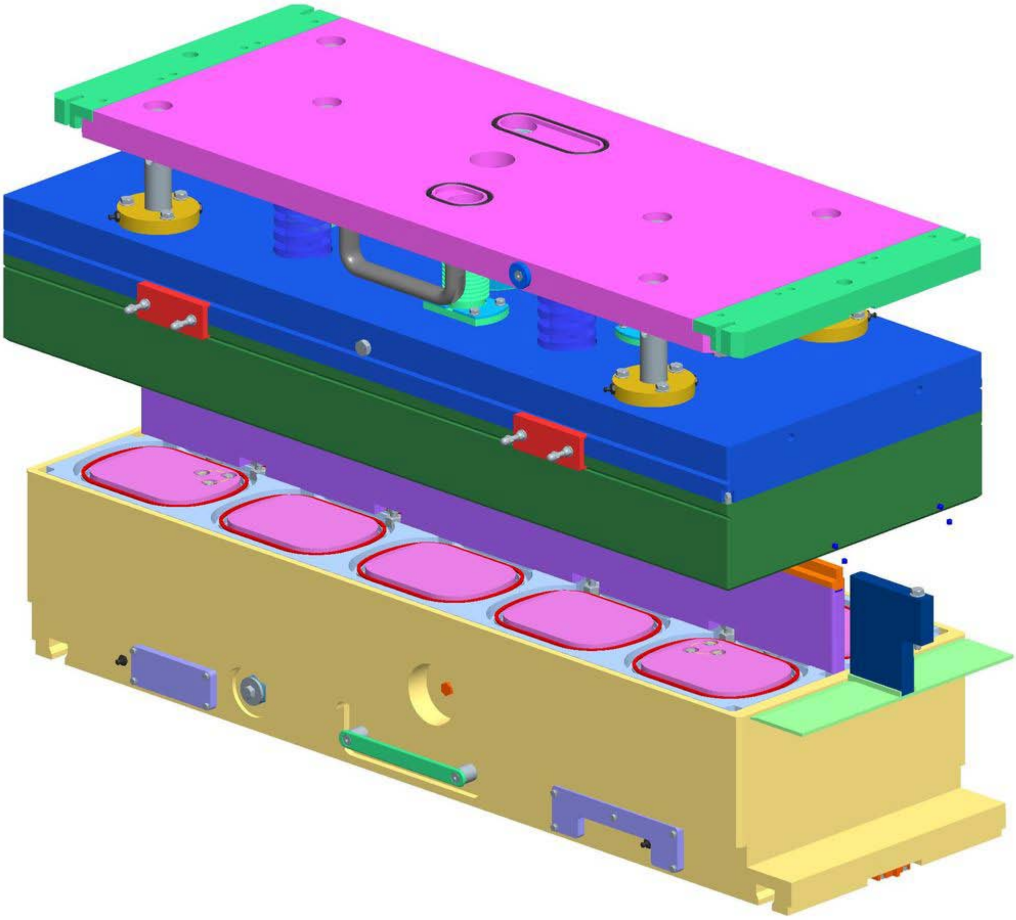
The sealing tool is another area of the machine that can easily get dirty and therefore requires special cleaning.

When entering the sealing tool, the tray is still open, filled with the metered product. During heat-sealing, the filling can spill out from the tray or even the tray can tip over in the tool.

In consideration of this and, more importantly, as a result of the fact that the tool comes into contact with the tray edge and the sealing film, it is extremely important to keep it clean and perfectly disinfected.

We recommend removing the sealing tool every day at the end of the production cycle. Leave it to cool down and clean it separately from the rest of the sealing machine.

After removing the various parts (counter-plate, intermediate plate, insulators) and accessing the internal structures, it is advisable to clean with a brush and remove any residue deposited on the tool. When all traces of product have been removed, blow the inside of the tool with compressed air at the lowest possible pressure, clean with a soaked soft cloth and sanitize with the recommended alcohol-based products. Then remount the tool back in position.



8.7 TIGHTENING TORQUES FOR NUTS AND BOLTS

For selection of the tightening torque class, consult table 5.2 and establish the bolt to be used as specified on table 5.3 and considering the traction force for each single bolt, as specified in table 5.4.

Table 5.2

TIGHTENING TORQUE CLASS	APPLICATIONS	TOLERANCES REFERRED TO NOMINAL TORQUE
I	EXTREMELY STRESSED: for extremely strained connections, subject to vibrations, impacts, high traction strengths or jarring movements, reducers, racks, connecting or piston rods, traverse motion stops, alternate loads and drive element components in general, dynamic hanging loads. INSPECTION TO BE PERFORMED EVERY 6 MONTHS USING A TORQUE WRENCH.	±5%
II	STRESSED: for uses not subject to excessive vibration, yet nevertheless requiring good locking safety (frames, static hanging loads, precision-locking, pallets, rollers, clamp and clench supports). INSPECTION REQUIRED ONCE A YEAR (USE OF A TORQUE WRENCH IS RECOMMENDED)	+5% -15%
III	LOW STRESS: for low stress applications or of secondary importance (blocks and plugs, references, dais and footboards, fencing) INSPECTION REQUIRED AT EVERY MACHINE INTERVENTION THAT MAY INFLUENCE TORQUE FEATURES, BASED ON THE TYPE OF OPERATION THAT IS PERFORMED.	+5% -35%

Table 5.3

THREAD	NUT AND/OR BOLT WRENCH SIZE	TIGHTENING TORQUE CLASS (Nm)		
		III	II	I
		SCREW STRENGTH CLASS		
		8.8	10.9	12.9
		NUT STRENGTH CLASS		
		8	10	12
M4	7	2.3	3.3	4
M5	8	4.8	6.8	8
M6	10	8	11.2	13.6
M8	13	20	28	32.8
M10	17	39.2	55.2	66.4
M12	19	68.8	96	116
M14	22	108	152	184
M16	24	168	236	284
M18	27	232	324	388
M20	30	328	464	552
M22	32	440	624	744
M24	36	568	800	960
M27	41	840	1200	1440
M30	46	1160	1600	1920

Table 5.4

THREAD	NUT AND/OR BOLT WRENCH SIZE	TIGHTENING TORQUE CLASS (Nm)		
		III	II	I
		SCREW STRENGTH CLASS		
		8.8	10.9	12.9
		NUT STRENGTH CLASS		
		8	10	12
M4	7	3120	4360	5240
M5	8	5080	7160	8560
M6	10	7200	10080	12080
M8	13	13200	18560	22320
M10	17	20960	29520	35440
M12	19	30640	43200	51600
M14	22	42000	59200	70800
M16	24	58400	81600	98400
M18	27	70400	99200	118400
M20	30	91200	128000	153600
M22	32	112800	159200	191200
M24	36	131200	184000	220800
M27	41	172000	241600	290400
M30	46	209600	294400	253600

8.8 COG BELT TENSIONING

The correct tensioning of the cog belts is essential, especially when servomotors are used.

If the belt is too taut. it causes early wear of the load-supporting bearings. If the belt is too loose, it causes incorrect positioning of the load being handled and irregular readings of the control system in the servomotors.

The belts can be tensioned in either of the following ways:

1. deflection is measured at the centre of the free section of the belt by applying a specific weight. The value should correspond exactly to the design values.
2. A properly calibrated tension-frequency metre is used to measure the oscillation of the belt and determine the desired tension.

The belts requiring a precise tensioning value are those making up:

- the pushers: forward and backward, open and close movements.
- Darfresh® on-tray / Zero system; forward and backward movements.

8.8.1 Belt tensioning tables

TYPE OF BELT AND SEALING MACHINE		Total force (N)	Single Force (N)	Type of belt	**WITH SONIC 507 C TENSION METER**				**WITH WEIGHT**	
					Mass (g/m)	Belt width (mm)	Pulley centre distance (mm)	**Frequency (Hz)**	Weight (Kg)	**Deflection (mm)**
Pushers Forward-Backward	TRAVE 340	-	-	-	-	-	-	**-**	-	**-**
	TRAVE 350	631	310	GT3-8	5,8	20	872	**30**	3,5	**17**
	TRAVE 367	631	315,5	GT3-8	5,8	20	1016	**25**	1,8	**10**
	TRAVE 384	631	315,5	GT3-8	5,8	20	1188	**22**	1,8	**12**
	TRAVE 1000	2209	1104,5	GT3-8	5,8	30	1396	**29**	10,5	**28**
	TRAVE 1200/590XL	2209	1104,5	GT3-8	5,8	30	1512	**27**	10,5	**30**
	TRAVE 1400	1894	947	Polychain 8	4,7	21	1872	**28**	8	**29**
Pushers Open - Close	TRAVE 1000/1200/590XL (stroke 100)	373	186,5	GT3-5	4,1	15	210	**130**	2	**4**
	TRAVE 1000/1200/590XL (stroke 160)	373	186,5	GT3-5	4,1	15	285	**185**	2	**6**
	TRAVE 1000/1200/590XL (stroke 200)	373	186,5	GT3-5	4,1	15	310	**90**	2	**6**
	TRAVE 1400 (stroke 160)	435	217,5	GT3-5	4,1	15	260	**115**	2,3	**5**
	TRAVE 1400 (stroke 200)	435	217,5	GT3-5	4,1	15	310	**75**	2,7	**6**
Darfresh® on Tray / Zero	TRAVE 340	631	315,5	GT3-8	5,8	20	624	**42**	3,6	**12**
	TRAVE 350	572	286	GT3-8	5,8	20	880	**28**	3,5	**18**
	TRAVE 367	600	300	GT3-8	5,8	20	1080	**24**	3,5	**22**
	TRAVE 384	600	300	GT3-8	5,8	20	1280	**20**	3,5	**26**
	TRAVE 1000	1578	789	GT3-8	5,8	20	1272	**33**	7,5	**25**
	TRAVE 1200/590XL	1578	789	GT3-8	5,8	20	1512	**28**	7,5	**30**
	TRAVE 1400	950	470	GT3-8	5,8	20	1672	**19**	5	**33**

8.9 PREVENTIVE MAINTENANCE

The following table lists the order of the preventive maintenance specifications designed for the **Closing machine TRAVE 350 VG**, complete with the time intervals for each intervention.

D	Daily
W	Weekly
M	Monthly
Q	Quarterly
Y	Yearly
WH	Working hours
WC	Work cycles
OR	On request

Frequency		Unit Assembly	Element	Maintenance	Description at section
First intervention	Repeated every				
M	D	Inlet grouping belt	Belts	Inspection of good belt working conditions and tensioning	8.11.1
M	W	Inlet grouping belt	Motor drums	Inspection of good working conditions	8.11.3
M	D	Timing Device	Cylinders	Inspection of good working conditions	8.12.1
M	D	Exit belt	Polycord belts	Inspection of good belt working conditions and tensioning	8.13.1
M	W	Exit belt	Motor drum and roller	Inspection of good working conditions	8.13.3

Frequency		Unit Assembly	Element	Maintenance	Description at section
First intervention	Repeated every				
M	W	Film unrolling roller	Servomotor and reducer	Inspection of good working conditions	8.14.1
Q	M	Film unrolling roller	Air shaft and bearings	Inspection of good working conditions	8.14.3
M	W	Film control and unroll rollers	Servomotor and reducer	Inspection of good working conditions	8.15.1
M	W	Film control and unroll rollers	Rubber coated shaft	Inspection of good working conditions	8.15.3
M	W	Film control and unroll rollers	Gearings	Inspection of good working conditions	8.15.5
M	W	Film control and unroll rollers	Cylinders	Inspection of good working conditions	8.15.7

Frequency		**Unit Assembly**	**Element**	**Maintenance**	**Description at section**
First intervention	Repeated every				
M	W	Winder Shaft	Motor and reducer	Inspection of good working conditions	8.16.1
M	W	Pusher arms	Servo motor and reducer	Inspection of good working conditions	8.17.1
M	D	Pusher arms	Cog belt	Inspection of good working conditions	8.17.3
M	W	Pusher arms	Cylinder	Inspection of good working conditions	8.17.5
M	W	Tool crank-handle	Motor and reducer	Inspection of good working conditions	8.18.1
W	D	Tool crank-handle	Connecting rod and bearings	Inspection of good working conditions	8.18.3
W	D	Tool crank-handle	Slider bushings	Inspection of good working conditions	8.18.5

Frequency		Unit Assembly	Element	Maintenance	Description at section
First intervention	Repeated every				
M	W	Upper toll	Intermediate cover, die cutter and sealer	Inspection of good working conditions	8.19.1
M	W	Upper tool	Upper plate and blower	Inspection of good working conditions	8.19.3
M	W	Upper tool	Internal plate and resistance	Inspection of good working conditions	8.19.5
M	W	Lower tool	Counter-tool assembly	Inspection of good working conditions	8.20.1
M	W	Lower tool	Plate and central guide	Inspection of good working conditions	8.20.3

8.10 REACTIVE MAINTENANCE

The following table lists the order of the reactive mechanical maintenance specifications designed for the **Closing machine TRAVE 350 VG**.

Unit Assembly	Element	Maintenance	Description at section
Inlet grouping belt	Belts	Replacement and tensioning	8.11.2
Inlet grouping belt	Motor drums	Replace	8.11.4
Timing Device	Cylinders	Replace	8.12.2
Exit belt	Polycord belts	Replacement and tensioning	8.13.2
Exit belt	Motor drum and roller	Replace	8.13.4
Film unrolling Roller	Servomotor and reducer	Replace	8.14.2
Film unrolling Roller	Air shaft and bearings	Replace	8.14.4

Unit Assembly	Element	Maintenance	Description at section
Film control and unroll rollers	Servomotor and reducer	Replace	8.15.2
Film control and unroll rollers	Rubber coated shaft	Replace	8.15.4
Film control and unroll rollers	Gearings	Replace	8.15.6
Film control and unroll rollers	Cylinders	Replace	8.15.8
Winder shaft	Motor and reducer	Replace	8.16.2

Unit Assembly	Element	Maintenance	Description at section
Pusher arms	Motor and reducer	Replace	8.17.2
Pusher arms	Cog belt	Replace	8.17.4
Pusher arms	Cylinder	Replace	8.17.6
Tool crank-handle	Motor and reducer	Replace	8.18.2
Tool crank-handle	Connecting rod and bearings	Replace	8.18.4
Tool crank-handle	Slider bushing	Replace	8.18.6

Unit Assembly	Element	Maintenance	Description at section
Upper toll	Intermediate cover, die cutter and sealer	Replace	8.19.2
Upper tool	Upper plate and blower	Replace	8.19.4
Upper tool	Internal plate and resistance	Replace	8.19.6
Lower tool	Counter-tool assembly	Replace	8.20.2
Lower tool	Plate and central guide	Replace	8.20.4

8.11 INLET GROUPING BELT

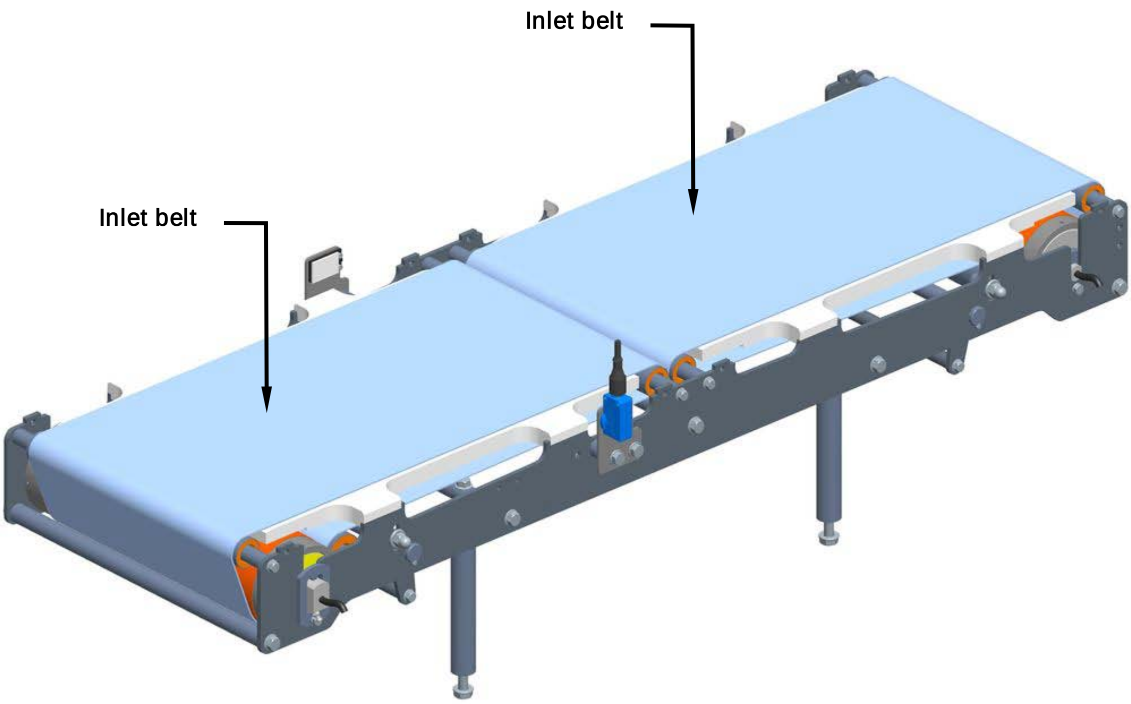
8.11.1 Checking on good inlet belts working conditions and tensioning

Machine status	Power ON.
Equipment	Standard workshop tools

During standard machine operation cycles, carry out a visual inspection to ensure that the inlet belts are in good working condition. Check that they are accurately centred, that there are no glitches whilst it feeds along and that it is not worn and/or torn.

Next, check that the belts are correctly tensioned, i.e. that the units being carried out are all moving correctly. Check too that the belts are well pressed down to contact the rollers during movement.

Should any anomalies be found, please warn the person responsible/people in charge of setting the unit group back to optimised working conditions.

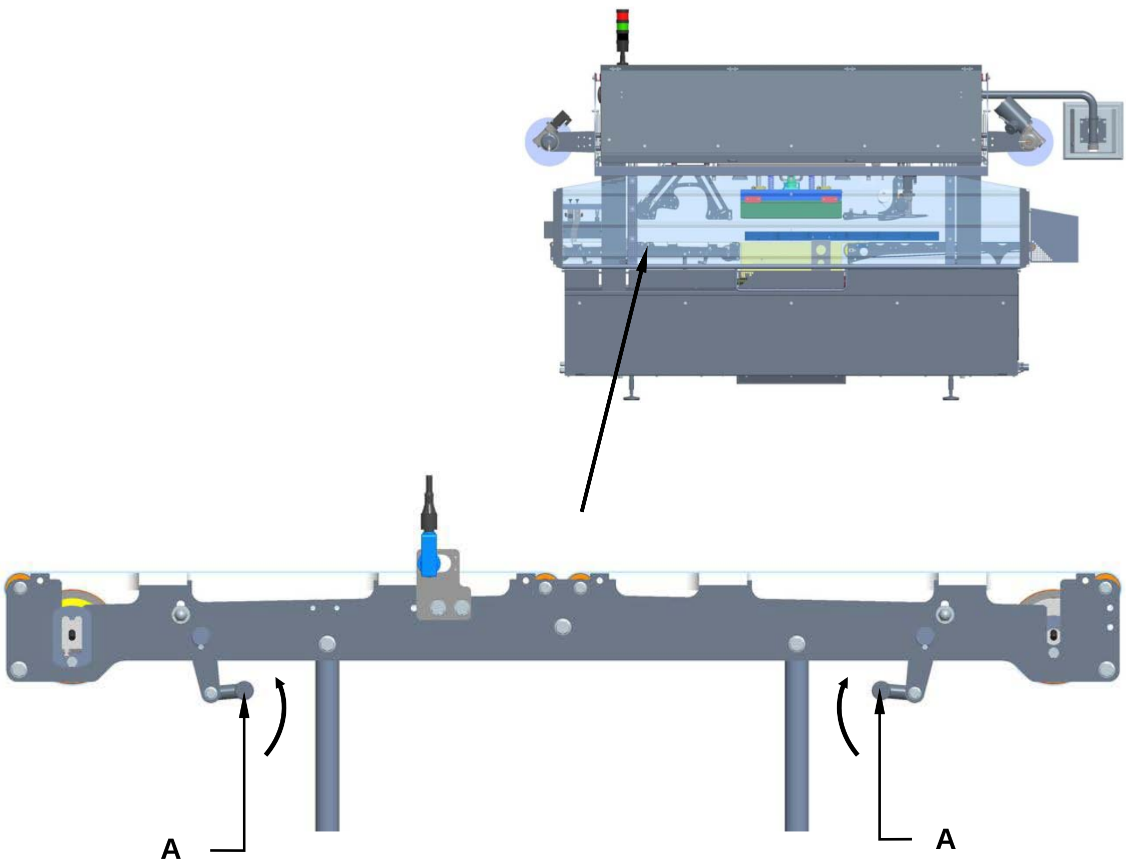


8.11.2 Replacement and tensioning of the inlet belts

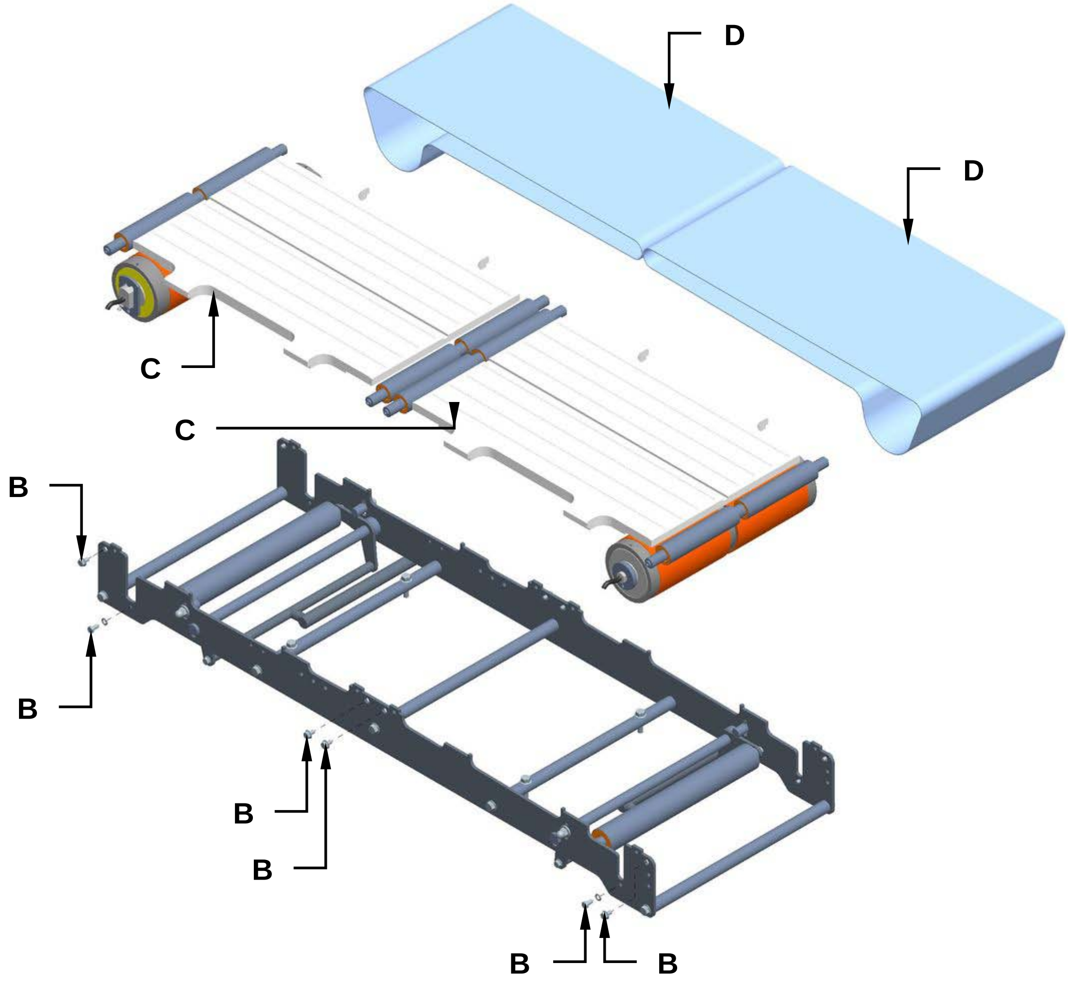
Machine status	Power totally disconnected
Equipment	Standard workshop tools

For inlet belts replacement, proceed as follows:

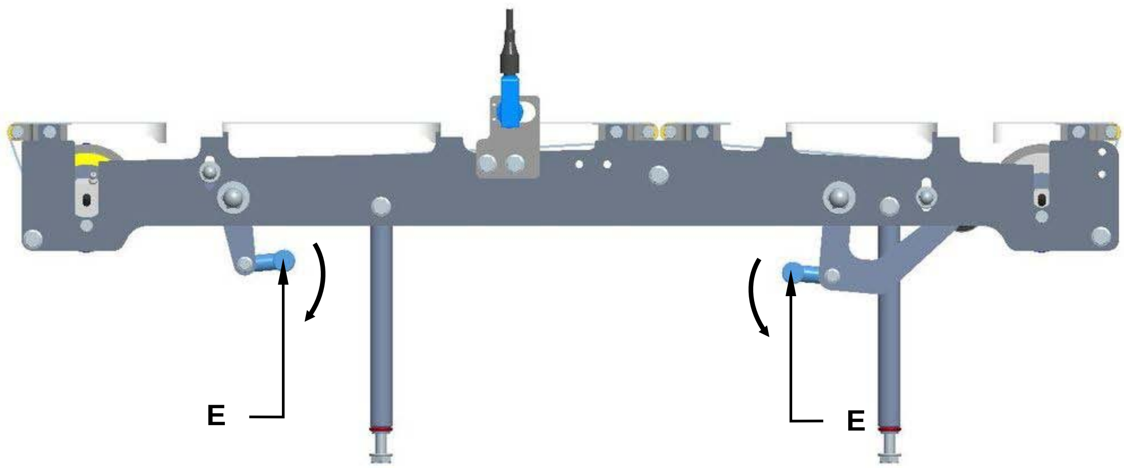
1. Disconnect the power supply of the machine.
2. Use the appropriate lever (**A**) to unlock the short infeed belts.



2. Unscrew the screws (**B**) and remove the belt supports (**C**).
3. Replace the belts (**D**) and reassemble all the components following the preceding procedure to bashful.



4. Extend the belts rotating the two levers (**E**).

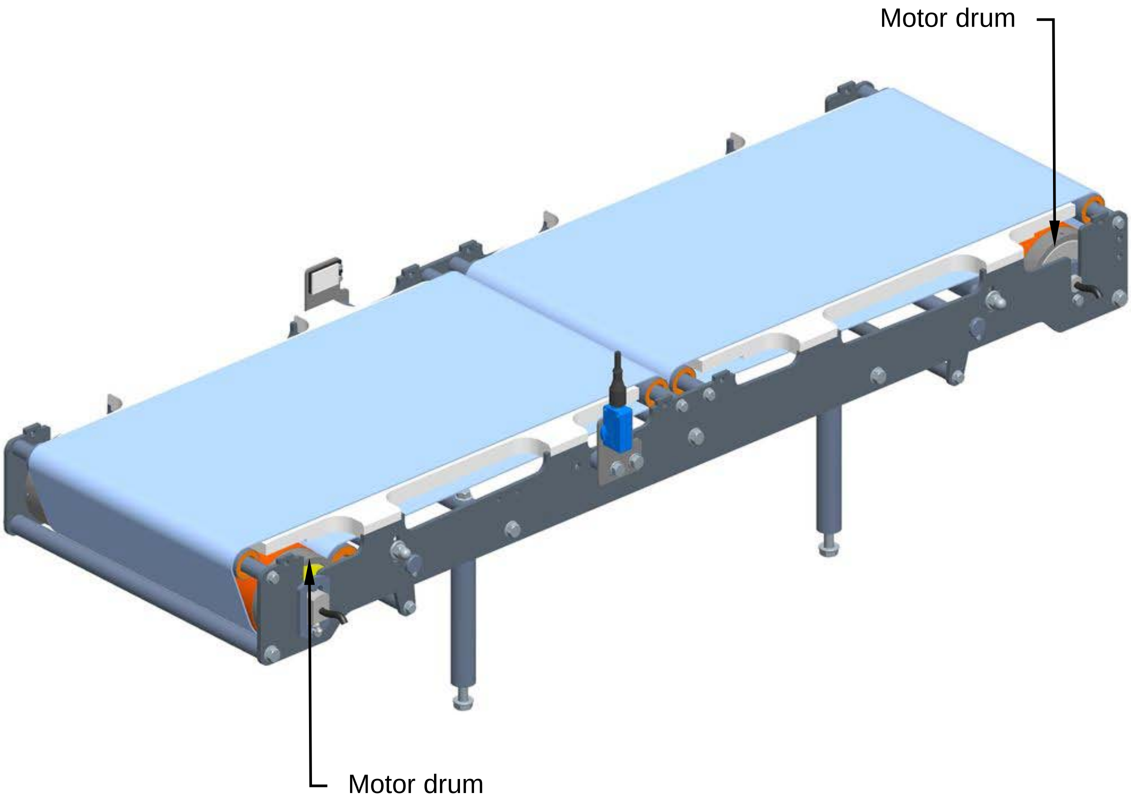


8.11.3 Checking the motor drums good working conditions

Machine status	Power ON.
Equipment	Standard workshop tools

During standard machine operation cycles, carry out a visual inspection to ensure that the motor drums are in good working conditions. Check to ensure that direction the motor drum is turning in, is correct.

Should any anomalies be found, please warn the person responsible/people in charge of setting the unit group back to optimised working conditions.

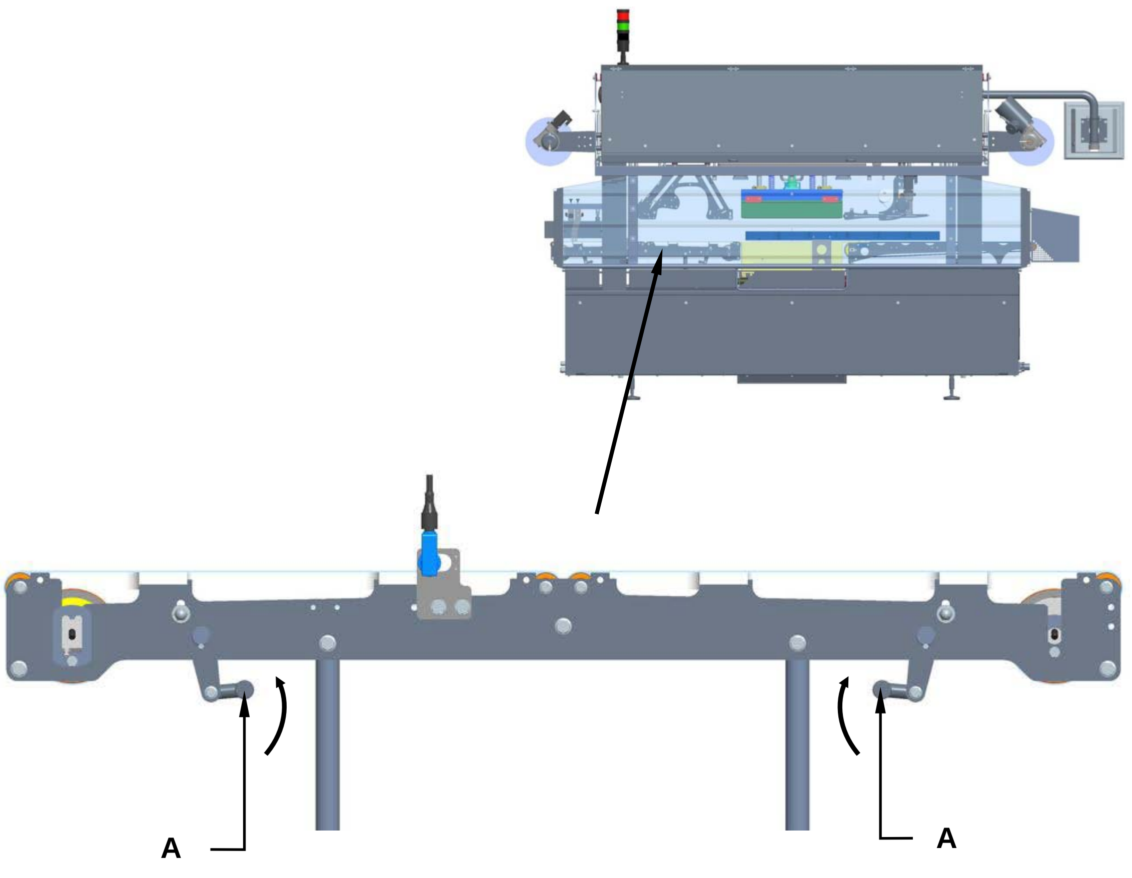


8.11.4 Motor drums replacement

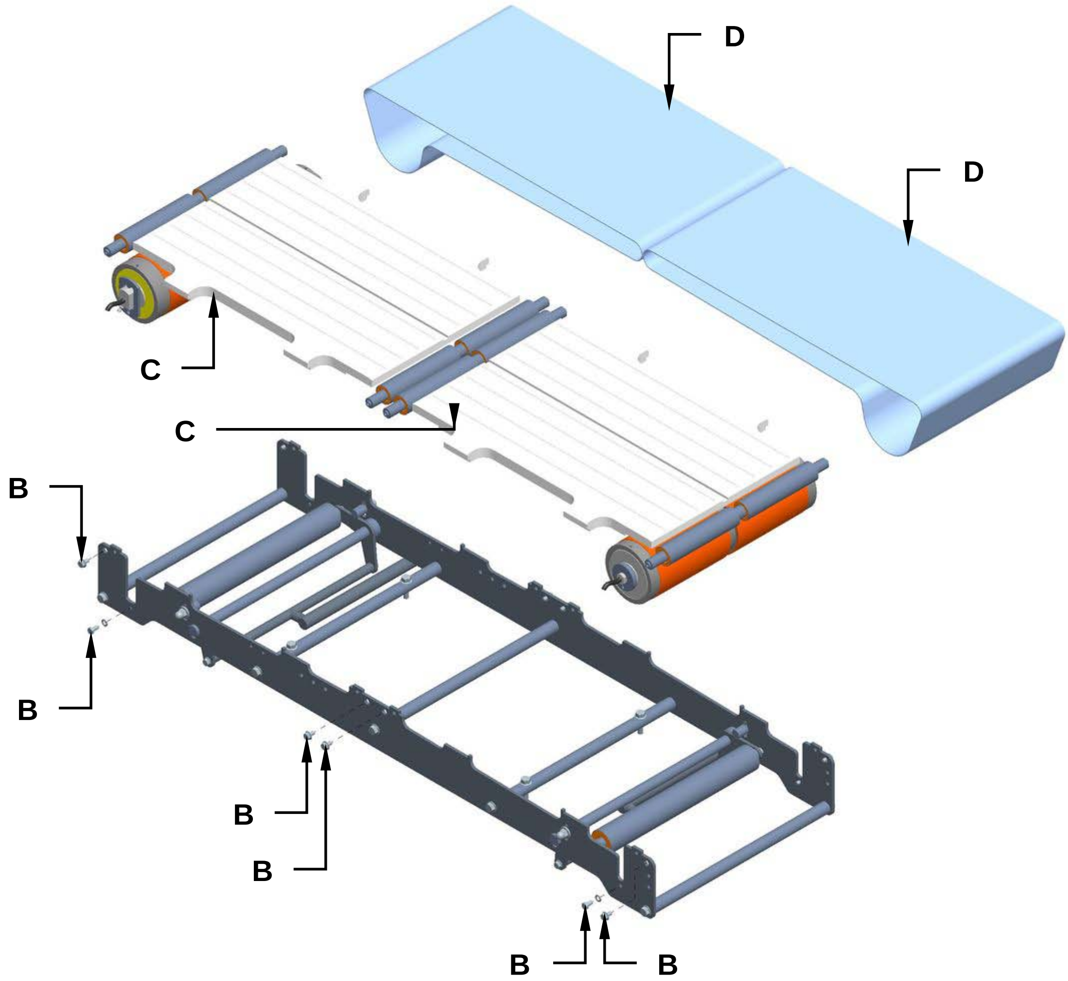
Machine status	Power ON.
Equipment	Standard workshop tools

For motor drums replacement, proceed as follows:

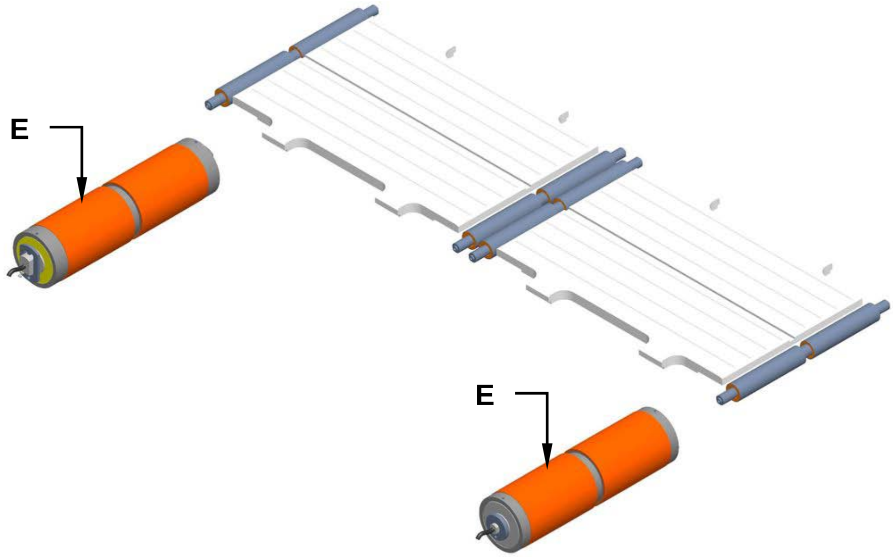
1. Disconnect the power supply of the machine.
2. Use the appropriate lever (**A**) to unlock the short infeed belts.



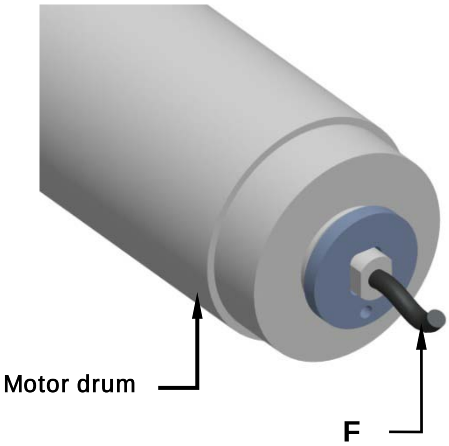
2. Unscrew the screws (**B**) and remove the belt supports (**C**).
3. Remove the belts (**D**).

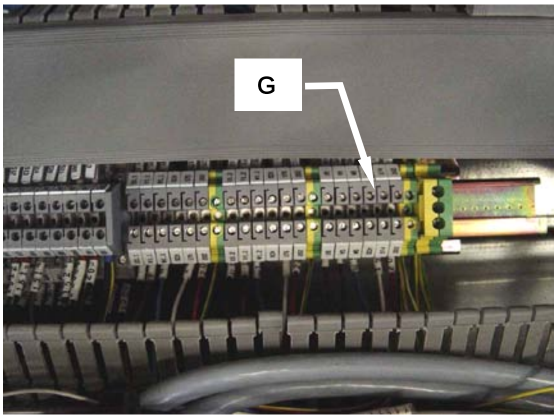


4. Replace the motor drums (**E**).



5. Disconnect the electric cable (**F**) from the terminal block (**G**) located in the electric control cabinet by consulting and using the code number specified both on the electric cable and on the terminal block. Slip the electrical cable out on the motor drum side and after having assembled on the new motor drum, slip its electric cable back on in the same way the replaced cable was on.





6. Reassemble all the parts back on again by reverse procedure.

8.12 TIMING DEVICE

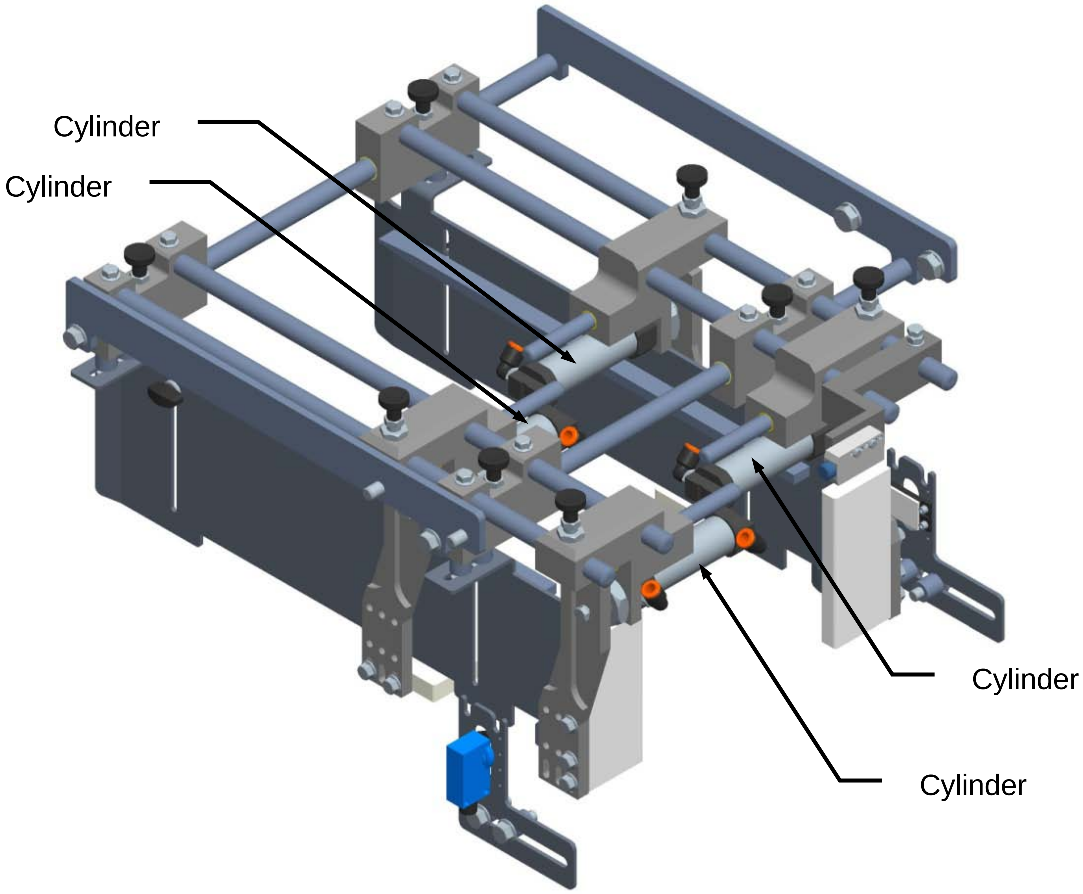
8.12.1 Checking the timing device cylinders

Machine status	Power sources activated
Equipment	Standard workshop tools

During standard machine operation cycles, carry out a visual inspection to ensure that the cylinders are in good working condition.

Check to make sure that there are no air leaks. This operation must be conducted by highly qualified service personnel with all the relative safety and accident prevention precautions and never during machine operation.

Should any anomalies be found, please warn the person responsible/people in charge of setting the unit group back to optimised working conditions.

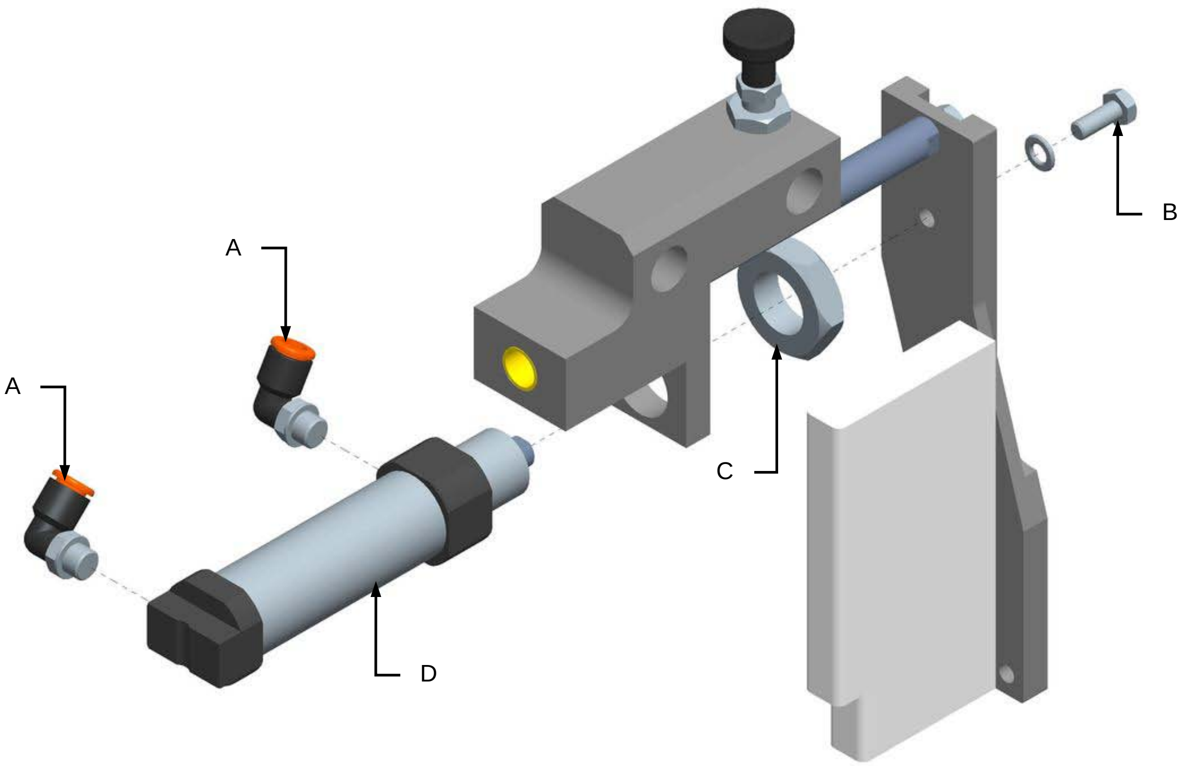


8.12.2 Replacing the timing device cylinders

Machine status	Power sources deactivated
Equipment	Standard workshop tools

Proceed as follows to replace the cylinders:

1. Disconnect the compressed air supply.
2. Remove the fittings (**A**).
3. Unscrew the screw (**B**) and the nut (**C**).
4. Replace the cylinder (**D**).



5. Remount everything, following the above procedure in the reverse order.

8.13 EXIT BELT

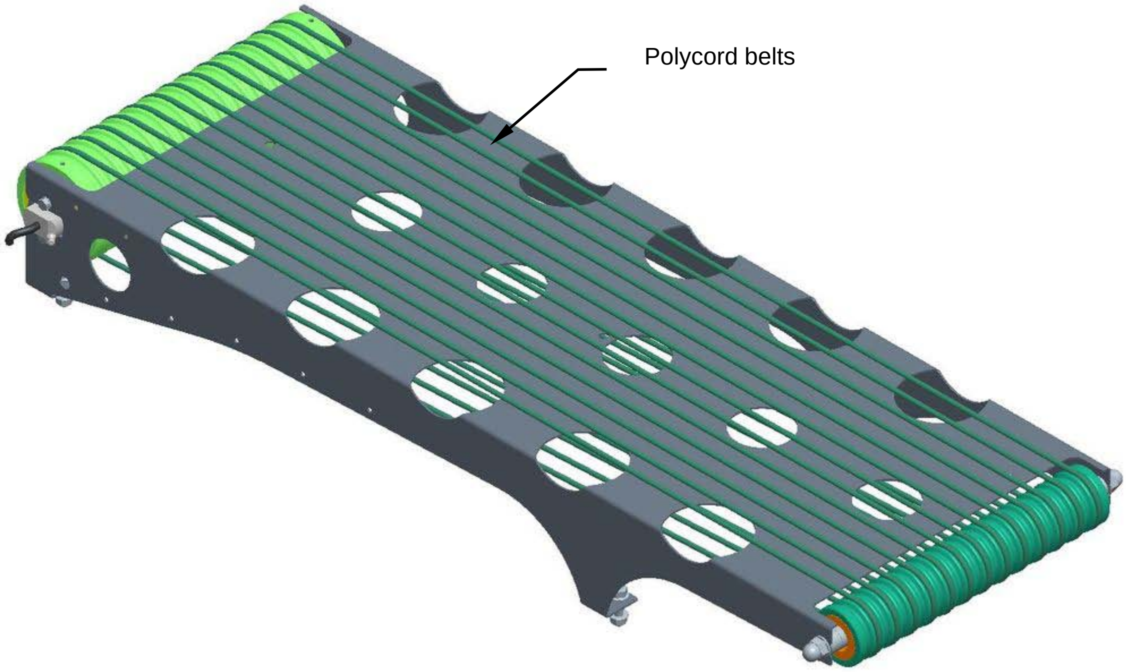
8.13.1 Inspection of the Polycord belt conditions

Machine status	Power ON.
Equipment	Standard workshop tools

During standard machine operation cycles, carry out a visual inspection to ensure that the Polycord belts are in good working condition. Check that they are accurately centred, that there are no glitches whilst they feed along and that they are not worn and torn.

Next, check that the belts are correctly tensioned, i.e. that the units being carried out are all moving correctly. Check too that the belts are well pressed down to contact the rollers during movement.

Should any anomalies be found, please warn the person responsible/people in charge of setting the unit group back to optimised working conditions.

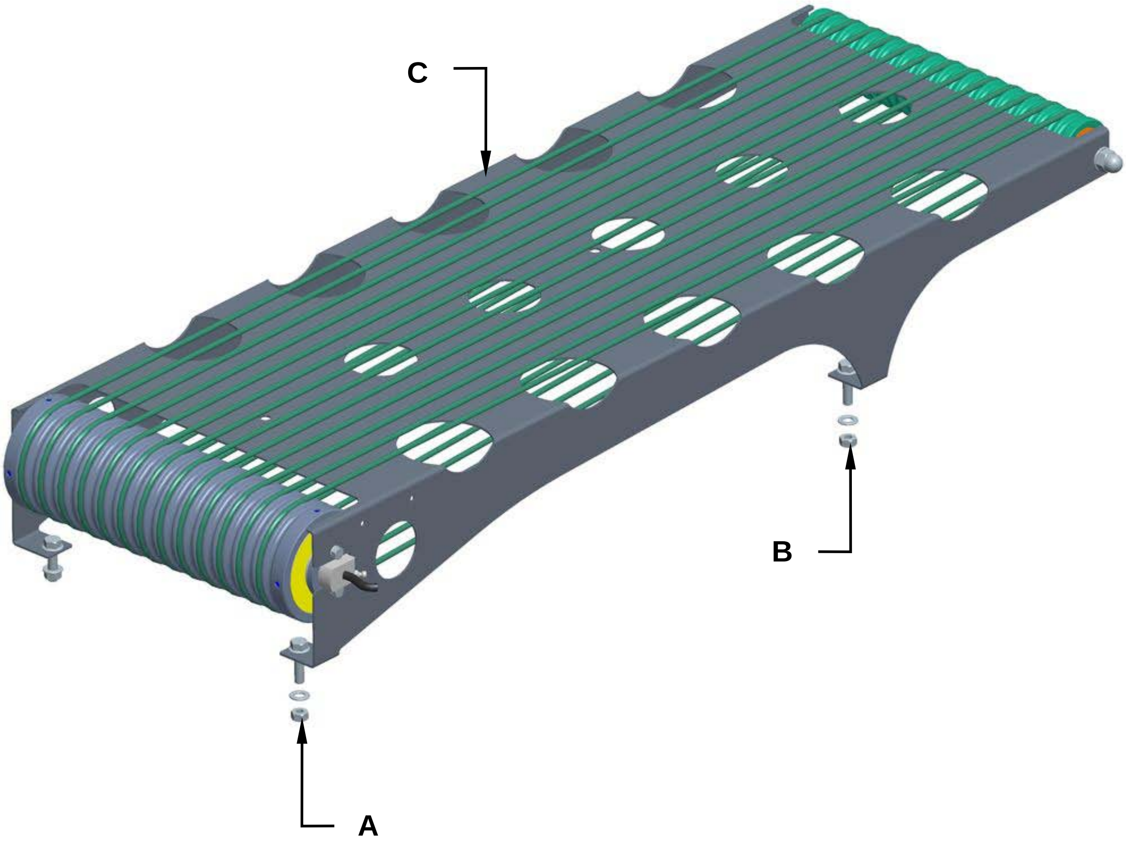


8.13.2 Replacement of the Polycord belts

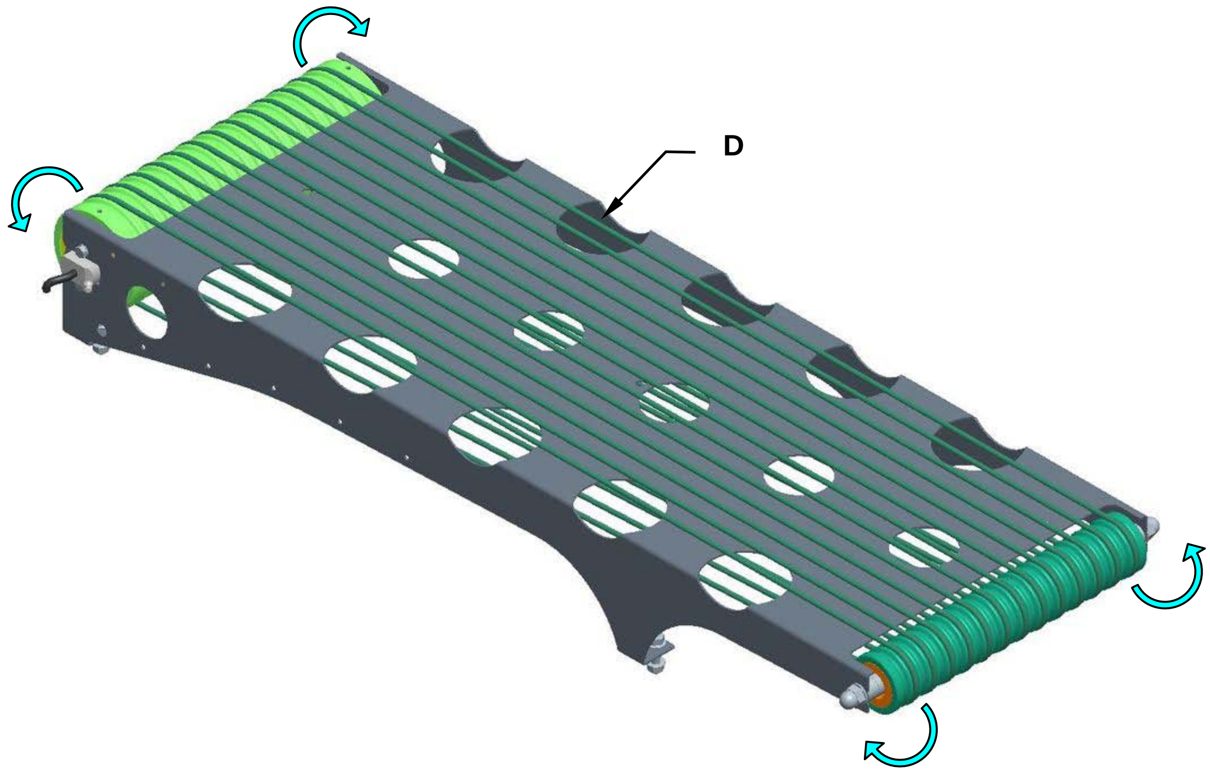
Machine status	Power ON.
Equipment	Standard workshop tools

For Polycord belts replacement, proceed as follows:

1. Unscrew the screws (**A**) and (**B**), remove the exit belt from the closing machine.



2. Remove the Polycord belts (**D**) from the drum motor and the roller.



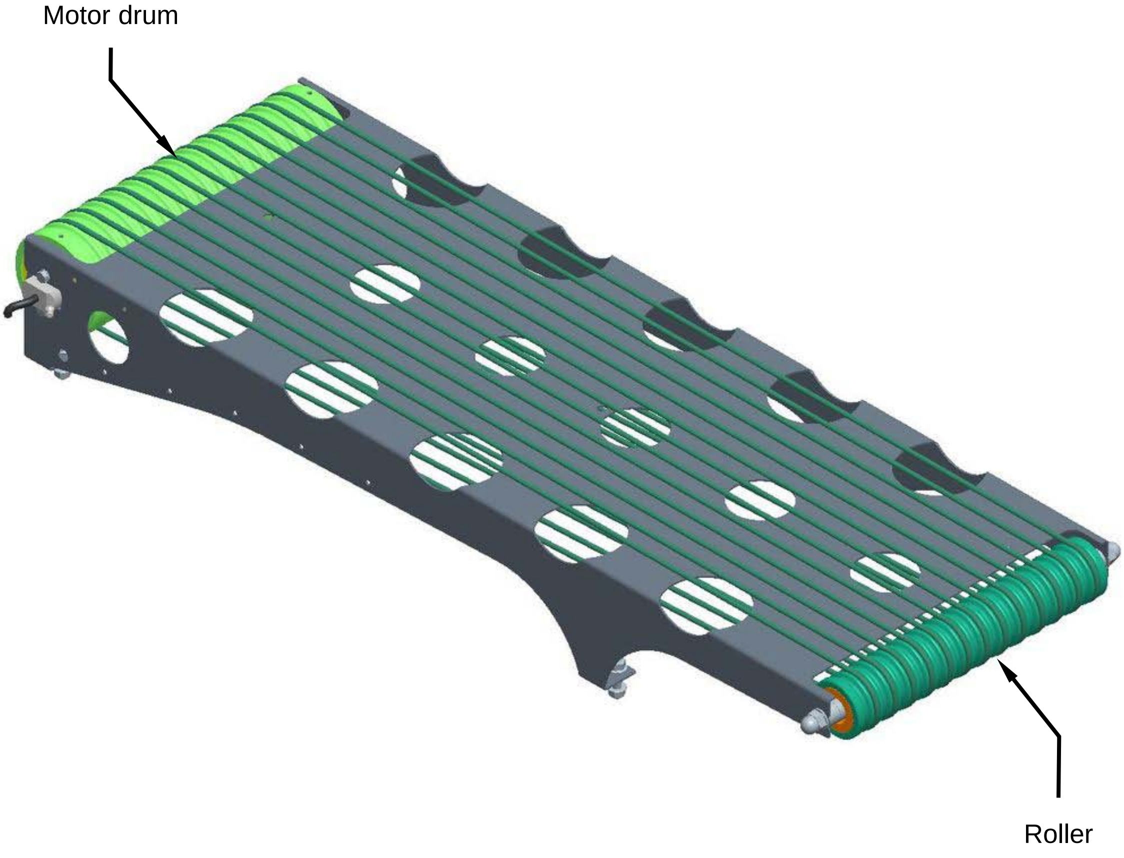
3. Replace the Polycord belts and assemble all the components following the preceding procedure to bashful.

8.13.3 Inspecting the motor drum and roller good working conditions

Machine status	Power ON.
Equipment	Standard workshop tools

During standard machine operation cycles, carry out a visual inspection to ensure that the motor drum and roller are in good working conditions. Check to ensure that direction the motor drum is turning in, is correct.

Should any anomalies be found, please warn the person responsible/people in charge of setting the unit group back to optimised working conditions.

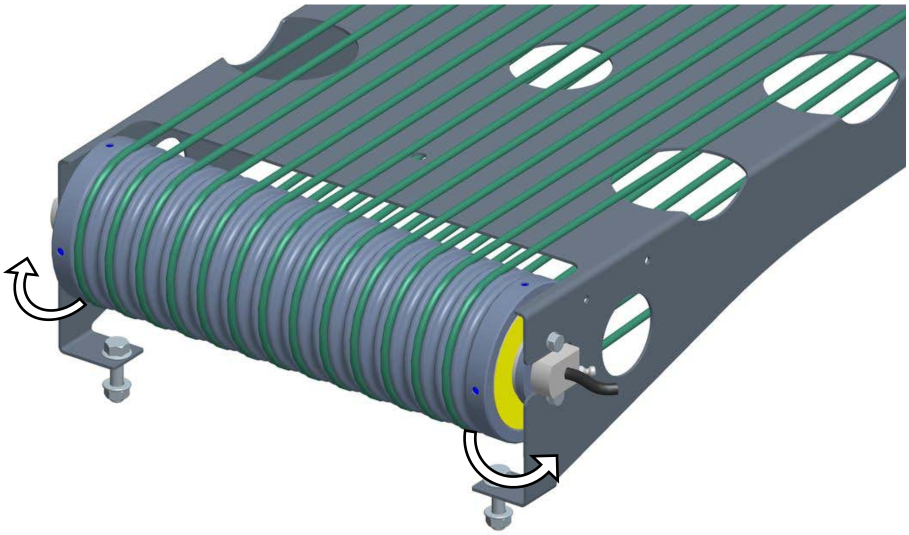


8.13.4 Motor drum and roller replacement

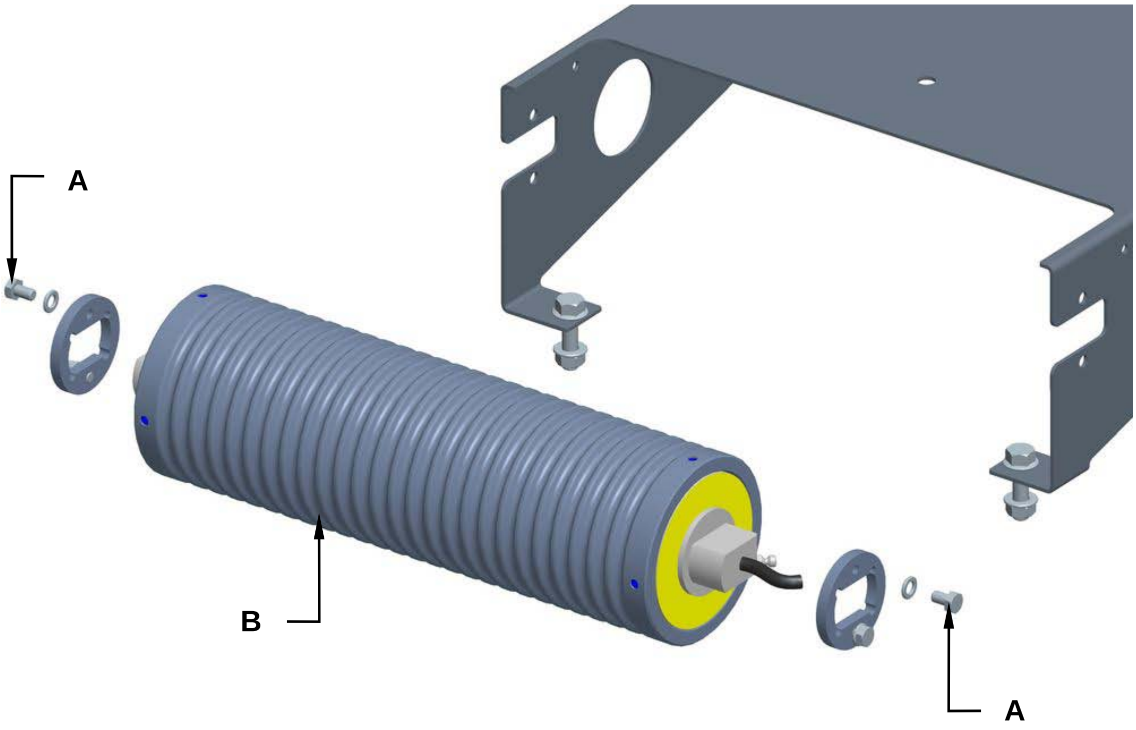
Machine status	Power ON.
Equipment	Standard workshop tools

For motor drum replacement, proceed as follows:

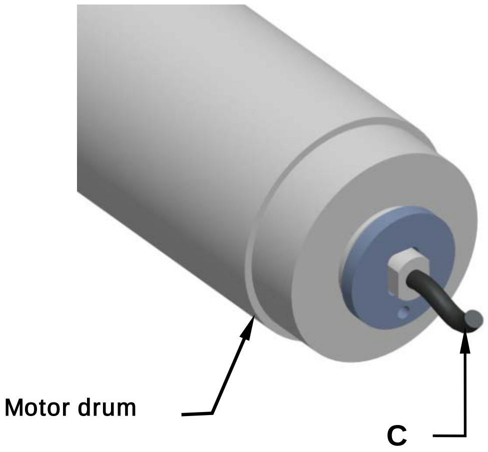
1. Remove the Polycord belts from the drum motor.

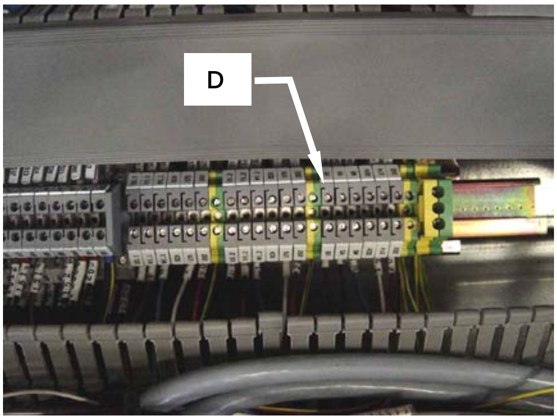


2. Unscrew the screws (**A**) and replace the drum motor (**B**).



3. Disconnect the electric cable (**C**) from the terminal block (**D**) located in the electric control cabinet by consulting and using the code number specified both on the electric cable and on the terminal block. Slip the electrical cable out on the motor drum side and after having assembled on the new motor drum, slip its electric cable back on in the same way the replaced cable was on.

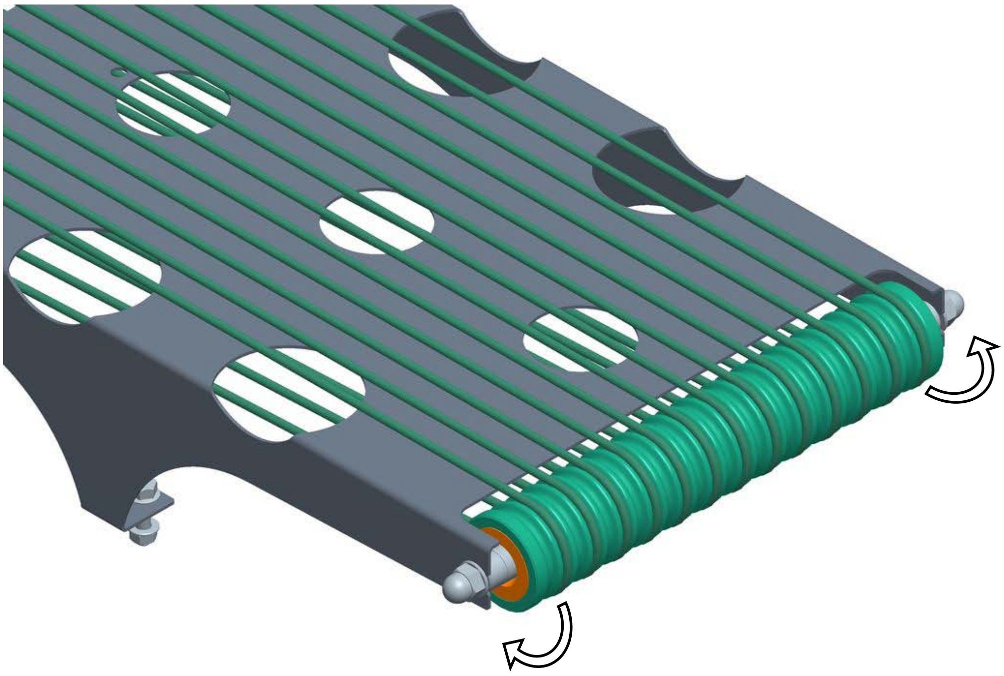




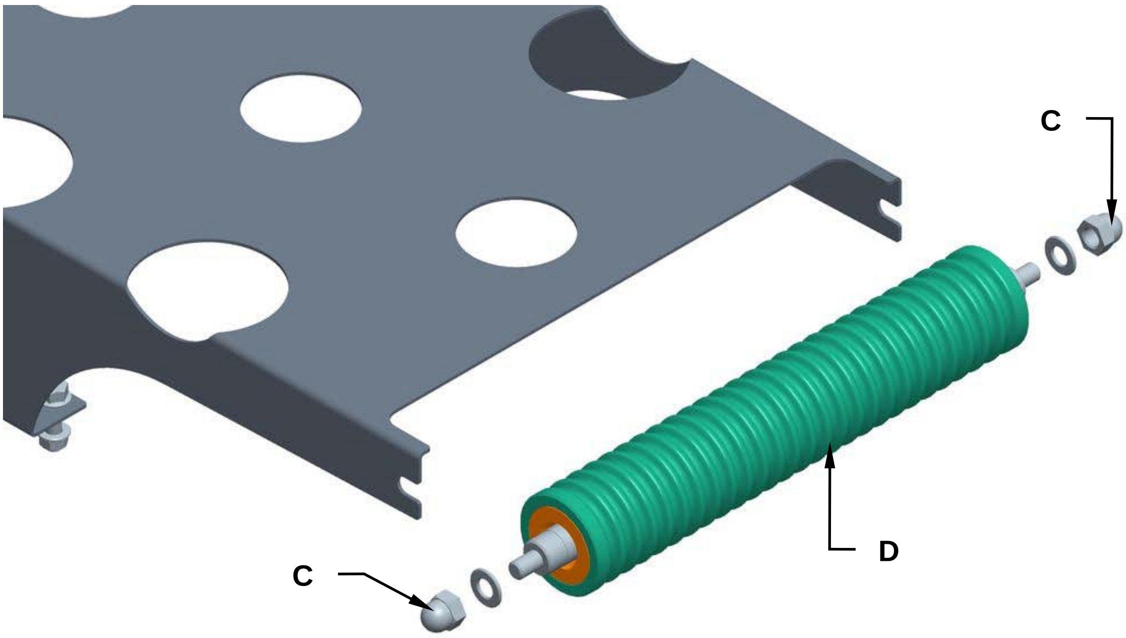
4. Reassemble all the parts back on again by reverse procedure.

To replace the roller, proceed as follows:

1. Remove the Polycord from the roller.



2. Unscrew the nuts (**C**) and replace the roller (**D**).



Assemble all the components following the preceding procedure to bashful.

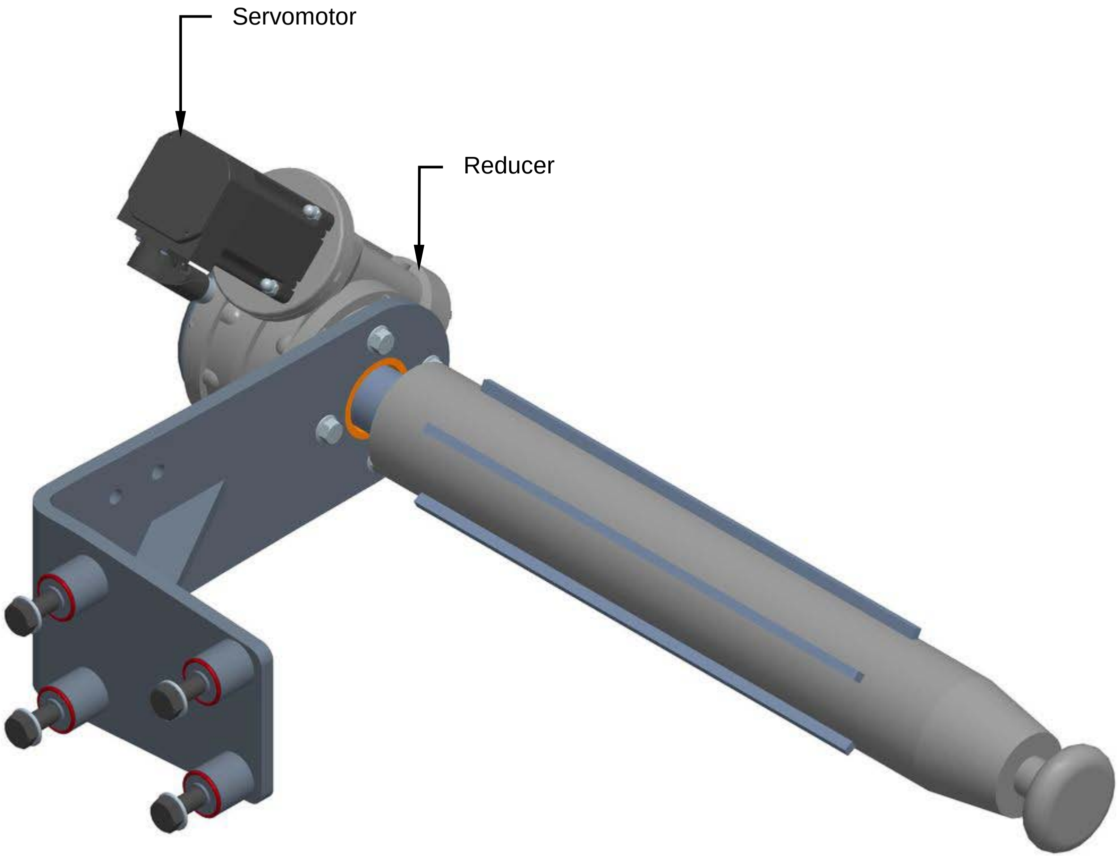
8.14 FILM UNROLLING ROLLERS

8.14.1 Inspection for servomotor and reducer good working conditions

Machine status	Power ON.
Equipment	Standard workshop tools

During standard machine operation cycles, carry out a visual inspection to ensure that the servomotor and reducer are in good working conditions. Check to ensure that direction the servomotor is turning in, is correct.

Should any anomalies be found, please warn the person responsible/people in charge of setting the unit group back to optimised working conditions.

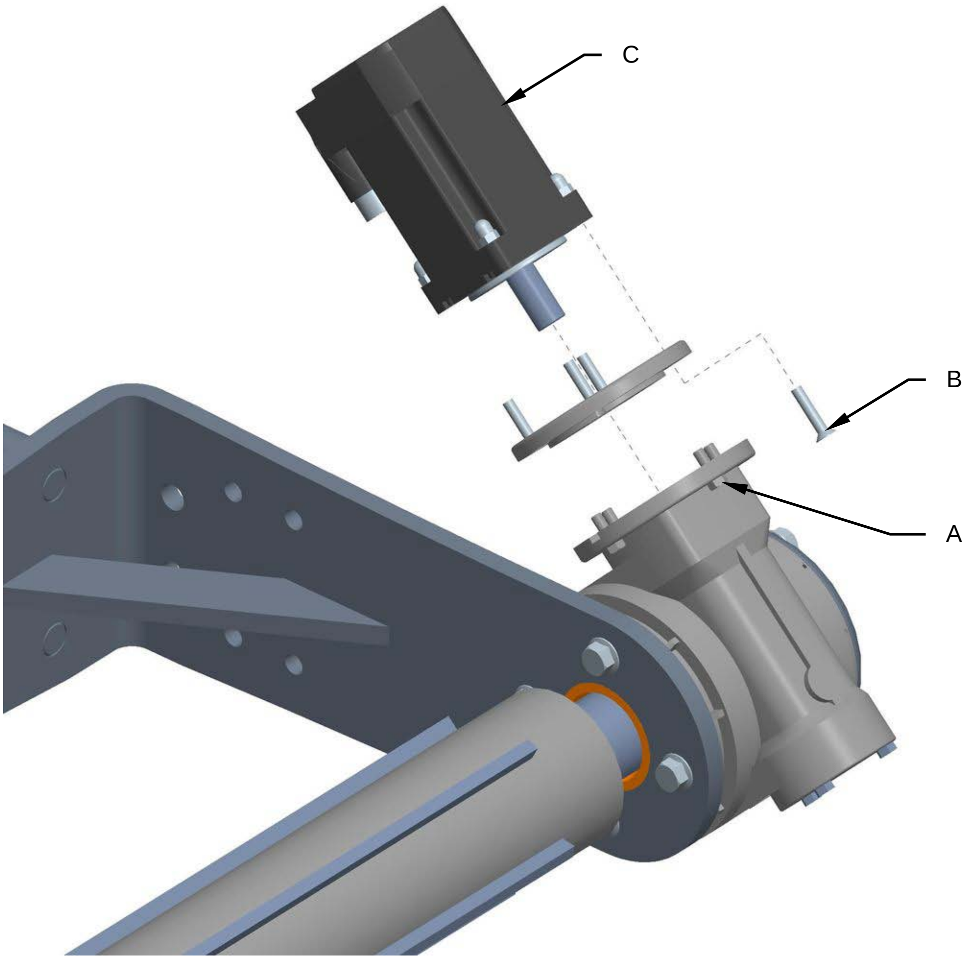


8.14.2 Servomotor and reducer replacement

Machine status	Power totally disconnected
Equipment	Standard workshop tools

For servomotor and reducer replacement proceed as follows:

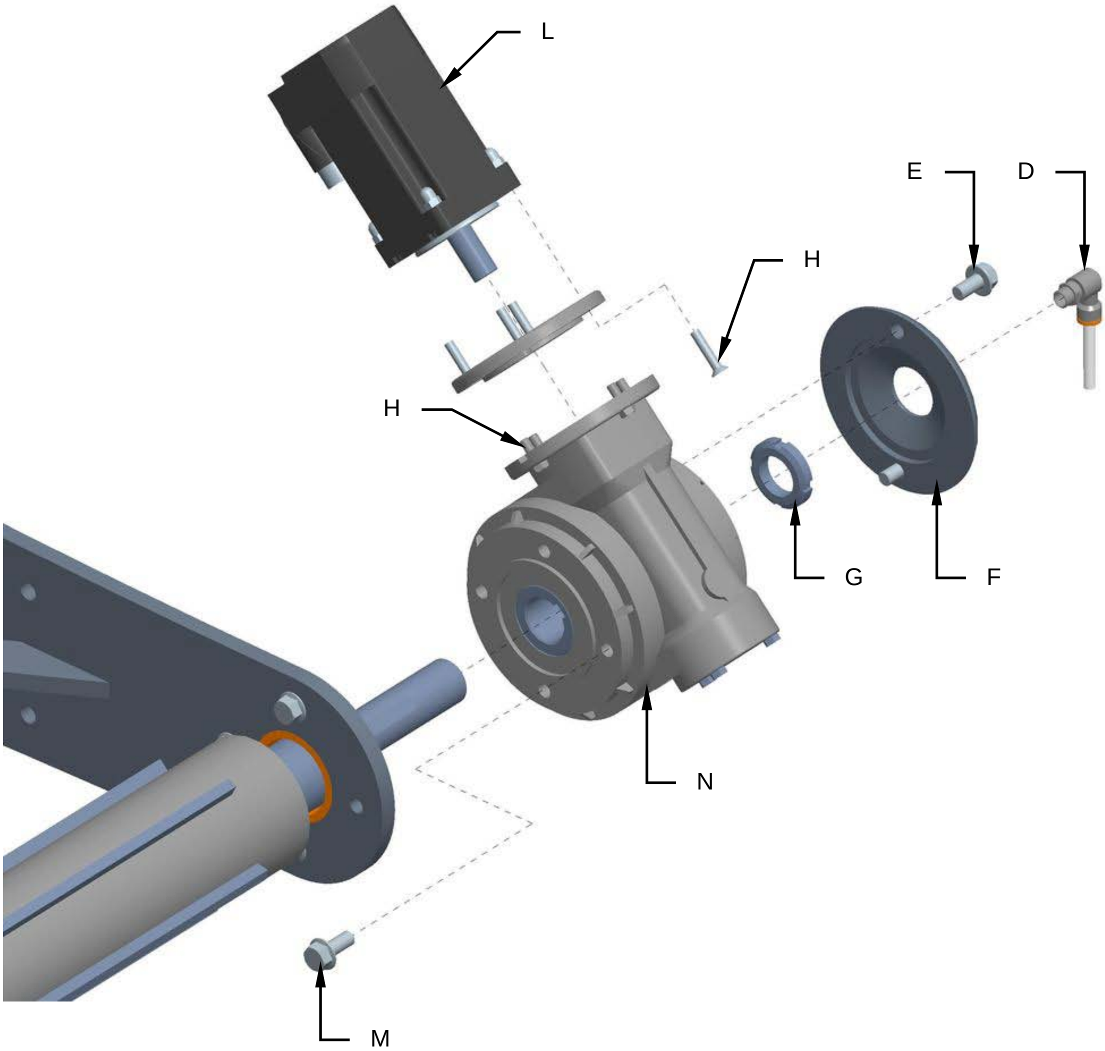
1. Disconnect the power supply of the machine.
2. Unscrew the screws (**A**) and (**B**).
3. Replace the servomotor (**C**).



4. Assemble all the components following the preceding procedure to bashful.

For reducer replacement proceed as follows:

1. Disconnect the power supply of the machine.
2. Close the pneumatic plant.
3. Unscrew the joint (**D**).
4. Unscrew the screws (**E**) and remove the cover (**F**).
5. Unscrew the gear (**G**).
6. Unscrew the screws (**H**) and remove the servomotor (**L**).
7. Unscrew the screws (**M**) and replace the reducer (**N**).



8. Assemble all the components following the preceding procedure to bashful.

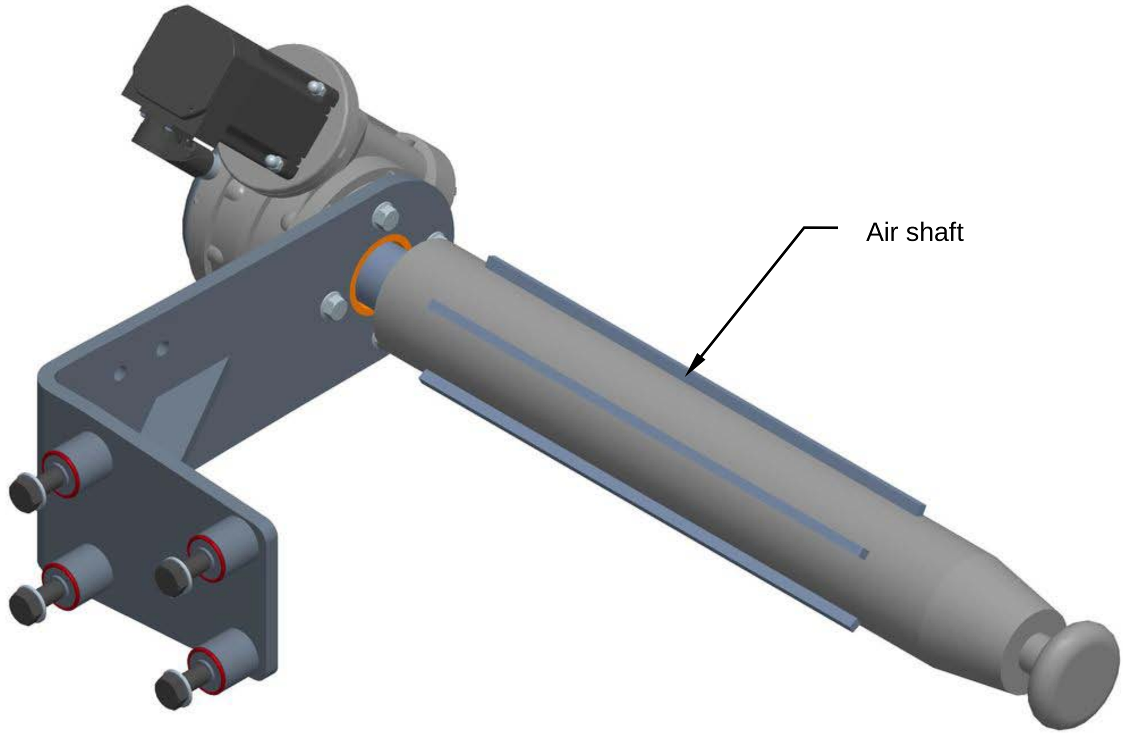
8.14.3 Air shaft and bearings good working order inspection

Machine status	Power ON.
Equipment	Standard workshop tools

During standard machine operation cycles, carry out a visual inspection to ensure that the air shaft and bearings are in good working conditions.

Check that the above components are not worn or torn enough to compromise the unit assembly’s good working order.

Should any anomalies be found, please warn the person responsible/people in charge of setting the unit group back to optimised working conditions.

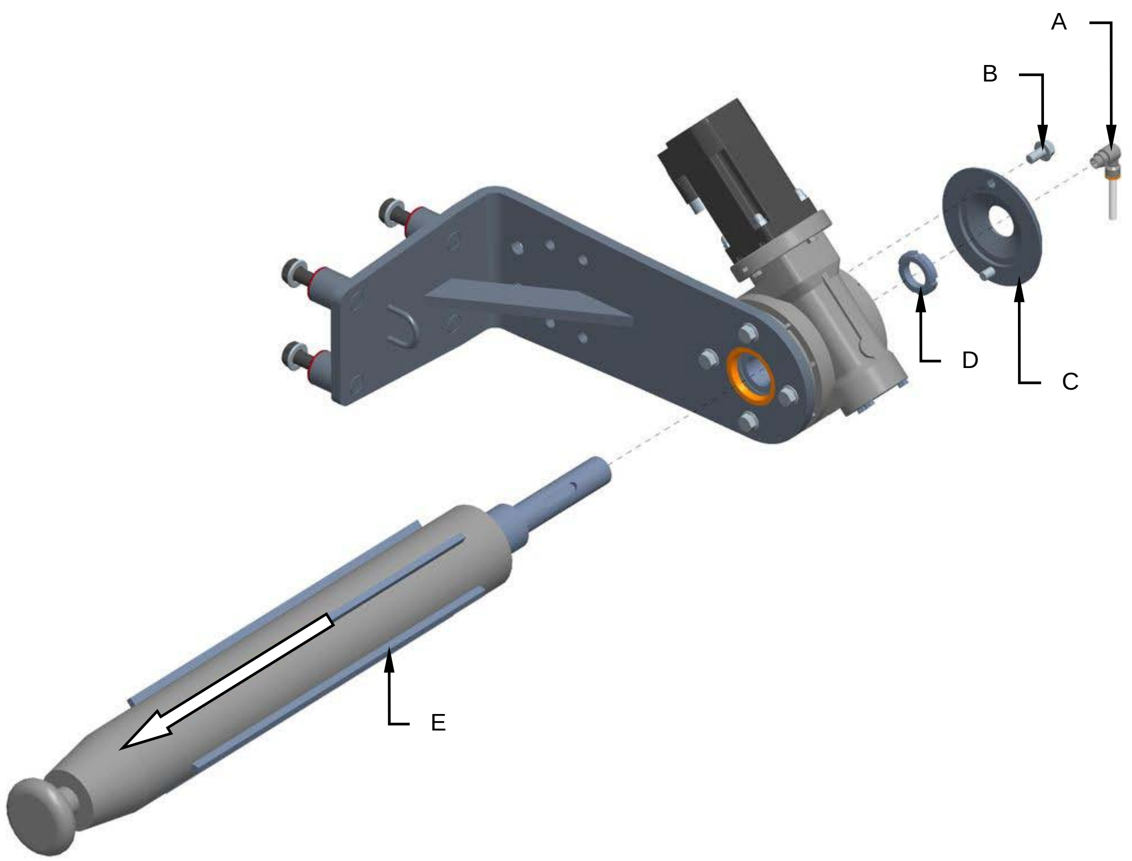


8.14.4 Replacement of the air shaft

Machine status	Power totally disconnected
Equipment	Standard workshop tools

For air shaft replacement, proceed as follows:

1. Disconnect the power supply of the machine.
2. Close the pneumatic plant.
3. Unscrew the joints (**A**) .
4. Unscrew the screws (**B**) and remove the cover (**C**).
5. Unscrew the gear (**D**).
6. Remove and replace the air shaft (**E**).



7. Assemble all the components following the preceding procedure to bashful.

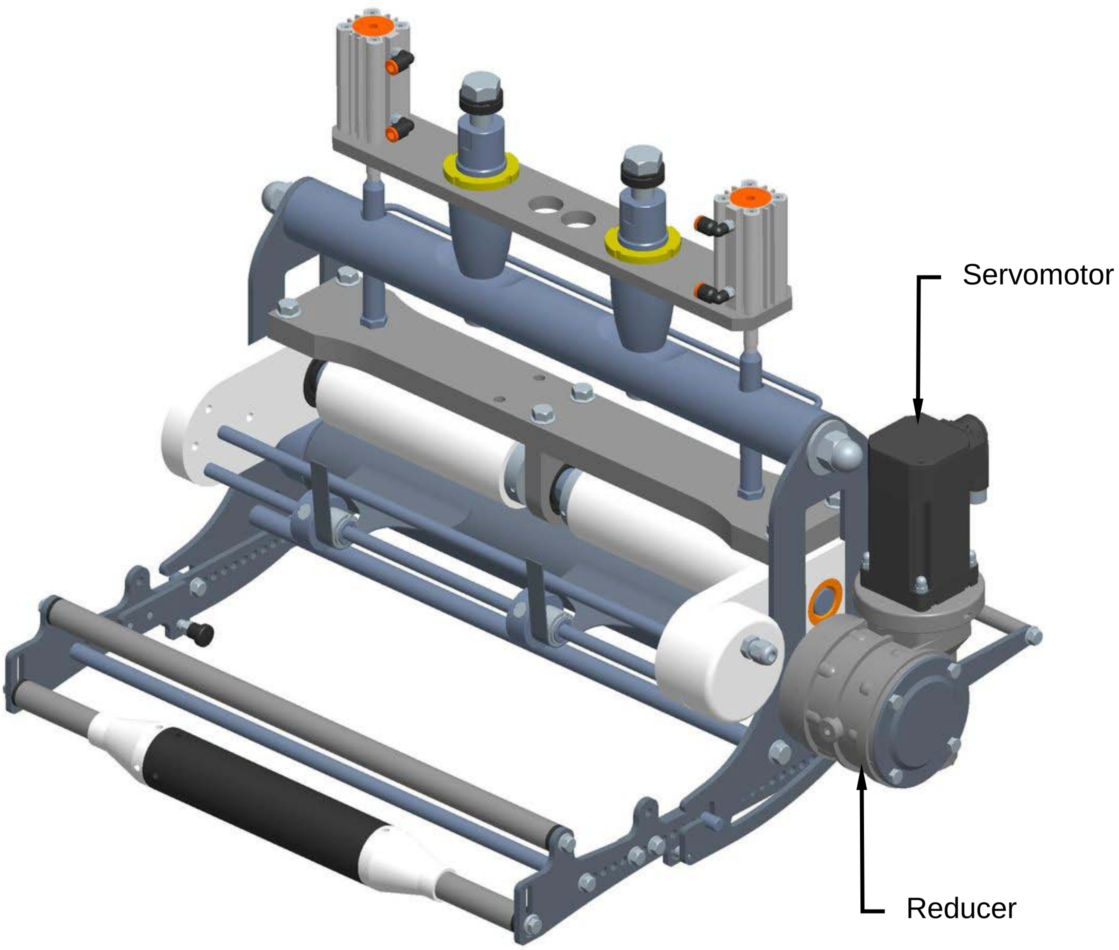
8.15 FILM CONTROL AND UNROLL ROLLERS

8.15.1 Servomotor and reducer good working order inspection

Machine status	Power ON.
Equipment	Standard workshop tools

During standard machine operation cycles, carry out a visual inspection to ensure that the servomotor and reducer are in good working conditions. Check to ensure that direction the servomotor is turning in, is correct.

Should any anomalies be found, please warn the person responsible/people in charge of setting the unit group back to optimised working conditions.

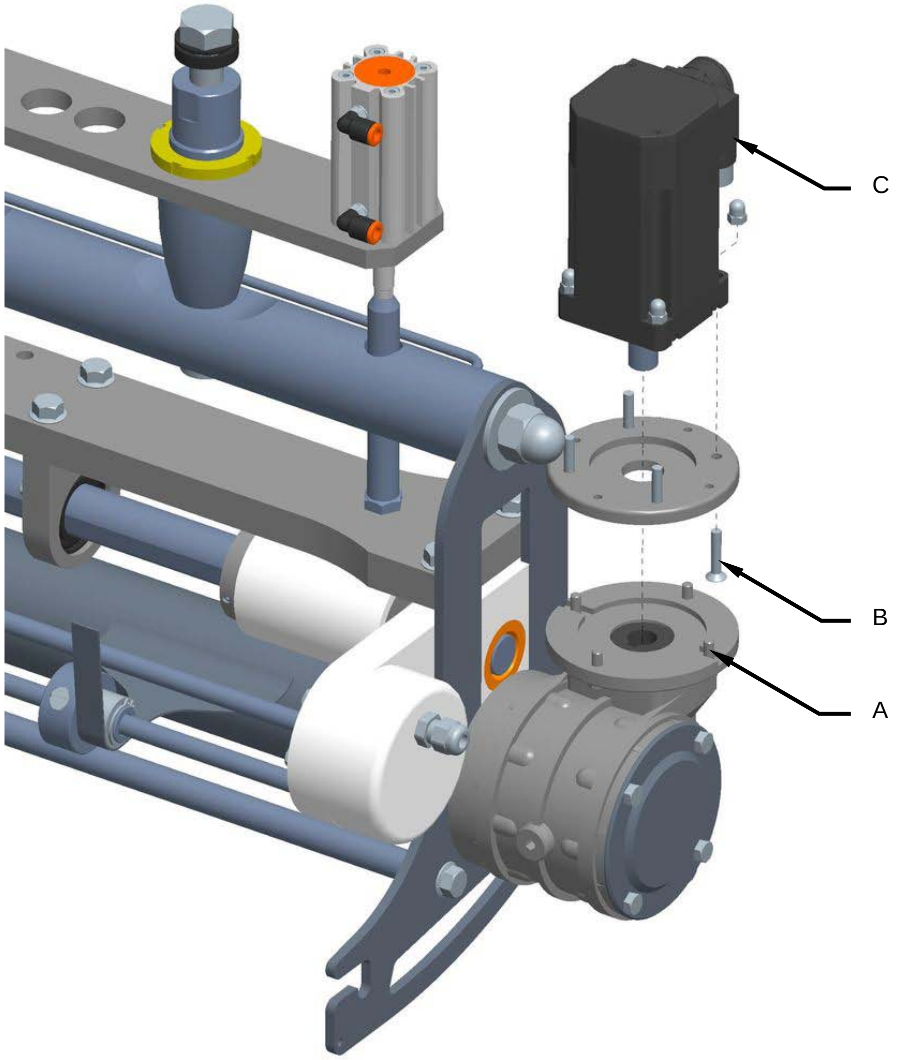


8.15.2 Servomotor and reducer replacement

Machine status	Power totally disconnected
Equipment	Standard workshop tools

For servomotor replacement, proceed as follows:

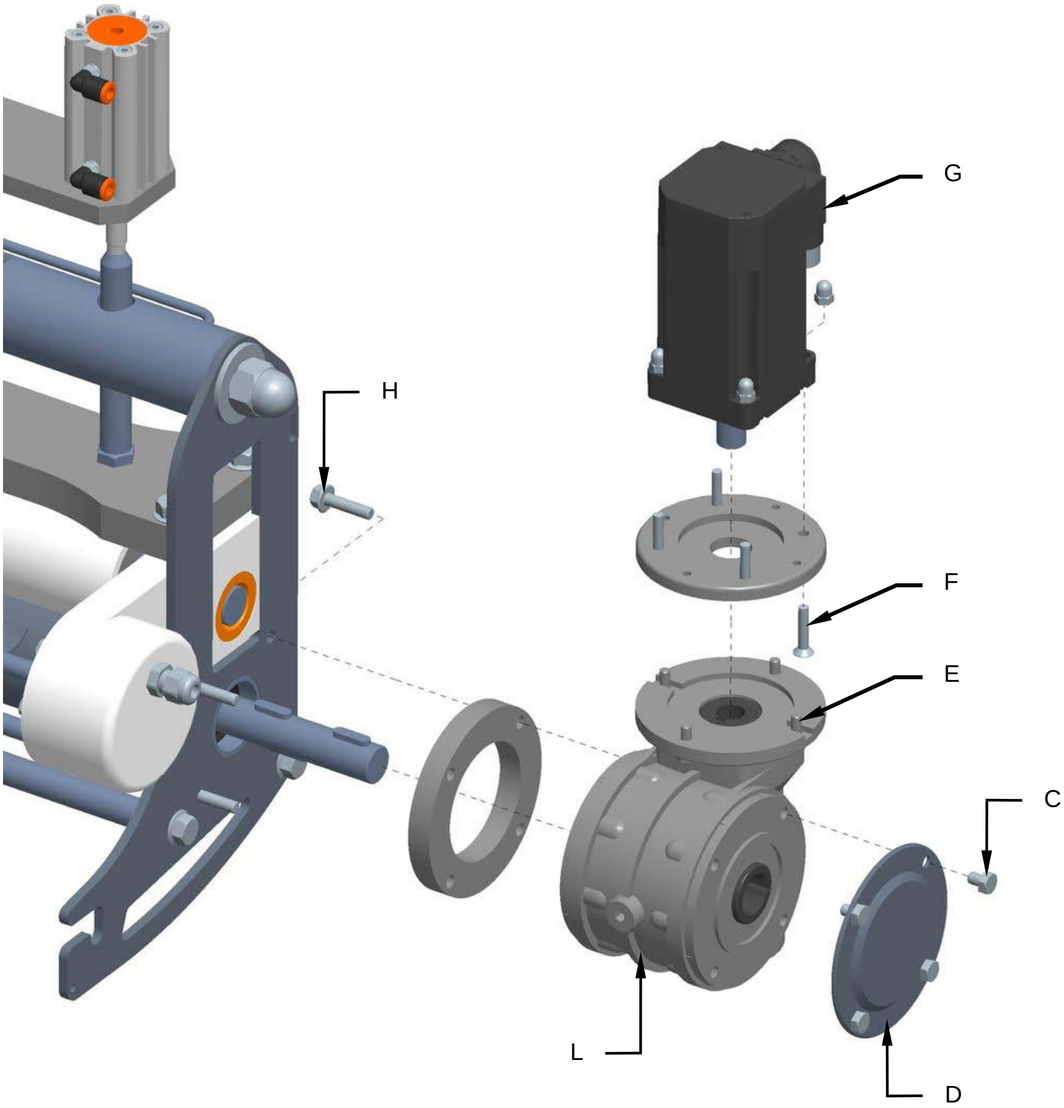
1. Disconnect the power supply of the machine.
2. Unscrew screws (**A**) and (**B**) and replace servomotor (**B**).



3. Reassemble all the parts back on again by reverse procedure.

For reducer replacement, proceed as follows:

1. Disconnect the power supply of the machine.
2. Unscrew the screws (**C**) and remove the cover (**D**).
3. Unscrew the screws (**E**) and (**F**) and remove the servomotor (**G**).
4. Unscrew the screws (**H**) and replace the reducer (**L**).



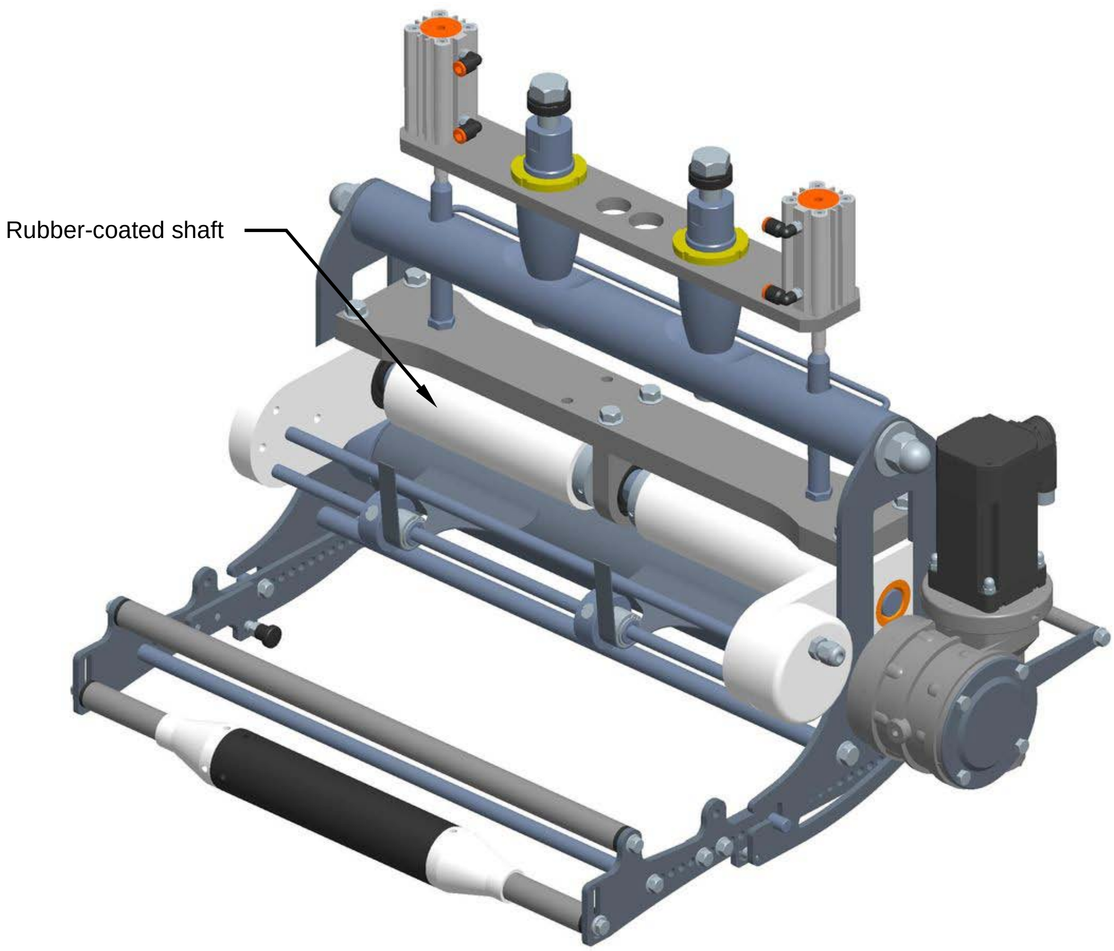
5. Reassemble all the parts back on again by reverse procedure.

8.15.3 Checking on good rubber coated shaft working order

Machine status	Power ON.
Equipment	Standard workshop tools

During standard machine operation cycles, carry out a visual inspection to ensure that the rubber coated shaft is in good working conditions. Check to ensure that they are not worn or torn enough to compromise film integrity.

Should any anomalies be found, please warn the person responsible/people in charge of setting the unit group back to optimized working conditions.

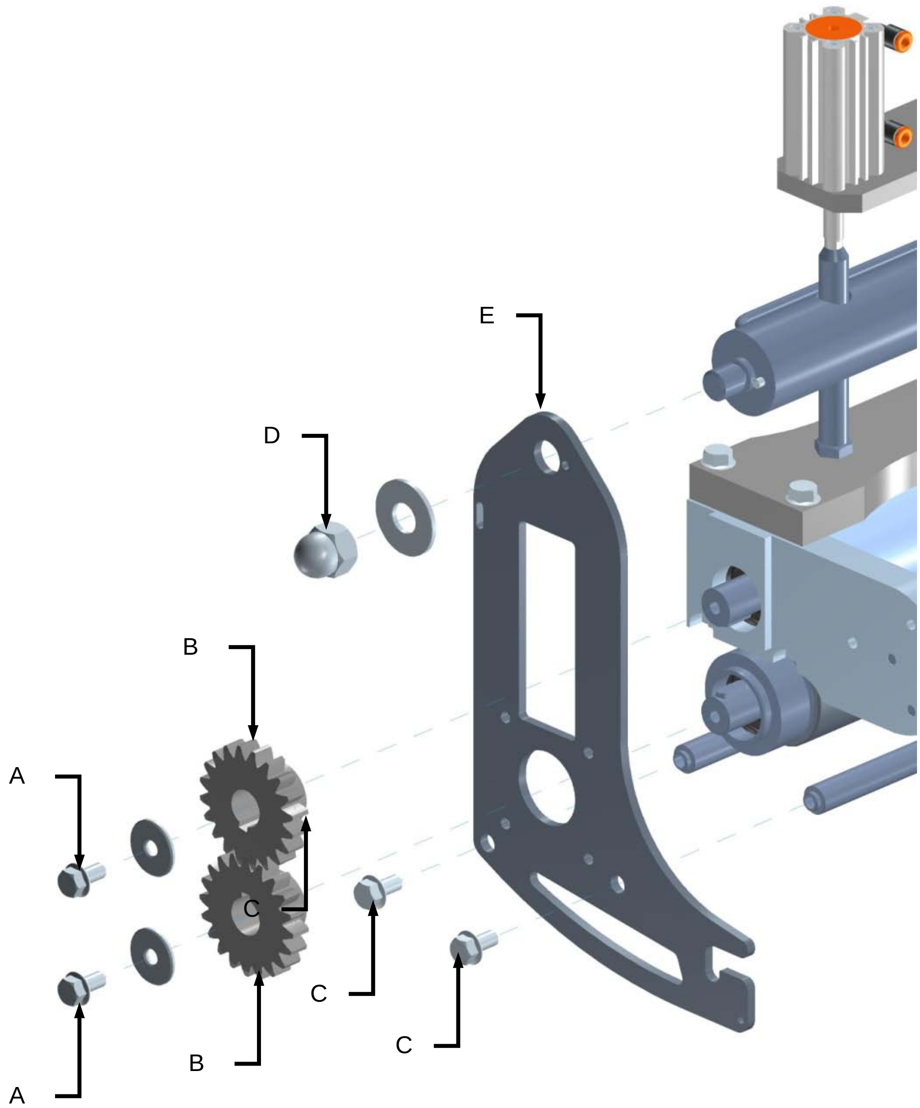


8.15.4 Rubber coated shaft replacement

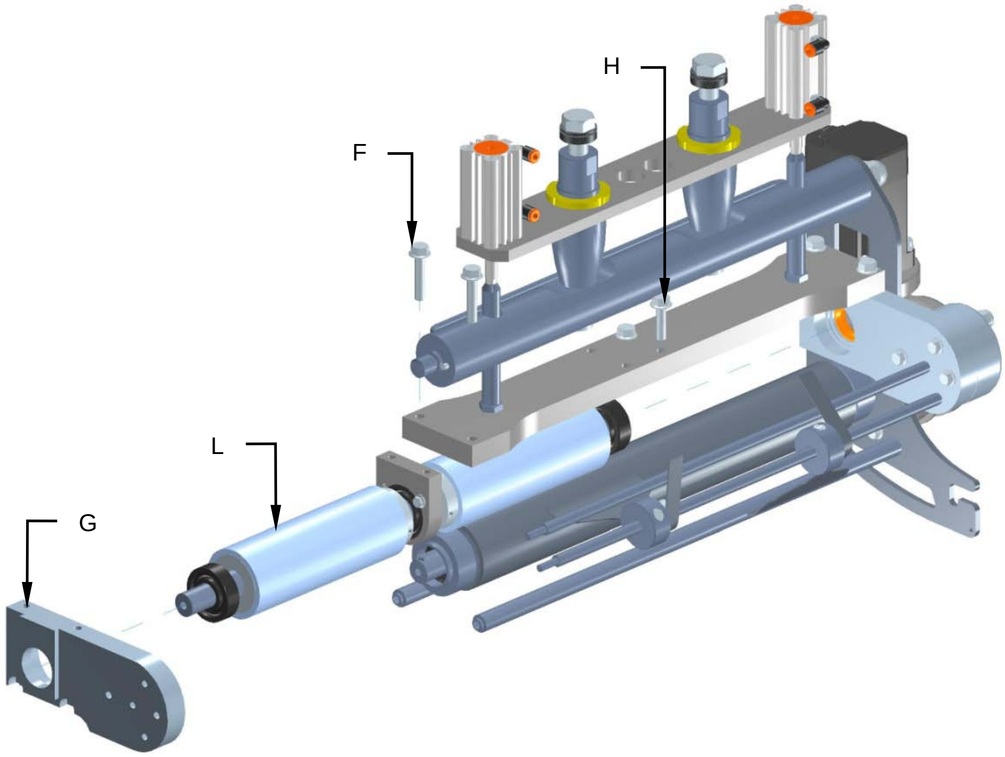
Machine status	Power totally disconnected
Equipment	Standard workshop tools

For rubber coated shaft replacement, proceed as follows:

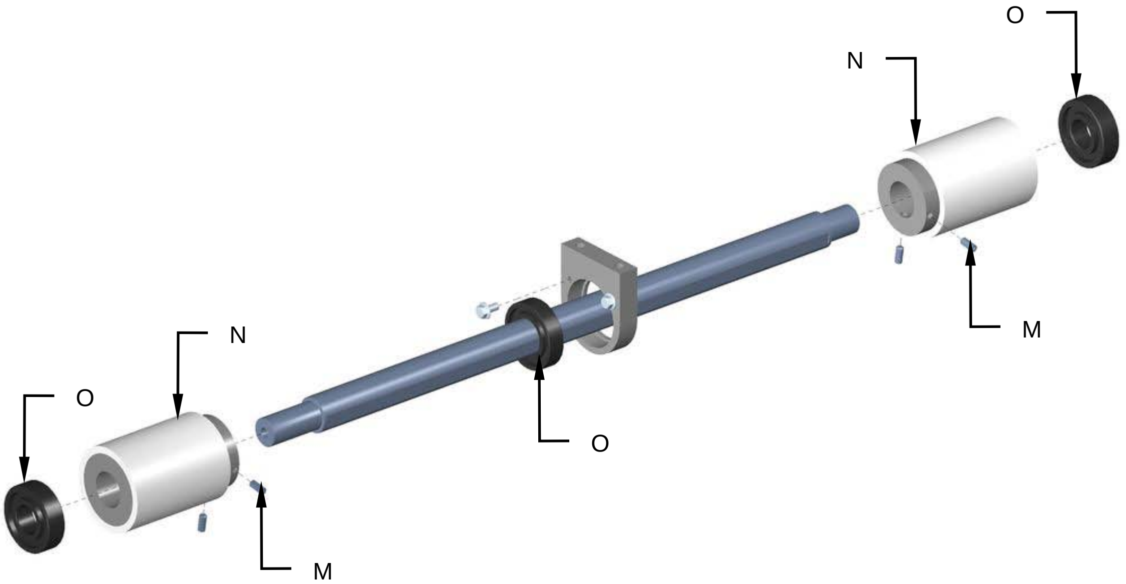
1. Disconnect the power supply of the machine.
2. Unscrew the screws (**A**) and remove the gearings (**B**).
3. Unscrew the screws (**C**), the nut (**D**) and remove the shoulder (**E**).



4. Unscrew the screws (**F**) and remove the support (**G**).
5. Unscrew the screws (**H**) and remove the rubber coated shaft group (**L**).



6. Unscrew the stifts (**M**) and replace the rubber coated rollers (**N**).
7. Replace the bearings (**O**).



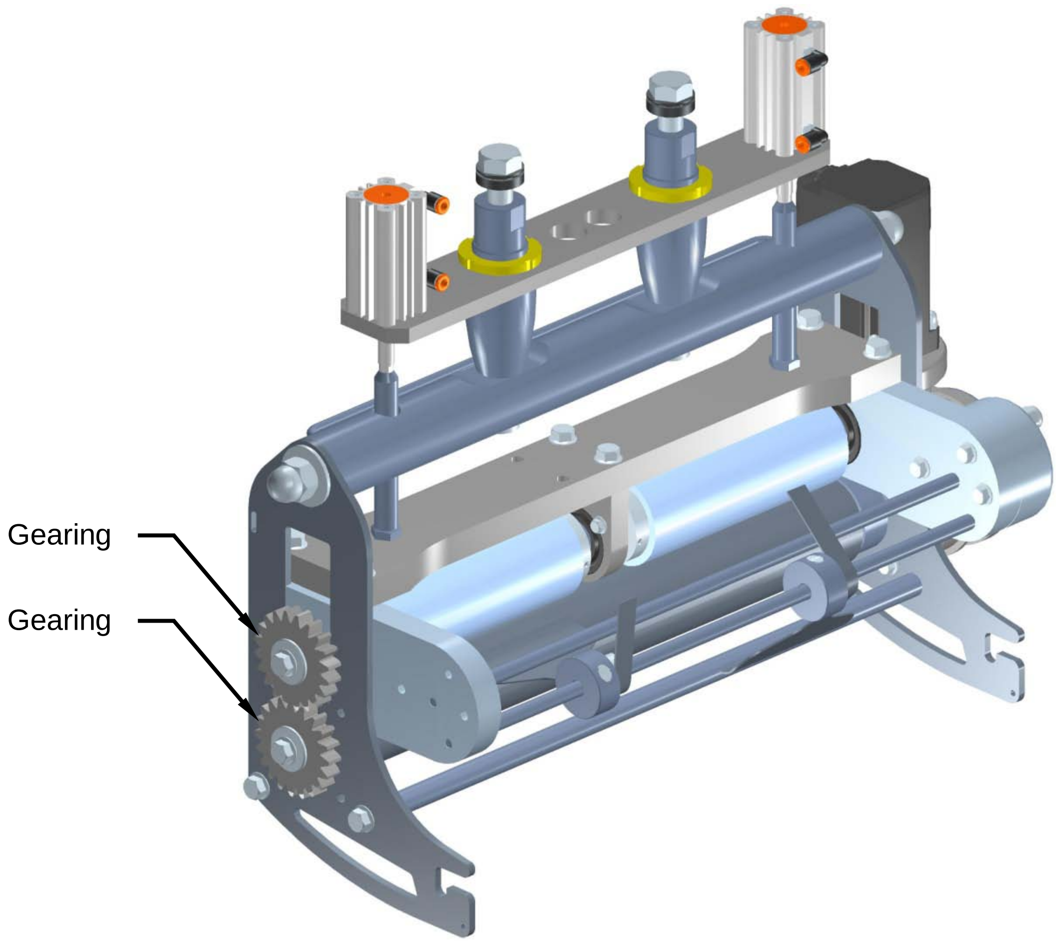
8. Reassemble all the parts back on again by reverse procedure.

8.15.5 Inspection of the gearings good working conditions

Machine status	Power ON.
Equipment	Standard workshop tools

During standard machine operation cycles, carry out a visual inspection to ensure that the gearings are in good working condition.

Should any anomalies be found, please warn the person responsible/people in charge of setting the unit group back to optimised working conditions.

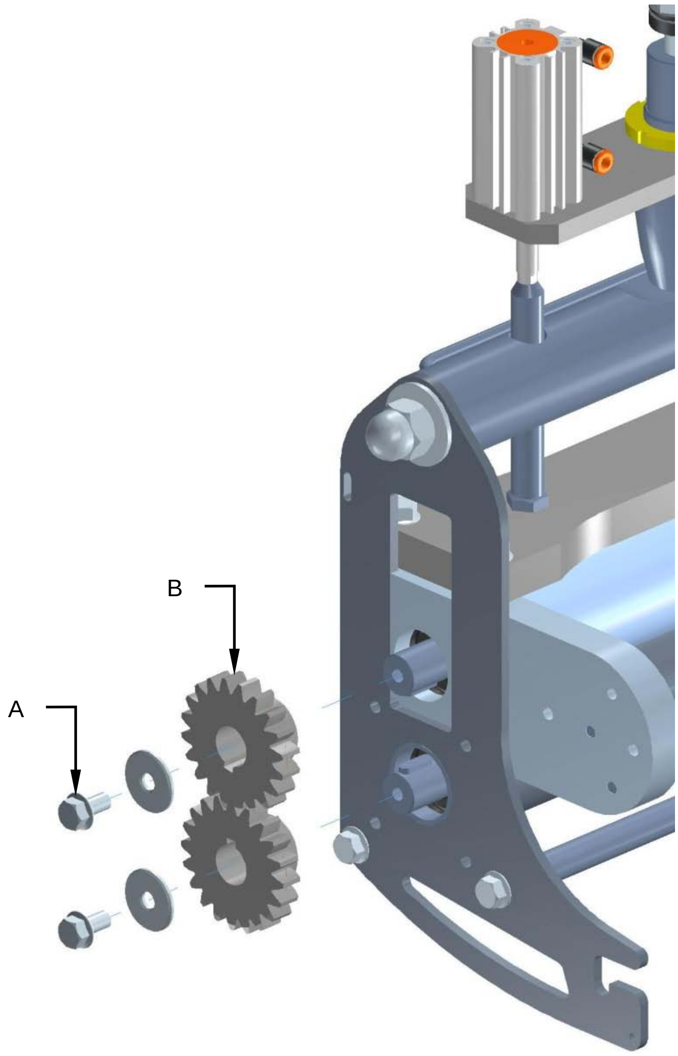


8.15.6 Gearings replacement

Machine status	Power totally disconnected
Equipment	Standard workshop tools

To replace the gearings, proceed as follows:

1. Unscrew the screws (**A**) and replace the gearings (**B**).



2. Reassemble all parts back on again by reverse procedure.

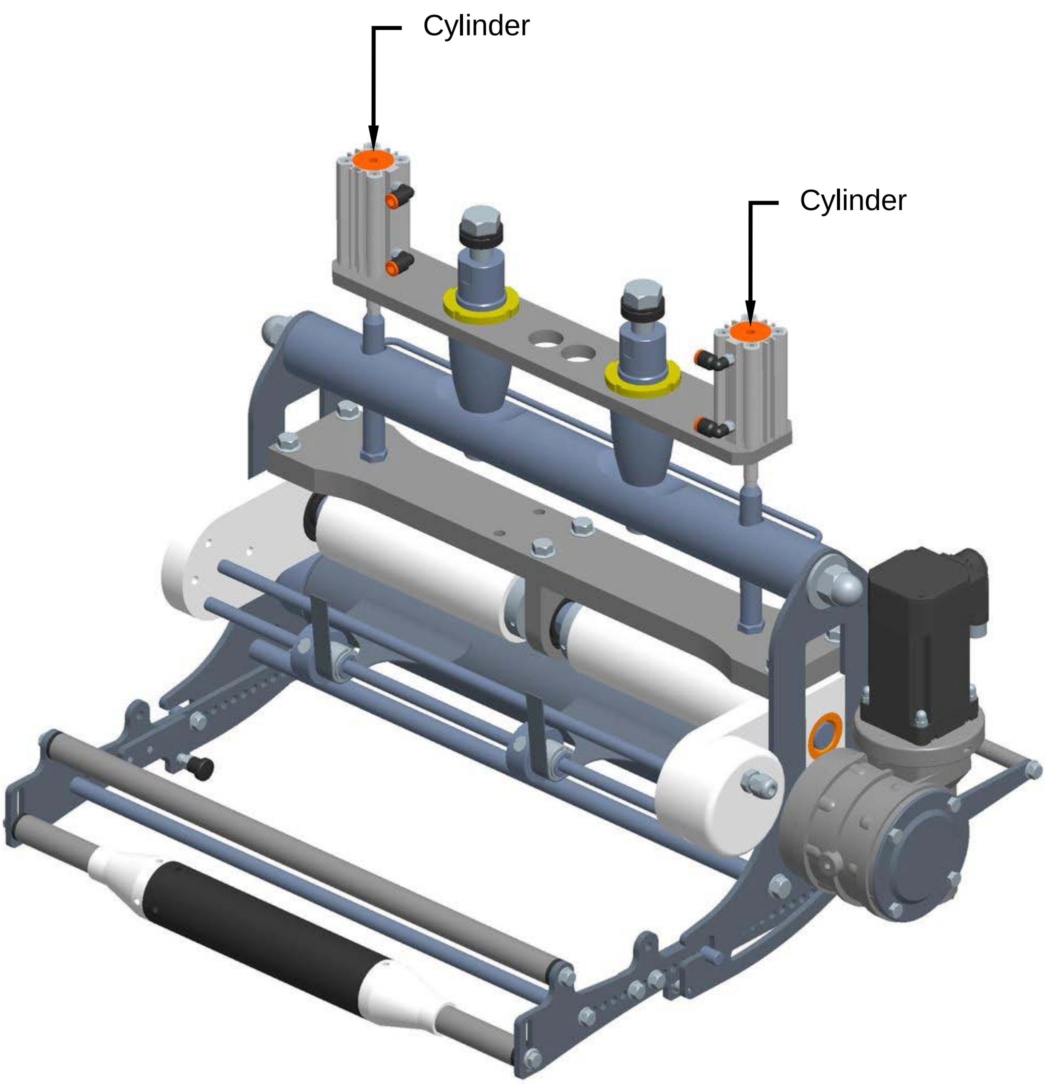
8.15.7 Inspection of the cylinders good working conditions

Machine status	Power ON.
Equipment	Standard workshop tools

During standard machine operation cycles, carry out a visual inspection to ensure that the cylinders are in good working condition.

Check to make sure that there are no air leaks. This operation must be conducted by highly qualified service personnel with all the relative safety and accident prevention precautions and never during machine operation.

Should any anomalies be found, please warn the person responsible/people in charge of setting the unit group back to optimised working conditions.

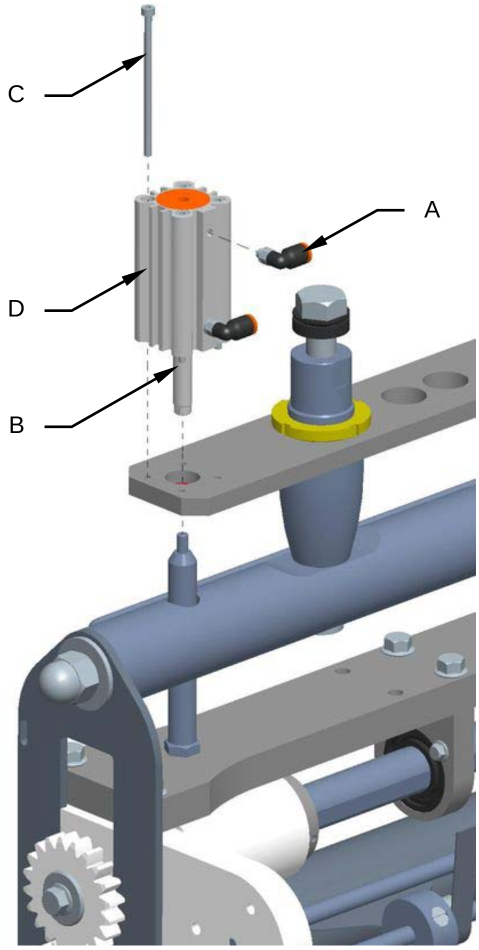


8.15.8 Cylinders replacement

Machine status	Power totally disconnected
Equipment	Standard workshop tools

To replace the cylinders, proceed as follows:

1. Disconnect the power supply of the machine.
2. Close the pneumatic plant.
3. Remove the joint fittings (**A**).
4. Unscrew the cylinder rod (**B**) and the screws (**C**).
5. Remove and replace the cylinder (**D**).



6. Reassemble all parts back on again by reverse procedure.

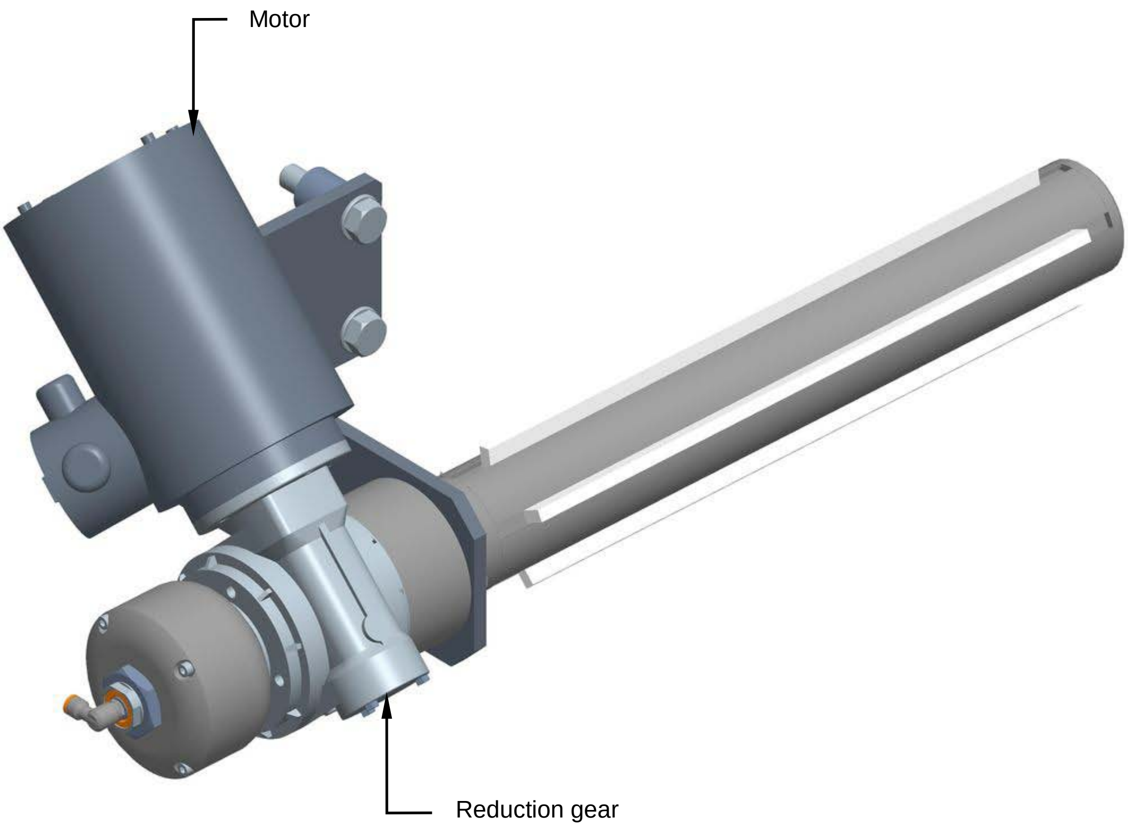
8.16 WINDER SHAFT

8.16.1 Checking the motor and reduction gear

Machine status	Power on
Tools	Normal workshop equipment

With the machine running a normal cycle, visually check the condition of the motor and Reduction gear. Make sure the motor runs in the correct direction.

If any problems are encountered, inform the person in charge, who will restore correct operation of the assembly.

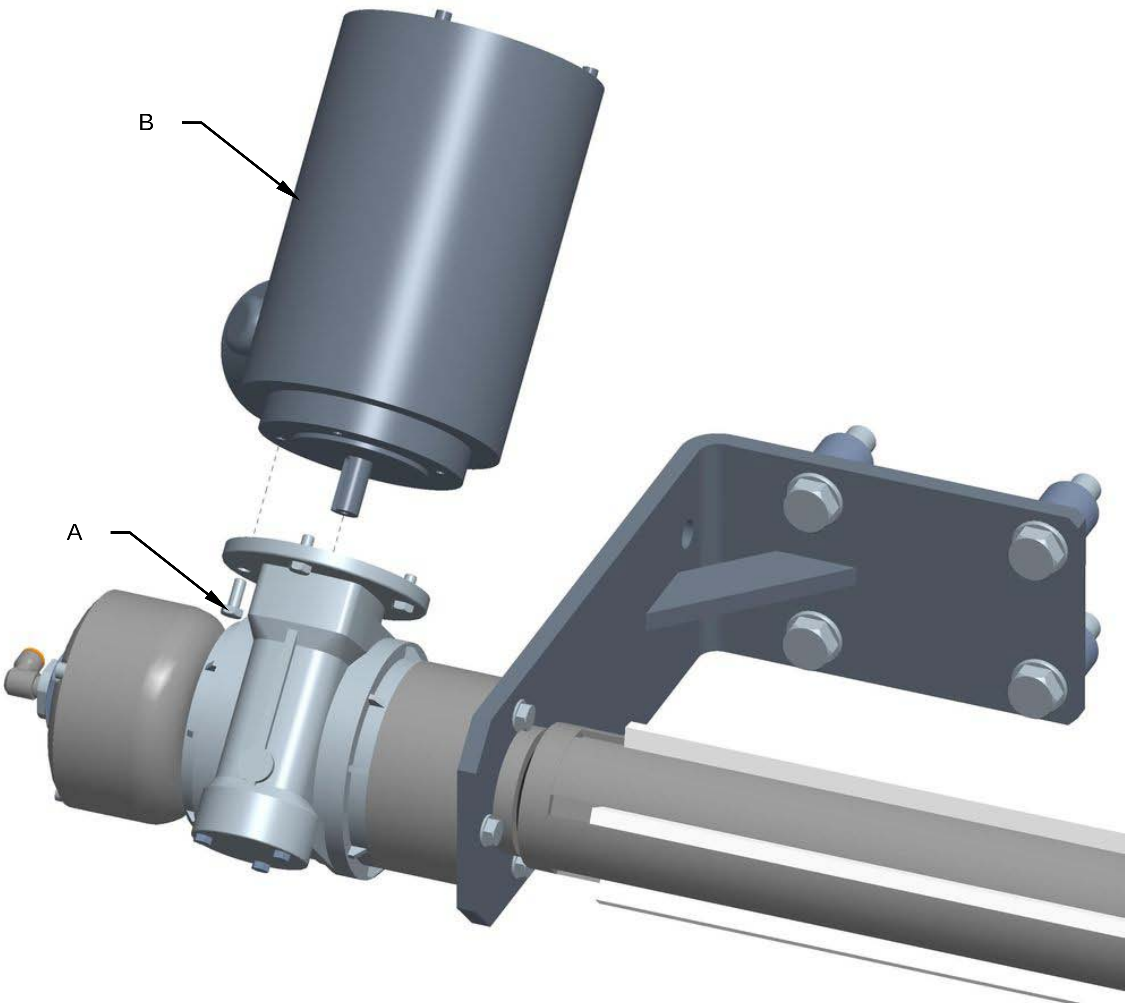


8.16.2 Replacing the motor and reduction gear

Machine status	Power off
Tools	Normal workshop equipment

Proceed as follows to replace the motor and reduction gear:

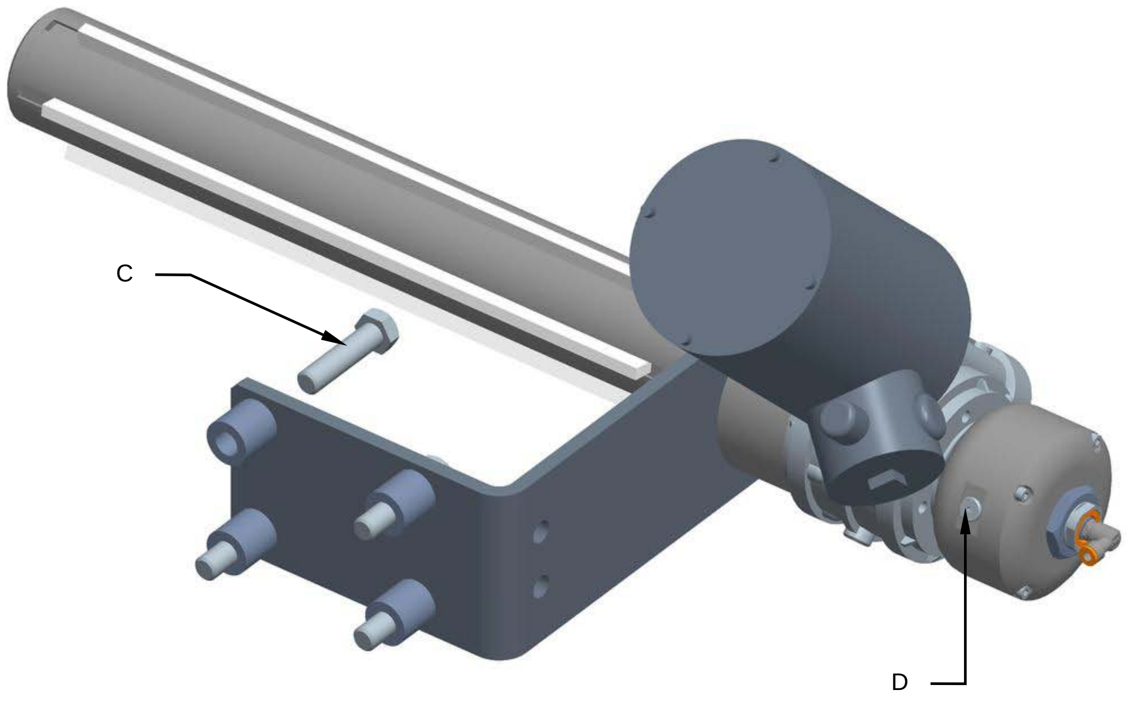
1. Disconnect the power supply of the machine.
2. Unscrew the screws (**A**) and replace the motor (**B**).



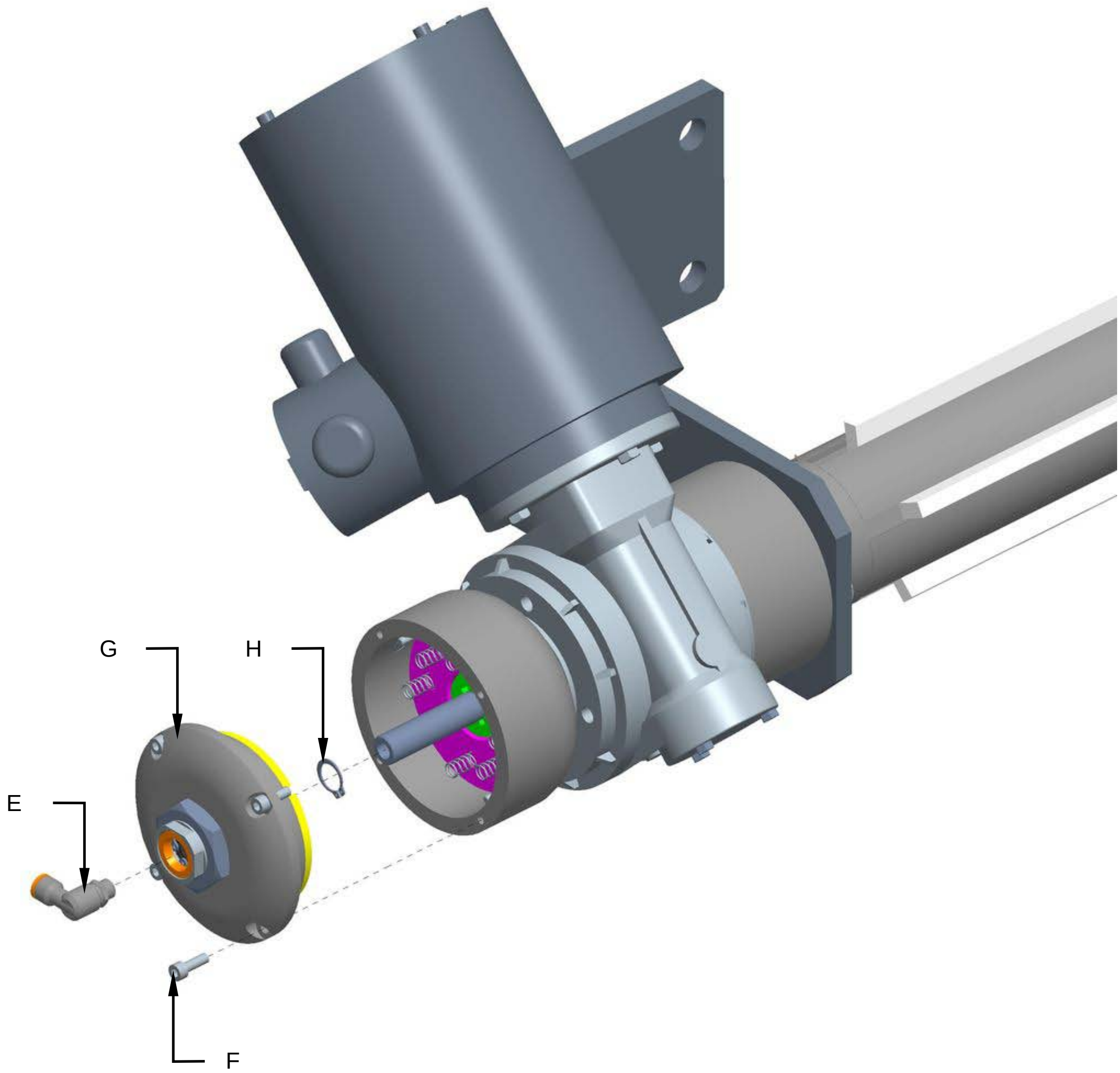
3. Remount all the components in the reverse order.

Proceed as follows to replace the reduction gear:

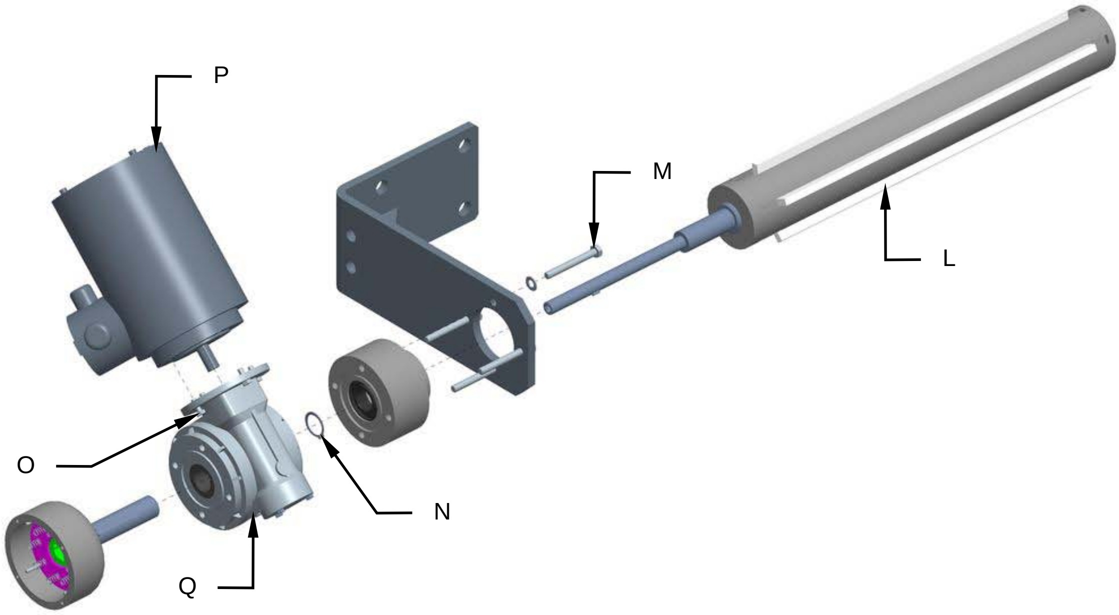
1. Disconnect the power supply of the machine.
2. Close the pneumatic plant.
3. Unscrew the screws (**C**) and remove the reduction gear from the machine.
4. Unscrew the cap (**D**) and drain the oil from the clutch.



5. Remove the fitting (**E**).
6. Unscrew the screws (**F**) and remove the clutch cover (**G**).
7. Remove the snap ring (**H**).



8. Remove the expanding shaft (**L**).
9. Unscrew the screws (**M**).
10. Remove the snap ring (**N**).
11. Unscrew the screws (**O**) and remove the motor (**P**).
12. Remove and replace the reduction gear (**Q**).



13. Remount all the components in the reverse order.

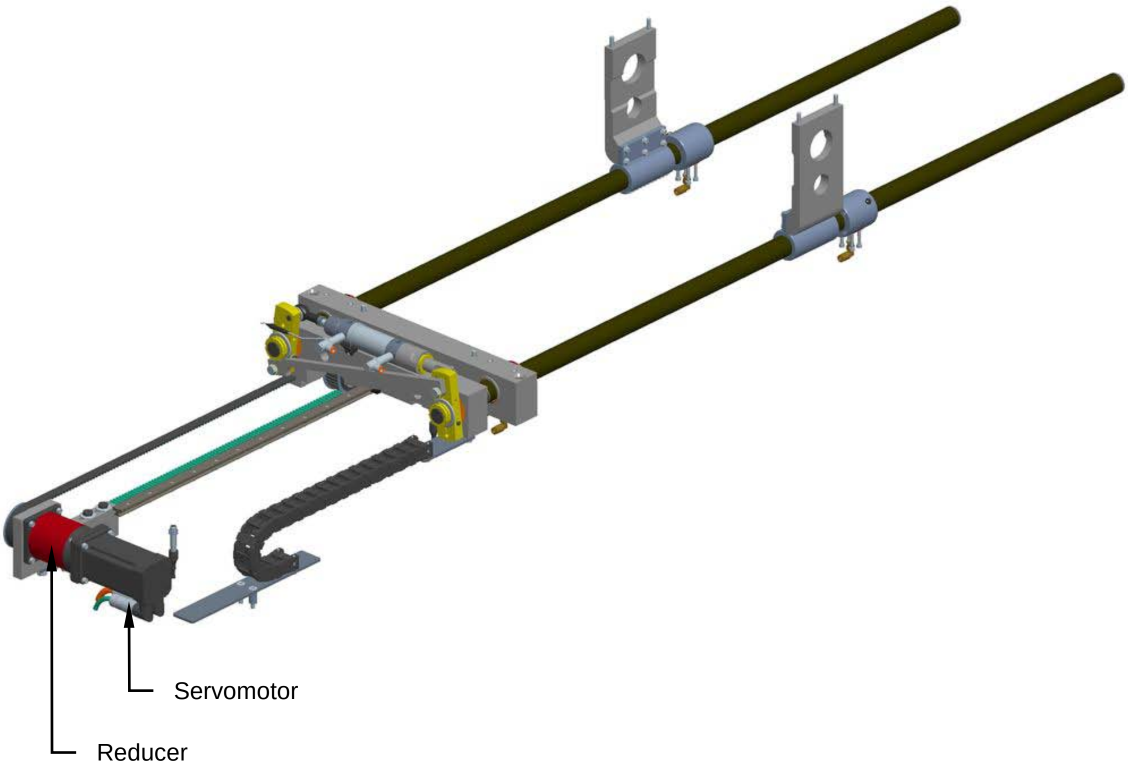
8.17 PUSHER ARMS

8.17.1 Checking on good servomotor and reducer working conditions

Machine status	Power ON.
Equipment	Standard workshop tools

During standard machine operation cycles, carry out a visual inspection to ensure that the servomotor and reducer are in good working conditions. Check to ensure that direction the servomotor is turning in, is correct.

Should any anomalies be found, please warn the person responsible/people in charge of setting the unit group back to optimised working conditions.

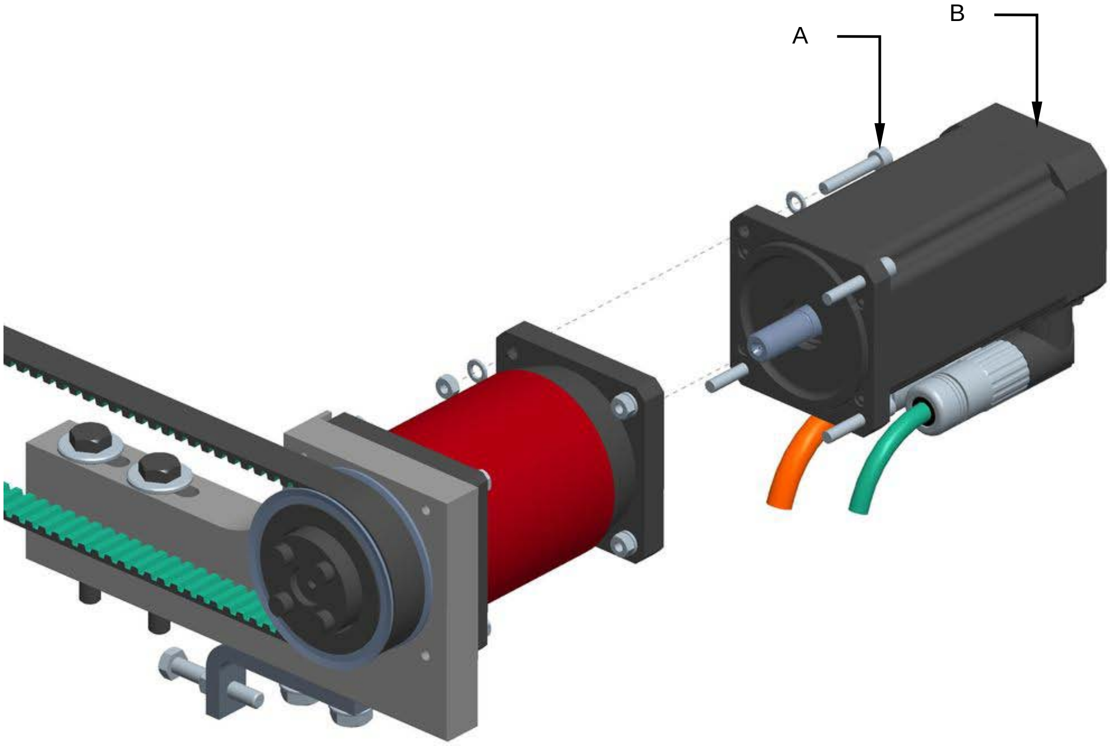


8.17.2 Servomotor and reducer replacement

Machine status	Power totally disconnected
Equipment	Standard workshop tools

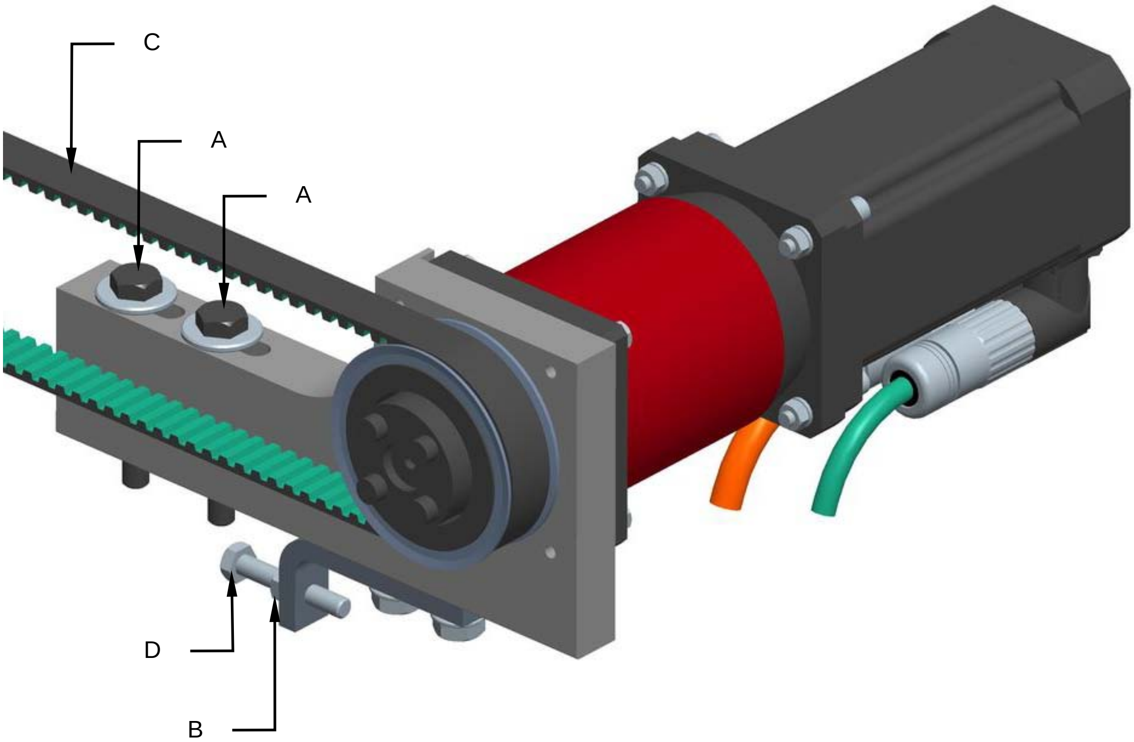
For servomotor replacement proceed as follows:

1. Disconnect the power supply of the machine.
2. Unscrew the screws (**A**) and replace the servomotor (**B**).

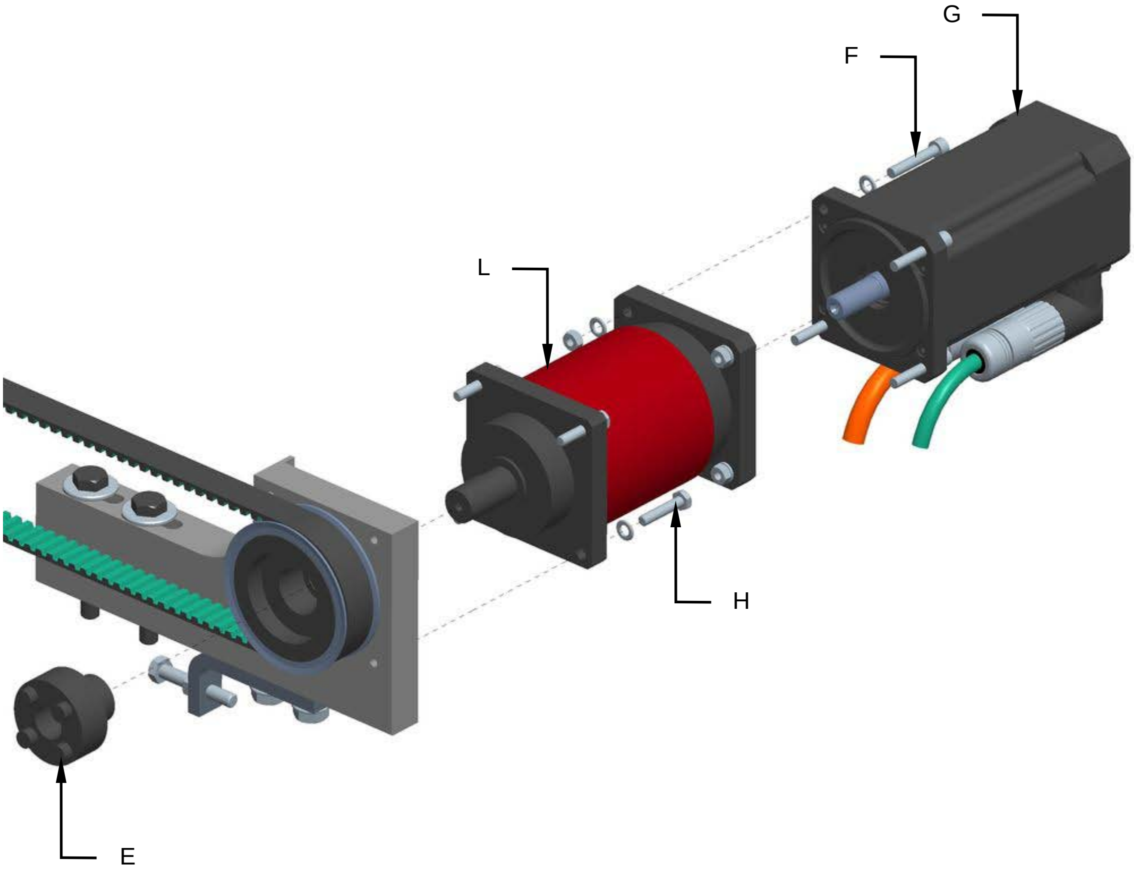


For reducer replacement proceed as follows:

1. Disconnect the power supply of the machine.
2. Unscrew the screws (**A**).
3. Loosen the nut (**B**) and reduce the cog-belt (**C**) unscrewing the screws (**D**).



4. Unscrew the tollok (**E**).
5. Unscrew the screws (**F**) and remove the servomotor (**G**).
6. Unscrew screws (**H**) and replace the reducer (**L**).



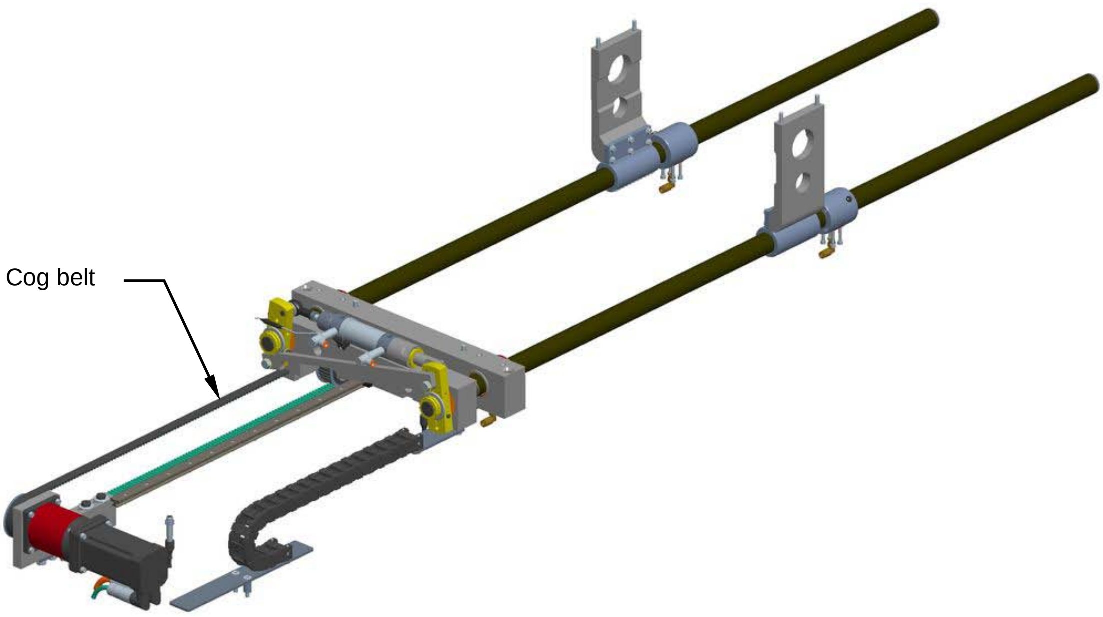
7. Reassemble all parts back on again by reverse procedure.

8.17.3 Checking the cog belt condition and tension

Machine status	Power off
Tools	Normal workshop equipment

With the machine running a normal cycle, visually check the condition of the cog belt.

If any problems are encountered, inform the person in charge, who will restore correct operation of the assembly.

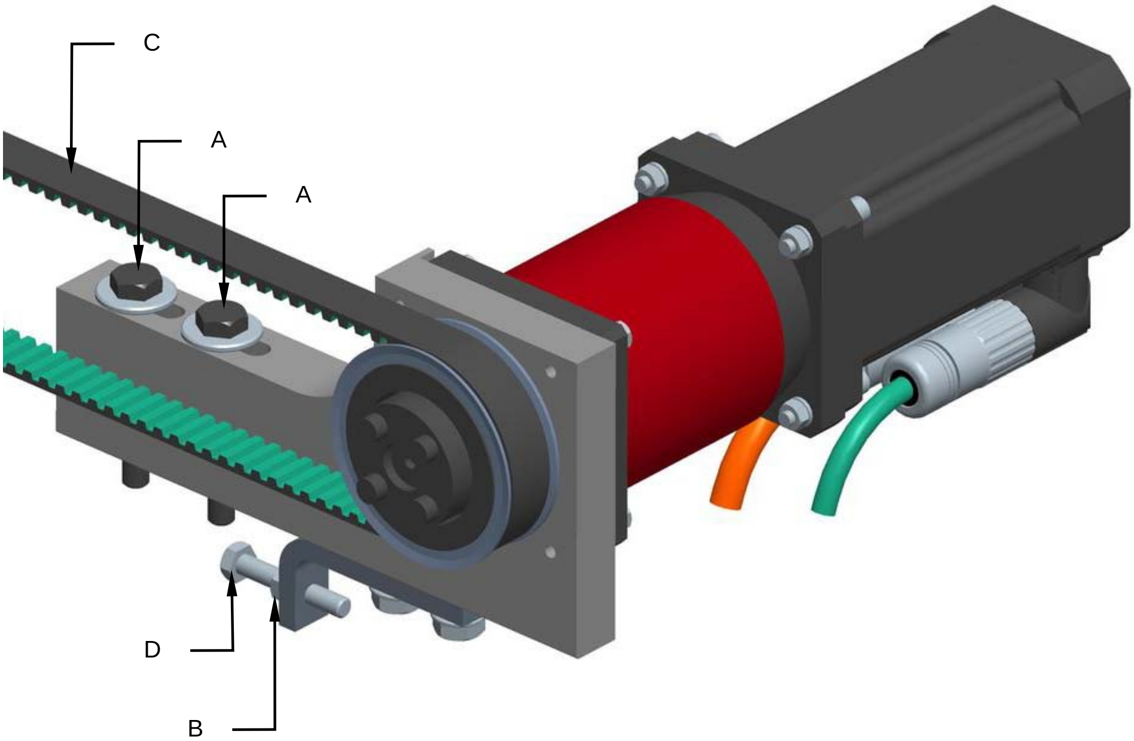


8.17.4 Replacing the cog belt

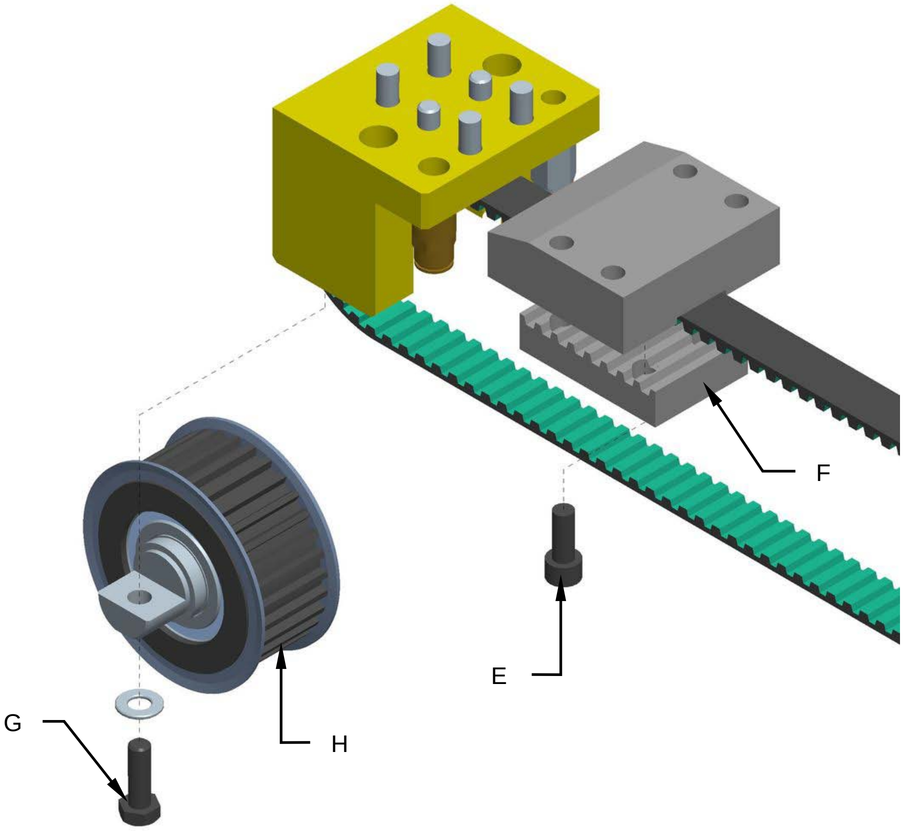
Machine status	Power off
Tools	Normal workshop equipment

Proceed as follows to replace the cog belt:

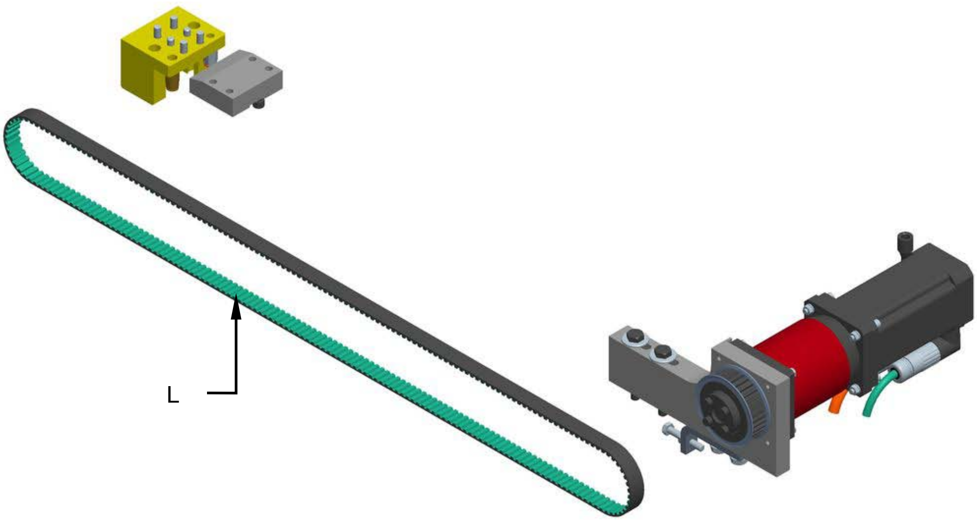
1. Disconnect the power supply of the machine.
2. Unscrew the screws (**A**).
3. Loosen the nut (**B**) and reduce the cog-belt (**C**) unscrewing the screws (**D**).



4. Unscrew the screws (**E**) and remove the bracket (**F**).
5. Unscrew the screws (**G**) and remove the pulley (**H**).



6. Remove and replace the cog belt (**L**).



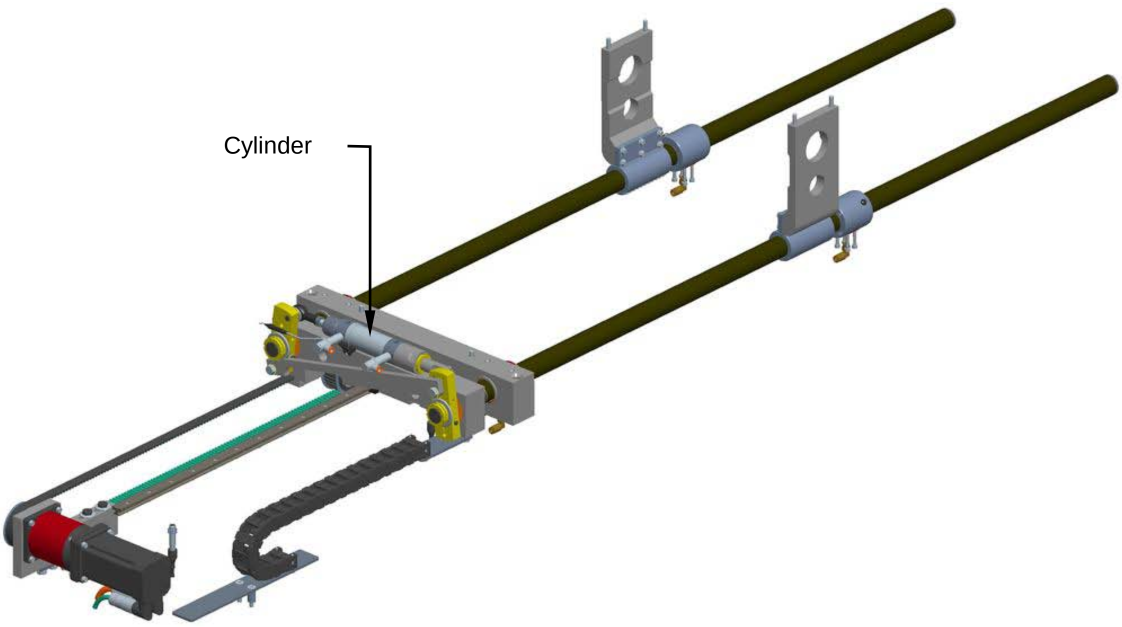
7. Reassemble all parts back on again by reverse procedure.

8.17.5 Inspection of good cylinder working conditions

Machine status	Power ON.
Equipment	Standard workshop tools

During standard machine operation cycles, carry out a visual inspection to ensure that the cylinder is in good working conditions.

Should any anomalies be found, please warn the person responsible/people in charge of setting the unit group back to optimised working conditions.

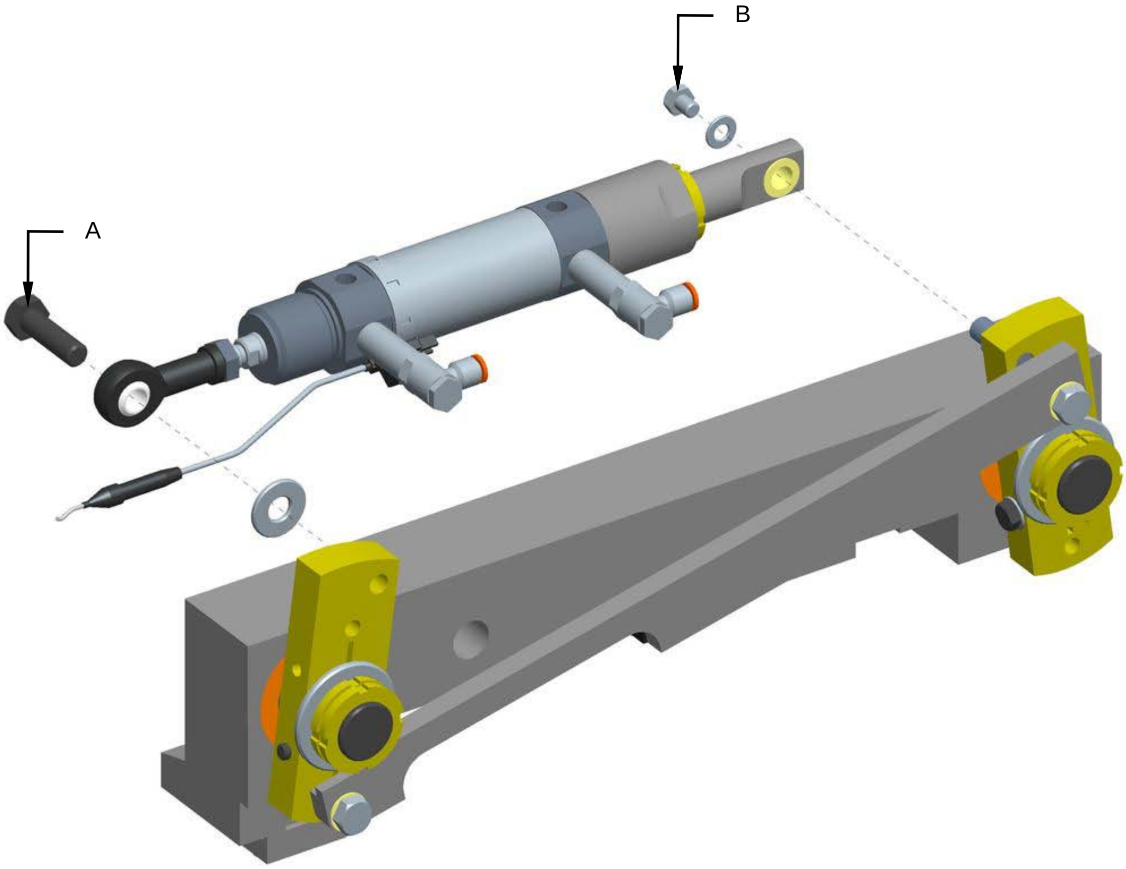


8.17.6 Replacement of the cylinder

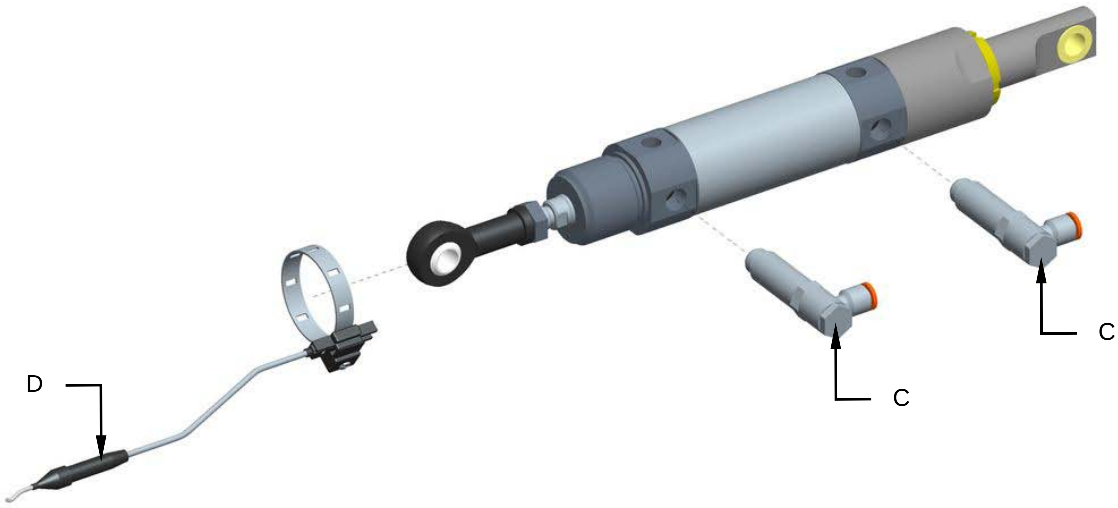
Machine status	Power totally disconnected
Equipment	Standard workshop tools

For cylinder replacement, proceed as follows:

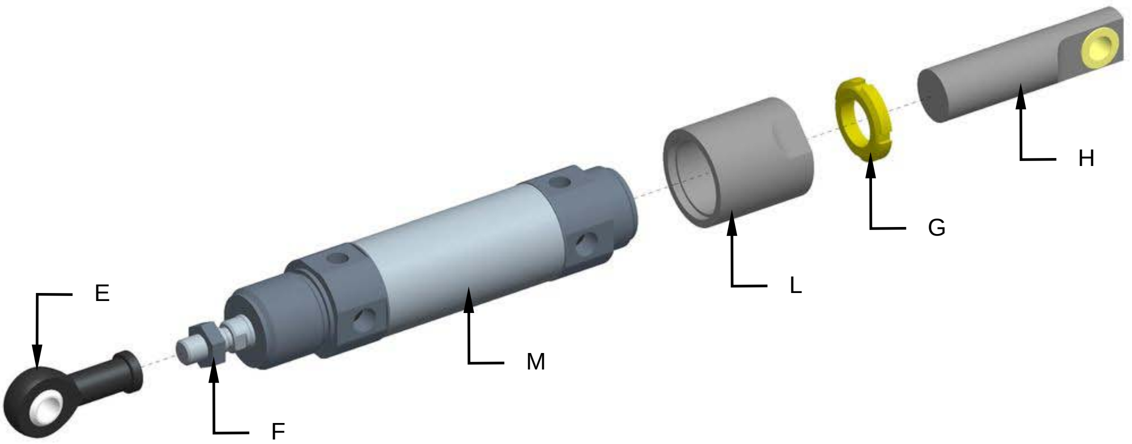
1. Disconnect the power supply of the machine.
2. Close the pneumatic plant.
3. Unscrew the screw (**A**) and (**B**).
4. Remove the cylinder group.



5. Remove the connections (**C**) and the sensor (**D**).



6. Unscrew the joint (**E**) and the nut (**F**).
7. Unscrew the ferrule (**G**) the prolongation (**H**) and the sleeve (**L**).
8. Replace the cylinder (**M**).



9. Reassemble all parts back on again by reverse procedure.

8.18 TOOL CRANK-HANDLE

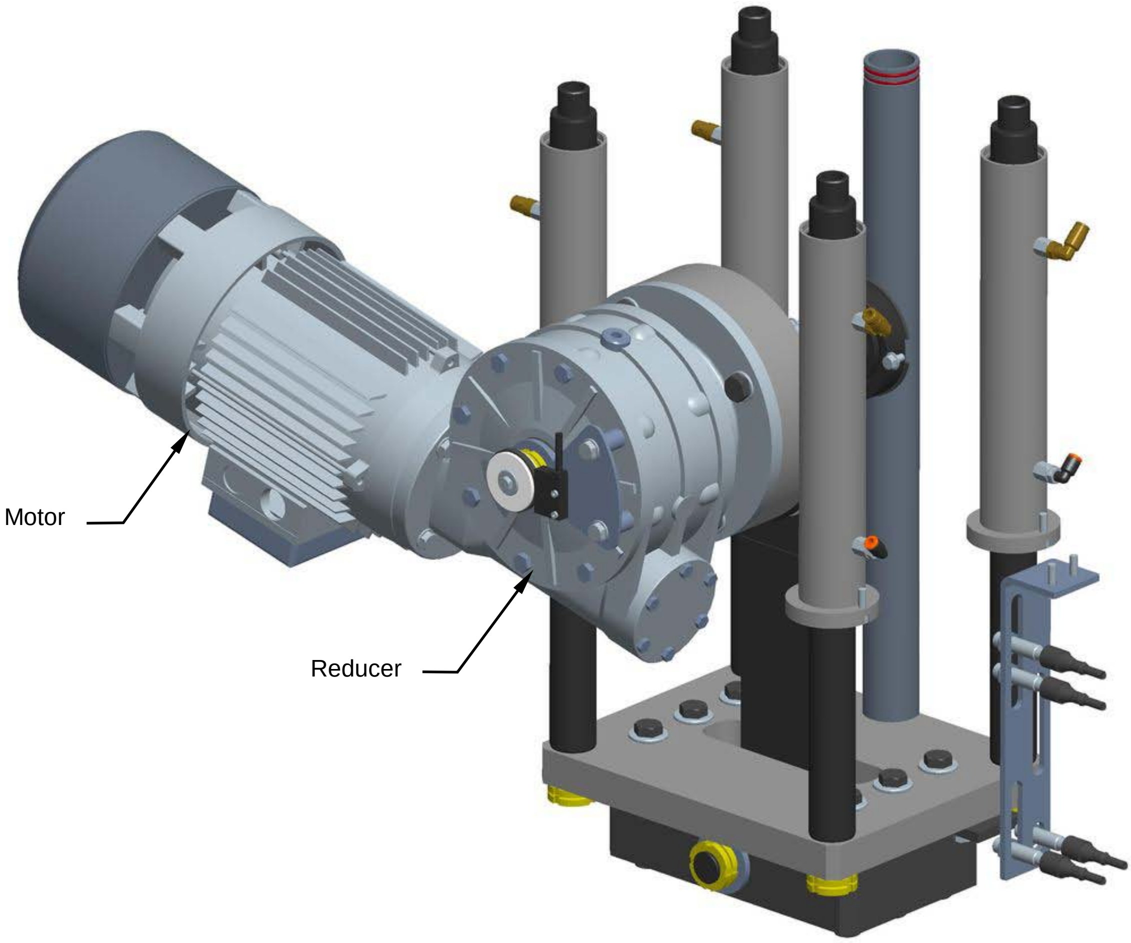
8.18.1 Inspection for motor and reducer good working conditions

Machine status	Power ON.
Equipment	Standard workshop tools

During standard machine operation cycles, carry out a visual inspection to ensure that the motor and reducer are in good working conditions.

Check to ensure that direction the motor is turning in, is correct.

Should any anomalies be found, please warn the person responsible/people in charge of setting the unit group back to optimised working conditions.

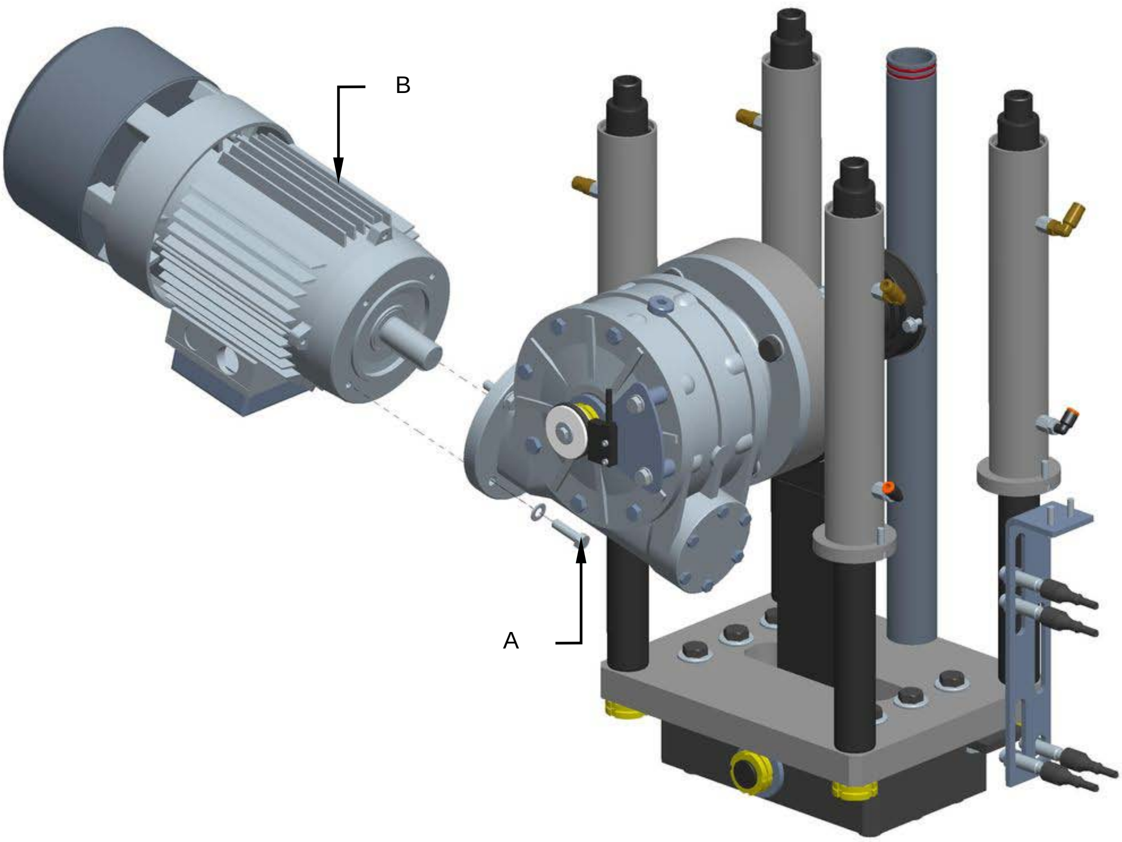


8.18.2 Motor and reducer replacement

Machine status	Power totally disconnected
Equipment	Standard workshop tools, steel support blocks.

For motor replacement, proceed as follows:

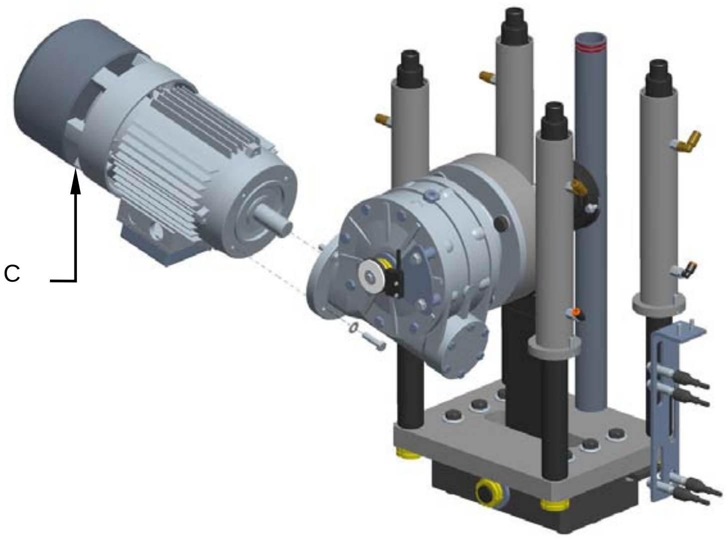
1. Disconnect the power supply of the motor.
2. Unscrew the screws (**A**) and replace the motor (**B**).



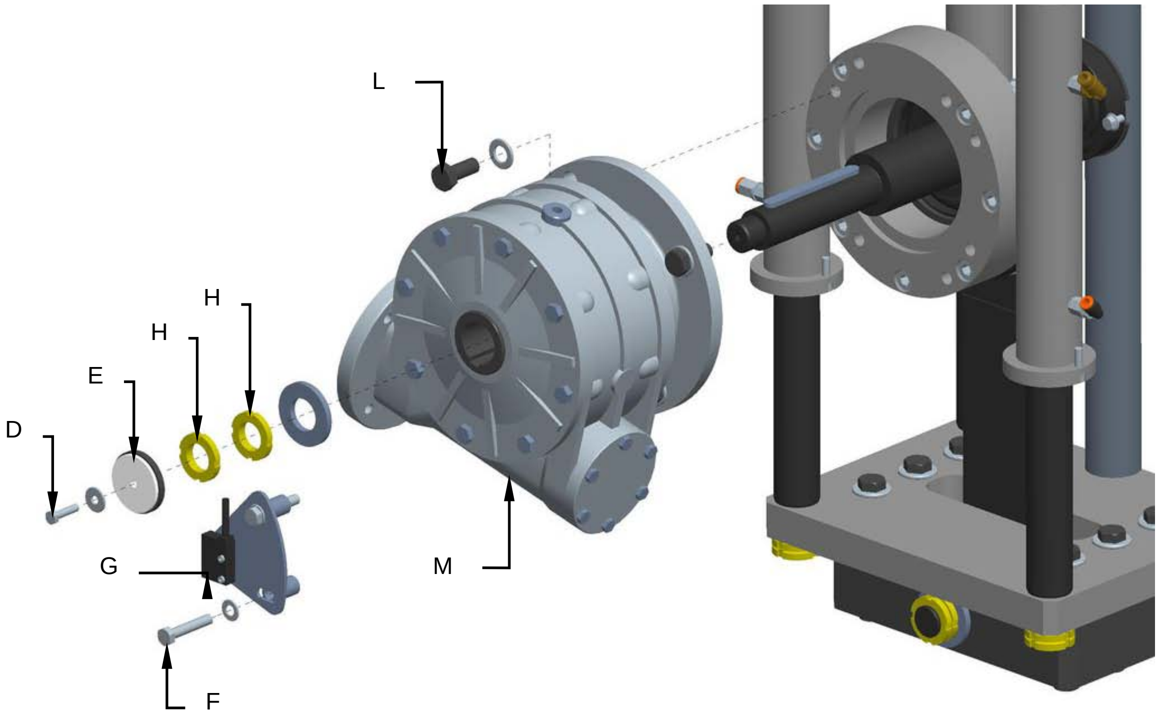
3. Reassemble all parts back on again by reverse procedure.

For reducer replacement, proceed as follows:

1. Disconnect the power supply of the motor.
2. Remove the motor (**C**).



3. Unscrew the screw (**D**) and remove the magnetic ring (**E**).
4. Unscrew the screws (**F**) and remove the encoder (**G**).
5. Unscrew the gears (**H**) and the screws (**L**).
6. Replace the reducer (**M**).



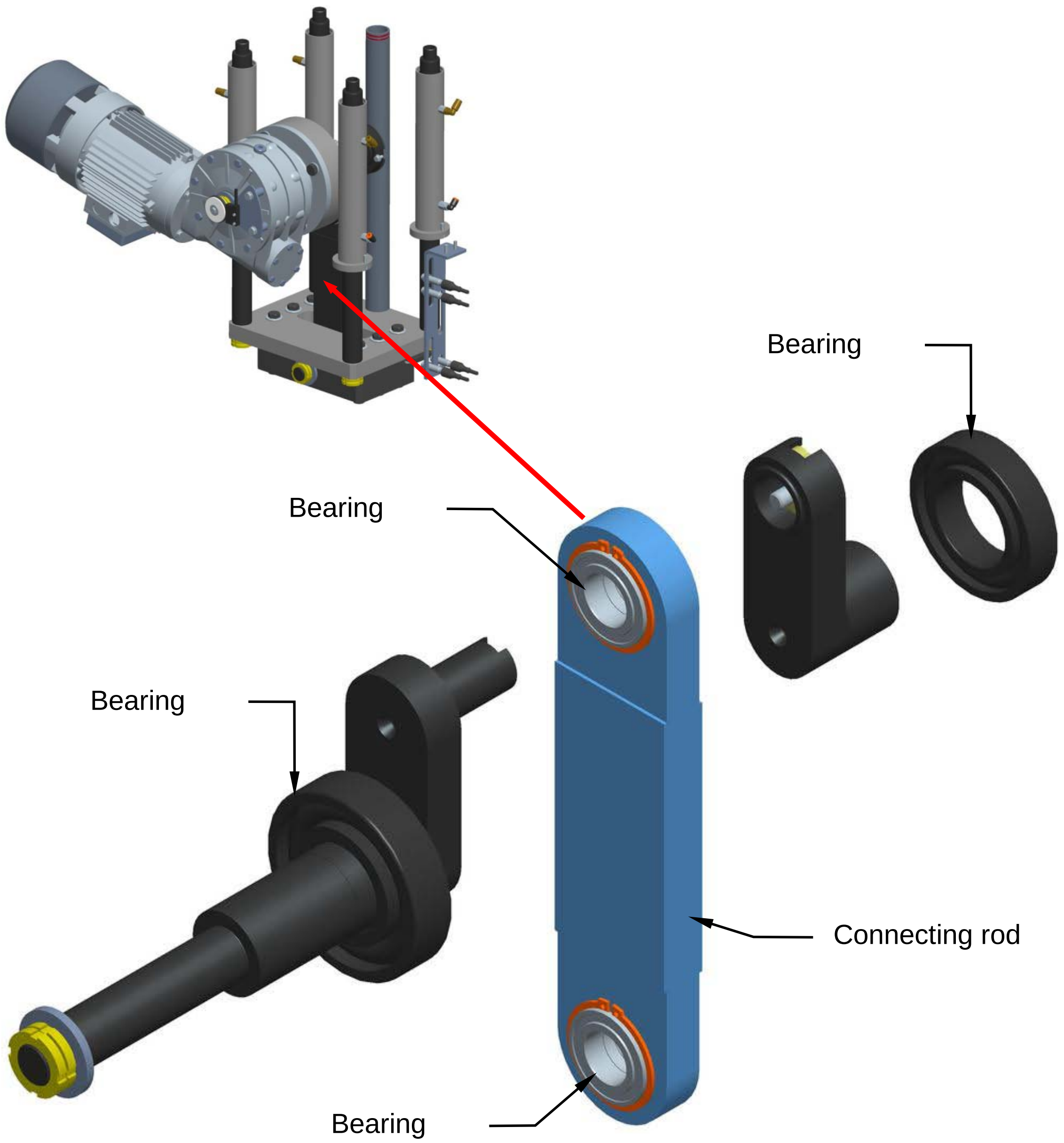
7. Reassemble all parts back on again by reverse procedure.

8.18.3 Checking on connecting rod and bearings good working conditions

Machine status	Power ON.
Equipment	Standard workshop tools

During standard machine operation cycles, carry out a visual inspection to ensure that the connecting rod is moving correctly.

Should any anomalies be found, please warn the person responsible/people in charge of setting the unit group back to optimised working conditions.

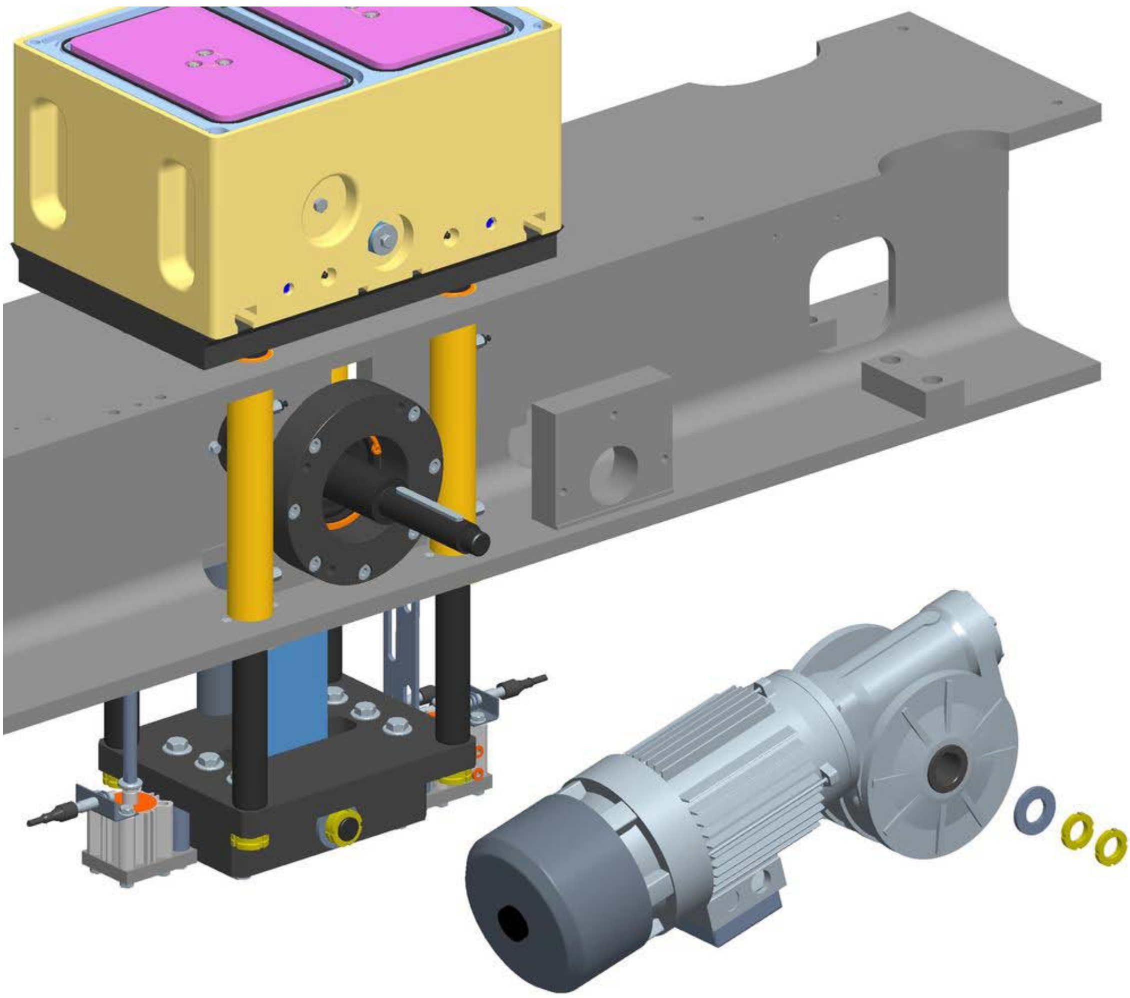


8.18.4 Connecting rod and bearings replacement

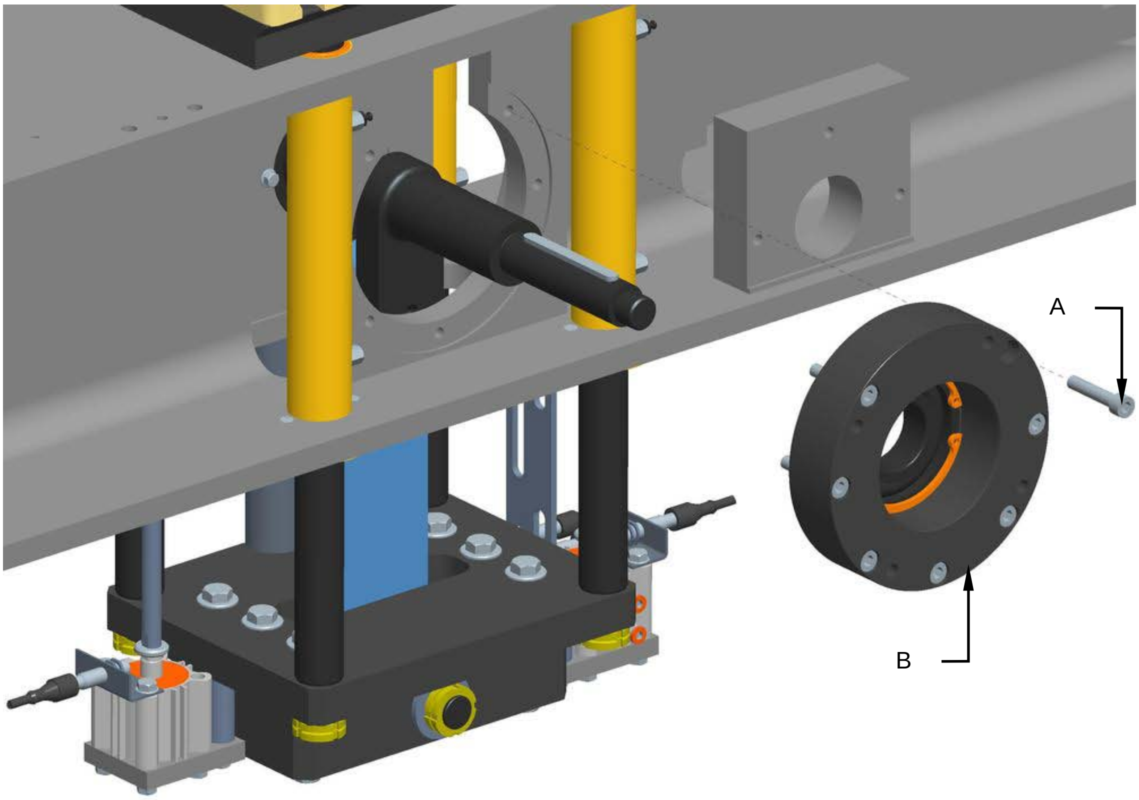
Machine status	Power totally disconnected
Equipment	Standard workshop tools, steel support blocks.

For connecting rod and bearings replacement operations, proceed as follows:

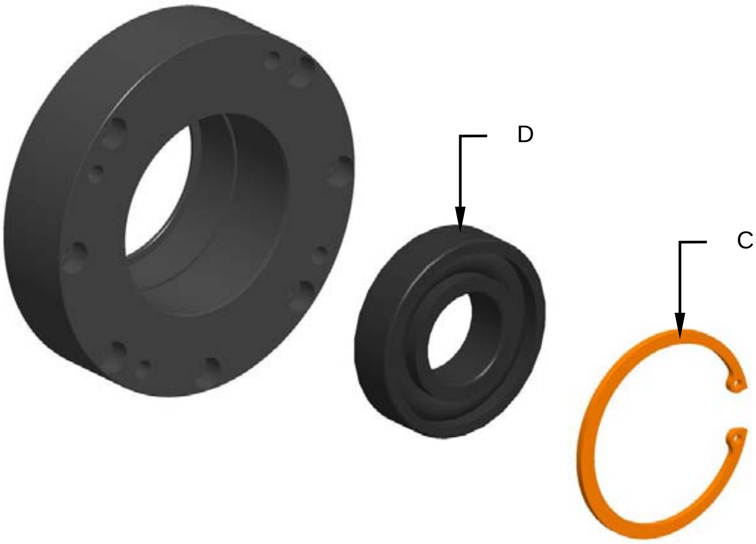
1. Press the stop emergency pushbutton.
2. Switch off the electrical plant.
3. Stop the pneumatic plant.
4. Remove the motor and reducer group.



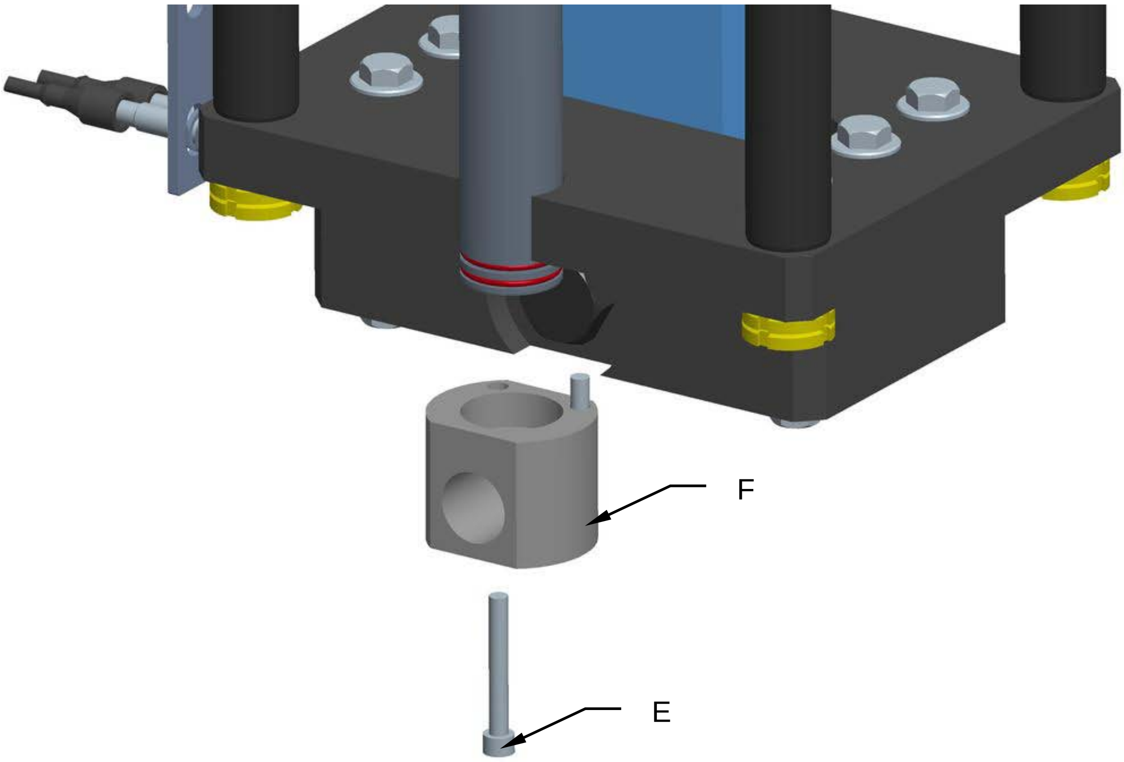
5. Unscrew the screws (**A**) and remove the stirrup (**B**).



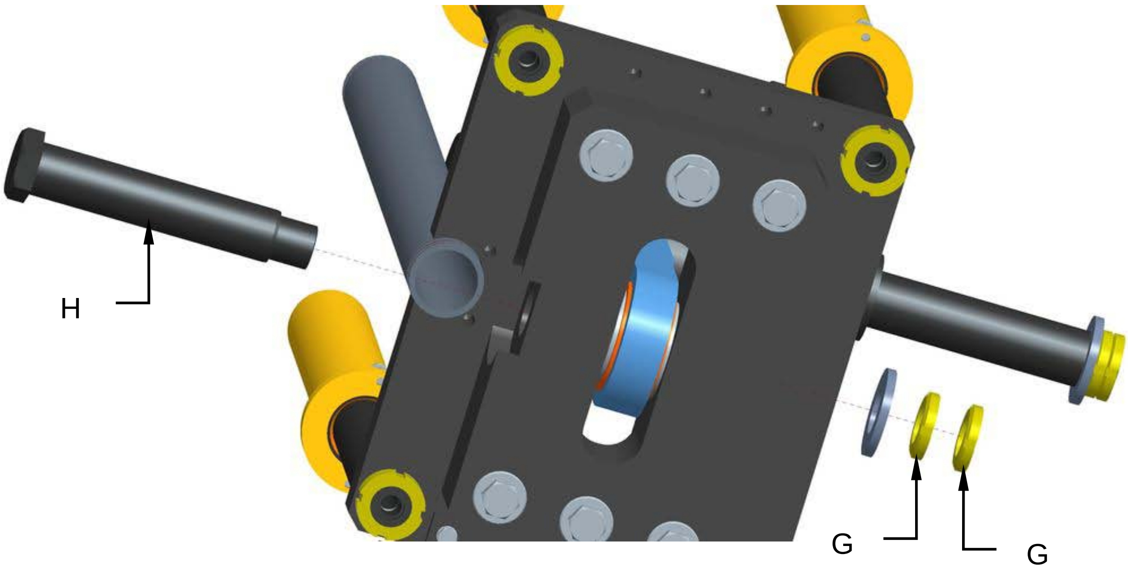
6. Remove the seeger (**C**) and replace the bearing (**D**).



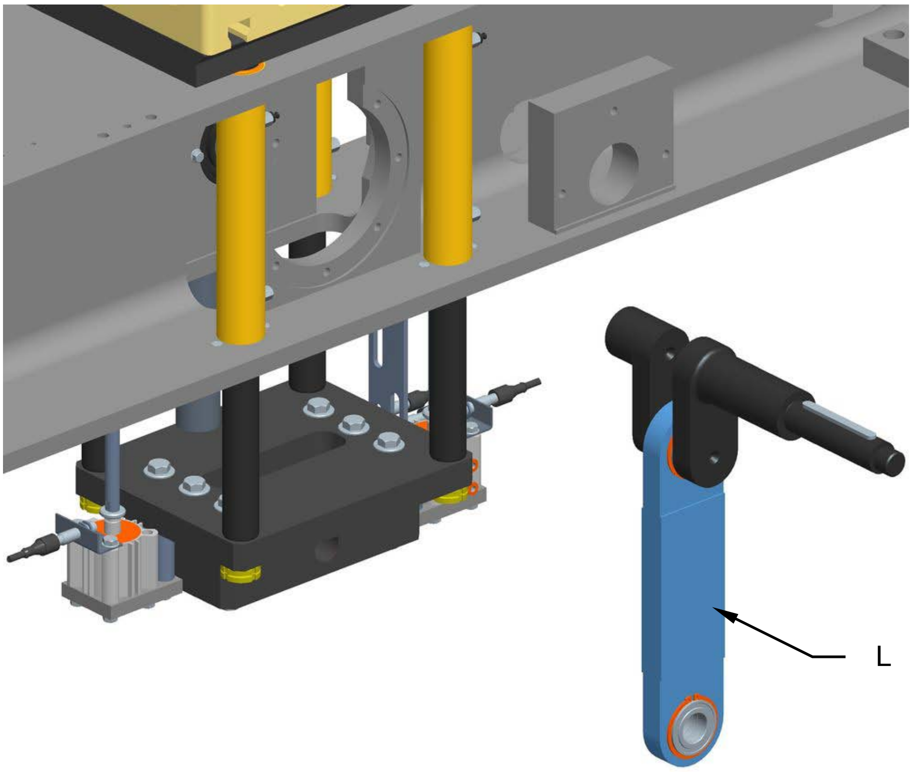
7. Unscrew the screws (**E**) and remove the vacuum connection (**F**).



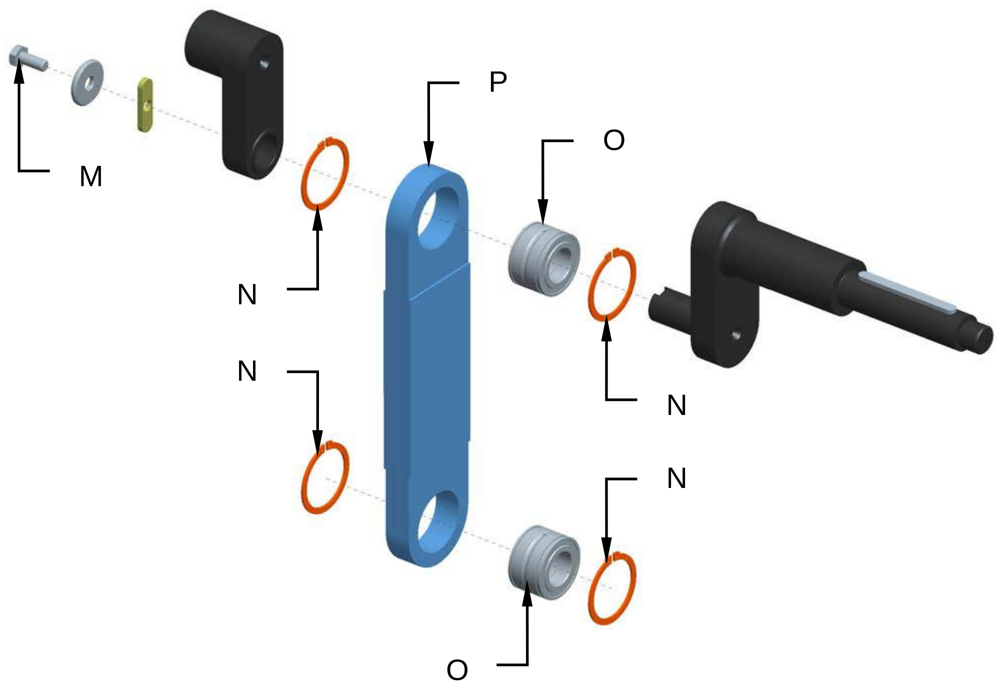
8. Unscrew the gears (**G**) and remove the pivot (**H**).



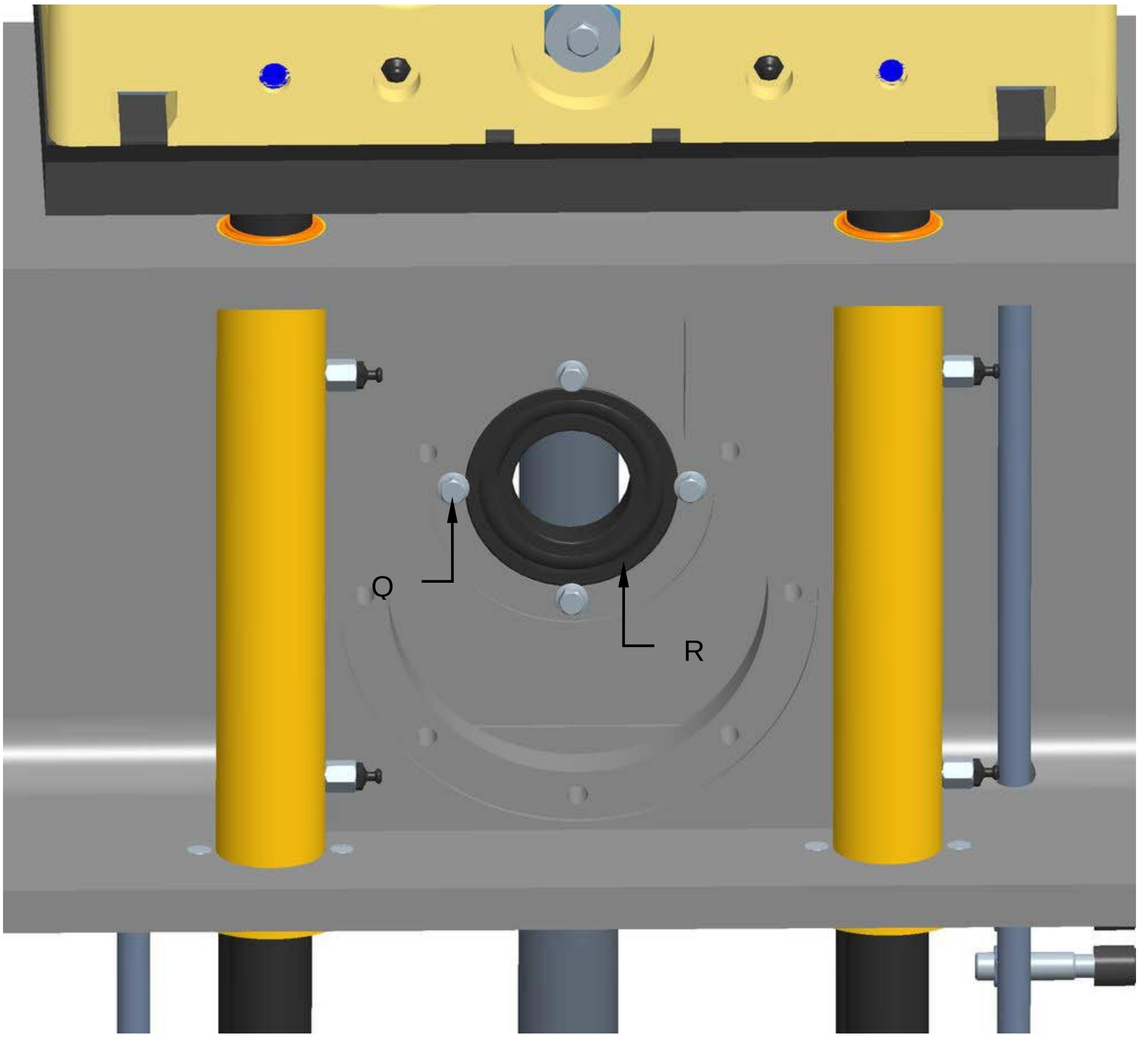
9. Remove the connecting rod group (**L**).



10. Unscrew the screw (**M**).
11. Remove the seegers (**N**).
12. Replace the bearings (**O**) and the connecting rod (**P**).



13. Unscrew the screws (**Q**) and replace the bearing (**R**).



14. Reassemble all parts back on again by reverse procedure.

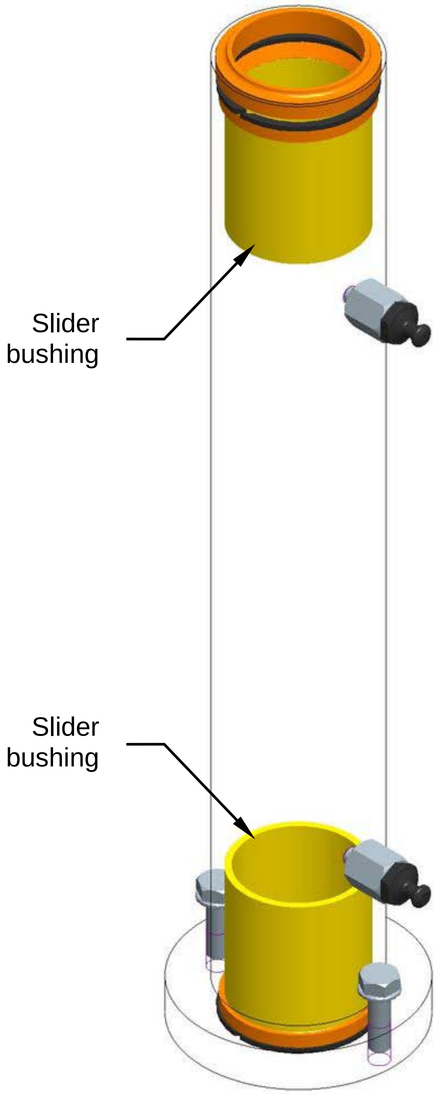
8.18.5 Slider bushing good working order inspection

Machine status	Power ON.
Equipment	Standard workshop tools

During standard machine operation cycles, carry out a visual inspection to ensure that the columns are gliding in good order and correctly.

Visually inspect the slider bushings and sleeves for any wear and tear.

Should any anomalies be found, please warn the person responsible/people in charge of setting the unit group back to optimised working conditions.

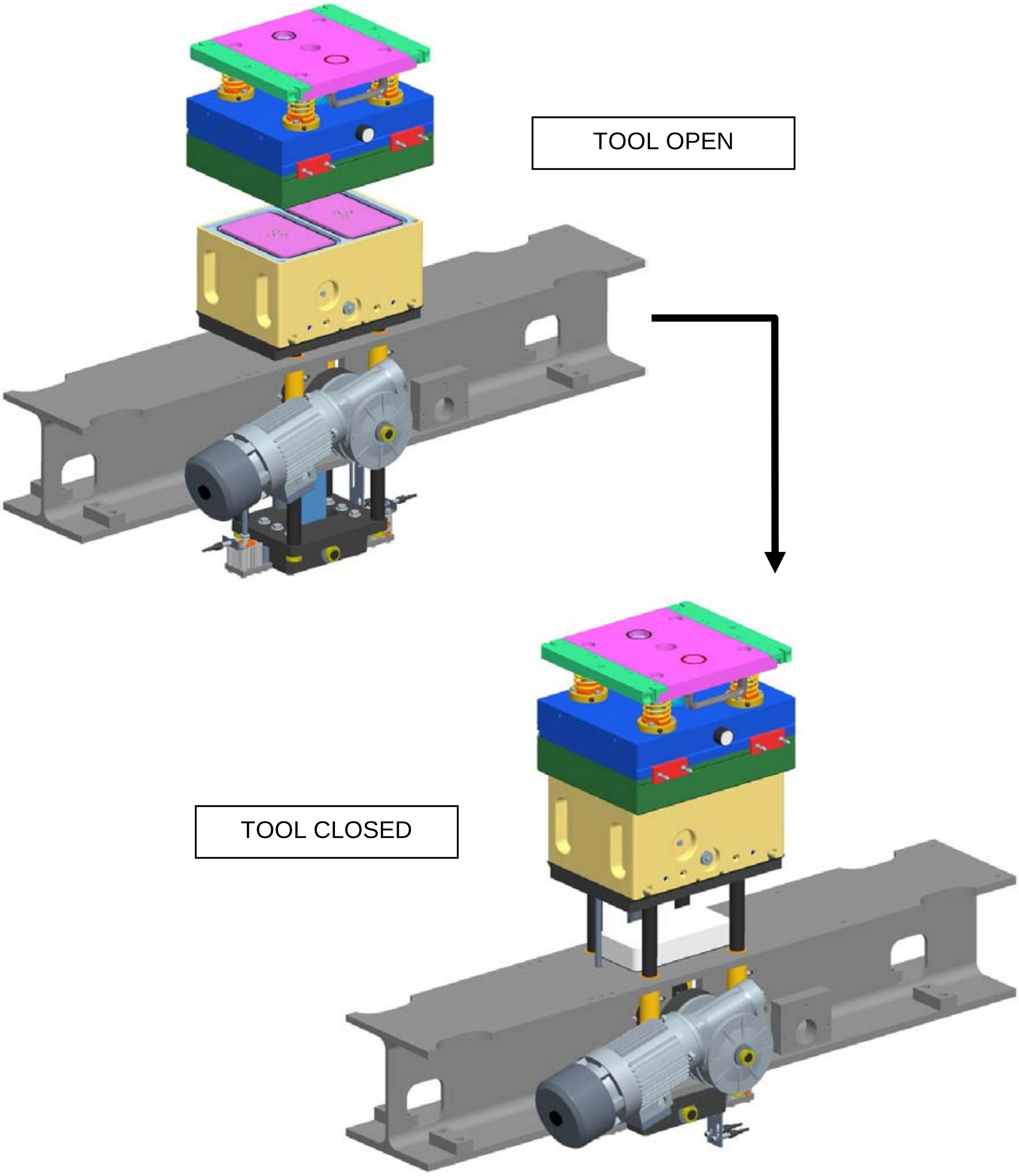


8.18.6 Slider bushing replacement

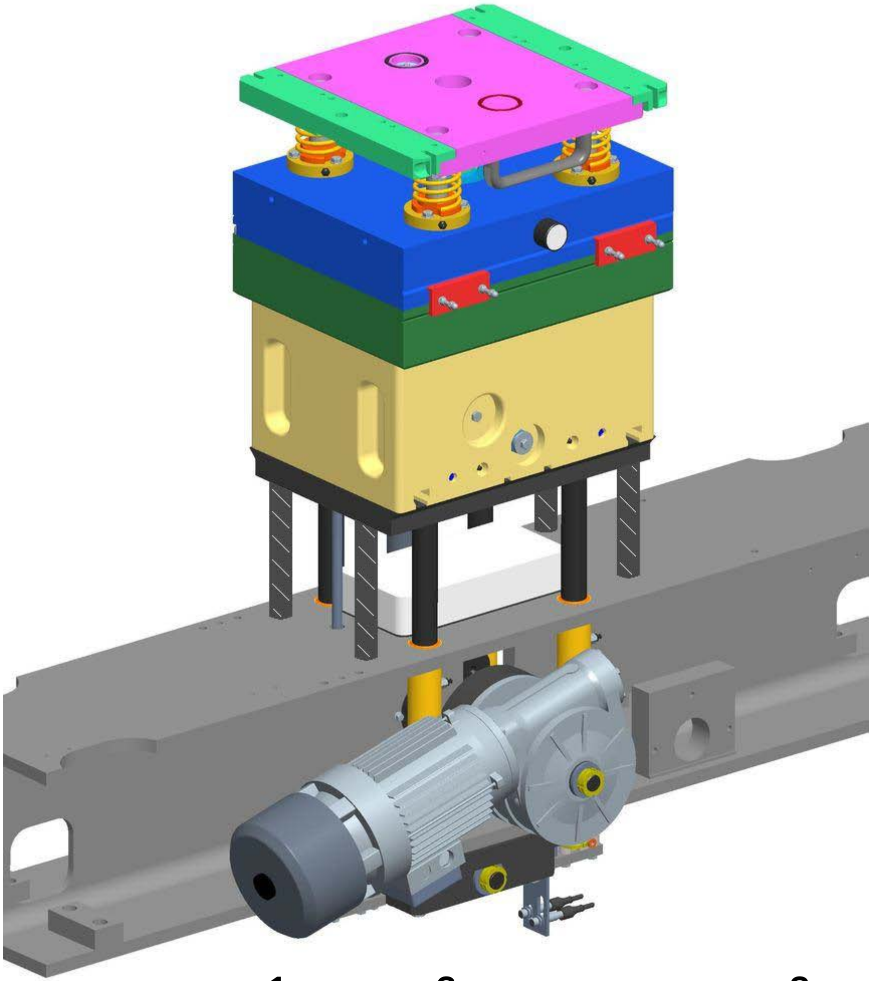
Machine status	Power totally disconnected
Equipment	Standard workshop tools, steel support blocks.

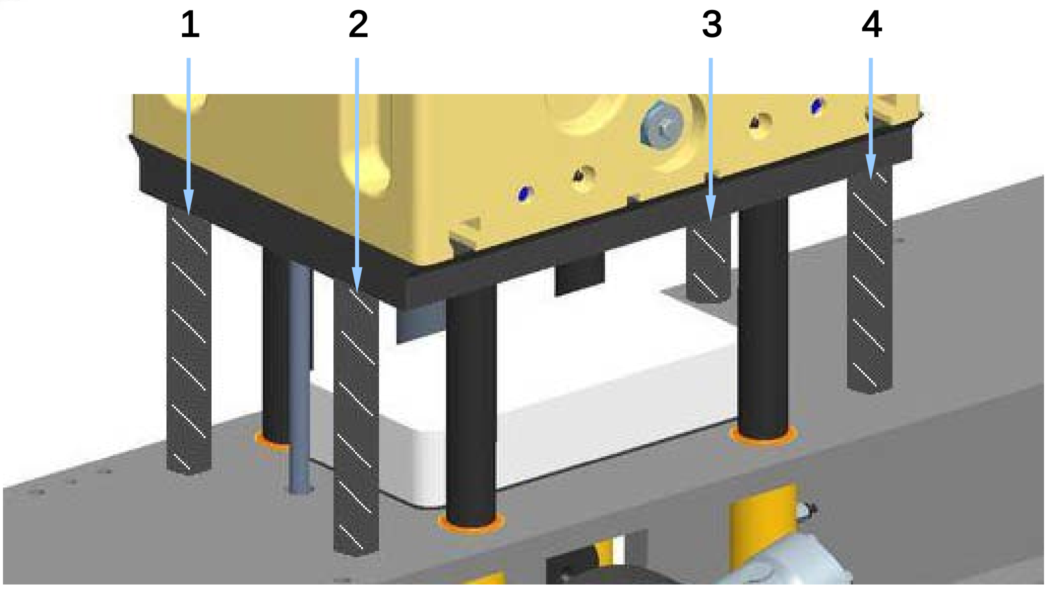
For slider bushing replacement, proceed as follows:

1. Lift the tool into its closed position.

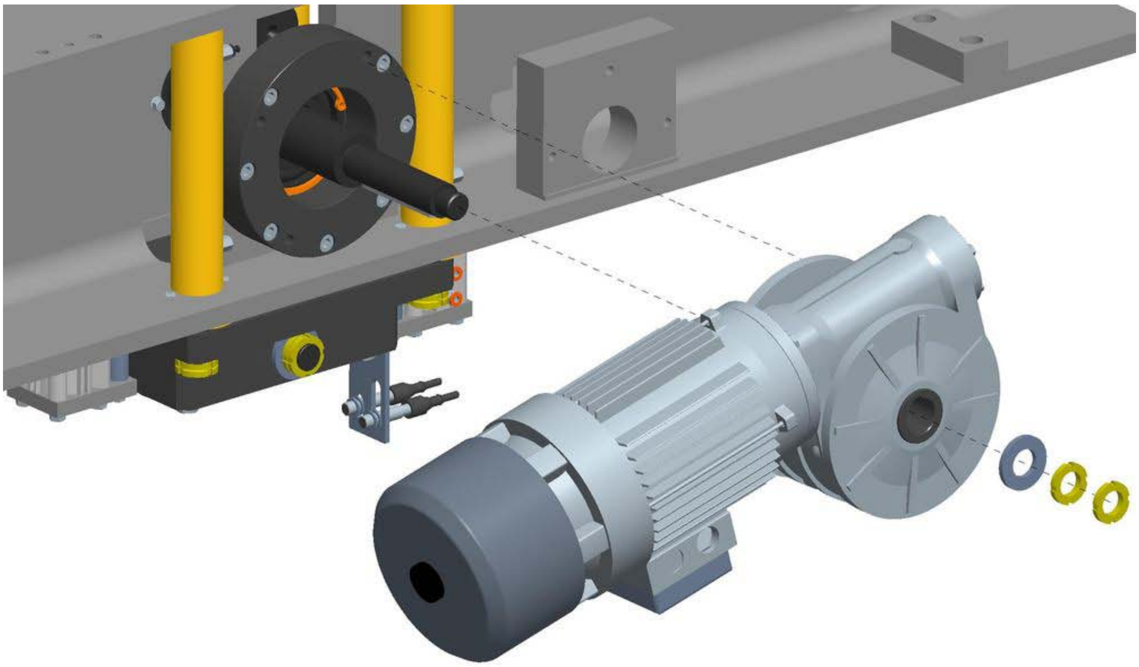


2. Insert **FOUR** shims in under the tool plate to make the following operation easier and to avoid serious injuries.

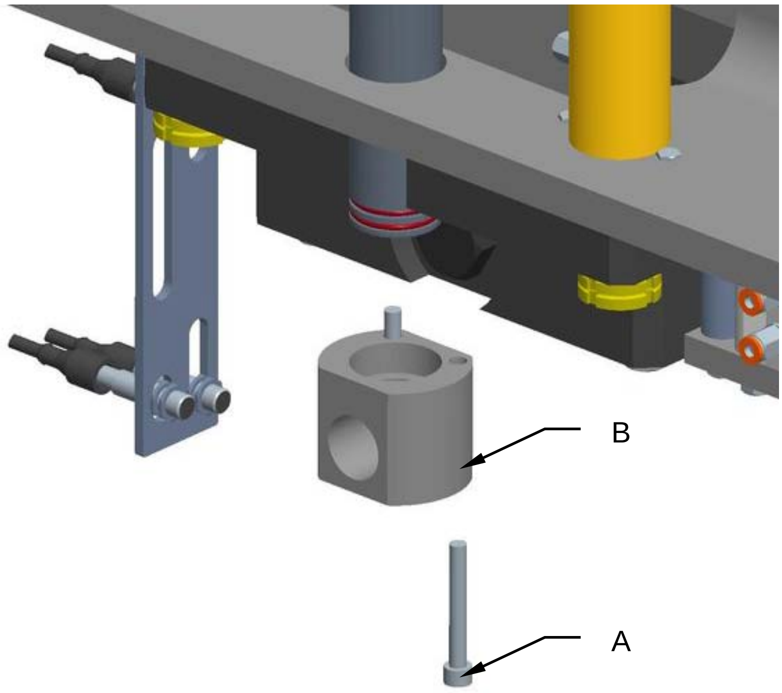




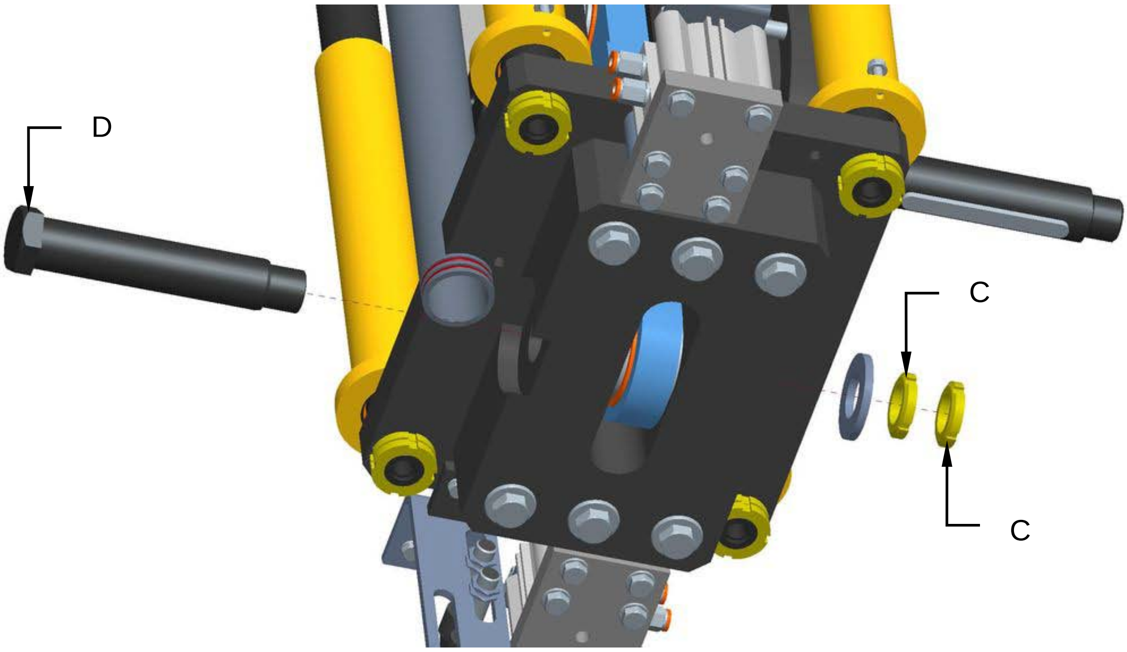
3. Activate the emergency pushbuttons.
4. Cut off the electrical system.
5. Cut off the pneumatic/air supply system.
6. Remove the motor and reducer assembly.



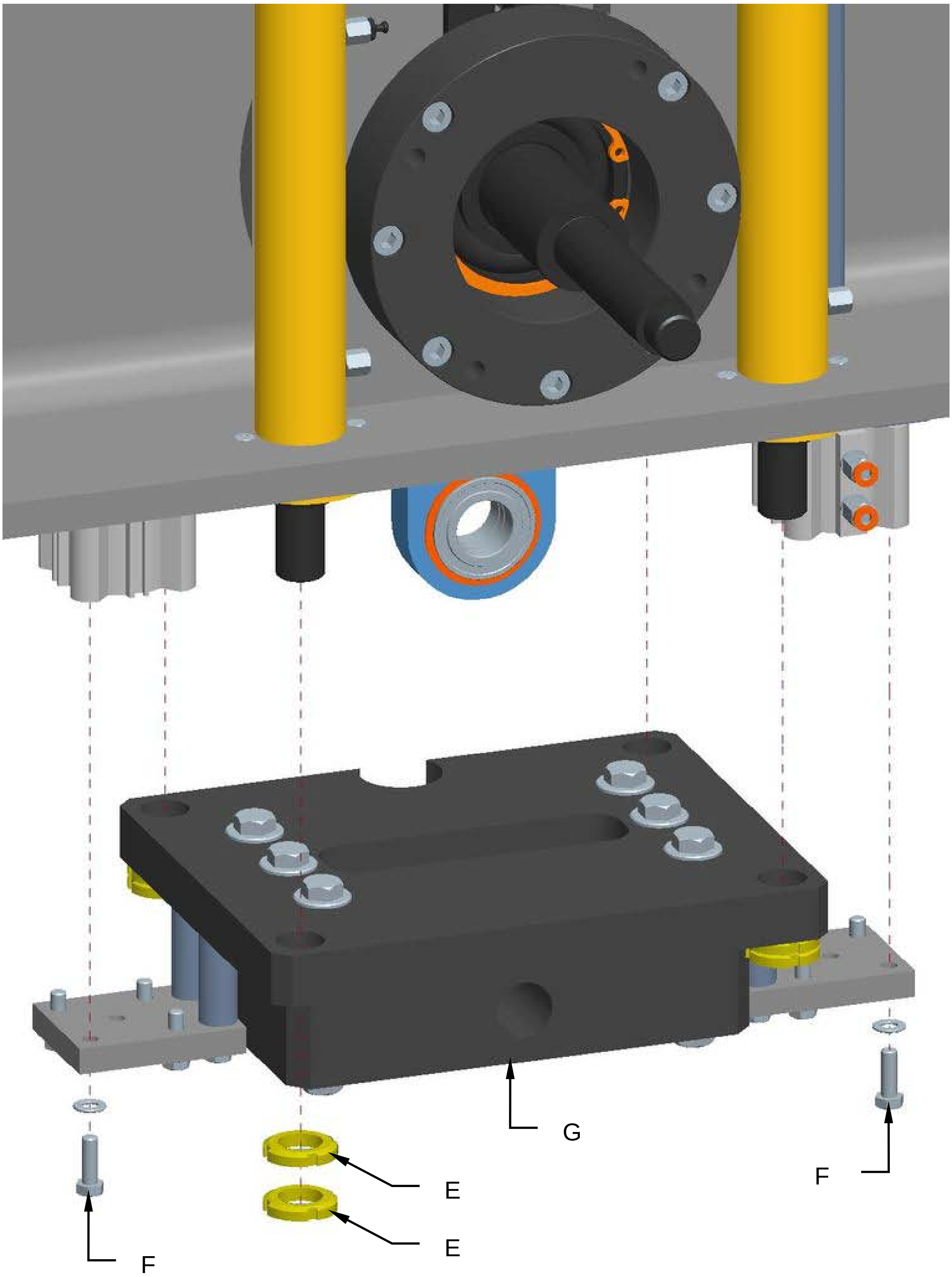
7. Unscrew screws (**A**) and remove the vacuum connection (**B**).



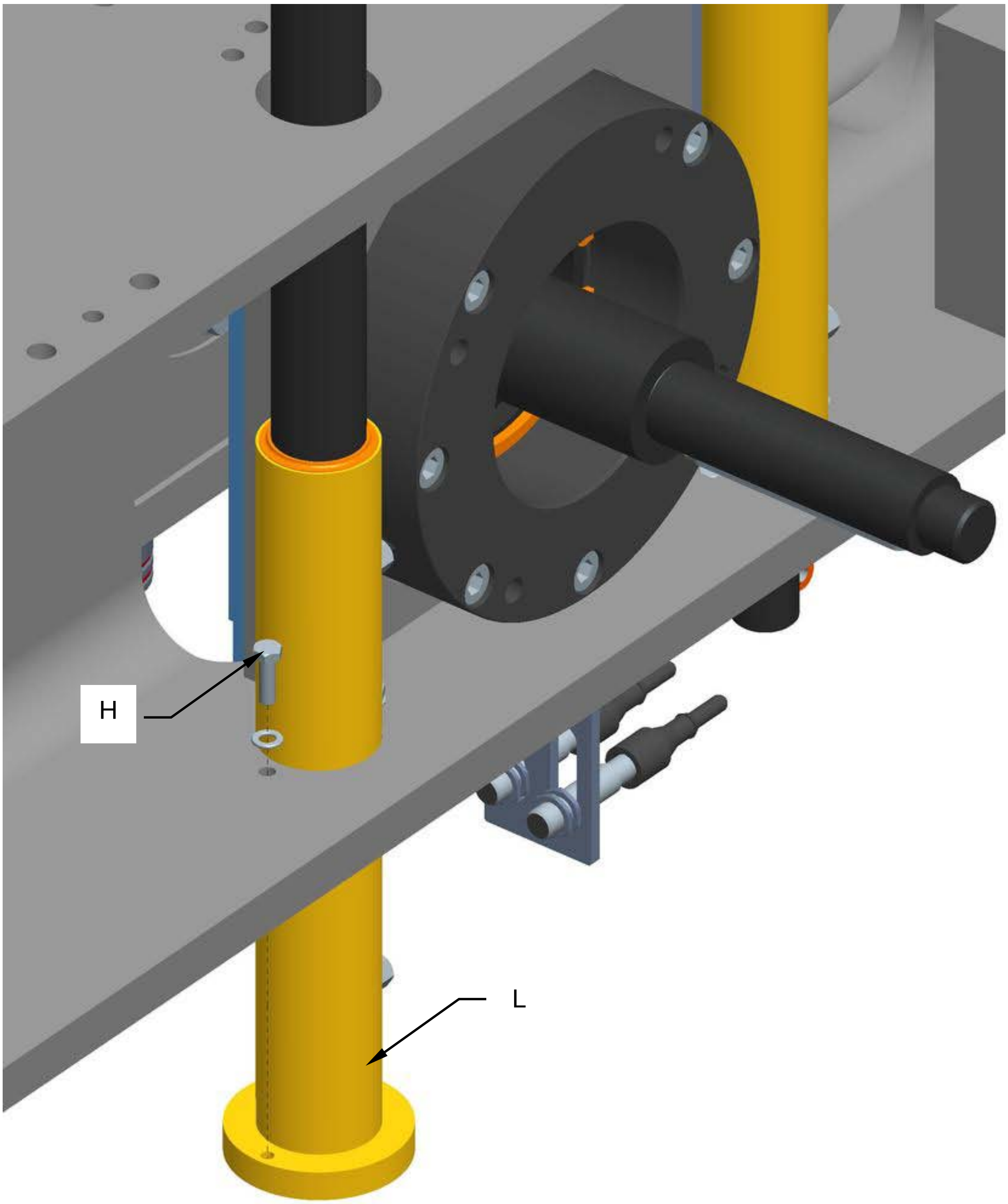
8. Unscrew the gears (**C**) and remove the lock pin (**D**).



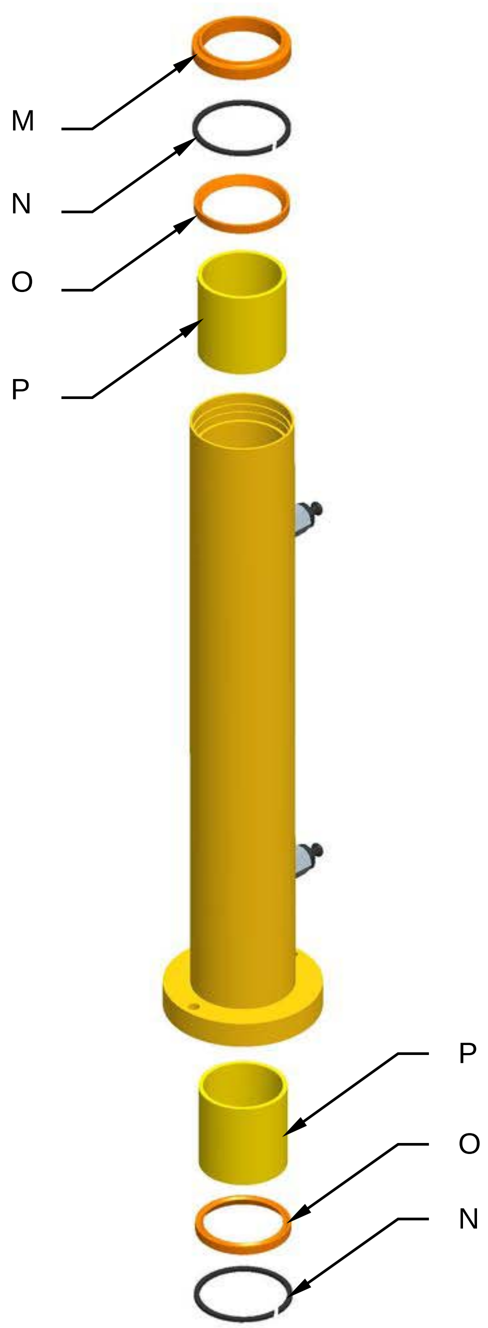
9. Unscrew the gears (**E**), the screws (**F**) and remove the plates (**G**).



10. Unscrew the screws (**H**) and remove the bushings (**L**).



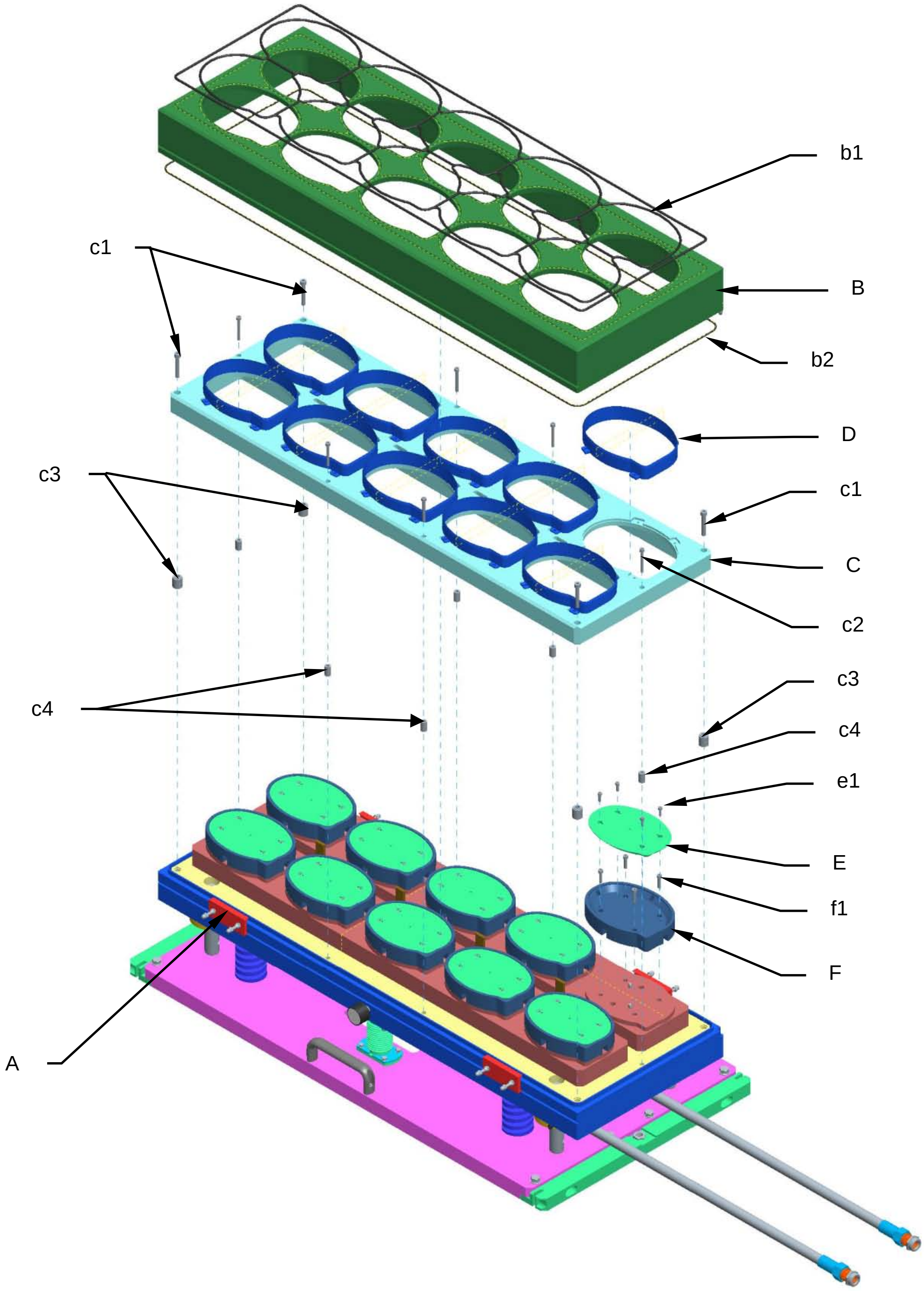
11. Remove the gaskets (**M**), the seegers (**N**) and the gaskets (**O**).
12. Replace the spider bushings (**P**).



13. Reassemble all parts back on again by reverse procedure.

8.19 UPPER TOOL MAINTENANCE

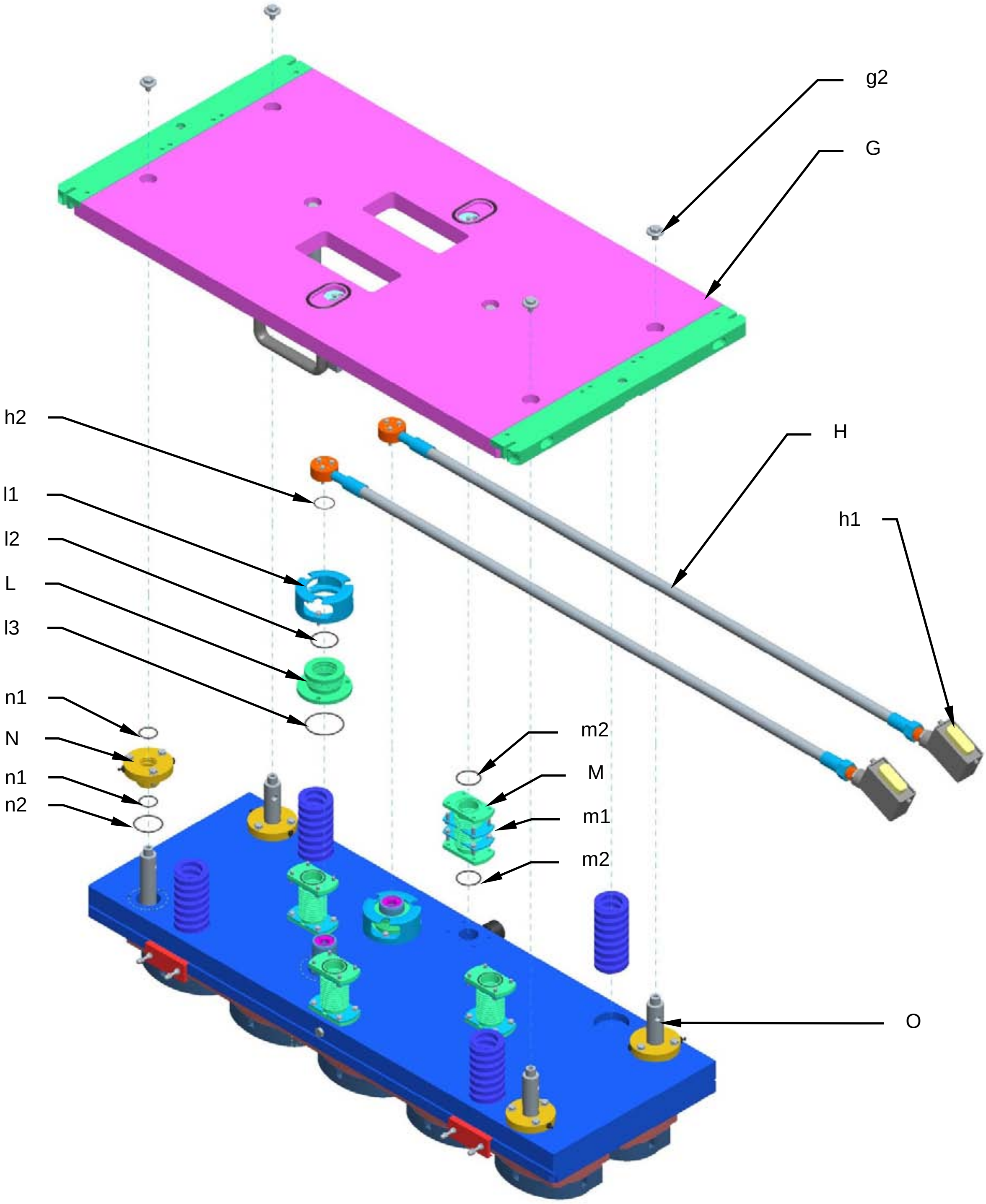
8.19.1 Intermediate cover, die cutter and sealer disassembly and cleaning sequence



1. Loosen the self-locking nuts in the bracket (**A**) until the bracket opens.
2. Remove the intermediate cover (**B**) and check the gaskets (**b1** and **b2**). Replace any damaged ones.
3. Unscrew the screws (**c1** and **c2**) to loosen the die cutter holders (**C**), taking care not to hurt yourself on the die cutters (**D**). Remove the spacers (**c3** and **c4**).
4. Clean and check the die cutters (**D**). If any damaged ones need to be replaced, unscrew them carefully from the plate (**C**).
5. Clean and check the insulation (**E**) and the sealer (**F**). Unscrew the screws (**e1**) if any damaged insulation needs to be replace, and screws (**f1**) if the sealer needs to be replaced.

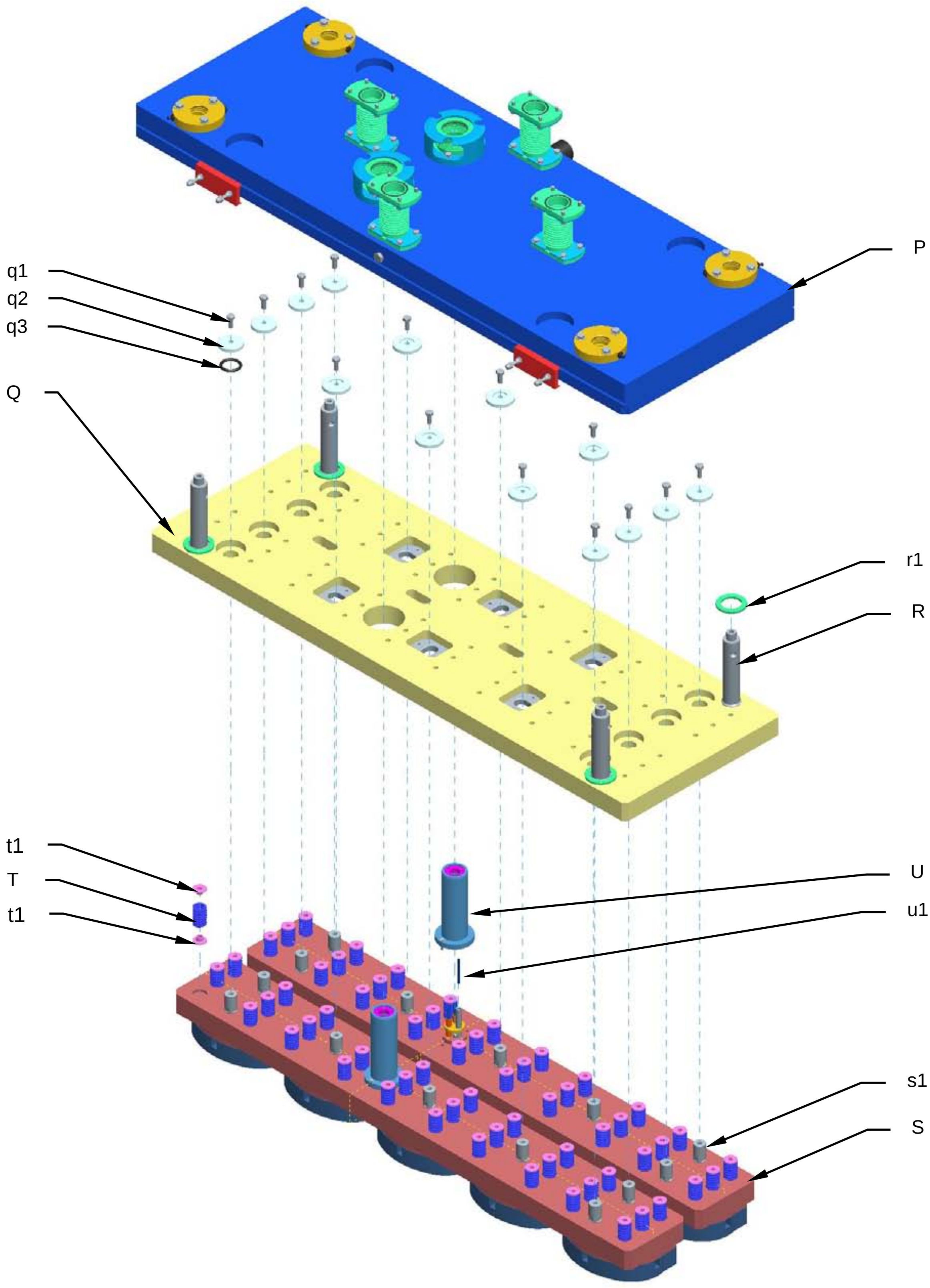
   Tighten all the screws using a torque wrench, keeping to the torques for each diameter, as shown in the attached table (**8.3**).

8.19.2 Upper plate and blower disassembly and cleaning sequence



1. Remove the plate (**G**) by unscrewing the fixing screw (**g1**) and the screws in the upper brackets (**m1**) that lock the blowers (**M**) to the plate.
2. Remove the sheaths (**H**) by disconnecting the power cables from the connectors (**h1**) and pulling them out of the sheaths. Check the O-ring (**h2**) and the sheath for wear and replace if necessary.
3. Check the blower (**L**) and the O-ring (**l2**). If they need to be replaced, remove the bracket (**l1**). Once the blower has been freed you can also check the O-ring (**l3**).
4. Check the blowers (**M**) and the O-rings (**m2**). If they need to be replaced, remove the lower brackets.
5. Check the grease nipples of the bushes (**N**) and make sure they are not clogged. They must allow the grease to lubricate the column. Also check the state of the O-rings (**n1**). If they need to be replaced, remove the bush and this will allow you to check the O-ring (**n2**).
6. Remove and check the springs (**O**).
7. Tighten all the screws using a torque wrench, keeping to the torques for each diameter, as shown in the attached table (**8.3**).

8.19.3 Internal plate and resistance disassembly and cleaning sequence

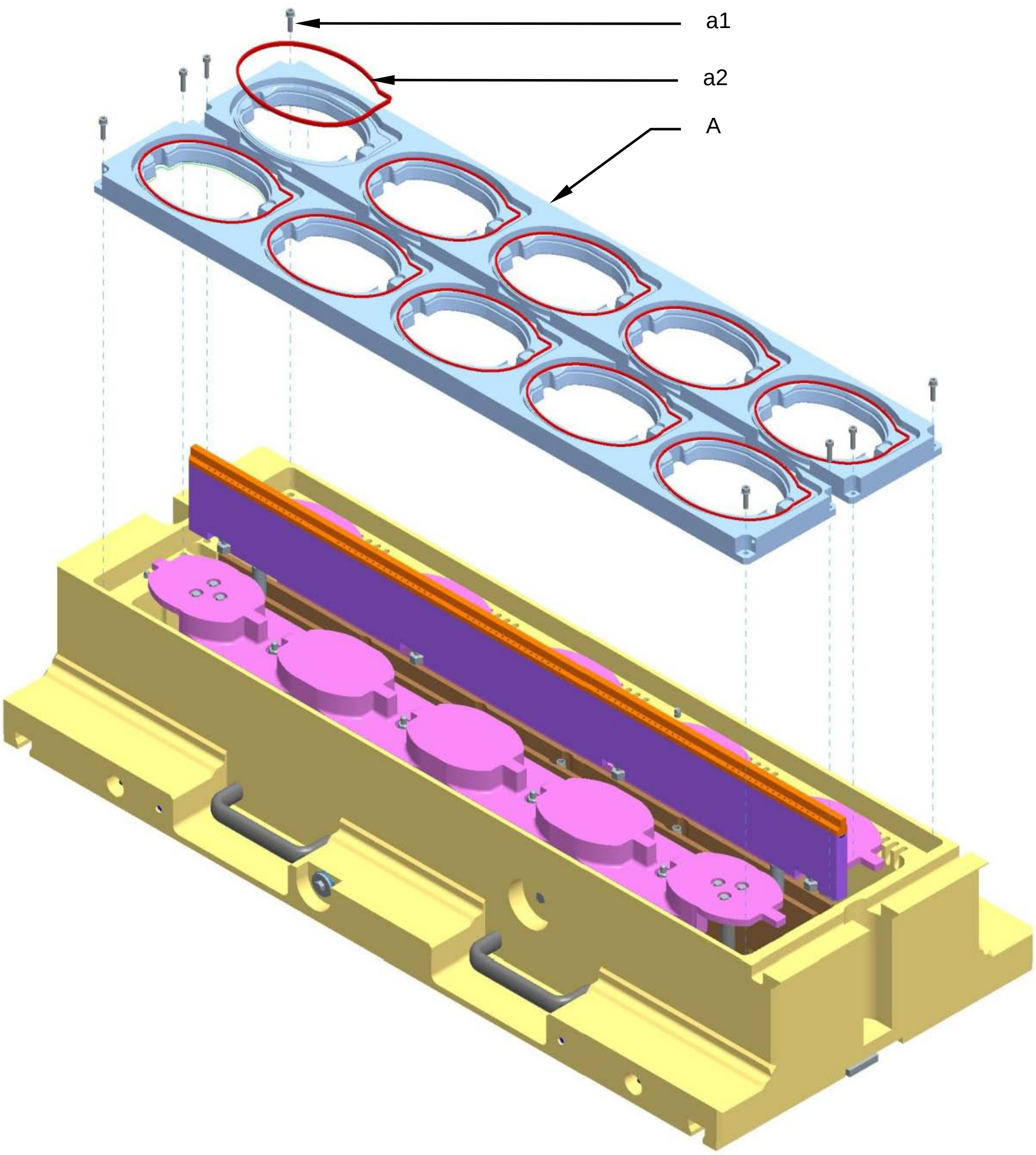


1. Remove the upper cover (**P**).
2. Check the columns (**R**) and the shock-absorbers (**r1**).
3. Remove the internal plate (**Q**) by unscrewing the screws (**q1**) in the washers (**q2**). Check the O-rings (**q3**), and replace if necessary.
4. Check the pins (**s1**) of the resistance (**S**) and make sure they are tight. If necessary, replace them or apply a threadlocker.
5. Check all the springs (**T**) and replace is necessary. Reposition with the flanges (**t1**) well lubricated with high-pressure/temperature grease.
6. To replace the PT100 probe (**u1**), remove the sleeve (**U**) and unscrew the probe-locking screw. Check the state of the sleeve (**U**) O-ring.
7. Tighten all the screws using a torque wrench, keeping to the torques for each diameter, as shown in the attached table (**8.3**).

It is good practice to check the tightness of the sealing tool screws at least once a month.

8.20 LOWER TOOL MAINTENANCE

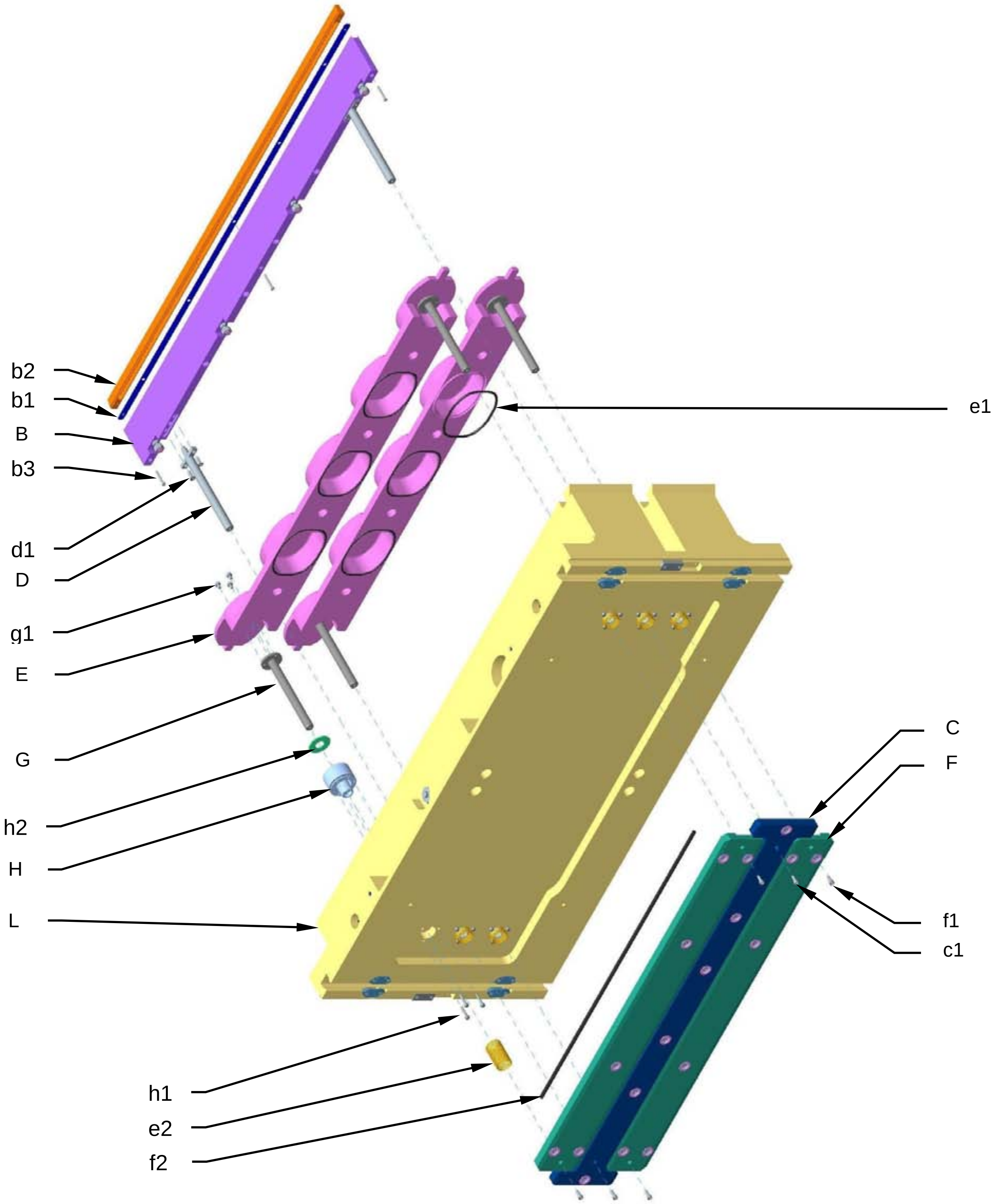
8.20.1 Counter-tool assembly and cleaning sequence



1. Unscrew the securing screws (**a1**) and remove the counter-tool (**A**). Check the gasket (**a2**) and slot and replace any worn parts.

2. Tighten all the screws using a torque wrench, keeping to the torques for each diameter, as shown in the attached table (**8.3**).

8.20.2 Plate and central guide assembly and cleaning sequence



1. Unscrew the screws (**c1**) of the pin plate (**C**) to remove the centre guide (**B**). Check that the holes in the central distributor (**b2**) are free. If necessary, unscrew the screws (**b3**) and clean the holes, checking the gasket (**b1**) as well.
2. Check the columns (**D**) in the centre guide. If they are not in perfect condition, unscrew the securing screws (**d1**) and replace them.
3. Unscrew the securing screws (**f1**) in the pin plate (**F**) and remove the tray-raising plate (**E**). Check the state of the gasket (**e1**) and the springs (**e2**).
4. Check the gaskets (**f2**) of the pin plates (**C** and **F**).
5. Check the state of the columns (**G**) of the tray-raising plate and replace if necessary by unscrewing the screws (**g1**).
6. Check the gaskets and guide bush (**H**). If they need to be replaced, unscrew the screws (**h1**) and remove the bush. Also check the state of the shock-absorber (**h2**).
7. Check the grease nipples of the lower cover (**L**) and make sure they are not clogged. They must allow the grease to lubricate the columns (**D** and **G**). Also make sure all the vacuum and gas passages are clear of obstruction and clean if necessary
8. Tighten all the screws using a torque wrench, keeping to the torques for each diameter, as shown in the attached table (**8.3**).

It is good practice to check the tightness of the sealing tool screws at least once a month.

8.21 TABLE OF PRE-LOADS AND TORQUES FOR STAINLESS STEEL SCREWS

		Pre-load in kN								Torque in Nm							
Friction coeff. >		0.1	0.12	0.14	0.16	0.18	0.20	0.30	0.40	0.1	0.12	0.14	0.16	0.18	0.20	0.30	0.40
M4	50	1.38	1.33	1.27	1.22	1.17	1.12	0.90	0.74	0.8	0.9	1.0	1.1	1.2	1.3	1.5	1.6
	70	2.97	2.85	2.73	2.62	2.50	2.40	1.94	1.60	1.7	2.0	2.2	2.3	2.5	2.6	3.0	3.3
	80	3.97	3.80	3.64	3.49	3.34	3.20	2.59	2.13	2.3	2.6	2.9	3.1	3.3	3.5	4.1	4.4
M5	50	2.26	2.18	2.09	2.00	1.92	1.83	1.49	1.22	1.6	1.8	2.0	2.1	2.2	2.4	2.8	3.2
	70	4.85	4.66	4.47	4.29	4.11	3.93	3.19	2.62	3.4	3.8	4.2	4.6	4.9	5.1	6.1	6.6
	80	6.47	6.22	5.96	5.72	5.48	5.24	4.25	3.50	4.6	5.1	5.6	6.1	6.5	6.9	8.0	8.8
M6	50	3.20	3.07	2.94	2.82	2.70	2.59	2.09	1.73	2.8	3.1	3.5	3.7	4.0	4.1	4.8	5.3
	70	6.85	6.57	6.31	6.05	5.79	5.54	4.49	3.70	5.9	6.7	7.4	7.9	8.4	8.8	10.4	11.3
	80	9.13	8.77	8.41	8.06	7.72	7.39	5.98	4.93	8.0	9.1	9.9	10.5	11.2	11.8	13.9	15.0
M8	50	5.86	5.63	5.40	5.18	4.96	4.75	3.85	3.17	6.8	7.6	8.4	9.0	9.6	10.1	11.9	12.9
	70	12.6	12.1	11.6	11.1	10.6	10.2	8.25	6.80	14.5	16.3	17.8	19.3	20.4	21.5	25.5	27.6
	80	16.7	16.1	15.4	14.8	14.2	13.6	11.0	9.1	19.3	21.7	23.8	25.7	27.3	28.7	33.9	36.8
M10	50	9.32	8.96	8.60	8.27	7.91	7.58	6.14	5.05	13.7	15.4	16.7	18.1	19.3	20.3	24.0	26.2
	70	20.0	19.2	18.4	17.7	16.9	16.2	13.1	10.8	30	33	36	39	41	44	51	56
	80	26.6	25.6	24.6	23.6	22.6	21.7	17.5	14.4	39.4	44	47.8	51.6	55.3	58	69	75
M12	50	13.6	13.1	12.6	12.0	11.6	11.1	9.00	7.38	23.3	26.0	28.9	30.8	32.8	34.8	41.0	44.6
	70	29.1	28.1	26.9	25.8	24.8	23.7	19.2	15.8	50	56	62	66	70	74	88	96
	80	38.8	37.4	35.9	34.4	33.0	31.6	25.6	21.1	67	74	82	88	94	100	117	128
M14	50	18.7	17.9	17.3	16.5	15.8	15.2	12.3	10.1	37.1	41.7	45.6	49	52	56	66	71
	70	40.6	38.5	37.0	35.4	34.0	32.6	26.4	21.7	79	89	98	105	112	119	141	152
	80	53.3	51.3	49.3	47.3	45.3	43.3	35.2	29.0	106	119	131	140	150	159	188	204
M16	50	25.7	24.7	23.8	22.8	21.9	20.9	17.0	14.0	56	63	70	75	81	86	102	110
	70	55.0	52.9	50.9	48.9	46.8	44.9	36.4	30.0	121	136	150	162	173	183	218	237
	80	73.3	70.6	67.9	65.2	62.4	59.8	48.6	40.0	161	181	198	217	231	245	291	316
M18	50	32.2	31.0	29.8	28.5	27.3	26.2	21.2	17.5	81	91	100	108	115	122	144	156
	70	69.0	66.4	63.8	61.2	58.6	56.2	45.5	37.5	174	196	213	232	246	260	308	334
	80	92.0	88.5	85.0	81.6	78.1	74.9	60.7	50.1	232	261	285	310	329	346	411	447
M20	50	41.3	39.8	38.3	36.7	35.2	33.8	27.4	22.6	114	128	142	153	164	173	205	223
	70	88.6	85.4	82.0	78.7	75.4	72.4	58.7	48.1	244	274	303	328	351	370	439	479
	80	118	114	109	105	101	96.5	78.3	64.6	325	366	404	438	467	494	586	639
M22	50	51.6	49.8	47.9	46.0	44.1	42.3	34.3	28.3	154	174	191	208	222	234	279	303
	70	61.5	59.3	57.0	54.7	52.5	50.3	40.9	33.7	182	206	227	247	263	279	332	361
	80	148	142	137	131	126	121	98.2	80.9	437	494	545	593	613	670	797	866
M24	50	59.6	57.4	55.1	52.9	50.7	48.6	39.4	32.6	197	222	243	264	282	298	354	385
	70	70.9	68.3	65.6	63.0	60.4	57.9	47.0	38.8	234	264	290	314	336	355	421	458
	80	170	170	157	151	145	139	113	93.1	561	634	696	754	806	852	1010	1099
M27	50	75.6	72.9	70.1	67.3	64.5	61.9	50.2	41.5	275	311	344	377	399	421	503	548
	70	90.0	86.8	83.4	80.1	76.9	73.7	59.8	49.4	328	371	410	444	475	502	599	652
M30	50	91.9	88.6	85.2	81.7	78.4	75.2	61.0	50.3	374	423	467	506	540	571	680	740
	70	104	105	101	97.3	93.3	89.5	72.6	59.9	445	503	556	602	643	680	809	881

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All the stainless steel screws, nuts and bolts used by G.MONDINI S.p.A. are class 70. We only guarantee this type. Use only spares obtained from G.MONDINI S.p.A.

8.22 TABLE OF PRE-LOADS AND TORQUES FOR STAINLESS STEEL SCREWS

		precar Pre-load in kN							momer Torque in Nm							
coeff.att	Friction coeff. >	12	0.14	0.16	0.18	0.20	0.30	0.40	0.1	0.12	0.14	0.16	0.18	0.20	0.30	0.40
M4	50	1.38	1.33	1.27	1.22	1.17	1.12	0.90	0.74	0.8	0.9	1.0	1.1	1.2	1.3	1.5
	70	2.97	2.85	2.73	2.62	2.50	2.40	1.94	1.60	1.7	2.0	2.2	2.3	2.5	2.6	3.0
	80	3.97	3.80	3.64	3.49	3.34	3.20	2.59	2.13	2.3	2.6	2.9	3.1	3.3	3.5	4.1
M5	50	2.26	2.18	2.09	2.00	1.92	1.83	1.49	1.22	1.6	1.8	2.0	2.1	2.2	2.4	2.8
	70	4.85	4.66	4.47	4.29	4.11	3.93	3.19	2.62	3.4	3.8	4.2	4.6	4.9	5.1	6.1
	80	6.47	6.22	5.96	5.72	5.48	5.24	4.25	3.50	4.6	5.1	5.6	6.1	6.5	6.9	8.0
M6	50	3.20	3.07	2.94	2.82	2.70	2.59	2.09	1.73	2.8	3.1	3.5	3.7	4.0	4.1	4.8
	70	6.85	6.57	6.31	6.05	5.79	5.54	4.49	3.70	5.9	6.7	7.4	7.9	8.4	8.8	10.4
	80	9.13	8.77	8.41	8.06	7.72	7.39	5.98	4.93	8.0	9.1	9.9	10.5	11.2	11.8	13.9
M8	50	5.86	5.63	5.40	5.18	4.96	4.75	3.85	3.17	6.8	7.6	8.4	9.0	9.6	10.1	11.9
	70	12.6	12.1	11.6	11.1	10.6	10.2	8.25	6.80	14.5	16.3	17.8	19.3	20.4	21.5	25.5
	80	16.7	16.1	15.4	14.8	14.2	13.6	11.0	9.1	19.3	21.7	23.8	25.7	27.3	28.7	33.9
M10	50	9.32	8.96	8.60	8.27	7.91	7.58	6.14	5.05	13.7	15.4	16.7	18.1	19.3	20.3	24.0
	70	20.0	19.2	18.4	17.7	16.9	16.2	13.1	10.8	30	33	36	39	41	44	51
	80	26.6	25.6	24.6	23.6	22.6	21.7	17.5	14.4	39.4	44	47.8	51.6	55.3	58	69
M12	50	13.6	13.1	12.6	12.0	11.6	11.1	9.00	7.38	23.3	26.0	28.9	30.8	32.8	34.8	41.0
	70	29.1	28.1	26.9	25.8	24.8	23.7	19.2	15.8	50	56	62	66	70	74	88
	80	38.8	37.4	35.9	34.4	33.0	31.6	25.6	21.1	67	74	82	88	94	100	117
M14	50	18.7	17.9	17.3	16.5	15.8	15.2	12.3	10.1	37.1	41.7	45.6	49	52	56	66
	70	40.6	38.5	37.0	35.4	34.0	32.6	26.4	21.7	79	89	98	105	112	119	141
	80	53.3	51.3	49.3	47.3	45.3	43.3	35.2	29.0	106	119	131	140	150	159	188
M16	50	25.7	24.7	23.8	22.8	21.9	20.9	17.0	14.0	56	63	70	75	81	86	102
	70	55.0	52.9	50.9	48.9	46.8	44.9	36.4	30.0	121	136	150	162	173	183	218
	80	73.3	70.6	67.9	65.2	62.4	59.8	48.6	40.0	161	181	198	217	231	245	291
M18	50	32.2	31.0	29.8	28.5	27.3	26.2	21.2	17.5	81	91	100	108	115	122	144
	70	69.0	66.4	63.8	61.2	58.6	56.2	45.5	37.5	174	196	213	232	246	260	308
	80	92.0	88.5	85.0	81.6	78.1	74.9	60.7	50.1	232	261	285	310	329	346	411
M20	50	41.3	39.8	38.3	36.7	35.2	33.8	27.4	22.6	114	128	142	153	164	173	205
	70	88.6	85.4	82.0	78.7	75.4	72.4	58.7	48.1	244	274	303	328	351	370	439
	80	118	114	109	105	101	96.5	78.3	64.6	325	366	404	438	467	494	586
M22	50	51.6	49.8	47.9	46.0	44.1	42.3	34.3	28.3	154	174	191	208	222	234	279
	70	61.5	59.3	57.0	54.7	52.5	50.3	40.9	33.7	182	206	227	247	263	279	332
	80	148	142	137	131	126	121	98.2	80.9	437	494	545	593	613	670	797
M24	50	59.6	57.4	55.1	52.9	50.7	48.6	39.4	32.6	197	222	243	264	282	298	354
	70	70.9	68.3	65.6	63.0	60.4	57.9	47.0	38.8	234	264	290	314	336	355	421
	80	170	170	157	151	145	139	113	93.1	561	634	696	754	806	852	1010
M27	50	75.6	72.9	70.1	67.3	64.5	61.9	50.2	41.5	275	311	344	377	399	421	503
	70	90.0	86.8	83.4	80.1	76.9	73.7	59.8	49.4	328	371	410	444	475	502	599
M30	50	91.9	88.6	85.2	81.7	78.4	75.2	61.0	50.3	374	423	467	506	540	571	680
	70	104	105	101	97.3	93.3	89.5	72.6	59.9	445	503	556	602	643	680	809

Size	Class	Torque 0.40
M4	50	1.6
M4	70	3.3
M4	80	4.4
M5	50	3.2
M5	70	6.6
M5	80	8.8
M6	50	5.3
M6	70	11.3
M6	80	15.0
M8	50	12.9
M8	70	27.6
M8	80	36.8
M10	50	26.2
M10	70	56
M10	80	75
M12	50	44.6
M12	70	96
M12	80	128
M14	50	71
M14	70	152
M14	80	204
M16	50	110
M16	70	237
M16	80	316
M18	50	156
M18	70	334
M18	80	447
M20	50	223
M20	70	479
M20	80	639
M22	50	303
M22	70	361
M22	80	866
M24	50	385
M24	70	458
M24	80	1099
M27	50	548
M27	70	652
M30	50	740
M30	70	881

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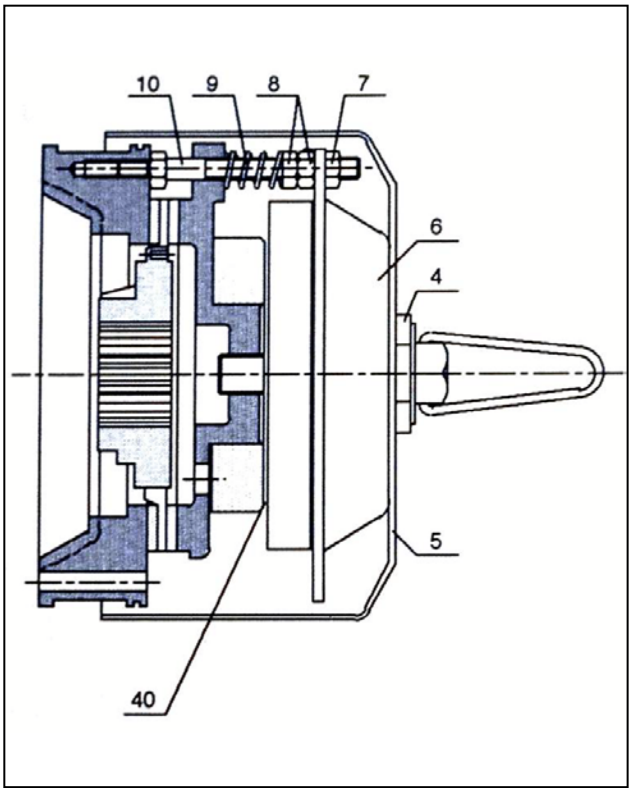
All the stainless steel screws, nuts and bolts used by G.MONDINI S.p.A. are class 70. We only guarantee this type. Use only spares obtained from G.MONDINI S.p.A.

8.23 CLUTCH-BRAKE UNIT ASSEMBLY (COEL)



- **Air gap adjustment**

Air gap 40 (i.e. the distance between the two magnetic nuclei of the electromagnet and of the mobile anchor) must amount to three tenths of a millimetre.
A first inspection after the machine's first week of work would be appropriate, with subsequent inspections after 30 days because, due to wear by the brake disk gaskets, the air gap tends to increase. It is very important that the air gap does not go over 0.5 mm.
To bring the air gap back to the size it should be, it is necessary to act on the couple of nuts (7-8) that lock on the electromagnet, in order to have it travel towards the mobile anchor. Once done, accurately check the nut torques.

Adjustment of the braking torque

The braking torque is proportional to the compression of springs 9, that can be adjusted by acting on nuts 8, i.e. unscrewing them to decrease and screwing them to increase torque. The spring compression must be uniform at all times.

Electromagnet replacement

Remove screw 4, remove the guard 5, remove the six magnet terminals, remove the three nuts 7, then slide the electromagnet 6 out of the columns 10. Rearrange the new electromagnet onto the columns, being very careful that when reconnecting the terminals, the relative colours all correspond with one another.
Lock in nuts 7-8 and check to make sure that the electromagnet works correctly.

Brake disk replacement

Remove screw 4, remove the guard 6, and remove the three nuts 7 without detaching the terminals. Remove nuts 8 and spring 9. Fix on the new brake disk.

9 DISPOSAL

9.1 DISPOSAL

The final user shall provide for the disposal of and elimination of the materials making up the component parts of the machine in accordance with EU and local applicable regulations.

If the machine or its parts are to be scrapped , take the necessary safety precautions to avoid any risks associated with the disposal of industrial equipment.

In particular, great care must be taken at the following stages:

- Removal of the machine from the working area
- Transportation and handling
- Dismantling
- Material sorting

IMPORTANT

For material sorting, recycling or disposal, reference must be made to National and regional Laws on industrial solid waste management.

If the machine is to be scrapped, the and user shall take particular care in disposing of the following materials in accordance with local regulations.

- Makrolon, PVC, Plexiglas or other material used for guards.
- Plastic used for compressed air pipes.
- Sheathed electric cables.
- Rubber belts
- Any toxic and corrosive substances.

If the materials used for normal operation, such as lubricants, grease and water condensate, are not disposed of properly in compliance with the applicable laws and regulations, there may be residual risks for:

- Environmental pollution
- Intoxication of personnel involved in disposal operations

9.2 DEMOLITION

Demolition may involve the same residual risks envisaged for installation of the system and connection to the power mains.

It is thus recommended to perform these operations with great care.

A SEALING TOOL TROLLEY CPS/R

INDEX

A.1 GENERAL DESCRIPTION

The CPS/S is a hand-operated sealing tool trolley used when fitting tools on the Evoluzione machines and removing them.

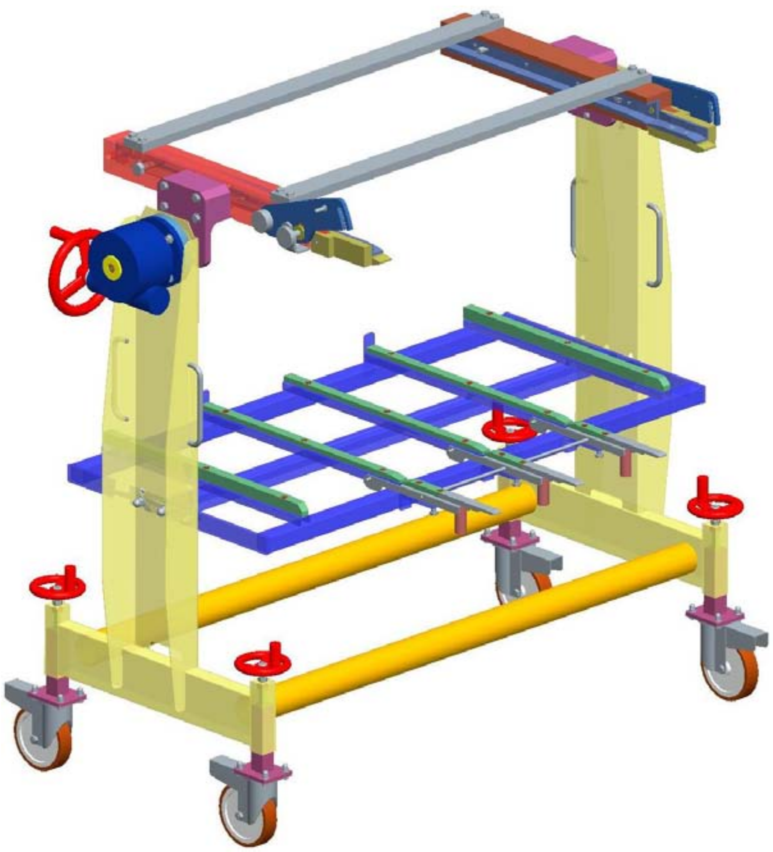
The trolley is made up of the following:

- A stainless steel frame with certain teflon parts to allow smooth running.

- A hand-operated tilting motor.

- A bottom stand.

This trolley is designed specifically for positioning and removing sealing tools from the work area. If used correctly, it ensures maximum safety when handling the sealing tools, which with certain trays can be very large and heavy.
The figure below shows the CPS/R trolley.



A.2 LIST OF HAZARDS

A.3.1) Upper section of the trolley

A.3.2) Rotation device

A.3.3) Locking pins

A.3.4) Castor wheels

A.3.5) Sealing tool

A.3 RISK ANALYSIS

A.3.1 Upper section of the trolley

This is the part of the trolley that is coupled to the machine so that the upper part of the sealing tool can be supported, removed and secured in position in complete safety.

- Solution: Make sure the trolley is secured and immobilized before releasing the tool from the pistons.

- Residual risks: None provided that the instructions are followed to the letter.

A.3.2 Rotation Device

The rotation device is connected to the upper part of the trolley. When the sealing tool has been removed from the work area, this device is used to support and remove the upper part of the tool safely and to invert it for maintenance purposes. If the sealing tool is not secured properly, it may fall off and cause the operator very serious injury.

- Solution: Make sure the trolley is secured and immobilized before releasing the tool and moving it.

- Residual risks: None provided that the instructions are followed to the letter.

A.3.3 Locking pins

The trolley is fitted with numerous mechanical locking devices allowing the sealing tool to be removed from the work area in complete safety. Before using the trolley to move sealing tools, it is very important to read this handbook so that you can identify the safety areas on the trolley.

- Solution: Follow these recommendations before attempting to move the sealing tool.

- Residual risks: None provided that the instructions are followed to the letter and don't approach the hands to the pivots.

A.3.4 Castor wheels

The trolley is on caster wheels to make it easier to move when loaded. They have mechanical brakes acting as an added safety measure while the sealing tool is being loaded. It is important to remember to apply the brakes because if the trolley moves unexpectedly during loading, the operator risks breaking a limb or other serious injury.

- Solution: Always apply the brakes before loading or unloading the trolley and put on special wear safety shoes.

- Residual risks: None provided that the instructions are followed to the letter.

A.3.5 Sealing tool

The sealing tool is used to apply a lid to each tray. This requires a very high temperature which is generated by an electric resistance. The actual temperature depends on the type of sealing.
With large trays, the sealing tool is extremely heavy and care must always be taken when moving it.

- Solution: The sealing tool stands on runners to make it easier to extract. The electric resistance supply must always be switched off before removing the sealing tool.

- Residual risks: The weight of the sealing tool must never be underestimated. IMPORTANT: Remember that after use, the sealing tool is still very hot and will cause very serious burns if touched. Insulated gloves should always be worn when changing the sealing tool.

A.4 SAFETY DEVICES

The sealing tool trolleys made by Mondini are designed and built to comply with the basic safety requirements and are fitted with a series of technical and mechanical safety devices.

All the safety devices on these machines have been thoroughly tested and fall into two categories:

1) Direct safety devices.

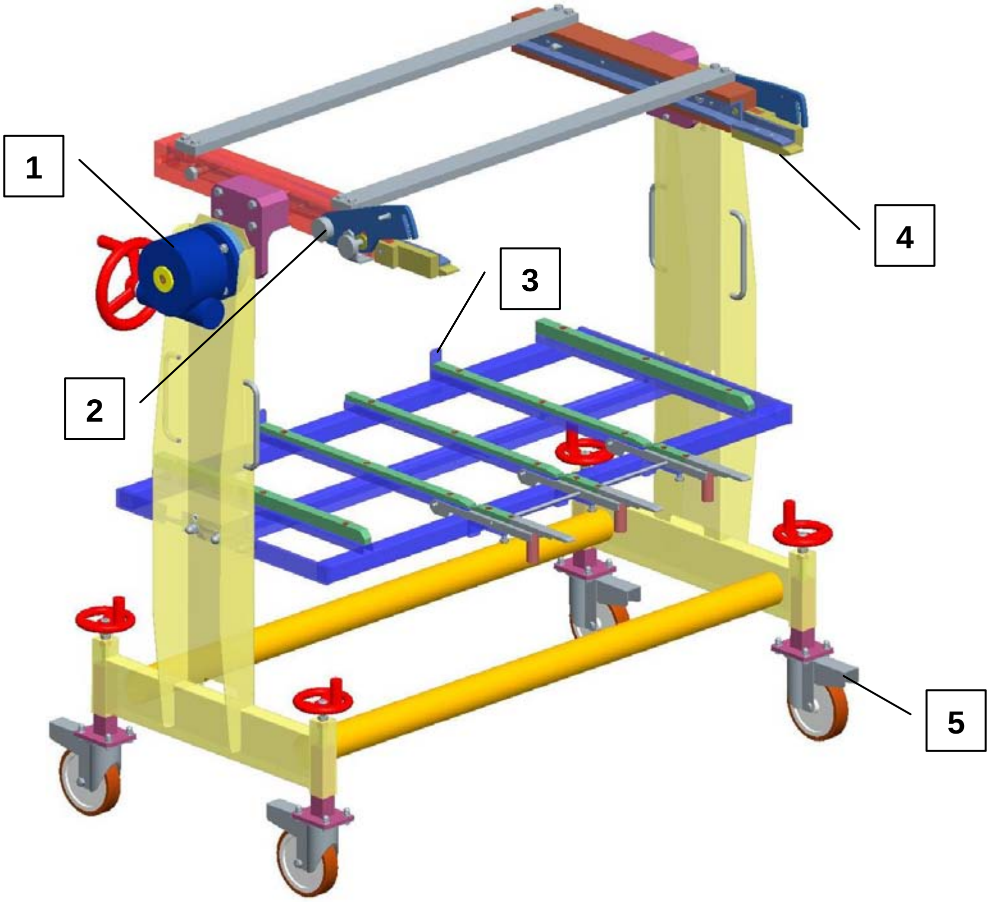
2) Indirect safety devices.

A.4.1 Direct safety devices

These are such devices as micro-switches, emergency stop buttons and master switches used specifically for industrial appliances.
As the trolley only operates mechanically, it is not equipped with any direct safety devices.

A.4.2 Indirect safety devices

These are devices which are not controlled by the operator but still prevent anyone from entering the ranger areas, even accidentally. These devices are called safety guards or panels and are made of stainless steel , metal mesh or plexiglass, depending on where they are used. The danger areas of all the mechanical parts of the machine are fitted with these safety devices. Certain areas where the operator carries out manual operation are also protected by indirect safety devices. The safety devices on the trolley are shown in the diagram and explained below:



A.4.2.1 Safety Device no. 1

This is a reducer which is connected to the device that rotates the system that connects the upper part of the sealing tool.

Once the tool has been secured, the hand-operated reducer guarantees total stability and prevents the top part of the trolley from rotating in an uncontrolled manner. If the wheel is free (not locked), the safety device keeps the trolley in position.

A.4.2.2 Safety Device no. 2

This safety system comprises locking knobs, which create a mechanical block between the front and the rear part of the upper tool holder plate.
This prevents the tool from slipping in the telescopic guides when it is in the horizontal or vertical position.

There is also a mechanical clip that must engage the upper plate of the machine to lock the trolley onto the machine while the upper part of the sealing tool is being removed.

A.4.2.3 Safety Device no. 3

This is a rear stop built into the trolley that holds the lower part of the sealing tool in position during removal.

A.4.2.4 Safety Device no. 4

These are telescopic guides used to remove the upper part of the sealing tool from the machine. They fit into slots in the support to ensure total safety when removing the upper part of the sealing tool from the machine.

A.4.2.5 Safety Device no. 5

This is a brake on the castors of the trolley. When the trolley has been secured to the upper plate of the machine, the brake must always be applied to prevent the weight of the sealing tool or wrong engaging of safety device no. 2 from causing the sealing tool to fall off during removal or from causing serious injury.

These safety devices must **never** be removed, changed or modified in any way.

Modification to any of the safety devices could also cause a malfunction and bring the entire system to a halt.

IMPORTANT NOTE

The system must not be modified in any way and the safety guards must not be removed or de-activated without first informing the Manufacturer. If these instructions are not complied with, the Manufacturer DECLINES ALL LIABILITY for accidents or malfunctions that might occur.

The Manufacturer cannot be held liable for :

A) THE USER'S SAFETY

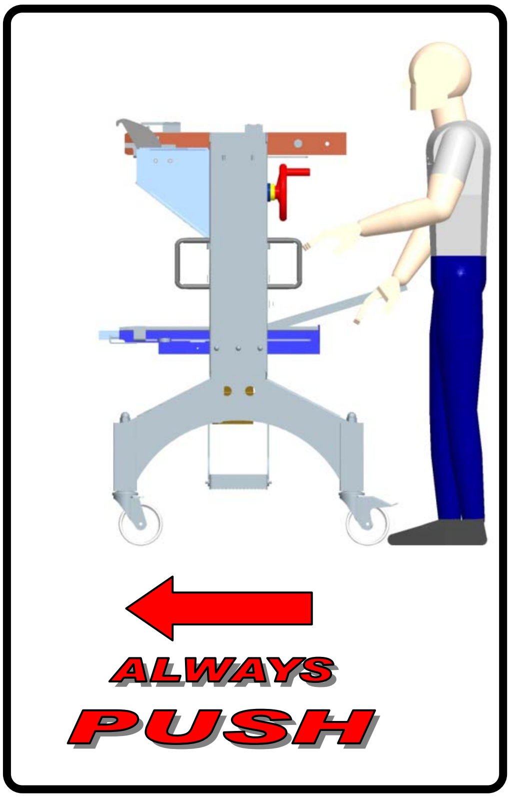
B) EFFICIENT OPERATION of the machine supplied.

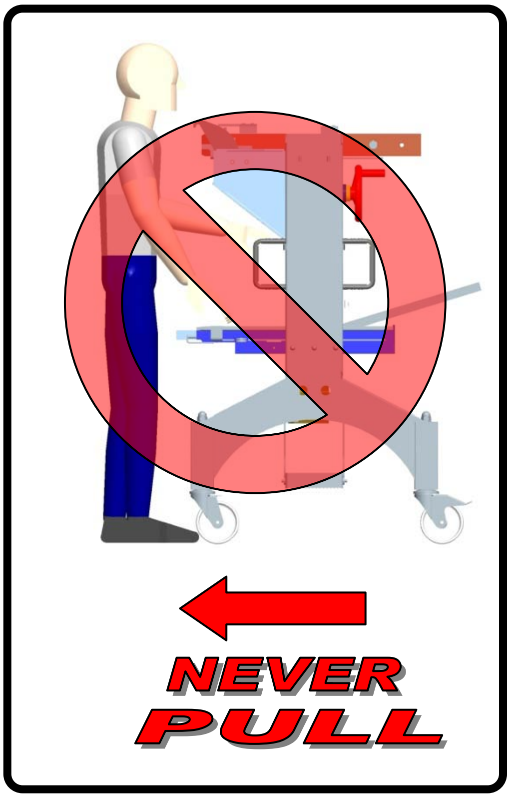
- IMPORTANT: If the machine is to be connected to an existing one, unless agreed otherwise, the user must provide adequate protection in the area of connection, Mondini S.p.A. declines all liability if failure to do so causes damage or injury.

WARNING!

When moving or shifting the tool carrier trolley, the operator must strictly PUSH it and NEVER pull it. This to prevent any possible crushing hazards occurring to the operator, due to the trolley possible capsizing and/or bucking.

WHEN MOVING OR SHIFTING THE TROLLEY, ALWAYS PUSH IT. NEVER PULL IT!





A.5 OPERATING INSTRUCTIONS

Like everything made by MONDINI, the CPS/R trolley is delivered fully tested, so only a few simple operations need to be carried out before it can be put into use.

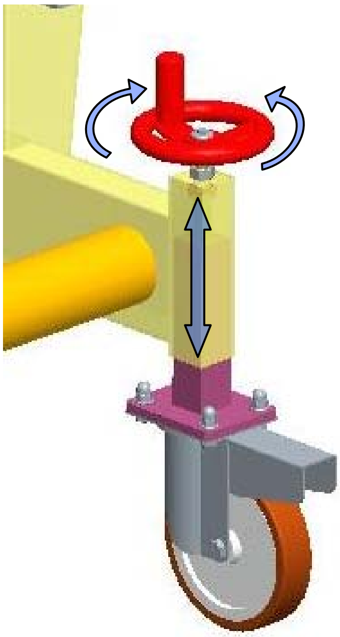
First of all unwrap the trolley. Using a spirit level check that the trolley is perfectly horizontal. If necessary, adjust the height controls on the castors.

This trolley is designed specifically for positioning and removing sealing tools from the work area in complete safety. Full instructions on how to remove and insert a sealing tool are provided below.

<table>
<tr>
<td>IMPORTANT!</td>
<td>Before beginning the sealing tool extraction/insertion procedure it is necessary to check that:

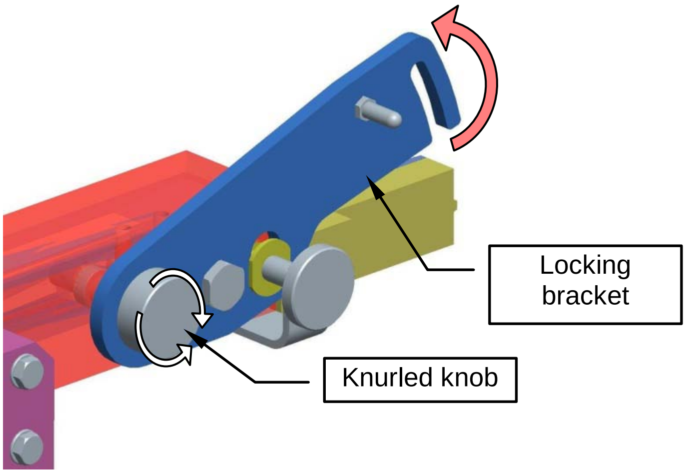
all the four wheels of the trolley are contemporarily on the floor;
the trolley slide guides are to the same height of the sealing machine slide guides;
the trolley has a correct horizontal position.

</td>
</tr>
</table>



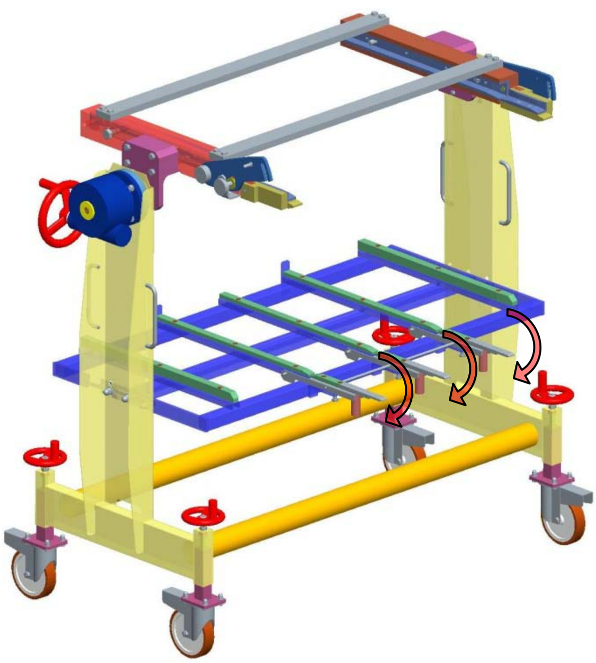
A.5.1 How to remove a tool

1. Loosen the knurled knob slightly to free the locking brackets. Raise the brackets and tighten the knob to lock them in position.

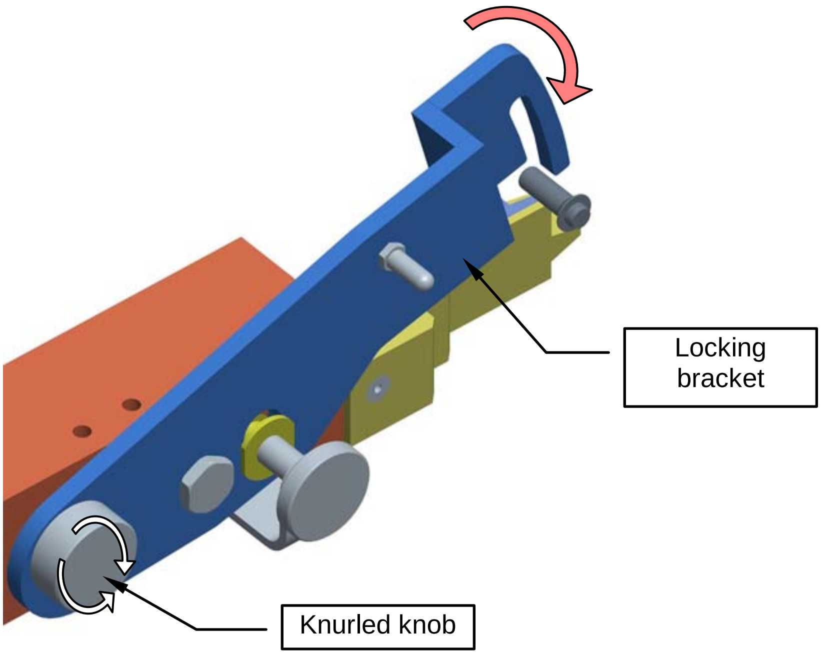


2. Switch off the machine and prepare it for tool replacement, as instructed in the operator manual.

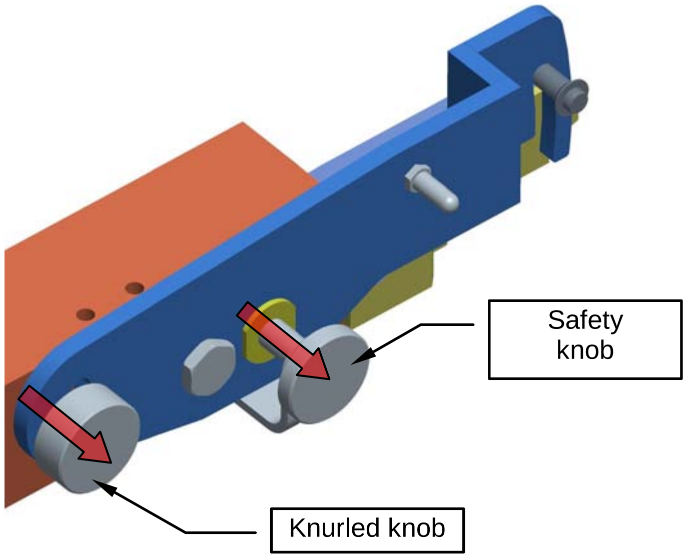
3. Lower the lower tool extraction guides until the part underneath rests correctly on the lower plate of the machine. The guides are correctly positioned when they are horizontally aligned with the direction of extraction of the lower tool. If necessary, adjust the position on the lower knobs.



4. Position the trolley next to the sealing machine. Loosen the knurled knob and couple the trolley, engaging the slots in the brackets.



5. Make sure the brackets are fully engaged. Unscrew the knurled knob completely until the bracket is right inside the guides.



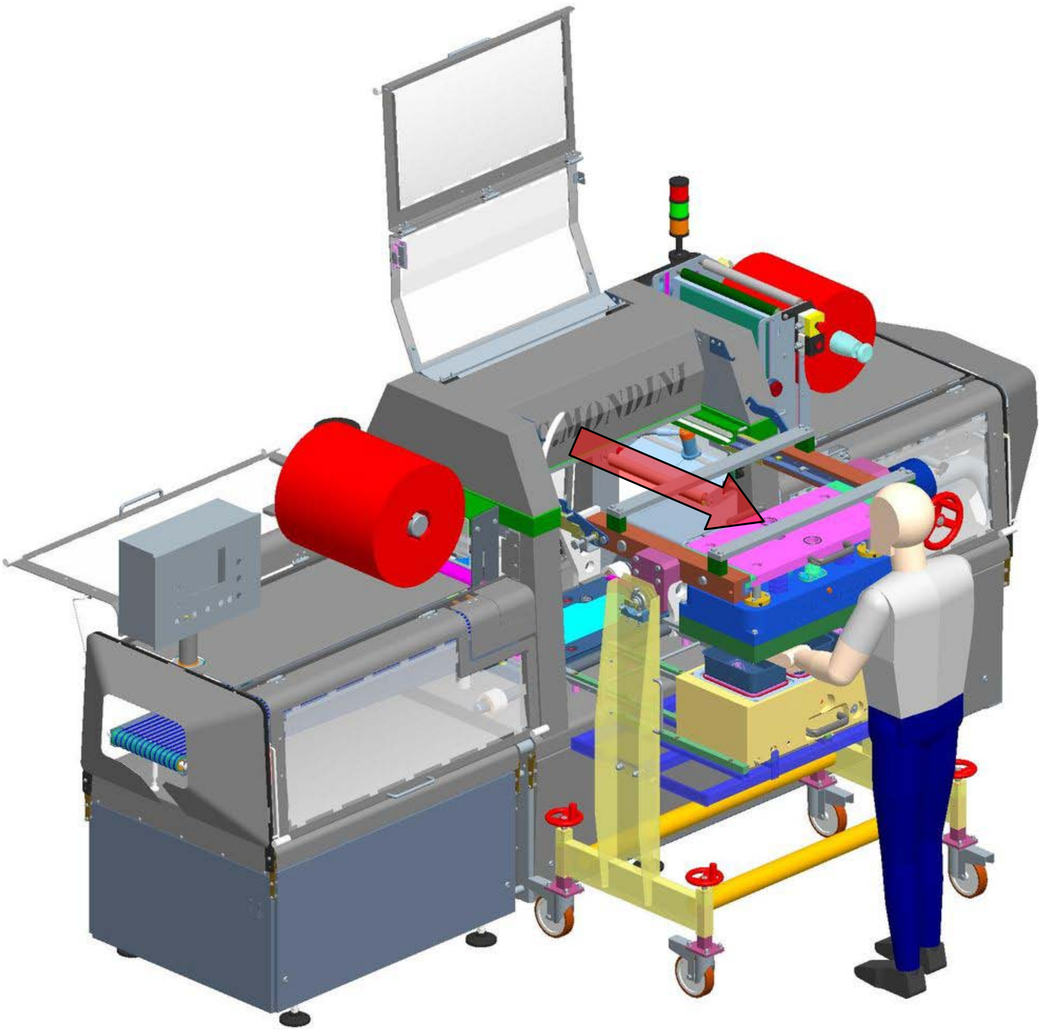
6. Apply the trolley brake and pull out the safety knob.

7. Now you can remove the tool. **Remember the machine must be switched off and the work cycle completed.**

   - Remove any residual film from inside the tool by hand.

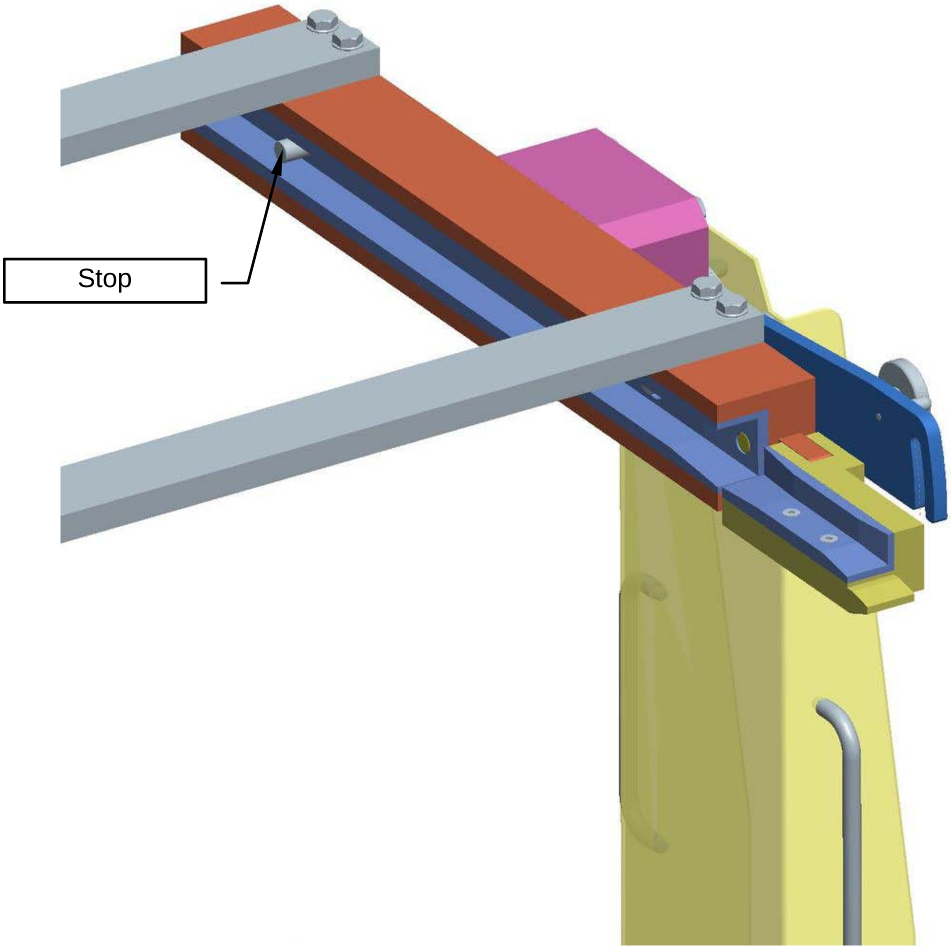
   - Pull out the electric connector of the heating resistance. IMPORTANT**!** If the tool has just been used, it will still be very hot, so handle with great care.

   - Operate the automatic tool release selector, which is pneumatically controlled. **IMPORTANT!** The selector uncouples the upper and lower parts of the tool, so proceed with extreme caution.



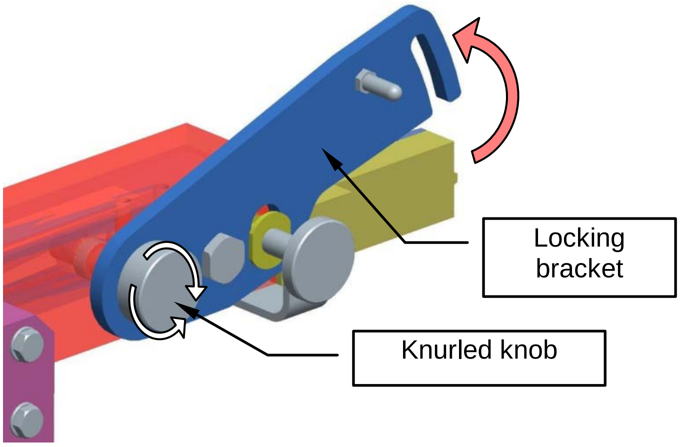
8. Take hold of the lower tool and pull it forcefully up against the stops on the guides.

9. Take hold of the upper tool and pull it forcefully up against the stops on the guides.



10. Tighten the knurled knob to prevent the upper tool from moving during transportation.

11. Raise the lower guides until they are perpendicular to the direction of movement. Lock them in position by inserting the safety pins in the holes.



12. Raise the upper plate anchoring brackets and tighten the knurled knob.

IMPORTANT! When the trolley is uncoupled from the machine, the operator is in complete safety only if the two knurled knobs have been tightened. If this is not done, the manufacturer disclaims all liability for any injuries occurring.

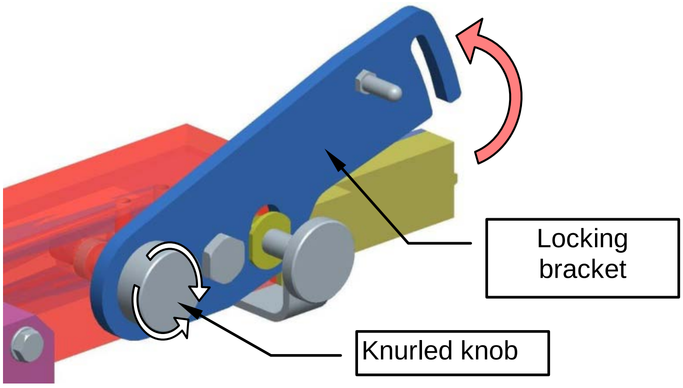
13. Release the trolley brake and take the tool back to its storage shelf.

A.5.2 How to insert a tool

The procedure below describes how to insert a new tool. It is very similar to the removal instructions.

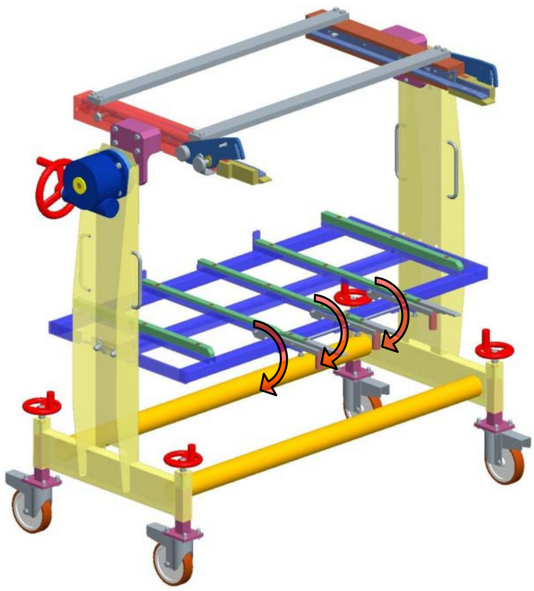
1. Push the trolley holding the new tool up against the sealing machine on the tool insertion side.

2. Loosen the knurled knob slightly to free the locking brackets. Raise the brackets and tighten the knob to lock them in position.

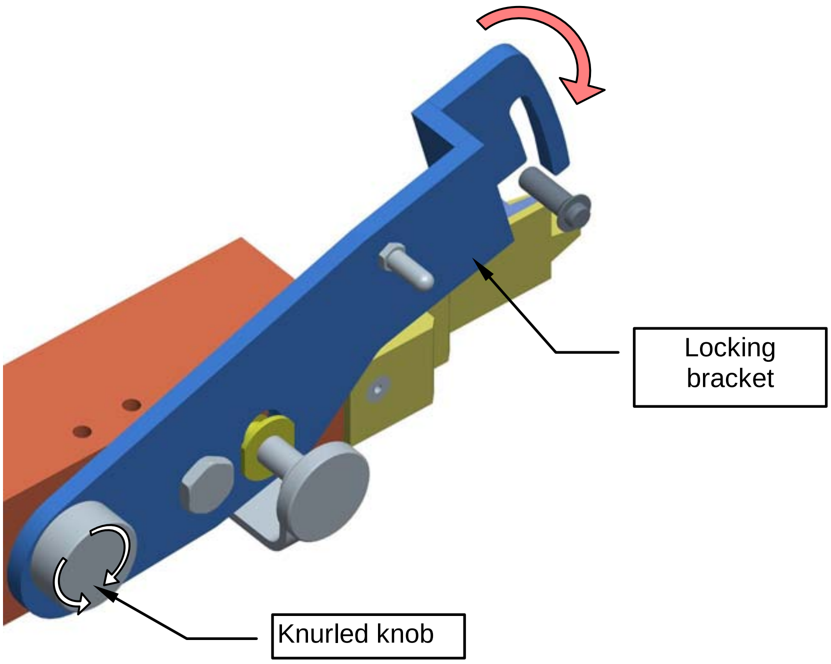


3. Switch off the machine and prepare it for tool replacement, as instructed in the operator manual.

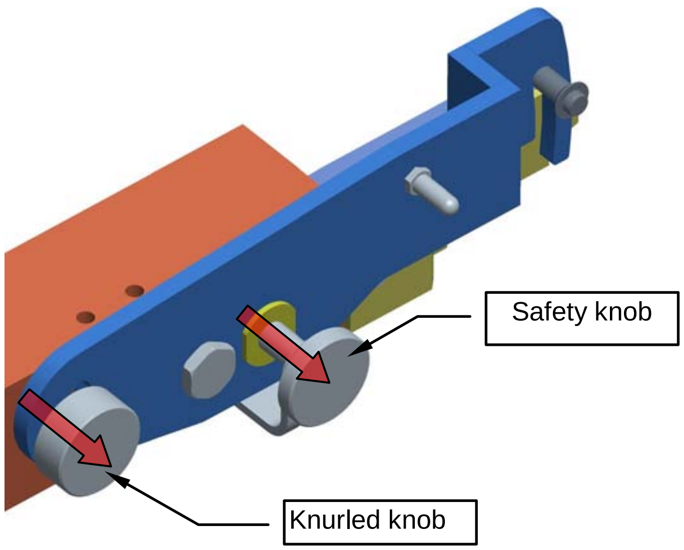
4. Lower the lower tool extraction guides until the part underneath rests correctly on the lower plate of the machine.



5. Position the trolley next to the sealing machine. Loosen the knurled knob and couple the trolley, engaging the slots in the brackets.



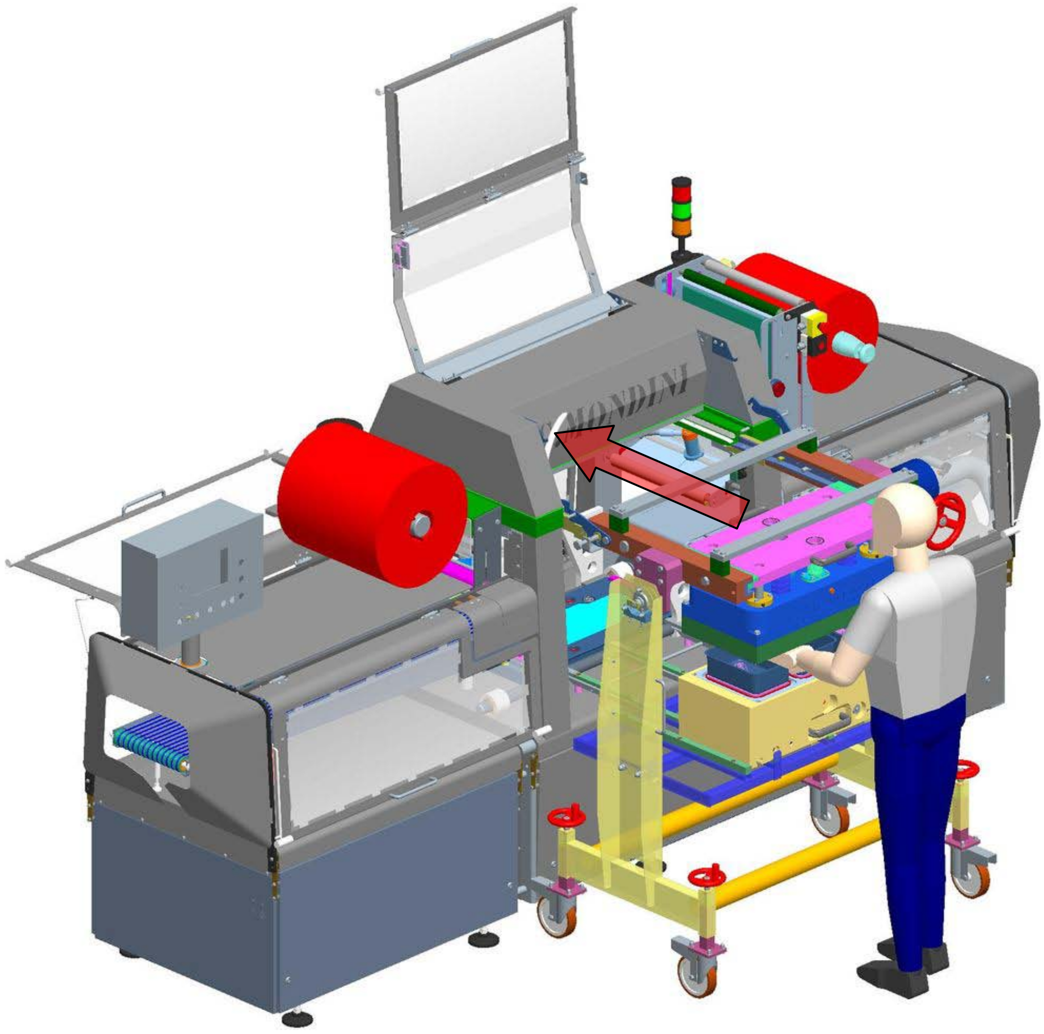
6. Make sure the brackets are fully engaged. Unscrew the knurled knob completely until the bracket is right inside the guides.



7. Apply the trolley brake and pull out the safety knob.

8. Take hold of the upper tool and push it forcefully into the forward position on the machine supporting guides. The correct position is marked by the stops.

9. Take hold of the lower tool and push it forcefully into the forward position on the machine supporting guides. The correct position is marked by the stops.



10. Operate the pneumatic selector to couple the tool with the sealing machine.

11. Insert the electric connector for tool heating.

12. Release the trolley from the machine and resume production as instructed in the operator handbook.

A.6 MALFUNCTIONS - SOLUTIONS

The sealing tool trolley will only malfunction if parts are wrongly adjusted or not kept in proper working order.

Malfunction	Possible Solutions
The safety pins are difficult to pull out	Check for dirt or encrustation on the outer surface of the pin. Remove any dirt and grease the safety pins.
The lower guides are not aligned	Check the position of the height adjusting pins below the guides and adjust as necessary. Check for dirt or other material accumulated under the guides. Remove any dirt and grease the guides.
The castors stick or do not move at all	Check for obstructions and clean thoroughly. Check that the brake is not engaged. Clean and lubricate the castors to allow the bearings to move freely.
The sealing tool does not slide on the guides	Check for obstructions and clean thoroughly. Check that the Teflon pads in the guides are not worn or out of position. Replace any worn pads. Make sure the pins have been removed correctly.

A.7 MAINTENANCE - CLEANING

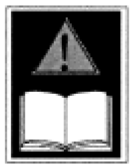
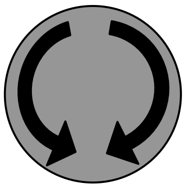
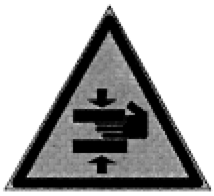
A.7.1 Maintenance

The trolley does not require any particular maintenance. The only point to bear in mind is to handle the guides and safety pins with care. Never hit them with a hammer or similar tool if they do not work.

Lubricate the bearings and moving parts periodically. Keep clean all the safety devices used when replacing and handling the sealing tools.

A.7.2 Cleaning

All traces of product or dirt deposited on the trolley when the sealing tools are being handled must be removed. The entire trolley is made of stainless steel and can be washed with water and detergent.

	Wear safety shoes.
	See the use and maintenance handbook.
	This symbol indicates the direction of rotation of the part .
	**Warning!** Danger of limbs being crushed.
	Constant hazard pay attention when handling.

B TOOL RACK C/PS

GENERAL INDEX

B.1 GENERAL DESCRIPTION

The tool rack with pre-heating cycle type C/PS is a manual device designed and built for the storage and pre-heating of the tools of the closing machines.

The tool rack is composed of:

- Stainless steel structure.
- Electric panel.
- Electrical connections for heating the tool resistance.
- "Thermocontrol" device.

The tool rack has the function of hosting the sealing tools that are not temporarily used in the production phase.
The correct use of the tool rack, associated with the correct use of the tool trolley CPS/R, ensures maximum safety in handling operations of the sealing tools both in the loading and in the removal phase.

The tool rack looks like this:

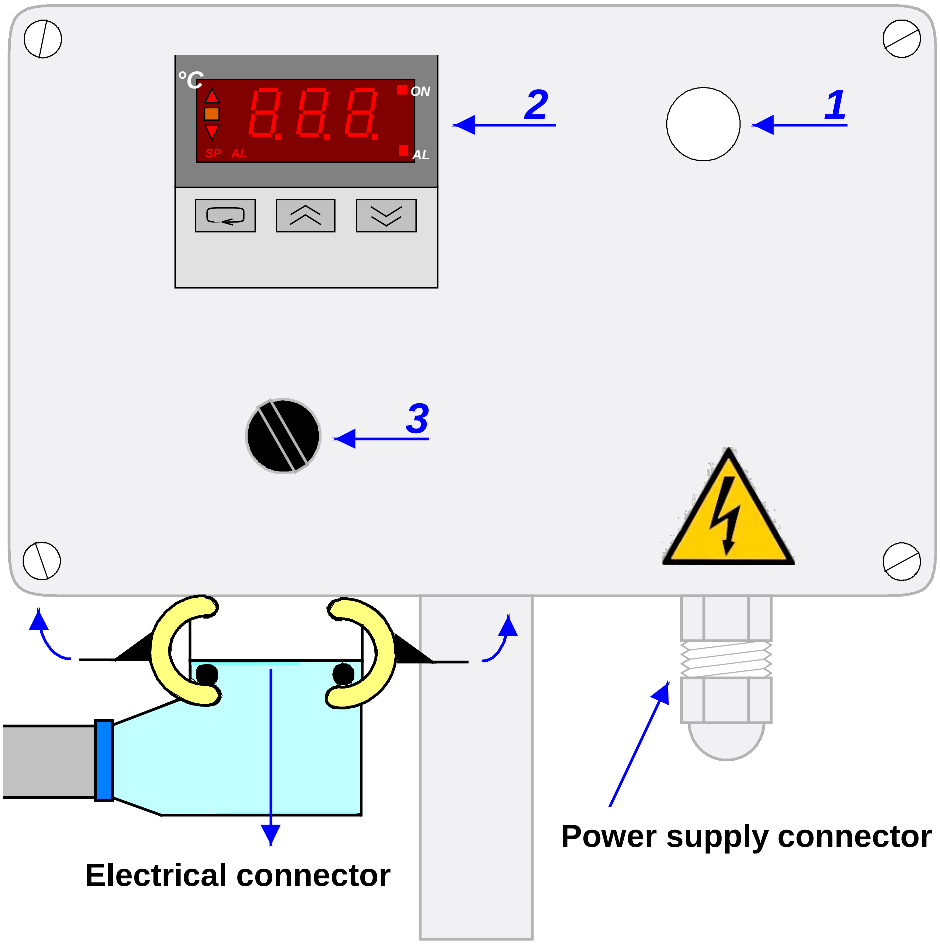
B.2 INSTRUCTIONS FOR USE

The tool rack C/PS, like every product of G.MONDINI S.p.a., is already tested and tested on delivery, so few and simple operations are required for immediate use.

Before considering the technical aspects, after unpacking, arrange the machine in optimal working conditions.

This device has the function of hosting the sealing tools related to those available tray formats not used in production and also has the function of pre-heating the tool before it is mounted on the closing machine.

The tool rack C/PS offers the possibility to heat the sealing tool if you need to change format during production.
The tool rack C/PS in fact has the same electrical connector as the closing machine that has the function of heating the electrical resistance located in the sealing tool and it is also equipped with a "thermocontrol" device that allows the operator to set the desired tool temperature during the production.



1)	**White lamp of "CURRENT INSERTED"**	When this lamp is on, it indicates that the machine is under electrical voltage.
2)	**"THERMOCONTROL" device**	This device controls and regulates the internal heat resistance of the sealing tool of the closing machine.
3)	**Black selector with two positions of "HEATING 0/1"**	This selector when in position "1", activates the resistance plate inside the sealing tool, in position "0" it deactivates it.

To pre-heat a sealing tool it is necessary to:

1) make sure that the rack is connected to an electric current source through the connector located in the lower right part of the electrical panel;

2) place the sealing tool to be pre-heated on the tool rack;

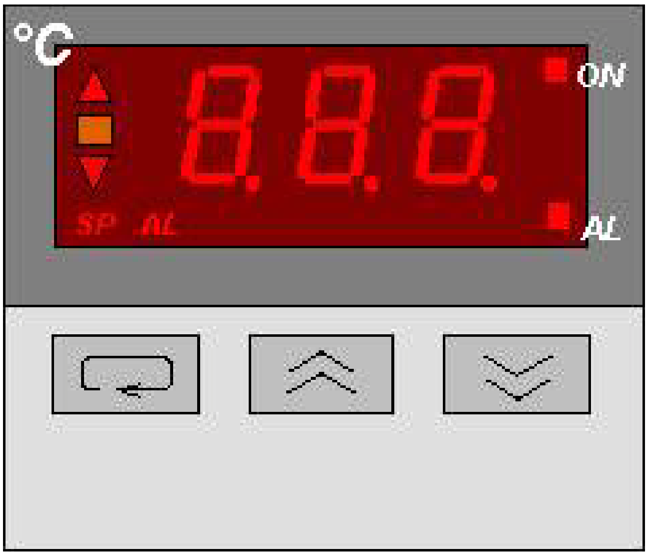
3) connect the electrical connector located in the lower left part of the electrical panel to the sealing tool;

4) turn the "HEATING 0/1" selector to position "1";

5) adjust on the display of the "thermocontrol" device the temperature to reach to pre-heat the sealing tool.

B.3 "THERMOCONTROL" DEVICE

B.3.1 Setting and operation of the "thermocontrol" device.



The thermocontrol device allows the operator to adjust the temperature of the heating resistance .

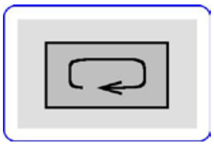
This easy-to-use panel has a visual part at the top with a display, and interactive or manual part with the function buttons at the bottom.

The display at the top enables the operator to monitor the process and check the various indications.

The bottom part contains the buttons used to modify the temperature control device parameters.

If no buttons are pressed, the display shows the current temperature of the controlled element.

B.3.2 Description of the buttons



- To modify parameters - called SP (Set Point).

- To increment the numerical value - called Arrow Up.

- To decrease the numerical value - called Arrow Down.

B.3.3 Setting the working temperature

The display shows the current temperature of the thermal resistance. To modify the working temperature, proceed as follows:

a) Press button SP once. The set temperature is displayed and SP lights up. Use the up and down arrows to enter the required value, then press SP again to confirm. The new value of the controlled element is now displayed.

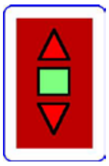
b) When SP is pressed twice, the admissible deviation (usually +/- 10°C) from the set temperature is displayed after the temperature value, and AL (Alarm) lights up. The numerical value of this deviation appears separately without a + or - sign, which indicates that it is valid for positive or negative deviations.

Example:

If 10 is displayed, it means the set temperature can vary 10°C more or less without the temperature control device intervening for an emergency machine stop.

If the deviation does not comply with the working requirements, the value can be altered by pressing the up and down arrows as indicated in point a).

c) When SP is pressed a third time, the current temperature of the controlled element is displayed again.



After the temperature has been altered by hand (see point a), one of the arrows at the side may be highlighted to indicate that the ideal working temperature has not yet been reached.

- If the illuminated arrow is the one pointing downwards (down arrow), this means that the temperature of the controlled element is lower than the set temperature.

- If the illuminated arrow is the upward arrow (arrow up), it means that the temperature of the controlled element is higher than the set temperature.

In both cases you must wait for the red arrow to go off and the green box between the arrows to light up.
After a machine setting, an alarm may be generated, in which case the machine stops.
This occurs when the correction made is outside the tolerance range set in function AL.

Example:

If after a correction the temperature is changed from 120° to 135°C, the thermoresistance must come on to increase the temperature of the controlled element by 15°C.In this case an alarm is generated and the machine stops because the usual deviation (see point b) is 10°, which means this is an unacceptable condition.

When the green light comes on, it means that the working temperature has been reached and production can start.
During operation, the red light on one of the arrows may come on because the temperature tends to drop with successive cycles.
This is not serious in itself, but if it is not borne in mind, a machine alarm will be generated and the machine will stop for the above reasons.

The tolerance range (point b) can be altered to solve this problem. If the problem persists, refer to table 6) PROBLEMS AND POSSIBLE SOLUTIONS in the machine instruction handbook. Lastly, there are two red lights on the display called ON and AL.

- ON = indicates that the thermoresistance is in operation.

- AL = indicates that there is a possible alarm condition due to a significant temperature variation.

When AL is on, it is good practice to wait a few moments for the temperature to return to the ideal value.

NOTE: *When AL is active, ON is usually active as well.*